nRF52832 Product Specification v1.4

Key features

- 2.4 GHz transceiver
  - -96 dBm sensitivity in *Bluetooth*® low energy mode
  - Supported data rates: 1 Mbps, 2 Mbps *Bluetooth*® low energy mode
  - -20 to +4 dBm TX power, configurable in 4 dB steps
  - On-chip balun (single-ended RF)
  - 5.3 mA peak current in TX (0 dBm)
  - 5.4 mA peak current in RX
  - RSSI (1 dB resolution)
- ARM® Cortex®-M4 32-bit processor with FPU, 64 MHz
  - 215 EEMBC CoreMark® score running from flash memory
  - 58 µA/MHz running from flash memory
  - 51.6 µA/MHz running from RAM
  - Data watchpoint and trace (DWT), embedded trace macrocell (ETM), and instrumentation trace macrocell (ITM)
  - Serial wire debug (SWD)
  - Trace port
- Flexible power management
  - 1.7 V–3.6 V supply voltage range
  - Fully automatic LDO and DC/DC regulator system
  - Fast wake-up using 64 MHz internal oscillator
  - 0.3 µA at 3 V in System OFF mode
  - 0.7 µA at 3 V in System OFF mode with full 64 kB RAM retention
  - 1.9 µA at 3 V in System ON mode, no RAM retention, wake on RTC
- Memory
  - 512 kB flash/64 kB RAM
  - 256 kB flash/32 kB RAM
- Nordic SoftDevice ready
- Support for concurrent multi-protocol
- Type 2 near field communication (NFC-A) tag with wakeup-on-field and touch-to-pair capabilities
- 12-bit, 200 ksps ADC - 8 configurable channels with programmable gain
- 64 level comparator
- 15 level low power comparator with wakeup from System OFF mode
- Temperature sensor
- 32 general purpose I/O pins
- 3x 4-channel pulse width modulator (PWM) unit with EasyDMA
- Digital microphone interface (PDM)
- 5x 32-bit timer with counter mode
- Up to 3x SPI master/slave with EasyDMA
- Up to 2x I2C compatible 2-wire master/slave
- I2S with EasyDMA
- UART (CTS/RTS) with EasyDMA
- Programmable peripheral interconnect (PPI)
- Quadrature decoder (QDEC)
- AES HW encryption with EasyDMA
- Autonomous peripheral operation without CPU intervention using PPI and EasyDMA
- 3x real-time counter (RTC)
- Single crystal operation
- Package variants
  - QFN48 package, 6 × 6 mm
  - WLCSP package, 3.0 × 3.2 mm

Applications

- Internet of Things (IoT)
  - Home automation
  - Sensor networks
  - Building automation
  - Industrial
  - Retail
- Personal area networks
  - Health/fitness sensor and monitor devices
  - Medical devices
  - Key fobs and wrist watches
- Interactive entertainment devices
  - Remote controls
  - Gaming controllers
- Beacons
- A4WP wireless chargers and devices
- Remote control toys
- Computer peripherals and I/O devices
  - Mouse
  - Keyboard
  - Multi-touch trackpad
  - Gaming

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2017-10-10

Contents

1 Revision history 9
2 About this document 10
2.1 Document naming and status 10
2.2 Peripheral naming and abbreviations 10
2.3 Register tables 10
2.4 Registers 11
3 Block diagram 12
4 Pin assignments 13
4.1 QFN48 pin assignments 13
4.2 WLCSP ball assignments 15
4.3 GPIO usage restrictions 17
5 Absolute maximum ratings 19
6 Recommended operating conditions 20
6.1 WLCSP light sensitivity 20
7 CPU 21
7.1 Floating point interrupt 21
7.2 Electrical specification 21
7.3 CPU and support module configuration 22
8 Memory 23
8.1 RAM - Random access memory 23
8.2 Flash - Non-volatile memory 24
8.3 Memory map 24
8.4 Instantiation 24
9 AHB multilayer 26
9.1 AHB multilayer priorities 26
10 EasyDMA 27
10.1 EasyDMA array list 28
11 NVMC — Non-volatile memory controller 29
11.1 Writing to Flash 29
11.2 Erasing a page in Flash 29
11.3 Writing to user information configuration registers (UICR) 29
11.4 Erasing user information configuration registers (UICR) 29
11.5 Erase all 30
11.6 Cache 30
11.7 Registers 30
11.8 Electrical specification 33
12 BPROT — Block protection 34
12.1 Registers 34
13 FICR — Factory information configuration registers 43
13.1 Registers 43
14 UICR — User information configuration registers 54
14.1 Registers 54
15 Peripheral interface 68
15.1 Peripheral ID 68
15.2 Peripherals with shared ID 68
15.3 Peripheral registers 69
15.4 Bit set and clear 69
15.5 Tasks 69
15.6 Events 70

15.7 Shortcuts ... 70
15.8 Interrupts ... 70
16 Debug and trace ... 72
16.1 DAP - Debug Access Port ... 72
16.2 CTRL-AP - Control Access Port ... 73
16.3 Debug interface mode ... 74
16.4 Real-time debug ... 74
16.5 Trace ... 75
17 Power and clock management ... 76
17.1 Current consumption scenarios ... 76
18 POWER — Power supply ... 78
18.1 Regulators ... 78
18.2 System OFF mode ... 79
18.3 System ON mode ... 80
18.4 Power supply supervisor ... 80
18.5 RAM sections ... 82
18.6 Reset ... 82
18.7 Retained registers ... 83
18.8 Reset behavior ... 83
18.9 Registers ... 83
18.10 Electrical specification ... 99
19 CLOCK — Clock control ... 101
19.1 HFCLK clock controller ... 101
19.2 LFCLK clock controller ... 103
19.3 Registers ... 105
19.4 Electrical specification ... 109
20 GPIO — General purpose input/output ... 111
20.1 Pin configuration ... 111
20.2 GPIO located near the RADIO ... 113
20.3 Registers ... 113
20.4 Electrical specification ... 154
21 GPIOTE — GPIO tasks and events ... 157
21.1 Pin events and tasks ... 157
21.2 Port event ... 158
21.3 Tasks and events pin configuration ... 158
21.4 Registers ... 158
21.5 Electrical specification ... 167
22 PPI — Programmable peripheral interconnect ... 168
22.1 Pre-programmed channels ... 169
22.2 Registers ... 169
23 RADIO — 2.4 GHz Radio ... 205
23.1 EasyDMA ... 205
23.2 Packet configuration ... 206
23.3 Maximum packet length ... 207
23.4 Address configuration ... 207
23.5 Data whitening ... 207
23.6 CRC ... 208
23.7 Radio states ... 209
23.8 Transmit sequence ... 209
23.9 Receive sequence ... 211
23.10 Received Signal Strength Indicator (RSSI) ... 212
23.11 Interframe spacing ... 212
23.12 Device address match ... 213
23.13 Bit counter ... 213
23.14 Registers ... 214
23.15 Electrical specification ... 230
24 TIMER — Timer/counter ... 234

24.1 Capture....235
24.2 Compare....235
24.3 Task delays....235
24.4 Task priority....235
24.5 Registers....235
24.6 Electrical specification....241
25 RTC — Real-time counter....242
25.1 Clock source....242
25.2 Resolution versus overflow and the PRESCALER....242
25.3 COUNTER register....243
25.4 Overflow features....243
25.5 TICK event....243
25.6 Event control feature....244
25.7 Compare feature....244
25.8 TASK and EVENT jitter/delay....246
25.9 Reading the COUNTER register....248
25.10 Registers....248
25.11 Electrical specification....254
26 RNG — Random number generator....255
26.1 Bias correction....255
26.2 Speed....255
26.3 Registers....255
26.4 Electrical specification....257
27 TEMP — Temperature sensor....258
27.1 Registers....258
27.2 Electrical specification....263
28 ECB — AES electronic codebook mode encryption....264
28.1 Shared resources....264
28.2 EasyDMA....264
28.3 ECB data structure....264
28.4 Registers....265
28.5 Electrical specification....266
29 CCM — AES CCM mode encryption....267
29.1 Shared resources....268
29.2 Encryption....268
29.3 Decryption....268
29.4 AES CCM and RADIO concurrent operation....269
29.5 Encrypting packets on-the-fly in radio transmit mode....269
29.6 Decrypting packets on-the-fly in radio receive mode....270
29.7 CCM data structure....271
29.8 EasyDMA and ERROR event....272
29.9 Registers....272
30 AAR — Accelerated address resolver....276
30.1 Shared resources....276
30.2 EasyDMA....276
30.3 Resolving a resolvable address....276
30.4 Use case example for chaining RADIO packet reception with address resolution using AAR....277
30.5 IRK data structure....277
30.6 Registers....278
30.7 Electrical specification....280
31 SPIM — Serial peripheral interface master with EasyDMA....281
31.1 Shared resources....281
31.2 EasyDMA....282
31.3 SPI master transaction sequence....283
31.4 Low power....284
31.5 Master mode pin configuration....284

31.6 Registers ... 285
31.7 Electrical specification ... 290
32 SPIS — Serial peripheral interface slave with EasyDMA ... 292
32.1 Shared resources ... 292
32.2 EasyDMA ... 292
32.3 SPI slave operation ... 293
32.4 Pin configuration ... 294
32.5 Registers ... 295
32.6 Electrical specification ... 303
33 TWIM — I²C compatible two-wire interface master with EasyDMA ... 305
33.1 Shared resources ... 306
33.2 EasyDMA ... 306
33.3 Master write sequence ... 307
33.4 Master read sequence ... 308
33.5 Master repeated start sequence ... 309
33.6 Low power ... 310
33.7 Master mode pin configuration ... 310
33.8 Registers ... 310
33.9 Electrical specification ... 317
34 TWIS — I²C compatible two-wire interface slave with EasyDMA ... 319
34.1 Shared resources ... 321
34.2 EasyDMA ... 321
34.3 TWI slave responding to a read command ... 321
34.4 TWI slave responding to a write command ... 322
34.5 Master repeated start sequence ... 323
34.6 Terminating an ongoing TWI transaction ... 324
34.7 Low power ... 324
34.8 Slave mode pin configuration ... 324
34.9 Registers ... 325
34.10 Electrical specification ... 331
35 UARTE — Universal asynchronous receiver/transmitter with EasyDMA ... 333
35.1 Shared resources ... 333
35.2 EasyDMA ... 333
35.3 Transmission ... 334
35.4 Reception ... 334
35.5 Error conditions ... 336
35.6 Using the UARTE without flow control ... 336
35.7 Parity configuration ... 336
35.8 Low power ... 336
35.9 Pin configuration ... 337
35.10 Registers ... 337
35.11 Electrical specification ... 345
36 QDEC — Quadrature decoder ... 347
36.1 Sampling and decoding ... 347
36.2 LED output ... 348
36.3 Debounce filters ... 348
36.4 Accumulators ... 349
36.5 Output/input pins ... 349
36.6 Pin configuration ... 349
36.7 Registers ... 350
36.8 Electrical specification ... 356
37 SAADC — Successive approximation analog-to-digital converter ... 357
37.1 Shared resources ... 357
37.2 Overview ... 357
37.3 Digital output ... 358
37.4 Analog inputs and channels ... 359
37.5 Operation modes ... 359

37.6 EasyDMA....361
37.7 Resistor ladder....362
37.8 Reference....363
37.9 Acquisition time....363
37.10 Limits event monitoring....364
37.11 Registers....365
37.12 Electrical specification....389
37.13 Performance factors....391
38 COMP — Comparator....392
38.1 Differential mode....393
38.2 Single-ended mode....394
38.3 Registers....396
38.4 Electrical specification....401
39 LPCOMP — Low power comparator....402
39.1 Shared resources....403
39.2 Pin configuration....403
39.3 Registers....404
39.4 Electrical specification....408
40 WDT — Watchdog timer....409
40.1 Reload criteria....409
40.2 Temporarily pausing the watchdog....409
40.3 Watchdog reset....409
40.4 Registers....410
40.5 Electrical specification....414
41 SWI — Software interrupts....415
41.1 Registers....415
42 NFCT — Near field communication tag....416
42.1 Overview....416
42.2 Pin configuration....418
42.3 EasyDMA....418
42.4 Collision resolution....419
42.5 Frame timing controller....420
42.6 Frame assembler....421
42.7 Frame disassembler....422
42.8 Antenna interface....423
42.9 NFCT antenna recommendations....423
42.10 Battery protection....423
42.11 References....424
42.12 Registers....424
42.13 Electrical specification....435
43 PDM — Pulse density modulation interface....436
43.1 Master clock generator....436
43.2 Module operation....436
43.3 Decimation filter....437
43.4 EasyDMA....437
43.5 Hardware example....438
43.6 Pin configuration....438
43.7 Registers....439
43.8 Electrical specification....443
44 I²S — Inter-IC sound interface....445
44.1 Mode....445
44.2 Transmitting and receiving....445
44.3 Left right clock (LRCK)....446
44.4 Serial clock (SCK)....446
44.5 Master clock (MCK)....447
44.6 Width, alignment and format....447
44.7 EasyDMA....449

44.8 Module operation ... 451
44.9 Pin configuration ... 452
44.10 Registers ... 453
44.11 Electrical specification ... 460
45 MWU — Memory watch unit ... 461
45.1 Registers ... 461
46 EGU — Event generator unit ... 488
46.1 Registers ... 488
46.2 Electrical specification ... 494
47 PWM — Pulse width modulation ... 495
47.1 Wave counter ... 495
47.2 Decoder with EasyDMA ... 498
47.3 Limitations ... 503
47.4 Pin configuration ... 503
47.5 Registers ... 504
47.6 Electrical specification ... 512
48 SPI — Serial peripheral interface master ... 513
48.1 Functional description ... 513
48.2 Registers ... 516
48.3 Electrical specification ... 519
49 TWI — I^2C compatible two-wire interface ... 521
49.1 Functional description ... 521
49.2 Master mode pin configuration ... 521
49.3 Shared resources ... 522
49.4 Master write sequence ... 522
49.5 Master read sequence ... 523
49.6 Master repeated start sequence ... 524
49.7 Low power ... 525
49.8 Registers ... 525
49.9 Electrical specification ... 529
50 UART — Universal asynchronous receiver/transmitter ... 531
50.1 Functional description ... 531
50.2 Pin configuration ... 531
50.3 Shared resources ... 532
50.4 Transmission ... 532
50.5 Reception ... 532
50.6 Suspending the UART ... 533
50.7 Error conditions ... 533
50.8 Using the UART without flow control ... 534
50.9 Parity configuration ... 534
50.10 Registers ... 534
50.11 Electrical specification ... 539
51 Mechanical specifications ... 540
51.1 QFN48 6 x 6 mm package ... 540
51.2 WLCSP package ... 541
52 Ordering information ... 542
52.1 IC marking ... 542
52.2 Box labels ... 542
52.3 Order code ... 543
52.4 Code ranges and values ... 543
52.5 Product options ... 544
53 Reference circuitry ... 545
53.1 Schematic QFAA and QFAB QFN48 with internal LDO setup ... 545
53.2 Schematic QFAA and QFAB QFN48 with DC/DC regulator setup ... 546
53.3 Schematic QFAA and QFAB QFN48 with DC/DC regulator and NFC setup ... 547
53.4 Schematic CIAA WLCSP with internal LDO setup ... 548
53.5 Schematic CIAA WLCSP with DC/DC regulator setup ... 549

53.6 Schematic CIAA WLCSP with DC/DC regulator and NFC setup....550
53.7 PCB guidelines....550
53.8 PCB layout example....551
54 Liability disclaimer....553
54.1 RoHS and REACH statement....553
54.2 Life support applications....553

1 Revision history

Date	Version	Description
October 2017	1.4	The following content has been added or updated: • *Recommended operating conditions* on page 20: Added WLCSP light sensitivity information. • *FICR — Factory information configuration registers* on page 43: Added registers PARTNO, HWREVISION and PRODUCTIONREVISION. • *UICR — User information configuration registers* on page 54: Changed width of PSELRESETn port fields. • *SPIM*: Polarity in SPI mode table corrected. • *COMP — Comparator* on page 392: Documentation structure improvements/changes. • Liability disclaimer updated: Directive 2011/65/EU (RoHS 2).
February 2017	1.3	The following content has been added or updated: • *RADIO — 2.4 GHz Radio* on page 205: Introduced 2 Mbps Bluetooth® low energy mode. • *FICR — Factory information configuration registers* on page 43: Updated INFO.PACKAGE register (new package added). • *UARTE*: Corrected the pin configuration table. • *PPI — Programmable peripheral interconnect* on page 168: Timing information corrected. • Updated the liability disclaimer.
September 2016	1.2	Updated the following: • Power and clock management, *Current consumption: Ultra-low power* on page 77. • Power, *Current consumption, sleep* on page 99
July 2016	1.1	Added documentation for nRF52832 CIAA WLCSP. Added or updated the following content: • Cover: Added *Key features*. • *Pin assignments* on page 13: Added *WLCSP ball assignments*. Moved *GPIO usage restrictions* here from *GPIO/Notes on usage and restrictions*. • *Absolute maximum ratings* on page 19: Added environmental information for WLCSP to the table. • *Memory* on page 23: Added QFAB and CIAA information to the table. • *FICR — Factory information configuration registers* on page 43: Updated INFO.PACKAGE register. • *UICR — User information configuration registers* on page 54: Updated APPROTECT register. • *Debug and trace* on page 72: Updated *DAP - Debug access port*. • *POWER — Power supply* on page 78: Updated *Pin reset*. • *CLOCK — Clock control* on page 101: Updated information on external 32 kHz clock support. • *GPIO — General purpose input/output* on page 111: Added *GPIO located near the RADIO*. • *RADIO — 2.4 GHz Radio* on page 205: Updated Figure 29 and *Interframe spacing*. • *CCM*: Updated SCRATCHPTR register. • *SPIM*: Updated *Master mode pin configuration*. • *UARTE*: Added RXDRDY and TXDRDY events. • *NFCT*: Updated *Electrical specifications*. • *PWM — Pulse width modulation* on page 495: Updated SEQ[1].REFRESH register. • *Mechanical specifications* on page 540: Added *WLCSP package*. • *Ordering information* on page 542: Updated with CIAA and QFAB information. • *Reference circuitry* on page 545: QFAB information added. CIAA WLCSP schematics added.
February 2016	1.0	First release.

2 About this document

This product specification is organized into chapters based on the modules and peripherals that are available in this IC.

The peripheral descriptions are divided into separate sections that include the following information:

- A detailed functional description of the peripheral
- Register configuration for the peripheral
- Electrical specification tables, containing performance data which apply for the operating conditions described in *Recommended operating conditions* on page 20.

2.1 Document naming and status

Nordic uses three distinct names for this document, which are reflecting the maturity and the status of the document and its content.

Table 1: Defined document names

Document name	Description
Objective Product Specification (OPS)	Applies to document versions up to 0.7.
Preliminary Product Specification (PPS)	This product specification contains target specifications for product development. Applies to document versions 0.7 and up to 1.0.
Product Specification (PS)	This product specification contains preliminary data. Supplementary data may be published from Nordic Semiconductor ASA later. Applies to document versions 1.0 and higher. This product specification contains final product specifications. Nordic Semiconductor ASA reserves the right to make changes at any time without notice in order to improve design and supply the best possible product.

2.2 Peripheral naming and abbreviations

Every peripheral has a unique capitalized name or an abbreviation of its name, e.g. TIMER, used for identification and reference. This name is used in chapter headings and references, and it will appear in the ARM® Cortex® Microcontroller Software Interface Standard (CMSIS) hardware abstraction layer to identify the peripheral.

The peripheral instance name, which is different from the peripheral name, is constructed using the peripheral name followed by a numbered postfix, starting with 0, for example, TIMER0. A postfix is normally only used if a peripheral can be instantiated more than once. The peripheral instance name is also used in the CMSIS to identify the peripheral instance.

2.3 Register tables

Individual registers are described using register tables. These tables are built up of two sections. The first three colored rows describe the position and size of the different fields in the register. The following rows describe the fields in more detail.

2.3.1 Fields and values

The **Id (Field Id)** row specifies the bits that belong to the different fields in the register. If a field has enumerated values, then every value will be identified with a unique value id in the **Value Id** column.

A blank space means that the field is reserved and read as undefined, and it also must be written as `0` to secure forward compatibility. If a register is divided into more than one field, a unique field name is specified for each field in the **Field** column. The **Value Id** may be omitted in the single-bit bit fields when values can be substituted with a Boolean type enumerator range, e.g. true/false, disable(d)/enable(d), on/off, and so on.

Values are usually provided as decimal or hexadecimal. Hexadecimal values have a `0x` prefix, decimal values have no prefix.

The **Value** column can be populated in the following ways:

- Individual enumerated values, for example 1, 3, 9.
- Range of values, e.g. [0..4], indicating all values from and including 0 and 4.
- Implicit values. If no values are indicated in the **Value** column, all bit combinations are supported, or alternatively the field's translation and limitations are described in the text instead.

If two or more fields are closely related, the **Value Id**, **Value**, and **Description** may be omitted for all but the first field. Subsequent fields will indicate inheritance with '..'.

A feature marked **Deprecated** should not be used for new designs.

2.4 Registers

Table 2: Register Overview

Register	Offset	Description
DUMMY	0x514	Example of a register controlling a dummy feature

2.4.1 DUMMY

Address offset: 0x514

Example of a register controlling a dummy feature

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								D	D	D	D						C	C	C								B							A	A
Reset 0x00050002				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	FIELD_A																																	
			Disabled	0																															
			NormalMode	1																															
			ExtendedMode	2																															
B	RW	FIELD_B																																	Deprecated
			Disabled	0																															
			Enabled	1																															
C	RW	FIELD_C																																	
			ValidRange	[2..7]																															
D	RW	FIELD_D																																	

3 Block diagram

This block diagram illustrates the overall system. Arrows with white heads indicate signals that share physical pins with other signals.

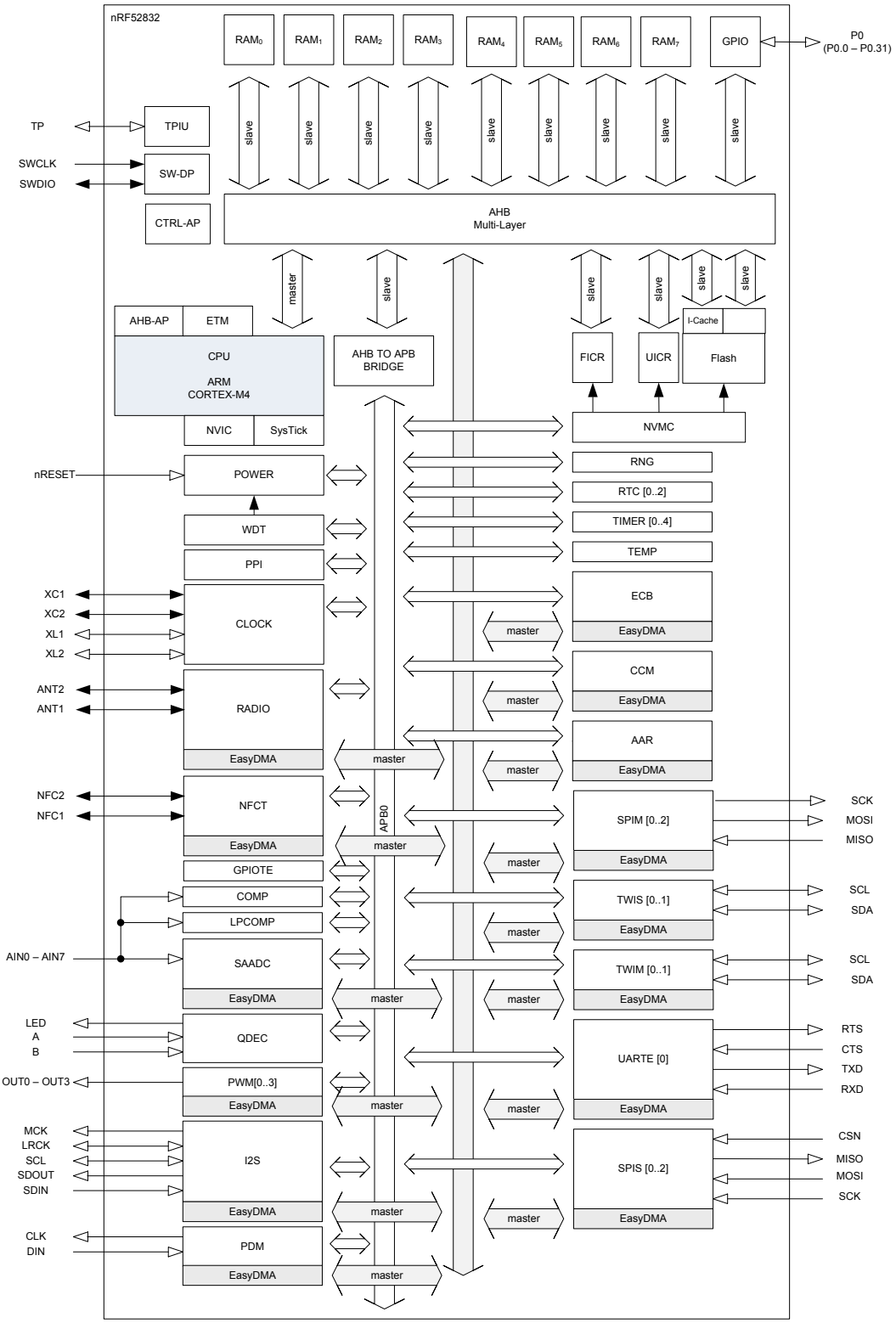


Figure 1: Block diagram

4 Pin assignments

Here we cover the pin assignments for each variant of the chip.

4.1 QFN48 pin assignments

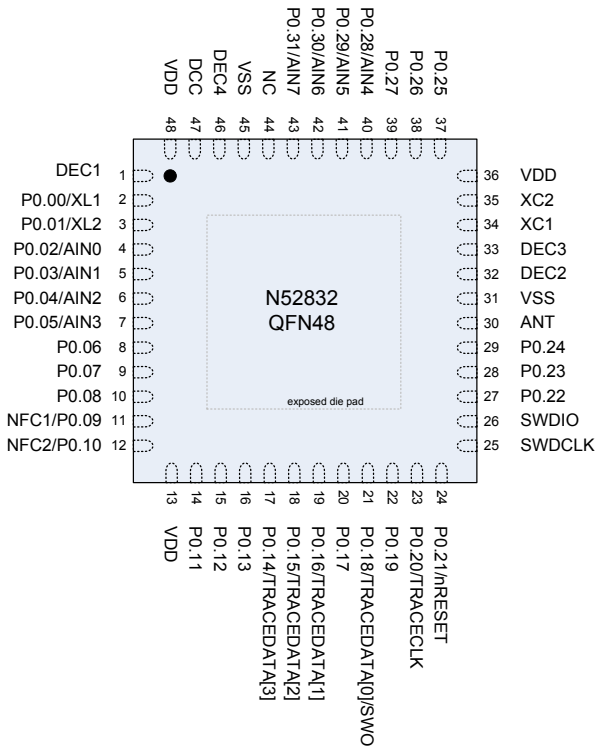


Figure 2: QFN48 pin assignments, top view

Table 3: QFN48 pin assignments

Pin	Name	Type	Description
Left Side of chip			
1	DEC1	Power	0.9 V regulator digital supply decoupling
2	P0.00	Digital I/O	General purpose I/O
	XL1	Analog input	Connection for 32.768 kHz crystal (LFXO)
3	P0.01	Digital I/O	General purpose I/O
	XL2	Analog input	Connection for 32.768 kHz crystal (LFXO)
4	P0.02	Digital I/O	General purpose I/O
	AIN0	Analog input	SAADC/COMP/LPCOMP input
5	P0.03	Digital I/O	General purpose I/O
	AIN1	Analog input	SAADC/COMP/LPCOMP input
6	P0.04	Digital I/O	General purpose I/O
	AIN2	Analog input	SAADC/COMP/LPCOMP input
7	P0.05	Digital I/O	General purpose I/O
	AIN3	Analog input	SAADC/COMP/LPCOMP input
8	P0.06	Digital I/O	General purpose I/O
9	P0.07	Digital I/O	General purpose I/O

Pin	Name	Type	Description
10	P0.08	Digital I/O	General purpose I/O
11	NFC1	NFC input	NFC antenna connection
	P0.09	Digital I/O	General purpose I/O[1]
12	NFC2	NFC input	NFC antenna connection
	P0.10	Digital I/O	General purpose I/O[1]
Bottom side of chip			
13	VDD	Power	Power supply
14	P0.11	Digital I/O	General purpose I/O
15	P0.12	Digital I/O	General purpose I/O
16	P0.13	Digital I/O	General purpose I/O
17	P0.14	Digital I/O	General purpose I/O
	TRACEDATA[3]		Trace port output
18	P0.15	Digital I/O	General purpose I/O
	TRACEDATA[2]		Trace port output
19	P0.16	Digital I/O	General purpose I/O
	TRACEDATA[1]		Trace port output
20	P0.17	Digital I/O	General purpose I/O
21	P0.18	Digital I/O	General purpose I/O
	TRACEDATA[0] / SWO		Single wire output Trace port output
22	P0.19	Digital I/O	General purpose I/O
23	P0.20	Digital I/O	General purpose I/O
	TRACECLK		Trace port clock output
24	P0.21	Digital I/O	General purpose I/O
	nRESET		Configurable as pin reset
Right Side of chip			
25	SWDCLK	Digital input	Serial wire debug clock input for debug and programming
26	SWDIO	Digital I/O	Serial wire debug I/O for debug and programming
27	P0.22	Digital I/O	General purpose I/O[2]
28	P0.23	Digital I/O	General purpose I/O[2]
29	P0.24	Digital I/O	General purpose I/O[2]
30	ANT	RF	Single-ended radio antenna connection
31	VSS	Power	Ground (Radio supply)
32	DEC2	Power	1.3 V regulator supply decoupling (Radio supply)
33	DEC3	Power	Power supply decoupling
34	XC1	Analog input	Connection for 32 MHz crystal
35	XC2	Analog input	Connection for 32 MHz crystal
36	VDD	Power	Power supply
Top side of chip			
37	P0.25	Digital I/O	General purpose I/O[2]
38	P0.26	Digital I/O	General purpose I/O[2]
39	P0.27	Digital I/O	General purpose I/O[2]
40	P0.28	Digital I/O	General purpose I/O[2]
	AIN4	Analog input	SAADC/COMP/LPCOMP input
41	P0.29	Digital I/O	General purpose I/O[2]
	AIN5	Analog input	SAADC/COMP/LPCOMP input
42	P0.30	Digital I/O	General purpose I/O[2]
	AIN6	Analog input	SAADC/COMP/LPCOMP input
43	P0.31	Digital I/O	General purpose I/O pin[2]
	AIN7	Analog input	SAADC/COMP/LPCOMP input

Pin	Name	Type	Description
44	NC		No connect Leave unconnected
45	VSS	Power	Ground
46	DEC4	Power	1.3 V regulator supply decoupling Input from DC/DC regulator Output from 1.3 V LDO
47	DCC	Power	DC/DC regulator output
48	VDD	Power	Power supply
Bottom of chip			
Die pad	VSS	Power	Ground pad Exposed die pad must be connected to ground (VSS) for proper device operation.

4.2 WLCSP ball assignments

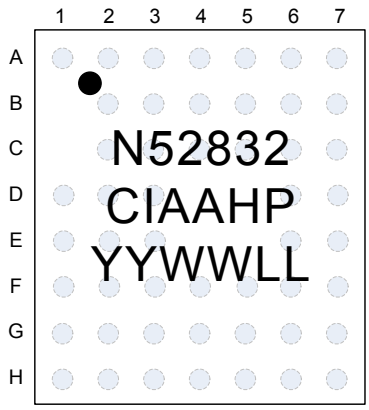


Figure 3: WLCSP ball assignments, top view

Table 4: WLCSP ball assignments

Ball	Name		Description
A1	XC2	Analog input	Connection for 32 MHz crystal
A2	DEC2	Power	1.3 V regulator supply decoupling (Radio supply)
A3	P0.28	Digital I/O	General purpose I/O[3]
	AIN4	Analog input	SAADC/COMP/LPCOMP input
A4	P0.29	Digital I/O	General purpose I/O[3]
	AIN5	Analog input	SAADC/COMP/LPCOMP input
A5	P0.30	Digital I/O	General purpose I/O[3]
	AIN6	Analog input	SAADC/COMP/LPCOMP input
A6	DEC4	Power	1.3 V regulator supply decoupling Input from DC/DC converter. Output from 1.3 V LDO
A7	VDD	Power	Power supply
B2	XC1	Analog input	Connection for 32 MHz crystal
B3	P0.25	Digital I/O	General purpose I/O[3]

[1] See *GPIO located near the radio* on page 17 for more information.
[2] See *NFC antenna pins* on page 17 for more information.

Ball	Name		Description
B4	P0.27	Digital I/O	General purpose I/O[3]
B5	P0.31	Digital I/O	General purpose I/O[3]
	AIN7	Analog input	SAADC/COMP/LPCOMP input
B6	DCC	Power	DC/DC converter output
B7	DEC1	Power	0.9 V regulator digital supply decoupling
C2	DEC3	Power	Power supply decoupling
C3	NC	N/A	Not connected
C4	VSS	Power	Ground
C5	VSS	Power	Ground
C6	P0.02	Digital I/O	General purpose I/O
	AIN0	Analog input	SAADC/COMP/LPCOMP input
C7	P0.01	Digital I/O	General purpose I/O
	XL2	Analog input	Connection for 32.768 kHz crystal (LFXO)
D1	ANT	RF	Single-ended radio antenna connection
D2	VSS_PA	Power	Ground (Radio supply)
D3	P0.26	Digital I/O	General purpose I/O [3]
D6	P0.03	Digital I/O	General purpose I/O
	AIN1	Analog input	SAADC/COMP/LPCOMP input
D7	P0.00	Digital I/O	General purpose I/O
	XL1	Analog input	Connection for 32.768 kHz crystal (LFXO)
E1	P0.24	Digital I/O	General purpose I/O[3]
E2	P0.23	Digital I/O	General purpose I/O[3]
E3	VSS	Power	Ground
E6	P0.04	Digital I/O	General purpose I/O
	AIN2	Analog input	SAADC/COMP/LPCOMP input
E7	P0.05	Digital I/O	General purpose I/O
	AIN3	Analog input	SAADC/COMP/LPCOMP input
F1	SWDCLK	Digital input	Serial wire debug clock input for debug and programming
F2	P0.22	Digital I/O	General purpose I/O[3]
F3	P0.19	Digital I/O	General purpose I/O
F4	P0.11	Digital I/O	General purpose I/O
F5	VSS	Power	Ground
F6	P0.07	Digital I/O	General purpose I/O
F7	P0.06	Digital I/O	General purpose I/O
G1	SWDIO	Digital I/O	Serial wire debug I/O for debug and programming
G2	P0.20	Digital I/O	General purpose I/O
	TRACECLK		Trace port clock output
G3	P0.17	Digital I/O	General purpose I/O
G4	P0.13	Digital I/O	General purpose I/O
G5	NFC2	NFC input	NFC antenna connection
	P0.10	Digital I/O	General purpose I/O[4]
G6	NFC1	NFC input	NFC antenna connection
	P0.09	Digital I/O	General purpose I/O[4]
G7	P0.08	Digital I/O	General purpose I/O
H1	P0.21	Digital I/O	General purpose I/O
	nRESET		Configurable as pin reset
H2	P0.18	Digital I/O	General purpose I/O
	TRACEDATA[0]		Trace port output
H3	P0.16	Digital I/O	General purpose I/O
	TRACEDATA[1]		Trace port output
H4	P0.15	Digital I/O	General purpose I/O

Ball	Name		Description
	TRACEDATA[2]		Trace port output
H5	P0.14	Digital I/O	General purpose I/O
	TRACEDATA[3]		Trace port output
H6	P0.12	Digital I/O	General purpose I/O
H7	VDD	Power	Power supply

4.3 GPIO usage restrictions

4.3.1 GPIO located near the radio

Radio performance parameters, such as sensitivity, may be affected by high frequency digital I/O with large sink/source current close to the Radio power supply and antenna pins.

Table 5: GPIO recommended usage for QFN48 package on page 17 and *Table 6: GPIO recommended usage for WLCSP package* on page 17 identify some GPIO that have recommended usage guidelines to maximize radio performance in an application.

Table 5: GPIO recommended usage for QFN48 package

Pin	GPIO	Recommended usage
27	P0.22	Low drive, low frequency I/O only.
28	P0.23	
29	P0.24	
37	P0.25	
38	P0.26	
39	P0.27	
40	P0.28	
41	P0.29	
42	P0.30	
43	P0.31	

Table 6: GPIO recommended usage for WLCSP package

Pin	GPIO	Recommended usage
F2	P0.22	Low drive, low frequency I/O only.
E2	P0.23	
E1	P0.24	
B3	P0.25	
D3	P0.26	
B4	P0.27	
A3	P0.28	
A4	P0.29	
A5	P0.30	
B5	P0.31	

4.3.2 NFC antenna pins

Two physical pins can be configured either as NFC antenna pins (factory default), or as GPIOs, as shown below.

Table 7: GPIO pins used by NFC

NFC pad name	GPIO
NFC1	P0.09
NFC2	P0.10

When configured as NFC antenna pins, the GPIOs on those pins will automatically be set to DISABLE state and a protection circuit will be enabled preventing the chip from being damaged in the presence of a strong NFC field. The protection circuit will short the two pins together if voltage difference exceeds approximately 2 V.

[3] See *GPIO located near the radio* on page 17 for more information.
[4] See *NFC antenna pins* on page 17 for more information.

For information on how to configure these pins as normal GPIOs, see *NFCT — Near field communication tag* on page 416 and *UICR — User information configuration registers* on page 54. Note that the device will not be protected against strong NFC field damage if the pins are configured as GPIO and an NFC antenna is connected to the device. The pins will always be configured as NFC pins during power-on reset until the configuration is set according to the UICR register.

These two pins will have some limitations when configured as GPIO. The pin capacitance will be higher on these pins, and there is some current leakage between the two pins if they are driven to different logical values. To avoid leakage between the pins when configured as GPIO, these GPIOs should always be at the same logical value whenever entering one of the device power saving modes. See *Electrical specification*.

5 Absolute maximum ratings

Maximum ratings are the extreme limits to which the chip can be exposed for a limited amount of time without permanently damaging it. Exposure to absolute maximum ratings for prolonged periods of time may affect the reliability of the device.

Table 8: Absolute maximum ratings

	Min.	Max.	Unit
Supply voltages			
VDD	-0.3	+3.9	V
VSS		0	V
I/O pin voltage			
$V_{I/O}$, VDD ≤3.6 V	-0.3	VDD + 0.3 V	V
$V_{I/O}$, VDD >3.6 V	-0.3	3.9 V	V
NFC antenna pin current			
$I_{NFC1/2}$		80	mA
Radio			
RF input level		10	dBm
Environmental QFN48, 6×6 mm package			
Storage temperature	-40	+125	°C
MSL (moisture sensitivity level)		2	
ESD HBM (human body model)		4	kV
ESD CDM (charged device model)		1000	V
Environmental WLCSP, 3.0×3.2 mm package			
Storage temperature	-40	+125	°C
MSL		1	
ESD HBM		2	kV
ESD CDM		500	V
Flash memory			
Endurance	10 000		Write/erase cycles
Retention	10 years at 40°C		



6 Recommended operating conditions

The operating conditions are the physical parameters that the chip can operate within.

Table 9: Recommended operating conditions

Symbol	Parameter	Notes	Min.	Nom.	Max.	Units
VDD	Supply voltage, independent of DCDC enable		1.7	3.0	3.6	V
t_{R_VDD}	Supply rise time (0 V to 1.7 V)				60	ms
TA	Operating temperature		-40	25	85	°C

Important: The on-chip power-on reset circuitry may not function properly for rise times longer than the specified maximum.

6.1 WLCSP light sensitivity

All WLCSP package variants are sensitive to visible and close-range infrared light. This means that a final product design must shield the chip properly, either by final product encapsulation or by shielding/coating of the WLCSP device.

7 CPU

The ARM® Cortex®-M4 processor with floating-point unit (FPU) has a 32-bit instruction set (Thumb®-2 technology) that implements a superset of 16 and 32-bit instructions to maximize code density and performance.

This processor implements several features that enable energy-efficient arithmetic and high-performance signal processing including:

- Digital signal processing (DSP) instructions
- Single-cycle multiply and accumulate (MAC) instructions
- Hardware divide
- 8 and 16-bit single instruction multiple data (SIMD) instructions
- Single-precision floating-point unit (FPU)

The ARM Cortex Microcontroller Software Interface Standard (CMSIS) hardware abstraction layer for the ARM Cortex processor series is implemented and available for the M4 CPU.

Real-time execution is highly deterministic in thread mode, to and from sleep modes, and when handling events at configurable priority levels via the Nested Vectored Interrupt Controller (NVIC).

Executing code from flash will have a wait state penalty on the nRF52 Series. An instruction cache can be enabled to minimize flash wait states when fetching instructions. For more information on cache, see *Cache* on page 30. The section *Electrical specification* on page 21 shows CPU performance parameters including wait states in different modes, CPU current and efficiency, and processing power and efficiency based on the CoreMark® benchmark.

7.1 Floating point interrupt

The floating point unit (FPU) may generate exceptions when used due to e.g. overflow or underflow. These exceptions will trigger the FPU interrupt (see *Instantiation* on page 24). To clear the IRQ line when an exception has occurred, the relevant exception bit within the FPSCR register needs to be cleared. For more information about the FPSCR or other FPU registers, see *Cortex-M4 Devices Generic User Guide*.

7.2 Electrical specification

7.2.1 CPU performance

The CPU clock speed is 64 MHz. Current and efficiency data is taken when in System ON and the CPU is executing the CoreMark™ benchmark. It includes power regulator and clock base currents. All other blocks are IDLE.

Symbol	Description	Min.	Typ.	Max.	Units
W_{FLASH}	CPU wait states, running from flash, cache disabled	0		2	
$W_{FLASHCACHE}$	CPU wait states, running from flash, cache enabled	0		3	
W_{RAM}	CPU wait states, running from RAM			0	
$I_{DDFLASHCACHE}$	CPU current, running from flash, cache enabled, LDO		7.4		mA
$I_{DDFLASHCACHEDCDC}$	CPU current, running from flash, cache enabled, DCDC 3V		3.7		mA
$I_{DDFLASH}$	CPU current, running from flash, cache disabled, LDO		8.0		mA
$I_{DDFLASHDCDC}$	CPU current, running from flash, cache disabled, DCDC 3V		3.9		mA
I_{DDRAM}	CPU current, running from RAM, LDO		6.7		mA
$I_{DDRAMDCDC}$	CPU current, running from RAM, DCDC 3V		3.3		mA
$I_{DDFLASH/MHz}$	CPU efficiency, running from flash, cache enabled, LDO		125		µA/MHz
$I_{DDFLASHDCDC/MHz}$	CPU efficiency, running from flash, cache enabled, DCDC 3V		58		µA/MHz

Symbol	Description	Min.	Typ.	Max.	Units
CM_{FLASH}	CoreMark[5], running from flash, cache enabled		215		CoreM
$CM_{FLASH/MHz}$	CoreMark per MHz, running from flash, cache enabled		3.36		CoreM/MHz
$CM_{FLASH/mA}$	CoreMark per mA, running from flash, cache enabled, DCDC 3V		58		CoreM/mA

7.3 CPU and support module configuration

The ARM® Cortex®-M4 processor has a number of CPU options and support modules implemented on the device.

Option / Module	Description	Implemented
Core options		
NVIC	Nested Vector Interrupt Controller	37 vectors
PRIORITIES	Priority bits	3
WIC	Wakeup Interrupt Controller	NO
Endianness	Memory system endianness	Little endian
Bit Banding	Bit banded memory	NO
DWT	Data Watchpoint and Trace	YES
SysTick	System tick timer	YES
Modules		
MPU	Memory protection unit	YES
FPU	Floating point unit	YES
DAP	Debug Access Port	YES
ETM	Embedded Trace Macrocell	YES
ITM	Instrumentation Trace Macrocell	YES
TPIU	Trace Port Interface Unit	YES
ETB	Embedded Trace Buffer	NO
FPB	Flash Patch and Breakpoint Unit	YES
HTM	AHB Trace Macrocell	NO

[5] Using IAR v6.50.1.4452 with flags --endian=little --cpu=Cortex-M4 -e --fpu=VFPv4_sp –Ohs --no_size_constraints

8 Memory

The nRF52832 contains flash and RAM that can be used for code and data storage.

The amount of RAM and flash will vary depending on variant, see *Table 10: Memory variants* on page 23.

Table 10: Memory variants

Device name	RAM	Flash	Comments
nRF52832-QFAA	64 kB	512 kB	
nRF52832-QFAB	32 kB	256 kB	
nRF52832-CIAA	64 kB	512 kB	

The CPU and the EasyDMA can access memory via the AHB multilayer interconnect. The CPU is also able to access peripherals via the AHB multilayer interconnect, as illustrated in *Figure 4: Memory layout* on page 23.

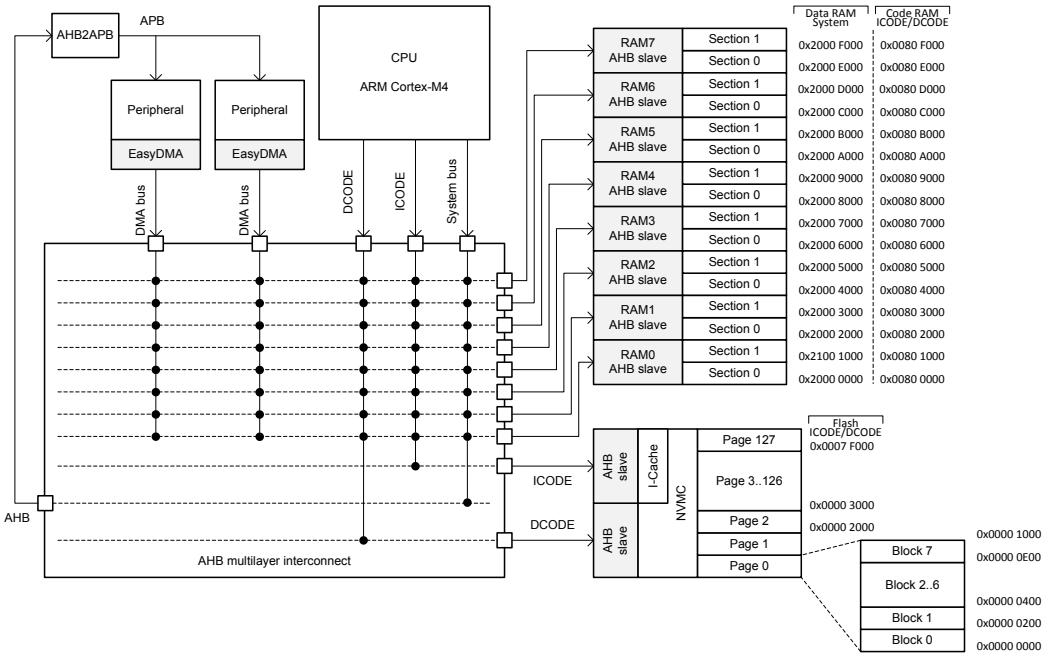


Figure 4: Memory layout

See *AHB multilayer* on page 26 and *EasyDMA* on page 27 for more information about the AHB multilayer interconnect and the EasyDMA.

The same physical RAM is mapped to both the Data RAM region and the Code RAM region. It is up to the application to partition the RAM within these regions so that one does not corrupt the other.

8.1 RAM - Random access memory

The RAM interface is divided into multiple RAM AHB slaves.

Each RAM AHB slave is connected to two 4-kilobyte RAM sections, see Section 0 and Section 1 in *Figure 4: Memory layout* on page 23.

Each of the RAM sections have separate power control for System ON and System OFF mode operation, which is configured via RAM register (see the *POWER — Power supply* on page 78).

8.2 Flash - Non-volatile memory

The Flash can be read an unlimited number of times by the CPU, but it has restrictions on the number of times it can be written and erased and also on how it can be written.

Writing to Flash is managed by the Non-volatile memory controller (NVMC), see *NVMC — Non-volatile memory controller* on page 29.

The Flash is divided into multiple pages that can be accessed by the CPU via both the ICODE and DCODE buses as shown in, *Figure 4: Memory layout* on page 23. Each page is divided into 8 blocks.

8.3 Memory map

The complete memory map is shown in *Figure 5: Memory map* on page 24. As described in *Memory* on page 23, Code RAM and the Data RAM are the same physical RAM.

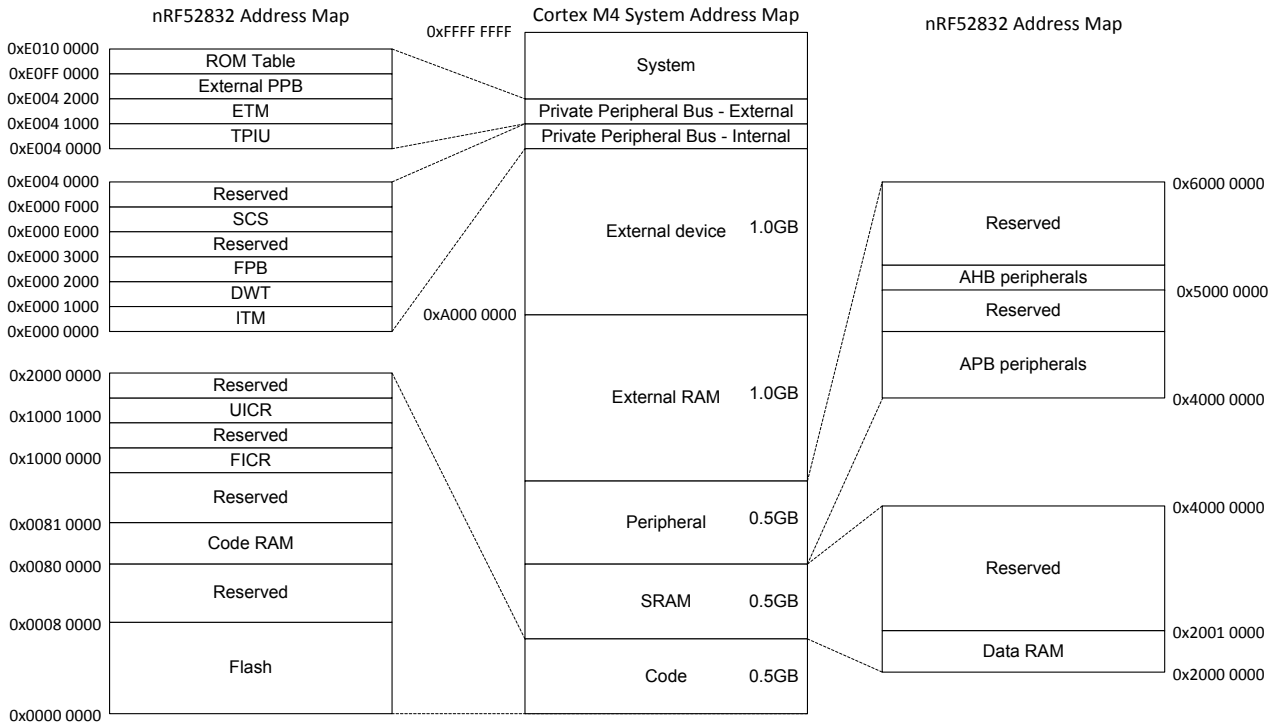


Figure 5: Memory map

8.4 Instantiation

Table 11: Instantiation table

ID	Base Address	Peripheral	Instance	Description	
0	0x40000000	CLOCK	CLOCK	Clock control	
0	0x40000000	POWER	POWER	Power control	
0	0x40000000	BPROT	BPROT	Block Protect	
1	0x40001000	RADIO	RADIO	2.4 GHz radio	
2	0x40002000	UARTE	UARTE0	Universal Asynchronous Receiver/Transmitter with EasyDMA	
2	0x40002000	UART	UART0	Universal Asynchronous Receiver/Transmitter	Deprecated
3	0x40003000	SPIM	SPIM0	SPI master 0	
3	0x40003000	SPIS	SPIS0	SPI slave 0	
3	0x40003000	TWIM	TWIM0	Two-wire interface master 0	
3	0x40003000	TWI	TWI0	Two-wire interface master 0	Deprecated
3	0x40003000	SPI	SPI0	SPI master 0	Deprecated

ID	Base Address	Peripheral	Instance	Description	
3	0x40003000	TWIS	TWIS0	Two-wire interface slave 0	
4	0x40004000	SPIM	SPIM1	SPI master 1	
4	0x40004000	TWI	TWI1	Two-wire interface master 1	Deprecated
4	0x40004000	SPIS	SPIS1	SPI slave 1	
4	0x40004000	TWIS	TWIS1	Two-wire interface slave 1	
4	0x40004000	TWIM	TWIM1	Two-wire interface master 1	
4	0x40004000	SPI	SPI1	SPI master 1	Deprecated
5	0x40005000	NFCT	NFCT	Near Field Communication Tag	
6	0x40006000	GPIOTE	GPIOTE	GPIO Tasks and Events	
7	0x40007000	SAADC	SAADC	Analog to digital converter	
8	0x40008000	TIMER	TIMER0	Timer 0	
9	0x40009000	TIMER	TIMER1	Timer 1	
10	0x4000A000	TIMER	TIMER2	Timer 2	
11	0x4000B000	RTC	RTC0	Real-time counter 0	
12	0x4000C000	TEMP	TEMP	Temperature sensor	
13	0x4000D000	RNG	RNG	Random number generator	
14	0x4000E000	ECB	ECB	AES Electronic Code Book (ECB) mode block encryption	
15	0x4000F000	CCM	CCM	AES CCM Mode Encryption	
15	0x4000F000	AAR	AAR	Acelerated Address Resolver	
16	0x40010000	WDT	WDT	Watchdog timer	
17	0x40011000	RTC	RTC1	Real-time counter 1	
18	0x40012000	QDEC	QDEC	Quadrature decoder	
19	0x40013000	LPCOMP	LPCOMP	Low power comparator	
19	0x40013000	COMP	COMP	General purpose comparator	
20	0x40014000	SWI	SWI0	Software interrupt 0	
20	0x40014000	EGU	EGU0	Event Generator Unit 0	
21	0x40015000	EGU	EGU1	Event Generator Unit 1	
21	0x40015000	SWI	SWI1	Software interrupt 1	
22	0x40016000	SWI	SWI2	Software interrupt 2	
22	0x40016000	EGU	EGU2	Event Generator Unit 2	
23	0x40017000	SWI	SWI3	Software interrupt 3	
23	0x40017000	EGU	EGU3	Event Generator Unit 3	
24	0x40018000	EGU	EGU4	Event Generator Unit 4	
24	0x40018000	SWI	SWI4	Software interrupt 4	
25	0x40019000	SWI	SWI5	Software interrupt 5	
25	0x40019000	EGU	EGU5	Event Generator Unit 5	
26	0x4001A000	TIMER	TIMER3	Timer 3	
27	0x4001B000	TIMER	TIMER4	Timer 4	
28	0x4001C000	PWM	PWM0	Pulse Width Modulation Unit 0	
29	0x4001D000	PDM	PDM	Pulse Density Modulation (Digital Microphone Interface)	
30	0x4001E000	NVMC	NVMC	Non-Volatile Memory Controller	
31	0x4001F000	PPI	PPI	Programmable Peripheral Interconnect	
32	0x40020000	MWU	MWU	Memory Watch Unit	
33	0x40021000	PWM	PWM1	Pulse Width Modulation Unit 1	
34	0x40022000	PWM	PWM2	Pulse Width Modulation Unit 2	
35	0x40023000	SPI	SPI2	SPI master 2	Deprecated
35	0x40023000	SPIS	SPIS2	SPI slave 2	
35	0x40023000	SPIM	SPIM2	SPI master 2	
36	0x40024000	RTC	RTC2	Real-time counter 2	
37	0x40025000	I2S	I2S	Inter-IC Sound Interface	
38	0x40026000	FPU	FPU	FPU interrupt	
0	0x50000000	GPIO	GPIO	General purpose input and output	Deprecated
0	0x50000000	GPIO	P0	General purpose input and output	
N/A	0x10000000	FICR	FICR	Factory Information Configuration	
N/A	0x10001000	UICR	UICR	User Information Configuration	

9 AHB multilayer

The CPU and all of the EasyDMAs are AHB bus masters on the AHB multilayer, while the RAM and various other modules are AHB slaves.

See *Block diagram* on page 12 for an overview of which peripherals implement EasyDMA.

The CPU has exclusive access to all AHB slaves except for the RAM that can also be accessed by the EasyDMA.

Access rights to each of the RAM AHB slaves are resolved using the priority of the different bus masters in the system

See *AHB multilayer priorities* on page 26 for information about the priority of the different AHB bus masters in the system. It is possible for two or more bus masters to have the same priority in cases where it is guaranteed by design that the related masters will never be able to access the same slave at the same time.

9.1 AHB multilayer priorities

Each master connected to the AHB multilayer is assigned a priority.

Table 12: AHB bus masters

Bus master name	Priority	Description
CPU	Highest priority	
SPIS1		Applies to SPIM1, SPIS1, TWIM1, TWIS1
RADIO		
CCM/ECB/AAR		
SAADC		
UARTE		
SERIAL0		Applies to SPIM0, SPIS0, TWIM0, TWIS0
SERIAL2		Applies to SPIM2, SPIS2
NFCT		
I2S		I2S
PDM		PDM
PWM	Lowest priority	Applies to PWM0, PWM1, PWM2

10 EasyDMA

EasyDMA is an easy-to-use direct memory access module that some peripherals implement to gain direct access to Data RAM.

The EasyDMA is an AHB bus master similar to the CPU and it is connected to the AHB multilayer interconnect for direct access to the Data RAM. The EasyDMA is not able to access the Flash.

A peripheral can implement multiple EasyDMA instances, for example to provide a dedicated channel for reading data from RAM into the peripheral at the same time as a second channel is dedicated for writing data to the RAM from the peripheral. This concept is illustrated in *Figure 6: EasyDMA example* on page 27

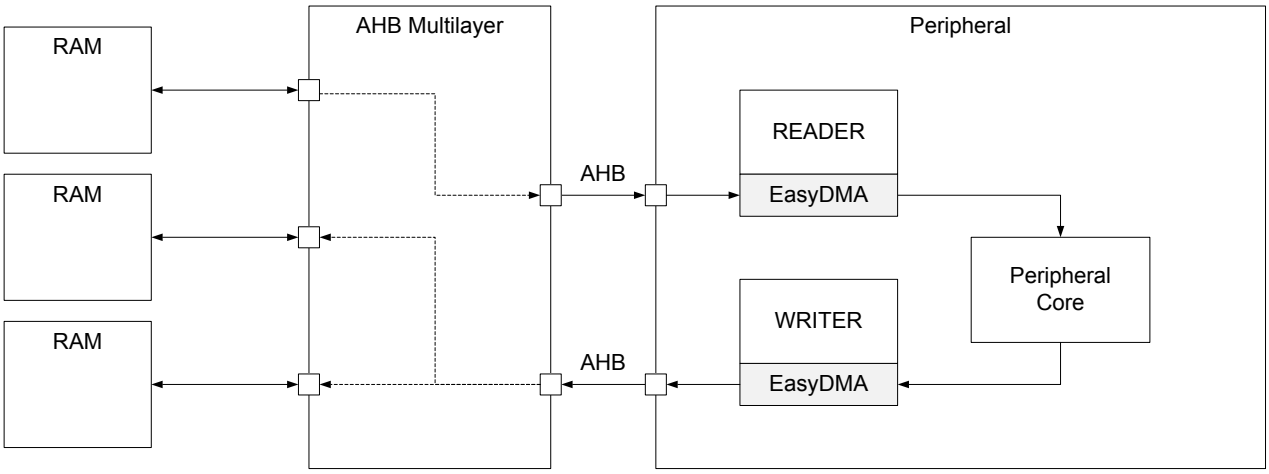


Figure 6: EasyDMA example

An EasyDMA channel is usually exposed to the user in the form illustrated below, but some variations may occur:

```
READERBUFFER_SIZE 5
WRITERBUFFER_SIZE 6

uint8_t readerBuffer[READERBUFFER_SIZE]  __at__ 0x20000000;
uint8_t writerBuffer[WRITERBUFFER_SIZE]  __at__ 0x20000005;

// Configuring the READER channel
MYPERIPHERAL->READER.MAXCNT = READERBUFFER_SIZE;
MYPERIPHERAL->READER.PTR = &readerBuffer;

// Configure the WRITER channel
MYPERIPHERAL->WRITER.MAXCNT = WRITEERBUFFER_SIZE;
MYPERIPHERAL->WRITER.PTR = &writerBuffer;
```

This example shows a peripheral called MYPERIPHERAL that implements two EasyDMA channels, one for reading, called READER, and one for writing, called WRITER. When the peripheral is started, it is here assumed that the peripheral will read 5 bytes from the readerBuffer located in RAM at address 0x20000000, process the data and then write no more than 6 bytes back to the writerBuffer located in RAM at address 0x20000005. The memory layout of these buffers is illustrated in *Figure 7: EasyDMA memory layout* on page 28.

0x20000000	readerBuffer[0]	readerBuffer[1]	readerBuffer[2]	readerBuffer[3]
0x20000004	readerBuffer[4]	writerBuffer[0]	writerBuffer[1]	writerBuffer[2]
0x20000008	writerBuffer[3]	writerBuffer[4]	writerBuffer[5]	

Figure 7: EasyDMA memory layout

The EasyDMA channel's MAXCNT register cannot be specified larger than the actual size of the buffer. If, for example, the WRITER.MAXCNT register is specified larger than the size of the writerBuffer, the WRITER EasyDMA channel may overflow the writerBuffer.

After the peripheral has completed the EasyDMA transfer, the CPU can read the EasyDMA channel's AMOUNT register to see how many bytes that were transferred, e.g. it is possible for the CPU to read the MYPERIPHERAL->WRITER.AMOUNT register to see how many bytes the WRITER wrote to RAM.

10.1 EasyDMA array list

The EasyDMA is able to operate in a mode called array list.

The EasyDMA array list can be represented by the data structure ArrayList_type illustrated in the code example below.

This data structure includes only a buffer with size equal to READER.MAXCNT. EasyDMA will use the READER.MAXCNT register to determine when the buffer is full.

This array list does not provide a mechanism to explicitly specify where the next item in the list is located. Instead, it assumes that the list is organized as a linear array where items are located one after the other in RAM.

```
#define BUFFER_SIZE  4

typedef struct ArrayList
{
  uint8_t buffer[BUFFER_SIZE];
} ArrayList_type;

ArrayList_type ReaderList[3];

READER.MAXCNT = BUFFER_SIZE;
READER.PTR = &ReaderList;
```

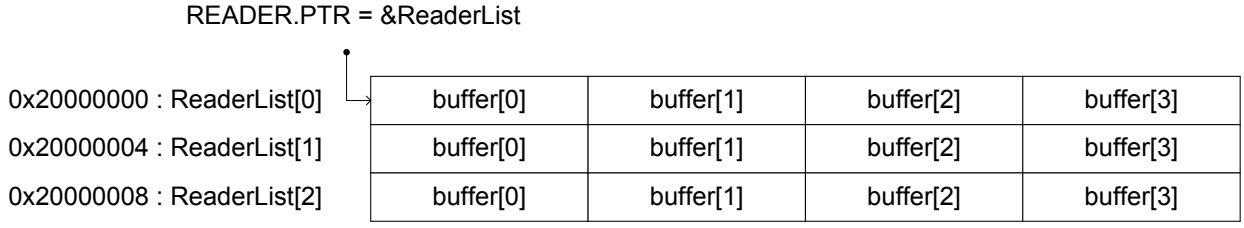


Figure 8: EasyDMA array list

11 NVMC — Non-volatile memory controller

The Non-volatile memory controller (NVMC) is used for writing and erasing the internal Flash memory and the UICR.

Before a write can be performed, the NVMC must be enabled for writing in CONFIG.WEN. Similarly, before an erase can be performed, the NVMC must be enabled for erasing in CONFIG.EEN, see *CONFIG* on page 31. The user must make sure that writing and erasing are not enabled at the same time. Failing to do so may result in unpredictable behavior.

11.1 Writing to Flash

When writing is enabled, the Flash is written by writing a full 32-bit word to a word-aligned address in the Flash.

The NVMC is only able to write '0' to bits in the Flash that are erased, that is, set to '1'. It cannot write back a bit to '1'.

As illustrated in *Memory* on page 23, the Flash is divided into multiple pages that are further divided into multiple blocks. The same block in the Flash can only be written n_{WRITE} number of times before an erase must be performed using *ERASEPAGE* or *ERASEALL*. See the memory size and organization in *Memory* on page 23 for block size.

Only full 32-bit words can be written to Flash using the NVMC interface. To write less than 32 bits to Flash, write the data as a word, and set all the bits that should remain unchanged in the word to '1'. Note that the restriction about the number of writes (see above) still applies in this case.

The time it takes to write a word to the Flash is specified by t_{WRITE}. The CPU is halted while the NVMC is writing to the Flash.

Only word-aligned writes are allowed. Byte or half-word-aligned writes will result in a hard fault.

11.2 Erasing a page in Flash

When erase is enabled, the Flash can be erased page by page using the ERASEPAGE register.

After erasing a Flash page, all bits in the page are set to '1'. The time it takes to erase a page is specified by $t_{ERASEPAGE}$. The CPU is halted while the NVMC performs the erase operation.

11.3 Writing to user information configuration registers (UICR)

User information configuration registers (UICR) are written in the same way as Flash. After UICR has been written, the new UICR configuration will only take effect after a reset.

UICR can only be written n_{WRITE} number of times before an erase must be performed using *ERASEUICR* or *ERASEALL*.

The time it takes to write a word to the UICR is specified by t_{WRITE}. The CPU is halted while the NVMC is writing to the UICR.

11.4 Erasing user information configuration registers (UICR)

When erase is enabled, UICR can be erased using the ERASEUICR register.

After erasing UICR all bits in UICR are set to '1'. The time it takes to erase UICR is specified by $t_{ERASEPAGE}$. The CPU is halted while the NVMC performs the erase operation.

11.5 Erase all

When erase is enabled, the whole Flash and UICR can be erased in one operation by using the ERASEALL register. ERASEALL will not erase the factory information configuration registers (FICR).

The time it takes to perform an ERASEALL command is specified by $t_{ERASEALL}$ The CPU is halted while the NVMC performs the erase operation.

11.6 Cache

An instruction cache (I-Cache) can be enabled for the ICODE bus in the NVMC.

See the Memory map in *Memory map* on page 24 for the location of Flash.

A cache hit is an instruction fetch from the cache, and it has a 0 wait-state delay. The number of wait-states for a cache miss, where the instruction is not available in the cache and needs to be fetched from Flash, depends on the processor frequency and is shown in *CPU* on page 21

Enabling the cache can increase CPU performance and reduce power consumption by reducing the number of wait cycles and the number of flash accesses. This will depend on the cache hit rate. Cache will use some current when enabled. If the reduction in average current due to reduced flash accesses is larger than the cache power requirement, the average current to execute the program code will reduce.

When disabled, the cache does not use current and does not retain its content.

It is possible to enable cache profiling to analyze the performance of the cache for your program using the *ICACHECNF* register. When profiling is enabled, the *IHIT* and *IMISS* registers are incremented for every instruction cache hit or miss respectively. The hit and miss profiling registers do not wrap around after reaching the maximum value. If the maximum value is reached, consider profiling for a shorter duration to get correct numbers.

11.7 Registers

Table 13: Instances

Base address	Peripheral	Instance	Description	Configuration
0x4001E000	NVMC	NVMC	Non-Volatile Memory Controller	

Table 14: Register Overview

Register	Offset	Description	
READY	0x400	Ready flag	
CONFIG	0x504	Configuration register	
ERASEPAGE	0x508	Register for erasing a page in Code area	
ERASEPCR1	0x508	Register for erasing a page in Code area. Equivalent to ERASEPAGE.	Deprecated
ERASEALL	0x50C	Register for erasing all non-volatile user memory	
ERASEPCR0	0x510	Register for erasing a page in Code area. Equivalent to ERASEPAGE.	Deprecated
ERASEUICR	0x514	Register for erasing User Information Configuration Registers	
ICACHECNF	0x540	I-Code cache configuration register.	
IHIT	0x548	I-Code cache hit counter.	
IMISS	0x54C	I-Code cache miss counter.	

11.7.1 READY

Address offset: 0x400

Ready flag

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	R	READY			NVMC is ready or busy																													
			Busy	0	NVMC is busy (on-going write or erase operation)																													
			Ready	1	NVMC is ready																													

11.7.2 CONFIG

Address offset: 0x504

Configuration register

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	WEN			Program memory access mode. It is strongly recommended to only activate erase and write modes when they are actively used. Enabling write or erase will invalidate the cache and keep it invalidated.																													
			Ren	0	Read only access																													
			Wen	1	Write Enabled																													
			Een	2	Erase enabled																													

11.7.3 ERASEPAGE

Address offset: 0x508

Register for erasing a page in Code area

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	ERASEPAGE			Register for starting erase of a page in Code area The value is the address to the page to be erased. (Addresses of first word in page). Note that code erase has to be enabled by CONFIG.EEN before the page can be erased. Attempts to erase pages that are outside the code area may result in undesirable behaviour, e.g. the wrong page may be erased.																											

11.7.4 ERASEPCR1 (Deprecated)

Address offset: 0x508

Register for erasing a page in Code area. Equivalent to ERASEPAGE.

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	ERASEPCR1			Register for erasing a page in Code area. Equivalent to ERASEPAGE.																													

11.7.5 ERASEALL

Address offset: 0x50C

Register for erasing all non-volatile user memory

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
A	RW	ERASEALL			Erase all non-volatile memory including UICR registers. Note that code erase has to be enabled by CONFIG.EEN before the UICR can be erased.
			NoOperation	0	No operation
			Erase	1	Start chip erase

11.7.6 ERASEPCR0 (Deprecated)

Address offset: 0x510

Register for erasing a page in Code area. Equivalent to ERASEPAGE.

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
A	RW	ERASEPCR0			Register for starting erase of a page in Code area. Equivalent to ERASEPAGE.

11.7.7 ERASEUICR

Address offset: 0x514

Register for erasing User Information Configuration Registers

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
A	RW	ERASEUICR			Register starting erase of all User Information Configuration Registers. Note that code erase has to be enabled by CONFIG.EEN before the UICR can be erased.
			NoOperation	0	No operation
			Erase	1	Start erase of UICR

11.7.8 ICACHECNF

Address offset: 0x540

I-Code cache configuration register.

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			B A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
A	RW	CACHEEN			Cache enable
			Disabled	0	Disable cache. Invalidates all cache entries.
			Enabled	1	Enable cache
B	RW	CACHEPROFEN			Cache profiling enable
			Disabled	0	Disable cache profiling
			Enabled	1	Enable cache profiling

11.7.9 IHIT

Address offset: 0x548

I-Code cache hit counter.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	HITS			Number of cache hits

11.7.10 IMISS

Address offset: 0x54C

I-Code cache miss counter.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	MISSES			Number of cache misses

11.8 Electrical specification

11.8.1 Flash programming

Symbol	Description	Min.	Typ.	Max.	Units
$n_{WRITE,BLOCK}$	Amount of writes allowed in a block between erase			181	
t_{WRITE}	Time to write one word	67.5		338	µs
$t_{ERASEPAGE}$	Time to erase one page	2.05		89.7	ms
$t_{ERASEALL}$	Time to erase all flash	6.72		295.3	ms

11.8.2 Cache size

Symbol	Description	Min.	Typ.	Max.	Units
$Size_{ICODE}$	I-Code cache size		2048		Bytes

12 BPROT — Block protection

The mechanism for protecting non-volatile memory can be used to prevent application code from erasing or writing to protected blocks.

Non-volatile memory can be protected from erases and writes depending on the settings in the CONFIG registers. One bit in a CONFIG register represents one protected block of 4 kB. There are four CONFIG registers of 32 bits, which means there are 128 protectable blocks in total.

> **Important:** If an erase or write to a protected block is detected, the CPU will hard fault. If an ERASEALL operation is attempted from the CPU while any block is protected, it will be blocked and the CPU will hard fault.

On reset, all the protection bits are cleared. To ensure safe operation, the first task after reset must be to set the protection bits. The only way of clearing protection bits is by resetting the device from any reset source.

The protection mechanism is turned off when in debug interface mode (a debugger is connected) and the DISABLEINDEBUG register is set to disable. For more information, see *Debug and trace* on page 72.

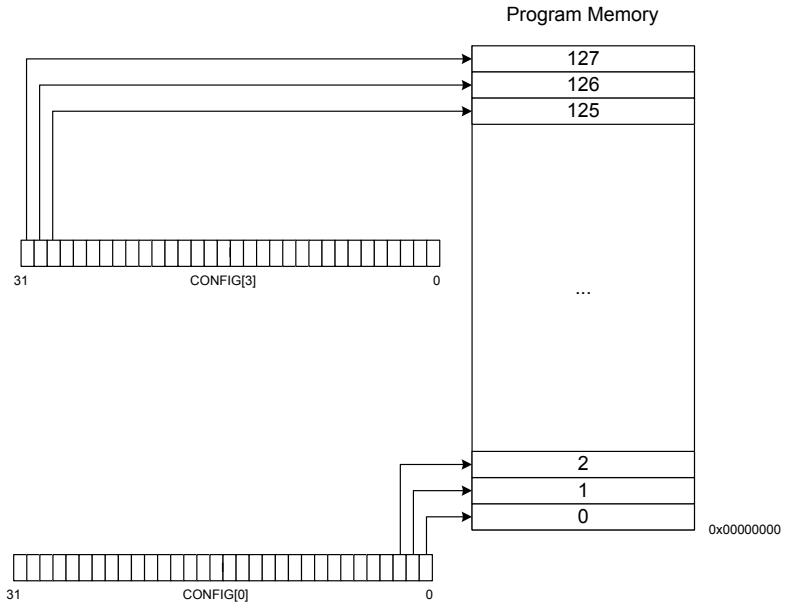


Figure 9: Protected regions of program memory

12.1 Registers

Table 15: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40000000	BPROT	BPROT	Block Protect	

Table 16: Register Overview

Register	Offset	Description	
CONFIG0	0x600	Block protect configuration register 0	
CONFIG1	0x604	Block protect configuration register 1	
DISABLEINDEBUG	0x608	Disable protection mechanism in debug interface mode	
	0x60C		Reserved
CONFIG2	0x610	Block protect configuration register 2	
CONFIG3	0x614	Block protect configuration register 3	

12.1.1 CONFIG0

Address offset: 0x600

Block protect configuration register 0

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	REGION0																																	
			Disabled	0																															
			Enabled	1																															
B	RW	REGION1																																	
			Disabled	0																															
			Enabled	1																															
C	RW	REGION2																																	
			Disabled	0																															
			Enabled	1																															
D	RW	REGION3																																	
			Disabled	0																															
			Enabled	1																															
E	RW	REGION4																																	
			Disabled	0																															
			Enabled	1																															
F	RW	REGION5																																	
			Disabled	0																															
			Enabled	1																															
G	RW	REGION6																																	
			Disabled	0																															
			Enabled	1																															
H	RW	REGION7																																	
			Disabled	0																															
			Enabled	1																															
I	RW	REGION8																																	
			Disabled	0																															
			Enabled	1																															
J	RW	REGION9																																	
			Disabled	0																															
			Enabled	1																															
K	RW	REGION10																																	
			Disabled	0																															
			Enabled	1																															
L	RW	REGION11																																	
			Disabled	0																															
			Enabled	1																															
M	RW	REGION12																																	
			Disabled	0																															
			Enabled	1																															
N	RW	REGION13																																	
			Disabled	0																															
			Enabled	1																															
O	RW	REGION14																																	
			Disabled	0																															
			Enabled	1																															
P	RW	REGION15																																	
			Disabled	0																															
			Enabled	1																															
Q	RW	REGION16																																	
			Disabled	0																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Enabled	1																															
R	RW	REGION17																																	
			Disabled	0																															
			Enabled	1																															
S	RW	REGION18																																	
			Disabled	0																															
			Enabled	1																															
T	RW	REGION19																																	
			Disabled	0																															
			Enabled	1																															
U	RW	REGION20																																	
			Disabled	0																															
			Enabled	1																															
V	RW	REGION21																																	
			Disabled	0																															
			Enabled	1																															
W	RW	REGION22																																	
			Disabled	0																															
			Enabled	1																															
X	RW	REGION23																																	
			Disabled	0																															
			Enabled	1																															
Y	RW	REGION24																																	
			Disabled	0																															
			Enabled	1																															
Z	RW	REGION25																																	
			Disabled	0																															
			Enabled	1																															
a	RW	REGION26																																	
			Disabled	0																															
			Enabled	1																															
b	RW	REGION27																																	
			Disabled	0																															
			Enabled	1																															
c	RW	REGION28																																	
			Disabled	0																															
			Enabled	1																															
d	RW	REGION29																																	
			Disabled	0																															
			Enabled	1																															
e	RW	REGION30																																	
			Disabled	0																															
			Enabled	1																															
f	RW	REGION31																																	
			Disabled	0																															
			Enabled	1																															

12.1.2 CONFIG1

Address offset: 0x604

Block protect configuration register 1

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	REGION32																																	
			Disabled	0																															
			Enabled	1																															
B	RW	REGION33																																	
			Disabled	0																															
			Enabled	1																															
C	RW	REGION34																																	
			Disabled	0																															
			Enabled	1																															
D	RW	REGION35																																	
			Disabled	0																															
			Enabled	1																															
E	RW	REGION36																																	
			Disabled	0																															
			Enabled	1																															
F	RW	REGION37																																	
			Disabled	0																															
			Enabled	1																															
G	RW	REGION38																																	
			Disabled	0																															
			Enabled	1																															
H	RW	REGION39																																	
			Disabled	0																															
			Enabled	1																															
I	RW	REGION40																																	
			Disabled	0																															
			Enabled	1																															
J	RW	REGION41																																	
			Disabled	0																															
			Enabled	1																															
K	RW	REGION42																																	
			Disabled	0																															
			Enabled	1																															
L	RW	REGION43																																	
			Disabled	0																															
			Enabled	1																															
M	RW	REGION44																																	
			Disabled	0																															
			Enabled	1																															
N	RW	REGION45																																	
			Disabled	0																															
			Enabled	1																															
O	RW	REGION46																																	
			Disabled	0																															
			Enabled	1																															
P	RW	REGION47																																	
			Disabled	0																															
			Enabled	1																															
Q	RW	REGION48																																	
			Disabled	0																															
			Enabled	1																															
R	RW	REGION49																																	
			Disabled	0																															
			Enabled	1																															
S	RW	REGION50																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Disabled	0																															
			Enabled	1																															
T	RW	REGION51																																	
			Disabled	0																															
			Enabled	1																															
U	RW	REGION52																																	
			Disabled	0																															
			Enabled	1																															
V	RW	REGION53																																	
			Disabled	0																															
			Enabled	1																															
W	RW	REGION54																																	
			Disabled	0																															
			Enabled	1																															
X	RW	REGION55																																	
			Disabled	0																															
			Enabled	1																															
Y	RW	REGION56																																	
			Disabled	0																															
			Enabled	1																															
Z	RW	REGION57																																	
			Disabled	0																															
			Enabled	1																															
a	RW	REGION58																																	
			Disabled	0																															
			Enabled	1																															
b	RW	REGION59																																	
			Disabled	0																															
			Enabled	1																															
c	RW	REGION60																																	
			Disabled	0																															
			Enabled	1																															
d	RW	REGION61																																	
			Disabled	0																															
			Enabled	1																															
e	RW	REGION62																																	
			Disabled	0																															
			Enabled	1																															
f	RW	REGION63																																	
			Disabled	0																															
			Enabled	1																															

12.1.3 DISABLEINDEBUG

Address offset: 0x608

Disable protection mechanism in debug interface mode

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000001				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	DISABLEINDEBUG																																	
			Disabled	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000001				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Enabled	0																															Enable in debug

12.1.4 CONFIG2

Address offset: 0x610

Block protect configuration register 2

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	REGION64																																	Enable protection for region 64. Write '0' has no effect.
			Disabled	0																															Protection disabled
			Enabled	1																															Protection enabled
B	RW	REGION65																																	Enable protection for region 65. Write '0' has no effect.
			Disabled	0																															Protection disabled
			Enabled	1																															Protection enabled
C	RW	REGION66																																	Enable protection for region 66. Write '0' has no effect.
			Disabled	0																															Protection disabled
			Enabled	1																															Protection enabled
D	RW	REGION67																																	Enable protection for region 67. Write '0' has no effect.
			Disabled	0																															Protection disabled
			Enabled	1																															Protection enabled
E	RW	REGION68																																	Enable protection for region 68. Write '0' has no effect.
			Disabled	0																															Protection disabled
			Enabled	1																															Protection enabled
F	RW	REGION69																																	Enable protection for region 69. Write '0' has no effect.
			Disabled	0																															Protection disabled
			Enabled	1																															Protection enabled
G	RW	REGION70																																	Enable protection for region 70. Write '0' has no effect.
			Disabled	0																															Protection disabled
			Enabled	1																															Protection enabled
H	RW	REGION71																																	Enable protection for region 71. Write '0' has no effect.
			Disabled	0																															Protection disabled
			Enabled	1																															Protection enabled
I	RW	REGION72																																	Enable protection for region 72. Write '0' has no effect.
			Disabled	0																															Protection disabled
			Enabled	1																															Protection enabled
J	RW	REGION73																																	Enable protection for region 73. Write '0' has no effect.
			Disabled	0																															Protection disabled
			Enabled	1																															Protection enabled
K	RW	REGION74																																	Enable protection for region 74. Write '0' has no effect.
			Disabled	0																															Protection disabled
			Enabled	1																															Protection enabled
L	RW	REGION75																																	Enable protection for region 75. Write '0' has no effect.
			Disabled	0																															Protection disabled
			Enabled	1																															Protection enabled
M	RW	REGION76																																	Enable protection for region 76. Write '0' has no effect.
			Disabled	0																															Protection disabled
			Enabled	1																															Protection enabled
N	RW	REGION77																																	Enable protection for region 77. Write '0' has no effect.
			Disabled	0																															Protection disabled
			Enabled	1																															Protection enabled
O	RW	REGION78																																	Enable protection for region 78. Write '0' has no effect.
			Disabled	0																															Protection disabled

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Enabled	1																															
P	RW	REGION79																																	
			Disabled	0																															
			Enabled	1																															
Q	RW	REGION80																																	
			Disabled	0																															
			Enabled	1																															
R	RW	REGION81																																	
			Disabled	0																															
			Enabled	1																															
S	RW	REGION82																																	
			Disabled	0																															
			Enabled	1																															
T	RW	REGION83																																	
			Disabled	0																															
			Enabled	1																															
U	RW	REGION84																																	
			Disabled	0																															
			Enabled	1																															
V	RW	REGION85																																	
			Disabled	0																															
			Enabled	1																															
W	RW	REGION86																																	
			Disabled	0																															
			Enabled	1																															
X	RW	REGION87																																	
			Disabled	0																															
			Enabled	1																															
Y	RW	REGION88																																	
			Disabled	0																															
			Enabled	1																															
Z	RW	REGION89																																	
			Disabled	0																															
			Enabled	1																															
a	RW	REGION90																																	
			Disabled	0																															
			Enabled	1																															
b	RW	REGION91																																	
			Disabled	0																															
			Enabled	1																															
c	RW	REGION92																																	
			Disabled	0																															
			Enabled	1																															
d	RW	REGION93																																	
			Disabled	0																															
			Enabled	1																															
e	RW	REGION94																																	
			Disabled	0																															
			Enabled	1																															
f	RW	REGION95																																	
			Disabled	0																															
			Enabled	1																															

12.1.5 CONFIG3

Address offset: 0x614

Block protect configuration register 3

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	REGION96																																	
			Disabled	0																															
			Enabled	1																															
B	RW	REGION97																																	
			Disabled	0																															
			Enabled	1																															
C	RW	REGION98																																	
			Disabled	0																															
			Enabled	1																															
D	RW	REGION99																																	
			Disabled	0																															
			Enabled	1																															
E	RW	REGION100																																	
			Disabled	0																															
			Enabled	1																															
F	RW	REGION101																																	
			Disabled	0																															
			Enabled	1																															
G	RW	REGION102																																	
			Disabled	0																															
			Enabled	1																															
H	RW	REGION103																																	
			Disabled	0																															
			Enabled	1																															
I	RW	REGION104																																	
			Disabled	0																															
			Enabled	1																															
J	RW	REGION105																																	
			Disabled	0																															
			Enabled	1																															
K	RW	REGION106																																	
			Disabled	0																															
			Enabled	1																															
L	RW	REGION107																																	
			Disabled	0																															
			Enabled	1																															
M	RW	REGION108																																	
			Disabled	0																															
			Enabled	1																															
N	RW	REGION109																																	
			Disabled	0																															
			Enabled	1																															
O	RW	REGION110																																	
			Disabled	0																															
			Enabled	1																															
P	RW	REGION111																																	
			Disabled	0																															
			Enabled	1																															
Q	RW	REGION112																																	
			Disabled	0																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Enabled	1																															
R	RW	REGION113																																	
			Disabled	0																															
			Enabled	1																															
S	RW	REGION114																																	
			Disabled	0																															
			Enabled	1																															
T	RW	REGION115																																	
			Disabled	0																															
			Enabled	1																															
U	RW	REGION116																																	
			Disabled	0																															
			Enabled	1																															
V	RW	REGION117																																	
			Disabled	0																															
			Enabled	1																															
W	RW	REGION118																																	
			Disabled	0																															
			Enabled	1																															
X	RW	REGION119																																	
			Disabled	0																															
			Enabled	1																															
Y	RW	REGION120																																	
			Disabled	0																															
			Enabled	1																															
Z	RW	REGION121																																	
			Disabled	0																															
			Enabled	1																															
a	RW	REGION122																																	
			Disabled	0																															
			Enabled	1																															
b	RW	REGION123																																	
			Disabled	0																															
			Enabled	1																															
c	RW	REGION124																																	
			Disabled	0																															
			Enabled	1																															
d	RW	REGION125																																	
			Disabled	0																															
			Enabled	1																															
e	RW	REGION126																																	
			Disabled	0																															
			Enabled	1																															
f	RW	REGION127																																	
			Disabled	0																															
			Enabled	1																															

13 FICR — Factory information configuration registers

Factory information configuration registers (FICR) are pre-programmed in factory and cannot be erased by the user. These registers contain chip-specific information and configuration.

13.1 Registers

Table 17: Instances

Base address	Peripheral	Instance	Description	Configuration
0x10000000	FICR	FICR	Factory Information Configuration	

Table 18: Register Overview

Register	Offset	Description	
CODEPAGESIZE	0x010	Code memory page size	
CODESIZE	0x014	Code memory size	
DEVICEID[0]	0x060	Device identifier	
DEVICEID[1]	0x064	Device identifier	
ER[0]	0x080	Encryption Root, word 0	
ER[1]	0x084	Encryption Root, word 1	
ER[2]	0x088	Encryption Root, word 2	
ER[3]	0x08C	Encryption Root, word 3	
IR[0]	0x090	Identity Root, word 0	
IR[1]	0x094	Identity Root, word 1	
IR[2]	0x098	Identity Root, word 2	
IR[3]	0x09C	Identity Root, word 3	
DEVICEADDRTYPE	0x0A0	Device address type	
DEVICEADDR[0]	0x0A4	Device address 0	
DEVICEADDR[1]	0x0A8	Device address 1	
INFO.PART	0x100	Part code	
INFO.VARIANT	0x104	Part Variant, Hardware version and Production configuration	
INFO.PACKAGE	0x108	Package option	
INFO.RAM	0x10C	RAM variant	
INFO.FLASH	0x110	Flash variant	
	0x114		Reserved
	0x118		Reserved
	0x11C		Reserved
TEMP.A0	0x404	Slope definition A0.	
TEMP.A1	0x408	Slope definition A1.	
TEMP.A2	0x40C	Slope definition A2.	
TEMP.A3	0x410	Slope definition A3.	
TEMP.A4	0x414	Slope definition A4.	
TEMP.A5	0x418	Slope definition A5.	
TEMP.B0	0x41C	y-intercept B0.	
TEMP.B1	0x420	y-intercept B1.	
TEMP.B2	0x424	y-intercept B2.	
TEMP.B3	0x428	y-intercept B3.	
TEMP.B4	0x42C	y-intercept B4.	
TEMP.B5	0x430	y-intercept B5.	
TEMP.T0	0x434	Segment end T0.	
TEMP.T1	0x438	Segment end T1.	
TEMP.T2	0x43C	Segment end T2.	
TEMP.T3	0x440	Segment end T3.	
TEMP.T4	0x444	Segment end T4.	

Register	Offset	Description
NFC.TAGHEADER0	0x450	Default header for NFC Tag. Software can read these values to populate NFCID1_3RD_LAST, NFCID1_2ND_LAST and NFCID1_LAST.
NFC.TAGHEADER1	0x454	Default header for NFC Tag. Software can read these values to populate NFCID1_3RD_LAST, NFCID1_2ND_LAST and NFCID1_LAST.
NFC.TAGHEADER2	0x458	Default header for NFC Tag. Software can read these values to populate NFCID1_3RD_LAST, NFCID1_2ND_LAST and NFCID1_LAST.
NFC.TAGHEADER3	0x45C	Default header for NFC Tag. Software can read these values to populate NFCID1_3RD_LAST, NFCID1_2ND_LAST and NFCID1_LAST.

13.1.1 CODEPAGESIZE

Address offset: 0x010

Code memory page size

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	R	CODEPAGESIZE																								Code memory page size									

13.1.2 CODESIZE

Address offset: 0x014

Code memory size

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	R	CODESIZE																								Code memory size in number of pages Total code space is: CODEPAGESIZE * CODESIZE									

13.1.3 DEVICEID[0]

Address offset: 0x060

Device identifier

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	R	DEVICEID																								64 bit unique device identifier DEVICEID[0] contains the least significant bits of the device identifier. DEVICEID[1] contains the most significant bits of the device identifier.									

13.1.4 DEVICEID[1]

Address offset: 0x064

Device identifier

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	R	DEVICEID																								64 bit unique device identifier									

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
					DEVICEID[0] contains the least significant bits of the device identifier. DEVICEID[1] contains the most significant bits of the device identifier.

13.1.5 ER[0]

Address offset: 0x080

Encryption Root, word 0

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	R	ER			Encryption Root, word n

13.1.6 ER[1]

Address offset: 0x084

Encryption Root, word 1

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	R	ER			Encryption Root, word n

13.1.7 ER[2]

Address offset: 0x088

Encryption Root, word 2

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	R	ER			Encryption Root, word n

13.1.8 ER[3]

Address offset: 0x08C

Encryption Root, word 3

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	R	ER			Encryption Root, word n

13.1.9 IR[0]

Address offset: 0x090

Identity Root, word 0

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	IR																																	
				Identity Root, word n																																	

13.1.10 IR[1]

Address offset: 0x094

Identity Root, word 1

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	IR																																	
				Identity Root, word n																																	

13.1.11 IR[2]

Address offset: 0x098

Identity Root, word 2

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	IR																																	
				Identity Root, word n																																	

13.1.12 IR[3]

Address offset: 0x09C

Identity Root, word 3

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	IR																																	
				Identity Root, word n																																	

13.1.13 DEVICEADDRTYPE

Address offset: 0x0A0

Device address type

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	DEVICEADDRTYPE																																	
			Public	0																															
			Random	1																															

13.1.14 DEVICEADDR[0]

Address offset: 0x0A4

Device address 0

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF			**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	R	DEVICEADDR			48 bit device address DEVICEADDR[0] contains the least significant bits of the device address. DEVICEADDR[1] contains the most significant bits of the device address. Only bits [15:0] of DEVICEADDR[1] are used.

13.1.15 DEVICEADDR[1]

Address offset: 0x0A8

Device address 1

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF			**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	R	DEVICEADDR			48 bit device address DEVICEADDR[0] contains the least significant bits of the device address. DEVICEADDR[1] contains the most significant bits of the device address. Only bits [15:0] of DEVICEADDR[1] are used.

13.1.16 INFO.PART

Address offset: 0x100

Part code

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00052832			**0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0 0 1 0 1 0 0 0 0 0 1 1 0 0 1 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	R	PART			Part code
			N52832	0x52832	nRF52832
			Unspecified	0xFFFFFFFF	Unspecified

13.1.17 INFO.VARIANT

Address offset: 0x104

Part Variant, Hardware version and Production configuration

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x41414142			**0 1 0 0 0 0 0 1 0 1 0 0 0 0 0 1 0 1 0 0 0 0 0 1 0 1 0 0 0 0 1 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	R	VARIANT			Part Variant, Hardware version and Production configuration, encoded as ASCII
			AAAA	0x41414141	AAAA
			AAAB	0x41414142	AAAB
			AABA	0x41414241	AABA
			AABB	0x41414242	AABB
			AAB0	0x41414230	AAB0
			AAE0	0x41414530	AAE0
			Unspecified	0xFFFFFFFF	Unspecified

13.1.18 INFO.PACKAGE

Address offset: 0x108

Package option

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00002000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	R	PACKAGE			Package option																													
			QF	0x2000	QFxx - 48-pin QFN																													
			CH	0x2001	CHxx - 7x8 WLCSP 56 balls																													
			CI	0x2002	CIxx - 7x8 WLCSP 56 balls																													
			CK	0x2005	CKxx - 7x8 WLCSP 56 balls with backside coating for light protection																													
			Unspecified	0xFFFFFFFF	Unspecified																													

13.1.19 INFO.RAM

Address offset: 0x10C

RAM variant

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000040			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	R	RAM			RAM variant																													
			K16	0x10	16 kByte RAM																													
			K32	0x20	32 kByte RAM																													
			K64	0x40	64 kByte RAM																													
			Unspecified	0xFFFFFFFF	Unspecified																													

13.1.20 INFO.FLASH

Address offset: 0x110

Flash variant

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000200			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	R	FLASH			Flash variant																													
			K128	0x80	128 kByte FLASH																													
			K256	0x100	256 kByte FLASH																													
			K512	0x200	512 kByte FLASH																													
			Unspecified	0xFFFFFFFF	Unspecified																													

13.1.21 TEMP.A0

Address offset: 0x404

Slope definition A0.

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																							A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000320			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	R	A			A (slope definition) register.																													

13.1.22 TEMP.A1

Address offset: 0x408

Slope definition A1.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																					A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000343	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**1**	**0**	**0**	**0**	**0**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	R	A			A (slope definition) register.

13.1.23 TEMP.A2

Address offset: 0x40C

Slope definition A2.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																					A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x0000035D	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**1**	**0**	**1**	**1**	**1**	**0**	**1**	

Id	RW	Field	Value Id	Value	Description
A	R	A			A (slope definition) register.

13.1.24 TEMP.A3

Address offset: 0x410

Slope definition A3.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																					A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000400	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	R	A			A (slope definition) register.

13.1.25 TEMP.A4

Address offset: 0x414

Slope definition A4.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																					A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000452	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**1**	**0**	**1**	**0**	**0**	**1**	**0**

Id	RW	Field	Value Id	Value	Description
A	R	A			A (slope definition) register.

13.1.26 TEMP.A5

Address offset: 0x418

Slope definition A5.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																					A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x0000037B	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**1**	**1**	**1**	**1**	**0**	**1**	**1**	

Id	RW	Field	Value Id	Value	Description
A	R	A			A (slope definition) register.

13.1.27 TEMP.B0

Address offset: 0x41C

y-intercept B0.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00003FCC	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**0**	**1**	**1**	**0**	**0**	

Id	RW	Field	Value Id	Value	Description
A	R	B			B (y-intercept)

13.1.28 TEMP.B1

Address offset: 0x420

y-intercept B1.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00003F98	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**0**	**1**	**1**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	R	B			B (y-intercept)

13.1.29 TEMP.B2

Address offset: 0x424

y-intercept B2.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00003F98	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**0**	**1**	**1**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	R	B			B (y-intercept)

13.1.30 TEMP.B3

Address offset: 0x428

y-intercept B3.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000012	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**1**	**0**

Id	RW	Field	Value Id	Value	Description
A	R	B			B (y-intercept)

13.1.31 TEMP.B4

Address offset: 0x42C

y-intercept B4.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x0000004D	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**1**	**1**	**0**	**1**

Id	RW	Field	Value Id	Value	Description
A	R	B			B (y-intercept)

13.1.32 TEMP.B5

Address offset: 0x430

y-intercept B5.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00003E10	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	R	B			B (y-intercept)

13.1.33 TEMP.T0

Address offset: 0x434

Segment end T0.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x000000E2	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**0**	**0**	**0**	**1**	**0**

Id	RW	Field	Value Id	Value	Description
A	R	T			T (segment end)register.

13.1.34 TEMP.T1

Address offset: 0x438

Segment end T1.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	R	T			T (segment end)register.

13.1.35 TEMP.T2

Address offset: 0x43C

Segment end T2.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x00000014	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	R	T			T (segment end)register.

13.1.36 TEMP.T3

Address offset: 0x440

Segment end T3.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x00000019	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**0**	**1**

Id	RW	Field	Value Id	Value	Description
A	R	T			T (segment end)register.

13.1.37 TEMP.T4

Address offset: 0x444

Segment end T4.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000050				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	R	T																										T (segment end)register.							

13.1.38 NFC.TAGHEADER0

Address offset: 0x450

Default header for NFC Tag. Software can read these values to populate NFCID1_3RD_LAST, NFCID1_2ND_LAST and NFCID1_LAST.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				D	D	D	D	D	D	D	D	C	C	C	C	C	C	C	C	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A
Reset 0xFFFFFF5F				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**1**	**0**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	R	MFGID																										Default Manufacturer ID: Nordic Semiconductor ASA has ICM 0x5F							
B	R	UD1																										Unique identifier byte 1							
C	R	UD2																										Unique identifier byte 2							
D	R	UD3																										Unique identifier byte 3							

13.1.39 NFC.TAGHEADER1

Address offset: 0x454

Default header for NFC Tag. Software can read these values to populate NFCID1_3RD_LAST, NFCID1_2ND_LAST and NFCID1_LAST.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				D	D	D	D	D	D	D	D	C	C	C	C	C	C	C	C	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	R	UD4																										Unique identifier byte 4							
B	R	UD5																										Unique identifier byte 5							
C	R	UD6																										Unique identifier byte 6							
D	R	UD7																										Unique identifier byte 7							

13.1.40 NFC.TAGHEADER2

Address offset: 0x458

Default header for NFC Tag. Software can read these values to populate NFCID1_3RD_LAST, NFCID1_2ND_LAST and NFCID1_LAST.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				D	D	D	D	D	D	D	D	C	C	C	C	C	C	C	C	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	R	UD8																										Unique identifier byte 8							
B	R	UD9																										Unique identifier byte 9							
C	R	UD10																										Unique identifier byte 10							
D	R	UD11																										Unique identifier byte 11							

13.1.41 NFC.TAGHEADER3

Address offset: 0x45C

Default header for NFC Tag. Software can read these values to populate NFCID1_3RD_LAST, NFCID1_2ND_LAST and NFCID1_LAST.

Bit number																															
Id																															
Reset 0xFFFFFFFF																															
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	R	UD12			Unique identifier byte 12
B	R	UD13			Unique identifier byte 13
C	R	UD14			Unique identifier byte 14
D	R	UD15			Unique identifier byte 15

14 UICR — User information configuration registers

The user information configuration registers (UICRs) are non-volatile memory (NVM) registers for configuring user specific settings.

For information on writing UICR registers, see the *NVMC — Non-volatile memory controller* on page 29 and *Memory* on page 23 chapters.

14.1 Registers

Table 19: Instances

Base address	Peripheral	Instance	Description	Configuration
0x10001000	UICR	UICR	User Information Configuration	

Table 20: Register Overview

Register	Offset	Description	
	0x000		Reserved
	0x004		Reserved
	0x008		Reserved
	0x010		Reserved
NRFFW[0]	0x014	Reserved for Nordic firmware design	
NRFFW[1]	0x018	Reserved for Nordic firmware design	
NRFFW[2]	0x01C	Reserved for Nordic firmware design	
NRFFW[3]	0x020	Reserved for Nordic firmware design	
NRFFW[4]	0x024	Reserved for Nordic firmware design	
NRFFW[5]	0x028	Reserved for Nordic firmware design	
NRFFW[6]	0x02C	Reserved for Nordic firmware design	
NRFFW[7]	0x030	Reserved for Nordic firmware design	
NRFFW[8]	0x034	Reserved for Nordic firmware design	
NRFFW[9]	0x038	Reserved for Nordic firmware design	
NRFFW[10]	0x03C	Reserved for Nordic firmware design	
NRFFW[11]	0x040	Reserved for Nordic firmware design	
NRFFW[12]	0x044	Reserved for Nordic firmware design	
NRFFW[13]	0x048	Reserved for Nordic firmware design	
NRFFW[14]	0x04C	Reserved for Nordic firmware design	
NRFHW[0]	0x050	Reserved for Nordic hardware design	
NRFHW[1]	0x054	Reserved for Nordic hardware design	
NRFHW[2]	0x058	Reserved for Nordic hardware design	
NRFHW[3]	0x05C	Reserved for Nordic hardware design	
NRFHW[4]	0x060	Reserved for Nordic hardware design	
NRFHW[5]	0x064	Reserved for Nordic hardware design	
NRFHW[6]	0x068	Reserved for Nordic hardware design	
NRFHW[7]	0x06C	Reserved for Nordic hardware design	
NRFHW[8]	0x070	Reserved for Nordic hardware design	
NRFHW[9]	0x074	Reserved for Nordic hardware design	
NRFHW[10]	0x078	Reserved for Nordic hardware design	
NRFHW[11]	0x07C	Reserved for Nordic hardware design	
CUSTOMER[0]	0x080	Reserved for customer	
CUSTOMER[1]	0x084	Reserved for customer	
CUSTOMER[2]	0x088	Reserved for customer	
CUSTOMER[3]	0x08C	Reserved for customer	
CUSTOMER[4]	0x090	Reserved for customer	
CUSTOMER[5]	0x094	Reserved for customer	
CUSTOMER[6]	0x098	Reserved for customer	

Register	Offset	Description
CUSTOMER[7]	0x09C	Reserved for customer
CUSTOMER[8]	0x0A0	Reserved for customer
CUSTOMER[9]	0x0A4	Reserved for customer
CUSTOMER[10]	0x0A8	Reserved for customer
CUSTOMER[11]	0x0AC	Reserved for customer
CUSTOMER[12]	0x0B0	Reserved for customer
CUSTOMER[13]	0x0B4	Reserved for customer
CUSTOMER[14]	0x0B8	Reserved for customer
CUSTOMER[15]	0x0BC	Reserved for customer
CUSTOMER[16]	0x0C0	Reserved for customer
CUSTOMER[17]	0x0C4	Reserved for customer
CUSTOMER[18]	0x0C8	Reserved for customer
CUSTOMER[19]	0x0CC	Reserved for customer
CUSTOMER[20]	0x0D0	Reserved for customer
CUSTOMER[21]	0x0D4	Reserved for customer
CUSTOMER[22]	0x0D8	Reserved for customer
CUSTOMER[23]	0x0DC	Reserved for customer
CUSTOMER[24]	0x0E0	Reserved for customer
CUSTOMER[25]	0x0E4	Reserved for customer
CUSTOMER[26]	0x0E8	Reserved for customer
CUSTOMER[27]	0x0EC	Reserved for customer
CUSTOMER[28]	0x0F0	Reserved for customer
CUSTOMER[29]	0x0F4	Reserved for customer
CUSTOMER[30]	0x0F8	Reserved for customer
CUSTOMER[31]	0x0FC	Reserved for customer
PSELRESET[0]	0x200	Mapping of the nRESET function (see POWER chapter for details)
PSELRESET[1]	0x204	Mapping of the nRESET function (see POWER chapter for details)
APPROTECT	0x208	Access Port protection
NFCPINS	0x20C	Setting of pins dedicated to NFC functionality: NFC antenna or GPIO

14.1.1 NRFFW[0]

Address offset: 0x014

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

14.1.2 NRFFW[1]

Address offset: 0x018

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

14.1.3 NRFFW[2]

Address offset: 0x01C

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

14.1.4 NRFFW[3]

Address offset: 0x020

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

14.1.5 NRFFW[4]

Address offset: 0x024

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

14.1.6 NRFFW[5]

Address offset: 0x028

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

14.1.7 NRFFW[6]

Address offset: 0x02C

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

14.1.8 NRFFW[7]

Address offset: 0x030

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

14.1.9 NRFFW[8]

Address offset: 0x034

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

14.1.10 NRFFW[9]

Address offset: 0x038

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

14.1.11 NRFFW[10]

Address offset: 0x03C

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

14.1.12 NRFFW[11]

Address offset: 0x040

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

14.1.13 NRFFW[12]

Address offset: 0x044

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

14.1.14 NRFFW[13]

Address offset: 0x048

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

14.1.15 NRFFW[14]

Address offset: 0x04C

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

14.1.16 NRFHW[0]

Address offset: 0x050

Reserved for Nordic hardware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

14.1.17 NRFHW[1]

Address offset: 0x054

Reserved for Nordic hardware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

14.1.18 NRFHW[2]

Address offset: 0x058

Reserved for Nordic hardware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

14.1.19 NRFHW[3]

Address offset: 0x05C

Reserved for Nordic hardware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

14.1.20 NRFHW[4]

Address offset: 0x060

Reserved for Nordic hardware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

14.1.21 NRFHW[5]

Address offset: 0x064

Reserved for Nordic hardware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

14.1.22 NRFHW[6]

Address offset: 0x068

Reserved for Nordic hardware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

14.1.23 NRFHW[7]

Address offset: 0x06C

Reserved for Nordic hardware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

14.1.24 NRFHW[8]

Address offset: 0x070

Reserved for Nordic hardware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

14.1.25 NRFHW[9]

Address offset: 0x074

Reserved for Nordic hardware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

14.1.26 NRFHW[10]

Address offset: 0x078

Reserved for Nordic hardware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

14.1.27 NRFHW[11]

Address offset: 0x07C

Reserved for Nordic hardware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

14.1.28 CUSTOMER[0]

Address offset: 0x080

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.29 CUSTOMER[1]

Address offset: 0x084

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.30 CUSTOMER[2]

Address offset: 0x088

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.31 CUSTOMER[3]

Address offset: 0x08C

Reserved for customer

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF					**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																															
A	RW	CUSTOMER			Reserved for customer																															

14.1.32 CUSTOMER[4]

Address offset: 0x090

Reserved for customer

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF					**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																															
A	RW	CUSTOMER			Reserved for customer																															

14.1.33 CUSTOMER[5]

Address offset: 0x094

Reserved for customer

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF					**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																															
A	RW	CUSTOMER			Reserved for customer																															

14.1.34 CUSTOMER[6]

Address offset: 0x098

Reserved for customer

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF					**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																															
A	RW	CUSTOMER			Reserved for customer																															

14.1.35 CUSTOMER[7]

Address offset: 0x09C

Reserved for customer

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF					**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																															
A	RW	CUSTOMER			Reserved for customer																															

14.1.36 CUSTOMER[8]

Address offset: 0x0A0

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.37 CUSTOMER[9]

Address offset: 0x0A4

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.38 CUSTOMER[10]

Address offset: 0x0A8

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.39 CUSTOMER[11]

Address offset: 0x0AC

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.40 CUSTOMER[12]

Address offset: 0x0B0

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.41 CUSTOMER[13]

Address offset: 0x0B4

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.42 CUSTOMER[14]

Address offset: 0x0B8

Reserved for customer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	RW	CUSTOMER																								Reserved for customer									

14.1.43 CUSTOMER[15]

Address offset: 0x0BC

Reserved for customer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	RW	CUSTOMER																								Reserved for customer									

14.1.44 CUSTOMER[16]

Address offset: 0x0C0

Reserved for customer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	RW	CUSTOMER																								Reserved for customer									

14.1.45 CUSTOMER[17]

Address offset: 0x0C4

Reserved for customer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	RW	CUSTOMER																								Reserved for customer									

14.1.46 CUSTOMER[18]

Address offset: 0x0C8

Reserved for customer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	RW	CUSTOMER																								Reserved for customer									

14.1.47 CUSTOMER[19]

Address offset: 0x0CC

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.48 CUSTOMER[20]

Address offset: 0x0D0

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.49 CUSTOMER[21]

Address offset: 0x0D4

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.50 CUSTOMER[22]

Address offset: 0x0D8

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.51 CUSTOMER[23]

Address offset: 0x0DC

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.52 CUSTOMER[24]

Address offset: 0x0E0

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.53 CUSTOMER[25]

Address offset: 0x0E4

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.54 CUSTOMER[26]

Address offset: 0x0E8

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.55 CUSTOMER[27]

Address offset: 0x0EC

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.56 CUSTOMER[28]

Address offset: 0x0F0

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.57 CUSTOMER[29]

Address offset: 0x0F4

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	CUSTOMER			Reserved for customer

14.1.58 CUSTOMER[30]

Address offset: 0x0F8

Reserved for customer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	CUSTOMER																																	

Reserved for customer

14.1.59 CUSTOMER[31]

Address offset: 0x0FC

Reserved for customer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	CUSTOMER																																	

Reserved for customer

14.1.60 PSELRESET[0]

Address offset: 0x200

Mapping of the nRESET function (see POWER chapter for details)

All PSELRESET registers have to contain the same value for a pin mapping to be valid. If they don't, there will be no nRESET function exposed on a GPIO, and the device will always start independently of the levels present on any of the GPIOs.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				B																										A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN		21																															
B	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

14.1.61 PSELRESET[1]

Address offset: 0x204

Mapping of the nRESET function (see POWER chapter for details)

All PSELRESET registers have to contain the same value for a pin mapping to be valid. If they don't, there will be no nRESET function exposed on a GPIO, and the device will always start independently of the levels present on any of the GPIOs.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				B																										A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN		21																															
B	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

14.1.62 APPROTECT

Address offset: 0x208

Access Port protection

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	RW	PALL																										Enable or disable Access Port protection. Any other value than 0xFF being written to this field will enable protection. See *Debug and trace* on page 72 for more information.							
			Disabled	0xFF																								Disable							
			Enabled	0x00																								Enable							

14.1.63 NFCPINS

Address offset: 0x20C

Setting of pins dedicated to NFC functionality: NFC antenna or GPIO

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	RW	PROTECT																										Setting of pins dedicated to NFC functionality							
			Disabled	0																								Operation as GPIO pins. Same protection as normal GPIO pins							
			NFC	1																								Operation as NFC antenna pins. Configures the protection for NFC operation							

15 Peripheral interface

Peripherals are controlled by the CPU by writing to configuration registers and task registers. Peripheral events are indicated to the CPU by event registers and interrupts if they are configured for a given event.

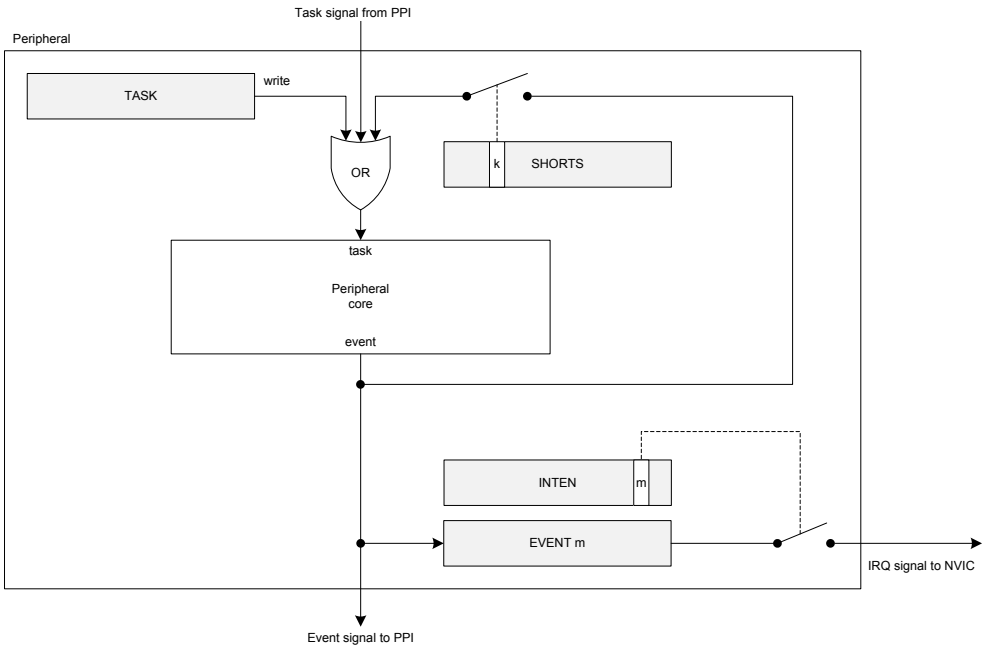


Figure 10: Tasks, events, shortcuts, and interrupts

15.1 Peripheral ID

Every peripheral is assigned a fixed block of 0x1000 bytes of address space, which is equal to 1024 x 32 bit registers.

See *Instantiation* on page 24 for more information about which peripherals are available and where they are located in the address map.

There is a direct relationship between the peripheral ID and base address. For example, a peripheral with base address 0x40000000 is assigned ID=0, a peripheral with base address 0x40001000 is assigned ID=1, and a peripheral with base address 0x4001F000 is assigned ID=31.

Peripherals may share the same ID, which may impose one or more of the following limitations:

- Some peripherals share some registers or other common resources.
- Operation is mutually exclusive. Only one of the peripherals can be used at a time.
- Switching from one peripheral to another must follow a specific pattern (disable the first, then enable the second peripheral).

15.2 Peripherals with shared ID

In general, and with the exception of ID 0, peripherals sharing an ID and base address may not be used simultaneously. The user can only enable one at the time on this specific ID.

When switching between two peripherals that share an ID, the user should do the following to prevent unwanted behavior:

- Disable the previously used peripheral

- Remove any PPI connections set up for the peripheral that is being disabled
- Clear all bits in the INTEN register, i.e. INTENCLR = 0xFFFFFFFF.
- Explicitly configure the peripheral that you enable and do not rely on configuration values that may be inherited from the peripheral that was disabled.
- Enable the now configured peripheral.

For each of the rows in the following table, the instance ID listed is shared by the peripherals in the same row.

Table 21: Peripherals sharing an ID

Instance						
ID 2 (0x40002000)	UARTE	UART				
-						
ID 3 (0x40003000)	SPIM	SPIS	SPI	TWIM	TWIS	TWI
ID 4 (0x40004000)	SPIM	SPIS	SPI	TWIM	TWIS	TWI
ID 35 (0x40023000)	SPIM	SPIS	SPI			
-						
ID 15 (0x4000F000)	AAR	CCM				
-						
ID 19 (0x40013000)	COMP	LPCOMP				
-						
ID 20 (0x40014000)	SWI	EGU				
ID 21 (0x40015000)	SWI	EGU				
ID 22 (0x40016000)	SWI	EGU				
ID 23 (0x40017000)	SWI	EGU				
ID 24 (0x40018000)	SWI	EGU				
ID 25 (0x40019000)	SWI	EGU				

15.3 Peripheral registers

Most peripherals feature an ENABLE register. Unless otherwise specified in the relevant chapter, the peripheral registers (in particular the PSEL registers) must be configured before enabling the peripheral.

Note that the peripheral must be enabled before tasks and events can be used.

15.4 Bit set and clear

Registers with multiple single-bit bit fields may implement the "set-and-clear" pattern. This pattern enables firmware to set and clear individual bits in a register without having to perform a read-modify-write operation on the main register.

This pattern is implemented using three consecutive addresses in the register map where the main register is followed by a dedicated SET and CLR register in that order.

The SET register is used to set individual bits in the main register while the CLR register is used to clear individual bits in the main register. Writing a '1' to a bit in the SET or CLR register will set or clear the same bit in the main register respectively. Writing a '0' to a bit in the SET or CLR register has no effect. Reading the SET or CLR registers returns the value of the main register.

Restriction: The main register may not be visible and hence not directly accessible in all cases.

15.5 Tasks

Tasks are used to trigger actions in a peripheral, for example, to start a particular behavior. A peripheral can implement multiple tasks with each task having a separate register in that peripheral's task register group.

A task is triggered when firmware writes a '1' to the task register or when the peripheral itself or another peripheral toggles the corresponding task signal. See *Figure 10: Tasks, events, shortcuts, and interrupts* on page 68.

15.6 Events

Events are used to notify peripherals and the CPU about events that have happened, for example, a state change in a peripheral. A peripheral may generate multiple events with each event having a separate register in that peripheral's event register group.

An event is generated when the peripheral itself toggles the corresponding event signal, and the event register is updated to reflect that the event has been generated. See *Figure 10: Tasks, events, shortcuts, and interrupts* on page 68. An event register is only cleared when firmware writes a '0' to it.

Events can be generated by the peripheral even when the event register is set to '1'.

15.7 Shortcuts

A shortcut is a direct connection between an event and a task within the same peripheral. If a shortcut is enabled, its associated task is automatically triggered when its associated event is generated.

Using a shortcut is the equivalent to making the same connection outside the peripheral and through the PPI. However, the propagation delay through the shortcut is usually shorter than the propagation delay through the PPI.

Shortcuts are predefined, which means their connections cannot be configured by firmware. Each shortcut can be individually enabled or disabled through the shortcut register, one bit per shortcut, giving a maximum of 32 shortcuts for each peripheral.

15.8 Interrupts

All peripherals support interrupts. Interrupts are generated by events.

A peripheral only occupies one interrupt, and the interrupt number follows the peripheral ID. For example, the peripheral with ID=4 is connected to interrupt number 4 in the Nested Vectored Interrupt Controller (NVIC).

Using the INTEN, INTENSET and INTENCLR registers, every event generated by a peripheral can be configured to generate that peripheral's interrupt. Multiple events can be enabled to generate interrupts simultaneously. To resolve the correct interrupt source, the event registers in the event group of peripheral registers will indicate the source.

Some peripherals implement only INTENSET and INTENCLR, and the INTEN register is not available on those peripherals. Refer to the individual chapters for details. In all cases, however, reading back the INTENSET or INTENCLR register returns the same information as in INTEN.

Each event implemented in the peripheral is associated with a specific bit position in the INTEN, INTENSET and INTENCLR registers.

The relationship between tasks, events, shortcuts, and interrupts is shown in *Figure 10: Tasks, events, shortcuts, and interrupts* on page 68.

15.8.1 Interrupt clearing

When clearing an interrupt by writing "0" to an event register, or disabling an interrupt using the INTENCLR register, it can take up to four CPU clock cycles to take effect. This means that an interrupt may reoccur immediatelly even if a new event has not come, if the program exits an interrupt handler after the interrupt is cleared or disabled, but before four clock cycles have passed.

> **Important:** To avoid an interrupt reoccurring before a new event has come, the program should perform a read from one of the peripheral registers, for example, the event register that has been cleared, or the INTENCLR register that has been used to disable the interrupt.

This will cause a one to three-cycle delay and ensure the interrupt is cleared before exiting the interrupt handler. Care should be taken to ensure the compiler does not remove the read operation as an optimization. If the program can guarantee a four-cycle delay after event clear or interrupt disable another way, then a read of a register is not required.

16 Debug and trace

The debug and trace system offers a flexible and powerful mechanism for non-intrusive debugging.

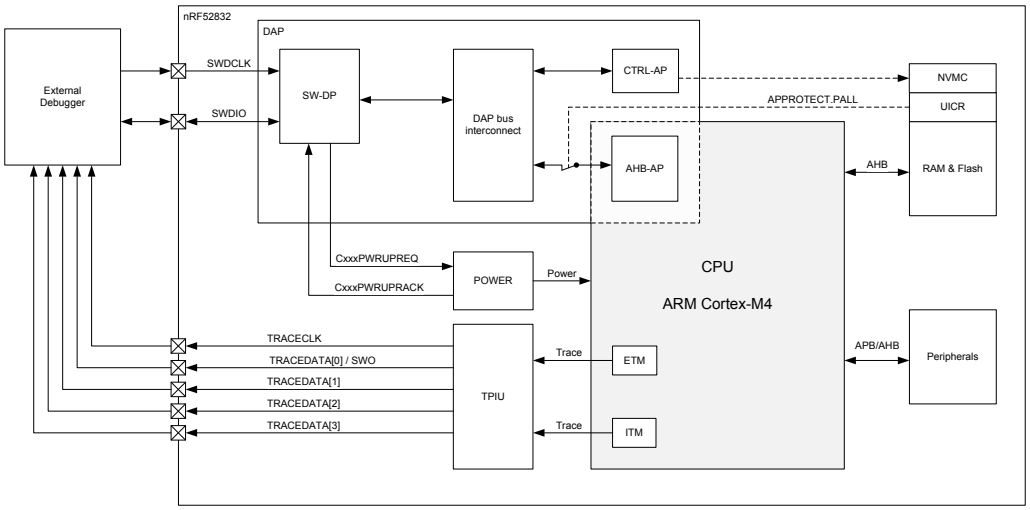


Figure 11: Debug and trace overview

The main features of the debug and trace system are:

- Two-pin Serial Wire Debug (SWD) interface
- Flash Patch and Breakpoint Unit (FPB) supports:
  - Two literal comparators
  - Six instruction comparators
- Data Watchpoint and Trace Unit (DWT)
  - Four comparators
- Instrumentation Trace Macrocell (ITM)
- Embedded Trace Macrocell (ETM)
- Trace Port Interface Unit (TPIU)
  - 4-bit parallel trace of ITM and ETM trace data
  - Serial Wire Output (SWO) trace of ITM data

16.1 DAP - Debug Access Port

An external debugger can access the device via the DAP.

The DAP implements a standard ARM® CoreSight™ Serial Wire Debug Port (SW-DP).

The SW-DP implements the Serial Wire Debug protocol (SWD) that is a two-pin serial interface, see SWDCLK and SWDIO in *Figure 11: Debug and trace overview* on page 72.

In addition to the default access port in the CPU (AHB-AP), the DAP includes a custom Control Access Port (CTRL-AP). The CTRL-AP is described in more detail in *CTRL-AP - Control Access Port* on page 73.

Important:

- The SWDIO line has an internal pull-up resistor.
- The SWDCLK line has an internal pull-down resistor.

16.2 CTRL-AP - Control Access Port

The Control Access Port (CTRL-AP) is a custom access port that enables control of the device even if the other access ports in the DAP are being disabled by the access port protection.

Access port protection blocks the debugger from read and write access to all CPU registers and memory-mapped addresses. See the UICR register *APPROTECT* on page 66 for more information about enabling access port protection.

This access port enables the following features:

- Soft reset, see *Reset* on page 82 for more information
- Disable access port protection

Access port protection can only be disabled by issuing an ERASEALL command via CTRL-AP. This command will erase the Flash, UICR, and RAM.

16.2.1 Registers

Table 22: Register Overview

Register	Offset	Description
RESET	0x000	Soft reset triggered through CTRL-AP
ERASEALL	0x004	Erase all
ERASEALLSTATUS	0x008	Status register for the ERASEALL operation
APPROTECTSTATUS	0x00C	Status register for access port protection
IDR	0x0FC	CTRL-AP Identification Register, IDR

RESET

Address offset: 0x000

Soft reset triggered through CTRL-AP

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	RESET																																	
			NoReset	0																															
			Reset	1																															

ERASEALL

Address offset: 0x004

Erase all

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	W	ERASEALL																																	
			NoOperation	0																															
			Erase	1																															

ERASEALLSTATUS

Address offset: 0x008

Status register for the ERASEALL operation

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															**Description**
A	R	ERASEALLSTATUS																																	Status register for the ERASEALL operation
			Ready	0																															ERASEALL is ready
			Busy	1																															ERASEALL is busy (on-going)

APPROTECTSTATUS

Address offset: 0x00C

Status register for access port protection

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															**Description**
A	R	APPROTECTSTATUS																																	Status register for access port protection
			Enabled	0																															Access port protection enabled
			Disabled	1																															Access port protection not enabled

IDR

Address offset: 0x0FC

CTRL-AP Identification Register, IDR

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				E	E	E	E	D	D	D	D	C	C	C	C	C	C	C	B	B	B	B						A	A	A	A	A	A	A	A
Reset 0x02880000				**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															**Description**
A	R	APID																																	AP Identification
B	R	CLASS																																	Access Port (AP) class
			NotDefined	0x0																															No defined class
			MEMAP	0x8																															Memory Access Port
C	R	JEP106ID																																	JEDEC JEP106 identity code
D	R	JEP106CONT																																	JEDEC JEP106 continuation code
E	R	REVISION																																	Revision

16.3 Debug interface mode

Before the external debugger can access the CPU's access port (AHB-AP) or the Control Access Port (CTRL-AP), the debugger must first request the device to power up via CxxxPWRUPREQ in the SWJ-DP.

As long as the debugger is requesting power via CxxxPWRUPREQ, the device will be in debug interface mode. If the debugger is not requesting power via CxxxPWRUPREQ, the device will be in normal mode.

Some peripherals will behave differently in debug interface mode compared to normal mode. These differences are described in more detail in the chapters of the peripherals that are affected.

When a debug session is over, the external debugger must make sure to put the device back into normal mode since the overall power consumption will be higher in debug interface mode compared to normal mode.

For details on how to use the debug capabilities please read the debug documentation of your IDE.

If the device is in System OFF when power is requested via CxxxPWRUPREQ, the system will wake up and the DIF flag in *RESETREAS* on page 85 will be set.

16.4 Real-time debug

The nRF52832 supports real-time debugging.

Real-time debugging will allow interrupts to execute to completion in real time when breakpoints are set in Thread mode or lower priority interrupts. This enables the developer to set a breakpoint and single-step through their code without a failure of the real-time event-driven threads running at higher priority. For example, this enables the device to continue to service the high-priority interrupts of an external controller or sensor without failure or loss of state synchronization while the developer steps through code in a low-priority thread.

16.5 Trace

The device supports ETM and ITM trace.

Trace data from the ETM and the ITM is sent to an external debugger via a 4-bit wide parallel trace port (TPIU), see TRACEDATA[0] through TRACEDATA[3] and TRACECLK in *Figure 11: Debug and trace overview* on page 72.

In addition to parallel trace, the TPIU supports serial trace via the Serial Wire Output (SWO) trace protocol.

Parallel and serial trace cannot be used at the same time.

ETM trace is only supported in parallel trace mode while ITM trace is supported in both parallel and serial trace modes.

For details on how to use the trace capabilities, please read the debug documentation of your IDE.

TPIU's trace pins are multiplexed with GPIOs, and SWO and TRACEDATA[0] use the same GPIO, see *Pin assignments* on page 13 for more information.

Trace speed is configured in the *TRACECONFIG* on page 108 register.

The speed of the trace pins depends on the DRIVE setting of the GPIOs that the trace pins are multiplexed with, see *PIN_CNF[14]* on page 142, *PIN_CNF[15]* on page 143, *PIN_CNF[16]* on page 144, *PIN_CNF[18]* on page 145 and *PIN_CNF[20]* on page 146. Only S0S1 and H0H1 drives are suitable for debugging. S0S1 is the default DRIVE at reset. If parallel or serial trace port signals are not fast enough in the debugging conditions, all GPIOs in use for tracing should be set to high drive (H0H1). The user shall make sure that these GPIOs' DRIVE is not overwritten by software during the debugging session.

16.5.1 Electrical specification

Trace port

Symbol	Description	Min.	Typ.	Max.	Units
T_{cyc}	Clock period, as defined by ARM (See ARM Infocenter, Embedded Trace Macrocell Architecture Specification, Trace Port Physical Interface, Timing specifications)	62.5		500	ns

17 Power and clock management

Power and clock management in nRF52832 is optimized for ultra-low power applications.

The core of the power and clock management system is the Power Management Unit (PMU) illustrated in *Figure 12: Power Management Unit* on page 76.

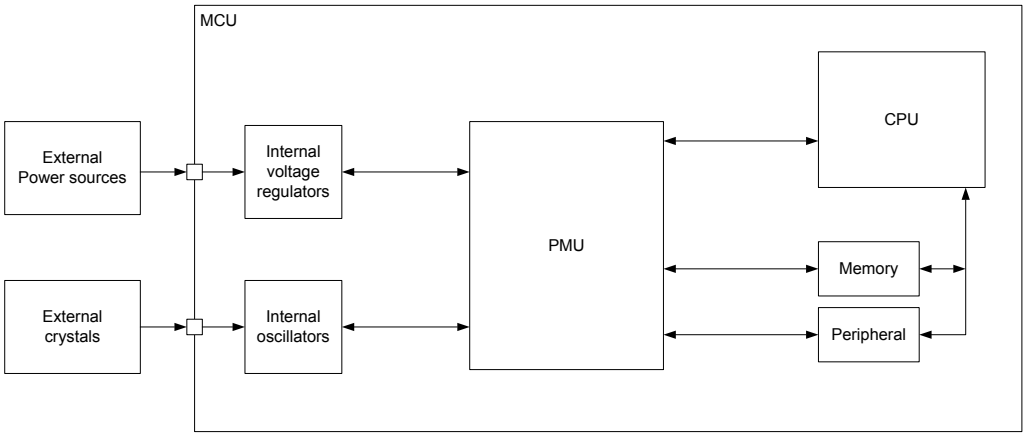


Figure 12: Power Management Unit

The user application is not required to actively control power and clock, since the PMU is able to automatically detect which resources are required by the different components in the system at any given time. The PMU will continuously optimize the system based on this information to achieve the lowest power consumption possible without user interaction.

17.1 Current consumption scenarios

As the system is being constantly tuned by the PMU, estimating the energy consumption of an application can be challenging if the designer is not able to do measurements on the hardware directly. See *Electrical specification* on page 76 for application scenarios showing average current drawn from the VDD supply.

Each scenario specifies a set of active operations and conditions applying to the given scenario. *Table 23: Current consumption scenarios, common conditions* on page 76 shows the conditions used for a scenario unless otherwise is stated in the scenario description.

Table 23: Current consumption scenarios, common conditions

Condition	Value
VDD	3 V
Temperature	25°C
CPU	WFI/WFE sleep
Peripherals	All idle
Clock	Not running
Regulator	DCDC

17.1.1 Electrical specification

Current consumption: Radio

Symbol	Description	Min.	Typ.	Max.	Units
I_{RADIO_TX0}	0 dBm TX @ 1 Mb/s Bluetooth Low Energy mode, Clock = HFXO		7.1		mA
I_{RADIO_TX1}	-40 dBm TX @ 1 Mb/s Bluetooth Low Energy mode, Clock = HFXO		4.1		mA
I_{RADIO_RX0}	Radio RX @ 1 Mb/s Bluetooth Low Energy mode, Clock = HFXO		6.5		mA

Current consumption: Radio protocol configurations

Symbol	Description	Min.	Typ.	Max.	Units
I_{S0}	CPU running CoreMark from Flash, Radio 0 dBm TX @ 1 Mb/s Bluetooth Low Energy mode, Clock = HFXO, Cache enabled		9.2		mA
I_{S1}	CPU running CoreMark from Flash, Radio RX @ 1 Mb/s Bluetooth Low Energy mode, Clock = HFXO, Cache enabled		9.2		mA

Current consumption: Ultra-low power

Symbol	Description	Min.	Typ.	Max.	Units
$I_{ON_RAMOFF_EVENT}$	System ON, No RAM retention, Wake on any event		1.2		µA
$I_{ON_RAMON_EVENT}$	System ON, Full RAM retention, Wake on any event		1.5		µA
$I_{ON_RAMOFF_RTC}$	System ON, No RAM retention, Wake on RTC		1.9		µA
$I_{OFF_RAMOFF_RESET}$	System OFF, No RAM retention, Wake on reset		0.3		µA
$I_{OFF_RAMOFF_GPIO}$	System OFF, No RAM retention, Wake on GPIO		0.3		µA
$I_{OFF_RAMOFF_LPCOMP}$	System OFF, No RAM retention, Wake on LPCOMP		1.9		µA
$I_{OFF_RAMOFF_NFC}$	System OFF, No RAM retention, Wake on NFC field		0.7		µA
$I_{OFF_RAMON_RESET}$	System OFF, Full 64 kB RAM retention, Wake on reset		0.7		µA

18 POWER — Power supply

This device has the following power supply features:

- On-chip LDO and DC/DC regulators
- Global System ON/OFF modes
- Individual RAM section power control for all system modes
- Analog or digital pin wakeup from System OFF
- Supervisor HW to manage power on reset, brownout, and power fail
- Auto-controlled refresh modes for LDO and DC/DC regulators to maximize efficiency
- Automatic switching between LDO and DC/DC regulator based on load to maximize efficiency

Note: Two additional external passive components are required to use the DC/DC regulator.

18.1 Regulators

The following internal power regulator alternatives are supported:

- Internal LDO regulator
- Internal DC/DC regulator

The LDO is the default regulator.

The DC/DC regulator can be used as an alternative to the LDO regulator and is enabled through the *DCDCEN* on page 88 register. Using the DC/DC regulator will reduce current consumption compared to when using the LDO regulator, but the DC/DC regulator requires an external LC filter to be connected, as shown in *Figure 14: DC/DC regulator setup* on page 79.

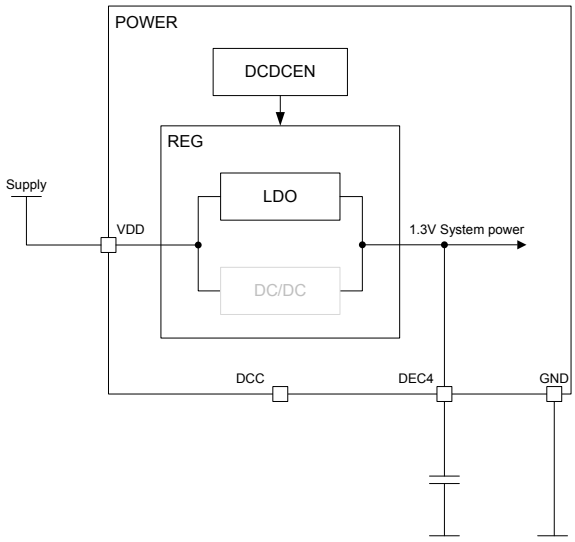


Figure 13: LDO regulator setup

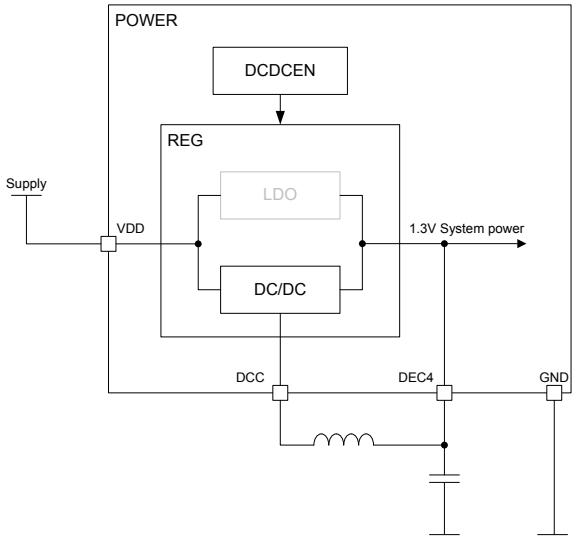


Figure 14: DC/DC regulator setup

18.2 System OFF mode

System OFF is the deepest power saving mode the system can enter. In this mode, the system's core functionality is powered down and all ongoing tasks are terminated.

The device can be put into System OFF mode using the POWER register interface. When in System OFF mode, the device can be woken up through one of the following signals:

1. The DETECT signal, optionally generated by the GPIO peripheral
2. The ANADETECT signal, optionally generated by the LPCOMP module
3. The SENSE signal, optionally generated by the NFC module to "wake-on-field"
4. A reset

When the system wakes up from System OFF mode, it gets reset. For more details, see *Reset behavior* on page 83.

One or more RAM sections can be retained in System OFF mode depending on the settings in the RAM[n].POWER registers.

RAM[n].POWER are retained registers, see *Reset behavior*. Note that these registers are usually overwritten by the startup code provided with the nRF application examples.

Before entering System OFF mode, the user must make sure that all on-going EasyDMA transactions have been completed. This is usually accomplished by making sure that the EasyDMA enabled peripheral is not active when entering System OFF.

18.2.1 Emulated System OFF mode

If the device is in debug interface mode, System OFF will be emulated to secure that all required resources needed for debugging are available during System OFF.

See *Debug and trace* on page 72 for more information. Required resources needed for debugging include the following key components: *Debug and trace* on page 72, *CLOCK — Clock control* on page 101, *POWER — Power supply* on page 78, *NVMC — Non-volatile memory controller* on page 29, CPU, Flash, and RAM. Since the CPU is kept on in an emulated System OFF mode, it is recommended to add an infinite loop directly after entering System OFF, to prevent the CPU from executing code that normally should not be executed.

18.3 System ON mode

System ON is the default state after power-on reset. In System ON, all functional blocks such as the CPU or peripherals, can be in IDLE or RUN mode, depending on the configuration set by the software and the state of the application executing.

Register *RESETREAS* on page 85 provides information about the source that caused the wakeup or reset.

The system can switch on and off the appropriate internal power sources, depending on how much power is needed at any given time. The power requirement of a peripheral is directly related to its activity level, and the activity level of a peripheral is usually raised and lowered when specific tasks are triggered or events are generated.

18.3.1 Sub power modes

In System ON mode, when both the CPU and all the peripherals are in IDLE mode, the system can reside in one of the two sub power modes.

The sub power modes are:

- Constant latency
- Low power

In constant latency mode the CPU wakeup latency and the PPI task response will be constant and kept at a minimum. This is secured by forcing a set of base resources on while in sleep. The advantage of having a constant and predictable latency will be at the cost of having increased power consumption. The constant latency mode is selected by triggering the CONSTLAT task.

In low power mode the automatic power management system, described in *System ON mode* on page 80, ensures the most efficient supply option is chosen to save the most power. The advantage of having the lowest power possible will be at the cost of having varying CPU wakeup latency and PPI task response. The low power mode is selected by triggering the LOWPWR task.

When the system enters System ON mode, it will, by default, reside in the low power sub-power mode.

18.4 Power supply supervisor

The power supply supervisor initializes the system at power-on and provides an early warning of impending power failure.

In addition, the power supply supervisor puts the system in a reset state if the supply voltage is too low for safe operation (brownout). The power supply supervisor is illustrated in *Figure 15: Power supply supervisor* on page 81.

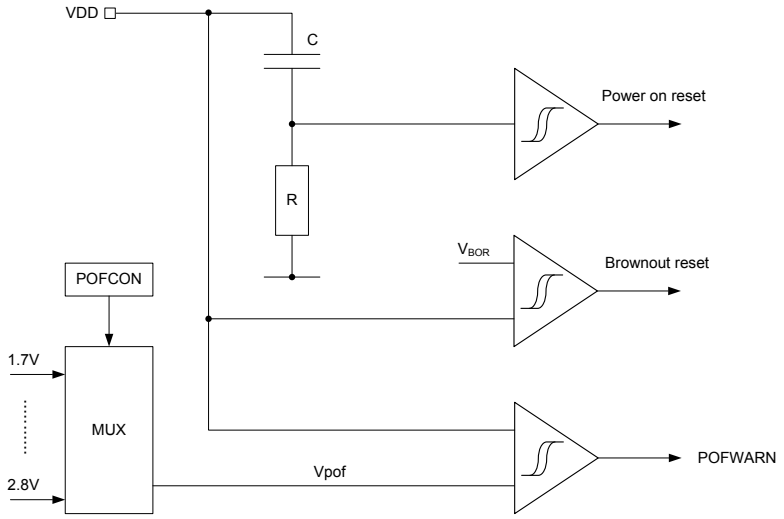


Figure 15: Power supply supervisor

18.4.1 Power-fail comparator

The power-fail comparator (POF) can provide the CPU with an early warning of impending power failure. It will not reset the system, but give the CPU time to prepare for an orderly power-down.

The comparator features a hysteresis of V_{HYST}, as illustrated in *Figure 16: Power-fail comparator (BOR = Brownout reset)* on page 81. The threshold V_{POF} is set in register *POFCON* on page 86. If the POF is enabled and the supply voltage falls below V_{POF}, the POFWARN event will be generated. This event will also be generated if the supply voltage is already below V_{POF} at the time the POF is enabled, or if V_{POF} is re-configured to a level above the supply voltage.

If power-fail warning is enabled and the supply voltage is below V_{POF} the power-fail comparator will prevent the NVMC from performing write operations to the NVM. See *NVMC — Non-volatile memory controller* on page 29 for more information about the NVMC.

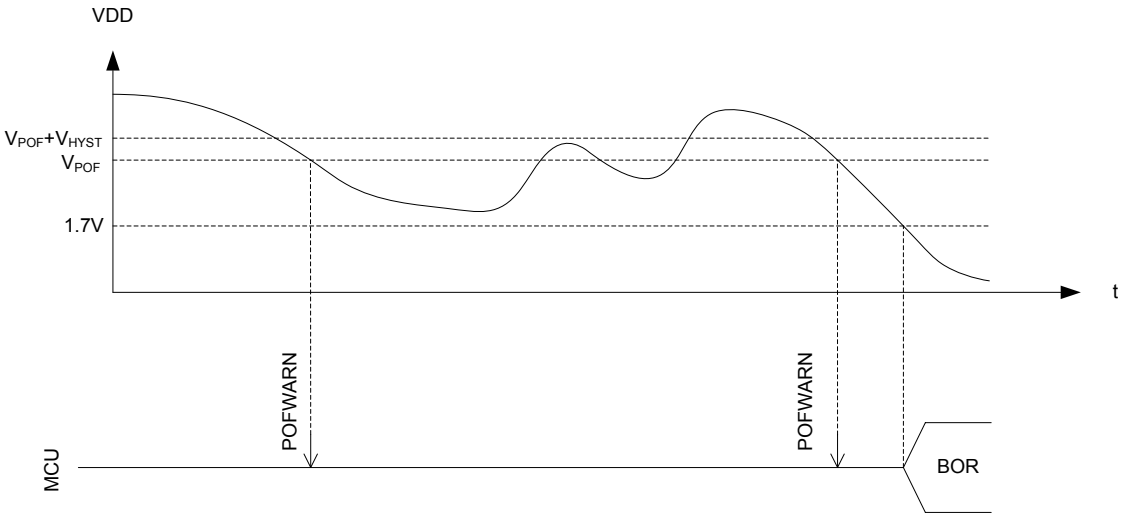


Figure 16: Power-fail comparator (BOR = Brownout reset)

To save power, the power-fail comparator is not active in System OFF or in System ON when HFCLK is not running.

18.5 RAM sections

RAM section power control is used for retention in System OFF mode and for powering down unused sections in System ON mode.

Each RAM section can power up and down independently in both System ON and System OFF mode. See chapter *Memory* on page 23 for more information on RAM sections.

18.6 Reset

There are multiple sources that may trigger a reset.

After a reset has occurred, register *RESETREAS* can be read to determine which source generated the reset.

18.6.1 Power-on reset

The power-on reset generator initializes the system at power-on.

The system is held in reset state until the supply has reached the minimum operating voltage and the internal voltage regulators have started.

A step increase in supply voltage of 300 mV or more, with rise time of 300 ms or less, within the valid supply range, may result in a system reset.

18.6.2 Pin reset

A pin reset is generated when the physical reset pin on the device is asserted.

Pin reset is configured via the *PSELRESET[0]* and *PSELRESET[1]* registers.

Note: Pin reset is not available on all pins.

18.6.3 Wakeup from System OFF mode reset

The device is reset when it wakes up from System OFF mode.

The DAP is not reset following a wake up from System OFF mode if the device is in debug interface mode. Refer to chapter *Debug and trace* on page 72 for more information.

18.6.4 Soft reset

A soft reset is generated when the SYSRESETREQ bit of the Application Interrupt and Reset Control Register (AIRCR register) in the ARM® core is set.

Refer to *ARM documentation* for more details.

A soft reset can also be generated via the *RESET* on page 73 register in the CTRL-AP.

18.6.5 Watchdog reset

A Watchdog reset is generated when the watchdog times out.

Refer to chapter *WDT — Watchdog timer* on page 409 for more information.

18.6.6 Brown-out reset

The brown-out reset generator puts the system in reset state if the supply voltage drops below the brownout reset (BOR) threshold.

Refer to section *Power fail comparator* on page 99 for more information.

18.7 Retained registers

A retained register is a register that will retain its value in System OFF mode and through a reset, depending on reset source. See individual peripheral chapters for information of which registers are retained for the various peripherals.

18.8 Reset behavior

Reset source	Reset target								
	CPU	Peripherals	GPIO	Debug[a]	SWJ-DP	RAM	WDT	Retained registers	RESETREAS
CPU lockup [6]	x	x	x						
Soft reset	x	x	x						
Wakeup from System OFF mode reset	x	x		x [7]		x [8]			
Watchdog reset [9]	x	x	x	x		x	x	x	
Pin reset	x	x	x	x		x	x	x	
Brownout reset	x	x	x	x	x	x	x	x	x
Power on reset	x	x	x	x	x	x	x	x	x

Note: The RAM is never reset, but depending on reset source, RAM content may be corrupted.

18.9 Registers

Table 24: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40000000	POWER	POWER	Power control	

Table 25: Register Overview

Register	Offset	Description	
TASKS_CONSTLAT	0x078	Enable constant latency mode	
TASKS_LOWPWR	0x07C	Enable low power mode (variable latency)	
EVENTS_POFWARN	0x108	Power failure warning	
EVENTS_SLEEPENTER	0x114	CPU entered WFI/WFE sleep	
EVENTS_SLEEPEXIT	0x118	CPU exited WFI/WFE sleep	
INTENSET	0x304	Enable interrupt	
INTENCLR	0x308	Disable interrupt	
RESETREAS	0x400	Reset reason	
RAMSTATUS	0x428	RAM status register	Deprecated
SYSTEMOFF	0x500	System OFF register	
POFCON	0x510	Power failure comparator configuration	
GPREGRET	0x51C	General purpose retention register	
GPREGRET2	0x520	General purpose retention register	
RAMON	0x524	RAM on/off register (this register is retained)	Deprecated
RAMONB	0x554	RAM on/off register (this register is retained)	Deprecated
DCDCEN	0x578	DC/DC enable register	
RAM[0].POWER	0x900	RAM0 power control register	

[a] All debug components excluding SWJ-DP. See *Debug and trace* on page 72 chapter for more information about the different debug components in the system.

[6] Reset from CPU lockup is disabled if the device is in debug interface mode. CPU lockup is not possible in System OFF.

[7] The Debug components will not be reset if the device is in debug interface mode.

[8] RAM is not reset on wakeup from OFF mode, but depending on settings in the RAM register parts, or the whole RAM, may not be retained after the device has entered System OFF mode.

[9] Watchdog reset is not available in System OFF.

Register	Offset	Description
RAM[0].POWERSET	0x904	RAM0 power control set register
RAM[0].POWERCLR	0x908	RAM0 power control clear register
RAM[1].POWER	0x910	RAM1 power control register
RAM[1].POWERSET	0x914	RAM1 power control set register
RAM[1].POWERCLR	0x918	RAM1 power control clear register
RAM[2].POWER	0x920	RAM2 power control register
RAM[2].POWERSET	0x924	RAM2 power control set register
RAM[2].POWERCLR	0x928	RAM2 power control clear register
RAM[3].POWER	0x930	RAM3 power control register
RAM[3].POWERSET	0x934	RAM3 power control set register
RAM[3].POWERCLR	0x938	RAM3 power control clear register
RAM[4].POWER	0x940	RAM4 power control register
RAM[4].POWERSET	0x944	RAM4 power control set register
RAM[4].POWERCLR	0x948	RAM4 power control clear register
RAM[5].POWER	0x950	RAM5 power control register
RAM[5].POWERSET	0x954	RAM5 power control set register
RAM[5].POWERCLR	0x958	RAM5 power control clear register
RAM[6].POWER	0x960	RAM6 power control register
RAM[6].POWERSET	0x964	RAM6 power control set register
RAM[6].POWERCLR	0x968	RAM6 power control clear register
RAM[7].POWER	0x970	RAM7 power control register
RAM[7].POWERSET	0x974	RAM7 power control set register
RAM[7].POWERCLR	0x978	RAM7 power control clear register

18.9.1 INTENSET

Address offset: 0x304

Enable interrupt

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												C	B			A		
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																	**Description**							
A	RW	POFWARN																										Write '1' to Enable interrupt for POFWARN event See *EVENTS_POFWARN*							
			Set								1																	Enable							
			Disabled								0																	Read: Disabled							
			Enabled								1																	Read: Enabled							
B	RW	SLEEPENTER																										Write '1' to Enable interrupt for SLEEPENTER event See *EVENTS_SLEEPENTER*							
			Set								1																	Enable							
			Disabled								0																	Read: Disabled							
			Enabled								1																	Read: Enabled							
C	RW	SLEEPEXIT																										Write '1' to Enable interrupt for SLEEPEXIT event See *EVENTS_SLEEPEXIT*							
			Set								1																	Enable							
			Disabled								0																	Read: Disabled							
			Enabled								1																	Read: Enabled							

18.9.2 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												C	B			A		
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																	**Description**							
A	RW	POFWARN																										Write '1' to Disable interrupt for POFWARN event							
																												See *EVENTS_POFWARN*							
			Clear								1																	Disable							
			Disabled								0																	Read: Disabled							
			Enabled								1																	Read: Enabled							
B	RW	SLEEPENTER																										Write '1' to Disable interrupt for SLEEPENTER event							
																												See *EVENTS_SLEEPENTER*							
			Clear								1																	Disable							
			Disabled								0																	Read: Disabled							
			Enabled								1																	Read: Enabled							
C	RW	SLEEPEXIT																										Write '1' to Disable interrupt for SLEEPEXIT event							
																												See *EVENTS_SLEEPEXIT*							
			Clear								1																	Disable							
			Disabled								0																	Read: Disabled							
			Enabled								1																	Read: Enabled							

18.9.3 RESETREAS

Address offset: 0x400

Reset reason

Unless cleared, the RESETREAS register will be cumulative. A field is cleared by writing '1' to it. If none of the reset sources are flagged, this indicates that the chip was reset from the on-chip reset generator, which will indicate a power-on-reset or a brownout reset.

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id															H	G	F	E													D	C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																	**Description**							
A	RW	RESETPIN																										Reset from pin-reset detected							
			NotDetected								0																	Not detected							
			Detected								1																	Detected							
B	RW	DOG																										Reset from watchdog detected							
			NotDetected								0																	Not detected							
			Detected								1																	Detected							
C	RW	SREQ																										Reset from soft reset detected							
			NotDetected								0																	Not detected							
			Detected								1																	Detected							
D	RW	LOCKUP																										Reset from CPU lock-up detected							
			NotDetected								0																	Not detected							
			Detected								1																	Detected							
E	RW	OFF																										Reset due to wake up from System OFF mode when wakeup is							
																												triggered from DETECT signal from GPIO							
			NotDetected								0																	Not detected							
			Detected								1																	Detected							
F	RW	LPCOMP																										Reset due to wake up from System OFF mode when wakeup is							
																												triggered from ANADETECT signal from LPCOMP							
			NotDetected								0																	Not detected							
			Detected								1																	Detected							
G	RW	DIF																										Reset due to wake up from System OFF mode when wakeup is							
																												triggered from entering into debug interface mode							
			NotDetected								0																	Not detected							
			Detected								1																	Detected							

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																H	G	F	E														D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
H	RW	NFC																																	
			NotDetected	0																															
			Detected	1																															

18.9.4 RAMSTATUS (Deprecated)

Address offset: 0x428

RAM status register

Since this register is deprecated the following substitutions have been made: RAM block 0 is equivalent to a block comprising RAM0.S0 and RAM1.S0, RAM block 1 is equivalent to a block comprising RAM2.S0 and RAM3.S0, RAM block 2 is equivalent to a block comprising RAM4.S0 and RAM5.S0 and RAM block 3 is equivalent to a block comprising RAM6.S0 and RAM7.S0. A RAM block field will indicate ON as long as any of the RAM sections associated with a block are on.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	RAMBLOCK0																																	
			Off	0																															
			On	1																															
B	R	RAMBLOCK1																																	
			Off	0																															
			On	1																															
C	R	RAMBLOCK2																																	
			Off	0																															
			On	1																															
D	R	RAMBLOCK3																																	
			Off	0																															
			On	1																															

18.9.5 SYSTEMOFF

Address offset: 0x500

System OFF register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	W	SYSTEMOFF																																	
			Enter	1																															

18.9.6 POFCON

Address offset: 0x510

Power failure comparator configuration

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															B	B	B	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	POF																																	
			Disabled	0																															

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																														B	B	B	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																							
			Enabled								1																							
B	RW	THRESHOLD																																
			V17								4																							
			V18								5																							
			V19								6																							
			V20								7																							
			V21								8																							
			V22								9																							
			V23								10																							
			V24								11																							
			V25								12																							
			V26								13																							
			V27								14																							
			V28								15																							

18.9.7 GPREGRET

Address offset: 0x51C

General purpose retention register

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																											A	A	A	A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																							
A	RW	GPREGRET																																

General purpose retention register

This register is a retained register

18.9.8 GPREGRET2

Address offset: 0x520

General purpose retention register

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																											A	A	A	A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																							
A	RW	GPREGRET																																

General purpose retention register

This register is a retained register

18.9.9 RAMON (Deprecated)

Address offset: 0x524

RAM on/off register (this register is retained)

Since this register is deprecated the following substitutions have been made: RAM block 0 is equivalent to a block comprising RAM0.S0 and RAM0.S1 and RAM block 1 is equivalent to a block comprising RAM1.S0 and RAM1.S1. For new designs it is recommended to use the POWER.RAM-0.POWER and its sibling registers instead.

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	D	C															B	A
Reset 0x00000003			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**
Id	**RW**	**Field**	**Value Id**								**Value**																							
A	RW	ONRAM0																																
			RAM0Off								0																							

Keep RAM block 0 on or off in system ON Mode

Off

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C															B	A
Reset 0x00000003				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			RAM0On	1																															
B	RW	ONRAM1																																	
			RAM1Off	0																															
			RAM1On	1																															
C	RW	OFFRAM0																																	
			RAM0Off	0																															
			RAM0On	1																															
D	RW	OFFRAM1																																	
			RAM1Off	0																															
			RAM1On	1																															

18.9.10 RAMONB (Deprecated)

Address offset: 0x554

RAM on/off register (this register is retained)

Since this register is deprecated the following substitutions have been made: RAM block 2 is equivalent to a block comprising RAM2.S0 and RAM2.S1 and RAM block 3 is equivalent to a block comprising RAM3.S0 and RAM3.S1. For new designs it is recommended to use the POWER.RAM-0.POWER and its sibling registers instead.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C															B	A
Reset 0x00000003				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	ONRAM2																																	
			RAM2Off	0																															
			RAM2On	1																															
B	RW	ONRAM3																																	
			RAM3Off	0																															
			RAM3On	1																															
C	RW	OFFRAM2																																	
			RAM2Off	0																															
			RAM2On	1																															
D	RW	OFFRAM3																																	
			RAM3Off	0																															
			RAM3On	1																															

18.9.11 DCDCEN

Address offset: 0x578

DC/DC enable register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	DCDCEN																																	
			Disabled	0																															
			Enabled	1																															

18.9.12 RAM[0].POWER

Address offset: 0x900

RAM0 power control register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C															B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	S0POWER																																	
			Off	0																															
			On	1																															
B	RW	S1POWER																																	
			Off	0																															
			On	1																															
C	RW	S0RETENTION																																	
			Off	0																															
			On	1																															
D	RW	S1RETENTION																																	
			Off	0																															
			On	1																															

18.9.13 RAM[0].POWERSET

Address offset: 0x904

RAM0 power control set register

When read, this register will return the value of the POWER register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C															B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	W	S0POWER																																	
			On	1																															
B	W	S1POWER																																	
			On	1																															
C	W	S0RETENTION																																	
			On	1																															
D	W	S1RETENTION																																	
			On	1																															

18.9.14 RAM[0].POWERCLR

Address offset: 0x908

RAM0 power control clear register

When read, this register will return the value of the POWER register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C															B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	W	S0POWER																																	
			Off	1																															
B	W	S1POWER																																	

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
			Off	1	Off
C	W	S0RETENTION			Keep retention on RAM section S0 when RAM section is switched off
			Off	1	Off
D	W	S1RETENTION			Keep retention on RAM section S1 when RAM section is switched off
			Off	1	Off

18.9.15 RAM[1].POWER

Address offset: 0x910

RAM1 power control register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	S0POWER			Keep RAM section S0 ON or OFF in System ON mode. RAM sections are always retained when ON, but can also be retained when OFF dependent on the settings in S0RETENTION. All RAM sections will be OFF in System OFF mode.
			Off	0	Off
			On	1	On
B	RW	S1POWER			Keep RAM section S1 ON or OFF in System ON mode. RAM sections are always retained when ON, but can also be retained when OFF dependent on the settings in S1RETENTION. All RAM sections will be OFF in System OFF mode.
			Off	0	Off
			On	1	On
C	RW	S0RETENTION			Keep retention on RAM section S0 when RAM section is in OFF
			Off	0	Off
			On	1	On
D	RW	S1RETENTION			Keep retention on RAM section S1 when RAM section is in OFF
			Off	0	Off
			On	1	On

18.9.16 RAM[1].POWERSET

Address offset: 0x914

RAM1 power control set register

When read, this register will return the value of the POWER register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	W	S0POWER			Keep RAM section S0 of RAM1 on or off in System ON mode
			On	1	On
B	W	S1POWER			Keep RAM section S1 of RAM1 on or off in System ON mode
			On	1	On
C	W	S0RETENTION			Keep retention on RAM section S0 when RAM section is switched off
			On	1	On

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C															B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
D	W	S1RETENTION																																	
			On	1																															

18.9.17 RAM[1].POWERCLR

Address offset: 0x918

RAM1 power control clear register

When read, this register will return the value of the POWER register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C															B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	W	S0POWER																																	
			Off	1																															
B	W	S1POWER																																	
			Off	1																															
C	W	S0RETENTION																																	
			Off	1																															
D	W	S1RETENTION																																	
			Off	1																															

18.9.18 RAM[2].POWER

Address offset: 0x920

RAM2 power control register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C															B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	S0POWER																																	
			Off	0																															
			On	1																															
B	RW	S1POWER																																	
			Off	0																															
			On	1																															
C	RW	S0RETENTION																																	
			Off	0																															
			On	1																															
D	RW	S1RETENTION																																	
			Off	0																															
			On	1																															

18.9.19 RAM[2].POWERSET

Address offset: 0x924

RAM2 power control set register

When read, this register will return the value of the POWER register.

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			D	C															B	A
Reset 0x0000FFFF					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																														
A	W	S0POWER			Keep RAM section S0 of RAM2 on or off in System ON mode																																
			On	1	On																														
B	W	S1POWER			Keep RAM section S1 of RAM2 on or off in System ON mode																																
			On	1	On																														
C	W	S0RETENTION			Keep retention on RAM section S0 when RAM section is switched off																																
			On	1	On																														
D	W	S1RETENTION			Keep retention on RAM section S1 when RAM section is switched off																																
			On	1	On																														

18.9.20 RAM[2].POWERCLR

Address offset: 0x928

RAM2 power control clear register

When read, this register will return the value of the POWER register.

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			D	C															B	A
Reset 0x0000FFFF					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																														
A	W	S0POWER			Keep RAM section S0 of RAM2 on or off in System ON mode																																
			Off	1	Off																														
B	W	S1POWER			Keep RAM section S1 of RAM2 on or off in System ON mode																																
			Off	1	Off																														
C	W	S0RETENTION			Keep retention on RAM section S0 when RAM section is switched off																																
			Off	1	Off																														
D	W	S1RETENTION			Keep retention on RAM section S1 when RAM section is switched off																																
			Off	1	Off																														

18.9.21 RAM[3].POWER

Address offset: 0x930

RAM3 power control register

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			D	C															B	A
Reset 0x0000FFFF					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																														
A	RW	S0POWER			Keep RAM section S0 ON or OFF in System ON mode. RAM sections are always retained when ON, but can also be retained when OFF dependent on the settings in S0RETENTION. All RAM sections will be OFF in System OFF mode.																																
			Off	0	Off																														
			On	1	On																														
B	RW	S1POWER			Keep RAM section S1 ON or OFF in System ON mode.																																

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C															B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
				RAM sections are always retained when ON, but can also be retained when OFF dependent on the settings in S1RETENTION. All RAM sections will be OFF in System OFF mode.																																	
			Off	0																																
			On	1																																
C	RW	S0RETENTION		Keep retention on RAM section S0 when RAM section is in OFF																																
			Off	0																																
			On	1																																
D	RW	S1RETENTION		Keep retention on RAM section S1 when RAM section is in OFF																																
			Off	0																																
			On	1																																

18.9.22 RAM[3].POWERSET

Address offset: 0x934

RAM3 power control set register

When read, this register will return the value of the POWER register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C															B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	W	S0POWER		Keep RAM section S0 of RAM3 on or off in System ON mode																																
			On	1																																
B	W	S1POWER		Keep RAM section S1 of RAM3 on or off in System ON mode																																
			On	1																																
C	W	S0RETENTION		Keep retention on RAM section S0 when RAM section is switched off																																
			On	1																																
D	W	S1RETENTION		Keep retention on RAM section S1 when RAM section is switched off																																
			On	1																																

18.9.23 RAM[3].POWERCLR

Address offset: 0x938

RAM3 power control clear register

When read, this register will return the value of the POWER register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C															B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	W	S0POWER		Keep RAM section S0 of RAM3 on or off in System ON mode																																
			Off	1																																
B	W	S1POWER		Keep RAM section S1 of RAM3 on or off in System ON mode																																
			Off	1																																
C	W	S0RETENTION		Keep retention on RAM section S0 when RAM section is switched off																																
			Off	1																																
D	W	S1RETENTION		Keep retention on RAM section S1 when RAM section is switched off																																
			Off	1																																

18.9.24 RAM[4].POWER

Address offset: 0x940

RAM4 power control register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C															B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	S0POWER																																	
			Off	0																															
			On	1																															
B	RW	S1POWER																																	
			Off	0																															
			On	1																															
C	RW	S0RETENTION																																	
			Off	0																															
			On	1																															
D	RW	S1RETENTION																																	
			Off	0																															
			On	1																															

18.9.25 RAM[4].POWERSET

Address offset: 0x944

RAM4 power control set register

When read, this register will return the value of the POWER register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C															B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	W	S0POWER																																	
			On	1																															
B	W	S1POWER																																	
			On	1																															
C	W	S0RETENTION																																	
			On	1																															
D	W	S1RETENTION																																	
			On	1																															

18.9.26 RAM[4].POWERCLR

Address offset: 0x948

RAM4 power control clear register

When read, this register will return the value of the POWER register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C																B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	W	S0POWER																																	
			Off	1																															
B	W	S1POWER																																	
			Off	1																															
C	W	S0RETENTION																																	
			Off	1																															
D	W	S1RETENTION																																	
			Off	1																															

18.9.27 RAM[5].POWER

Address offset: 0x950

RAM5 power control register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C																B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	S0POWER																																	
			Off	0																															
			On	1																															
B	RW	S1POWER																																	
			Off	0																															
			On	1																															
C	RW	S0RETENTION																																	
			Off	0																															
			On	1																															
D	RW	S1RETENTION																																	
			Off	0																															
			On	1																															

18.9.28 RAM[5].POWERSET

Address offset: 0x954

RAM5 power control set register

When read, this register will return the value of the POWER register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C																B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	W	S0POWER																																	
			On	1																															
B	W	S1POWER																																	
			On	1																															

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
C	W	S0RETENTION			Keep retention on RAM section S0 when RAM section is switched off
			On	1	On
D	W	S1RETENTION			Keep retention on RAM section S1 when RAM section is switched off
			On	1	On

18.9.29 RAM[5].POWERCLR

Address offset: 0x958

RAM5 power control clear register

When read, this register will return the value of the POWER register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	W	S0POWER			Keep RAM section S0 of RAM5 on or off in System ON mode
			Off	1	Off
B	W	S1POWER			Keep RAM section S1 of RAM5 on or off in System ON mode
			Off	1	Off
C	W	S0RETENTION			Keep retention on RAM section S0 when RAM section is switched off
			Off	1	Off
D	W	S1RETENTION			Keep retention on RAM section S1 when RAM section is switched off
			Off	1	Off

18.9.30 RAM[6].POWER

Address offset: 0x960

RAM6 power control register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	S0POWER			Keep RAM section S0 ON or OFF in System ON mode. RAM sections are always retained when ON, but can also be retained when OFF dependent on the settings in S0RETENTION. All RAM sections will be OFF in System OFF mode.
			Off	0	Off
			On	1	On
B	RW	S1POWER			Keep RAM section S1 ON or OFF in System ON mode. RAM sections are always retained when ON, but can also be retained when OFF dependent on the settings in S1RETENTION. All RAM sections will be OFF in System OFF mode.
			Off	0	Off
			On	1	On
C	RW	S0RETENTION			Keep retention on RAM section S0 when RAM section is in OFF
			Off	0	Off
			On	1	On
D	RW	S1RETENTION			Keep retention on RAM section S1 when RAM section is in OFF
			Off	0	Off

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C															B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			On	1																															
				On																															

18.9.31 RAM[6].POWERSET

Address offset: 0x964

RAM6 power control set register

When read, this register will return the value of the POWER register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C															B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	W	S0POWER																																	
			On	1																															
				On																															
B	W	S1POWER																																	
			On	1																															
				On																															
C	W	S0RETENTION																																	
			On	1																															
				On																															
D	W	S1RETENTION																																	
			On	1																															
				On																															

18.9.32 RAM[6].POWERCLR

Address offset: 0x968

RAM6 power control clear register

When read, this register will return the value of the POWER register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C															B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	W	S0POWER																																	
			Off	1																															
				Off																															
B	W	S1POWER																																	
			Off	1																															
				Off																															
C	W	S0RETENTION																																	
			Off	1																															
				Off																															
D	W	S1RETENTION																																	
			Off	1																															
				Off																															

18.9.33 RAM[7].POWER

Address offset: 0x970

RAM7 power control register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C															B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	S0POWER																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C																B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
				RAM sections are always retained when ON, but can also be retained when OFF dependent on the settings in S0RETENTION. All RAM sections will be OFF in System OFF mode.																																	
			Off	0																																
			On	1																																
B	RW	S1POWER																																		
				RAM sections are always retained when ON, but can also be retained when OFF dependent on the settings in S1RETENTION. All RAM sections will be OFF in System OFF mode.																																	
			Off	0																																
			On	1																																
C	RW	S0RETENTION																																		
				Keep retention on RAM section S0 when RAM section is in OFF																																
			Off	0																																
			On	1																																
D	RW	S1RETENTION																																		
				Keep retention on RAM section S1 when RAM section is in OFF																																
			Off	0																																
			On	1																																

18.9.34 RAM[7].POWERSET

Address offset: 0x974

RAM7 power control set register

When read, this register will return the value of the POWER register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C																B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																																
A	W	S0POWER																																		
				Keep RAM section S0 of RAM7 on or off in System ON mode																																
			On	1																																
B	W	S1POWER																																		
				Keep RAM section S1 of RAM7 on or off in System ON mode																																
			On	1																																
C	W	S0RETENTION																																		
				Keep retention on RAM section S0 when RAM section is switched off																																
			On	1																																
D	W	S1RETENTION																																		
				Keep retention on RAM section S1 when RAM section is switched off																																
			On	1																																

18.9.35 RAM[7].POWERCLR

Address offset: 0x978

RAM7 power control clear register

When read, this register will return the value of the POWER register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C																B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																																
A	W	S0POWER																																		
				Keep RAM section S0 of RAM7 on or off in System ON mode																																
			Off	1																																
B	W	S1POWER																																		
				Keep RAM section S1 of RAM7 on or off in System ON mode																																
			Off	1																																

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		D	C															B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																														
C	W	S0RETENTION			Keep retention on RAM section S0 when RAM section is switched off																																
			Off	1	Off																														
D	W	S1RETENTION			Keep retention on RAM section S1 when RAM section is switched off																																
			Off	1	Off																														

18.10 Electrical specification

18.10.1 Current consumption, sleep

Symbol	Description	Min.	Typ.	Max.	Units
I_{OFF}	System OFF current, no RAM retention		0.3		µA
I_{ON}	System ON base current, no RAM retention		1.2		µA
I_{RAM}	Additional RAM retention current per 4 KB RAM section		20		nA

18.10.2 Device startup times

Symbol	Description	Min.	Typ.	Max.	Units
t_{POR}	Time in Power on Reset after VDD reaches 1.7 V for all supply voltages and temperatures. Dependent on supply rise time. [10]				
$t_{POR,10us}$	VDD rise time 10us		1		ms
$t_{POR,10ms}$	VDD rise time 10ms		9		ms
$t_{POR,60ms}$	VDD rise time 60ms		23		ms
t_{PINR}	If a GPIO pin is configured as reset, the maximum time taken to pull up the pin and release reset after power on reset. Dependent on the pin capacitive load (C)[11]: t=5RC, R = 13kOhm				
$t_{PINR,500nF}$	C = 500nF			32.5	ms
$t_{PINR,10uF}$	C = 10uF			650	ms
t_{R2ON}	Time from reset to ON (CPU execute)				
$t_{R2ON,NOTCONF}$	If reset pin not configured	tPOR			ms
$t_{R2ON,CONF}$	If reset pin configured	tPOR + tPINR			ms
t_{OFF2ON}	Time from OFF to CPU execute		16.5		µs
$t_{IDLE2CPU}$	Time from IDLE to CPU execute		3.0		µs
$t_{EVTSET,CL1}$	Time from HW event to PPI event in Constant Latency System ON mode		0.0625		µs
$t_{EVTSET,CL0}$	Time from HW event to PPI event in Low Power System ON mode		0.0625		µs

18.10.3 Power fail comparator

Symbol	Description	Min.	Typ.	Max.	Units
I_{POF}	Current consumption when enabled[12]		<4		µA
V_{POF}	Nominal power level warning thresholds (falling supply voltage). Levels are configurable between Min. and Max. in 100mV increments.	1.7		2.8	V
V_{POFTOL}	Threshold voltage tolerance		±1	±5	%

10 A step increase in supply voltage of 300 mV or more, with rise time of 300 ms or less, within the valid supply range, may result in a system reset.

11 To decrease maximum time a device could hold in reset, a strong external pullup resistor can be used.

12 To save power, POF will not operate nor consume in System OFF, or while HFCLK is not running, even if left enabled by software

Symbol	Description	Min.	Typ.	Max.	Units
$V_{POFHYST}$	Threshold voltage hysteresis		50		mV
$V_{BOR,OFF}$	Brown out reset voltage range SYSTEM OFF mode	1.2		1.7	V
$V_{BOR,ON}$	Brown out reset voltage range SYSTEM ON mode	1.5		1.7	V

19 CLOCK — Clock control

The clock control system can source the system clocks from a range of internal or external high and low frequency oscillators and distribute them to modules based upon a module's individual requirements. Clock distribution is automated and grouped independently by module to limit current consumption in unused branches of the clock tree.

Listed here are the main features for CLOCK:

- 64 MHz on-chip oscillator
- 64 MHz crystal oscillator, using external 32 MHz crystal
- 32.768 kHz +/-250 ppm RC oscillator
- 32.768 kHz crystal oscillator, using external 32.768 kHz crystal
- 32.768 kHz oscillator synthesized from 64 MHz oscillator
- Firmware (FW) override control of oscillator activity for low latency start up
- Automatic oscillator and clock control, and distribution for ultra-low power

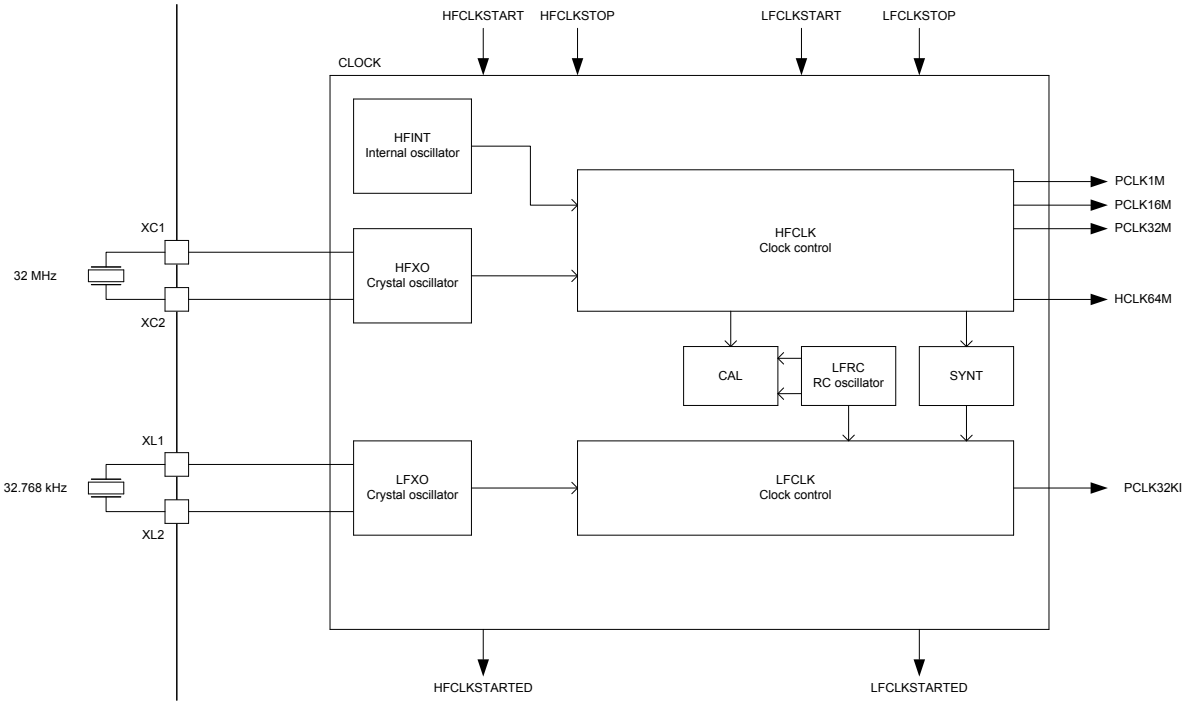


Figure 17: Clock control

19.1 HFCLK clock controller

The HFCLK clock controller provides the following clocks to the system.

- HCLK64M: 64 MHz CPU clock
- PCLK1M: 1 MHz peripheral clock
- PCLK16M: 16 MHz peripheral clock
- PCLK32M: 32 MHz peripheral clock

The HFCLK controller supports the following high frequency clock (HFCLK) sources:

- 64 MHz internal oscillator (HFINT)
- 64 MHz crystal oscillator (HFXO)

For illustration, see *Figure 17: Clock control* on page 101.

When the system requests one or more clocks from the HFCLK controller, the HFCLK controller will automatically provide them. If the system does not request any clocks provided by the HFCLK controller, the controller will enter a power saving mode.

These clocks are only available when the system is in ON mode. When the system enters ON mode, the internal oscillator (HFINT) clock source will automatically start to be able to provide the required HFCLK clock(s) for the system.

The HFINT will be used when HFCLK is requested and HFXO has not been started. The HFXO is started by triggering the HFCLKSTART task and stopped using the HFCLKSTOP task. A HFCLKSTARTED event will be generated when the HFXO has started and its frequency is stable.

The HFXO must be running to use the RADIO, NFC module or the calibration mechanism associated with the 32.768 kHz RC oscillator.

19.1.1 64 MHz crystal oscillator (HFXO)

The 64 MHz crystal oscillator (HFXO) is controlled by a 32 MHz external crystal

The crystal oscillator is designed for use with an AT-cut quartz crystal in parallel resonant mode. To achieve correct oscillation frequency, the load capacitance must match the specification in the crystal data sheet.

Figure 18: Circuit diagram of the 64 MHz crystal oscillator on page 102 shows how the 32 MHz crystal is connected to the 64 MHz crystal oscillator.

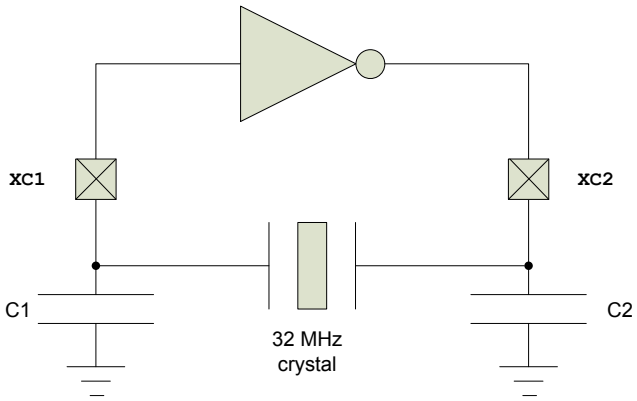


Figure 18: Circuit diagram of the 64 MHz crystal oscillator

The load capacitance (CL) is the total capacitance seen by the crystal across its terminals and is given by:

$$CL = \frac{(C1' \cdot C2')}{(C1' + C2')}$$

$$C1' = C1 + C_{pcb1} + C_{pin}$$
$$C2' = C2 + C_{pcb2} + C_{pin}$$

C1 and C2 are ceramic SMD capacitors connected between each crystal terminal and ground. For more information, see *Reference circuitry* on page 545. C_{pcb1} and C_{pcb2} are stray capacitances on the PCB. C_{pin} is the pin input capacitance on the `XC1` and `XC2` pins. See table *64 MHz crystal oscillator (HFXO)* on page 109. The load capacitors C1 and C2 should have the same value.

For reliable operation, the crystal load capacitance, shunt capacitance, equivalent series resistance, and drive level must comply with the specifications in table *64 MHz crystal oscillator (HFXO)* on page 109. It is recommended to use a crystal with lower than maximum load capacitance and/or shunt capacitance. A low load capacitance will reduce both start up time and current consumption.

19.2 LFCLK clock controller

The system supports several low frequency clock sources.

As illustrated in *Figure 17: Clock control* on page 101, the system supports the following low frequency clock sources:

- 32.768 kHz RC oscillator (LFRC)
- 32.768 kHz crystal oscillator (LFXO)
- 32.768 kHz synthesized from HFCLK (LFSYNT)

The LFCLK clock is started by first selecting the preferred clock source in register *LFCLKSRC* on page 108 and then triggering the LFCLKSTART task. If the LFXO is selected as the clock source, the LFCLK will initially start running from the 32.768 kHz LFRC while the LFXO is starting up and automatically switch to using the LFXO once this oscillator is running. The LFCLKSTARTED event will be generated when the LFXO has been started.

The LFCLK clock is stopped by triggering the LFCLKSTOP task.

It is not allowed to write to register *LFCLKSRC* on page 108 when the LFCLK is running.

A LFCLKSTOP task will stop the LFCLK oscillator. However, the LFCLKSTOP task can only be triggered after the STATE field in register *LFCLKSTAT* on page 107 indicates a 'LFCLK running' state.

The LFCLK clock controller and all of the LFCLK clock sources are always switched off when in OFF mode.

19.2.1 32.768 kHz RC oscillator (LFRC)

The default source of the low frequency clock (LFCLK) is the 32.768 kHz RC oscillator (LFRC).

The LFRC frequency will be affected by variation in temperature. The LFRC oscillator can be calibrated to improve accuracy by using the HFXO as a reference oscillator during calibration. See Table *32.768 kHz RC oscillator (LFRC)* on page 109 for details on the default and calibrated accuracy of the LFRC oscillator. The LFRC oscillator does not require additional external components.

19.2.2 Calibrating the 32.768 kHz RC oscillator

After the 32.768 kHz RC oscillator is started and running, it can be calibrated by triggering the CAL task. In this case, the HFCLK will be temporarily switched on and used as a reference.

A DONE event will be generated when calibration has finished. The calibration mechanism will only work as long as HFCLK is generated from the HFCLK crystal oscillator, it is therefore necessary to explicitly start this crystal oscillator before calibration can be started, see HFCLKSTART task.

19.2.3 Calibration timer

The calibration timer can be used to time the calibration interval of the 32.768 kHz RC oscillator.

The calibration timer is started by triggering the CTSTART task and stopped by triggering the CTSTOP task. The calibration timer will always start counting down from the value specified in CTIV and generate a CTTO timeout event when it reaches 0. The Calibration timer will stop by itself when it reaches 0.

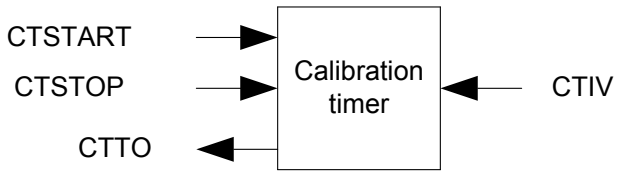


Figure 19: Calibration timer

Due to limitations in the calibration timer, only one task related to calibration, that is, CAL, CTSTART and CTSTOP, can be triggered for every period of LFCLK.

19.2.4 32.768 kHz crystal oscillator (LFXO)

For higher LFCLK accuracy (when better than +/- 250 ppm accuracy is required), the low frequency crystal oscillator (LFXO) must be used.

The following external clock sources are supported:

- Low swing clock signal applied to the *XL1* pin. The *XL2* pin shall then be grounded.
- Rail-to-rail clock signal applied to the *XL1* pin. The *XL2* pin shall then be grounded or left unconnected.

The *LFCLKSRC* on page 108 register controls the clock source, and its allowed swing. The truth table for various situations is as follows:

Table 26: LFCLKSRC configuration depending on clock source

SRC	EXTERNAL	BYPASS	Comment
0	0	0	Normal operation, RC is source
0	0	1	DO NOT USE
0	1	X	DO NOT USE
1	0	0	Normal XTAL operation
1	1	0	Apply external low swing signal to XL1, ground XL2
1	1	1	Apply external full swing signal to XL1, leave XL2 grounded or unconnected
1	0	1	DO NOT USE
2	0	0	Normal operation, synth is source
2	0	1	DO NOT USE
2	1	X	DO NOT USE

To achieve correct oscillation frequency, the load capacitance must match the specification in the crystal data sheet. *Figure 20: Circuit diagram of the 32.768 kHz crystal oscillator* on page 104 shows the LFXO circuitry.

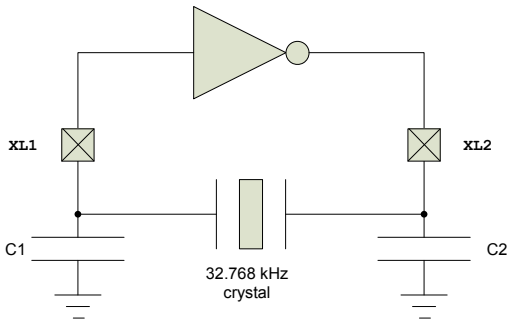


Figure 20: Circuit diagram of the 32.768 kHz crystal oscillator

The load capacitance (CL) is the total capacitance seen by the crystal across its terminals and is given by:

$$CL = \frac{(C1' \cdot C2')}{(C1' + C2')}$$

$$C1' = C1 + C_{pcb1} + C_{pin}$$
$$C2' = C2 + C_{pcb2} + C_{pin}$$

C1 and C2 are ceramic SMD capacitors connected between each crystal terminal and ground. C_{pcb1} and C_{pcb2} are stray capacitances on the PCB. C_{pin} is the pin input capacitance on the XC1 and XC2 pins (see *32.768 kHz crystal oscillator (LFXO)* on page 109). The load capacitors C1 and C2 should have the same value.

For more information, see *Reference circuitry* on page 545.

19.2.5 32.768 kHz synthesized from HFCLK (LFSYNT)

LFCLK can also be synthesized from the HFCLK clock source. The accuracy of LFCLK will then be the accuracy of the HFCLK.

Using the LFSYNT clock avoids the requirement for a 32.768 kHz crystal, but increases average power consumption as the HFCLK will need to be requested in the system.

19.3 Registers

Table 27: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40000000	CLOCK	CLOCK	Clock control	

Table 28: Register Overview

Register	Offset	Description	
TASKS_HFCLKSTART	0x000	Start HFCLK crystal oscillator	
TASKS_HFCLKSTOP	0x004	Stop HFCLK crystal oscillator	
TASKS_LFCLKSTART	0x008	Start LFCLK source	
TASKS_LFCLKSTOP	0x00C	Stop LFCLK source	
TASKS_CAL	0x010	Start calibration of LFRC oscillator	
TASKS_CTSTART	0x014	Start calibration timer	
TASKS_CTSTOP	0x018	Stop calibration timer	
EVENTS_HFCLKSTARTED	0x100	HFCLK oscillator started	
EVENTS_LFCLKSTARTED	0x104	LFCLK started	
EVENTS_DONE	0x10C	Calibration of LFCLK RC oscillator complete event	
EVENTS_CTTO	0x110	Calibration timer timeout	
INTENSET	0x304	Enable interrupt	
INTENCLR	0x308	Disable interrupt	
HFCLKRUN	0x408	Status indicating that HFCLKSTART task has been triggered	
HFCLKSTAT	0x40C	HFCLK status	
LFCLKRUN	0x414	Status indicating that LFCLKSTART task has been triggered	
LFCLKSTAT	0x418	LFCLK status	
LFCLKSRCCOPY	0x41C	Copy of LFCLKSRC register, set when LFCLKSTART task was triggered	
LFCLKSRC	0x518	Clock source for the LFCLK	
CTIV	0x538	Calibration timer interval	Retained
TRACECONFIG	0x55C	Clocking options for the Trace Port debug interface	

19.3.1 INTENSET

Address offset: 0x304

Enable interrupt

Bit number																																
																												D	C		B	A
Reset 0x00000000																												**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	HFCLKSTARTED			Write '1' to Enable interrupt for HFCLKSTARTED event See *EVENTS_HFCLKSTARTED*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	LFCLKSTARTED			Write '1' to Enable interrupt for LFCLKSTARTED event See *EVENTS_LFCLKSTARTED*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															D	C			B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
C	RW	DONE																																	
D	RW	CTTO																																	

19.3.2 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															D	C			B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	HFCLKSTARTED																																	
B	RW	LFCLKSTARTED																																	
C	RW	DONE																																	
D	RW	CTTO																																	

19.3.3 HFCLKRUN

Address offset: 0x408

Status indicating that HFCLKSTART task has been triggered

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	STATUS																																	
			NotTriggered	0																															
			Triggered	1																															

19.3.4 HFCLKSTAT

Address offset: 0x40C

HFCLK status

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			B																A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															**Description**
A	R	SRC																																	Source of HFCLK
			RC	0																															64 MHz internal oscillator (HFINT)
			Xtal	1																															64 MHz crystal oscillator (HFXO)
B	R	STATE																																	HFCLK state
			NotRunning	0																															HFCLK not running
			Running	1																															HFCLK running

19.3.5 LFCLKRUN

Address offset: 0x414

Status indicating that LFCLKSTART task has been triggered

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															**Description**
A	R	STATUS																																	LFCLKSTART task triggered or not
			NotTriggered	0																															Task not triggered
			Triggered	1																															Task triggered

19.3.6 LFCLKSTAT

Address offset: 0x418

LFCLK status

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			B														A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															**Description**
A	R	SRC																																	Source of LFCLK
			RC	0																															32.768 kHz RC oscillator
			Xtal	1																															32.768 kHz crystal oscillator
			Synth	2																															32.768 kHz synthesized from HFCLK
B	R	STATE																																	LFCLK state
			NotRunning	0																															LFCLK not running
			Running	1																															LFCLK running

19.3.7 LFCLKSRCCOPY

Address offset: 0x41C

Copy of LFCLKSRC register, set when LFCLKSTART task was triggered

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															**Description**
A	R	SRC																																	Clock source
			RC	0																															32.768 kHz RC oscillator
			Xtal	1																															32.768 kHz crystal oscillator
			Synth	2																															32.768 kHz synthesized from HFCLK

19.3.8 LFCLKSRC

Address offset: 0x518

Clock source for the LFCLK

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		C	B																A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	SRC																																	
			RC	0																															
			Xtal	1																															
			Synth	2																															
B	RW	BYPASS																																	
			Disabled	0																															
			Enabled	1																															
C	RW	EXTERNAL																																	
			Disabled	0																															
			Enabled	1																															

19.3.9 CTIV (Retained)

Address offset: 0x538

This register is a retained register

Calibration timer interval

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																													A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	CTIV																																	

19.3.10 TRACECONFIG

Address offset: 0x55C

Clocking options for the Trace Port debug interface

This register is a retained register. Reset behavior is the same as debug components.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			B	B															A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	TRACEPORTSPEED																																	
			32MHz	0																															
			16MHz	1																															
			8MHz	2																															
			4MHz	3																															
B	RW	TRACEMUX																																	
			GPIO	0																															
			Serial	1																															
			Parallel	2																															

19.4 Electrical specification

19.4.1 64 MHz internal oscillator (HFINT)

Symbol	Description	Min.	Typ.	Max.	Units
f_{NOM_HFINT}	Nominal output frequency		64		MHz
f_{TOL_HFINT}	Frequency tolerance		<±1.5	<±6	%
I_{HFINT}	Run current		60		µA
I_{START_HFINT}	Average startup current		I_HFINT		µA
t_{START_HFINT}	Startup time		3		us

19.4.2 64 MHz crystal oscillator (HFXO)

Symbol	Description	Min.	Typ.	Max.	Units
f_{NOM_HFXO}	Nominal output frequency		64		MHz
f_{XTAL_HFXO}	External crystal frequency		32		MHz
f_{TOL_HFXO}	Frequency tolerance requirement for 2.4 GHz proprietary radio applications			±60	ppm
$f_{TOL_HFXO_BLE}$	Frequency tolerance requirement, Bluetooth low energy applications			±40	ppm
C_{L_HFXO}	Load capacitance			12	pF
C_{0_HFXO}	Shunt capacitance			7	pF
$R_{S_HFXO_7PF}$	Equivalent series resistance C0 = 7 pF			60	ohm
$R_{S_HFXO_5PF}$	Equivalent series resistance C0 = 5 pF			80	ohm
$R_{S_HFXO_3PF}$	Equivalent series resistance C0 = 3 pF			100	ohm
P_{D_HFXO}	Drive level			100	uW
C_{PIN_HFXO}	Input capacitance XC1 and XC2		4		pF
I_{STBY_X32M}	Core standby current[13]		50		µA
I_{HFXO}	Run current		250		µA
I_{START_HFXO}	Average startup current, first 1 ms		0.4		mA
t_{START_HFXO}	Startup time		0.36		ms

19.4.3 32.768 kHz RC oscillator (LFRC)

Symbol	Description	Min.	Typ.	Max.	Units
f_{NOM_LFRC}	Nominal frequency		32.768		kHz
f_{TOL_LFRC}	Frequency tolerance			±2	%
$f_{TOL_CAL_LFRC}$	Frequency tolerance for LFRC after calibration[14]			±250	ppm
I_{LFRC}	Run current for 32.768 kHz RC oscillator		0.6	1	µA
t_{START_LFRC}	Startup time for 32.768 kHz RC oscillator		600		us

19.4.4 32.768 kHz crystal oscillator (LFXO)

Symbol	Description	Min.	Typ.	Max.	Units
f_{NOM_LFXO}	Crystal frequency		32.768		kHz
$f_{TOL_LFXO_BLE}$	Frequency tolerance requirement for BLE stack			±250	ppm
$f_{TOL_LFXO_ANT}$	Frequency tolerance requirement for ANT stack			±50	ppm
C_{L_LFXO}	Load capacitance			12.5	pF
C_{0_LFXO}	Shunt capacitance			2	pF
R_{S_LFXO}	Equivalent series resistance			100	kohm
P_{D_LFXO}	Drive level			1	uW
C_{pin}	Input capacitance on XL1 and XL2 pads		4		pF
I_{LFXO}	Run current for 32.768 kHz crystal oscillator		0.25		µA

[13] Current drawn if HFXO is forced on through for instance using the low latency power mode.
[14] Constant temperature within ±0.5 °C and calibration performed at least every 8 seconds

Symbol	Description	Min.	Typ.	Max.	Units
t_{START_LFXO}	Startup time for 32.768 kHz crystal oscillator		0.25		s
$V_{AMP_IN_XO_LOW}$	Peak to peak amplitude for external low swing clock. Input signal must not swing outside supply rails.	200		1000	mV

19.4.5 32.768 kHz synthesized from HFCLK (LFSYNT)

Symbol	Description	Min.	Typ.	Max.	Units
f_{NOM_LFSYNT}	Nominal frequency		32.768		kHz
f_{TOL_LFSYNT}	Frequency tolerance in addition to HFLCK tolerance[15]		8		ppm
I_{LFSYNT}	Run current for synthesized 32.768 kHz		100		µA
t_{START_LFSYNT}	Startup time for synthesized 32.768 kHz		100		us

[15] Frequency tolerance will be derived from the HFCLK source clock plus the LFSYNT tolerance

20 GPIO — General purpose input/output

The general purpose input/output (GPIO) is organized as one port with up to 32 I/Os (dependent on package) enabling access and control of up to 32 pins through one port. Each GPIO can be accessed individually.

GPIO has the following user-configurable features:

- Up to 32 GPIO
- 8 GPIO with Analog channels for SAADC, COMP or LPCOMP inputs
- Configurable output drive strength
- Internal pull-up and pull-down resistors
- Wake-up from high or low level triggers on all pins
- Trigger interrupt on state changes on any pin
- All pins can be used by the PPI task/event system
- One or more GPIO outputs can be controlled through PPI and GPIOTE channels
- All pins can be individually mapped to interface blocks for layout flexibility
- GPIO state changes captured on SENSE signal can be stored by LATCH register

The GPIO Port peripheral implements up to 32 pins, `PIN0` through `PIN31`. Each of these pins can be individually configured in the PIN_CNF[n] registers (n=0..31).

The following parameters can be configured through these registers:

- Direction
- Drive strength
- Enabling of pull-up and pull-down resistors
- Pin sensing
- Input buffer disconnect
- Analog input (for selected pins)

The PIN_CNF registers are retained registers. See *POWER — Power supply* on page 78 chapter for more information about retained registers.

20.1 Pin configuration

Pins can be individually configured, through the SENSE field in the PIN_CNF[n] register, to detect either a high level or a low level on their input.

When the correct level is detected on any such configured pin, the sense mechanism will set the DETECT signal high. Each pin has a separate DETECT signal, and the default behaviour, as defined by the DETECTMODE register, is that the DETECT signal from all pins in the GPIO Port are combined into a common DETECT signal that is routed throughout the system, which then can be utilized by other peripherals, see *Figure 21: GPIO Port and the GPIO pin details* on page 112. This mechanism is functional in both ON and OFF mode.

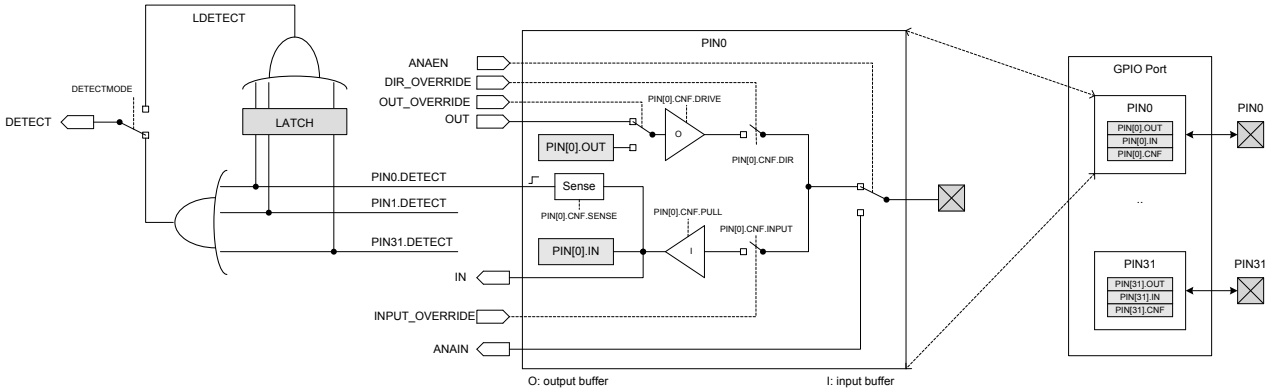


Figure 21: GPIO Port and the GPIO pin details

Figure 21: GPIO Port and the GPIO pin details on page 112 illustrates the GPIO port containing 32 individual pins, where `PIN0` is illustrated in more detail as a reference. All the signals on the left side of the illustration are used by other peripherals in the system, and therefore, are not directly available to the CPU.

Make sure that a pin is in a level that cannot trigger the sense mechanism before enabling it. Detect will go high immediately if the sense condition configured in the PIN_CNF registers is met when the sense mechanism is enabled. This will trigger a PORT event if the DETECT signal was low before enabling the sense mechanism. See *GPIOTE — GPIO tasks and events* on page 157.

See the following peripherals for more information about how the DETECT signal is used:

- POWER: uses the DETECT signal to exit from System OFF.
- GPIOTE: uses the DETECT signal to generate the PORT event.

When a pin's PINx.DETECT signal goes high, a flag will be set in the LATCH register, e.g. when the PIN0.DETECT signal goes high, bit 0 in the LATCH register will be set to '1'.

The LATCH register will only be cleared if the CPU explicitly clears it by writing a '1' to the bit that shall be cleared, i.e. the LATCH register will not be affected by a PINx.DETECT signal being set low.

If the CPU performs a clear operation on a bit in the LATCH register when the associated PINx.DETECT signal is high, the bit in the LATCH register will not be cleared.

The LDETECT signal will be set high when one or more bits in the LATCH register are '1'. The LDETECT signal will be set low when all bits in the LATCH register are successfully cleared to '0'.

If one or more bits in the LATCH register are '1' after the CPU has performed a clear operation on the LATCH registers, a rising edge will be generated on the LDETECT signal, this is illustrated in *Figure 22: DETECT signal behavior* on page 113.

> **Important:** The CPU can read the LATCH register at any time to check if a SENSE condition has been met on one or more of the the GPIO pins even if that condition is no longer met at the time the CPU queries the LATCH register. This mechanism will work even if the LDETECT signal is not used as the DETECT signal.

The LDETECT signal is by default not connected to the GPIO port's DETECT signal, but via the DETECTMODE register it is possible to change the behaviour of the GPIO port's DETECT signal from the default behaviour described above to instead be derived directly from the LDETECT signal, see *Figure 21: GPIO Port and the GPIO pin details* on page 112. *Figure 22: DETECT signal behavior* on page 113 illustrates the DETECT signals behaviour for these two alternatives.

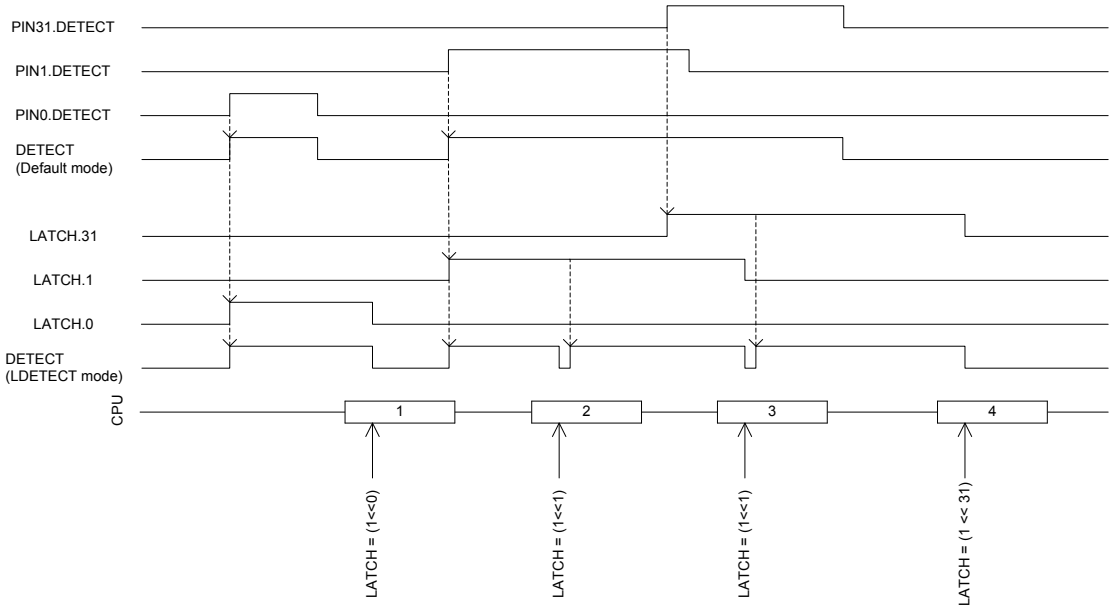


Figure 22: DETECT signal behavior

The input buffer of a GPIO pin can be disconnected from the pin to enable power savings when the pin is not used as an input, see *Figure 21: GPIO Port and the GPIO pin details* on page 112. Inputs must be connected in order to get a valid input value in the IN register and for the sense mechanism to get access to the pin.

Other peripherals in the system can attach themselves to GPIO pins and override their output value and configuration, or read their analog or digital input value, see *Figure 21: GPIO Port and the GPIO pin details* on page 112.

Selected pins also support analog input signals, see ANAIN in *Figure 21: GPIO Port and the GPIO pin details* on page 112. The assignment of the analog pins can be found in *Pin assignments* on page 13.

> **Important:** When a pin is configured as digital input, care has been taken in the nRF52832 design to minimize increased current consumption when the input voltage is between V_{IL} and V_{IH}. However, it is a good practice to ensure that the external circuitry does not drive that pin to levels between V_{IL} and V_{IH} for a long period of time.

20.2 GPIO located near the RADIO

Radio performance parameters, such as sensitivity, may be affected by high frequency digital I/O with large sink/source current close to the radio power supply and antenna pins.

Refer to *Pin assignments* on page 13 for recommended usage guidelines to maximize radio performance in an application.

20.3 Registers

Table 29: Instances

Base address	Peripheral	Instance	Description	Configuration	
0x50000000	GPIO	GPIO	General purpose input and output		Deprecated
0x50000000	GPIO	P0	General purpose input and output		

Table 30: Register Overview

Register	Offset	Description
OUT	0x504	Write GPIO port
OUTSET	0x508	Set individual bits in GPIO port
OUTCLR	0x50C	Clear individual bits in GPIO port
IN	0x510	Read GPIO port
DIR	0x514	Direction of GPIO pins
DIRSET	0x518	DIR set register
DIRCLR	0x51C	DIR clear register
LATCH	0x520	Latch register indicating what GPIO pins that have met the criteria set in the PIN_CNF[n].SENSE registers
DETECTMODE	0x524	Select between default DETECT signal behaviour and LDETECT mode
PIN_CNF[0]	0x700	Configuration of GPIO pins
PIN_CNF[1]	0x704	Configuration of GPIO pins
PIN_CNF[2]	0x708	Configuration of GPIO pins
PIN_CNF[3]	0x70C	Configuration of GPIO pins
PIN_CNF[4]	0x710	Configuration of GPIO pins
PIN_CNF[5]	0x714	Configuration of GPIO pins
PIN_CNF[6]	0x718	Configuration of GPIO pins
PIN_CNF[7]	0x71C	Configuration of GPIO pins
PIN_CNF[8]	0x720	Configuration of GPIO pins
PIN_CNF[9]	0x724	Configuration of GPIO pins
PIN_CNF[10]	0x728	Configuration of GPIO pins
PIN_CNF[11]	0x72C	Configuration of GPIO pins
PIN_CNF[12]	0x730	Configuration of GPIO pins
PIN_CNF[13]	0x734	Configuration of GPIO pins
PIN_CNF[14]	0x738	Configuration of GPIO pins
PIN_CNF[15]	0x73C	Configuration of GPIO pins
PIN_CNF[16]	0x740	Configuration of GPIO pins
PIN_CNF[17]	0x744	Configuration of GPIO pins
PIN_CNF[18]	0x748	Configuration of GPIO pins
PIN_CNF[19]	0x74C	Configuration of GPIO pins
PIN_CNF[20]	0x750	Configuration of GPIO pins
PIN_CNF[21]	0x754	Configuration of GPIO pins
PIN_CNF[22]	0x758	Configuration of GPIO pins
PIN_CNF[23]	0x75C	Configuration of GPIO pins
PIN_CNF[24]	0x760	Configuration of GPIO pins
PIN_CNF[25]	0x764	Configuration of GPIO pins
PIN_CNF[26]	0x768	Configuration of GPIO pins
PIN_CNF[27]	0x76C	Configuration of GPIO pins
PIN_CNF[28]	0x770	Configuration of GPIO pins
PIN_CNF[29]	0x774	Configuration of GPIO pins
PIN_CNF[30]	0x778	Configuration of GPIO pins
PIN_CNF[31]	0x77C	Configuration of GPIO pins

20.3.1 OUT

Address offset: 0x504

Write GPIO port

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN0																																	
			Low	0																															
			High	1																															
B	RW	PIN1																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Low	0																															
			High	1																															
C	RW	PIN2																																	
			Low	0																															
			High	1																															
D	RW	PIN3																																	
			Low	0																															
			High	1																															
E	RW	PIN4																																	
			Low	0																															
			High	1																															
F	RW	PIN5																																	
			Low	0																															
			High	1																															
G	RW	PIN6																																	
			Low	0																															
			High	1																															
H	RW	PIN7																																	
			Low	0																															
			High	1																															
I	RW	PIN8																																	
			Low	0																															
			High	1																															
J	RW	PIN9																																	
			Low	0																															
			High	1																															
K	RW	PIN10																																	
			Low	0																															
			High	1																															
L	RW	PIN11																																	
			Low	0																															
			High	1																															
M	RW	PIN12																																	
			Low	0																															
			High	1																															
N	RW	PIN13																																	
			Low	0																															
			High	1																															
O	RW	PIN14																																	
			Low	0																															
			High	1																															
P	RW	PIN15																																	
			Low	0																															
			High	1																															
Q	RW	PIN16																																	
			Low	0																															
			High	1																															
R	RW	PIN17																																	
			Low	0																															
			High	1																															
S	RW	PIN18																																	
			Low	0																															
			High	1																															
T	RW	PIN19																																	
			Low	0																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			High	1																															
U	RW	PIN20																																	
			Low	0																															
			High	1																															
V	RW	PIN21																																	
			Low	0																															
			High	1																															
W	RW	PIN22																																	
			Low	0																															
			High	1																															
X	RW	PIN23																																	
			Low	0																															
			High	1																															
Y	RW	PIN24																																	
			Low	0																															
			High	1																															
Z	RW	PIN25																																	
			Low	0																															
			High	1																															
a	RW	PIN26																																	
			Low	0																															
			High	1																															
b	RW	PIN27																																	
			Low	0																															
			High	1																															
c	RW	PIN28																																	
			Low	0																															
			High	1																															
d	RW	PIN29																																	
			Low	0																															
			High	1																															
e	RW	PIN30																																	
			Low	0																															
			High	1																															
f	RW	PIN31																																	
			Low	0																															
			High	1																															

20.3.2 OUTSET

Address offset: 0x508

Set individual bits in GPIO port

Read: reads value of OUT register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN0																																	
			Low	0																															
			High	1																															
			Set	1																															
B	RW	PIN1																																	
			Low	0																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			High	1																															
			Set	1																															
C	RW	PIN2																																	
			Low	0																															
			High	1																															
			Set	1																															
D	RW	PIN3																																	
			Low	0																															
			High	1																															
			Set	1																															
E	RW	PIN4																																	
			Low	0																															
			High	1																															
			Set	1																															
F	RW	PIN5																																	
			Low	0																															
			High	1																															
			Set	1																															
G	RW	PIN6																																	
			Low	0																															
			High	1																															
			Set	1																															
H	RW	PIN7																																	
			Low	0																															
			High	1																															
			Set	1																															
I	RW	PIN8																																	
			Low	0																															
			High	1																															
			Set	1																															
J	RW	PIN9																																	
			Low	0																															
			High	1																															
			Set	1																															
K	RW	PIN10																																	
			Low	0																															
			High	1																															
			Set	1																															
L	RW	PIN11																																	
			Low	0																															
			High	1																															
			Set	1																															
M	RW	PIN12																																	
			Low	0																															
			High	1																															
			Set	1																															
N	RW	PIN13																																	
			Low	0																															
			High	1																															
			Set	1																															
O	RW	PIN14																																	
			Low	0																															
			High	1																															
			Set	1																															
P	RW	PIN15																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Low	0																															
			High	1																															
			Set	1																															
Q	RW	PIN16																																	
			Low	0																															
			High	1																															
			Set	1																															
R	RW	PIN17																																	
			Low	0																															
			High	1																															
			Set	1																															
S	RW	PIN18																																	
			Low	0																															
			High	1																															
			Set	1																															
T	RW	PIN19																																	
			Low	0																															
			High	1																															
			Set	1																															
U	RW	PIN20																																	
			Low	0																															
			High	1																															
			Set	1																															
V	RW	PIN21																																	
			Low	0																															
			High	1																															
			Set	1																															
W	RW	PIN22																																	
			Low	0																															
			High	1																															
			Set	1																															
X	RW	PIN23																																	
			Low	0																															
			High	1																															
			Set	1																															
Y	RW	PIN24																																	
			Low	0																															
			High	1																															
			Set	1																															
Z	RW	PIN25																																	
			Low	0																															
			High	1																															
			Set	1																															
a	RW	PIN26																																	
			Low	0																															
			High	1																															
			Set	1																															
b	RW	PIN27																																	
			Low	0																															
			High	1																															
			Set	1																															
c	RW	PIN28																																	
			Low	0																															
			High	1																															
			Set	1																															

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
d	RW	PIN29			Pin 29
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Set	1	Write: writing a '1' sets the pin high; writing a '0' has no effect
e	RW	PIN30			Pin 30
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Set	1	Write: writing a '1' sets the pin high; writing a '0' has no effect
f	RW	PIN31			Pin 31
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Set	1	Write: writing a '1' sets the pin high; writing a '0' has no effect

20.3.3 OUTCLR

Address offset: 0x50C

Clear individual bits in GPIO port

Read: reads value of OUT register.

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	PIN0			Pin 0
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
B	RW	PIN1			Pin 1
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
C	RW	PIN2			Pin 2
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
D	RW	PIN3			Pin 3
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
E	RW	PIN4			Pin 4
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
F	RW	PIN5			Pin 5
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
G	RW	PIN6			Pin 6
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
H	RW	PIN7			Pin 7
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
I	RW	PIN8			Pin 8
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
J	RW	PIN9			Pin 9
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
K	RW	PIN10			Pin 10
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
L	RW	PIN11			Pin 11
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
M	RW	PIN12			Pin 12
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
N	RW	PIN13			Pin 13
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
O	RW	PIN14			Pin 14
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
P	RW	PIN15			Pin 15
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
Q	RW	PIN16			Pin 16
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
R	RW	PIN17			Pin 17
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
S	RW	PIN18			Pin 18
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
T	RW	PIN19			Pin 19
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
U	RW	PIN20			Pin 20
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: writing a '1' sets the pin low; writing a '0' has no effect
V	RW	PIN21			Pin 21
			Low	0	Read: pin driver is low

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			High	1																															
			Clear	1																															
W	RW	PIN22																																	
			Low	0																															
			High	1																															
			Clear	1																															
X	RW	PIN23																																	
			Low	0																															
			High	1																															
			Clear	1																															
Y	RW	PIN24																																	
			Low	0																															
			High	1																															
			Clear	1																															
Z	RW	PIN25																																	
			Low	0																															
			High	1																															
			Clear	1																															
a	RW	PIN26																																	
			Low	0																															
			High	1																															
			Clear	1																															
b	RW	PIN27																																	
			Low	0																															
			High	1																															
			Clear	1																															
c	RW	PIN28																																	
			Low	0																															
			High	1																															
			Clear	1																															
d	RW	PIN29																																	
			Low	0																															
			High	1																															
			Clear	1																															
e	RW	PIN30																																	
			Low	0																															
			High	1																															
			Clear	1																															
f	RW	PIN31																																	
			Low	0																															
			High	1																															
			Clear	1																															

20.3.4 IN

Address offset: 0x510

Read GPIO port

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	PIN0																																	
			Low	0																															
			High	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
B	R	PIN1																																	
			Low	0																															
			High	1																															
C	R	PIN2																																	
			Low	0																															
			High	1																															
D	R	PIN3																																	
			Low	0																															
			High	1																															
E	R	PIN4																																	
			Low	0																															
			High	1																															
F	R	PIN5																																	
			Low	0																															
			High	1																															
G	R	PIN6																																	
			Low	0																															
			High	1																															
H	R	PIN7																																	
			Low	0																															
			High	1																															
I	R	PIN8																																	
			Low	0																															
			High	1																															
J	R	PIN9																																	
			Low	0																															
			High	1																															
K	R	PIN10																																	
			Low	0																															
			High	1																															
L	R	PIN11																																	
			Low	0																															
			High	1																															
M	R	PIN12																																	
			Low	0																															
			High	1																															
N	R	PIN13																																	
			Low	0																															
			High	1																															
O	R	PIN14																																	
			Low	0																															
			High	1																															
P	R	PIN15																																	
			Low	0																															
			High	1																															
Q	R	PIN16																																	
			Low	0																															
			High	1																															
R	R	PIN17																																	
			Low	0																															
			High	1																															
S	R	PIN18																																	
			Low	0																															
			High	1																															
T	R	PIN19																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Low	0																															
			High	1																															
U	R	PIN20																																	
			Low	0																															
			High	1																															
V	R	PIN21																																	
			Low	0																															
			High	1																															
W	R	PIN22																																	
			Low	0																															
			High	1																															
X	R	PIN23																																	
			Low	0																															
			High	1																															
Y	R	PIN24																																	
			Low	0																															
			High	1																															
Z	R	PIN25																																	
			Low	0																															
			High	1																															
a	R	PIN26																																	
			Low	0																															
			High	1																															
b	R	PIN27																																	
			Low	0																															
			High	1																															
c	R	PIN28																																	
			Low	0																															
			High	1																															
d	R	PIN29																																	
			Low	0																															
			High	1																															
e	R	PIN30																																	
			Low	0																															
			High	1																															
f	R	PIN31																																	
			Low	0																															
			High	1																															

20.3.5 DIR

Address offset: 0x514

Direction of GPIO pins

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN0																																	
			Input	0																															
			Output	1																															
B	RW	PIN1																																	
			Input	0																															
			Output	1																															
C	RW	PIN2																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Input	0																															
			Output	1																															
D	RW	PIN3																																	
			Input	0																															
			Output	1																															
E	RW	PIN4																																	
			Input	0																															
			Output	1																															
F	RW	PIN5																																	
			Input	0																															
			Output	1																															
G	RW	PIN6																																	
			Input	0																															
			Output	1																															
H	RW	PIN7																																	
			Input	0																															
			Output	1																															
I	RW	PIN8																																	
			Input	0																															
			Output	1																															
J	RW	PIN9																																	
			Input	0																															
			Output	1																															
K	RW	PIN10																																	
			Input	0																															
			Output	1																															
L	RW	PIN11																																	
			Input	0																															
			Output	1																															
M	RW	PIN12																																	
			Input	0																															
			Output	1																															
N	RW	PIN13																																	
			Input	0																															
			Output	1																															
O	RW	PIN14																																	
			Input	0																															
			Output	1																															
P	RW	PIN15																																	
			Input	0																															
			Output	1																															
Q	RW	PIN16																																	
			Input	0																															
			Output	1																															
R	RW	PIN17																																	
			Input	0																															
			Output	1																															
S	RW	PIN18																																	
			Input	0																															
			Output	1																															
T	RW	PIN19																																	
			Input	0																															
			Output	1																															
U	RW	PIN20																																	
			Input	0																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Output	1																															
				Pin set as output																															
V	RW	PIN21																																	
				Pin 21																															
			Input	0																															
				Pin set as input																															
			Output	1																															
				Pin set as output																															
W	RW	PIN22																																	
				Pin 22																															
			Input	0																															
				Pin set as input																															
			Output	1																															
				Pin set as output																															
X	RW	PIN23																																	
				Pin 23																															
			Input	0																															
				Pin set as input																															
			Output	1																															
				Pin set as output																															
Y	RW	PIN24																																	
				Pin 24																															
			Input	0																															
				Pin set as input																															
			Output	1																															
				Pin set as output																															
Z	RW	PIN25																																	
				Pin 25																															
			Input	0																															
				Pin set as input																															
			Output	1																															
				Pin set as output																															
a	RW	PIN26																																	
				Pin 26																															
			Input	0																															
				Pin set as input																															
			Output	1																															
				Pin set as output																															
b	RW	PIN27																																	
				Pin 27																															
			Input	0																															
				Pin set as input																															
			Output	1																															
				Pin set as output																															
c	RW	PIN28																																	
				Pin 28																															
			Input	0																															
				Pin set as input																															
			Output	1																															
				Pin set as output																															
d	RW	PIN29																																	
				Pin 29																															
			Input	0																															
				Pin set as input																															
			Output	1																															
				Pin set as output																															
e	RW	PIN30																																	
				Pin 30																															
			Input	0																															
				Pin set as input																															
			Output	1																															
				Pin set as output																															
f	RW	PIN31																																	
				Pin 31																															
			Input	0																															
				Pin set as input																															
			Output	1																															
				Pin set as output																															

20.3.6 DIRSET

Address offset: 0x518

DIR set register

Read: reads value of DIR register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN0																																	
				Set as output pin 0																															
			Input	0																															
				Read: pin set as input																															
			Output	1																															
				Read: pin set as output																															
			Set	1																															
				Write: writing a '1' sets pin to output; writing a '0' has no effect																															
B	RW	PIN1																																	
				Set as output pin 1																															
			Input	0																															
				Read: pin set as input																															
			Output	1																															
				Read: pin set as output																															
			Set	1																															
				Write: writing a '1' sets pin to output; writing a '0' has no effect																															
C	RW	PIN2																																	
				Set as output pin 2																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Input	0																															
			Output	1																															
			Set	1																															
D	RW	PIN3																																	
			Input	0																															
			Output	1																															
			Set	1																															
E	RW	PIN4																																	
			Input	0																															
			Output	1																															
			Set	1																															
F	RW	PIN5																																	
			Input	0																															
			Output	1																															
			Set	1																															
G	RW	PIN6																																	
			Input	0																															
			Output	1																															
			Set	1																															
H	RW	PIN7																																	
			Input	0																															
			Output	1																															
			Set	1																															
I	RW	PIN8																																	
			Input	0																															
			Output	1																															
			Set	1																															
J	RW	PIN9																																	
			Input	0																															
			Output	1																															
			Set	1																															
K	RW	PIN10																																	
			Input	0																															
			Output	1																															
			Set	1																															
L	RW	PIN11																																	
			Input	0																															
			Output	1																															
			Set	1																															
M	RW	PIN12																																	
			Input	0																															
			Output	1																															
			Set	1																															
N	RW	PIN13																																	
			Input	0																															
			Output	1																															
			Set	1																															
O	RW	PIN14																																	
			Input	0																															
			Output	1																															
			Set	1																															
P	RW	PIN15																																	
			Input	0																															
			Output	1																															
			Set	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
Q	RW	PIN16																																	
			Input	0																															
			Output	1																															
			Set	1																															
R	RW	PIN17																																	
			Input	0																															
			Output	1																															
			Set	1																															
S	RW	PIN18																																	
			Input	0																															
			Output	1																															
			Set	1																															
T	RW	PIN19																																	
			Input	0																															
			Output	1																															
			Set	1																															
U	RW	PIN20																																	
			Input	0																															
			Output	1																															
			Set	1																															
V	RW	PIN21																																	
			Input	0																															
			Output	1																															
			Set	1																															
W	RW	PIN22																																	
			Input	0																															
			Output	1																															
			Set	1																															
X	RW	PIN23																																	
			Input	0																															
			Output	1																															
			Set	1																															
Y	RW	PIN24																																	
			Input	0																															
			Output	1																															
			Set	1																															
Z	RW	PIN25																																	
			Input	0																															
			Output	1																															
			Set	1																															
a	RW	PIN26																																	
			Input	0																															
			Output	1																															
			Set	1																															
b	RW	PIN27																																	
			Input	0																															
			Output	1																															
			Set	1																															
c	RW	PIN28																																	
			Input	0																															
			Output	1																															
			Set	1																															
d	RW	PIN29																																	
			Input	0																															
			Output	1																															

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
			Set	1	Write: writing a '1' sets pin to output; writing a '0' has no effect
e	RW	PIN30			Set as output pin 30
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Set	1	Write: writing a '1' sets pin to output; writing a '0' has no effect
f	RW	PIN31			Set as output pin 31
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Set	1	Write: writing a '1' sets pin to output; writing a '0' has no effect

20.3.7 DIRCLR

Address offset: 0x51C

DIR clear register

Read: reads value of DIR register.

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	PIN0			Set as input pin 0
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: writing a '1' sets pin to input; writing a '0' has no effect
B	RW	PIN1			Set as input pin 1
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: writing a '1' sets pin to input; writing a '0' has no effect
C	RW	PIN2			Set as input pin 2
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: writing a '1' sets pin to input; writing a '0' has no effect
D	RW	PIN3			Set as input pin 3
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: writing a '1' sets pin to input; writing a '0' has no effect
E	RW	PIN4			Set as input pin 4
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: writing a '1' sets pin to input; writing a '0' has no effect
F	RW	PIN5			Set as input pin 5
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: writing a '1' sets pin to input; writing a '0' has no effect
G	RW	PIN6			Set as input pin 6
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: writing a '1' sets pin to input; writing a '0' has no effect
H	RW	PIN7			Set as input pin 7
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: writing a '1' sets pin to input; writing a '0' has no effect
I	RW	PIN8			Set as input pin 8
			Input	0	Read: pin set as input

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Output	1																															
			Clear	1																															
J	RW	PIN9																																	
			Input	0																															
			Output	1																															
			Clear	1																															
K	RW	PIN10																																	
			Input	0																															
			Output	1																															
			Clear	1																															
L	RW	PIN11																																	
			Input	0																															
			Output	1																															
			Clear	1																															
M	RW	PIN12																																	
			Input	0																															
			Output	1																															
			Clear	1																															
N	RW	PIN13																																	
			Input	0																															
			Output	1																															
			Clear	1																															
O	RW	PIN14																																	
			Input	0																															
			Output	1																															
			Clear	1																															
P	RW	PIN15																																	
			Input	0																															
			Output	1																															
			Clear	1																															
Q	RW	PIN16																																	
			Input	0																															
			Output	1																															
			Clear	1																															
R	RW	PIN17																																	
			Input	0																															
			Output	1																															
			Clear	1																															
S	RW	PIN18																																	
			Input	0																															
			Output	1																															
			Clear	1																															
T	RW	PIN19																																	
			Input	0																															
			Output	1																															
			Clear	1																															
U	RW	PIN20																																	
			Input	0																															
			Output	1																															
			Clear	1																															
V	RW	PIN21																																	
			Input	0																															
			Output	1																															
			Clear	1																															
W	RW	PIN22																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Input	0																															
			Output	1																															
			Clear	1																															
X	RW	PIN23																																	
			Input	0																															
			Output	1																															
			Clear	1																															
Y	RW	PIN24																																	
			Input	0																															
			Output	1																															
			Clear	1																															
Z	RW	PIN25																																	
			Input	0																															
			Output	1																															
			Clear	1																															
a	RW	PIN26																																	
			Input	0																															
			Output	1																															
			Clear	1																															
b	RW	PIN27																																	
			Input	0																															
			Output	1																															
			Clear	1																															
c	RW	PIN28																																	
			Input	0																															
			Output	1																															
			Clear	1																															
d	RW	PIN29																																	
			Input	0																															
			Output	1																															
			Clear	1																															
e	RW	PIN30																																	
			Input	0																															
			Output	1																															
			Clear	1																															
f	RW	PIN31																																	
			Input	0																															
			Output	1																															
			Clear	1																															

20.3.8 LATCH

Address offset: 0x520

Latch register indicating what GPIO pins that have met the criteria set in the PIN_CNF[n].SENSE registers

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN0																																	
			NotLatched	0																															
			Latched	1																															
B	RW	PIN1																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			NotLatched	0																															
			Latched	1																															
C	RW	PIN2																																	
			NotLatched	0																															
			Latched	1																															
D	RW	PIN3																																	
			NotLatched	0																															
			Latched	1																															
E	RW	PIN4																																	
			NotLatched	0																															
			Latched	1																															
F	RW	PIN5																																	
			NotLatched	0																															
			Latched	1																															
G	RW	PIN6																																	
			NotLatched	0																															
			Latched	1																															
H	RW	PIN7																																	
			NotLatched	0																															
			Latched	1																															
I	RW	PIN8																																	
			NotLatched	0																															
			Latched	1																															
J	RW	PIN9																																	
			NotLatched	0																															
			Latched	1																															
K	RW	PIN10																																	
			NotLatched	0																															
			Latched	1																															
L	RW	PIN11																																	
			NotLatched	0																															
			Latched	1																															
M	RW	PIN12																																	
			NotLatched	0																															
			Latched	1																															
N	RW	PIN13																																	
			NotLatched	0																															
			Latched	1																															
O	RW	PIN14																																	
			NotLatched	0																															
			Latched	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
P	RW	PIN15																																	
			NotLatched	0																															
			Latched	1																															
Q	RW	PIN16																																	
			NotLatched	0																															
			Latched	1																															
R	RW	PIN17																																	
			NotLatched	0																															
			Latched	1																															
S	RW	PIN18																																	
			NotLatched	0																															
			Latched	1																															
T	RW	PIN19																																	
			NotLatched	0																															
			Latched	1																															
U	RW	PIN20																																	
			NotLatched	0																															
			Latched	1																															
V	RW	PIN21																																	
			NotLatched	0																															
			Latched	1																															
W	RW	PIN22																																	
			NotLatched	0																															
			Latched	1																															
X	RW	PIN23																																	
			NotLatched	0																															
			Latched	1																															
Y	RW	PIN24																																	
			NotLatched	0																															
			Latched	1																															
Z	RW	PIN25																																	
			NotLatched	0																															
			Latched	1																															
a	RW	PIN26																																	
			NotLatched	0																															
			Latched	1																															
b	RW	PIN27																																	
			NotLatched	0																															
			Latched	1																															
c	RW	PIN28																																	
			NotLatched	0																															

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
			Latched	1	Criteria has been met																													
d	RW	PIN29			Status on whether PIN29 has met criteria set in PIN_CNF29.SENSE register. Write '1' to clear.																													
			NotLatched	0	Criteria has not been met																													
			Latched	1	Criteria has been met																													
e	RW	PIN30			Status on whether PIN30 has met criteria set in PIN_CNF30.SENSE register. Write '1' to clear.																													
			NotLatched	0	Criteria has not been met																													
			Latched	1	Criteria has been met																													
f	RW	PIN31			Status on whether PIN31 has met criteria set in PIN_CNF31.SENSE register. Write '1' to clear.																													
			NotLatched	0	Criteria has not been met																													
			Latched	1	Criteria has been met																													

20.3.9 DETECTMODE

Address offset: 0x524

Select between default DETECT signal behaviour and LDETECT mode

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	DETECTMODE			Select between default DETECT signal behaviour and LDETECT mode																													
			Default	0	DETECT directly connected to PIN DETECT signals																													
			LDETECT	1	Use the latched LDETECT behaviour																													

20.3.10 PIN_CNF[0]

Address offset: 0x700

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D				C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	DIR			Pin direction. Same physical register as DIR register																													
			Input	0	Configure pin as an input pin																													
			Output	1	Configure pin as an output pin																													
B	RW	INPUT			Connect or disconnect input buffer																													
			Connect	0	Connect input buffer																													
			Disconnect	1	Disconnect input buffer																													
C	RW	PULL			Pull configuration																													
			Disabled	0	No pull																													
			Pulldown	1	Pull down on pin																													
			Pullup	3	Pull up on pin																													
D	RW	DRIVE			Drive configuration																													
			S0S1	0	Standard '0', standard '1'																													
			H0S1	1	High drive '0', standard '1'																													
			S0H1	2	Standard '0', high drive '1'																													
			H0H1	3	High drive '0', high 'drive '1''																													
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)																													

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
			D0H1								5																Disconnect '0', high drive '1' (normally used for wired-or connections)							
			S0D1								6																Standard '0'. disconnect '1' (normally used for wired-and connections)							
			H0D1								7																High drive '0', disconnect '1' (normally used for wired-and connections)							
E	RW	SENSE																									Pin sensing mechanism							
			Disabled								0																Disabled							
			High								2																Sense for high level							
			Low								3																Sense for low level							

20.3.11 PIN_CNF[1]

Address offset: 0x704

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
A	RW	DIR																									Pin direction. Same physical register as DIR register							
			Input								0																Configure pin as an input pin							
			Output								1																Configure pin as an output pin							
B	RW	INPUT																									Connect or disconnect input buffer							
			Connect								0																Connect input buffer							
			Disconnect								1																Disconnect input buffer							
C	RW	PULL																									Pull configuration							
			Disabled								0																No pull							
			Pulldown								1																Pull down on pin							
			Pullup								3																Pull up on pin							
D	RW	DRIVE																									Drive configuration							
			S0S1								0																Standard '0', standard '1'							
			H0S1								1																High drive '0', standard '1'							
			S0H1								2																Standard '0', high drive '1'							
			H0H1								3																High drive '0', high 'drive '1''							
			D0S1								4																Disconnect '0' standard '1' (normally used for wired-or connections)							
			D0H1								5																Disconnect '0', high drive '1' (normally used for wired-or connections)							
			S0D1								6																Standard '0'. disconnect '1' (normally used for wired-and connections)							
			H0D1								7																High drive '0', disconnect '1' (normally used for wired-and connections)							
E	RW	SENSE																									Pin sensing mechanism							
			Disabled								0																Disabled							
			High								2																Sense for high level							
			Low								3																Sense for low level							

20.3.12 PIN_CNF[2]

Address offset: 0x708

Configuration of GPIO pins

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		E	E						D	D	D						C	C	B	A
Reset 0x00000002				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	DIR																																	
			Input	0																															
			Output	1																															
B	RW	INPUT																																	
			Connect	0																															
			Disconnect	1																															
C	RW	PULL																																	
			Disabled	0																															
			Pulldown	1																															
			Pullup	3																															
D	RW	DRIVE																																	
			S0S1	0																															
			H0S1	1																															
			S0H1	2																															
			H0H1	3																															
			D0S1	4																															
			D0H1	5																															
			S0D1	6																															
			H0D1	7																															
E	RW	SENSE																																	
			Disabled	0																															
			High	2																															
			Low	3																															

20.3.13 PIN_CNF[3]

Address offset: 0x70C

Configuration of GPIO pins

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		E	E						D	D	D						C	C	B	A
Reset 0x00000002				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	DIR																																	
			Input	0																															
			Output	1																															
B	RW	INPUT																																	
			Connect	0																															
			Disconnect	1																															
C	RW	PULL																																	
			Disabled	0																															
			Pulldown	1																															
			Pullup	3																															
D	RW	DRIVE																																	
			S0S1	0																															
			H0S1	1																															
			S0H1	2																															
			H0H1	3																															
			D0S1	4																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		E	E						D	D	D						C	C	B	A
Reset 0x00000002				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			D0H1	5																															
				Disconnect '0', high drive '1' (normally used for wired-or connections)																															
			S0D1	6																															
				Standard '0'. disconnect '1' (normally used for wired-and connections)																															
			H0D1	7																															
				High drive '0', disconnect '1' (normally used for wired-and connections)																															
E	RW	SENSE																																	
			Disconnected	0																															
				Disabled																															
			High	2																															
				Sense for high level																															
			Low	3																															
				Sense for low level																															

20.3.14 PIN_CNF[4]

Address offset: 0x710

Configuration of GPIO pins

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		E	E						D	D	D						C	C	B	A
Reset 0x00000002				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	DIR																																	
			Input	0																															
				Configure pin as an input pin																															
			Output	1																															
				Configure pin as an output pin																															
B	RW	INPUT																																	
			Connect	0																															
				Connect input buffer																															
			Disconnect	1																															
				Disconnect input buffer																															
C	RW	PULL																																	
			Disabled	0																															
				No pull																															
			Pulldown	1																															
				Pull down on pin																															
			Pullup	3																															
				Pull up on pin																															
D	RW	DRIVE																																	
			S0S1	0																															
				Standard '0', standard '1'																															
			H0S1	1																															
				High drive '0', standard '1'																															
			S0H1	2																															
				Standard '0', high drive '1'																															
			H0H1	3																															
				High drive '0', high 'drive '1''																															
			D0S1	4																															
				Disconnect '0' standard '1' (normally used for wired-or connections)																															
			D0H1	5																															
				Disconnect '0', high drive '1' (normally used for wired-or connections)																															
			S0D1	6																															
				Standard '0'. disconnect '1' (normally used for wired-and connections)																															
			H0D1	7																															
				High drive '0', disconnect '1' (normally used for wired-and connections)																															
E	RW	SENSE																																	
			Disabled	0																															
				Disabled																															
			High	2																															
				Sense for high level																															
			Low	3																															
				Sense for low level																															

20.3.15 PIN_CNF[5]

Address offset: 0x714

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	DIR																																
			Input	0																														
			Output	1																														
B	RW	INPUT																																
			Connect	0																														
			Disconnect	1																														
C	RW	PULL																																
			Disabled	0																														
			Pulldown	1																														
			Pullup	3																														
D	RW	DRIVE																																
			S0S1	0																														
			H0S1	1																														
			S0H1	2																														
			H0H1	3																														
			D0S1	4																														
			D0H1	5																														
			S0D1	6																														
			H0D1	7																														
E	RW	SENSE																																
			Disabled	0																														
			High	2																														
			Low	3																														

20.3.16 PIN_CNF[6]

Address offset: 0x718

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	DIR																																
			Input	0																														
			Output	1																														
B	RW	INPUT																																
			Connect	0																														
			Disconnect	1																														
C	RW	PULL																																
			Disabled	0																														
			Pulldown	1																														
			Pullup	3																														
D	RW	DRIVE																																
			S0S1	0																														
			H0S1	1																														
			S0H1	2																														
			H0H1	3																														
			D0S1	4																														

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
			D0H1								5																Disconnect '0', high drive '1' (normally used for wired-or connections)							
			S0D1								6																Standard '0'. disconnect '1' (normally used for wired-and connections)							
			H0D1								7																High drive '0', disconnect '1' (normally used for wired-and connections)							
E	RW	SENSE																									Pin sensing mechanism							
			Disabled								0																Disabled							
			High								2																Sense for high level							
			Low								3																Sense for low level							

20.3.17 PIN_CNF[7]

Address offset: 0x71C

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
A	RW	DIR																									Pin direction. Same physical register as DIR register							
			Input								0																Configure pin as an input pin							
			Output								1																Configure pin as an output pin							
B	RW	INPUT																									Connect or disconnect input buffer							
			Connect								0																Connect input buffer							
			Disconnect								1																Disconnect input buffer							
C	RW	PULL																									Pull configuration							
			Disabled								0																No pull							
			Pulldown								1																Pull down on pin							
			Pullup								3																Pull up on pin							
D	RW	DRIVE																									Drive configuration							
			S0S1								0																Standard '0', standard '1'							
			H0S1								1																High drive '0', standard '1'							
			S0H1								2																Standard '0', high drive '1'							
			H0H1								3																High drive '0', high 'drive '1''							
			D0S1								4																Disconnect '0' standard '1' (normally used for wired-or connections)							
			D0H1								5																Disconnect '0', high drive '1' (normally used for wired-or connections)							
			S0D1								6																Standard '0'. disconnect '1' (normally used for wired-and connections)							
			H0D1								7																High drive '0', disconnect '1' (normally used for wired-and connections)							
E	RW	SENSE																									Pin sensing mechanism							
			Disabled								0																Disabled							
			High								2																Sense for high level							
			Low								3																Sense for low level							

20.3.18 PIN_CNF[8]

Address offset: 0x720

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	DIR																																
			Input	0																														
			Output	1																														
B	RW	INPUT																																
			Connect	0																														
			Disconnect	1																														
C	RW	PULL																																
			Disabled	0																														
			Pulldown	1																														
			Pullup	3																														
D	RW	DRIVE																																
			S0S1	0																														
			H0S1	1																														
			S0H1	2																														
			H0H1	3																														
			D0S1	4																														
			D0H1	5																														
			S0D1	6																														
			H0D1	7																														
E	RW	SENSE																																
			Disabled	0																														
			High	2																														
			Low	3																														

20.3.19 PIN_CNF[9]

Address offset: 0x724

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	DIR																																
			Input	0																														
			Output	1																														
B	RW	INPUT																																
			Connect	0																														
			Disconnect	1																														
C	RW	PULL																																
			Disabled	0																														
			Pulldown	1																														
			Pullup	3																														
D	RW	DRIVE																																
			S0S1	0																														
			H0S1	1																														
			S0H1	2																														
			H0H1	3																														
			D0S1	4																														

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
			D0H1								5																Disconnect '0', high drive '1' (normally used for wired-or connections)							
			S0D1								6																Standard '0'. disconnect '1' (normally used for wired-and connections)							
			H0D1								7																High drive '0', disconnect '1' (normally used for wired-and connections)							
E	RW	SENSE																									Pin sensing mechanism							
			Disabled								0																Disabled							
			High								2																Sense for high level							
			Low								3																Sense for low level							

20.3.20 PIN_CNF[10]

Address offset: 0x728

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
A	RW	DIR																									Pin direction. Same physical register as DIR register							
			Input								0																Configure pin as an input pin							
			Output								1																Configure pin as an output pin							
B	RW	INPUT																									Connect or disconnect input buffer							
			Connect								0																Connect input buffer							
			Disconnect								1																Disconnect input buffer							
C	RW	PULL																									Pull configuration							
			Disabled								0																No pull							
			Pulldown								1																Pull down on pin							
			Pullup								3																Pull up on pin							
D	RW	DRIVE																									Drive configuration							
			S0S1								0																Standard '0', standard '1'							
			H0S1								1																High drive '0', standard '1'							
			S0H1								2																Standard '0', high drive '1'							
			H0H1								3																High drive '0', high 'drive '1''							
			D0S1								4																Disconnect '0' standard '1' (normally used for wired-or connections)							
			D0H1								5																Disconnect '0', high drive '1' (normally used for wired-or connections)							
			S0D1								6																Standard '0'. disconnect '1' (normally used for wired-and connections)							
			H0D1								7																High drive '0', disconnect '1' (normally used for wired-and connections)							
E	RW	SENSE																									Pin sensing mechanism							
			Disabled								0																Disabled							
			High								2																Sense for high level							
			Low								3																Sense for low level							

20.3.21 PIN_CNF[11]

Address offset: 0x72C

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														**Description**
A	RW	DIR																																Pin direction. Same physical register as DIR register
			Input	0																														Configure pin as an input pin
			Output	1																														Configure pin as an output pin
B	RW	INPUT																																Connect or disconnect input buffer
			Connect	0																														Connect input buffer
			Disconnect	1																														Disconnect input buffer
C	RW	PULL																																Pull configuration
			Disabled	0																														No pull
			Pulldown	1																														Pull down on pin
			Pullup	3																														Pull up on pin
D	RW	DRIVE																																Drive configuration
			S0S1	0																														Standard '0', standard '1'
			H0S1	1																														High drive '0', standard '1'
			S0H1	2																														Standard '0', high drive '1'
			H0H1	3																														High drive '0', high 'drive '1''
			D0S1	4																														Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5																														Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6																														Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7																														High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE																																Pin sensing mechanism
			Disabled	0																														Disabled
			High	2																														Sense for high level
			Low	3																														Sense for low level

20.3.22 PIN_CNF[12]

Address offset: 0x730

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														**Description**
A	RW	DIR																																Pin direction. Same physical register as DIR register
			Input	0																														Configure pin as an input pin
			Output	1																														Configure pin as an output pin
B	RW	INPUT																																Connect or disconnect input buffer
			Connect	0																														Connect input buffer
			Disconnect	1																														Disconnect input buffer
C	RW	PULL																																Pull configuration
			Disabled	0																														No pull
			Pulldown	1																														Pull down on pin
			Pullup	3																														Pull up on pin
D	RW	DRIVE																																Drive configuration
			S0S1	0																														Standard '0', standard '1'
			H0S1	1																														High drive '0', standard '1'
			S0H1	2																														Standard '0', high drive '1'
			H0H1	3																														High drive '0', high 'drive '1''
			D0S1	4																														Disconnect '0' standard '1' (normally used for wired-or connections)

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		E	E						D	D	D						C	C	B	A
Reset 0x00000002				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			D0H1	5																															
				Disconnect '0', high drive '1' (normally used for wired-or connections)																															
			S0D1	6																															
				Standard '0'. disconnect '1' (normally used for wired-and connections)																															
			H0D1	7																															
				High drive '0', disconnect '1' (normally used for wired-and connections)																															
E	RW	SENSE																																	
			Disconnected	0																															
				Disabled																															
			High	2																															
				Sense for high level																															
			Low	3																															
				Sense for low level																															

20.3.23 PIN_CNF[13]

Address offset: 0x734

Configuration of GPIO pins

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		E	E						D	D	D						C	C	B	A
Reset 0x00000002				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	DIR																																	
			Input	0																															
				Configure pin as an input pin																															
			Output	1																															
				Configure pin as an output pin																															
B	RW	INPUT																																	
			Connect	0																															
				Connect input buffer																															
			Disconnect	1																															
				Disconnect input buffer																															
C	RW	PULL																																	
			Disabled	0																															
				No pull																															
			Pulldown	1																															
				Pull down on pin																															
			Pullup	3																															
				Pull up on pin																															
D	RW	DRIVE																																	
			S0S1	0																															
				Standard '0', standard '1'																															
			H0S1	1																															
				High drive '0', standard '1'																															
			S0H1	2																															
				Standard '0', high drive '1'																															
			H0H1	3																															
				High drive '0', high 'drive '1''																															
			D0S1	4																															
				Disconnect '0' standard '1' (normally used for wired-or connections)																															
			D0H1	5																															
				Disconnect '0', high drive '1' (normally used for wired-or connections)																															
			S0D1	6																															
				Standard '0'. disconnect '1' (normally used for wired-and connections)																															
			H0D1	7																															
				High drive '0', disconnect '1' (normally used for wired-and connections)																															
E	RW	SENSE																																	
			Disabled	0																															
				Disabled																															
			High	2																															
				Sense for high level																															
			Low	3																															
				Sense for low level																															

20.3.24 PIN_CNF[14]

Address offset: 0x738

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	DIR																																
			Input	0																														
			Output	1																														
B	RW	INPUT																																
			Connect	0																														
			Disconnect	1																														
C	RW	PULL																																
			Disabled	0																														
			Pulldown	1																														
			Pullup	3																														
D	RW	DRIVE																																
			S0S1	0																														
			H0S1	1																														
			S0H1	2																														
			H0H1	3																														
			D0S1	4																														
			D0H1	5																														
			S0D1	6																														
			H0D1	7																														
E	RW	SENSE																																
			Disabled	0																														
			High	2																														
			Low	3																														

20.3.25 PIN_CNF[15]

Address offset: 0x73C

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	DIR																																
			Input	0																														
			Output	1																														
B	RW	INPUT																																
			Connect	0																														
			Disconnect	1																														
C	RW	PULL																																
			Disabled	0																														
			Pulldown	1																														
			Pullup	3																														
D	RW	DRIVE																																
			S0S1	0																														
			H0S1	1																														
			S0H1	2																														
			H0H1	3																														
			D0S1	4																														

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		E	E						D	D	D						C	C	B	A
Reset 0x00000002				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			D0H1	5																															
				Disconnect '0', high drive '1' (normally used for wired-or connections)																															
			S0D1	6																															
				Standard '0'. disconnect '1' (normally used for wired-and connections)																															
			H0D1	7																															
				High drive '0', disconnect '1' (normally used for wired-and connections)																															
E	RW	SENSE																																	
			Disconnect	0																															
				Disabled																															
			High	2																															
				Sense for high level																															
			Low	3																															
				Sense for low level																															

20.3.26 PIN_CNF[16]

Address offset: 0x740

Configuration of GPIO pins

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		E	E						D	D	D						C	C	B	A
Reset 0x00000002				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	DIR																																	
			Input	0																															
				Configure pin as an input pin																															
			Output	1																															
				Configure pin as an output pin																															
B	RW	INPUT																																	
			Connect	0																															
				Connect input buffer																															
			Disconnect	1																															
				Disconnect input buffer																															
C	RW	PULL																																	
			Disabled	0																															
				No pull																															
			Pulldown	1																															
				Pull down on pin																															
			Pullup	3																															
				Pull up on pin																															
D	RW	DRIVE																																	
			S0S1	0																															
				Standard '0', standard '1'																															
			H0S1	1																															
				High drive '0', standard '1'																															
			S0H1	2																															
				Standard '0', high drive '1'																															
			H0H1	3																															
				High drive '0', high 'drive '1''																															
			D0S1	4																															
				Disconnect '0' standard '1' (normally used for wired-or connections)																															
			D0H1	5																															
				Disconnect '0', high drive '1' (normally used for wired-or connections)																															
			S0D1	6																															
				Standard '0'. disconnect '1' (normally used for wired-and connections)																															
			H0D1	7																															
				High drive '0', disconnect '1' (normally used for wired-and connections)																															
E	RW	SENSE																																	
			Disconnect	0																															
				Disabled																															
			High	2																															
				Sense for high level																															
			Low	3																															
				Sense for low level																															

20.3.27 PIN_CNF[17]

Address offset: 0x744

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	DIR																																
			Input	0																														
			Output	1																														
B	RW	INPUT																																
			Connect	0																														
			Disconnect	1																														
C	RW	PULL																																
			Disabled	0																														
			Pulldown	1																														
			Pullup	3																														
D	RW	DRIVE																																
			S0S1	0																														
			H0S1	1																														
			S0H1	2																														
			H0H1	3																														
			D0S1	4																														
			D0H1	5																														
			S0D1	6																														
			H0D1	7																														
E	RW	SENSE																																
			Disabled	0																														
			High	2																														
			Low	3																														

20.3.28 PIN_CNF[18]

Address offset: 0x748

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	DIR																																
			Input	0																														
			Output	1																														
B	RW	INPUT																																
			Connect	0																														
			Disconnect	1																														
C	RW	PULL																																
			Disabled	0																														
			Pulldown	1																														
			Pullup	3																														
D	RW	DRIVE																																
			S0S1	0																														
			H0S1	1																														
			S0H1	2																														
			H0H1	3																														
			D0S1	4																														

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
			D0H1								5																Disconnect '0', high drive '1' (normally used for wired-or connections)							
			S0D1								6																Standard '0'. disconnect '1' (normally used for wired-and connections)							
			H0D1								7																High drive '0', disconnect '1' (normally used for wired-and connections)							
E	RW	SENSE																									Pin sensing mechanism							
			Disabled								0																Disabled							
			High								2																Sense for high level							
			Low								3																Sense for low level							

20.3.29 PIN_CNF[19]

Address offset: 0x74C

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
A	RW	DIR																									Pin direction. Same physical register as DIR register							
			Input								0																Configure pin as an input pin							
			Output								1																Configure pin as an output pin							
B	RW	INPUT																									Connect or disconnect input buffer							
			Connect								0																Connect input buffer							
			Disconnect								1																Disconnect input buffer							
C	RW	PULL																									Pull configuration							
			Disabled								0																No pull							
			Pulldown								1																Pull down on pin							
			Pullup								3																Pull up on pin							
D	RW	DRIVE																									Drive configuration							
			S0S1								0																Standard '0', standard '1'							
			H0S1								1																High drive '0', standard '1'							
			S0H1								2																Standard '0', high drive '1'							
			H0H1								3																High drive '0', high 'drive '1''							
			D0S1								4																Disconnect '0' standard '1' (normally used for wired-or connections)							
			D0H1								5																Disconnect '0', high drive '1' (normally used for wired-or connections)							
			S0D1								6																Standard '0'. disconnect '1' (normally used for wired-and connections)							
			H0D1								7																High drive '0', disconnect '1' (normally used for wired-and connections)							
E	RW	SENSE																									Pin sensing mechanism							
			Disabled								0																Disabled							
			High								2																Sense for high level							
			Low								3																Sense for low level							

20.3.30 PIN_CNF[20]

Address offset: 0x750

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	DIR																																
			Input	0																														
			Output	1																														
B	RW	INPUT																																
			Connect	0																														
			Disconnect	1																														
C	RW	PULL																																
			Disabled	0																														
			Pulldown	1																														
			Pullup	3																														
D	RW	DRIVE																																
			S0S1	0																														
			H0S1	1																														
			S0H1	2																														
			H0H1	3																														
			D0S1	4																														
			D0H1	5																														
			S0D1	6																														
			H0D1	7																														
E	RW	SENSE																																
			Disabled	0																														
			High	2																														
			Low	3																														

20.3.31 PIN_CNF[21]

Address offset: 0x754

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	DIR																																
			Input	0																														
			Output	1																														
B	RW	INPUT																																
			Connect	0																														
			Disconnect	1																														
C	RW	PULL																																
			Disabled	0																														
			Pulldown	1																														
			Pullup	3																														
D	RW	DRIVE																																
			S0S1	0																														
			H0S1	1																														
			S0H1	2																														
			H0H1	3																														
			D0S1	4																														

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
			D0H1								5																Disconnect '0', high drive '1' (normally used for wired-or connections)							
			S0D1								6																Standard '0'. disconnect '1' (normally used for wired-and connections)							
			H0D1								7																High drive '0', disconnect '1' (normally used for wired-and connections)							
E	RW	SENSE																									Pin sensing mechanism							
			Disabled								0																Disabled							
			High								2																Sense for high level							
			Low								3																Sense for low level							

20.3.32 PIN_CNF[22]

Address offset: 0x758

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
A	RW	DIR																									Pin direction. Same physical register as DIR register							
			Input								0																Configure pin as an input pin							
			Output								1																Configure pin as an output pin							
B	RW	INPUT																									Connect or disconnect input buffer							
			Connect								0																Connect input buffer							
			Disconnect								1																Disconnect input buffer							
C	RW	PULL																									Pull configuration							
			Disabled								0																No pull							
			Pulldown								1																Pull down on pin							
			Pullup								3																Pull up on pin							
D	RW	DRIVE																									Drive configuration							
			S0S1								0																Standard '0', standard '1'							
			H0S1								1																High drive '0', standard '1'							
			S0H1								2																Standard '0', high drive '1'							
			H0H1								3																High drive '0', high 'drive '1''							
			D0S1								4																Disconnect '0' standard '1' (normally used for wired-or connections)							
			D0H1								5																Disconnect '0', high drive '1' (normally used for wired-or connections)							
			S0D1								6																Standard '0'. disconnect '1' (normally used for wired-and connections)							
			H0D1								7																High drive '0', disconnect '1' (normally used for wired-and connections)							
E	RW	SENSE																									Pin sensing mechanism							
			Disabled								0																Disabled							
			High								2																Sense for high level							
			Low								3																Sense for low level							

20.3.33 PIN_CNF[23]

Address offset: 0x75C

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	DIR			Pin direction. Same physical register as DIR register																															
			Input	0	Configure pin as an input pin																															
			Output	1	Configure pin as an output pin																															
B	RW	INPUT			Connect or disconnect input buffer																															
			Connect	0	Connect input buffer																															
			Disconnect	1	Disconnect input buffer																															
C	RW	PULL			Pull configuration																															
			Disabled	0	No pull																															
			Pulldown	1	Pull down on pin																															
			Pullup	3	Pull up on pin																															
D	RW	DRIVE			Drive configuration																															
			S0S1	0	Standard '0', standard '1'																															
			H0S1	1	High drive '0', standard '1'																															
			S0H1	2	Standard '0', high drive '1'																															
			H0H1	3	High drive '0', high 'drive '1''																															
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)																															
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)																															
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)																															
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)																															
E	RW	SENSE			Pin sensing mechanism																															
			Disabled	0	Disabled																															
			High	2	Sense for high level																															
			Low	3	Sense for low level																															

20.3.34 PIN_CNF[24]

Address offset: 0x760

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	DIR			Pin direction. Same physical register as DIR register																															
			Input	0	Configure pin as an input pin																															
			Output	1	Configure pin as an output pin																															
B	RW	INPUT			Connect or disconnect input buffer																															
			Connect	0	Connect input buffer																															
			Disconnect	1	Disconnect input buffer																															
C	RW	PULL			Pull configuration																															
			Disabled	0	No pull																															
			Pulldown	1	Pull down on pin																															
			Pullup	3	Pull up on pin																															
D	RW	DRIVE			Drive configuration																															
			S0S1	0	Standard '0', standard '1'																															
			H0S1	1	High drive '0', standard '1'																															
			S0H1	2	Standard '0', high drive '1'																															
			H0H1	3	High drive '0', high 'drive '1''																															
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)																															

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
			D0H1								5																Disconnect '0', high drive '1' (normally used for wired-or connections)							
			S0D1								6																Standard '0'. disconnect '1' (normally used for wired-and connections)							
			H0D1								7																High drive '0', disconnect '1' (normally used for wired-and connections)							
E	RW	SENSE																									Pin sensing mechanism							
			Disabled								0																Disabled							
			High								2																Sense for high level							
			Low								3																Sense for low level							

20.3.35 PIN_CNF[25]

Address offset: 0x764

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
A	RW	DIR																									Pin direction. Same physical register as DIR register							
			Input								0																Configure pin as an input pin							
			Output								1																Configure pin as an output pin							
B	RW	INPUT																									Connect or disconnect input buffer							
			Connect								0																Connect input buffer							
			Disconnect								1																Disconnect input buffer							
C	RW	PULL																									Pull configuration							
			Disabled								0																No pull							
			Pulldown								1																Pull down on pin							
			Pullup								3																Pull up on pin							
D	RW	DRIVE																									Drive configuration							
			S0S1								0																Standard '0', standard '1'							
			H0S1								1																High drive '0', standard '1'							
			S0H1								2																Standard '0', high drive '1'							
			H0H1								3																High drive '0', high 'drive '1''							
			D0S1								4																Disconnect '0' standard '1' (normally used for wired-or connections)							
			D0H1								5																Disconnect '0', high drive '1' (normally used for wired-or connections)							
			S0D1								6																Standard '0'. disconnect '1' (normally used for wired-and connections)							
			H0D1								7																High drive '0', disconnect '1' (normally used for wired-and connections)							
E	RW	SENSE																									Pin sensing mechanism							
			Disabled								0																Disabled							
			High								2																Sense for high level							
			Low								3																Sense for low level							

20.3.36 PIN_CNF[26]

Address offset: 0x768

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	DIR			Pin direction. Same physical register as DIR register																															
			Input	0	Configure pin as an input pin																															
			Output	1	Configure pin as an output pin																															
B	RW	INPUT			Connect or disconnect input buffer																															
			Connect	0	Connect input buffer																															
			Disconnect	1	Disconnect input buffer																															
C	RW	PULL			Pull configuration																															
			Disabled	0	No pull																															
			Pulldown	1	Pull down on pin																															
			Pullup	3	Pull up on pin																															
D	RW	DRIVE			Drive configuration																															
			S0S1	0	Standard '0', standard '1'																															
			H0S1	1	High drive '0', standard '1'																															
			S0H1	2	Standard '0', high drive '1'																															
			H0H1	3	High drive '0', high 'drive '1''																															
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)																															
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)																															
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)																															
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)																															
E	RW	SENSE			Pin sensing mechanism																															
			Disabled	0	Disabled																															
			High	2	Sense for high level																															
			Low	3	Sense for low level																															

20.3.37 PIN_CNF[27]

Address offset: 0x76C

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	DIR			Pin direction. Same physical register as DIR register																															
			Input	0	Configure pin as an input pin																															
			Output	1	Configure pin as an output pin																															
B	RW	INPUT			Connect or disconnect input buffer																															
			Connect	0	Connect input buffer																															
			Disconnect	1	Disconnect input buffer																															
C	RW	PULL			Pull configuration																															
			Disabled	0	No pull																															
			Pulldown	1	Pull down on pin																															
			Pullup	3	Pull up on pin																															
D	RW	DRIVE			Drive configuration																															
			S0S1	0	Standard '0', standard '1'																															
			H0S1	1	High drive '0', standard '1'																															
			S0H1	2	Standard '0', high drive '1'																															
			H0H1	3	High drive '0', high 'drive '1''																															
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)																															

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
			D0H1								5																Disconnect '0', high drive '1' (normally used for wired-or connections)							
			S0D1								6																Standard '0'. disconnect '1' (normally used for wired-and connections)							
			H0D1								7																High drive '0', disconnect '1' (normally used for wired-and connections)							
E	RW	SENSE																									Pin sensing mechanism							
			Disabled								0																Disabled							
			High								2																Sense for high level							
			Low								3																Sense for low level							

20.3.38 PIN_CNF[28]

Address offset: 0x770

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
A	RW	DIR																									Pin direction. Same physical register as DIR register							
			Input								0																Configure pin as an input pin							
			Output								1																Configure pin as an output pin							
B	RW	INPUT																									Connect or disconnect input buffer							
			Connect								0																Connect input buffer							
			Disconnect								1																Disconnect input buffer							
C	RW	PULL																									Pull configuration							
			Disabled								0																No pull							
			Pulldown								1																Pull down on pin							
			Pullup								3																Pull up on pin							
D	RW	DRIVE																									Drive configuration							
			S0S1								0																Standard '0', standard '1'							
			H0S1								1																High drive '0', standard '1'							
			S0H1								2																Standard '0', high drive '1'							
			H0H1								3																High drive '0', high 'drive '1''							
			D0S1								4																Disconnect '0' standard '1' (normally used for wired-or connections)							
			D0H1								5																Disconnect '0', high drive '1' (normally used for wired-or connections)							
			S0D1								6																Standard '0'. disconnect '1' (normally used for wired-and connections)							
			H0D1								7																High drive '0', disconnect '1' (normally used for wired-and connections)							
E	RW	SENSE																									Pin sensing mechanism							
			Disabled								0																Disabled							
			High								2																Sense for high level							
			Low								3																Sense for low level							

20.3.39 PIN_CNF[29]

Address offset: 0x774

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	DIR			Pin direction. Same physical register as DIR register																															
			Input	0	Configure pin as an input pin																															
			Output	1	Configure pin as an output pin																															
B	RW	INPUT			Connect or disconnect input buffer																															
			Connect	0	Connect input buffer																															
			Disconnect	1	Disconnect input buffer																															
C	RW	PULL			Pull configuration																															
			Disabled	0	No pull																															
			Pulldown	1	Pull down on pin																															
			Pullup	3	Pull up on pin																															
D	RW	DRIVE			Drive configuration																															
			S0S1	0	Standard '0', standard '1'																															
			H0S1	1	High drive '0', standard '1'																															
			S0H1	2	Standard '0', high drive '1'																															
			H0H1	3	High drive '0', high 'drive '1''																															
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)																															
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)																															
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)																															
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)																															
E	RW	SENSE			Pin sensing mechanism																															
			Disabled	0	Disabled																															
			High	2	Sense for high level																															
			Low	3	Sense for low level																															

20.3.40 PIN_CNF[30]

Address offset: 0x778

Configuration of GPIO pins

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	E	E						D	D	D					C	C	B	A
Reset 0x00000002			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	DIR			Pin direction. Same physical register as DIR register																															
			Input	0	Configure pin as an input pin																															
			Output	1	Configure pin as an output pin																															
B	RW	INPUT			Connect or disconnect input buffer																															
			Connect	0	Connect input buffer																															
			Disconnect	1	Disconnect input buffer																															
C	RW	PULL			Pull configuration																															
			Disabled	0	No pull																															
			Pulldown	1	Pull down on pin																															
			Pullup	3	Pull up on pin																															
D	RW	DRIVE			Drive configuration																															
			S0S1	0	Standard '0', standard '1'																															
			H0S1	1	High drive '0', standard '1'																															
			S0H1	2	Standard '0', high drive '1'																															
			H0H1	3	High drive '0', high 'drive '1''																															
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		E	E						D	D	D						C	C	B	A
Reset 0x00000002				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			D0H1	5																															
				Disconnect '0', high drive '1' (normally used for wired-or connections)																															
			S0D1	6																															
				Standard '0'. disconnect '1' (normally used for wired-and connections)																															
			H0D1	7																															
				High drive '0', disconnect '1' (normally used for wired-and connections)																															
E	RW	SENSE																																	
				Pin sensing mechanism																															
			Disabled	0																															
				Disabled																															
			High	2																															
				Sense for high level																															
			Low	3																															
				Sense for low level																															

20.3.41 PIN_CNF[31]

Address offset: 0x77C

Configuration of GPIO pins

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		E	E						D	D	D						C	C	B	A
Reset 0x00000002				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	DIR																																	
				Pin direction. Same physical register as DIR register																															
			Input	0																															
				Configure pin as an input pin																															
			Output	1																															
				Configure pin as an output pin																															
B	RW	INPUT																																	
				Connect or disconnect input buffer																															
			Connect	0																															
				Connect input buffer																															
			Disconnect	1																															
				Disconnect input buffer																															
C	RW	PULL																																	
				Pull configuration																															
			Disabled	0																															
				No pull																															
			Pulldown	1																															
				Pull down on pin																															
			Pullup	3																															
				Pull up on pin																															
D	RW	DRIVE																																	
				Drive configuration																															
			S0S1	0																															
				Standard '0', standard '1'																															
			H0S1	1																															
				High drive '0', standard '1'																															
			S0H1	2																															
				Standard '0', high drive '1'																															
			H0H1	3																															
				High drive '0', high 'drive '1''																															
			D0S1	4																															
				Disconnect '0' standard '1' (normally used for wired-or connections)																															
			D0H1	5																															
				Disconnect '0', high drive '1' (normally used for wired-or connections)																															
			S0D1	6																															
				Standard '0'. disconnect '1' (normally used for wired-and connections)																															
			H0D1	7																															
				High drive '0', disconnect '1' (normally used for wired-and connections)																															
E	RW	SENSE																																	
				Pin sensing mechanism																															
			Disabled	0																															
				Disabled																															
			High	2																															
				Sense for high level																															
			Low	3																															
				Sense for low level																															

20.4 Electrical specification

20.4.1 GPIO Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
V_{IH}	Input high voltage	0.7 x VDD		VDD	V

Symbol	Description	Min.	Typ.	Max.	Units
V_{IL}	Input low voltage	VSS		0.3 x VDD	V
$V_{OH,SD}$	Output high voltage, standard drive, 0.5 mA, VDD ≥1.7	VDD-0.4		VDD	V
$V_{OH,HDH}$	Output high voltage, high drive, 5 mA, VDD >= 2.7 V	VDD-0.4		VDD	V
$V_{OH,HDL}$	Output high voltage, high drive, 3 mA, VDD >= 1.7 V	VDD-0.4		VDD	V
$V_{OL,SD}$	Output low voltage, standard drive, 0.5 mA, VDD ≥1.7	VSS		VSS+0.4	V
$V_{OL,HDH}$	Output low voltage, high drive, 5 mA, VDD >= 2.7 V	VSS		VSS+0.4	V
$V_{OL,HDL}$	Output low voltage, high drive, 3 mA, VDD >= 1.7 V	VSS		VSS+0.4	V
$I_{OL,SD}$	Current at VSS+0.4 V, output set low, standard drive, VDD ≥1.7	1	2	4	mA
$I_{OL,HDH}$	Current at VSS+0.4 V, output set low, high drive, VDD >= 2.7 V	6	10	15	mA
$I_{OL,HDL}$	Current at VSS+0.4 V, output set low, high drive, VDD >= 1.7 V	3			mA
$I_{OH,SD}$	Current at VDD-0.4 V, output set high, standard drive, VDD ≥1.7	1	2	4	mA
$I_{OH,HDH}$	Current at VDD-0.4 V, output set high, high drive, VDD >= 2.7 V	6	9	14	mA
$I_{OH,HDL}$	Current at VDD-0.4 V, output set high, high drive, VDD >= 1.7 V	3			mA
$t_{RF,15pF}$	Rise/fall time, low drive mode, 10-90%, 15 pF load[1]		9		ns
$t_{RF,25pF}$	Rise/fall time, low drive mode, 10-90%, 25 pF load[1]		13		ns
$t_{RF,50pF}$	Rise/fall time, low drive mode, 10-90%, 50 pF load[1]		25		ns
$t_{HRF,15pF}$	Rise/Fall time, high drive mode, 10-90%, 15 pF load[1]		4		ns
$t_{HRF,25pF}$	Rise/Fall time, high drive mode, 10-90%, 25 pF load[1]		5		ns
$t_{HRF,50pF}$	Rise/Fall time, high drive mode, 10-90%, 50 pF load[1]		8		ns
R_{PU}	Pull-up resistance	11	13	16	kΩ
R_{PD}	Pull-down resistance	11	13	16	kΩ
C_{PAD}	Pad capacitance		3		pF
C_{PAD_NFC}	Pad capacitance on NFC pads		4		pF
I_{NFC_LEAK}	Leakage current between NFC pads when driven to different states		2	10	µA

The current drawn from the battery when GPIO is active as an output is calculated as follows:

$I_{GPIO}=V_{DD}\ C_{load}\ f$

C_{load} being the load capacitance and “f” is the switching frequency.

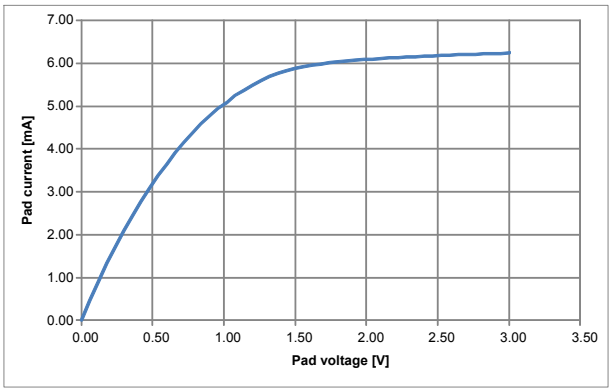


Figure 23: GPIO drive strength vs Voltage, standard drive, VDD = 3.0 V

[1] Rise and fall times based on simulations

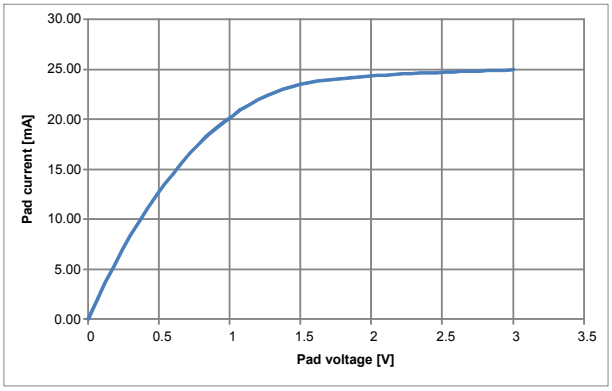


Figure 24: GPIO drive strength vs Voltage, high drive, VDD = 3.0 V

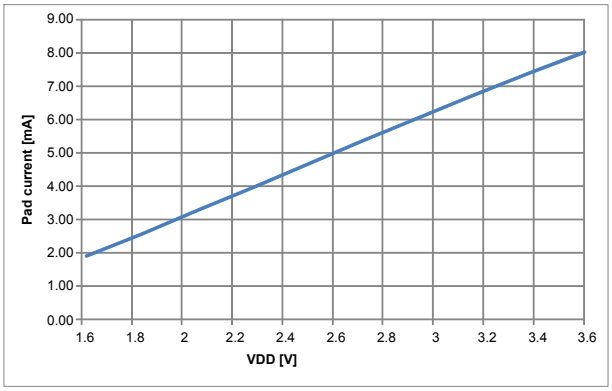


Figure 25: Max sink current vs Voltage, standard drive

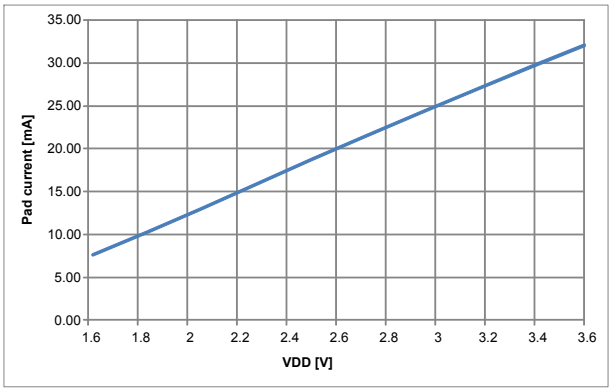


Figure 26: Max sink current vs Voltage, high drive

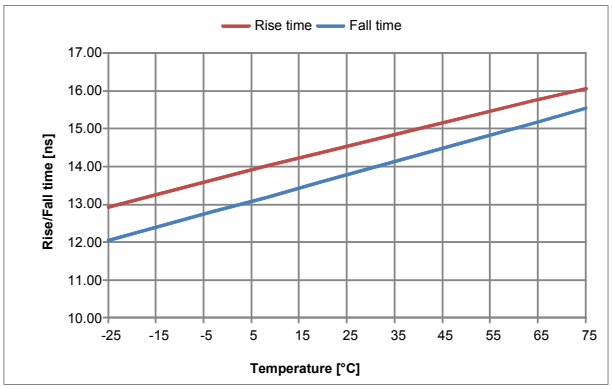


Figure 27: Rise and fall time vs Temperature, 10%-90%, 25pF load capacitance, VDD = 3.0 V

21 GPIOTE — GPIO tasks and events

The GPIO tasks and events (GPIOTE) module provides functionality for accessing GPIO pins using tasks and events. Each GPIOTE channel can be assigned to one pin.

A GPIOTE block enables GPIOs to generate events on pin state change which can be used to carry out tasks through the PPI system. A GPIO can also be driven to change state on system events using the PPI system. Low power detection of pin state changes is possible when in System ON or System OFF.

Table 31: GPIOTE properties

Instance	Number of GPIOTE channels
GPIOTE	8

Up to three tasks can be used in each GPIOTE channel for performing write operations to a pin. Two tasks are fixed (SET and CLR), and one (OUT) is configurable to perform following operations:

- Set
- Clear
- Toggle

An event can be generated in each GPIOTE channel from one of the following input conditions:

- Rising edge
- Falling edge
- Any change

21.1 Pin events and tasks

The GPIOTE module has a number of tasks and events that can be configured to operate on individual GPIO pins.

The tasks (SET[n], CLR[n] and OUT[n]) can be used for writing to individual pins, and the events (IN[n]) can be generated from changes occurring at the inputs of individual pins.

The SET task will set the pin selected in CONFIG[n].PSEL to high.

The CLR task will set the pin low.

The effect of the OUT task on the pin is configurable in CONFIG[n].POLARITY , and can either set the pin high, set it low, or toggle it.

The tasks and events are configured using the CONFIG[n] registers. Every set of SET, CLR and OUT[n] tasks and IN[n] events has one CONFIG[n] register associated with it.

As long as a SET[n], CLR[n] and OUT[n] task or an IN[n] event is configured to control a pin **n**, the pin's output value will only be updated by the GPIOTE module. The pin's output value as specified in the GPIO will therefore be ignored as long as the pin is controlled by GPIOTE. Attempting to write a pin as a normal GPIO pin will have no effect. When the GPIOTE is disconnected from a pin, see MODE field in CONFIG[n] register, the associated pin will get the output and configuration values specified in the GPIO module.

When conflicting tasks are triggered simultaneously (i.e. during the same clock cycle) in one channel, the precedence of the tasks will be as described in *Table 32: Task priorities* on page 157.

Table 32: Task priorities

Priority	Task
1	OUT
2	CLR
3	SET

When setting the CONFIG[n] registers, MODE=Disabled does not have the same effect as MODE=Task and POLARITY=None. In the latter case, a CLR or SET task occurring at the exact same time as OUT will end up with no change on the pin, according to the priorities described in the table above.

When a GPIOTE channel is configured to operate on a pin as a task, the initial value of that pin is configured in the OUTINIT field of CONFIG[n].

21.2 Port event

PORT is an event that can be generated from multiple input pins using the GPIO DETECT signal.

The event will be generated on the rising edge of the DETECT signal. See *GPIO — General purpose input/output* on page 111 for more information about the DETECT signal.

Putting the system into System ON IDLE while DETECT is high will not cause DETECT to wake the system up again. Make sure to clear all DETECT sources before entering sleep. If the LATCH register is used as a source, if any bit in LATCH is still high after clearing all or part of the register (for instance due to one of the PINx.DETECT signal still high), a new rising edge will be generated on DETECT, see *Pin configuration* on page 111.

Trying to put the system to System OFF while DETECT is high will cause a wakeup from System OFF reset.

This feature is always enabled although the peripheral itself appears to be IDLE, that is, no clocks or other power intensive infrastructure have to be requested to keep this feature enabled. This feature can therefore be used to wake up the CPU from a WFI or WFE type sleep in System ON with all peripherals and the CPU idle, that is, lowest power consumption in System ON mode.

In order to prevent spurious interrupts from the PORT event while configuring the sources, the user shall first disable interrupts on the PORT event (through INTENCLR.PORT), then configure the sources (PIN_CNF[n].SENSE), clear any potential event that could have occurred during configuration (write '1' to EVENTS_PORT), and finally enable interrupts (through INTENSET.PORT).

21.3 Tasks and events pin configuration

Each GPIOTE channel is associated with one physical GPIO pin through the CONFIG.PSEL field.

When Event mode is selected in CONFIG.MODE, the pin specified by CONFIG.PSEL will be configured as an input, overriding the DIR setting in GPIO. Similarly, when Task mode is selected in CONFIG.MODE, the pin specified by CONFIG.PSEL will be configured as an output overriding the DIR setting and OUT value in GPIO. When Disabled is selected in CONFIG.MODE, the pin specified by CONFIG.PSEL will use its configuration from the PIN[n].CNF registers in GPIO.

Only one GPIOTE channel can be assigned to one physical pin. Failing to do so may result in unpredictable behavior.

21.4 Registers

Table 33: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40006000	GPIOTE	GPIOTE	GPIO Tasks and Events	

Table 34: Register Overview

Register	Offset	Description
TASKS_OUT[0]	0x000	Task for writing to pin specified in CONFIG[0].PSEL. Action on pin is configured in CONFIG[0].POLARITY.
TASKS_OUT[1]	0x004	Task for writing to pin specified in CONFIG[1].PSEL. Action on pin is configured in CONFIG[1].POLARITY.

Register	Offset	Description
TASKS_OUT[2]	0x008	Task for writing to pin specified in CONFIG[2].PSEL. Action on pin is configured in CONFIG[2].POLARITY.
TASKS_OUT[3]	0x00C	Task for writing to pin specified in CONFIG[3].PSEL. Action on pin is configured in CONFIG[3].POLARITY.
TASKS_OUT[4]	0x010	Task for writing to pin specified in CONFIG[4].PSEL. Action on pin is configured in CONFIG[4].POLARITY.
TASKS_OUT[5]	0x014	Task for writing to pin specified in CONFIG[5].PSEL. Action on pin is configured in CONFIG[5].POLARITY.
TASKS_OUT[6]	0x018	Task for writing to pin specified in CONFIG[6].PSEL. Action on pin is configured in CONFIG[6].POLARITY.
TASKS_OUT[7]	0x01C	Task for writing to pin specified in CONFIG[7].PSEL. Action on pin is configured in CONFIG[7].POLARITY.
TASKS_SET[0]	0x030	Task for writing to pin specified in CONFIG[0].PSEL. Action on pin is to set it high.
TASKS_SET[1]	0x034	Task for writing to pin specified in CONFIG[1].PSEL. Action on pin is to set it high.
TASKS_SET[2]	0x038	Task for writing to pin specified in CONFIG[2].PSEL. Action on pin is to set it high.
TASKS_SET[3]	0x03C	Task for writing to pin specified in CONFIG[3].PSEL. Action on pin is to set it high.
TASKS_SET[4]	0x040	Task for writing to pin specified in CONFIG[4].PSEL. Action on pin is to set it high.
TASKS_SET[5]	0x044	Task for writing to pin specified in CONFIG[5].PSEL. Action on pin is to set it high.
TASKS_SET[6]	0x048	Task for writing to pin specified in CONFIG[6].PSEL. Action on pin is to set it high.
TASKS_SET[7]	0x04C	Task for writing to pin specified in CONFIG[7].PSEL. Action on pin is to set it high.
TASKS_CLR[0]	0x060	Task for writing to pin specified in CONFIG[0].PSEL. Action on pin is to set it low.
TASKS_CLR[1]	0x064	Task for writing to pin specified in CONFIG[1].PSEL. Action on pin is to set it low.
TASKS_CLR[2]	0x068	Task for writing to pin specified in CONFIG[2].PSEL. Action on pin is to set it low.
TASKS_CLR[3]	0x06C	Task for writing to pin specified in CONFIG[3].PSEL. Action on pin is to set it low.
TASKS_CLR[4]	0x070	Task for writing to pin specified in CONFIG[4].PSEL. Action on pin is to set it low.
TASKS_CLR[5]	0x074	Task for writing to pin specified in CONFIG[5].PSEL. Action on pin is to set it low.
TASKS_CLR[6]	0x078	Task for writing to pin specified in CONFIG[6].PSEL. Action on pin is to set it low.
TASKS_CLR[7]	0x07C	Task for writing to pin specified in CONFIG[7].PSEL. Action on pin is to set it low.
EVENTS_IN[0]	0x100	Event generated from pin specified in CONFIG[0].PSEL
EVENTS_IN[1]	0x104	Event generated from pin specified in CONFIG[1].PSEL
EVENTS_IN[2]	0x108	Event generated from pin specified in CONFIG[2].PSEL
EVENTS_IN[3]	0x10C	Event generated from pin specified in CONFIG[3].PSEL
EVENTS_IN[4]	0x110	Event generated from pin specified in CONFIG[4].PSEL
EVENTS_IN[5]	0x114	Event generated from pin specified in CONFIG[5].PSEL
EVENTS_IN[6]	0x118	Event generated from pin specified in CONFIG[6].PSEL
EVENTS_IN[7]	0x11C	Event generated from pin specified in CONFIG[7].PSEL
EVENTS_PORT	0x17C	Event generated from multiple input GPIO pins with SENSE mechanism enabled
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
CONFIG[0]	0x510	Configuration for OUT[n], SET[n] and CLR[n] tasks and IN[n] event
CONFIG[1]	0x514	Configuration for OUT[n], SET[n] and CLR[n] tasks and IN[n] event
CONFIG[2]	0x518	Configuration for OUT[n], SET[n] and CLR[n] tasks and IN[n] event
CONFIG[3]	0x51C	Configuration for OUT[n], SET[n] and CLR[n] tasks and IN[n] event
CONFIG[4]	0x520	Configuration for OUT[n], SET[n] and CLR[n] tasks and IN[n] event
CONFIG[5]	0x524	Configuration for OUT[n], SET[n] and CLR[n] tasks and IN[n] event
CONFIG[6]	0x528	Configuration for OUT[n], SET[n] and CLR[n] tasks and IN[n] event
CONFIG[7]	0x52C	Configuration for OUT[n], SET[n] and CLR[n] tasks and IN[n] event

21.4.1 INTENSET

Address offset: 0x304

Enable interrupt

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									H	G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	IN0			Write '1' to Enable interrupt for IN[0] event

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				I																								H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			I																								H	G	F	E	D	C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	IN0			Write '1' to Disable interrupt for IN[0] event See *EVENTS_IN[0]*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	IN1			Write '1' to Disable interrupt for IN[1] event See *EVENTS_IN[1]*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	IN2			Write '1' to Disable interrupt for IN[2] event See *EVENTS_IN[2]*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	IN3			Write '1' to Disable interrupt for IN[3] event See *EVENTS_IN[3]*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	IN4			Write '1' to Disable interrupt for IN[4] event See *EVENTS_IN[4]*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	IN5			Write '1' to Disable interrupt for IN[5] event See *EVENTS_IN[5]*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	IN6			Write '1' to Disable interrupt for IN[6] event See *EVENTS_IN[6]*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	IN7			Write '1' to Disable interrupt for IN[7] event See *EVENTS_IN[7]*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
I	RW	PORT			Write '1' to Disable interrupt for PORT event See *EVENTS_PORT*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

21.4.3 CONFIG[0]

Address offset: 0x510

Configuration for OUT[n], SET[n] and CLR[n] tasks and IN[n] event

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id															D			C	C				B	B	B	B	B							A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	MODE																																	
			Disabled	0																															
			Event	1																															
			Task	3																															
B	RW	PSEL		[0..31]																															
C	RW	POLARITY																																	
			None	0																															
			LoToHi	1																															
			HiToLo	2																															
			Toggle	3																															
D	RW	OUTINIT																																	
			Low	0																															
			High	1																															

21.4.4 CONFIG[1]

Address offset: 0x514

Configuration for OUT[n], SET[n] and CLR[n] tasks and IN[n] event

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id															D			C	C				B	B	B	B	B							A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	MODE																																	
			Disabled	0																															
			Event	1																															
			Task	3																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id															D			C	C				B	B	B	B	B							A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
B	RW	PSEL		[0..31]																															
				GPIO number associated with SET[n], CLR[n] and OUT[n] tasks and IN[n] event																															
C	RW	POLARITY																																	
				When In task mode: Operation to be performed on output when OUT[n] task is triggered. When In event mode: Operation on input that shall trigger IN[n] event.																															
			None	0																															
				Task mode: No effect on pin from OUT[n] task. Event mode: no IN[n] event generated on pin activity.																															
			LoToHi	1																															
				Task mode: Set pin from OUT[n] task. Event mode: Generate IN[n] event when rising edge on pin.																															
			HiToLo	2																															
				Task mode: Clear pin from OUT[n] task. Event mode: Generate IN[n] event when falling edge on pin.																															
			Toggle	3																															
				Task mode: Toggle pin from OUT[n]. Event mode: Generate IN[n] when any change on pin.																															
D	RW	OUTINIT																																	
				When in task mode: Initial value of the output when the GPIOTE channel is configured. When in event mode: No effect.																															
			Low	0																															
				Task mode: Initial value of pin before task triggering is low																															
			High	1																															
				Task mode: Initial value of pin before task triggering is high																															

21.4.5 CONFIG[2]

Address offset: 0x518

Configuration for OUT[n], SET[n] and CLR[n] tasks and IN[n] event

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id															D			C	C				B	B	B	B	B							A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	MODE																																	
				Mode																															
			Disabled	0																															
				Disabled. Pin specified by PSEL will not be acquired by the GPIOTE module.																															
			Event	1																															
				Event mode																															
				The pin specified by PSEL will be configured as an input and the IN[n] event will be generated if operation specified in POLARITY occurs on the pin.																															
			Task	3																															
				Task mode																															
				The GPIO specified by PSEL will be configured as an output and triggering the SET[n], CLR[n] or OUT[n] task will perform the operation specified by POLARITY on the pin. When enabled as a task the GPIOTE module will acquire the pin and the pin can no longer be written as a regular output pin from the GPIO module.																															
B	RW	PSEL		[0..31]																															
				GPIO number associated with SET[n], CLR[n] and OUT[n] tasks and IN[n] event																															
C	RW	POLARITY																																	
				When In task mode: Operation to be performed on output when OUT[n] task is triggered. When In event mode: Operation on input that shall trigger IN[n] event.																															
			None	0																															
				Task mode: No effect on pin from OUT[n] task. Event mode: no IN[n] event generated on pin activity.																															
			LoToHi	1																															
				Task mode: Set pin from OUT[n] task. Event mode: Generate IN[n] event when rising edge on pin.																															
			HiToLo	2																															
				Task mode: Clear pin from OUT[n] task. Event mode: Generate IN[n] event when falling edge on pin.																															
			Toggle	3																															
				Task mode: Toggle pin from OUT[n]. Event mode: Generate IN[n] when any change on pin.																															

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id														D			C	C				B	B	B	B	B							A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
D	RW	OUTINIT																									When in task mode: Initial value of the output when the GPIOTE channel is configured. When in event mode: No effect.							
			Low								0																Task mode: Initial value of pin before task triggering is low							
			High								1																Task mode: Initial value of pin before task triggering is high							

21.4.6 CONFIG[3]

Address offset: 0x51C

Configuration for OUT[n], SET[n] and CLR[n] tasks and IN[n] event

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id														D			C	C				B	B	B	B	B							A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
A	RW	MODE																									Mode							
			Disabled								0																Disabled. Pin specified by PSEL will not be acquired by the GPIOTE module.							
			Event								1																Event mode The pin specified by PSEL will be configured as an input and the IN[n] event will be generated if operation specified in POLARITY occurs on the pin.							
			Task								3																Task mode The GPIO specified by PSEL will be configured as an output and triggering the SET[n], CLR[n] or OUT[n] task will perform the operation specified by POLARITY on the pin. When enabled as a task the GPIOTE module will acquire the pin and the pin can no longer be written as a regular output pin from the GPIO module.							
B	RW	PSEL									[0..31]																GPIO number associated with SET[n], CLR[n] and OUT[n] tasks and IN[n] event							
C	RW	POLARITY																									When In task mode: Operation to be performed on output when OUT[n] task is triggered. When In event mode: Operation on input that shall trigger IN[n] event.							
			None								0																Task mode: No effect on pin from OUT[n] task. Event mode: no IN[n] event generated on pin activity.							
			LoToHi								1																Task mode: Set pin from OUT[n] task. Event mode: Generate IN[n] event when rising edge on pin.							
			HiToLo								2																Task mode: Clear pin from OUT[n] task. Event mode: Generate IN[n] event when falling edge on pin.							
			Toggle								3																Task mode: Toggle pin from OUT[n]. Event mode: Generate IN[n] when any change on pin.							
D	RW	OUTINIT																									When in task mode: Initial value of the output when the GPIOTE channel is configured. When in event mode: No effect.							
			Low								0																Task mode: Initial value of pin before task triggering is low							
			High								1																Task mode: Initial value of pin before task triggering is high							

21.4.7 CONFIG[4]

Address offset: 0x520

Configuration for OUT[n], SET[n] and CLR[n] tasks and IN[n] event

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id															D			C	C				B	B	B	B	B							A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	MODE																																	
			Disabled	0																															
			Event	1																															
			Task	3																															
B	RW	PSEL		[0..31]																															
C	RW	POLARITY																																	
			None	0																															
			LoToHi	1																															
			HiToLo	2																															
			Toggle	3																															
D	RW	OUTINIT																																	
			Low	0																															
			High	1																															

21.4.8 CONFIG[5]

Address offset: 0x524

Configuration for OUT[n], SET[n] and CLR[n] tasks and IN[n] event

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id															D			C	C				B	B	B	B	B							A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	MODE																																	
			Disabled	0																															
			Event	1																															
			Task	3																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id															D			C	C				B	B	B	B	B							A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
B	RW	PSEL		[0..31]																															
				GPIO number associated with SET[n], CLR[n] and OUT[n] tasks and IN[n] event																															
C	RW	POLARITY																																	
				When In task mode: Operation to be performed on output when OUT[n] task is triggered. When In event mode: Operation on input that shall trigger IN[n] event.																															
			None	0																															
				Task mode: No effect on pin from OUT[n] task. Event mode: no IN[n] event generated on pin activity.																															
			LoToHi	1																															
				Task mode: Set pin from OUT[n] task. Event mode: Generate IN[n] event when rising edge on pin.																															
			HiToLo	2																															
				Task mode: Clear pin from OUT[n] task. Event mode: Generate IN[n] event when falling edge on pin.																															
			Toggle	3																															
				Task mode: Toggle pin from OUT[n]. Event mode: Generate IN[n] when any change on pin.																															
D	RW	OUTINIT																																	
				When in task mode: Initial value of the output when the GPIOTE channel is configured. When in event mode: No effect.																															
			Low	0																															
				Task mode: Initial value of pin before task triggering is low																															
			High	1																															
				Task mode: Initial value of pin before task triggering is high																															

21.4.9 CONFIG[6]

Address offset: 0x528

Configuration for OUT[n], SET[n] and CLR[n] tasks and IN[n] event

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id															D			C	C				B	B	B	B	B							A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	MODE																																	
				Mode																															
			Disabled	0																															
				Disabled. Pin specified by PSEL will not be acquired by the GPIOTE module.																															
			Event	1																															
				Event mode The pin specified by PSEL will be configured as an input and the IN[n] event will be generated if operation specified in POLARITY occurs on the pin.																															
			Task	3																															
				Task mode The GPIO specified by PSEL will be configured as an output and triggering the SET[n], CLR[n] or OUT[n] task will perform the operation specified by POLARITY on the pin. When enabled as a task the GPIOTE module will acquire the pin and the pin can no longer be written as a regular output pin from the GPIO module.																															
B	RW	PSEL		[0..31]																															
				GPIO number associated with SET[n], CLR[n] and OUT[n] tasks and IN[n] event																															
C	RW	POLARITY																																	
				When In task mode: Operation to be performed on output when OUT[n] task is triggered. When In event mode: Operation on input that shall trigger IN[n] event.																															
			None	0																															
				Task mode: No effect on pin from OUT[n] task. Event mode: no IN[n] event generated on pin activity.																															
			LoToHi	1																															
				Task mode: Set pin from OUT[n] task. Event mode: Generate IN[n] event when rising edge on pin.																															
			HiToLo	2																															
				Task mode: Clear pin from OUT[n] task. Event mode: Generate IN[n] event when falling edge on pin.																															
			Toggle	3																															
				Task mode: Toggle pin from OUT[n]. Event mode: Generate IN[n] when any change on pin.																															

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id														D			C	C				B	B	B	B	B							A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
D	RW	OUTINIT																									When in task mode: Initial value of the output when the GPIOTE channel is configured. When in event mode: No effect.							
			Low								0																Task mode: Initial value of pin before task triggering is low							
			High								1																Task mode: Initial value of pin before task triggering is high							

21.4.10 CONFIG[7]

Address offset: 0x52C

Configuration for OUT[n], SET[n] and CLR[n] tasks and IN[n] event

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id														D			C	C				B	B	B	B	B							A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
A	RW	MODE																									Mode							
			Disabled								0																Disabled. Pin specified by PSEL will not be acquired by the GPIOTE module.							
			Event								1																Event mode The pin specified by PSEL will be configured as an input and the IN[n] event will be generated if operation specified in POLARITY occurs on the pin.							
			Task								3																Task mode The GPIO specified by PSEL will be configured as an output and triggering the SET[n], CLR[n] or OUT[n] task will perform the operation specified by POLARITY on the pin. When enabled as a task the GPIOTE module will acquire the pin and the pin can no longer be written as a regular output pin from the GPIO module.							
B	RW	PSEL									[0..31]																GPIO number associated with SET[n], CLR[n] and OUT[n] tasks and IN[n] event							
C	RW	POLARITY																									When In task mode: Operation to be performed on output when OUT[n] task is triggered. When In event mode: Operation on input that shall trigger IN[n] event.							
			None								0																Task mode: No effect on pin from OUT[n] task. Event mode: no IN[n] event generated on pin activity.							
			LoToHi								1																Task mode: Set pin from OUT[n] task. Event mode: Generate IN[n] event when rising edge on pin.							
			HiToLo								2																Task mode: Clear pin from OUT[n] task. Event mode: Generate IN[n] event when falling edge on pin.							
			Toggle								3																Task mode: Toggle pin from OUT[n]. Event mode: Generate IN[n] when any change on pin.							
D	RW	OUTINIT																									When in task mode: Initial value of the output when the GPIOTE channel is configured. When in event mode: No effect.							
			Low								0																Task mode: Initial value of pin before task triggering is low							
			High								1																Task mode: Initial value of pin before task triggering is high							

21.5 Electrical specification

21.5.1 GPIOTE Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
$I_{GPIOTE,IN}$	Run current with 1 or more GPIOTE active channels in Input mode		0.1	0.5	µA

22 PPI — Programmable peripheral interconnect

The Programmable peripheral interconnect (PPI) enables peripherals to interact autonomously with each other using tasks and events independent of the CPU. The PPI allows precise synchronization between peripherals when real-time application constraints exist and eliminates the need for CPU activity to implement behavior which can be predefined using PPI.

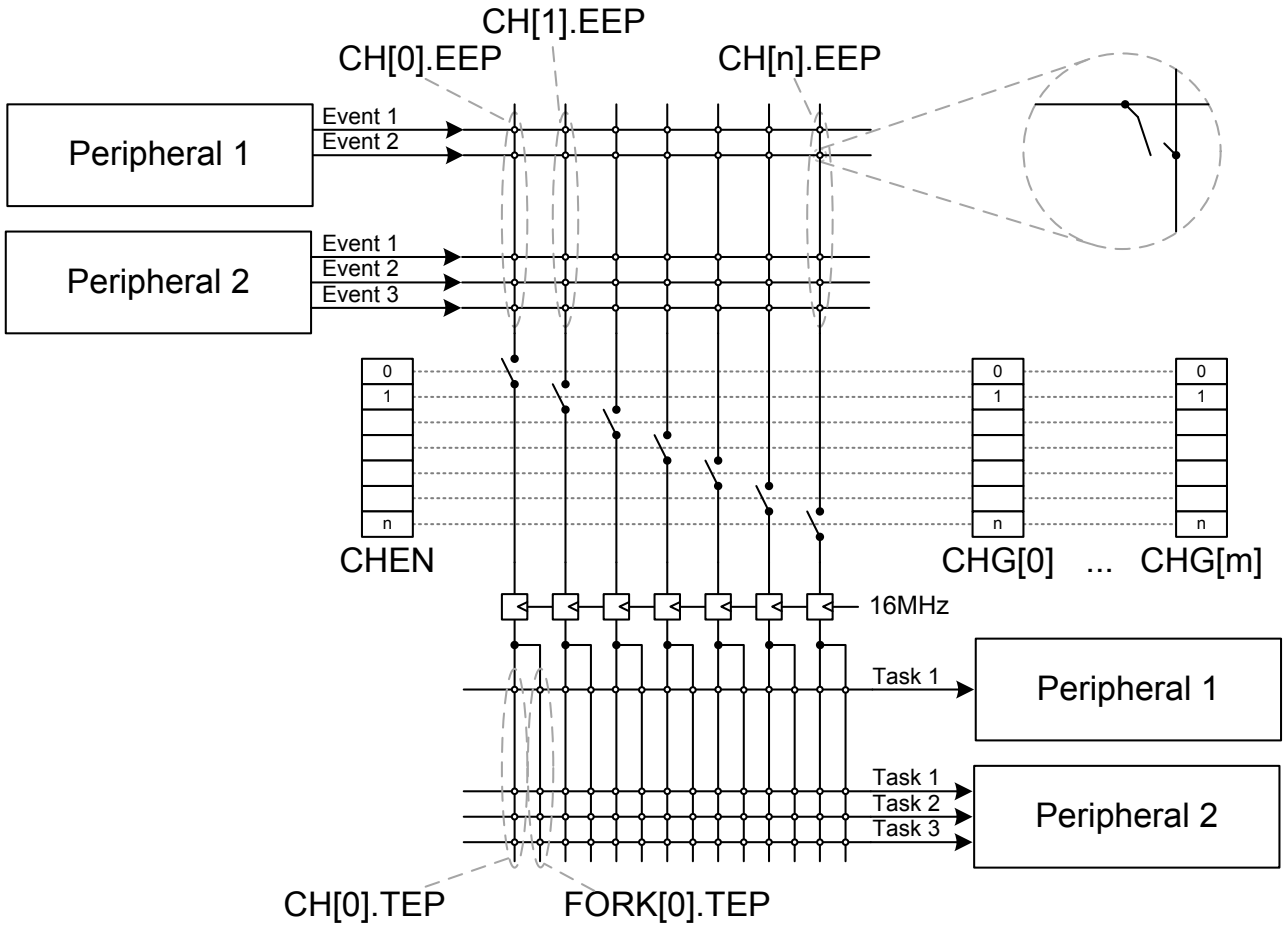


Figure 28: PPI block diagram

The PPI system has, in addition to the fully programmable peripheral interconnections, a set of channels where the event end point (EEP) and task end points (TEP) are fixed in hardware. These fixed channels can be individually enabled, disabled, or added to PPI channel groups in the same way as ordinary PPI channels.

Table 35: Configurable and fixed PPI channels

Instance	Channel	Number of channels	Number of groups
PPI	0-19	20	6
PPI (fixed)	20-31	12	

The PPI provides a mechanism to automatically trigger a task in one peripheral as a result of an event occurring in another peripheral. A task is connected to an event through a PPI channel. The PPI channel is composed of three end point registers, one EEP and two TEPs. A peripheral task is connected to a TEP using the address of the task register associated with the task. Similarly, a peripheral event is connected to an EEP using the address of the event register associated with the event.

On each PPI channel, the signals are synchronized to the 16 MHz clock, to avoid any internal violation of setup and hold timings. As a consequence, events that are synchronous to the 16 MHz clock will be delayed by one clock period, while other asynchronous events will be delayed by up to one 16 MHz clock period.

Note that shortcuts (as defined in the SHORTS register in each peripheral) are not affected by this 16 MHz synchronization, and are therefore not delayed.

Each TEP implements a fork mechanism that enables a second task to be triggered at the same time as the task specified in the TEP is triggered. This second task is configured in the task end point register in the FORK registers groups, e.g. FORK.TEP[0] is associated with PPI channel CH[0].

There are two ways of enabling and disabling PPI channels:

- Enable or disable PPI channels individually using the CHEN, CHENSET, and CHENCLR registers.
- Enable or disable PPI channels in PPI channel groups through the groups' ENABLE and DISABLE tasks. Prior to these tasks being triggered, the PPI channel group must be configured to define which PPI channels belongs to which groups.

Note that when a channel belongs to two groups m and n, and CHG[m].EN and CHG[n].DIS occur simultaneously (m and n can be equal or different), EN on that channel has priority.

PPI tasks (for example, CHG[0].EN) can be triggered through the PPI like any other task, which means they can be hooked up to a PPI channel as a TEP. One event can trigger multiple tasks by using multiple channels and one task can be triggered by multiple events in the same way.

22.1 Pre-programmed channels

Some of the PPI channels are pre-programmed. These channels cannot be configured by the CPU, but can be added to groups and enabled and disabled like the general purpose PPI channels. The FORK TEP for these channels are still programmable and can be used by the application.

For a list of pre-programmed PPI channels, see the table below.

Table 36: Pre-programmed channels

Channel	EEP	TEP
20	TIMER0->EVENTS_COMPARE[0]	RADIO->TASKS_TXEN
21	TIMER0->EVENTS_COMPARE[0]	RADIO->TASKS_RXEN
22	TIMER0->EVENTS_COMPARE[1]	RADIO->TASKS_DISABLE
23	RADIO->EVENTS_BCMATCH	AAR->TASKS_START
24	RADIO->EVENTS_READY	CCM->TASKS_KSGEN
25	RADIO->EVENTS_ADDRESS	CCM->TASKS_CRYPT
26	RADIO->EVENTS_ADDRESS	TIMER0->TASKS_CAPTURE[1]
27	RADIO->EVENTS_END	TIMER0->TASKS_CAPTURE[2]
28	RTC0->EVENTS_COMPARE[0]	RADIO->TASKS_TXEN
29	RTC0->EVENTS_COMPARE[0]	RADIO->TASKS_RXEN
30	RTC0->EVENTS_COMPARE[0]	TIMER0->TASKS_CLEAR
31	RTC0->EVENTS_COMPARE[0]	TIMER0->TASKS_START

22.2 Registers

Table 37: Instances

Base address	Peripheral	Instance	Description	Configuration
0x4001F000	PPI	PPI	Programmable Peripheral Interconnect	

Table 38: Register Overview

Register	Offset	Description
TASKS_CHG[0].EN	0x000	Enable channel group 0
TASKS_CHG[0].DIS	0x004	Disable channel group 0
TASKS_CHG[1].EN	0x008	Enable channel group 1
TASKS_CHG[1].DIS	0x00C	Disable channel group 1
TASKS_CHG[2].EN	0x010	Enable channel group 2
TASKS_CHG[2].DIS	0x014	Disable channel group 2
TASKS_CHG[3].EN	0x018	Enable channel group 3
TASKS_CHG[3].DIS	0x01C	Disable channel group 3
TASKS_CHG[4].EN	0x020	Enable channel group 4

Register	Offset	Description
TASKS_CHG[4].DIS	0x024	Disable channel group 4
TASKS_CHG[5].EN	0x028	Enable channel group 5
TASKS_CHG[5].DIS	0x02C	Disable channel group 5
CHEN	0x500	Channel enable register
CHENSET	0x504	Channel enable set register
CHENCLR	0x508	Channel enable clear register
CH[0].EEP	0x510	Channel 0 event end-point
CH[0].TEP	0x514	Channel 0 task end-point
CH[1].EEP	0x518	Channel 1 event end-point
CH[1].TEP	0x51C	Channel 1 task end-point
CH[2].EEP	0x520	Channel 2 event end-point
CH[2].TEP	0x524	Channel 2 task end-point
CH[3].EEP	0x528	Channel 3 event end-point
CH[3].TEP	0x52C	Channel 3 task end-point
CH[4].EEP	0x530	Channel 4 event end-point
CH[4].TEP	0x534	Channel 4 task end-point
CH[5].EEP	0x538	Channel 5 event end-point
CH[5].TEP	0x53C	Channel 5 task end-point
CH[6].EEP	0x540	Channel 6 event end-point
CH[6].TEP	0x544	Channel 6 task end-point
CH[7].EEP	0x548	Channel 7 event end-point
CH[7].TEP	0x54C	Channel 7 task end-point
CH[8].EEP	0x550	Channel 8 event end-point
CH[8].TEP	0x554	Channel 8 task end-point
CH[9].EEP	0x558	Channel 9 event end-point
CH[9].TEP	0x55C	Channel 9 task end-point
CH[10].EEP	0x560	Channel 10 event end-point
CH[10].TEP	0x564	Channel 10 task end-point
CH[11].EEP	0x568	Channel 11 event end-point
CH[11].TEP	0x56C	Channel 11 task end-point
CH[12].EEP	0x570	Channel 12 event end-point
CH[12].TEP	0x574	Channel 12 task end-point
CH[13].EEP	0x578	Channel 13 event end-point
CH[13].TEP	0x57C	Channel 13 task end-point
CH[14].EEP	0x580	Channel 14 event end-point
CH[14].TEP	0x584	Channel 14 task end-point
CH[15].EEP	0x588	Channel 15 event end-point
CH[15].TEP	0x58C	Channel 15 task end-point
CH[16].EEP	0x590	Channel 16 event end-point
CH[16].TEP	0x594	Channel 16 task end-point
CH[17].EEP	0x598	Channel 17 event end-point
CH[17].TEP	0x59C	Channel 17 task end-point
CH[18].EEP	0x5A0	Channel 18 event end-point
CH[18].TEP	0x5A4	Channel 18 task end-point
CH[19].EEP	0x5A8	Channel 19 event end-point
CH[19].TEP	0x5AC	Channel 19 task end-point
CHG[0]	0x800	Channel group 0
CHG[1]	0x804	Channel group 1
CHG[2]	0x808	Channel group 2
CHG[3]	0x80C	Channel group 3
CHG[4]	0x810	Channel group 4
CHG[5]	0x814	Channel group 5
FORK[0].TEP	0x910	Channel 0 task end-point
FORK[1].TEP	0x914	Channel 1 task end-point
FORK[2].TEP	0x918	Channel 2 task end-point
FORK[3].TEP	0x91C	Channel 3 task end-point
FORK[4].TEP	0x920	Channel 4 task end-point
FORK[5].TEP	0x924	Channel 5 task end-point

Register	Offset	Description
FORK[6].TEP	0x928	Channel 6 task end-point
FORK[7].TEP	0x92C	Channel 7 task end-point
FORK[8].TEP	0x930	Channel 8 task end-point
FORK[9].TEP	0x934	Channel 9 task end-point
FORK[10].TEP	0x938	Channel 10 task end-point
FORK[11].TEP	0x93C	Channel 11 task end-point
FORK[12].TEP	0x940	Channel 12 task end-point
FORK[13].TEP	0x944	Channel 13 task end-point
FORK[14].TEP	0x948	Channel 14 task end-point
FORK[15].TEP	0x94C	Channel 15 task end-point
FORK[16].TEP	0x950	Channel 16 task end-point
FORK[17].TEP	0x954	Channel 17 task end-point
FORK[18].TEP	0x958	Channel 18 task end-point
FORK[19].TEP	0x95C	Channel 19 task end-point
FORK[20].TEP	0x960	Channel 20 task end-point
FORK[21].TEP	0x964	Channel 21 task end-point
FORK[22].TEP	0x968	Channel 22 task end-point
FORK[23].TEP	0x96C	Channel 23 task end-point
FORK[24].TEP	0x970	Channel 24 task end-point
FORK[25].TEP	0x974	Channel 25 task end-point
FORK[26].TEP	0x978	Channel 26 task end-point
FORK[27].TEP	0x97C	Channel 27 task end-point
FORK[28].TEP	0x980	Channel 28 task end-point
FORK[29].TEP	0x984	Channel 29 task end-point
FORK[30].TEP	0x988	Channel 30 task end-point
FORK[31].TEP	0x98C	Channel 31 task end-point

22.2.1 CHEN

Address offset: 0x500

Channel enable register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																														
A	RW	CH0			Enable or disable channel 0																																
			Disabled	0	Disable channel																														
			Enabled	1	Enable channel																														
B	RW	CH1			Enable or disable channel 1																														
			Disabled	0	Disable channel																														
			Enabled	1	Enable channel																														
C	RW	CH2			Enable or disable channel 2																														
			Disabled	0	Disable channel																														
			Enabled	1	Enable channel																														
D	RW	CH3			Enable or disable channel 3																														
			Disabled	0	Disable channel																														
			Enabled	1	Enable channel																														
E	RW	CH4			Enable or disable channel 4																														
			Disabled	0	Disable channel																														
			Enabled	1	Enable channel																														
F	RW	CH5			Enable or disable channel 5																														
			Disabled	0	Disable channel																														
			Enabled	1	Enable channel																														
G	RW	CH6			Enable or disable channel 6																														
			Disabled	0	Disable channel																														
			Enabled	1	Enable channel																														
H	RW	CH7			Enable or disable channel 7																														

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Disabled	0																															
			Enabled	1																															
I	RW	CH8																																	
			Disabled	0																															
			Enabled	1																															
J	RW	CH9																																	
			Disabled	0																															
			Enabled	1																															
K	RW	CH10																																	
			Disabled	0																															
			Enabled	1																															
L	RW	CH11																																	
			Disabled	0																															
			Enabled	1																															
M	RW	CH12																																	
			Disabled	0																															
			Enabled	1																															
N	RW	CH13																																	
			Disabled	0																															
			Enabled	1																															
O	RW	CH14																																	
			Disabled	0																															
			Enabled	1																															
P	RW	CH15																																	
			Disabled	0																															
			Enabled	1																															
Q	RW	CH16																																	
			Disabled	0																															
			Enabled	1																															
R	RW	CH17																																	
			Disabled	0																															
			Enabled	1																															
S	RW	CH18																																	
			Disabled	0																															
			Enabled	1																															
T	RW	CH19																																	
			Disabled	0																															
			Enabled	1																															
U	RW	CH20																																	
			Disabled	0																															
			Enabled	1																															
V	RW	CH21																																	
			Disabled	0																															
			Enabled	1																															
W	RW	CH22																																	
			Disabled	0																															
			Enabled	1																															
X	RW	CH23																																	
			Disabled	0																															
			Enabled	1																															
Y	RW	CH24																																	
			Disabled	0																															
			Enabled	1																															
Z	RW	CH25																																	
			Disabled	0																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Enabled	1																															
a	RW	CH26																																	
			Disabled	0																															
			Enabled	1																															
b	RW	CH27																																	
			Disabled	0																															
			Enabled	1																															
c	RW	CH28																																	
			Disabled	0																															
			Enabled	1																															
d	RW	CH29																																	
			Disabled	0																															
			Enabled	1																															
e	RW	CH30																																	
			Disabled	0																															
			Enabled	1																															
f	RW	CH31																																	
			Disabled	0																															
			Enabled	1																															

22.2.2 CHENSET

Address offset: 0x504

Channel enable set register

Read: reads value of CH{i} field in CHEN register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	CH0																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
B	RW	CH1																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
C	RW	CH2																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
D	RW	CH3																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
E	RW	CH4																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
F	RW	CH5																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
G	RW	CH6																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
H	RW	CH7																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
I	RW	CH8																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
J	RW	CH9																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
K	RW	CH10																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
L	RW	CH11																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
M	RW	CH12																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
N	RW	CH13																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
O	RW	CH14																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
P	RW	CH15																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
Q	RW	CH16																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
R	RW	CH17																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
S	RW	CH18																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
T	RW	CH19																																	
			Disabled	0																															
			Enabled	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Set	1																															
U	RW	CH20																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
V	RW	CH21																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
W	RW	CH22																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
X	RW	CH23																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
Y	RW	CH24																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
Z	RW	CH25																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
a	RW	CH26																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
b	RW	CH27																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
c	RW	CH28																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
d	RW	CH29																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
e	RW	CH30																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
f	RW	CH31																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															

22.2.3 CHENCLR

Address offset: 0x508

Channel enable clear register

Read: reads value of CH{i} field in CHEN register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	CH0																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
B	RW	CH1																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
C	RW	CH2																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
D	RW	CH3																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
E	RW	CH4																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
F	RW	CH5																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
G	RW	CH6																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
H	RW	CH7																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
I	RW	CH8																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
J	RW	CH9																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
K	RW	CH10																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
L	RW	CH11																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
M	RW	CH12																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
N	RW	CH13																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
O	RW	CH14																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
P	RW	CH15																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
Q	RW	CH16																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
R	RW	CH17																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
S	RW	CH18																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
T	RW	CH19																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
U	RW	CH20																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
V	RW	CH21																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
W	RW	CH22																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
X	RW	CH23																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
Y	RW	CH24																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
Z	RW	CH25																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
a	RW	CH26																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
b	RW	CH27																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
c	RW	CH28																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
d	RW	CH29																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
e	RW	CH30																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
f	RW	CH31																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															

22.2.4 CH[0].EEP

Address offset: 0x510

Channel 0 event end-point

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	EEP																																	

Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.5 CH[0].TEP

Address offset: 0x514

Channel 0 task end-point

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	TEP																																	

Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.6 CH[1].EEP

Address offset: 0x518

Channel 1 event end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.7 CH[1].TEP

Address offset: 0x51C

Channel 1 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.8 CH[2].EEP

Address offset: 0x520

Channel 2 event end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.9 CH[2].TEP

Address offset: 0x524

Channel 2 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.10 CH[3].EEP

Address offset: 0x528

Channel 3 event end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.11 CH[3].TEP

Address offset: 0x52C

Channel 3 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.12 CH[4].EEP

Address offset: 0x530

Channel 4 event end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.13 CH[4].TEP

Address offset: 0x534

Channel 4 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.14 CH[5].EEP

Address offset: 0x538

Channel 5 event end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.15 CH[5].TEP

Address offset: 0x53C

Channel 5 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.16 CH[6].EEP

Address offset: 0x540

Channel 6 event end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.17 CH[6].TEP

Address offset: 0x544

Channel 6 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.18 CH[7].EEP

Address offset: 0x548

Channel 7 event end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.19 CH[7].TEP

Address offset: 0x54C

Channel 7 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.20 CH[8].EEP

Address offset: 0x550

Channel 8 event end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.21 CH[8].TEP

Address offset: 0x554

Channel 8 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.22 CH[9].EEP

Address offset: 0x558

Channel 9 event end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.23 CH[9].TEP

Address offset: 0x55C

Channel 9 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.24 CH[10].EEP

Address offset: 0x560

Channel 10 event end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.25 CH[10].TEP

Address offset: 0x564

Channel 10 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.26 CH[11].EEP

Address offset: 0x568

Channel 11 event end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.27 CH[11].TEP

Address offset: 0x56C

Channel 11 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.28 CH[12].EEP

Address offset: 0x570

Channel 12 event end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.29 CH[12].TEP

Address offset: 0x574

Channel 12 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.30 CH[13].EEP

Address offset: 0x578

Channel 13 event end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.31 CH[13].TEP

Address offset: 0x57C

Channel 13 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.32 CH[14].EEP

Address offset: 0x580

Channel 14 event end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.33 CH[14].TEP

Address offset: 0x584

Channel 14 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.34 CH[15].EEP

Address offset: 0x588

Channel 15 event end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.35 CH[15].TEP

Address offset: 0x58C

Channel 15 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.36 CH[16].EEP

Address offset: 0x590

Channel 16 event end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.37 CH[16].TEP

Address offset: 0x594

Channel 16 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.38 CH[17].EEP

Address offset: 0x598

Channel 17 event end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.39 CH[17].TEP

Address offset: 0x59C

Channel 17 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.40 CH[18].EEP

Address offset: 0x5A0

Channel 18 event end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.41 CH[18].TEP

Address offset: 0x5A4

Channel 18 task end-point

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	TEP																																	

Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.42 CH[19].EEP

Address offset: 0x5A8

Channel 19 event end-point

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	EEP																																	

Pointer to event register. Accepts only addresses to registers from the Event group.

22.2.43 CH[19].TEP

Address offset: 0x5AC

Channel 19 task end-point

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	TEP																																	

Pointer to task register. Accepts only addresses to registers from the Task group.

22.2.44 CHG[0]

Address offset: 0x800

Channel group 0

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	CH0																																	
			Excluded	0																															
			Included	1																															
B	RW	CH1																																	
			Excluded	0																															
			Included	1																															
C	RW	CH2																																	
			Excluded	0																															
			Included	1																															
D	RW	CH3																																	
			Excluded	0																															
			Included	1																															
E	RW	CH4																																	
			Excluded	0																															
			Included	1																															
F	RW	CH5																																	
			Excluded	0																															
			Included	1																															
G	RW	CH6																																	

Include or exclude channel 0

Include or exclude channel 1

Include or exclude channel 2

Include or exclude channel 3

Include or exclude channel 4

Include or exclude channel 5

Include or exclude channel 6

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Excluded	0																															
			Included	1																															
H	RW	CH7																																	
			Excluded	0																															
			Included	1																															
I	RW	CH8																																	
			Excluded	0																															
			Included	1																															
J	RW	CH9																																	
			Excluded	0																															
			Included	1																															
K	RW	CH10																																	
			Excluded	0																															
			Included	1																															
L	RW	CH11																																	
			Excluded	0																															
			Included	1																															
M	RW	CH12																																	
			Excluded	0																															
			Included	1																															
N	RW	CH13																																	
			Excluded	0																															
			Included	1																															
O	RW	CH14																																	
			Excluded	0																															
			Included	1																															
P	RW	CH15																																	
			Excluded	0																															
			Included	1																															
Q	RW	CH16																																	
			Excluded	0																															
			Included	1																															
R	RW	CH17																																	
			Excluded	0																															
			Included	1																															
S	RW	CH18																																	
			Excluded	0																															
			Included	1																															
T	RW	CH19																																	
			Excluded	0																															
			Included	1																															
U	RW	CH20																																	
			Excluded	0																															
			Included	1																															
V	RW	CH21																																	
			Excluded	0																															
			Included	1																															
W	RW	CH22																																	
			Excluded	0																															
			Included	1																															
X	RW	CH23																																	
			Excluded	0																															
			Included	1																															
Y	RW	CH24																																	
			Excluded	0																															

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
			Included	1																														
Z	RW	CH25																																
			Excluded	0																														
			Included	1																														
a	RW	CH26																																
			Excluded	0																														
			Included	1																														
b	RW	CH27																																
			Excluded	0																														
			Included	1																														
c	RW	CH28																																
			Excluded	0																														
			Included	1																														
d	RW	CH29																																
			Excluded	0																														
			Included	1																														
e	RW	CH30																																
			Excluded	0																														
			Included	1																														
f	RW	CH31																																
			Excluded	0																														
			Included	1																														

22.2.45 CHG[1]

Address offset: 0x804

Channel group 1

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	CH0																																
			Excluded	0																														
			Included	1																														
B	RW	CH1																																
			Excluded	0																														
			Included	1																														
C	RW	CH2																																
			Excluded	0																														
			Included	1																														
D	RW	CH3																																
			Excluded	0																														
			Included	1																														
E	RW	CH4																																
			Excluded	0																														
			Included	1																														
F	RW	CH5																																
			Excluded	0																														
			Included	1																														
G	RW	CH6																																
			Excluded	0																														
			Included	1																														
H	RW	CH7																																
			Excluded	0																														

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
			Included	1	Include
I	RW	CH8			Include or exclude channel 8
			Excluded	0	Exclude
			Included	1	Include
J	RW	CH9			Include or exclude channel 9
			Excluded	0	Exclude
			Included	1	Include
K	RW	CH10			Include or exclude channel 10
			Excluded	0	Exclude
			Included	1	Include
L	RW	CH11			Include or exclude channel 11
			Excluded	0	Exclude
			Included	1	Include
M	RW	CH12			Include or exclude channel 12
			Excluded	0	Exclude
			Included	1	Include
N	RW	CH13			Include or exclude channel 13
			Excluded	0	Exclude
			Included	1	Include
O	RW	CH14			Include or exclude channel 14
			Excluded	0	Exclude
			Included	1	Include
P	RW	CH15			Include or exclude channel 15
			Excluded	0	Exclude
			Included	1	Include
Q	RW	CH16			Include or exclude channel 16
			Excluded	0	Exclude
			Included	1	Include
R	RW	CH17			Include or exclude channel 17
			Excluded	0	Exclude
			Included	1	Include
S	RW	CH18			Include or exclude channel 18
			Excluded	0	Exclude
			Included	1	Include
T	RW	CH19			Include or exclude channel 19
			Excluded	0	Exclude
			Included	1	Include
U	RW	CH20			Include or exclude channel 20
			Excluded	0	Exclude
			Included	1	Include
V	RW	CH21			Include or exclude channel 21
			Excluded	0	Exclude
			Included	1	Include
W	RW	CH22			Include or exclude channel 22
			Excluded	0	Exclude
			Included	1	Include
X	RW	CH23			Include or exclude channel 23
			Excluded	0	Exclude
			Included	1	Include
Y	RW	CH24			Include or exclude channel 24
			Excluded	0	Exclude
			Included	1	Include
Z	RW	CH25			Include or exclude channel 25
			Excluded	0	Exclude
			Included	1	Include

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														**Description**
a	RW	CH26																																Include or exclude channel 26
			Excluded	0																														Exclude
			Included	1																														Include
b	RW	CH27																																Include or exclude channel 27
			Excluded	0																														Exclude
			Included	1																														Include
c	RW	CH28																																Include or exclude channel 28
			Excluded	0																														Exclude
			Included	1																														Include
d	RW	CH29																																Include or exclude channel 29
			Excluded	0																														Exclude
			Included	1																														Include
e	RW	CH30																																Include or exclude channel 30
			Excluded	0																														Exclude
			Included	1																														Include
f	RW	CH31																																Include or exclude channel 31
			Excluded	0																														Exclude
			Included	1																														Include

22.2.46 CHG[2]

Address offset: 0x808

Channel group 2

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														**Description**
A	RW	CH0																																Include or exclude channel 0
			Excluded	0																														Exclude
			Included	1																														Include
B	RW	CH1																																Include or exclude channel 1
			Excluded	0																														Exclude
			Included	1																														Include
C	RW	CH2																																Include or exclude channel 2
			Excluded	0																														Exclude
			Included	1																														Include
D	RW	CH3																																Include or exclude channel 3
			Excluded	0																														Exclude
			Included	1																														Include
E	RW	CH4																																Include or exclude channel 4
			Excluded	0																														Exclude
			Included	1																														Include
F	RW	CH5																																Include or exclude channel 5
			Excluded	0																														Exclude
			Included	1																														Include
G	RW	CH6																																Include or exclude channel 6
			Excluded	0																														Exclude
			Included	1																														Include
H	RW	CH7																																Include or exclude channel 7
			Excluded	0																														Exclude
			Included	1																														Include
I	RW	CH8																																Include or exclude channel 8
			Excluded	0																														Exclude
			Included	1																														Include

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
J	RW	CH9			Include or exclude channel 9
			Excluded	0	Exclude
			Included	1	Include
K	RW	CH10			Include or exclude channel 10
			Excluded	0	Exclude
			Included	1	Include
L	RW	CH11			Include or exclude channel 11
			Excluded	0	Exclude
			Included	1	Include
M	RW	CH12			Include or exclude channel 12
			Excluded	0	Exclude
			Included	1	Include
N	RW	CH13			Include or exclude channel 13
			Excluded	0	Exclude
			Included	1	Include
O	RW	CH14			Include or exclude channel 14
			Excluded	0	Exclude
			Included	1	Include
P	RW	CH15			Include or exclude channel 15
			Excluded	0	Exclude
			Included	1	Include
Q	RW	CH16			Include or exclude channel 16
			Excluded	0	Exclude
			Included	1	Include
R	RW	CH17			Include or exclude channel 17
			Excluded	0	Exclude
			Included	1	Include
S	RW	CH18			Include or exclude channel 18
			Excluded	0	Exclude
			Included	1	Include
T	RW	CH19			Include or exclude channel 19
			Excluded	0	Exclude
			Included	1	Include
U	RW	CH20			Include or exclude channel 20
			Excluded	0	Exclude
			Included	1	Include
V	RW	CH21			Include or exclude channel 21
			Excluded	0	Exclude
			Included	1	Include
W	RW	CH22			Include or exclude channel 22
			Excluded	0	Exclude
			Included	1	Include
X	RW	CH23			Include or exclude channel 23
			Excluded	0	Exclude
			Included	1	Include
Y	RW	CH24			Include or exclude channel 24
			Excluded	0	Exclude
			Included	1	Include
Z	RW	CH25			Include or exclude channel 25
			Excluded	0	Exclude
			Included	1	Include
a	RW	CH26			Include or exclude channel 26
			Excluded	0	Exclude
			Included	1	Include
b	RW	CH27			Include or exclude channel 27

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Excluded	0																															
			Included	1																															
c	RW	CH28																																	
			Excluded	0																															
			Included	1																															
d	RW	CH29																																	
			Excluded	0																															
			Included	1																															
e	RW	CH30																																	
			Excluded	0																															
			Included	1																															
f	RW	CH31																																	
			Excluded	0																															
			Included	1																															

22.2.47 CHG[3]

Address offset: 0x80C

Channel group 3

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	CH0																																	
			Excluded	0																															
			Included	1																															
B	RW	CH1																																	
			Excluded	0																															
			Included	1																															
C	RW	CH2																																	
			Excluded	0																															
			Included	1																															
D	RW	CH3																																	
			Excluded	0																															
			Included	1																															
E	RW	CH4																																	
			Excluded	0																															
			Included	1																															
F	RW	CH5																																	
			Excluded	0																															
			Included	1																															
G	RW	CH6																																	
			Excluded	0																															
			Included	1																															
H	RW	CH7																																	
			Excluded	0																															
			Included	1																															
I	RW	CH8																																	
			Excluded	0																															
			Included	1																															
J	RW	CH9																																	
			Excluded	0																															
			Included	1																															
K	RW	CH10																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Excluded	0																															
			Included	1																															
L	RW	CH11																																	
			Excluded	0																															
			Included	1																															
M	RW	CH12																																	
			Excluded	0																															
			Included	1																															
N	RW	CH13																																	
			Excluded	0																															
			Included	1																															
O	RW	CH14																																	
			Excluded	0																															
			Included	1																															
P	RW	CH15																																	
			Excluded	0																															
			Included	1																															
Q	RW	CH16																																	
			Excluded	0																															
			Included	1																															
R	RW	CH17																																	
			Excluded	0																															
			Included	1																															
S	RW	CH18																																	
			Excluded	0																															
			Included	1																															
T	RW	CH19																																	
			Excluded	0																															
			Included	1																															
U	RW	CH20																																	
			Excluded	0																															
			Included	1																															
V	RW	CH21																																	
			Excluded	0																															
			Included	1																															
W	RW	CH22																																	
			Excluded	0																															
			Included	1																															
X	RW	CH23																																	
			Excluded	0																															
			Included	1																															
Y	RW	CH24																																	
			Excluded	0																															
			Included	1																															
Z	RW	CH25																																	
			Excluded	0																															
			Included	1																															
a	RW	CH26																																	
			Excluded	0																															
			Included	1																															
b	RW	CH27																																	
			Excluded	0																															
			Included	1																															
c	RW	CH28																																	
			Excluded	0																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Included	1																															
d	RW	CH29																																	
			Excluded	0																															
			Included	1																															
e	RW	CH30																																	
			Excluded	0																															
			Included	1																															
f	RW	CH31																																	
			Excluded	0																															
			Included	1																															

22.2.48 CHG[4]

Address offset: 0x810

Channel group 4

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	CH0																																	
			Excluded	0																															
			Included	1																															
B	RW	CH1																																	
			Excluded	0																															
			Included	1																															
C	RW	CH2																																	
			Excluded	0																															
			Included	1																															
D	RW	CH3																																	
			Excluded	0																															
			Included	1																															
E	RW	CH4																																	
			Excluded	0																															
			Included	1																															
F	RW	CH5																																	
			Excluded	0																															
			Included	1																															
G	RW	CH6																																	
			Excluded	0																															
			Included	1																															
H	RW	CH7																																	
			Excluded	0																															
			Included	1																															
I	RW	CH8																																	
			Excluded	0																															
			Included	1																															
J	RW	CH9																																	
			Excluded	0																															
			Included	1																															
K	RW	CH10																																	
			Excluded	0																															
			Included	1																															
L	RW	CH11																																	
			Excluded	0																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
			Included	1	Include
M	RW	CH12			Include or exclude channel 12
			Excluded	0	Exclude
			Included	1	Include
N	RW	CH13			Include or exclude channel 13
			Excluded	0	Exclude
			Included	1	Include
O	RW	CH14			Include or exclude channel 14
			Excluded	0	Exclude
			Included	1	Include
P	RW	CH15			Include or exclude channel 15
			Excluded	0	Exclude
			Included	1	Include
Q	RW	CH16			Include or exclude channel 16
			Excluded	0	Exclude
			Included	1	Include
R	RW	CH17			Include or exclude channel 17
			Excluded	0	Exclude
			Included	1	Include
S	RW	CH18			Include or exclude channel 18
			Excluded	0	Exclude
			Included	1	Include
T	RW	CH19			Include or exclude channel 19
			Excluded	0	Exclude
			Included	1	Include
U	RW	CH20			Include or exclude channel 20
			Excluded	0	Exclude
			Included	1	Include
V	RW	CH21			Include or exclude channel 21
			Excluded	0	Exclude
			Included	1	Include
W	RW	CH22			Include or exclude channel 22
			Excluded	0	Exclude
			Included	1	Include
X	RW	CH23			Include or exclude channel 23
			Excluded	0	Exclude
			Included	1	Include
Y	RW	CH24			Include or exclude channel 24
			Excluded	0	Exclude
			Included	1	Include
Z	RW	CH25			Include or exclude channel 25
			Excluded	0	Exclude
			Included	1	Include
a	RW	CH26			Include or exclude channel 26
			Excluded	0	Exclude
			Included	1	Include
b	RW	CH27			Include or exclude channel 27
			Excluded	0	Exclude
			Included	1	Include
c	RW	CH28			Include or exclude channel 28
			Excluded	0	Exclude
			Included	1	Include
d	RW	CH29			Include or exclude channel 29
			Excluded	0	Exclude
			Included	1	Include

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
e	RW	CH30			Include or exclude channel 30																															
			Excluded	0	Exclude																															
			Included	1	Include																															
f	RW	CH31			Include or exclude channel 31																															
			Excluded	0	Exclude																															
			Included	1	Include																															

22.2.49 CHG[5]

Address offset: 0x814

Channel group 5

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	CH0			Include or exclude channel 0																															
			Excluded	0	Exclude																															
			Included	1	Include																															
B	RW	CH1			Include or exclude channel 1																															
			Excluded	0	Exclude																															
			Included	1	Include																															
C	RW	CH2			Include or exclude channel 2																															
			Excluded	0	Exclude																															
			Included	1	Include																															
D	RW	CH3			Include or exclude channel 3																															
			Excluded	0	Exclude																															
			Included	1	Include																															
E	RW	CH4			Include or exclude channel 4																															
			Excluded	0	Exclude																															
			Included	1	Include																															
F	RW	CH5			Include or exclude channel 5																															
			Excluded	0	Exclude																															
			Included	1	Include																															
G	RW	CH6			Include or exclude channel 6																															
			Excluded	0	Exclude																															
			Included	1	Include																															
H	RW	CH7			Include or exclude channel 7																															
			Excluded	0	Exclude																															
			Included	1	Include																															
I	RW	CH8			Include or exclude channel 8																															
			Excluded	0	Exclude																															
			Included	1	Include																															
J	RW	CH9			Include or exclude channel 9																															
			Excluded	0	Exclude																															
			Included	1	Include																															
K	RW	CH10			Include or exclude channel 10																															
			Excluded	0	Exclude																															
			Included	1	Include																															
L	RW	CH11			Include or exclude channel 11																															
			Excluded	0	Exclude																															
			Included	1	Include																															
M	RW	CH12			Include or exclude channel 12																															
			Excluded	0	Exclude																															
			Included	1	Include																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
N	RW	CH13			Include or exclude channel 13
			Excluded	0	Exclude
			Included	1	Include
O	RW	CH14			Include or exclude channel 14
			Excluded	0	Exclude
			Included	1	Include
P	RW	CH15			Include or exclude channel 15
			Excluded	0	Exclude
			Included	1	Include
Q	RW	CH16			Include or exclude channel 16
			Excluded	0	Exclude
			Included	1	Include
R	RW	CH17			Include or exclude channel 17
			Excluded	0	Exclude
			Included	1	Include
S	RW	CH18			Include or exclude channel 18
			Excluded	0	Exclude
			Included	1	Include
T	RW	CH19			Include or exclude channel 19
			Excluded	0	Exclude
			Included	1	Include
U	RW	CH20			Include or exclude channel 20
			Excluded	0	Exclude
			Included	1	Include
V	RW	CH21			Include or exclude channel 21
			Excluded	0	Exclude
			Included	1	Include
W	RW	CH22			Include or exclude channel 22
			Excluded	0	Exclude
			Included	1	Include
X	RW	CH23			Include or exclude channel 23
			Excluded	0	Exclude
			Included	1	Include
Y	RW	CH24			Include or exclude channel 24
			Excluded	0	Exclude
			Included	1	Include
Z	RW	CH25			Include or exclude channel 25
			Excluded	0	Exclude
			Included	1	Include
a	RW	CH26			Include or exclude channel 26
			Excluded	0	Exclude
			Included	1	Include
b	RW	CH27			Include or exclude channel 27
			Excluded	0	Exclude
			Included	1	Include
c	RW	CH28			Include or exclude channel 28
			Excluded	0	Exclude
			Included	1	Include
d	RW	CH29			Include or exclude channel 29
			Excluded	0	Exclude
			Included	1	Include
e	RW	CH30			Include or exclude channel 30
			Excluded	0	Exclude
			Included	1	Include
f	RW	CH31			Include or exclude channel 31

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
			Excluded	0	Exclude																													
			Included	1	Include																													

22.2.50 FORK[0].TEP

Address offset: 0x910

Channel 0 task end-point

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	TEP			Pointer to task register																													

22.2.51 FORK[1].TEP

Address offset: 0x914

Channel 1 task end-point

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	TEP			Pointer to task register																													

22.2.52 FORK[2].TEP

Address offset: 0x918

Channel 2 task end-point

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	TEP			Pointer to task register																													

22.2.53 FORK[3].TEP

Address offset: 0x91C

Channel 3 task end-point

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	TEP			Pointer to task register																													

22.2.54 FORK[4].TEP

Address offset: 0x920

Channel 4 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.55 FORK[5].TEP

Address offset: 0x924

Channel 5 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.56 FORK[6].TEP

Address offset: 0x928

Channel 6 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.57 FORK[7].TEP

Address offset: 0x92C

Channel 7 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.58 FORK[8].TEP

Address offset: 0x930

Channel 8 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.59 FORK[9].TEP

Address offset: 0x934

Channel 9 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.60 FORK[10].TEP

Address offset: 0x938

Channel 10 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.61 FORK[11].TEP

Address offset: 0x93C

Channel 11 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.62 FORK[12].TEP

Address offset: 0x940

Channel 12 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.63 FORK[13].TEP

Address offset: 0x944

Channel 13 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.64 FORK[14].TEP

Address offset: 0x948

Channel 14 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.65 FORK[15].TEP

Address offset: 0x94C

Channel 15 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.66 FORK[16].TEP

Address offset: 0x950

Channel 16 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.67 FORK[17].TEP

Address offset: 0x954

Channel 17 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.68 FORK[18].TEP

Address offset: 0x958

Channel 18 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.69 FORK[19].TEP

Address offset: 0x95C

Channel 19 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.70 FORK[20].TEP

Address offset: 0x960

Channel 20 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.71 FORK[21].TEP

Address offset: 0x964

Channel 21 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.72 FORK[22].TEP

Address offset: 0x968

Channel 22 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.73 FORK[23].TEP

Address offset: 0x96C

Channel 23 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.74 FORK[24].TEP

Address offset: 0x970

Channel 24 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.75 FORK[25].TEP

Address offset: 0x974

Channel 25 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.76 FORK[26].TEP

Address offset: 0x978

Channel 26 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.77 FORK[27].TEP

Address offset: 0x97C

Channel 27 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.78 FORK[28].TEP

Address offset: 0x980

Channel 28 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.79 FORK[29].TEP

Address offset: 0x984

Channel 29 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.80 FORK[30].TEP

Address offset: 0x988

Channel 30 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

22.2.81 FORK[31].TEP

Address offset: 0x98C

Channel 31 task end-point

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TEP			Pointer to task register

23 RADIO — 2.4 GHz Radio

The RADIO contains a 2.4 GHz radio receiver and a 2.4 GHz radio transmitter that is compatible with Nordic's proprietary 1 Mbps and 2 Mbps radio modes in addition to 1 Mbps and 2 Mbps *Bluetooth*® low energy mode.

EasyDMA in combination with an automated packet assembler and packet disassembler, and an automated CRC generator and CRC checker, makes it very easy to configure and use the RADIO. See *Figure 29: RADIO block diagram* on page 205 for details.

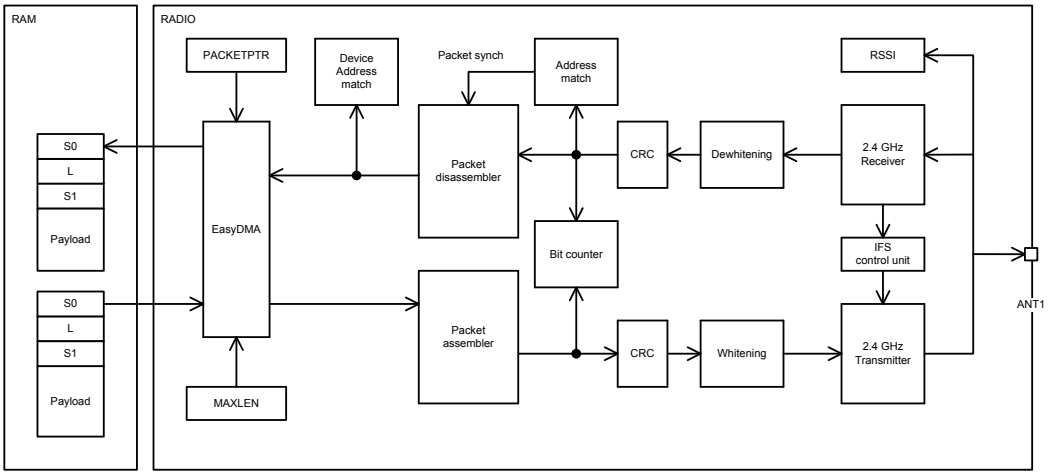


Figure 29: RADIO block diagram

The RADIO includes a Device Address Match unit and an interframe spacing control unit that can be utilized to simplify address white listing and interframe spacing respectively, in *Bluetooth* Smart and similar applications.

The RADIO also includes a Received Signal Strength Indicator (RSSI) and a bit counter. The bit counter generates events when a preconfigured number of bits have been sent or received by the RADIO.

23.1 EasyDMA

The RADIO use EasyDMA for reading and writing of data packets from and to the RAM without CPU involvement.

As illustrated in *Figure 29: RADIO block diagram* on page 205, the RADIO's EasyDMA utilizes the same PACKETPTR for receiving and transmitting packets. The CPU should reconfigure this pointer every time before the RADIO is started via the START task.

The structure of a radio packet is described in detail in *Packet configuration* on page 206. The data that is stored in Data RAM and transported by EasyDMA consists of S0, LENGTH, S1, the payload itself, and a static add-on sent immediately after the payload.

The size of each of the above elements in the frame is configurable (see *Packet configuration* on page 206), and the space occupied in RAM depends on these settings. A size of zero is possible for any of the fields, it is up to the user to make sure that the resulting frame complies with the RF protocol chosen.

For the field sizes defined in bits, the occupation in RAM will always be rounded up to the next full byte size (for instance 3 bit length will allocate 1 byte in RAM, 9 bit length will allocate 2 bytes, etc.).

In addition, the S0INCL field in PCNF0 determines if S0 is present in RAM at all if its length is zero. If present, one byte is allocated in RAM.

The size of S0 is configured through the S0LEN field in PCNF0. The size of LENGTH is configured through the LFLEN field in PCNF0. The size of S1 is configured through the S1LEN field in PCNF0. The size of the payload is configured through the value in RAM corresponding to the LENGTH field. The size of the static add-on to the payload is configured through the STATLEN field in PCNF1.

The MAXLEN field in the PCNF1 register configures the maximum packet payload plus add-on size in number of bytes that can be transmitted or received by the RADIO. This feature can be used to ensure that the RADIO does not overwrite, or read beyond, the RAM assigned to the packet payload. This means that if the packet payload length defined by PCNF1.STATLEN and the LENGTH field in the packet specifies a packet larger than MAXLEN, the payload will be truncated at MAXLEN.

Note that MAXLEN includes the payload and the add-on, but excludes the size occupied by the S0, LENGTH and S1 fields. This has to be taken into account when allocating RAM.

If the payload plus add-on length is specified larger than MAXLEN, the RADIO will still transmit or receive in the same way as before except the payload is now truncated to MAXLEN. The packet's LENGTH field will not be altered when the payload is truncated. The RADIO will calculate CRC as if the packet length is equal to MAXLEN.

If the PACKETPTR is not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See *Memory* on page 23 for more information about the different memory regions.

The DISABLED event indicates that the EasyDMA has finished accessing the RAM.

23.2 Packet configuration

A Radio packet contains the following fields: PREAMBLE, ADDRESS, S0, LENGTH, S1, PAYLOAD and CRC.

See *Figure 30: On-air packet layout* on page 206. Not shown in the figure is the static payload add-on (the length of which is defined in STATLEN, and which is 0 bytes long in a standard BLE packet), and would be sent between PAYLOAD and CRC. The Radio sends the different fields in the packet in the order they are illustrated below, from left to right. The preamble will be sent least significant bit first on-air.

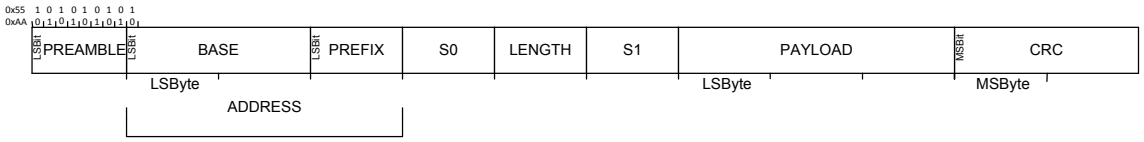


Figure 30: On-air packet layout

For all modes, except for 2 Mbit/s Bluetooth Low Energy mode, the preamble is one byte long. For 2 Mbit/s Bluetooth Low Energy mode the preamble is 2 bytes long. If the first bit of the ADDRESS is 0 the preamble will be set to 0xAA otherwise the PREAMBLE will be set to 0x55.

Radio packets are stored in memory inside instances of a radio packet data structure as illustrated in *Figure 31: In-RAM representation of radio packet, S0, LENGTH and S1 are optional* on page 206. The PREAMBLE, ADDRESS and CRC fields are omitted in this data structure.

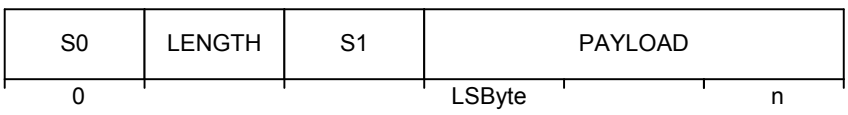


Figure 31: In-RAM representation of radio packet, S0, LENGTH and S1 are optional

The byte ordering on air is always Least Significant Byte First for the ADDRESS and PAYLOAD fields and Most Significant Byte First for the CRC field. The ADDRESS fields are always transmitted and received least significant bit first on-air. The CRC field is always transmitted and received Most Significant Bit first. The bit-endian, i.e. which order the bits are sent and received in, of the S0, LENGTH, S1 and PAYLOAD fields can be configured via the ENDIAN in PCNF1.

The S0INCL field in PCNF0 determines if S0 is present in RAM at all if its length is zero. If present, one byte is allocated in RAM.

The sizes of the S0, LENGTH and S1 fields can be individually configured via S0LEN, LFLEN and S1LEN in PCNF0 respectively. If any of these fields are configured to be less than 8 bit long the, the least significant bits of the fields, as seen from the RAM representation, are used.

If S0, LENGTH or S1 are specified with zero length their fields will be omitted in memory, otherwise each field will be represented as a separate byte, regardless of the number of bits in their on-air counterpart.

23.3 Maximum packet length

Independent of the configuration of MAXLEN, the combined length of S0, LENGTH, S1 and PAYLOAD cannot exceed 258 bytes.

23.4 Address configuration

The on-air radio ADDRESS field is composed of two parts, the base address field and the address prefix field.

The size of the base address field is configurable via BALEN in PCNF1. The base address is truncated from LSByte if the BALEN is less than 4. See *Table 39: Definition of logical addresses* on page 207.

The on-air addresses are defined in the BASEn and PREFIXn registers, and it is only when writing these registers the user will have to relate to actual on-air addresses. For other radio address registers such as the TXADDRESS, RXADDRESSES and RXMATCH registers, logical radio addresses ranging from 0 to 7 are being used. The relationship between the on-air radio addresses and the logical addresses is described in *Table 39: Definition of logical addresses* on page 207.

Table 39: Definition of logical addresses

Logical address	Base address	Prefix byte
0	BASE0	PREFIX0.AP0
1	BASE1	PREFIX0.AP1
2	BASE1	PREFIX0.AP2
3	BASE1	PREFIX0.AP3
4	BASE1	PREFIX1.AP4
5	BASE1	PREFIX1.AP5
6	BASE1	PREFIX1.AP6
7	BASE1	PREFIX1.AP7

23.5 Data whitening

The RADIO is able to do packet whitening and de-whitening.

See WHITEEN in PCNF1 register for how to enable whitening. When enabled, whitening and de-whitening will be handled by the RADIO automatically as packets are sent and received.

The whitening word is generated using polynomial $g(D) = D^7 + D^4 + 1$, which then is XORed with the data packet that is to be whitened, or de-whitened. See the figure below.

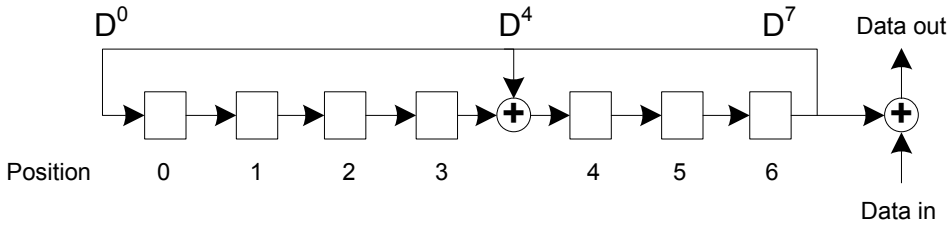


Figure 32: Data whitening and de-whitening

Whitening and de-whitening will be performed over the whole packet (except for the preamble and the address field).

The linear feedback shift register, illustrated in *Figure 32: Data whitening and de-whitening* on page 208 can be initialised via the DATAWHITEIV register.

23.6 CRC

The CRC generator in the RADIO calculates the CRC over the whole packet excluding the preamble. If desirable, the address field can be excluded from the CRC calculation as well

See CRCCNF register for more information.

The CRC polynomial is configurable as illustrated in *Figure 33: CRC generation of an n bit CRC* on page 208 where bit 0 in the CRCPOLY register corresponds to X^0 and bit 1 corresponds to X^1 etc. See CRCPOLY for more information.

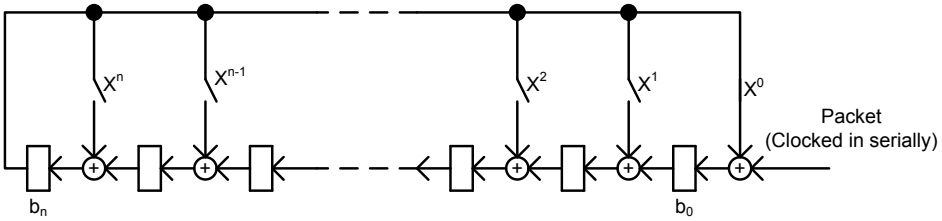


Figure 33: CRC generation of an n bit CRC

As illustrated in *Figure 33: CRC generation of an n bit CRC* on page 208, the CRC is calculated by feeding the packet serially through the CRC generator. Before the packet is clocked through the CRC generator, the CRC generator's latches b_0 through b_n will be initialized with a predefined value specified in the CRCINIT register. When the whole packet is clocked through the CRC generator, latches b_0 through b_n will hold the resulting CRC. This value will be used by the RADIO during both transmission and reception but it is not available to be read by the CPU at any time. A received CRC can however be read by the CPU via the RXCRC register independent of whether or not it has passed the CRC check.

The length (n) of the CRC is configurable, see CRCCNF for more information.

After the whole packet including the CRC has been received, the RADIO will generate a CRCOK event if no CRC errors were detected, or alternatively generate a CRCERROR event if CRC errors were detected.

The status of the CRC check can be read from the CRCSTATUS register after a packet has been received.

23.7 Radio states

The RADIO can enter a number of states.

The RADIO can enter the states described the table below. An overview state diagram for the RADIO is illustrated in *Figure 34: Radio states* on page 209. This figure shows how the tasks and events relate to the RADIO's operation. The RADIO does not prevent a task from being triggered from the wrong state. If a task is triggered from the wrong state, for example if the RXEN task is triggered from the RXDISABLE state, this may lead to incorrect behaviour. As illustrated in *Figure 34: Radio states* on page 209, the PAYLOAD event is always generated even if the payload is zero.

Table 40: RADIO state diagram

State	Description
DISABLED	No operations are going on inside the radio and the power consumption is at a minimum
RXRU	The radio is ramping up and preparing for reception
RXIDLE	The radio is ready for reception to start
RX	Reception has been started and the addresses enabled in the RXADDRESSES register are being monitored
TXRU	The radio is ramping up and preparing for transmission
TXIDLE	The radio is ready for transmission to start
TX	The radio is transmitting a packet
RXDISABLE	The radio is disabling the receiver
TXDISABLE	The radio is disabling the transmitter

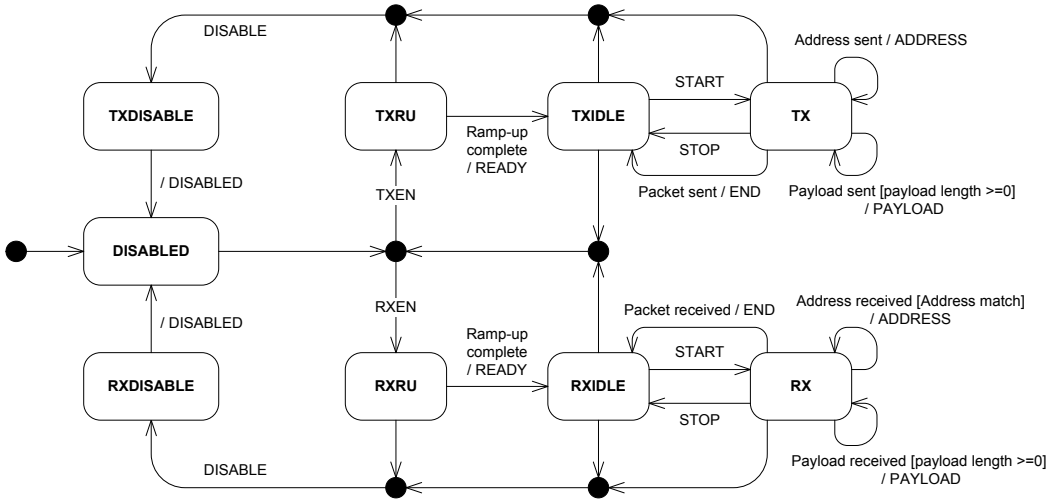


Figure 34: Radio states

23.8 Transmit sequence

Before the RADIO is able to transmit a packet, it must first ramp-up in TX mode.

See TXRU in *Figure 34: Radio states* on page 209 and *Figure 35: Transmit sequence* on page 210.
A TXRU ramp-up sequence is initiated when the TXEN task is triggered. After the radio has successfully ramped up it will generate the READY event indicating that a packet transmission can be initiate. A packet transmission is initiated by triggering the START task. As illustrated in *Figure 34: Radio states* on page 209 the START task can first be triggered after the RADIO has entered into the TXIDLE state.

Figure 35: Transmit sequence on page 210 illustrates a single packet transmission where the CPU manually triggers the different tasks needed to control the flow of the RADIO, i.e. no shortcuts are used. If shortcuts are not used, a certain amount of delay caused by CPU execution is expected between READY and START, and between END and DISABLE. As illustrated in *Figure 35: Transmit sequence* on page 210

the RADIO will by default transmit '1's between READY and START, and between END and DISABLED. What is transmitted can be programmed through the DTX field in the MODECNF0 register.

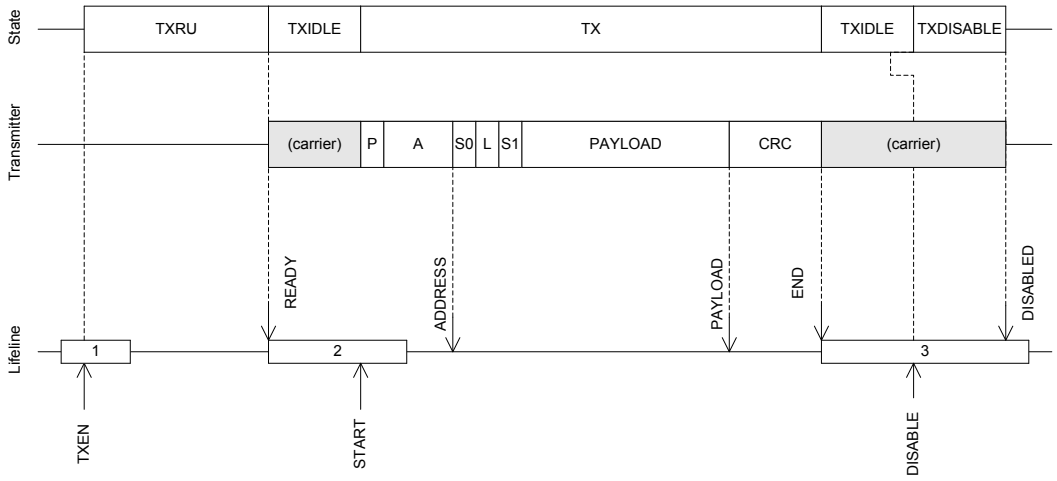


Figure 35: Transmit sequence

A slightly modified version of the transmit sequence from *Figure 35: Transmit sequence* on page 210 is illustrated in *Figure 36: Transmit sequence using shortcuts to avoid delays* on page 210 where the RADIO is configured to use shortcuts between READY and START, and between END and DISABLE, which means that no delay is introduced.

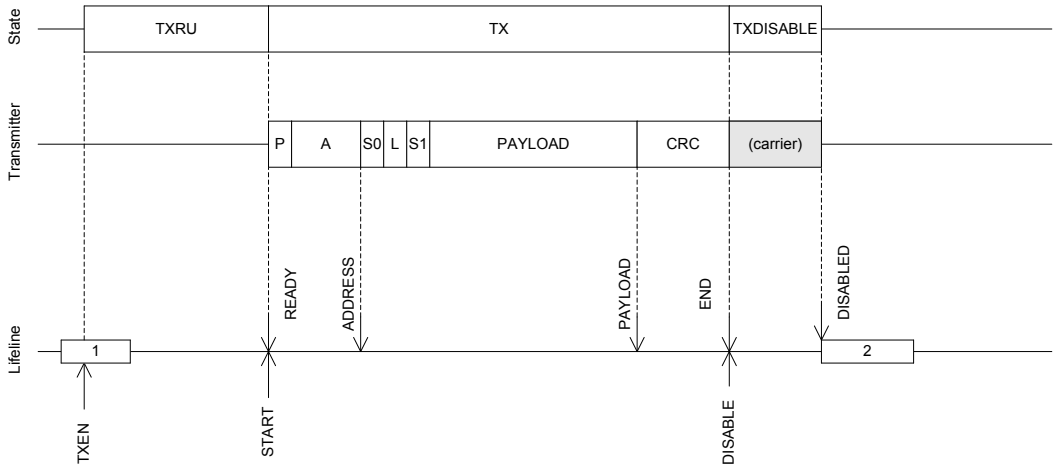


Figure 36: Transmit sequence using shortcuts to avoid delays

The RADIO is able to send multiple packets one after the other without having to disable and re-enable the RADIO between packets, this is illustrated in *Figure 37: Transmission of multiple packets* on page 211.

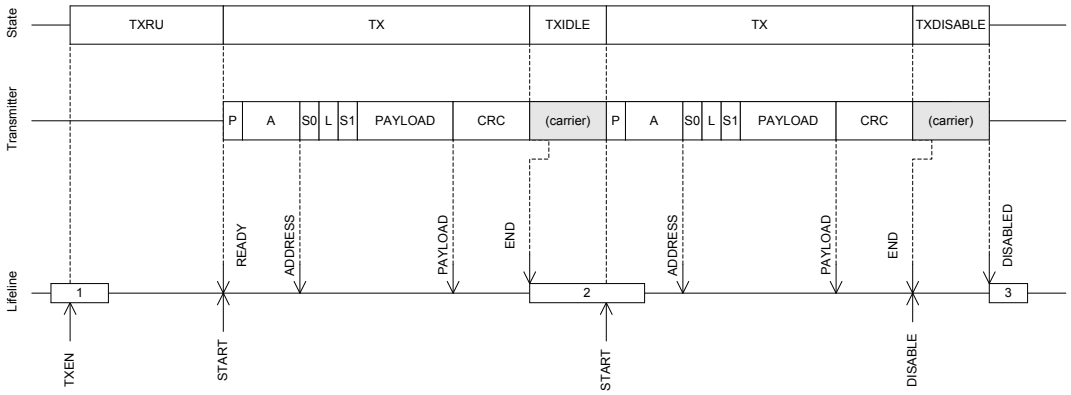


Figure 37: Transmission of multiple packets

23.9 Receive sequence

Before the RADIO is able to receive a packet, it must first ramp up in RX mode

See RXRU in *Figure 34: Radio states* on page 209 and *Figure 38: Receive sequence* on page 211.
An RXRU ramp-up sequence is initiated when the RXEN task is triggered. After the radio has successfully ramped up it will generate the READY event indicating that a packet reception can be initiated. A packet reception is initiated by triggering the START task. As illustrated in *Figure 34: Radio states* on page 209 the START task can, first be triggered after the RADIO has entered into the RXIDLE state.

Figure 38: Receive sequence on page 211 illustrates a single packet reception where the CPU manually triggers the different tasks needed to control the flow of the RADIO, i.e. no shortcuts are used. If shortcuts are not used, a certain amount of delay, caused by CPU execution, is expected between READY and START, and between END and DISABLE. As illustrated *Figure 38: Receive sequence* on page 211 the RADIO will be listening and possibly receiving undefined data, illustrated with an 'X', from START and until a packet with valid preamble (P) is received.

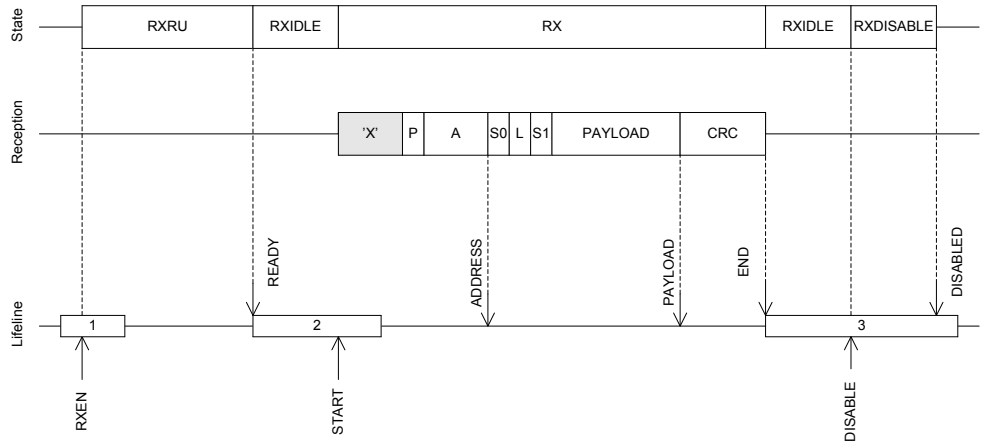


Figure 38: Receive sequence

A slightly modified version of the receive sequence from *Figure 38: Receive sequence* on page 211 is illustrated in *Figure 39: Receive sequence using shortcuts to avoid delays* on page 212 where the the RADIO is configured to use shortcuts between READY and START, and between END and DISABLE, which means that no delay is introduced.

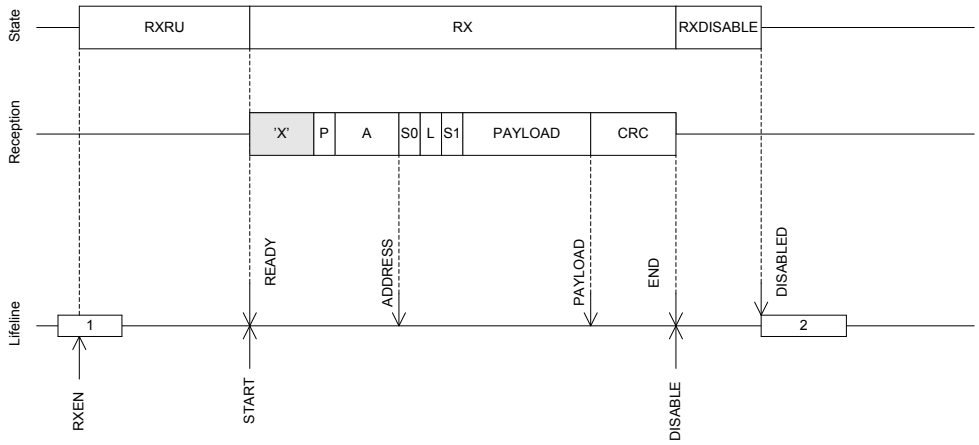


Figure 39: Receive sequence using shortcuts to avoid delays

The RADIO is able to receive multiple packets one after the other without having to disable and re-enable the RADIO between packets, this is illustrated *Figure 40: Reception of multiple packets* on page 212.

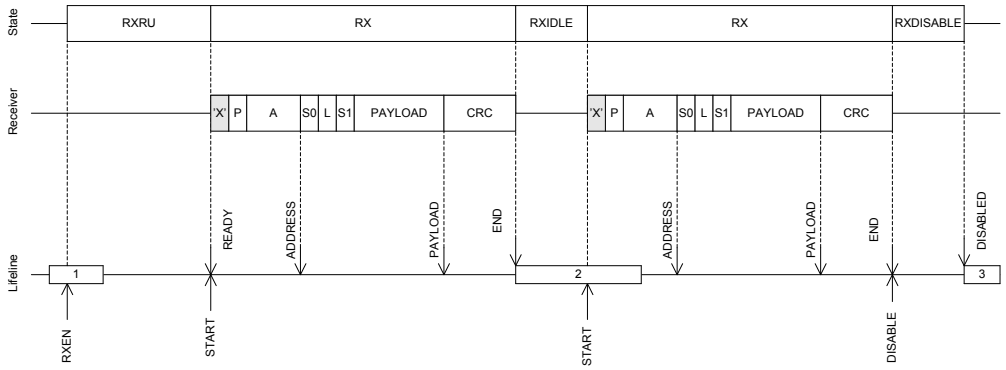


Figure 40: Reception of multiple packets

23.10 Received Signal Strength Indicator (RSSI)

The radio implements a mechanism for measuring the power in the received radio signal. This feature is called Received Signal Strength Indicator (RSSI).

Sampling of the received signal strength is started by using the RSSISTART task. The sample can be read from the RSSISAMPLE register.

The sample period of the RSSI is defined by $RSSI_{PERIOD}$, see the device product specification for details. The RSSI sample will hold the average received signal strength during this sample period.

For the RSSI sample to be valid the radio has to be enabled in receive mode (RXEN task) and the reception has to be started (READY event followed by START task).

23.11 Interframe spacing

Interframe spacing is the time interval between two consecutive packets.

It is defined as the time, in micro seconds, from the end of the last bit of the previous packet received and to the start of the first bit of the subsequent packet that is transmitted. The RADIO is able to enforce this

interval as specified in the TIFS register as long as TIFS is not specified to be shorter than the RADIO's turn-around time, i.e. the time needed to switch off the receiver, and switch back on the transmitter.

TIFS is only enforced if END_DISABLE and DISABLED_TXEN or END_DISABLE and DISABLED_RXEN shortcuts are enabled. TIFS is only qualified for use in BLE_1MBIT mode, and default ramp-up mode.

23.12 Device address match

The device address match feature is tailored for address white listing in a Bluetooth Smart and similar implementations.

This feature enables on-the-fly device address matching while receiving a packet on air. This feature only works in receive mode and as long as RADIO is configured for little endian, see PCNF1.ENDIAN.

The Device Address match unit assumes that the 48 first bits of the payload is the device address and that bit number 6 in S0 is the TxAdd bit. See the Bluetooth Core Specification for more information about device addresses, TxAdd and whitelisting.

The RADIO is able to listen for eight different device addresses at the same time. These addresses are specified in a DAB/DAP register pair, one pair per address, in addition to a TxAdd bit configured in the DACNF register. The DAB register specifies the 32 least significant bits of the device address, while the DAP register specifies the 16 most significant bits of the device address.

Each of the device addresses can be individually included or excluded from the matching mechanism. This is configured in the DACNF register.

23.13 Bit counter

The RADIO implements a simple counter that can be configured to generate an event after a specific number of bits have been transmitted or received.

By using shortcuts, this counter can be started from different events generated by the RADIO and hence count relative to these.

The bit counter is started by triggering the BCSTART task, and stopped by triggering the BCSTOP task. A BCMATCH event will be generated when the bit counter has counted the number of bits specified in the BCC register. The bit counter will continue to count bits until the DISABLED event is generated or until the BCSTOP task is triggered. The CPU can therefore, after a BCMATCH event, reconfigure the BCC value for new BCMATCH events within the same packet.

The bit counter can only be started after the RADIO has received the ADDRESS event.

The bit counter will stop and reset on BCSTOP, STOP, END and DISABLE tasks.

The figure below illustrates how the bit counter can be used to generate a BCMATCH event in the beginning of the packet payload, and again generate a second BCMATCH event after sending 2 bytes (16 bits) of the payload.

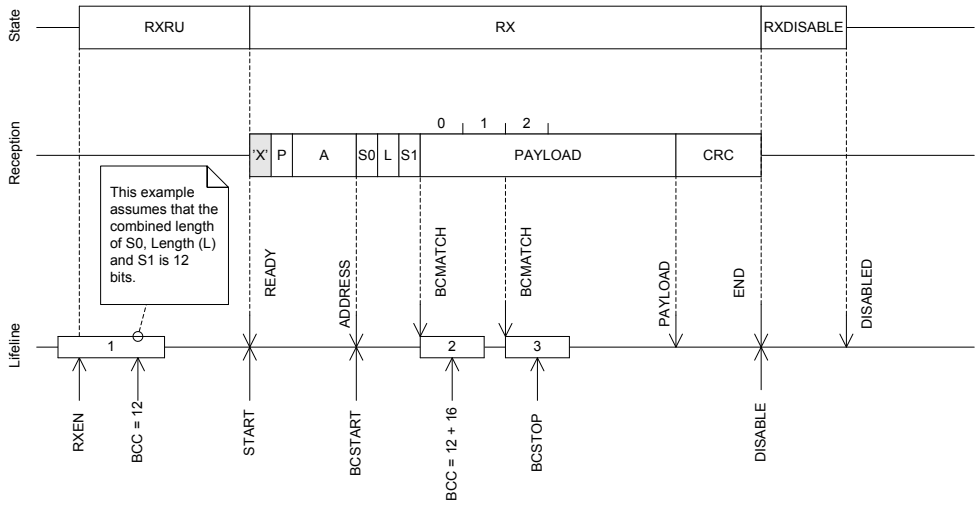


Figure 41: Bit counter example

23.14 Registers

Table 41: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40001000	RADIO	RADIO	2.4 GHz radio	

Table 42: Register Overview

Register	Offset	Description
TASKS_TXEN	0x000	Enable RADIO in TX mode
TASKS_RXEN	0x004	Enable RADIO in RX mode
TASKS_START	0x008	Start RADIO
TASKS_STOP	0x00C	Stop RADIO
TASKS_DISABLE	0x010	Disable RADIO
TASKS_RSSISTART	0x014	Start the RSSI and take one single sample of the receive signal strength.
TASKS_RSSISTOP	0x018	Stop the RSSI measurement
TASKS_BCSTART	0x01C	Start the bit counter
TASKS_BCSTOP	0x020	Stop the bit counter
EVENTS_READY	0x100	RADIO has ramped up and is ready to be started
EVENTS_ADDRESS	0x104	Address sent or received
EVENTS_PAYLOAD	0x108	Packet payload sent or received
EVENTS_END	0x10C	Packet sent or received
EVENTS_DISABLED	0x110	RADIO has been disabled
EVENTS_DEVMATCH	0x114	A device address match occurred on the last received packet
EVENTS_DEVMISS	0x118	No device address match occurred on the last received packet
EVENTS_RSSIEND	0x11C	Sampling of receive signal strength complete.
EVENTS_BCMATCH	0x128	Bit counter reached bit count value.
EVENTS_CRCOK	0x130	Packet received with CRC ok
EVENTS_CRCERROR	0x134	Packet received with CRC error
SHORTS	0x200	Shortcut register
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
CRCSTATUS	0x400	CRC status
RXMATCH	0x408	Received address
RXCRC	0x40C	CRC field of previously received packet

Register	Offset	Description
DAI	0x410	Device address match index
PACKETPTR	0x504	Packet pointer
FREQUENCY	0x508	Frequency
TXPOWER	0x50C	Output power
MODE	0x510	Data rate and modulation
PCNF0	0x514	Packet configuration register 0
PCNF1	0x518	Packet configuration register 1
BASE0	0x51C	Base address 0
BASE1	0x520	Base address 1
PREFIX0	0x524	Prefixes bytes for logical addresses 0-3
PREFIX1	0x528	Prefixes bytes for logical addresses 4-7
TXADDRESS	0x52C	Transmit address select
RXADDRESSES	0x530	Receive address select
CRCCNF	0x534	CRC configuration
CRCPOLY	0x538	CRC polynomial
CRCINIT	0x53C	CRC initial value
	0x540	Reserved
TIFS	0x544	Inter Frame Spacing in us
RSSISAMPLE	0x548	RSSI sample
STATE	0x550	Current radio state
DATAWHITEIV	0x554	Data whitening initial value
BCC	0x560	Bit counter compare
DAB[0]	0x600	Device address base segment 0
DAB[1]	0x604	Device address base segment 1
DAB[2]	0x608	Device address base segment 2
DAB[3]	0x60C	Device address base segment 3
DAB[4]	0x610	Device address base segment 4
DAB[5]	0x614	Device address base segment 5
DAB[6]	0x618	Device address base segment 6
DAB[7]	0x61C	Device address base segment 7
DAP[0]	0x620	Device address prefix 0
DAP[1]	0x624	Device address prefix 1
DAP[2]	0x628	Device address prefix 2
DAP[3]	0x62C	Device address prefix 3
DAP[4]	0x630	Device address prefix 4
DAP[5]	0x634	Device address prefix 5
DAP[6]	0x638	Device address prefix 6
DAP[7]	0x63C	Device address prefix 7
DACNF	0x640	Device address match configuration
MODECNF0	0x650	Radio mode configuration register 0
POWER	0xFFC	Peripheral power control

23.14.1 SHORTS

Address offset: 0x200

Shortcut register

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																								H		G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	READY_START			Shortcut between READY event and START task See EVENTS_READY and TASKS_START
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
B	RW	END_DISABLE			Shortcut between END event and DISABLE task See EVENTS_END and TASKS_DISABLE

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																											H		G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Disabled	0																															
			Enabled	1																															
C	RW	DISABLED_TXEN																																	
				See *EVENTS_DISABLED* and *TASKS_TXEN*																																	
			Disabled	0																															
			Enabled	1																															
D	RW	DISABLED_RXEN																																	
				See *EVENTS_DISABLED* and *TASKS_RXEN*																																	
			Disabled	0																															
			Enabled	1																															
E	RW	ADDRESS_RSSISTART																																	
				See *EVENTS_ADDRESS* and *TASKS_RSSISTART*																																	
			Disabled	0																															
			Enabled	1																															
F	RW	END_START																																	
				See *EVENTS_END* and *TASKS_START*																																	
			Disabled	0																															
			Enabled	1																															
G	RW	ADDRESS_BCSTART																																	
				See *EVENTS_ADDRESS* and *TASKS_BCSTART*																																	
			Disabled	0																															
			Enabled	1																															
H	RW	DISABLED_RSSISTOP																																	
				See *EVENTS_DISABLED* and *TASKS_RSSISTOP*																																	
			Disabled	0																															
			Enabled	1																															

23.14.2 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																						L	K		I			H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	READY																																	
				See *EVENTS_READY*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	ADDRESS																																	
				See *EVENTS_ADDRESS*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	PAYLOAD																																	
				See *EVENTS_PAYLOAD*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	END																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																						L	K		I			H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																						L	K		I			H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Disabled	0																															
			Enabled	1																															
B	RW	ADDRESS																																	
				See *EVENTS_ADDRESS*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	PAYLOAD																																	
				See *EVENTS_PAYLOAD*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	END																																	
				See *EVENTS_END*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	DISABLED																																	
				See *EVENTS_DISABLED*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
F	RW	DEVMATCH																																	
				See *EVENTS_DEVMATCH*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
G	RW	DEVMISS																																	
				See *EVENTS_DEVMISS*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
H	RW	RSSIEND																																	
				See *EVENTS_RSSIEND*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
I	RW	BCMATCH																																	
				See *EVENTS_BCMATCH*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
K	RW	CRCOK																																	
				See *EVENTS_CRCOK*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
L	RW	CRCERROR																																	
				See *EVENTS_CRCERROR*																															
			Clear	1																															
			Disabled	0																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																						L	K		I			H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
			Enabled	1																								Read: Enabled							

23.14.4 CRCSTATUS

Address offset: 0x400

CRC status

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	R	CRCSTATUS																										CRC status of packet received							
			CRCError	0																								Packet received with CRC error							
			CRCOk	1																								Packet received with CRC ok							

23.14.5 RXMATCH

Address offset: 0x408

Received address

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	R	RXMATCH																										Received address							
																												Logical address of which previous packet was received							

23.14.6 RXCRC

Address offset: 0x40C

CRC field of previously received packet

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id												A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	R	RXCRC																										CRC field of previously received packet							
																												CRC field of previously received packet							

23.14.7 DAI

Address offset: 0x410

Device address match index

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	R	DAI																										Device address match index							
																												Index (n) of device address, see DAB[n] and DAP[n], that got an address match.							

23.14.8 PACKETPTR

Address offset: 0x504

Packet pointer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PACKETPTR																																	
				Packet pointer Packet address to be used for the next transmission or reception. When transmitting, the packet pointed to by this address will be transmitted and when receiving, the received packet will be written to this address. This address is a byte aligned ram address.																																		

23.14.9 FREQUENCY

Address offset: 0x508

Frequency

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																										B		A	A	A	A	A	A	A
Reset 0x00000002				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	FREQUENCY		[0..100]																															
				Radio channel frequency Frequency = 2400 + FREQUENCY (MHz).																																	
B	RW	MAP																																	
				Channel map selection.																																	
			Default	0																																	
				Channel map between 2400 MHZ .. 2500 MHz Frequency = 2400 + FREQUENCY (MHz)																																	
			Low	1																																	
				Channel map between 2360 MHZ .. 2460 MHz Frequency = 2360 + FREQUENCY (MHz)																																	

23.14.10 TXPOWER

Address offset: 0x50C

Output power

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	TXPOWER																																	
				RADIO output power. Output power in number of dBm, i.e. if the value -20 is specified the output power will be set to -20dBm.																																	
			Pos4dBm	0x04																																	
				+4 dBm																																	
			Pos3dBm	0x03																																	
				+3 dBm																																	
			0dBm	0x00																																	
				0 dBm																																	
			Neg4dBm	0xFC																																	
				-4 dBm																																	
			Neg8dBm	0xF8																																	
				-8 dBm																																	
			Neg12dBm	0xF4																																	
				-12 dBm																																	
			Neg16dBm	0xF0																																	
				-16 dBm																																	
			Neg20dBm	0xEC																																	
				-20 dBm																																	
			Neg30dBm	0xD8																																	
				-40 dBm																																	Deprecated
			Neg40dBm	0xD8																																	
				-40 dBm																																	

23.14.11 MODE

Address offset: 0x510

Data rate and modulation

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	MODE																																	
				Radio data rate and modulation setting. The radio supports Frequency-shift Keying (FSK) modulation.																																	
			Nrf_1Mbit	0																																	
				1 Mbit/s Nordic proprietary radio mode																																	
			Nrf_2Mbit	1																																	
				2 Mbit/s Nordic proprietary radio mode																																	
			Nrf_250Kbit	2																																	
				250 kbit/s Nordic proprietary radio mode																																	Deprecated
			Ble_1Mbit	3																																	
				1 Mbit/s Bluetooth Low Energy																																	
			Ble_2Mbit	4																																	
				2 Mbit/s Bluetooth Low Energy																																	

23.14.12 PCNF0

Address offset: 0x514

Packet configuration register 0

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											G				F	E	E	E	E								C					A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	LFLEN																																	
				Length on air of LENGTH field in number of bits.																																	
C	RW	S0LEN																																	
				Length on air of S0 field in number of bytes.																																	
E	RW	S1LEN																																	
				Length on air of S1 field in number of bits.																																	
F	RW	S1INCL																																	
				Include or exclude S1 field in RAM																																	
			Automatic	0																																	
				Include S1 field in RAM only if S1LEN > 0																																	
			Include	1																																	
				Always include S1 field in RAM independent of S1LEN																																	
G	RW	PLEN																																	
				Length of preamble on air. Decision point: TASKS_START task																																	
			8bit	0																																	
				8-bit preamble																																	
			16bit	1																																	
				16-bit preamble																																	

23.14.13 PCNF1

Address offset: 0x518

Packet configuration register 1

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id										E	D						C	C	C	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	MAXLEN		[0..255]																															
				Maximum length of packet payload. If the packet payload is larger than MAXLEN, the radio will truncate the payload to MAXLEN.																																	
B	RW	STATLEN		[0..255]																															
				Static length in number of bytes The static length parameter is added to the total length of the payload when sending and receiving packets, e.g. if the static length is set to N the radio will receive or send N bytes more than what is defined in the LENGTH field of the packet.																																	
C	RW	BALEN		[2..4]																															
				Base address length in number of bytes The address field is composed of the base address and the one byte long address prefix, e.g. set BALEN=2 to get a total address of 3 bytes.																																	
D	RW	ENDIAN																																	
				On air endianness of packet, this applies to the S0, LENGTH, S1 and the PAYLOAD fields.																																	
			Little	0																																	
				Least Significant bit on air first																																	
			Big	1																																	
				Most significant bit on air first																																	
E	RW	WHITEEN																																	
				Enable or disable packet whitening																																	
			Disabled	0																																	
				Disable																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id										E	D						C	C	C	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**								**Description**																							
			Enabled	1								Enable																							

23.14.14 BASE0

Address offset: 0x51C

Base address 0

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**								**Description**																							
A	RW	BASE0										Base address 0 Radio base address 0.																							

23.14.15 BASE1

Address offset: 0x520

Base address 1

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**								**Description**																							
A	RW	BASE1										Base address 1 Radio base address 1.																							

23.14.16 PREFIX0

Address offset: 0x524

Prefixes bytes for logical addresses 0-3

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				D	D	D	D	D	D	D	D	C	C	C	C	C	C	C	C	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**								**Description**																							
A	RW	AP0										Address prefix 0.																							
B	RW	AP1										Address prefix 1.																							
C	RW	AP2										Address prefix 2.																							
D	RW	AP3										Address prefix 3.																							

23.14.17 PREFIX1

Address offset: 0x528

Prefixes bytes for logical addresses 4-7

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				D	D	D	D	D	D	D	D	C	C	C	C	C	C	C	C	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**								**Description**																							
A	RW	AP4										Address prefix 4.																							
B	RW	AP5										Address prefix 5.																							
C	RW	AP6										Address prefix 6.																							
D	RW	AP7										Address prefix 7.																							

23.14.18 TXADDRESS

Address offset: 0x52C

Transmit address select

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		A	A	A
Reset 0x00000000					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																														
A	RW	TXADDRESS			Transmit address select Logical address to be used when transmitting a packet.																																

23.14.19 RXADDRESSES

Address offset: 0x530

Receive address select

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																													H	G	F	E	D	C	B	A
Reset 0x00000000					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																														
A	RW	ADDR0			Enable or disable reception on logical address 0.																																
			Disabled	0	Disable																																
			Enabled	1	Enable																																
B	RW	ADDR1			Enable or disable reception on logical address 1.																																
			Disabled	0	Disable																																
			Enabled	1	Enable																																
C	RW	ADDR2			Enable or disable reception on logical address 2.																																
			Disabled	0	Disable																																
			Enabled	1	Enable																																
D	RW	ADDR3			Enable or disable reception on logical address 3.																																
			Disabled	0	Disable																																
			Enabled	1	Enable																																
E	RW	ADDR4			Enable or disable reception on logical address 4.																																
			Disabled	0	Disable																																
			Enabled	1	Enable																																
F	RW	ADDR5			Enable or disable reception on logical address 5.																																
			Disabled	0	Disable																																
			Enabled	1	Enable																																
G	RW	ADDR6			Enable or disable reception on logical address 6.																																
			Disabled	0	Disable																																
			Enabled	1	Enable																																
H	RW	ADDR7			Enable or disable reception on logical address 7.																																
			Disabled	0	Disable																																
			Enabled	1	Enable																																

23.14.20 CRCCNF

Address offset: 0x534

CRC configuration

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												B							A	A
Reset 0x00000000					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																														
A	RW	LEN		[1..3]	CRC length in number of bytes.																																
			Disabled	0	CRC length is zero and CRC calculation is disabled																																
			One	1	CRC length is one byte and CRC calculation is enabled																																
			Two	2	CRC length is two bytes and CRC calculation is enabled																																

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																											B							A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
			Three	3																						CRC length is three bytes and CRC calculation is enabled									
B	RW	SKIPADDR																								Include or exclude packet address field out of CRC calculation.									
			Include	0																						CRC calculation includes address field									
			Skip	1																						CRC calculation does not include address field. The CRC calculation will start at the first byte after the address.									

23.14.21 CRCPOLY

Address offset: 0x538

CRC polynomial

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id												A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	RW	CRCPOLY																								CRC polynomial									
																										Each term in the CRC polynomial is mapped to a bit in this register which index corresponds to the term's exponent. The least significant term/bit is hard-wired internally to 1, and bit number 0 of the register content is ignored by the hardware. The following example is for an 8 bit CRC polynomial: x8 + x7 + x3 + x2 + 1 = 1 1000 1101 .									

23.14.22 CRCINIT

Address offset: 0x53C

CRC initial value

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id												A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	RW	CRCINIT																								CRC initial value									
																										Initial value for CRC calculation.									

23.14.23 TIFS

Address offset: 0x544

Inter Frame Spacing in us

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	RW	TIFS																								Inter Frame Spacing in us									
																										Inter frame space is the time interval between two consecutive packets. It is defined as the time, in micro seconds, from the end of the last bit of the previous packet to the start of the first bit of the subsequent packet.									

23.14.24 RSSISAMPLE

Address offset: 0x548

RSSI sample

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																													A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																									**Description**						
A	R	RSSISAMPLE		[0..127]																									RSSI sample						
																													RSSI sample result. The value of this register is read as a positive value while the actual received signal strength is a negative value. Actual received signal strength is therefore as follows: received signal strength = -A dBm						

23.14.25 STATE

Address offset: 0x550

Current radio state

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																									**Description**						
A	R	STATE																											Current radio state						
			Disabled	0																									RADIO is in the Disabled state						
			RxRu	1																									RADIO is in the RXRU state						
			RxIdle	2																									RADIO is in the RXIDLE state						
			Rx	3																									RADIO is in the RX state						
			RxDisable	4																									RADIO is in the RXDISABLED state						
			TxRu	9																									RADIO is in the TXRU state						
			TxIdle	10																									RADIO is in the TXIDLE state						
			Tx	11																									RADIO is in the TX state						
			TxDisable	12																									RADIO is in the TXDISABLED state						

23.14.26 DATAWHITEIV

Address offset: 0x554

Data whitening initial value

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																													A	A	A	A	A	A	A
Reset 0x00000040				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																									**Description**						
A	RW	DATAWHITEIV																											Data whitening initial value. Bit 6 is hard-wired to '1', writing '0' to it has no effect, and it will always be read back and used by the device as '1'.						
																													Bit 0 corresponds to Position 6 of the LSFR, Bit 1 to Position 5, etc.						

23.14.27 BCC

Address offset: 0x560

Bit counter compare

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																									**Description**						
A	RW	BCC																											Bit counter compare						
																													Bit counter compare register						

23.14.28 DAB[0]

Address offset: 0x600

Device address base segment 0

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	DAB			Device address base segment 0

23.14.29 DAB[1]

Address offset: 0x604

Device address base segment 1

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	DAB			Device address base segment 1

23.14.30 DAB[2]

Address offset: 0x608

Device address base segment 2

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	DAB			Device address base segment 2

23.14.31 DAB[3]

Address offset: 0x60C

Device address base segment 3

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	DAB			Device address base segment 3

23.14.32 DAB[4]

Address offset: 0x610

Device address base segment 4

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	DAB			Device address base segment 4

23.14.33 DAB[5]

Address offset: 0x614

Device address base segment 5

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	DAB			Device address base segment 5

23.14.34 DAB[6]

Address offset: 0x618

Device address base segment 6

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	DAB			Device address base segment 6

23.14.35 DAB[7]

Address offset: 0x61C

Device address base segment 7

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	DAB			Device address base segment 7

23.14.36 DAP[0]

Address offset: 0x620

Device address prefix 0

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	DAP			Device address prefix 0

23.14.37 DAP[1]

Address offset: 0x624

Device address prefix 1

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	DAP			Device address prefix 1

23.14.38 DAP[2]

Address offset: 0x628

Device address prefix 2

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	DAP			Device address prefix 2

23.14.39 DAP[3]

Address offset: 0x62C

Device address prefix 3

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	DAP			Device address prefix 3

23.14.40 DAP[4]

Address offset: 0x630

Device address prefix 4

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	DAP			Device address prefix 4

23.14.41 DAP[5]

Address offset: 0x634

Device address prefix 5

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	DAP			Device address prefix 5

23.14.42 DAP[6]

Address offset: 0x638

Device address prefix 6

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	DAP			Device address prefix 6

23.14.43 DAP[7]

Address offset: 0x63C

Device address prefix 7

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	DAP			Device address prefix 7

23.14.44 DACNF

Address offset: 0x640

Device address match configuration

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																					P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																															
A	RW	ENA0			Enable or disable device address matching using device address 0																																	
			Disabled	0	Disabled																															
			Enabled	1	Enabled																															
B	RW	ENA1			Enable or disable device address matching using device address 1																																	
			Disabled	0	Disabled																															
			Enabled	1	Enabled																															
C	RW	ENA2			Enable or disable device address matching using device address 2																																	
			Disabled	0	Disabled																															
			Enabled	1	Enabled																															
D	RW	ENA3			Enable or disable device address matching using device address 3																																	
			Disabled	0	Disabled																															
			Enabled	1	Enabled																															
E	RW	ENA4			Enable or disable device address matching using device address 4																																	
			Disabled	0	Disabled																															
			Enabled	1	Enabled																															
F	RW	ENA5			Enable or disable device address matching using device address 5																																	
			Disabled	0	Disabled																															
			Enabled	1	Enabled																															
G	RW	ENA6			Enable or disable device address matching using device address 6																																	
			Disabled	0	Disabled																															
			Enabled	1	Enabled																															
H	RW	ENA7			Enable or disable device address matching using device address 7																																	
			Disabled	0	Disabled																															
			Enabled	1	Enabled																															
I	RW	TXADD0			TxAdd for device address 0																																	
J	RW	TXADD1			TxAdd for device address 1																																	
K	RW	TXADD2			TxAdd for device address 2																																	
L	RW	TXADD3			TxAdd for device address 3																																	
M	RW	TXADD4			TxAdd for device address 4																																	
N	RW	TXADD5			TxAdd for device address 5																																	
O	RW	TXADD6			TxAdd for device address 6																																	
P	RW	TXADD7			TxAdd for device address 7																																	

23.14.45 MODECNF0

Address offset: 0x650

Radio mode configuration register 0

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																											C	C								A
Reset 0x00000200					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																															
A	RW	RU			Radio ramp-up time																																	
			Default	0	Default ramp-up time (tRXEN), compatible with firmware written for nRF51																															
			Fast	1	Fast ramp-up (tRXEN,FAST), see electrical specification for more information																															

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			C C A
Reset 0x00000200			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
C	RW	DTX			Default TX value Specifies what the RADIO will transmit when it is not started, i.e. between: RADIO.EVENTS_READY and RADIO.TASKS_START RADIO.EVENTS_END and RADIO.TASKS_START RADIO.EVENTS_END and RADIO.EVENTS_DISABLED
			B1	0	Transmit '1'
			B0	1	Transmit '0'
			Center	2	Transmit center frequency When tuning the crystal for centre frequency, the RADIO must be set in DTX = Center mode to be able to achieve the expected accuracy.

23.14.46 POWER

Address offset: 0xFFC

Peripheral power control

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A
Reset 0x00000001			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	POWER			Peripheral power control. The peripheral and its registers will be reset to its initial state by switching the peripheral off and then back on again.
			Disabled	0	Peripheral is powered off
			Enabled	1	Peripheral is powered on

23.15 Electrical specification

23.15.1 General Radio Characteristics

Symbol	Description	Min.	Typ.	Max.	Units
f_{OP}	Operating frequencies	2360		2500	MHz
$f_{PLL,PROG,RES}$	PLL programming resolution		2		kHz
$f_{PLL,CH,SP}$	PLL channel spacing		1		MHz
$f_{DELTA,1M}$	Frequency deviation @ 1 Msps		±170		kHz
$f_{DELTA,BLE,1M}$	Frequency deviation @ BLE 1Msps		±250		kHz
$f_{DELTA,2M}$	Frequency deviation @ 2 Msps		±320		kHz
$f_{DELTA,BLE,2M}$	Frequency deviation @ BLE 2 Msps		±500		kHz
fsk_{SPS}	On-the-air data rate	1		2	Msps

23.15.2 Radio current consumption (Transmitter)

Symbol	Description	Min.	Typ.	Max.	Units
$I_{TX,PLUS4dBM,DCDC}$	TX only run current (DCDC, 3V) P_{RF} =+4 dBm		7.5		mA
$I_{TX,PLUS4dBM}$	TX only run current P_{RF} = +4 dBm		16.6		mA
$I_{TX,0dBM,DCDC}$	TX only run current (DCDC, 3V)P_{RF} = 0dBm		5.3		mA
$I_{TX,0dBM}$	TX only run current P_{RF} = 0dBm		11.6		mA
$I_{TX,MINUS4dBM,DCDC}$	TX only run current DCDC, 3V P_{RF} = -4dBm		4.2		mA
$I_{TX,MINUS4dBM}$	TX only run current P_{RF} = -4 dBm		9.3		mA
$I_{TX,MINUS8dBM,DCDC}$	TX only run current DCDC, 3V P_{RF} = -8 dBm		3.8		mA

Symbol	Description	Min.	Typ.	Max.	Units
$I_{TX,MINUS8dBM}$	TX only run current P_{RF} = -8 dBm		8.4		mA
$I_{TX,MINUS12dBM,DCDC}$	TX only run current DCDC, 3V P_{RF} = -12 dBm		3.5		mA
$I_{TX,MINUS12dBM}$	TX only run current P_{RF} = -12 dBm		7.7		mA
$I_{TX,MINUS16dBM,DCDC}$	TX only run current DCDC, 3V P_{RF} = -16 dBm		3.3		mA
$I_{TX,MINUS16dBM}$	TX only run current P_{RF} = -16 dBm		7.3		mA
$I_{TX,MINUS20dBM,DCDC}$	TX only run current DCDC, 3V P_{RF} = -20 dBm		3.2		mA
$I_{TX,MINUS20dBM}$	TX only run current P_{RF} = -20 dBm		7.0		mA
$I_{TX,MINUS40dBM,DCDC}$	TX only run current DCDC, 3V P_{RF} = -40 dBm		2.7		mA
$I_{TX,MINUS40dBM}$	TX only run current P_{RF} = -40 dBm		5.9		mA
$I_{START,TX,DCDC}$	TX start-up current DCDC, 3V, P_{RF} = 4 dBm		4.0		mA
$I_{START,TX}$	TX start-up current, P_{RF} = 4 dBm		8.8		mA

23.15.3 Radio current consumption (Receiver)

Symbol	Description	Min.	Typ.	Max.	Units
$I_{RX,1M,DCDC}$	RX only run current (DCDC, 3V) 1Msps / 1Msps BLE		5.4		mA
$I_{RX,1M}$	RX only run current 1Msps / 1Msps BLE		11.7		mA
$I_{RX,2M,DCDC}$	RX only run current (DCDC, 3V) 2Msps / 2Msps BLE		5.8		mA
$I_{RX,2M}$	RX only run current 2Msps / 2Msps BLE		12.9		mA
$I_{START,RX,DCDC}$	RX start-up current (DCDC 3V)		3.5		mA
$I_{START,RX,LDO}$	RX start-up current (LDO 3V)		7.5		mA

23.15.4 Transmitter specification

Symbol	Description	Min.	Typ.	Max.	Units
P_{RF}	Maximum output power		4	6	dBm
P_{RFC}	RF power control range		24		dB
P_{RFCR}	RF power accuracy			±4	dB
$P_{RF1,1}$	1st Adjacent Channel Transmit Power 1 MHz (1 Msps Nordic proprietary mode)		-25		dBc
$P_{RF2,1}$	2nd Adjacent Channel Transmit Power 2 MHz (1 Msps Nordic proprietary mode)		-50		dBc
$P_{RF1,2}$	1st Adjacent Channel Transmit Power 2 MHz (2 Msps Nordic proprietary mode)		-25		dBc
$P_{RF2,2}$	2nd Adjacent Channel Transmit Power 4 MHz (2 Msps Nordic proprietary mode)		-50		dBc
$P_{RF1,2,BLE}$	1st Adjacent Channel Transmit Power 2 MHz (2 Msps BLE mode)		-20		dBc
$P_{RF2,2,BLE}$	2nd Adjacent Channel Transmit Power 4 MHz (2 Msps BLE mode)		-50		dBc

23.15.5 Receiver operation

Symbol	Description	Min.	Typ.	Max.	Units
$P_{RX,MAX}$	Maximum received signal strength at < 0.1% BER		0		dBm
$P_{SENS,IT,1M}$	Sensitivity, 1Msps nRF mode[16]		-93		dBm
$P_{SENS,IT,SP,1M,BLE}$	Sensitivity, 1Msps BLE ideal transmitter, <=37 bytes BER=1E-3[17]		-96		dBm
$P_{SENS,IT,LP,1M,BLE}$	Sensitivity, 1Msps BLE ideal transmitter >=128 bytes BER=1E-4[18]		-95		dBm
$P_{SENS,IT,2M}$	Sensitivity, 2Msps nRF mode[19]		-89		dBm

16 Typical sensitivity applies when ADDR0 is used for receiver address correlation. When ADDR[1...7] are used for receiver address correlation, the typical sensitivity for this mode is degraded by 3dB.

17 As defined in the Bluetooth Core Specification v4.0 Volume 6: Core System Package (Low Energy Controller Volume)

18 Equivalent BER limit < 10E-04

19 Typical sensitivity applies when ADDR0 is used for receiver address correlation. When ADDR[1...7] are used for receiver address correlation, the typical sensitivity for this mode is degraded by 3dB.

Symbol	Description	Min.	Typ.	Max.	Units
$P_{SENS,IT,SP,2M,BLE}$	Sensitivity, 2Msps BLE ideal transmitter, Packet length <=37bytes		-93		dBm
$P_{SENS,DT,SP,2M,BLE}$	Sensitivity, 2Msps BLE dirty transmitter, Packet length <=37bytes		-93		dBm
$P_{SENS,IT,LP,2M,BLE}$	Sensitivity, 2Msps BLE ideal transmitter >= 128bytes		-92		dBm
$P_{SENS,DT,LP,2M,BLE}$	Sensitivity, 2Msps BLE dirty transmitter, Packet length >= 128bytes		-92		dBm

23.15.6 RX selectivity

RX selectivity with equal modulation on interfering signal[20]

Symbol	Description	Min.	Typ.	Max.	Units
$C/I_{1M,co\text{-}channel}$	1Msps mode, Co-Channel interference		9		dB
$C/I_{1M,-1MHz}$	1 Msps mode, Adjacent (-1 MHz) interference		-2		dB
$C/I_{1M,+1MHz}$	1 Msps mode, Adjacent (+1 MHz) interference		-10		dB
$C/I_{1M,-2MHz}$	1 Msps mode, Adjacent (-2 MHz) interference		-19		dB
$C/I_{1M,+2MHz}$	1 Msps mode, Adjacent (+2 MHz) interference		-42		dB
$C/I_{1M,-3MHz}$	1 Msps mode, Adjacent (-3 MHz) interference		-38		dB
$C/I_{1M,+3MHz}$	1 Msps mode, Adjacent (+3 MHz) interference		-48		dB
$C/I_{1M,\pm6MHz}$	1 Msps mode, Adjacent (≥6 MHz) interference		-50		dB
$C/I_{1MBLE,co\text{-}channel}$	1 Msps BLE mode, Co-Channel interference		6		dB
$C/I_{1MBLE,-1MHz}$	1 Msps BLE mode, Adjacent (-1 MHz) interference		-2		dB
$C/I_{1MBLE,+1MHz}$	1 Msps BLE mode, Adjacent (+1 MHz) interference		-9		dB
$C/I_{1MBLE,-2MHz}$	1 Msps BLE mode, Adjacent (-2 MHz) interference		-22		dB
$C/I_{1MBLE,+2MHz}$	1 Msps BLE mode, Adjacent (+2 MHz) interference		-46		dB
$C/I_{1MBLE,>3MHz}$	1 Msps BLE mode, Adjacent (≥3 MHz) interference		-50		dB
$C/I_{1MBLE,image}$	Image frequency Interference		-22		dB
$C/I_{1MBLE,image,1MHz}$	Adjacent (1 MHz) interference to in-band image frequency		-35		dB
$C/I_{2M,co\text{-}channel}$	2Msps mode, Co-Channel interference		10		dB
$C/I_{2M,-2MHz}$	2 Msps mode, Adjacent (-2 MHz) interference		6		dB
$C/I_{2M,+2MHz}$	2 Msps mode, Adjacent (+2 MHz) interference		-14		dB
$C/I_{2M,-4MHz}$	2 Msps mode, Adjacent (-4 MHz) interference		-20		dB
$C/I_{2M,+4MHz}$	2 Msps mode, Adjacent (+4 MHz) interference		-44		dB
$C/I_{2M,-6MHz}$	2 Msps mode, Adjacent (-6 MHz) interference		-42		dB
$C/I_{2M,+6MHz}$	2 Msps mode, Adjacent (+6 MHz) interference		-47		dB
$C/I_{2M,\geq12MHz}$	2 Msps mode, Adjacent (≥12 MHz) interference		-52		dB
$C/I_{2MBLE,co\text{-}channel}$	2 Msps BLE mode, Co-Channel interference		7		dB
$C/I_{2MBLE,\pm2MHz}$	2 Msps BLE mode, Adjacent (±2 MHz) interference		0		dB
$C/I_{2MBLE,\pm4MHz}$	2 Msps BLE mode, Adjacent (±4 MHz) interference		-47		dB
$C/I_{2MBLE,\geq6MHz}$	2 Msps BLE mode, Adjacent (≥6 MHz) interference		-49		dB
$C/I_{2MBLE,image}$	Image frequency Interference		-21		dB
$C/I_{2MBLE,image,\ 2MHz}$	Adjacent (2 MHz) interference to in-band image frequency		-36		dB

23.15.7 RX intermodulation

RX intermodulation[21]

Symbol	Description	Min.	Typ.	Max.	Units
$P_{IMD,1M}$	IMD performance, 1 Msps (3 MHz, 4 MHz, and 5 MHz offset)		-33		dBm
$P_{IMD,1M,BLE}$	IMD performance, BLE 1 Msps (3 MHz, 4 MHz, and 5 MHz offset)		-30		dBm
$P_{IMD,2M}$	IMD performance, 2 Msps (6 MHz, 8 MHz, and 10 MHz offset)		-33		dBm

[20] Wanted signal level at PIN = -67 dBm. One interferer is used, having equal modulation as the wanted signal. The input power of the interferer where the sensitivity equals BER = 0.1% is presented

[21] Wanted signal level at PIN = -64 dBm. Two interferers with equal input power are used. The interferer closest in frequency is not modulated, the other interferer is modulated equal with the wanted signal. The input power of the interferers where the sensitivity equals BER = 0.1% is presented.

Symbol	Description	Min.	Typ.	Max.	Units
$P_{IMD,2M,BLE}$	IMD performance, BLE 2 Msps (6 MHz, 8 MHz, and 10 MHz offset)		-32		dBm

23.15.8 Radio timing

Symbol	Description	Min.	Typ.	Max.	Units
t_{TXEN}	Time between TXEN task and READY event after channel FREQUENCY configured		140		us
$t_{TXEN,FAST}$	Time between TXEN task and READY event after channel FREQUENCY configured (Fast Mode)		40		us
$t_{TXDISABLE}$	Time between DISABLE task and DISABLED event when the radio was in TX and mode is set to 1Msps		6		us
$t_{TXDISABLE,2M}$	Time between DISABLE task and DISABLED event when the radio was in TX and mode is set to 2Msps		4		us
t_{RXEN}	Time between the RXEN task and READY event after channel FREQUENCY configured in default mode		140		us
$t_{RXEN,FAST}$	Time between the RXEN task and READY event after channel FREQUENCY configured in fast mode		40		us
t_{SWITCH}	The minimum time taken to switch from RX to TX or TX to RX (channel FREQUENCY unchanged)		20		us
$t_{RXDISABLE}$	Time between DISABLE task and DISABLED event when the radio was in RX		0		us
$t_{TXCHAIN}$	TX chain delay		0.6		us
$t_{RXCHAIN}$	RX chain delay		9.4		us
$t_{RXCHAIN,2M}$	RX chain delay in 2Msps mode		5		us

23.15.9 Received Signal Strength Indicator (RSSI) specifications

Symbol	Description	Min.	Typ.	Max.	Units
$RSSI_{ACC}$	RSSI Accuracy Valid range -90 to -20 dBm		±2		dB
$RSSI_{RESOLUTION}$	RSSI resolution		1		dB
$RSSI_{PERIOD}$	Sample period		0.25		us

23.15.10 Jitter

Symbol	Description	Min.	Typ.	Max.	Units
$t_{DISABLEDJITTER}$	Jitter on DISABLED event relative to END event when shortcut between END and DISABLE is enabled.		0.25		us
$t_{READYJITTER}$	Jitter on READY event relative to TXEN and RXEN task.		0.25		us

23.15.11 Delay when disabling the RADIO

Symbol	Description	Min.	Typ.	Max.	Units
$t_{TXDISABLE,1M}$	Disable delay from TX. Delay between DISABLE and DISABLED for MODE = Nrf_1Mbit and MODE = Ble_1Mbit		6		us
$t_{RXDISABLE,1M}$	Disable delay from RX. Delay between DISABLE and DISABLED for MODE = Nrf_1Mbit and MODE = Ble_1Mbit		0		us

24 TIMER — Timer/counter

The TIMER can operate in two modes: timer and counter.

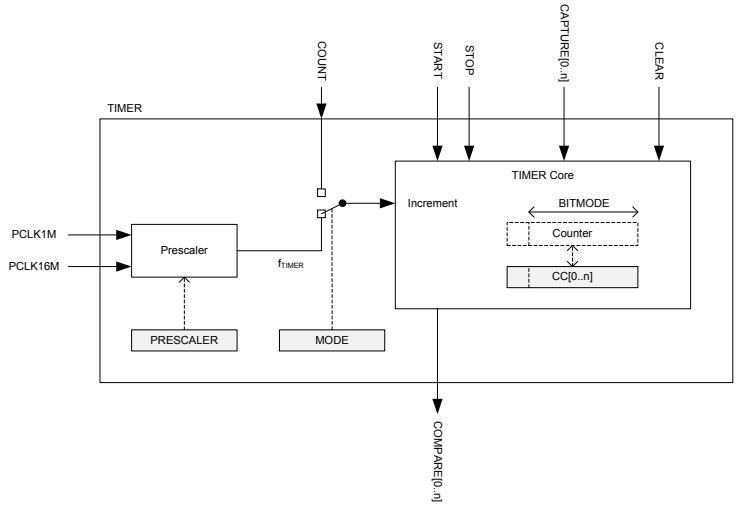


Figure 42: Block schematic for timer/counter

The timer/counter runs on the high-frequency clock source (HFCLK) and includes a four-bit (1/2X) prescaler that can divide the timer input clock from the HFCLK controller. Clock source selection between PCLK16M and PCLK1M is automatic according to TIMER base frequency set by the prescaler. The TIMER base frequency is always given as 16 MHz divided by the prescaler value.

The PPI system allows a TIMER event to trigger a task of any other system peripheral of the device. The PPI system also enables the TIMER task/event features to generate periodic output and PWM signals to any GPIO. The number of input/outputs used at the same time is limited by the number of GPIOTE channels.

The TIMER can operate in two modes, Timer mode and Counter mode. In both modes, the TIMER is started by triggering the START task, and stopped by triggering the STOP task. After the timer is stopped the timer can resume timing/counting by triggering the START task again. When timing/counting is resumed, the timer will continue from the value it had prior to being stopped.

In Timer mode, the TIMER's internal Counter register is incremented by one for every tick of the timer frequency f_{TIMER} as illustrated in *Figure 42: Block schematic for timer/counter* on page 234. The timer frequency is derived from PCLK16M as shown below, using the values specified in the PRESCALER register:

$$f_{TIMER} = 16\ MHz\ /\ (2^{PRESCALER})$$

When f_{TIMER} <= 1 MHz the TIMER will use PCLK1M instead of PCLK16M for reduced power consumption.

In counter mode, the TIMER's internal Counter register is incremented by one each time the COUNT task is triggered, that is, the timer frequency and the prescaler are not utilized in counter mode. Similarly, the COUNT task has no effect in Timer mode.

The TIMER's maximum value is configured by changing the bit-width of the timer in the *BITMODE* on page 239 register.

PRESCALER on page 239 and the *BITMODE* on page 239 must only be updated when the timer is stopped. If these registers are updated while the TIMER is started then this may result in unpredictable behavior.

When the timer is incremented beyond its maximum value the Counter register will overflow and the TIMER will automatically start over from zero.

The Counter register can be cleared, that is, its internal value set to zero explicitly, by triggering the CLEAR task.

The TIMER implements multiple capture/compare registers.

Independent of prescaler setting the accuracy of the TIMER is equivalent to one tick of the timer frequency f_{TIMER} as illustrated in *Figure 42: Block schematic for timer/counter* on page 234.

24.1 Capture

The TIMER implements one capture task for every available capture/compare register.

Every time the CAPTURE[n] task is triggered, the Counter value is copied to the CC[n] register.

24.2 Compare

The TIMER implements one COMPARE event for every available capture/compare register.

A COMPARE event is generated when the Counter is incremented and then becomes equal to the value specified in one of the capture compare registers. When the Counter value becomes equal to the value specified in a capture compare register CC[n], the corresponding compare event COMPARE[n] is generated.

BITMODE on page 239 specifies how many bits of the Counter register and the capture/compare register that are used when the comparison is performed. Other bits will be ignored.

24.3 Task delays

After the TIMER is started, the CLEAR task, COUNT task and the STOP task will guarantee to take effect within one clock cycle of the PCLK16M.

24.4 Task priority

If the START task and the STOP task are triggered at the same time, that is, within the same period of PCLK16M, the STOP task will be prioritized.

24.5 Registers

Table 43: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40008000	TIMER	TIMER0	Timer 0	This timer instance has 4 CC registers (CC[0..3])
0x40009000	TIMER	TIMER1	Timer 1	This timer instance has 4 CC registers (CC[0..3])
0x4000A000	TIMER	TIMER2	Timer 2	This timer instance has 4 CC registers (CC[0..3])
0x4001A000	TIMER	TIMER3	Timer 3	This timer instance has 6 CC registers (CC[0..5])
0x4001B000	TIMER	TIMER4	Timer 4	This timer instance has 6 CC registers (CC[0..5])

Table 44: Register Overview

Register	Offset	Description
TASKS_START	0x000	Start Timer

Register	Offset	Description	
TASKS_STOP	0x004	Stop Timer	
TASKS_COUNT	0x008	Increment Timer (Counter mode only)	
TASKS_CLEAR	0x00C	Clear time	
TASKS_SHUTDOWN	0x010	Shut down timer	Deprecated
TASKS_CAPTURE[0]	0x040	Capture Timer value to CC[0] register	
TASKS_CAPTURE[1]	0x044	Capture Timer value to CC[1] register	
TASKS_CAPTURE[2]	0x048	Capture Timer value to CC[2] register	
TASKS_CAPTURE[3]	0x04C	Capture Timer value to CC[3] register	
TASKS_CAPTURE[4]	0x050	Capture Timer value to CC[4] register	
TASKS_CAPTURE[5]	0x054	Capture Timer value to CC[5] register	
EVENTS_COMPARE[0]	0x140	Compare event on CC[0] match	
EVENTS_COMPARE[1]	0x144	Compare event on CC[1] match	
EVENTS_COMPARE[2]	0x148	Compare event on CC[2] match	
EVENTS_COMPARE[3]	0x14C	Compare event on CC[3] match	
EVENTS_COMPARE[4]	0x150	Compare event on CC[4] match	
EVENTS_COMPARE[5]	0x154	Compare event on CC[5] match	
SHORTS	0x200	Shortcut register	
INTENSET	0x304	Enable interrupt	
INTENCLR	0x308	Disable interrupt	
MODE	0x504	Timer mode selection	
BITMODE	0x508	Configure the number of bits used by the TIMER	
PRESCALER	0x510	Timer prescaler register	
CC[0]	0x540	Capture/Compare register 0	
CC[1]	0x544	Capture/Compare register 1	
CC[2]	0x548	Capture/Compare register 2	
CC[3]	0x54C	Capture/Compare register 3	
CC[4]	0x550	Capture/Compare register 4	
CC[5]	0x554	Capture/Compare register 5	

24.5.1 SHORTS

Address offset: 0x200

Shortcut register

Bit number																															
Id																				L	K	J	I	H	G			F	E	D	C	B	A
Reset 0x00000000																				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	COMPARE0_CLEAR			Shortcut between COMPARE[0] event and CLEAR task See *EVENTS_COMPARE[0]* and *TASKS_CLEAR*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
B	RW	COMPARE1_CLEAR			Shortcut between COMPARE[1] event and CLEAR task See *EVENTS_COMPARE[1]* and *TASKS_CLEAR*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
C	RW	COMPARE2_CLEAR			Shortcut between COMPARE[2] event and CLEAR task See *EVENTS_COMPARE[2]* and *TASKS_CLEAR*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
D	RW	COMPARE3_CLEAR			Shortcut between COMPARE[3] event and CLEAR task See *EVENTS_COMPARE[3]* and *TASKS_CLEAR*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
E	RW	COMPARE4_CLEAR			Shortcut between COMPARE[4] event and CLEAR task See *EVENTS_COMPARE[4]* and *TASKS_CLEAR*

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
F	RW	COMPARE5_CLEAR			Shortcut between COMPARE[5] event and CLEAR task See *EVENTS_COMPARE[5]* and *TASKS_CLEAR*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
G	RW	COMPARE0_STOP			Shortcut between COMPARE[0] event and STOP task See *EVENTS_COMPARE[0]* and *TASKS_STOP*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
H	RW	COMPARE1_STOP			Shortcut between COMPARE[1] event and STOP task See *EVENTS_COMPARE[1]* and *TASKS_STOP*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
I	RW	COMPARE2_STOP			Shortcut between COMPARE[2] event and STOP task See *EVENTS_COMPARE[2]* and *TASKS_STOP*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
J	RW	COMPARE3_STOP			Shortcut between COMPARE[3] event and STOP task See *EVENTS_COMPARE[3]* and *TASKS_STOP*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
K	RW	COMPARE4_STOP			Shortcut between COMPARE[4] event and STOP task See *EVENTS_COMPARE[4]* and *TASKS_STOP*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
L	RW	COMPARE5_STOP			Shortcut between COMPARE[5] event and STOP task See *EVENTS_COMPARE[5]* and *TASKS_STOP*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

24.5.2 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	COMPARE0			Write '1' to Enable interrupt for COMPARE[0] event See *EVENTS_COMPARE[0]*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	COMPARE1			Write '1' to Enable interrupt for COMPARE[1] event See *EVENTS_COMPARE[1]*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	COMPARE2			Write '1' to Enable interrupt for COMPARE[2] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
					See *EVENTS_COMPARE[2]*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	COMPARE3			Write '1' to Enable interrupt for COMPARE[3] event See *EVENTS_COMPARE[3]*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	COMPARE4			Write '1' to Enable interrupt for COMPARE[4] event See *EVENTS_COMPARE[4]*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	COMPARE5			Write '1' to Enable interrupt for COMPARE[5] event See *EVENTS_COMPARE[5]*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

24.5.3 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	COMPARE0			Write '1' to Disable interrupt for COMPARE[0] event See *EVENTS_COMPARE[0]*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	COMPARE1			Write '1' to Disable interrupt for COMPARE[1] event See *EVENTS_COMPARE[1]*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	COMPARE2			Write '1' to Disable interrupt for COMPARE[2] event See *EVENTS_COMPARE[2]*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	COMPARE3			Write '1' to Disable interrupt for COMPARE[3] event See *EVENTS_COMPARE[3]*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	COMPARE4			Write '1' to Disable interrupt for COMPARE[4] event See *EVENTS_COMPARE[4]*
			Clear	1	Disable

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			F E D C B A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	COMPARE5			Write '1' to Disable interrupt for COMPARE[5] event See *EVENTS_COMPARE[5]*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

24.5.4 MODE

Address offset: 0x504

Timer mode selection

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	MODE			Timer mode
			Timer	0	Select Timer mode
			Counter	1	Select Counter mode
			LowPowerCounter	2	Select Low Power Counter mode

Deprecated

24.5.5 BITMODE

Address offset: 0x508

Configure the number of bits used by the TIMER

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	BITMODE			Timer bit width
			16Bit	0	16 bit timer bit width
			08Bit	1	8 bit timer bit width
			24Bit	2	24 bit timer bit width
			32Bit	3	32 bit timer bit width

24.5.6 PRESCALER

Address offset: 0x510

Timer prescaler register

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A
Reset 0x00000004			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	PRESCALER		[0..9]	Prescaler value

24.5.7 CC[0]

Address offset: 0x540

Capture/Compare register 0

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	CC			Capture/Compare value Only the number of bits indicated by BITMODE will be used by the TIMER.

24.5.8 CC[1]

Address offset: 0x544

Capture/Compare register 1

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	CC			Capture/Compare value Only the number of bits indicated by BITMODE will be used by the TIMER.

24.5.9 CC[2]

Address offset: 0x548

Capture/Compare register 2

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	CC			Capture/Compare value Only the number of bits indicated by BITMODE will be used by the TIMER.

24.5.10 CC[3]

Address offset: 0x54C

Capture/Compare register 3

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	CC			Capture/Compare value Only the number of bits indicated by BITMODE will be used by the TIMER.

24.5.11 CC[4]

Address offset: 0x550

Capture/Compare register 4

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	CC			Capture/Compare value

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
					Only the number of bits indicated by BITMODE will be used by the TIMER.

24.5.12 CC[5]

Address offset: 0x554

Capture/Compare register 5

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	CC			Capture/Compare value Only the number of bits indicated by BITMODE will be used by the TIMER.

24.6 Electrical specification

24.6.1 Timers Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
I_{TIMER_1M}	Run current with 1 MHz clock input (PCLK1M)	3	5	8	µA
I_{TIMER_16M}	Run current with 16 MHz clock input (PCLK16M)	50	70	120	µA
$t_{TIMER,START}$	Time from START task is given until timer starts counting		0.25		µs

25 RTC — Real-time counter

The Real-time counter (RTC) module provides a generic, low power timer on the low-frequency clock source (LFCLK).

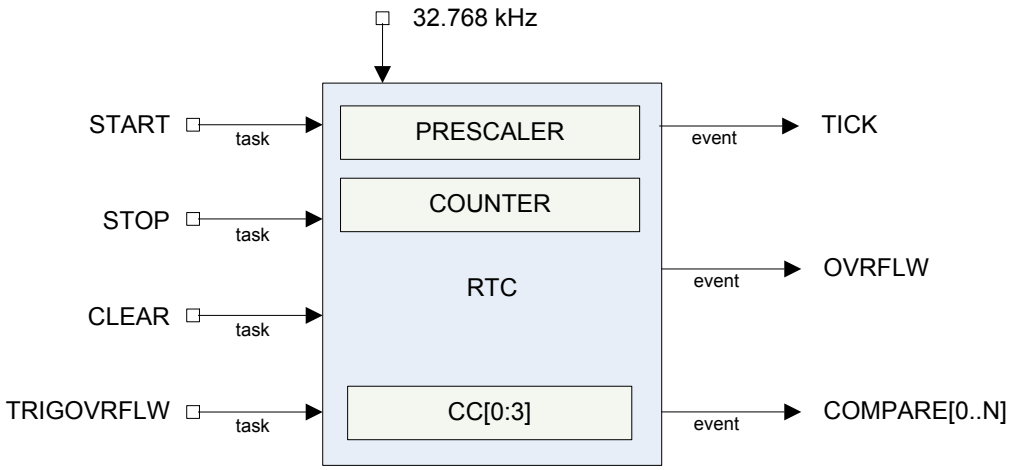


Figure 43: RTC block schematic

The RTC module features a 24-bit COUNTER, a 12-bit (1/X) prescaler, capture/compare registers, and a tick event generator for low power, tickless RTOS implementation.

25.1 Clock source

The RTC will run off the LFCLK.

The COUNTER resolution will therefore be 30.517 µs. Depending on the source, the RTC is able to run while the HFCLK is OFF and PCLK16M is not available.

The software has to explicitely start LFCLK before using the RTC.

See *CLOCK — Clock control* on page 101 for more information about clock sources.

25.2 Resolution versus overflow and the PRESCALER

Counter increment frequency:

$$f_{RTC}\ [kHz] = 32.768\ /\ (PRESCALER + 1\ )$$

The PRESCALER register is read/write when the RTC is stopped. The PRESCALER register is read-only once the RTC is STARTed. Writing to the PRESCALER register when the RTC is started has no effect.

The PRESCALER is restarted on START, CLEAR and TRIGOVRFLW, that is, the prescaler value is latched to an internal register (<<PRESC>>) on these tasks.

Examples:

1. Desired COUNTER frequency 100 Hz (10 ms counter period)

   PRESCALER = round(32.768 kHz / 100 Hz) - 1 = 327

   f_{RTC} = 99.9 Hz

10009.576 µs counter period

2. Desired COUNTER frequency 8 Hz (125 ms counter period)

   PRESCALER = round(32.768 kHz / 8 Hz) – 1 = 4095

   f_{RTC} = 8 Hz

   125 ms counter period

Table 45: RTC resolution versus overflow

Prescaler	Counter resolution	Overflow
0	30.517 µs	512 seconds
2^{8}-1	7812.5 µs	131072 seconds
2^{12}-1	125 ms	582.542 hours

25.3 COUNTER register

The COUNTER increments on LFCLK when the internal PRESCALER register (<<PRESC>>) is 0x00. <<PRESC>> is reloaded from the PRESCALER register. If enabled, the TICK event occurs on each increment of the COUNTER. The TICK event is disabled by default.

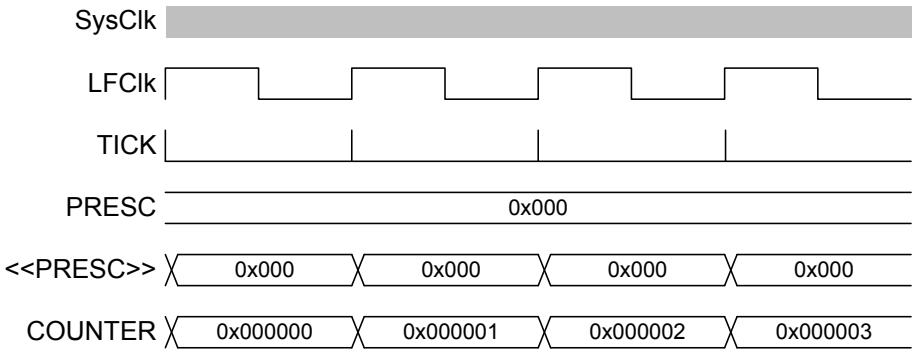


Figure 44: Timing diagram - COUNTER_PRESCALER_0

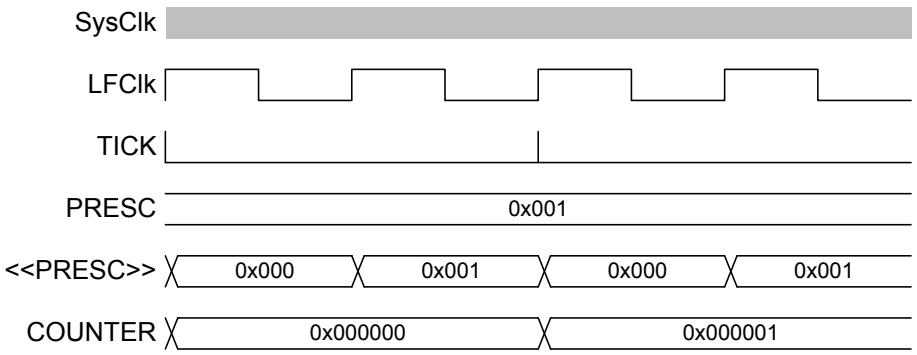


Figure 45: Timing diagram - COUNTER_PRESCALER_1

25.4 Overflow features

The TRIGOVRFLW task sets the COUNTER value to 0xFFFFF0 to allow SW test of the overflow condition.

OVRFLW occurs when COUNTER overflows from 0xFFFFFF to 0.

Important: The OVRFLW event is disabled by default.

25.5 TICK event

The TICK event enables low power "tick-less" RTOS implementation as it optionally provides a regular interrupt source for a RTOS without the need to use the ARM® SysTick feature.

Using the RTC TICK event rather than the SysTick allows the CPU to be powered down while still keeping RTOS scheduling active.

Important: The TICK event is disabled by default.

25.6 Event control feature

To optimize RTC power consumption, events in the RTC can be individually disabled to prevent PCLK16M and HFCLK being requested when those events are triggered. This is managed using the EVTEN register.

For example, if the TICK event is not required for an application, this event should be disabled as it is frequently occurring and may increase power consumption if HFCLK otherwise could be powered down for long durations.

This means that the RTC implements a slightly different task and event system compared to the standard system described in *Peripheral interface* on page 68. The RTC task and event system is illustrated in *Figure 46: Tasks, events and interrupts in the RTC* on page 244.

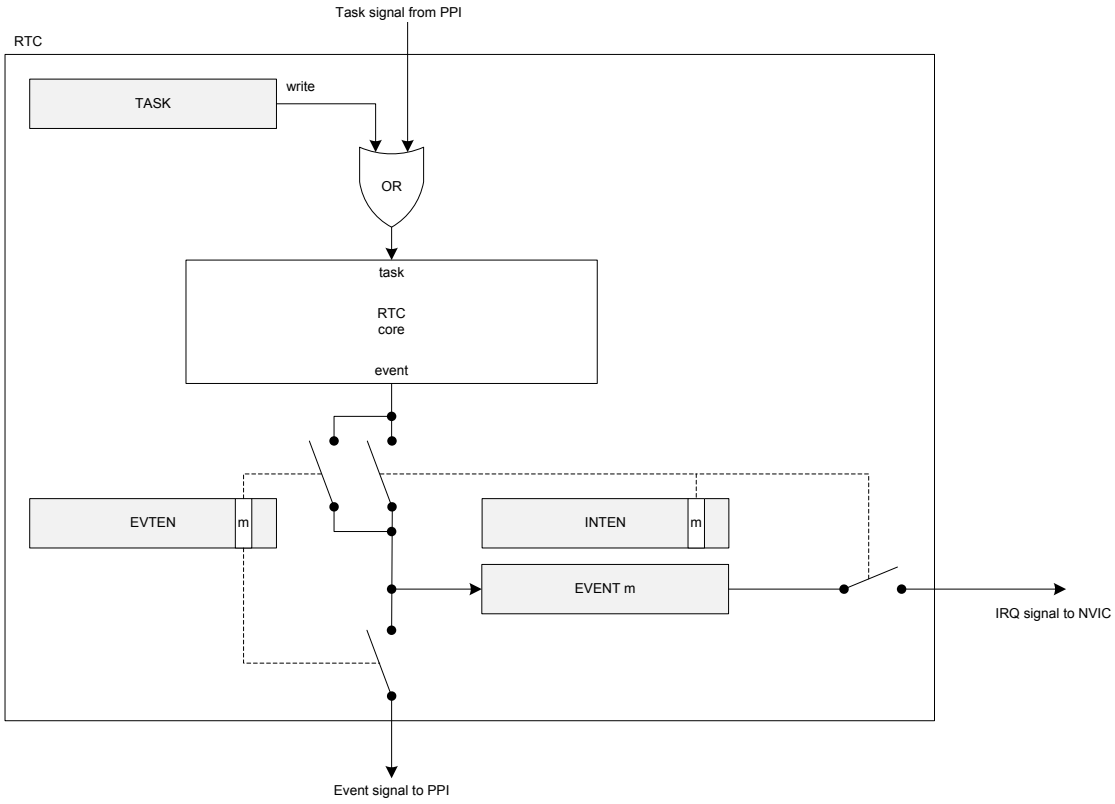


Figure 46: Tasks, events and interrupts in the RTC

25.7 Compare feature

There are a number of Compare registers.

For more information, see *Registers* on page 248.

When setting a compare register, the following behavior of the RTC compare event should be noted:

- If a CC register value is 0 when a CLEAR task is set, this will not trigger a COMPARE event.

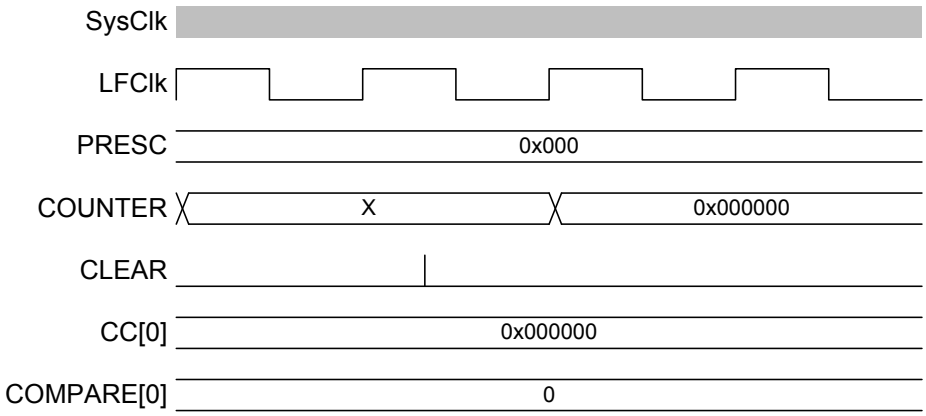


Figure 47: Timing diagram - COMPARE_CLEAR

- If a CC register is N and the COUNTER value is N when the START task is set, this will not trigger a COMPARE event.

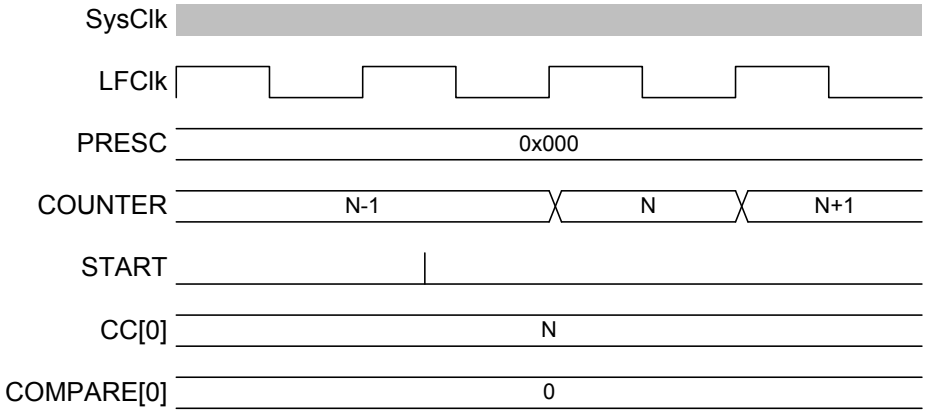


Figure 48: Timing diagram - COMPARE_START

- COMPARE occurs when a CC register is N and the COUNTER value transitions from N-1 to N.

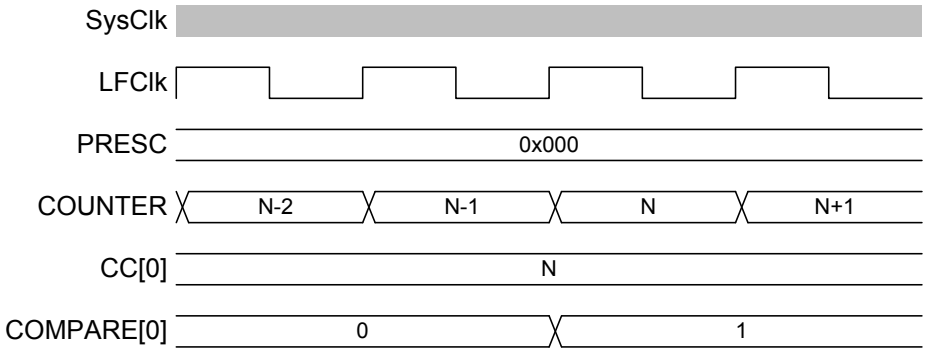


Figure 49: Timing diagram - COMPARE

- If the COUNTER is N, writing N+2 to a CC register is guaranteed to trigger a COMPARE event at N+2.

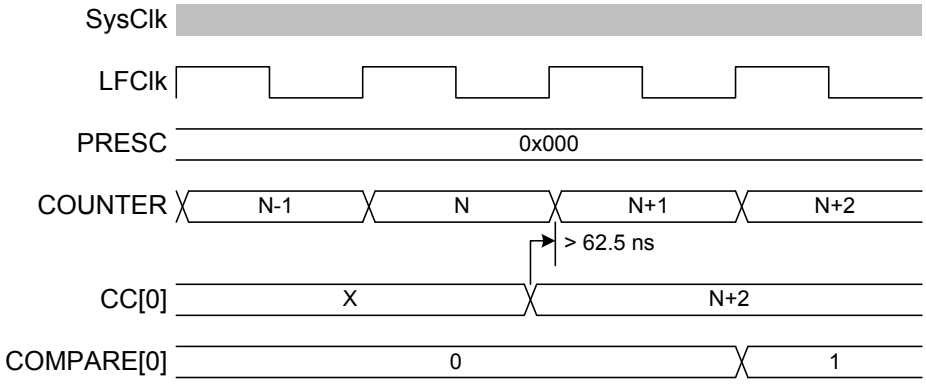


Figure 50: Timing diagram - COMPARE_N+2

- If the COUNTER is N, writing N or N+1 to a CC register may not trigger a COMPARE event.

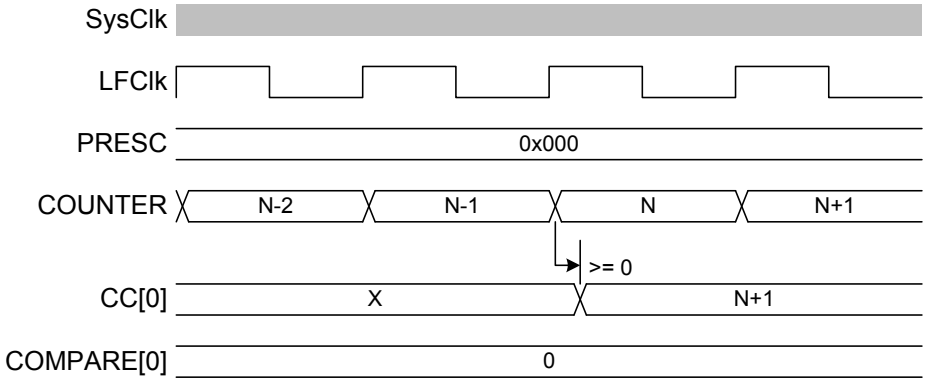


Figure 51: Timing diagram - COMPARE_N+1

- If the COUNTER is N and the current CC register value is N+1 or N+2 when a new CC value is written, a match may trigger on the previous CC value before the new value takes effect. If the current CC value greater than N+2 when the new value is written, there will be no event due to the old value.

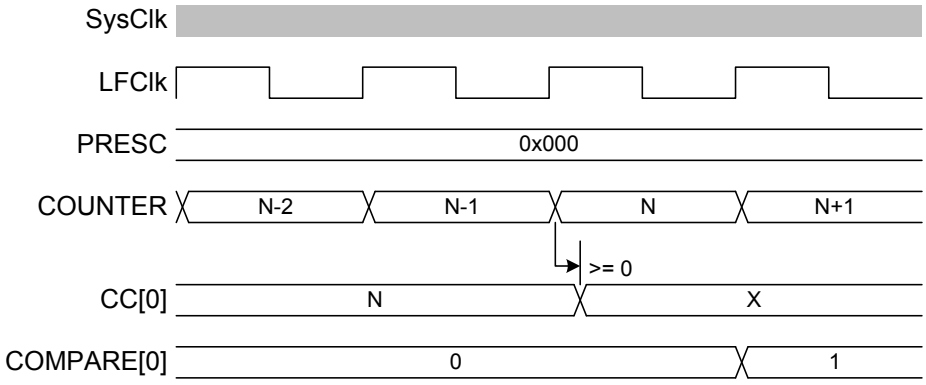


Figure 52: Timing diagram - COMPARE_N-1

25.8 TASK and EVENT jitter/delay

Jitter or delay in the RTC is due to the peripheral clock being a low frequency clock (LFCLK) which is not synchronous to the faster PCLK16M.

Registers in the peripheral interface, part of the PCLK16M domain, have a set of mirrored registers in the LFCLK domain. For example, the COUNTER value accessible from the CPU is in the PCLK16M domain and is latched on read from an internal register called COUNTER in the LFCLK domain. COUNTER is the register which is actually modified each time the RTC ticks. These registers must be synchronised between clock domains (PCLK16M and LFCLK).

The following is a summary of the jitter introduced on tasks and events. Figures illustrating jitter follow.

Table 46: RTC jitter magnitudes on tasks

Task	Delay
CLEAR, STOP, START, TRIGOVRFLOW	+15 to 46 µs

Table 47: RTC jitter magnitudes on events

Operation/Function	Jitter
START to COUNTER increment	+/- 15 µs
COMPARE to COMPARE [22]	+/- 62.5 ns

1. CLEAR and STOP (and TRIGOVRFLW; not shown) will be delayed as long as it takes for the peripheral to clock a falling edge and rising of the LFCLK. This is between 15.2585 µs and 45.7755 µs – rounded to 15 µs and 46 µs for the remainder of the section.

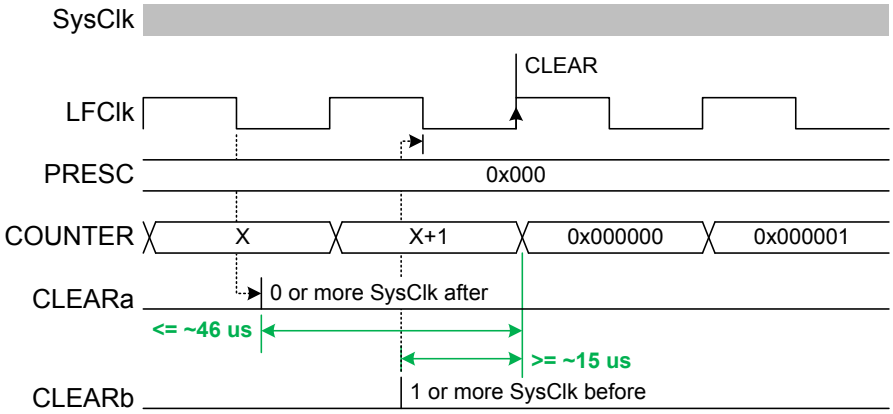


Figure 53: Timing diagram - DELAY_CLEAR

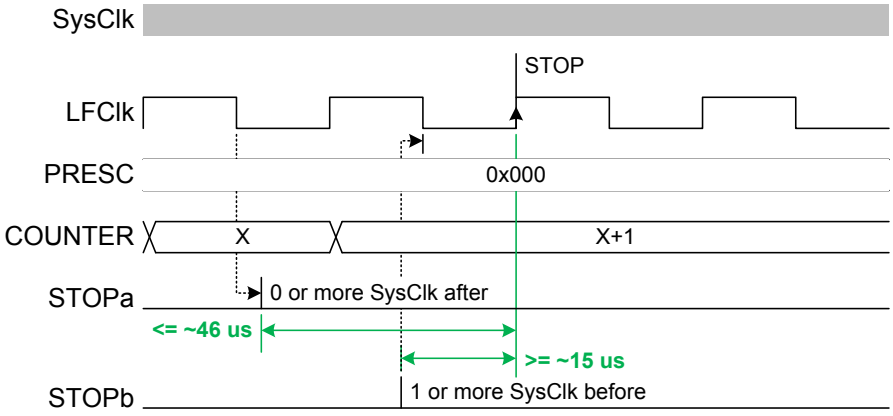


Figure 54: Timing diagram - DELAY_STOP

2. The START task will start the RTC. Assuming that the LFCLK was previously running and stable, the first increment of COUNTER (and instance of TICK event) will be typically after 30.5 µs +/-15 µs. In some cases, in particular if the RTC is STARTed before the LFCLK is running, that timing can be up to ~250 µs. The software should therefore wait for the first TICK if it has to make sure the RTC is running. Sending a TRIGOVRFLW task sets the COUNTER to a value close to overflow. However, since the update of COUNTER relies on a stable LFCLK, sending this task while LFCLK is not running will start LFCLK, but the update will then be delayed by the same amount of time of up to ~250 us. The figures show the smallest and largest delays to on the START task which appears as a +/-15 µs jitter on the first COUNTER increment.

[22] Assumes RTC runs continuously between these events.

Note: 32.768 kHz clock jitter is additional to the numbers provided above.

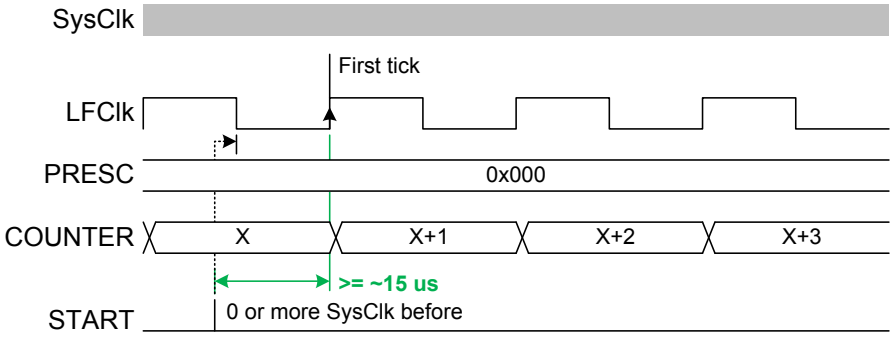


Figure 55: Timing diagram - JITTER_START-

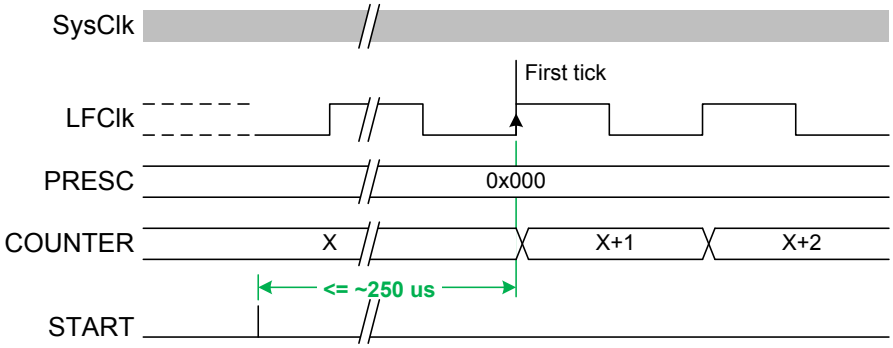


Figure 56: Timing diagram - JITTER_START+

25.9 Reading the COUNTER register

To read the COUNTER register, the internal <<COUNTER>> value is sampled.

To ensure that the <<COUNTER>> is safely sampled (considering an LFCLK transition may occur during a read), the CPU and core memory bus are halted for three cycles by lowering the core PREADY signal. The Read takes the CPU 2 cycles in addition resulting in the COUNTER register read taking a fixed five PCLK16M clock cycles.

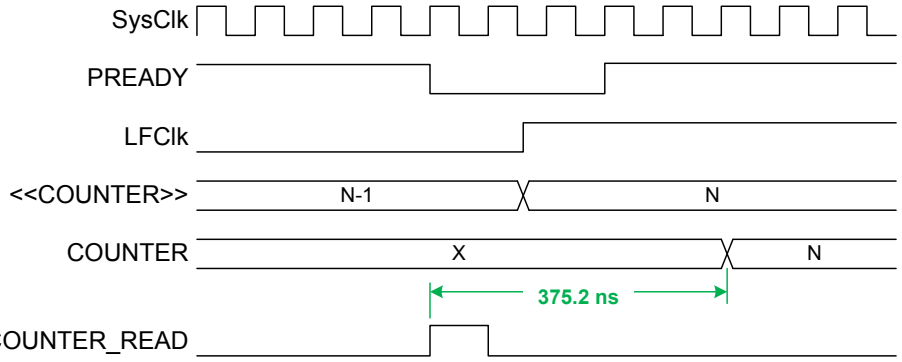


Figure 57: Timing diagram - COUNTER_READ

25.10 Registers

Table 48: Instances

Base address	Peripheral	Instance	Description	Configuration
0x4000B000	RTC	RTC0	Real-time counter 0	CC[0..2] implemented, CC[3] not implemented
0x40011000	RTC	RTC1	Real-time counter 1	CC[0..3] implemented

Base address	Peripheral	Instance	Description	Configuration
0x40024000	RTC	RTC2	Real-time counter 2	CC[0..3] implemented

Table 49: Register Overview

Register	Offset	Description
TASKS_START	0x000	Start RTC COUNTER
TASKS_STOP	0x004	Stop RTC COUNTER
TASKS_CLEAR	0x008	Clear RTC COUNTER
TASKS_TRIGOVRFLW	0x00C	Set COUNTER to 0xFFFFF0
EVENTS_TICK	0x100	Event on COUNTER increment
EVENTS_OVRFLW	0x104	Event on COUNTER overflow
EVENTS_COMPARE[0]	0x140	Compare event on CC[0] match
EVENTS_COMPARE[1]	0x144	Compare event on CC[1] match
EVENTS_COMPARE[2]	0x148	Compare event on CC[2] match
EVENTS_COMPARE[3]	0x14C	Compare event on CC[3] match
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
EVTEN	0x340	Enable or disable event routing
EVTENSET	0x344	Enable event routing
EVTENCLR	0x348	Disable event routing
COUNTER	0x504	Current COUNTER value
PRESCALER	0x508	12 bit prescaler for COUNTER frequency (32768/(PRESCALER+1)).Must be written when RTC is stopped
CC[0]	0x540	Compare register 0
CC[1]	0x544	Compare register 1
CC[2]	0x548	Compare register 2
CC[3]	0x54C	Compare register 3

25.10.1 INTENSET

Address offset: 0x304

Enable interrupt

Bit number																															
Id																		F	E	D	C										B	A
Reset 0x00000000																		**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TICK			Write '1' to Enable interrupt for TICK event See *EVENTS_TICK*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	OVRFLW			Write '1' to Enable interrupt for OVRFLW event See *EVENTS_OVRFLW*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	COMPARE0			Write '1' to Enable interrupt for COMPARE[0] event See *EVENTS_COMPARE[0]*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	COMPARE1			Write '1' to Enable interrupt for COMPARE[1] event See *EVENTS_COMPARE[1]*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																F	E	D	C																B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
E	RW	COMPARE2																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
F	RW	COMPARE3																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															

25.10.2 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																F	E	D	C																B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	TICK																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	OVRFLW																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	COMPARE0																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	COMPARE1																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	COMPARE2																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
F	RW	COMPARE3																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															

25.10.3 EVTEN

Address offset: 0x340

Enable or disable event routing

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																F	E	D	C																B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	TICK																																	
			Disabled	0																															
			Enabled	1																															
B	RW	OVRFLW																																	
			Disabled	0																															
			Enabled	1																															
C	RW	COMPARE0																																	
			Disabled	0																															
			Enabled	1																															
D	RW	COMPARE1																																	
			Disabled	0																															
			Enabled	1																															
E	RW	COMPARE2																																	
			Disabled	0																															
			Enabled	1																															
F	RW	COMPARE3																																	
			Disabled	0																															
			Enabled	1																															

25.10.4 EVTENSET

Address offset: 0x344

Enable event routing

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																F	E	D	C																B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	TICK																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	OVRFLW																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	COMPARE0																																	
			Set	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																F	E	D	C																B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Disabled	0																															
			Enabled	1																															
D	RW	COMPARE1																																	
				See *EVENTS_COMPARE[1]*																															
			Set	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	COMPARE2																																	
				See *EVENTS_COMPARE[2]*																															
			Set	1																															
			Disabled	0																															
			Enabled	1																															
F	RW	COMPARE3																																	
				See *EVENTS_COMPARE[3]*																															
			Set	1																															
			Disabled	0																															
			Enabled	1																															

25.10.5 EVTENCLR

Address offset: 0x348

Disable event routing

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																F	E	D	C																B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	TICK																																	
				See *EVENTS_TICK*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	OVRFLW																																	
				See *EVENTS_OVRFLW*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	COMPARE0																																	
				See *EVENTS_COMPARE[0]*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	COMPARE1																																	
				See *EVENTS_COMPARE[1]*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	COMPARE2																																	
				See *EVENTS_COMPARE[2]*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
F	RW	COMPARE3			Write '1' to Disable event routing for COMPARE[3] event See *EVENTS_COMPARE[3]*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

25.10.6 COUNTER

Address offset: 0x504

Current COUNTER value

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	R	COUNTER			Counter value

25.10.7 PRESCALER

Address offset: 0x508

12 bit prescaler for COUNTER frequency (32768/(PRESCALER+1)).Must be written when RTC is stopped

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	PRESCALER			Prescaler value

25.10.8 CC[0]

Address offset: 0x540

Compare register 0

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	COMPARE			Compare value

25.10.9 CC[1]

Address offset: 0x544

Compare register 1

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	COMPARE			Compare value

25.10.10 CC[2]

Address offset: 0x548

Compare register 2

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id									A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	COMPARE			Compare value

25.10.11 CC[3]

Address offset: 0x54C

Compare register 3

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id									A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	COMPARE			Compare value

25.11 Electrical specification

25.11.1 RTC Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
I_{RTC}	Run current Real Time Counter (LFCLK source)		0.1		µA

26 RNG — Random number generator

The Random number generator (RNG) generates true non-deterministic random numbers based on internal thermal noise that are suitable for cryptographic purposes. The RNG does not require a seed value.

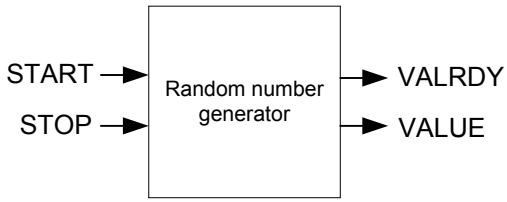


Figure 58: Random number generator

The RNG is started by triggering the START task and stopped by triggering the STOP task. When started, new random numbers are generated continuously and written to the VALUE register when ready. A VALRDY event is generated for every new random number that is written to the VALUE register. This means that after a VALRDY event is generated the CPU has the time until the next VALRDY event to read out the random number from the VALUE register before it is overwritten by a new random number.

26.1 Bias correction

A bias correction algorithm is employed on the internal bit stream to remove any bias toward '1' or '0'. The bits are then queued into an eight-bit register for parallel readout from the VALUE register.

It is possible to enable bias correction in the CONFIG register. This will result in slower value generation, but will ensure a statistically uniform distribution of the random values.

26.2 Speed

The time needed to generate one random byte of data is unpredictable, and may vary from one byte to the next. This is especially true when bias correction is enabled.

26.3 Registers

Table 50: Instances

Base address	Peripheral	Instance	Description	Configuration
0x4000D000	RNG	RNG	Random number generator	

Table 51: Register Overview

Register	Offset	Description
TASKS_START	0x000	Task starting the random number generator
TASKS_STOP	0x004	Task stopping the random number generator
EVENTS_VALRDY	0x100	Event being generated for every new random number written to the VALUE register
SHORTS	0x200	Shortcut register
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
CONFIG	0x504	Configuration register
VALUE	0x508	Output random number

26.3.1 SHORTS

Address offset: 0x200

Shortcut register

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
A	RW	VALRDY_STOP			Shortcut between VALRDY event and STOP task
					See *EVENTS_VALRDY* and *TASKS_STOP*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

26.3.2 INTENSET

Address offset: 0x304

Enable interrupt

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
A	RW	VALRDY			Write '1' to Enable interrupt for VALRDY event
					See *EVENTS_VALRDY*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

26.3.3 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
A	RW	VALRDY			Write '1' to Disable interrupt for VALRDY event
					See *EVENTS_VALRDY*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

26.3.4 CONFIG

Address offset: 0x504

Configuration register

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
A	RW	DERCEN			Bias correction
			Disabled	0	Disabled
			Enabled	1	Enabled

26.3.5 VALUE

Address offset: 0x508

Output random number

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	R	VALUE		[0..255]	Generated random number

26.4 Electrical specification

26.4.1 RNG Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
I_{RNG}	Run current, CPU sleeping.		500		µA
$t_{RNG,START}$	Time from setting the START task to generation begins. This is a one-time delay on START signal and does not apply between samples.		128		µs
$t_{RNG,RAW}$	Run time per byte without bias correction. Uniform distribution of 0 and 1 is not guaranteed.		30		µs
$t_{RNG,BC}$	Run time per byte with bias correction. Uniform distribution of 0 and 1 is guaranteed. Time to generate a byte cannot be guaranteed.		120		µs

27 TEMP — Temperature sensor

The temperature sensor measures die temperature over the temperature range of the device. Linearity compensation can be implemented if required by the application.

Listed here are the main features for TEMP:

- Temperature range is greater than or equal to operating temperature of the device
- Resolution is 0.25 degrees

TEMP is started by triggering the START task.

When the temperature measurement is completed, a DATARDY event will be generated and the result of the measurement can be read from the TEMP register.

To achieve the measurement accuracy stated in the electrical specification, the crystal oscillator must be selected as the HFCLK source, see *CLOCK — Clock control* on page 101 for more information.

When the temperature measurement is completed, TEMP analog electronics power down to save power.

TEMP only supports one-shot operation, meaning that every TEMP measurement has to be explicitly started using the START task.

27.1 Registers

Table 52: Instances

Base address	Peripheral	Instance	Description	Configuration
0x4000C000	TEMP	TEMP	Temperature sensor	

Table 53: Register Overview

Register	Offset	Description
TASKS_START	0x000	Start temperature measurement
TASKS_STOP	0x004	Stop temperature measurement
EVENTS_DATARDY	0x100	Temperature measurement complete, data ready
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
TEMP	0x508	Temperature in °C (0.25° steps)
A0	0x520	Slope of 1st piece wise linear function
A1	0x524	Slope of 2nd piece wise linear function
A2	0x528	Slope of 3rd piece wise linear function
A3	0x52C	Slope of 4th piece wise linear function
A4	0x530	Slope of 5th piece wise linear function
A5	0x534	Slope of 6th piece wise linear function
B0	0x540	y-intercept of 1st piece wise linear function
B1	0x544	y-intercept of 2nd piece wise linear function
B2	0x548	y-intercept of 3rd piece wise linear function
B3	0x54C	y-intercept of 4th piece wise linear function
B4	0x550	y-intercept of 5th piece wise linear function
B5	0x554	y-intercept of 6th piece wise linear function
T0	0x560	End point of 1st piece wise linear function
T1	0x564	End point of 2nd piece wise linear function
T2	0x568	End point of 3rd piece wise linear function
T3	0x56C	End point of 4th piece wise linear function
T4	0x570	End point of 5th piece wise linear function

27.1.1 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															**Description**
A	RW	DATARDY																																	Write '1' to Enable interrupt for DATARDY event See *EVENTS_DATARDY*
			Set	1																															Enable
			Disabled	0																															Read: Disabled
			Enabled	1																															Read: Enabled

27.1.2 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															**Description**
A	RW	DATARDY																																	Write '1' to Disable interrupt for DATARDY event See *EVENTS_DATARDY*
			Clear	1																															Disable
			Disabled	0																															Read: Disabled
			Enabled	1																															Read: Enabled

27.1.3 TEMP

Address offset: 0x508

Temperature in °C (0.25° steps)

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															**Description**
A	R	TEMP																																	Temperature in °C (0.25° steps) Result of temperature measurement. Die temperature in °C, 2's complement format, 0.25 °C steps Decision point: DATARDY

27.1.4 A0

Address offset: 0x520

Slope of 1st piece wise linear function

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																								A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000320				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															**Description**
A	RW	A0																																	Slope of 1st piece wise linear function

27.1.5 A1

Address offset: 0x524

Slope of 2nd piece wise linear function

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																								A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000343				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**1**	**0**	**0**	**0**	**0**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																				**Description**											
A	RW	A1																						Slope of 2nd piece wise linear function											

27.1.6 A2

Address offset: 0x528

Slope of 3rd piece wise linear function

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																								A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x0000035D				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**1**	**0**	**1**	**1**	**1**	**0**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																				**Description**											
A	RW	A2																						Slope of 3rd piece wise linear function											

27.1.7 A3

Address offset: 0x52C

Slope of 4th piece wise linear function

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																								A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000400				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																				**Description**											
A	RW	A3																						Slope of 4th piece wise linear function											

27.1.8 A4

Address offset: 0x530

Slope of 5th piece wise linear function

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																								A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x0000047F				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																				**Description**											
A	RW	A4																						Slope of 5th piece wise linear function											

27.1.9 A5

Address offset: 0x534

Slope of 6th piece wise linear function

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																								A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x0000037B				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**1**	**1**	**1**	**1**	**0**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																				**Description**											
A	RW	A5																						Slope of 6th piece wise linear function											

27.1.10 B0

Address offset: 0x540

y-intercept of 1st piece wise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00003FCC	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**0**	**1**	**1**	**0**	**0**	

Id	RW	Field	Value Id	Value	Description
A	RW	B0			y-intercept of 1st piece wise linear function

27.1.11 B1

Address offset: 0x544

y-intercept of 2nd piece wise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00003F98	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**0**	**1**	**1**	**0**	**0**		

Id	RW	Field	Value Id	Value	Description
A	RW	B1			y-intercept of 2nd piece wise linear function

27.1.12 B2

Address offset: 0x548

y-intercept of 3rd piece wise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00003F98	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**0**	**1**	**1**	**0**	**0**		

Id	RW	Field	Value Id	Value	Description
A	RW	B2			y-intercept of 3rd piece wise linear function

27.1.13 B3

Address offset: 0x54C

y-intercept of 4th piece wise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000012	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**1**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	B3			y-intercept of 4th piece wise linear function

27.1.14 B4

Address offset: 0x550

y-intercept of 5th piece wise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x0000006A	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**1**	**0**	**1**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	B4			y-intercept of 5th piece wise linear function

27.1.15 B5

Address offset: 0x554

y-intercept of 6th piece wise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00003DD0	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**0**	**1**	**1**	**1**	**0**	**1**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	B5			y-intercept of 6th piece wise linear function

27.1.16 T0

Address offset: 0x560

End point of 1st piece wise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x000000E2	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**0**	**0**	**0**	**1**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	T0			End point of 1st piece wise linear function

27.1.17 T1

Address offset: 0x564

End point of 2nd piece wise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	T1			End point of 2nd piece wise linear function

27.1.18 T2

Address offset: 0x568

End point of 3rd piece wise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x00000014	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	T2			End point of 3rd piece wise linear function

27.1.19 T3

Address offset: 0x56C

End point of 4th piece wise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x00000019	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**0**	**1**

Id	RW	Field	Value Id	Value	Description
A	RW	T3			End point of 4th piece wise linear function

27.1.20 T4

Address offset: 0x570

End point of 5th piece wise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x00000050	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	T4			End point of 5th piece wise linear function

27.2 Electrical specification

27.2.1 Temperature Sensor Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
t_{TEMP}	Time required for temperature measurement		36		µs
$T_{TEMP,RANGE}$	Temperature sensor range	-40		85	°C
$T_{TEMP,ACC}$	Temperature sensor accuracy	-5		5	°C
$T_{TEMP,RES}$	Temperature sensor resolution		0.25		°C
$T_{TEMP,STB}$	Sample to sample stability at constant device temperature		+/-0.25		°C
$T_{TEMP,OFFST}$	Sample offset at 25°C	-2.5		2.5	°C

28 ECB — AES electronic codebook mode encryption

The AES electronic codebook mode encryption (ECB) can be used for a range of cryptographic functions like hash generation, digital signatures, and keystream generation for data encryption/decryption. The ECB encryption block supports 128 bit AES encryption (encryption only, not decryption).

AES ECB operates with EasyDMA access to system Data RAM for in-place operations on cleartext and ciphertext during encryption. ECB uses the same AES core as the CCM and AAR blocks and is an asynchronous operation which may not complete if the AES core is busy.

AES ECB features:

- 128 bit AES encryption
- Supports standard AES ECB block encryption
- Memory pointer support
- DMA data transfer

AES ECB performs a 128 bit AES block encrypt. At the STARTECB task, data and key is loaded into the algorithm by EasyDMA. When output data has been written back to memory, the ENDECB event is triggered.

AES ECB can be stopped by triggering the STOPECB task.

28.1 Shared resources

The ECB, CCM, and AAR share the same AES module. The ECB will always have lowest priority and if there is a sharing conflict during encryption, the ECB operation will be aborted and an ERRORECB event will be generated.

28.2 EasyDMA

The ECB implements an EasyDMA mechanism for reading and writing to the Data RAM. This DMA cannot access the program memory or any other parts of the memory area except RAM.

If the ECBDATAPTR is not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See *Memory* on page 23 for more information about the different memory regions.

The EasyDMA will have finished accessing the Data RAM when the ENDECB or ERRORECB is generated.

28.3 ECB data structure

Input to the block encrypt and output from the block encrypt are stored in the same data structure. ECBDATAPTR should point to this data structure before STARTECB is initiated.

Table 54: ECB data structure overview

Property	Address offset	Description
KEY	0	16 byte AES key
CLEARTEXT	16	16 byte AES cleartext input block
CIPHERTEXT	32	16 byte AES ciphertext output block

28.4 Registers

Table 55: Instances

Base address	Peripheral	Instance	Description	Configuration
0x4000E000	ECB	ECB	AES Electronic Code Book (ECB) mode block encryption	

Table 56: Register Overview

Register	Offset	Description
TASKS_STARTECB	0x000	Start ECB block encrypt
TASKS_STOPECB	0x004	Abort a possible executing ECB operation
EVENTS_ENDECB	0x100	ECB block encrypt complete
EVENTS_ERRORECB	0x104	ECB block encrypt aborted because of a STOPECB task or due to an error
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ECBDATAPTR	0x504	ECB block encrypt memory pointers

28.4.1 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	ENDECB			Write '1' to Enable interrupt for ENDECB event See *EVENTS_ENDECB*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	ERRORECB			Write '1' to Enable interrupt for ERRORECB event See *EVENTS_ERRORECB*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

28.4.2 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	ENDECB			Write '1' to Disable interrupt for ENDECB event See *EVENTS_ENDECB*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	ERRORECB			Write '1' to Disable interrupt for ERRORECB event See *EVENTS_ERRORECB*
			Clear	1	Disable
			Disabled	0	Read: Disabled

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Enabled	1																															
				Read: Enabled																															

28.4.3 ECBDATAPTR

Address offset: 0x504

ECB block encrypt memory pointers

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	ECBDATAPTR																																	
				Pointer to the ECB data structure (see Table 1 ECB data structure overview)																																	

28.5 Electrical specification

28.5.1 ECB Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
t_{ECB}	Run time per 16 byte block in all modes		6		µs

29 CCM — AES CCM mode encryption

Cipher block chaining - message authentication code (CCM) mode is an authenticated encryption algorithm designed to provide both authentication and confidentiality during data transfer. CCM combines counter mode encryption and CBC-MAC authentication. The CCM terminology "Message authentication code (MAC)" is called the "Message integrity check (MIC)" in '*Bluetooth* terminology and also in this document.

The CCM block generates an encrypted keystream that is applied to input data using the XOR operation and generates the 4 byte MIC field in one operation. The CCM and radio can be configured to work synchronously. The CCM will encrypt in time for transmission and decrypt after receiving bytes into memory from the Radio. All operations can complete within the packet RX or TX time. CCM on this device is implemented according to *Bluetooth* requirements and the algorithm as defined in IETF *RFC3610*, and depends on the AES-128 block cipher. A description of the CCM algorithm can also be found in *NIST Special Publication 800-38C*. The *Bluetooth* specification describes the configuration of counter mode blocks and encryption blocks to implement compliant encryption for BLE.

The CCM block uses EasyDMA to load key, counter mode blocks (including the nonce required), and to read/write plain text and cipher text.

The AES CCM supports three operations: key-stream generation, packet encryption, and packet decryption. All these operations are done in compliance with the *Bluetooth* specification.[23] A new key-stream must be generated before a new packet encryption or packet decryption operation can be started.

A key-stream is generated by triggering the KSGEN task. An ENDKSGEN event will be generated when the new key-stream has been generated. The key-stream will be stored in the AES CCM's temporary memory area, specified by the SCRATCHPTR, where it will be used in subsequent encryption and decryption operations.

Encryption is started by triggering the CRYPT task with the MODE register set to ENCRYPTION. Similarly, decryption is started by triggering the same task with MODE set to DECRYPTION. An ENDCRYPT event will be generated when packet encryption is completed as well as when packet decryption is completed, see *Figure 59: Key-stream generation followed by encryption or decryption. The shortcut is optional.* on page 267.

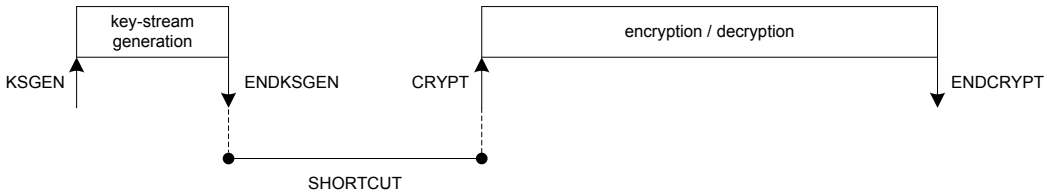


Figure 59: Key-stream generation followed by encryption or decryption. The shortcut is optional.

Key-stream generation, packet encryption, and packet decryption operations utilize the configuration specified in the data structure pointed to by the CNFPTR pointer. It is necessary to configure this pointer and its underlying data structure, and the MODE register before the KSGEN task is triggered. It is also necessary to configure the INPTR pointer and the OUTPTR pointer before the CRYPT task is triggered.

If a shortcut is used between ENDKSGEN event and CRYPT task, the INPTR pointer and the OUTPTR pointer must be configured before the KSGEN task is triggered.

The AES CCM supports different packet lengths, this is configured via the PACKETLENGTH field in the MODE register.

[23] *Bluetooth* AES CCM 128 bit block encryption, see *Bluetooth* Core specification Version 4.0.

29.1 Shared resources

The CCM shares registers and other resources with other peripherals that have the same ID as the CCM. The user must therefore disable all peripherals that have the same ID as the CCM before the CCM can be configured and used.

Disabling a peripheral that have the same ID as the CCM will not reset any of the registers that are shared with the CCM. It is therefore important to configure all relevant CCM registers explicitly to secure that it operates correctly.

See the Instantiation table in *Instantiation* on page 24 for details on peripherals and their IDs.

29.2 Encryption

During packet encryption, the AES CCM will read the unencrypted packet located in RAM at the address specified in the INPTR pointer, encrypt the packet and append a four byte long Message Integrity Check (MIC) field to the packet.

The AES CCM will also modify the length field of the packet to adjust for the appended MIC field, that is, add four bytes to the length, and store the resulting packet back into RAM at the address specified in the OUTPTR pointer, see *Figure 60: Encryption* on page 268.

Empty packets (length field is set to 0) will not be encrypted but instead moved unmodified through the AES CCM.

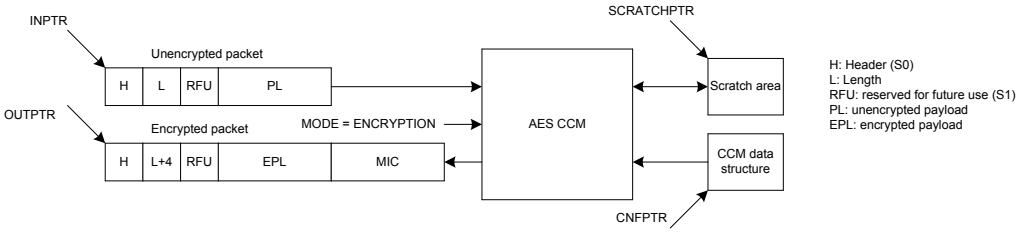


Figure 60: Encryption

29.3 Decryption

During packet decryption, the AES CCM will read the encrypted packet located in RAM at the address specified in the INPTR pointer, decrypt the packet, authenticate the packet's MIC field and generate the appropriate MIC status.

The AES CCM will also modify the length field of the packet to adjust for the MIC field, that is, subtract four bytes from the length, and then store the decrypted packet into RAM at the address pointed to by the OUTPTR pointer, see *Figure 61: Decryption* on page 269.

The CCM is only able to decrypt packets that are at least 5 bytes long, that is, 1 byte or more encrypted payload (EPL) and 4 bytes of MIC. The CCM will therefore generate a MIC error for packets where the length field is set to 1, 2, 3 or 4.

Empty packets (length field is set to 0) will not be decrypted but instead moved unmodified through the AES CCM, these packets will always pass the MIC check.

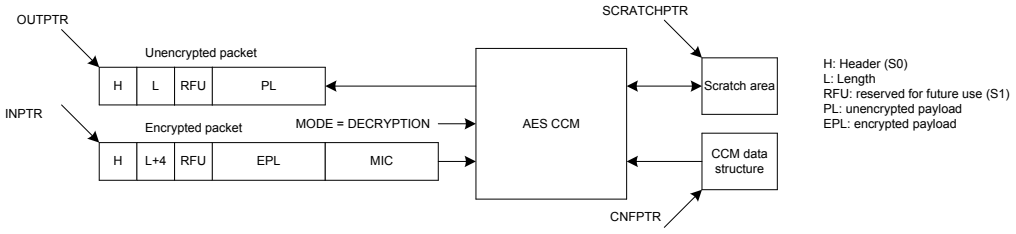


Figure 61: Decryption

29.4 AES CCM and RADIO concurrent operation

The AES CCM is designed to run in parallel with the RADIO to enable on-the-fly encryption and decryption of RADIO packets without CPU involvement. To facilitate this, the RADIO has to be configured with specific settings.

Table 57: Radio configuration settings

Radio parameter	Value	Description
PCNF0.S0LEN	1	Length of HEADER field in: *Table 59: Data structure for unencrypted packet* on page 271 and *Table 60: Data structure for encrypted packet* on page 271.
PCNF0.LFLEN	5 or 8	Length of LENGTH field in: *Table 59: Data structure for unencrypted packet* on page 271 and *Table 60: Data structure for encrypted packet* on page 271.
PCNF0.S1LEN	3 or 0	Length of the RFU field in: *Table 59: Data structure for unencrypted packet* on page 271 and *Table 60: Data structure for encrypted packet* on page 271. The combined length of LENGTH and RFU must always be 8 bit.
PCNF0.S1	Include	Always include the S1 field (RFU field) in RAM to secure that the same data structure can be used for PCNF0.S1LEN = 3 and PCNF0.S1LEN = 0: *Table 59: Data structure for unencrypted packet* on page 271 and *Table 60: Data structure for encrypted packet* on page 271.
MODE	Ble_1Mbit	Data rate. Must match CCM->MODE.DATARATE
PCNF1.BALEN	3	Length of address (32 bit)
CRCCNF.LEN	3	Length of CRC (24 bit)

29.5 Encrypting packets on-the-fly in radio transmit mode

When the AES CCM is encrypting a packet on-the-fly at the same time as the RADIO is transmitting it, the RADIO must read the encrypted packet from the same memory location as the AES CCM is writing to.

The OUTPTR pointer in the AES CCM must therefore point to the same memory location as the PACKETPTR pointer in the RADIO, see *Figure 62: Configuration of on-the-fly encryption* on page 269.

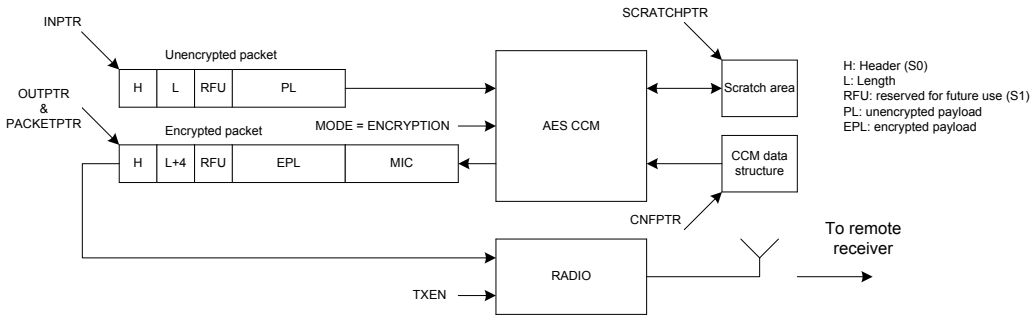


Figure 62: Configuration of on-the-fly encryption

In order to match the RADIO's timing, the KSGEN task must be triggered no later than when the START task in the RADIO is triggered, in addition the shortcut between the ENDKSGEN event and the CRYPT task must be enabled. This use-case is illustrated in *Figure 63: On-the-fly encryption using a PPI connection* on page 270 using a PPI connection between the READY event in the RADIO and the KSGEN task in the AES CCM.

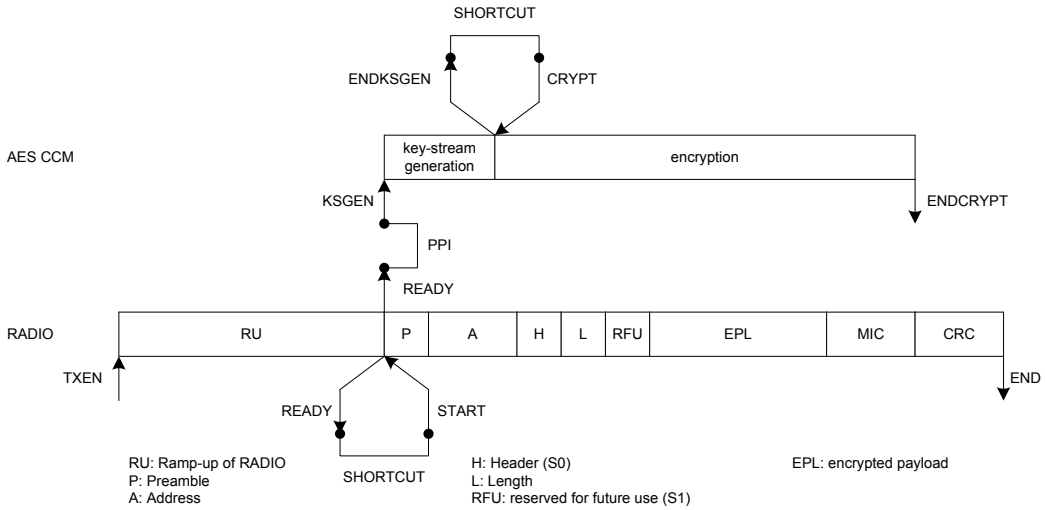


Figure 63: On-the-fly encryption using a PPI connection

29.6 Decrypting packets on-the-fly in radio receive mode

When the AES CCM is decrypting a packet on-the-fly at the same time as the RADIO is receiving it, the AES CCM must read the encrypted packet from the same memory location as the RADIO is writing to.

The INPTR pointer in the AES CCM must therefore point to the same memory location as the PACKETPTR pointer in the RADIO, see *Figure 64: Configuration of on-the-fly decryption* on page 270.

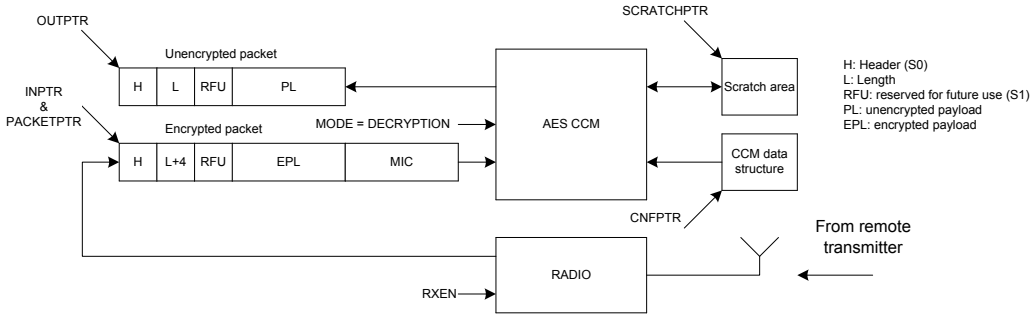


Figure 64: Configuration of on-the-fly decryption

In order to match the RADIO's timing, the KSGEN task must be triggered no later than when the START task in the RADIO is triggered. In addition, the CRYPT task must be triggered no earlier than when the ADDRESS event is generated by the RADIO.

If the CRYPT task is triggered exactly at the same time as the ADDRESS event is generated by the RADIO, the AES CCM will guarantee that the decryption is completed no later than when the END event in the RADIO is generated.

This use-case is illustrated in *Figure 65: On-the-fly decryption using a PPI connection between the READY event in the RADIO and the KSGEN task in the AES CCM* on page 271 using a PPI connection between the ADDRESS event in the RADIO and the CRYPT task in the AES CCM. The KSGEN task is triggered from the READY event in the RADIO through a PPI connection.

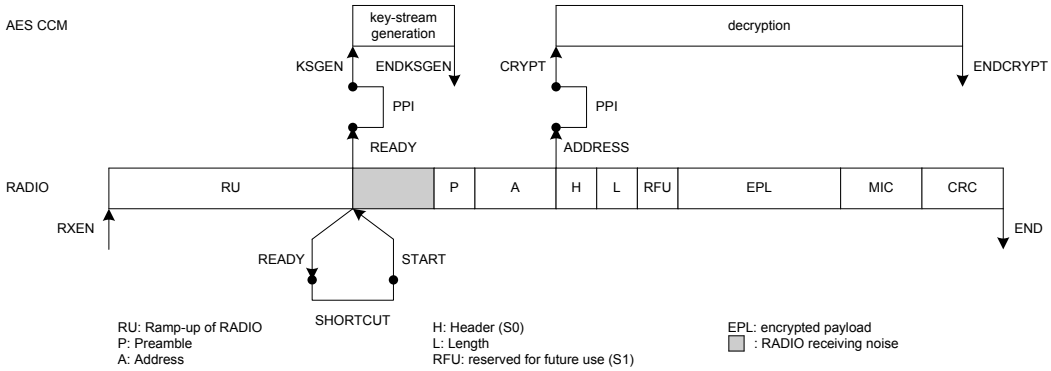


Figure 65: On-the-fly decryption using a PPI connection between the READY event in the RADIO and the KSGEN task in the AES CCM

29.7 CCM data structure

The CCM data structure is located in Data RAM at the memory location specified by the CNFPTR pointer register.

Table 58: CCM data structure overview

Property	Address offset	Description
KEY	0	16 byte AES key
PKTCTR	16	Octet0 (LSO) of packet counter
	17	Octet1 of packet counter
	18	Octet2 of packet counter
	19	Octet3 of packet counter
	20	Bit 6 – Bit 0: Octet4 (7 most significant bits of packet counter, with Bit 6 being the most significant bit) Bit7: Ignored
	21	Ignored
	22	Ignored
	23	Ignored
	24	Bit 0: Direction bit Bit 7 – Bit 1: Zero padded
IV	25	8 byte initialization vector (IV) Octet0 (LSO) of IV, Octet1 of IV, … , Octet7 (MSO) of IV

The NONCE vector (as specified by the *Bluetooth* Core Specification) will be generated by hardware based on the information specified in the CCM data structure from *Table 58: CCM data structure overview* on page 271 .

Table 59: Data structure for unencrypted packet

Property	Address offset	Description
HEADER	0	Packet Header
LENGTH	1	Number of bytes in unencrypted payload
RFU	2	Reserved Future Use
PAYLOAD	3	Unencrypted payload

Table 60: Data structure for encrypted packet

Property	Address offset	Description
HEADER	0	Packet Header
LENGTH	1	Number of bytes in encrypted payload including length of MIC **Important:** LENGTH will be 0 for empty packets since the MIC is not added to empty packets
RFU	2	Reserved Future Use
PAYLOAD	3	Encrypted payload
MIC	3 + payload length	ENCRYPT: 4 bytes encrypted MIC

Property	Address offset	Description
		Important: MIC is not added to empty packets

29.8 EasyDMA and ERROR event

The CCM implements an EasyDMA mechanism for reading and writing to the RAM.

In some scenarios where the CPU and other DMA enabled peripherals are accessing the RAM at the same time, the CCM DMA could experience some bus conflicts which may also result in an error during encryption. If this happens, the ERROR event will be generated.

The EasyDMA will have finished accessing the RAM when the ENDKSGEN and ENDCRYPT events are generated.

If the CNFPTR, SCRATCHPTR, INPTR and the OUTPTR are not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See *Memory* on page 23 for more information about the different memory regions.

29.9 Registers

Table 61: Instances

Base address	Peripheral	Instance	Description	Configuration
0x4000F000	CCM	CCM	AES CCM Mode Encryption	

Table 62: Register Overview

Register	Offset	Description
TASKS_KSGEN	0x000	Start generation of key-stream. This operation will stop by itself when completed.
TASKS_CRYPT	0x004	Start encryption/decryption. This operation will stop by itself when completed.
TASKS_STOP	0x008	Stop encryption/decryption
EVENTS_ENDKSGEN	0x100	Key-stream generation complete
EVENTS_ENDCRYPT	0x104	Encrypt/decrypt complete
EVENTS_ERROR	0x108	CCM error event
SHORTS	0x200	Shortcut register
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
MICSTATUS	0x400	MIC check result
ENABLE	0x500	Enable
MODE	0x504	Operation mode
CNFPTR	0x508	Pointer to data structure holding AES key and NONCE vector
INPTR	0x50C	Input pointer
OUTPTR	0x510	Output pointer
SCRATCHPTR	0x514	Pointer to data area used for temporary storage

29.9.1 SHORTS

Address offset: 0x200

Shortcut register

Bit number																																
Id																																A
Reset 0x00000000																																**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																											
A	RW	ENDKSGEN_CRYPT			Shortcut between ENDKSGEN event and CRYPT task																											
					See *EVENTS_ENDKSGEN* and *TASKS_CRYPT*																											
			Disabled	0	Disable shortcut																											
			Enabled	1	Enable shortcut																											

29.9.2 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	ENDKSGEN																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	ENDCRYPT																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	ERROR																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															

29.9.3 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	ENDKSGEN																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	ENDCRYPT																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	ERROR																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															

29.9.4 MICSTATUS

Address offset: 0x400

MIC check result

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	MICSTATUS																																	
				The result of the MIC check performed during the previous decryption operation																															
			CheckFailed	0																															
				MIC check failed																															
			CheckPassed	1																															
				MIC check passed																															

29.9.5 ENABLE

Address offset: 0x500

Enable

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	ENABLE																																	
				Enable or disable CCM																															
			Disabled	0																															
				Disable																															
			Enabled	2																															
				Enable																															

29.9.6 MODE

Address offset: 0x504

Operation mode

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											C								B															A
Reset 0x00000001				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	MODE																																	
				The mode of operation to be used																															
			Encryption	0																															
				AES CCM packet encryption mode																															
			Decryption	1																															
				AES CCM packet decryption mode																															
B	RW	DATARATE																																	
				Data rate that the CCM shall run in synch with																															
			1Mbit	0																															
				In synch with 1 Mbit data rate																															
			2Mbit	1																															
				In synch with 2 Mbit data rate																															
C	RW	LENGTH																																	
				Packet length configuration																															
			Default	0																															
				Default length. Effective length of LENGTH field is 5-bit																															
			Extended	1																															
				Extended length. Effective length of LENGTH field is 8-bit																															

29.9.7 CNFPTR

Address offset: 0x508

Pointer to data structure holding AES key and NONCE vector

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	CNFPTR																																	
				Pointer to the data structure holding the AES key and the CCM NONCE vector (see Table 1 CCM data structure overview)																															

29.9.8 INPTR

Address offset: 0x50C

Input pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	INPTR			Input pointer

29.9.9 OUTPTR

Address offset: 0x510

Output pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	OUTPTR			Output pointer

29.9.10 SCRATCHPTR

Address offset: 0x514

Pointer to data area used for temporary storage

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	SCRATCHPTR			Pointer to a scratch data area used for temporary storage during key-stream generation, MIC generation and encryption/ decryption. The scratch area is used for temporary storage of data during key-stream generation and encryption. A space of 43 bytes, or (16 + MAXPACKETSIZE) bytes, whatever is largest, must be reserved in RAM.

30 AAR — Accelerated address resolver

Accelerated address resolver is a cryptographic support function for implementing the "Resolvable Private Address Resolution Procedure" described in the *Bluetooth Core specification* v4.0. "Resolvable private address generation" should be achieved using ECB and is not supported by AAR.

The procedure allows two devices that share a secret key to generate and resolve a hash based on their device address. The AAR block enables real-time address resolution on incoming packets when configured as described in this chapter. This allows real-time packet filtering (whitelisting) using a list of known shared keys (Identity Resolving Keys (IRK) in *Bluetooth*).

30.1 Shared resources

The AAR shares registers and other resources with the peripherals that have the same ID as the AAR. The user must therefore disable all peripherals that have the same ID as the AAR before the AAR can be configured and used.

Disabling a peripheral that have the same ID as the AAR will not reset any of the registers that are shared with the AAR. It is therefore important to configure all relevant AAR registers explicitly to secure that it operates correctly.

See the Instantiation table in *Instantiation* on page 24 for details on peripherals and their IDs.

30.2 EasyDMA

The AAR implements EasyDMA for reading and writing to the RAM. The EasyDMA will have finished accessing the RAM when the END, RESOLVED, and NOTRESOLVED events are generated.

If the IRKPTR, ADDRPTR and the SCRATCHPTR is not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See *Memory* on page 23 for more information about the different memory regions.

30.3 Resolving a resolvable address

As per *Bluetooth* specification, a private resolvable address is composed of six bytes.

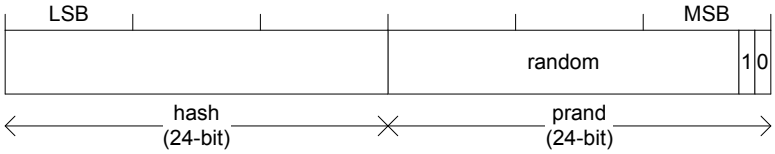


Figure 66: Resolvable address

To resolve an address the ADDRPTR register must point to the start of packet. The resolver is started by triggering the START task. A RESOLVED event is generated when the AAR manages to resolve the address using one of the Identity Resolving Keys (IRK) found in the IRK data structure. The AAR will use the IRK specified in the register IRK0 to IRK15 starting from IRK0. How many to be used is specified by the NIRK register. The AAR module will generate a NOTRESOLVED event if it is not able to resolve the address using the specified list of IRKs.

The AAR will go through the list of available IRKs in the IRK data structure and for each IRK try to resolve the address according to the Resolvable Private Address Resolution Procedure described in the *Bluetooth* Specification[24]. The time it takes to resolve an address may vary depending on where in the list the

[24] *Bluetooth* Specification Version 4.0 [Vol 3] chapter 10.8.2.3.

resolvable address is located. The resolution time will also be affected by RAM accesses performed by other peripherals and the CPU. See the *Electrical specifications* for more information about resolution time.

The AAR will only do a comparison of the received address to those programmed in the module. And not check what type of address it actually is.

The AAR will stop as soon as it has managed to resolve the address, or after trying to resolve the address using NIRK number of IRKs from the IRK data structure. The AAR will generate an END event after it has stopped.

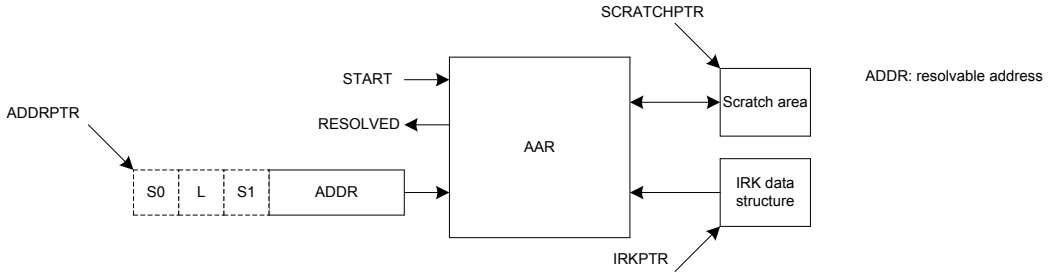


Figure 67: Address resolution with packet preloaded into RAM

30.4 Use case example for chaining RADIO packet reception with address resolution using AAR

The AAR may be started as soon as the 6 bytes required by the AAR have been received by the RADIO and stored in RAM. The ADDRPTR pointer must point to the start of packet.

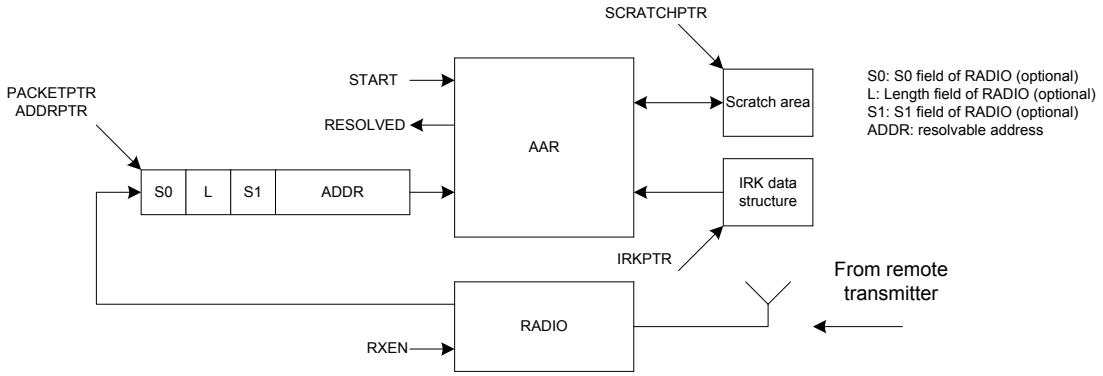


Figure 68: Address resolution with packet loaded into RAM by the RADIO

30.5 IRK data structure

The IRK data structure is located in RAM at the memory location specified by the CNFPTR pointer register.

Table 63: IRK data structure overview

Property	Address offset	Description
IRK0	0	IRK number 0 (16 - byte)
IRK1	16	IRK number 1 (16 - byte)
..	..	..
IRK15	240	IRK number 15 (16 - byte)

30.6 Registers

Table 64: Instances

Base address	Peripheral	Instance	Description	Configuration
0x4000F000	AAR	AAR	Acelerated Address Resolver	

Table 65: Register Overview

Register	Offset	Description
TASKS_START	0x000	Start resolving addresses based on IRKs specified in the IRK data structure
TASKS_STOP	0x008	Stop resolving addresses
EVENTS_END	0x100	Address resolution procedure complete
EVENTS_RESOLVED	0x104	Address resolved
EVENTS_NOTRESOLVED	0x108	Address not resolved
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
STATUS	0x400	Resolution status
ENABLE	0x500	Enable AAR
NIRK	0x504	Number of IRKs
IRKPTR	0x508	Pointer to IRK data structure
ADDRPTR	0x510	Pointer to the resolvable address
SCRATCHPTR	0x514	Pointer to data area used for temporary storage

30.6.1 INTENSET

Address offset: 0x304

Enable interrupt

Bit number																																
																														C	B	A
Reset 0x00000000																														**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	END			Write '1' to Enable interrupt for END event See *EVENTS_END*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	RESOLVED			Write '1' to Enable interrupt for RESOLVED event See *EVENTS_RESOLVED*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	NOTRESOLVED			Write '1' to Enable interrupt for NOTRESOLVED event See *EVENTS_NOTRESOLVED*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

30.6.2 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	END			Write '1' to Disable interrupt for END event See *EVENTS_END*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	RESOLVED			Write '1' to Disable interrupt for RESOLVED event See *EVENTS_RESOLVED*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	NOTRESOLVED			Write '1' to Disable interrupt for NOTRESOLVED event See *EVENTS_NOTRESOLVED*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

30.6.3 STATUS

Address offset: 0x400

Resolution status

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	R	STATUS		[0..15]	The IRK that was used last time an address was resolved

30.6.4 ENABLE

Address offset: 0x500

Enable AAR

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	ENABLE			Enable or disable AAR
			Disabled	0	Disable
			Enabled	3	Enable

30.6.5 NIRK

Address offset: 0x504

Number of IRKs

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A A A A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	NIRK		[1..16]	Number of Identity root keys available in the IRK data structure

30.6.6 IRKPTR

Address offset: 0x508

Pointer to IRK data structure

Bit number																															
Id																															
Reset 0x00000000																															
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																										
A	RW	IRKPTR			Pointer to the IRK data structure																										

30.6.7 ADDRPTR

Address offset: 0x510

Pointer to the resolvable address

Bit number																															
Id																															
Reset 0x00000000																															
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																										
A	RW	ADDRPTR			Pointer to the resolvable address (6-bytes)																										

30.6.8 SCRATCHPTR

Address offset: 0x514

Pointer to data area used for temporary storage

Bit number																															
Id																															
Reset 0x00000000																															
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																										
A	RW	SCRATCHPTR			Pointer to a scratch data area used for temporary storage during resolution.A space of minimum 3 bytes must be reserved.																										

30.7 Electrical specification

30.7.1 AAR Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
$t_{AAR,8}$	Time for address resolution of 8 IRKs		48		µs

31 SPIM — Serial peripheral interface master with EasyDMA

The SPI master can communicate with multiple slaves using individual chip select signals for each of the slave devices attached to a bus.

Listed here are the main features for the SPIM

- Three SPIM instances
- SPI mode 0-3
- EasyDMA direct transfer to/from RAM for both SPI Slave and SPI Master
- Individual selection of IO pin for each SPI signal

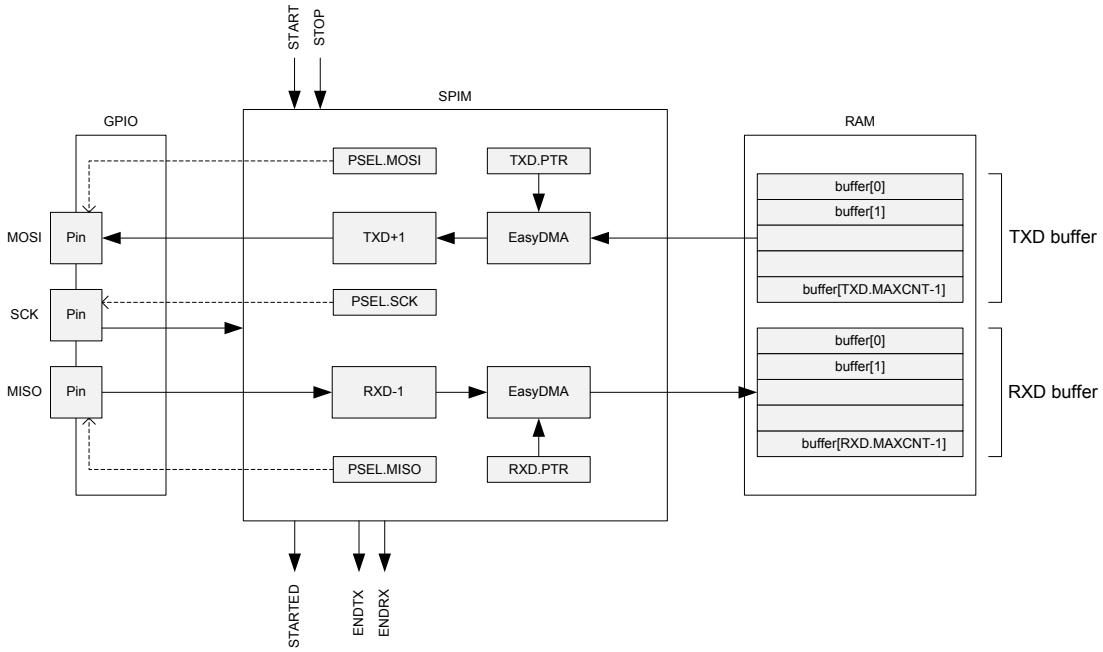


Figure 69: SPIM — SPI master with EasyDMA

The SPIM does not implement support for chip select directly. Therefore, the CPU must use available GPIOs to select the correct slave and control this independently of the SPI master. The SPIM supports SPI modes 0 through 3. The CONFIG register allows setting CPOL and CPHA appropriately.

Table 66: SPI modes

Mode	Clock polarity CPOL	Clock phase CPHA
SPI_MODE	0 (Active High)	0 (Leading)
SPI_MODE	0 (Active High)	1 (Trailing)
SPI_MODE	1 (Active Low)	0 (Leading)
SPI_MODE	1 (Active Low)	1 (Trailing)

31.1 Shared resources

The SPI shares registers and other resources with other peripherals that have the same ID as the SPI. Therefore, the user must disable all peripherals that have the same ID as the SPI before the SPI can be configured and used.

Disabling a peripheral that has the same ID as the SPI will not reset any of the registers that are shared with the SPI. It is therefore important to configure all relevant SPI registers explicitly to secure that it operates correctly.

See the Instantiation table in *Instantiation* on page 24 for details on peripherals and their IDs.

31.2 EasyDMA

The SPI master implements EasyDMA for reading and writing of data packets from and to the DATA RAM without CPU involvement.

The RXD.PTR and TXD.PTR point to the RXD buffer (receive buffer) and TXD buffer (transmit buffer) respectively, see *Figure 69: SPIM — SPI master with EasyDMA* on page 281. RXD.MAXCNT and TXD.MAXCNT specify the maximum number of bytes allocated to the buffers.

The SPI master will automatically stop transmitting after TXD.MAXCNT bytes have been transmitted and RXD.MAXCNT bytes have been received. If TXD.MAXCNT is larger than RXD.MAXCNT, the superfluous received bytes will be ignored. If RXD.MAXCNT is larger than TXD.MAXCNT, the remaining transmitted bytes will contain the value defined in the ORC register.

If the RXD.PTR and the TXD.PTR are not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See *Memory* on page 23 for more information about the different memory regions.

The .PTR and .MAXCNT registers are double-buffered. They can be updated and prepared for the next transmission immediately after having received the STARTED event.

The ENDRX/ENDTX event indicate that EasyDMA has finished accessing respectively the RX/TX buffer in RAM. The END event gets generated when both RX and TX are finished accessing the buffers in RAM.

31.2.1 EasyDMA list

EasyDMA supports one list type.

The supported list type is:

- Array list

EasyDMA array list

The EasyDMA array list can be represented by the data structure ArrayList_type.

For illustration, see the code example below. This data structure includes only a buffer with size equal to Channel.MAXCNT. EasyDMA will use the Channel.MAXCNT register to determine when the buffer is full. Replace 'Channel' by the specific data channel you want to use, for instance 'NRF_SPIM->RXD', 'NRF_SPIM->TXD', 'NRF_TWIM->RXD', etc.

The Channel.MAXCNT register cannot be specified larger than the actual size of the buffer. If Channel.MAXCNT is specified larger than the size of the buffer, the EasyDMA channel may overflow the buffer.

This array list does not provide a mechanism to explicitly specify where the next item in the list is located. Instead, it assumes that the list is organized as a linear array where items are located one after the other in RAM.

```
#define BUFFER_SIZE  4

typedef struct ArrayList
{
  uint8_t buffer[BUFFER_SIZE];
} ArrayList_type;

ArrayList_type MyArrayList[3];
```

```
//replace 'Channel' below by the specific data channel you want to use,
//         for instance 'NRF_SPIM->RXD', 'NRF_TWIM->RXD', etc.
Channel.MAXCNT = BUFFER_SIZE;
Channel.PTR = &MyArrayList;
```

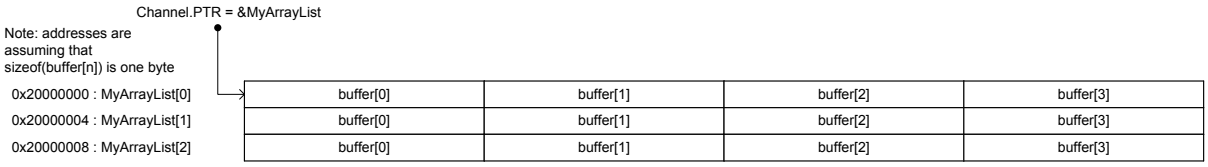


Figure 70: EasyDMA array list

31.3 SPI master transaction sequence

An SPI master transaction consists of a sequence started by the START task followed by a number of events, and finally the STOP task.

An SPI master transaction is started by triggering the START task. The ENDTX event will be generated when the transmitter has transmitted all bytes in the TXD buffer as specified in the TXD.MAXCNT register. The ENDRX event will be generated when the receiver has filled the RXD buffer, i.e. received the last possible byte as specified in the RXD.MAXCNT register.

Following a START task, the SPI master will generate an END event when both ENDRX and ENDTX have been generated.

The SPI master is stopped by triggering the STOP task. A STOPPED event is generated when the SPI master has stopped.

If the ENDRX event has not already been generated when the SPI master has come to a stop, the SPI master will generate the ENDRX event explicitly even though the RX buffer is not full.

If the ENDTX event has not already been generated when the SPI master has come to a stop, the SPI master will generate the ENDTX event explicitly even though all bytes in the TXD buffer, as specified in the TXD.MAXCNT register, have not been transmitted.

The SPI master is a synchronous interface, and for every byte that is sent, a different byte will be received at the same time; this is illustrated in *Figure 71: SPI master transaction* on page 284.

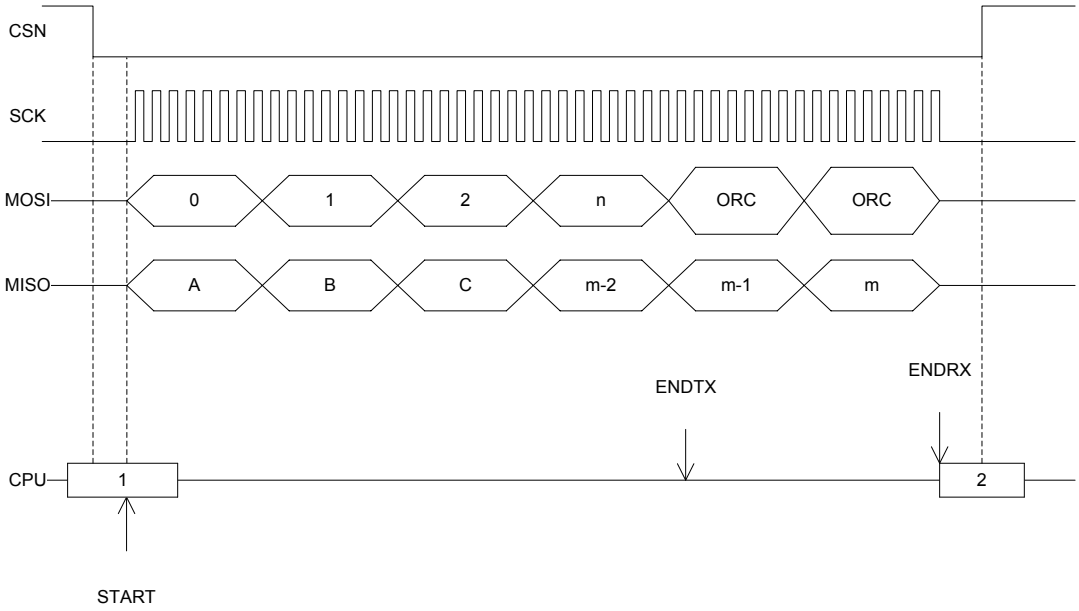


Figure 71: SPI master transaction

31.4 Low power

When putting the system in low power and the peripheral is not needed, lowest possible power consumption is achieved by stopping, and then disabling the peripheral.

The STOP task may not be always needed (the peripheral might already be stopped), but if it is sent, software shall wait until the STOPPED event was received as a response before disabling the peripheral through the ENABLE register.

31.5 Master mode pin configuration

The SCK, MOSI, and MISO signals associated with the SPI master are mapped to physical pins according to the configuration specified in the PSEL.SCK, PSEL.MOSI, and PSEL.MISO registers respectively.

The PSEL.SCK, PSEL.MOSI, and PSEL.MISO registers and their configurations are only used as long as the SPI master is enabled, and retained only as long as the device is in ON mode. PSEL.SCK, PSEL.MOSI and PSEL.MISO must only be configured when the SPI master is disabled.

To secure correct behavior in the SPI, the pins used by the SPI must be configured in the GPIO peripheral as described in *Table 67: GPIO configuration* on page 284 prior to enabling the SPI. This configuration must be retained in the GPIO for the selected IOs as long as the SPI is enabled.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behavior.

Table 67: GPIO configuration

SPI master signal	SPI master pin	Direction	Output value	Comments
SCK	As specified in PSEL.SCK	Output	Same as CONFIG.CPOL	
MOSI	As specified in PSEL.MOSI	Output	0	
MISO	As specified in PSEL.MISO	Input	Not applicable	

31.6 Registers

Table 68: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40003000	SPIM	SPIM0	SPI master 0	
0x40004000	SPIM	SPIM1	SPI master 1	
0x40023000	SPIM	SPIM2	SPI master 2	

Table 69: Register Overview

Register	Offset	Description
TASKS_START	0x010	Start SPI transaction
TASKS_STOP	0x014	Stop SPI transaction
TASKS_SUSPEND	0x01C	Suspend SPI transaction
TASKS_RESUME	0x020	Resume SPI transaction
EVENTS_STOPPED	0x104	SPI transaction has stopped
EVENTS_ENDRX	0x110	End of RXD buffer reached
EVENTS_END	0x118	End of RXD buffer and TXD buffer reached
EVENTS_ENDTX	0x120	End of TXD buffer reached
EVENTS_STARTED	0x14C	Transaction started
SHORTS	0x200	Shortcut register
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ENABLE	0x500	Enable SPIM
PSEL.SCK	0x508	Pin select for SCK
PSEL.MOSI	0x50C	Pin select for MOSI signal
PSEL.MISO	0x510	Pin select for MISO signal
FREQUENCY	0x524	SPI frequency. Accuracy depends on the HFCLK source selected.
RXD.PTR	0x534	Data pointer
RXD.MAXCNT	0x538	Maximum number of bytes in receive buffer
RXD.AMOUNT	0x53C	Number of bytes transferred in the last transaction
RXD.LIST	0x540	EasyDMA list type
TXD.PTR	0x544	Data pointer
TXD.MAXCNT	0x548	Maximum number of bytes in transmit buffer
TXD.AMOUNT	0x54C	Number of bytes transferred in the last transaction
TXD.LIST	0x550	EasyDMA list type
CONFIG	0x554	Configuration register
ORC	0x5C0	Over-read character. Character clocked out in case and over-read of the TXD buffer.

31.6.1 SHORTS

Address offset: 0x200

Shortcut register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		A																	
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	END_START																																	
				See *EVENTS_END* and *TASKS_START*																																	
			Disabled	0																															
			Enabled	1																															

31.6.2 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																E											D		C		B			A	
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	STOPPED																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	ENDRX																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	END																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	ENDTX																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	STARTED																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															

31.6.3 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																E											D		C		B			A	
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	STOPPED																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	ENDRX																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	END																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	ENDTX																																	

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id															E											D		C		B			A	
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**															**Description**									
			Clear								1															Disable									
			Disabled								0															Read: Disabled									
			Enabled								1															Read: Enabled									
E	RW	STARTED																								Write '1' to Disable interrupt for STARTED event See *EVENTS_STARTED*									
			Clear								1															Disable									
			Disabled								0															Read: Disabled									
			Enabled								1															Read: Enabled									

31.6.4 ENABLE

Address offset: 0x500

Enable SPIM

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**															**Description**									
A	RW	ENABLE																								Enable or disable SPIM									
			Disabled								0															Disable SPIM									
			Enabled								7															Enable SPIM									

31.6.5 PSEL.SCK

Address offset: 0x508

Pin select for SCK

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id															C															A	A	A	A	A
Reset 0xFFFFFFFF			**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**								**Value**															**Description**									
A	RW	PIN									[0..31]															Pin number									
C	RW	CONNECT																								Connection									
			Disconnected								1															Disconnect									
			Connected								0															Connect									

31.6.6 PSEL.MOSI

Address offset: 0x50C

Pin select for MOSI signal

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id															C															A	A	A	A	A
Reset 0xFFFFFFFF			**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**								**Value**															**Description**									
A	RW	PIN									[0..31]															Pin number									
C	RW	CONNECT																								Connection									
			Disconnected								1															Disconnect									
			Connected								0															Connect									

31.6.7 PSEL.MISO

Address offset: 0x510

Pin select for MISO signal

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				C																												A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN		[0..31]																															
C	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

31.6.8 FREQUENCY

Address offset: 0x524

SPI frequency. Accuracy depends on the HFCLK source selected.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x04000000				**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	FREQUENCY																																	
			K125	0x02000000																															
			K250	0x04000000																															
			K500	0x08000000																															
			M1	0x10000000																															
			M2	0x20000000																															
			M4	0x40000000																															
			M8	0x80000000																															

31.6.9 RXD.PTR

Address offset: 0x534

Data pointer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PTR																																	

31.6.10 RXD.MAXCNT

Address offset: 0x538

Maximum number of bytes in receive buffer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	MAXCNT																																	

31.6.11 RXD.AMOUNT

Address offset: 0x53C

Number of bytes transferred in the last transaction

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	AMOUNT																																	

31.6.12 RXD.LIST

Address offset: 0x540

EasyDMA list type

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	LIST																																	
			Disabled	0																															
			ArrayList	1																															

31.6.13 TXD.PTR

Address offset: 0x544

Data pointer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PTR																																	

31.6.14 TXD.MAXCNT

Address offset: 0x548

Maximum number of bytes in transmit buffer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	MAXCNT																																	

31.6.15 TXD.AMOUNT

Address offset: 0x54C

Number of bytes transferred in the last transaction

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	AMOUNT																																	

31.6.16 TXD.LIST

Address offset: 0x550

EasyDMA list type

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	LIST																																	
			Disabled	0																															
			ArrayList	1																															

31.6.17 CONFIG

Address offset: 0x554

Configuration register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																														
A	RW	ORDER			Bit order																														
			MsbFirst	0	Most significant bit shifted out first																														
			LsbFirst	1	Least significant bit shifted out first																														
B	RW	CPHA			Serial clock (SCK) phase																														
			Leading	0	Sample on leading edge of clock, shift serial data on trailing edge																														
			Trailing	1	Sample on trailing edge of clock, shift serial data on leading edge																														
C	RW	CPOL			Serial clock (SCK) polarity																														
			ActiveHigh	0	Active high																														
			ActiveLow	1	Active low																														

31.6.18 ORC

Address offset: 0x5C0

Over-read character. Character clocked out in case and over-read of the TXD buffer.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																														
A	RW	ORC			Over-read character. Character clocked out in case and over-read of the TXD buffer.																														

31.7 Electrical specification

31.7.1 SPIM master interface electrical specifications

Symbol	Description	Min.	Typ.	Max.	Units
f_{SPIM}	Bit rates for SPIM[25]			8[26]	Mbps
$I_{SPIM,2Mbps}$	Run current for SPIM, 2 Mbps		50		µA
$I_{SPIM,8Mbps}$	Run current for SPIM, 8 Mbps		50		µA
$I_{SPIM,IDLE}$	Idle current for SPIM (STARTed, no CSN activity)		1		µA
$t_{SPIM,START}$	Time from START task to transmission started	..	..	..	µs

31.7.2 Serial Peripheral Interface Master (SPIM) timing specifications

Symbol	Description	Min.	Typ.	Max.	Units
$t_{SPIM,CSCK}$	SCK period	..	..	..	ns
$t_{SPIM,RSCK,LD}$	SCK rise time, standard drive[a]			$t_{RF,25pF}$	
$t_{SPIM,RSCK,HD}$	SCK rise time, high drive[a]			$t_{HRF,25pF}$	
$t_{SPIM,FSCK,LD}$	SCK fall time, standard drive[a]			$t_{RF,25pF}$	
$t_{SPIM,FSCK,HD}$	SCK fall time, high drive[a]			$t_{HRF,25pF}$	
$t_{SPIM,WHSCK}$	SCK high time[a]	$(0.5*t_{CSCK})$ – t_{RSCK}			

[25] High bit rates may require GPIOs to be set as High Drive, see GPIO chapter for more details.

[26] The actual maximum data rate depends on the slave's CLK to MISO and MOSI setup and hold timings.

[a] At 25pF load, including GPIO pin capacitance, see GPIO spec.

Symbol	Description	Min.	Typ.	Max.	Units
$t_{SPIM,WLSCK}$	SCK low time[a]	$(0.5*t_{CSCK})$ – t_{FSCK}			
$t_{SPIM,SUMI}$	MISO to CLK edge setup time	19			ns
$t_{SPIM,HMI}$	CLK edge to MISO hold time	18			ns
$t_{SPIM,VMO}$	CLK edge to MOSI valid			59	ns
$t_{SPIM,HMO}$	MOSI hold time after CLK edge	20			ns

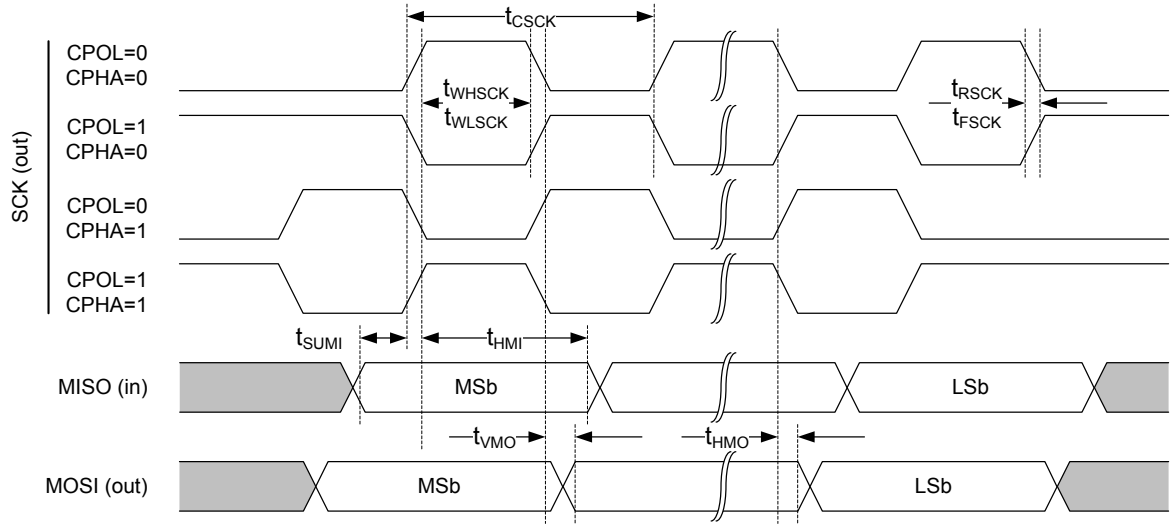


Figure 72: SPIM timing diagram

32 SPIS — Serial peripheral interface slave with EasyDMA

SPI slave (SPIS) is implemented with EasyDMA support for ultra low power serial communication from an external SPI master. EasyDMA in conjunction with hardware-based semaphore mechanisms removes all real-time requirements associated with controlling the SPI slave from a low priority CPU execution context.

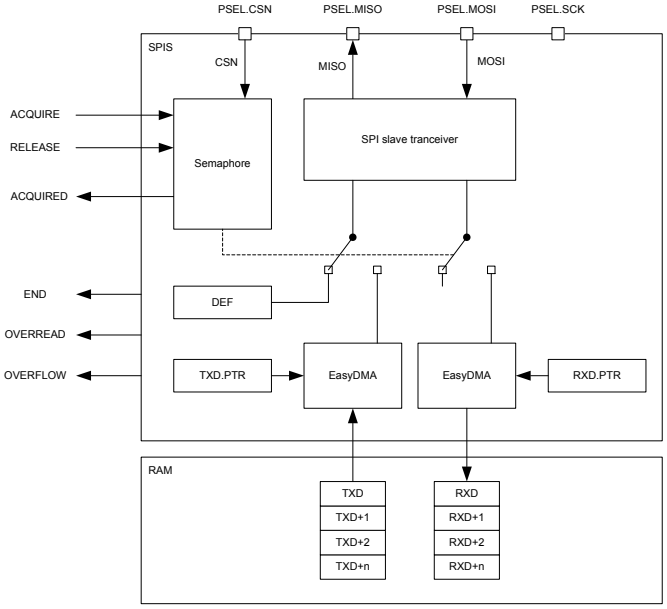


Figure 73: SPI slave

The SPIS supports SPI modes 0 through 3. The CONFIG register allows setting CPOL and CPHA appropriately.

Table 70: SPI modes

Mode	Clock polarity CPOL	Clock phase CPHA
SPI_MODE0	0 (Leading)	0 (Active High)
SPI_MODE1	0 (Leading)	1 (Active Low)
SPI_MODE2	1 (Trailing)	0 (Active High)
SPI_MODE3	1 (Trailing)	1 (Active Low)

32.1 Shared resources

The SPI slave shares registers and other resources with other peripherals that have the same ID as the SPI slave. Therefore, you must disable all peripherals that have the same ID as the SPI slave before the SPI slave can be configured and used.

Disabling a peripheral that has the same ID as the SPI slave will not reset any of the registers that are shared with the SPI slave. It is important to configure all relevant SPI slave registers explicitly to secure that it operates correctly.

The Instantiation table in *Instantiation* on page 24 shows which peripherals have the same ID as the SPI slave.

32.2 EasyDMA

The SPI slave implements EasyDMA for reading and writing to and from the RAM. The END event indicates that EasyDMA has finished accessing the buffer in RAM.

If the TXD.PTR and the RXD.PTR are not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See *Memory* on page 23 for more information about the different memory regions.

32.3 SPI slave operation

SPI slave uses two memory pointers, RXD.PTR and TXD.PTR, that point to the RXD buffer (receive buffer) and TXD buffer (transmit buffer) respectively. Since these buffers are located in RAM, which can be accessed by both the SPI slave and the CPU, a hardware based semaphore mechanism is implemented to enable safe sharing.

See *Figure 74: SPI transaction when shortcut between END and ACQUIRE is enabled* on page 294.

Before the CPU can safely update the RXD.PTR and TXD.PTR pointers it must first acquire the SPI semaphore. The CPU can acquire the semaphore by triggering the ACQUIRE task and then receiving the ACQUIRED event. When the CPU has updated the RXD.PTR and TXD.PTR pointers the CPU must release the semaphore before the SPI slave will be able to acquire it. The CPU releases the semaphore by triggering the RELEASE task. This is illustrated in *Figure 74: SPI transaction when shortcut between END and ACQUIRE is enabled* on page 294. Triggering the RELEASE task when the semaphore is not granted to the CPU will have no effect.

The semaphore mechanism does not, at any time, prevent the CPU from performing read or write access to the RXD.PTR register, the TXD.PTR registers, or the RAM that these pointers are pointing to. The semaphore is only telling when these can be updated by the CPU so that safe sharing is achieved.

The semaphore is by default assigned to the CPU after the SPI slave is enabled. No ACQUIRED event will be generated for this initial semaphore handover. An ACQUIRED event will be generated immediately if the ACQUIRE task is triggered while the semaphore is assigned to the CPU.

The SPI slave will try to acquire the semaphore when CSN goes low. If the SPI slave does not manage to acquire the semaphore at this point, the transaction will be ignored. This means that all incoming data on MOSI will be discarded, and the DEF (default) character will be clocked out on the MISO line throughout the whole transaction. This will also be the case even if the semaphore is released by the CPU during the transaction. In case of a race condition where the CPU and the SPI slave try to acquire the semaphore at the same time, as illustrated in lifeline item 2 in *Figure 74: SPI transaction when shortcut between END and ACQUIRE is enabled* on page 294, the semaphore will be granted to the CPU.

If the SPI slave acquires the semaphore, the transaction will be granted. The incoming data on MOSI will be stored in the RXD buffer and the data in the TXD buffer will be clocked out on MISO.

When a granted transaction is completed and CSN goes high, the SPI slave will automatically release the semaphore and generate the END event.

As long as the semaphore is available the SPI slave can be granted multiple transactions one after the other. If the CPU is not able to reconfigure the TXD.PTR and RXD.PTR between granted transactions, the same TX data will be clocked out and the RX buffers will be overwritten. To prevent this from happening, the END_ACQUIRE shortcut can be used. With this shortcut enabled the semaphore will be handed over to the CPU automatically after the granted transaction has completed, giving the CPU the ability to update the TXPTR and RXPTR between every granted transaction.

If the CPU tries to acquire the semaphore while it is assigned to the SPI slave, an immediate handover will not be granted. However, the semaphore will be handed over to the CPU as soon as the SPI slave has released the semaphore after the granted transaction is completed. If the END_ACQUIRE shortcut is enabled and the CPU has triggered the ACQUIRE task during a granted transaction, only one ACQUIRE request will be served following the END event.

The MAXRX register specifies the maximum number of bytes the SPI slave can receive in one granted transaction. If the SPI slave receives more than MAXRX number of bytes, an OVERFLOW will be indicated in the STATUS register and the incoming bytes will be discarded.

The MAXTX parameter specifies the maximum number of bytes the SPI slave can transmit in one granted transaction. If the SPI slave is forced to transmit more than MAXTX number of bytes, an OVERREAD will be indicated in the STATUS register and the ORC character will be clocked out.

The RXD.AMOUNT and TXD.AMOUNT registers are updated when a granted transaction is completed. The TXD.AMOUNT register indicates how many bytes were read from the TX buffer in the last transaction, that is, ORC (over-read) characters are not included in this number. Similarly, the RXD.AMOUNT register indicates how many bytes were written into the RX buffer in the last transaction.

The ENDRX event is generated when the RX buffer has been filled.

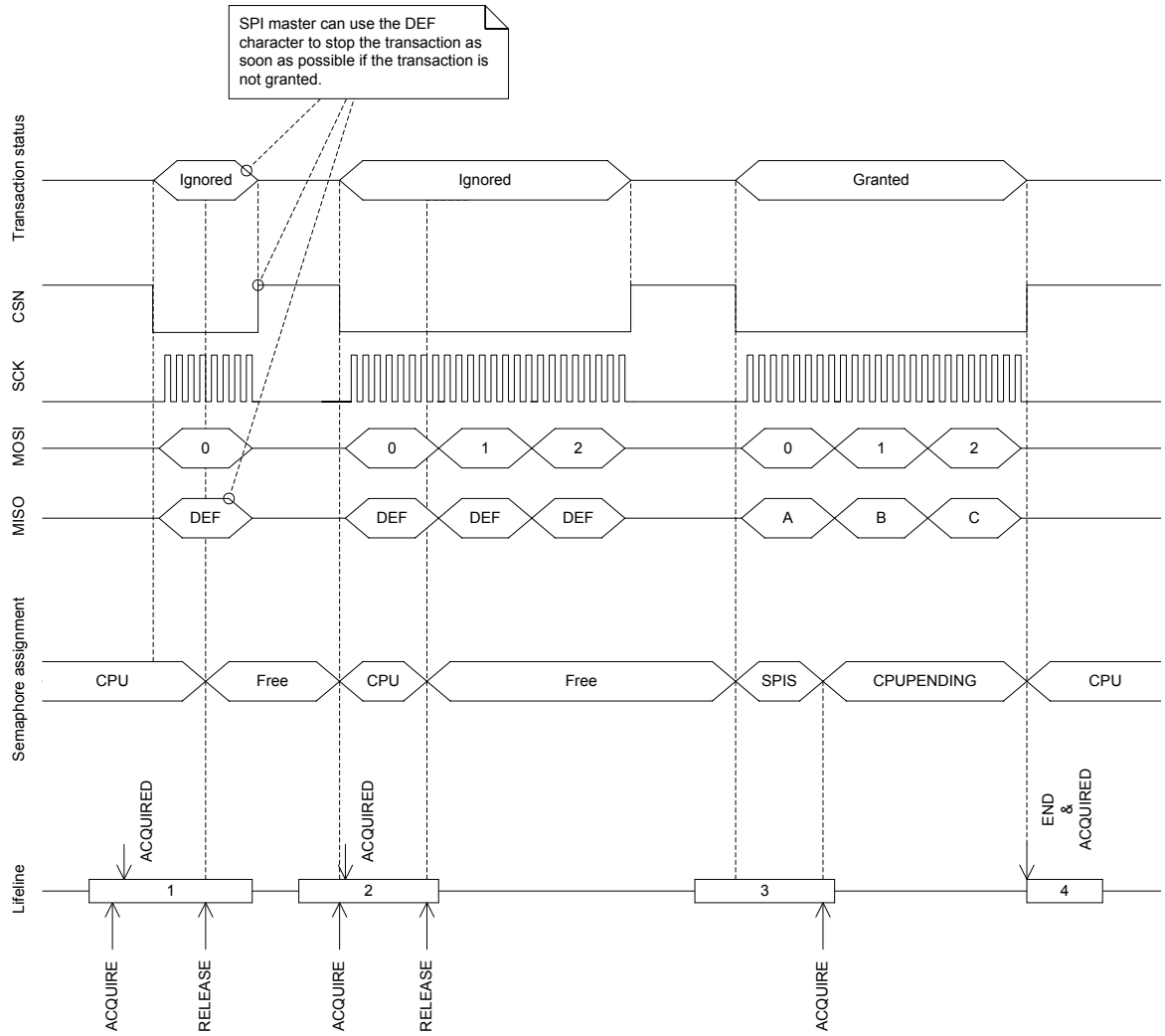


Figure 74: SPI transaction when shortcut between END and ACQUIRE is enabled

32.4 Pin configuration

The CSN, SCK, MOSI, and MISO signals associated with the SPI slave are mapped to physical pins according to the configuration specified in the PSEL.CSN, PSEL.SCK, PSEL.MOSI, and PSEL.MISO registers respectively. If the CONNECT field of any of these registers is set to Disconnected, the associated SPI slave signal will not be connected to any physical pins.

The PSEL.CSN, PSEL.SCK, PSEL.MOSI, and PSEL.MISO registers and their configurations are only used as long as the SPI slave is enabled, and retained only as long as the device is in System ON mode, see *POWER — Power supply* on page 78 chapter for more information about power modes. When the peripheral is disabled, the pins will behave as regular GPIOs, and use the configuration in their respective OUT bit field and PIN_CNF[n] register. PSEL.CSN, PSEL.SCK, PSEL.MOSI, and PSEL.MISO must only be configured when the SPI slave is disabled.

To secure correct behavior in the SPI slave, the pins used by the SPI slave must be configured in the GPIO peripheral as described in *Table 71: GPIO configuration before enabling peripheral* on page 295 before enabling the SPI slave. This is to secure that the pins used by the SPI slave are driven correctly if the SPI

slave itself is temporarily disabled, or if the device temporarily enters System OFF. This configuration must be retained in the GPIO for the selected I/Os as long as the SPI slave is to be recognized by an external SPI master.

The MISO line is set in high impedance as long as the SPI slave is not selected with CSN.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behavior.

Table 71: GPIO configuration before enabling peripheral

SPI signal	SPI pin	Direction	Output value	Comment
CSN	As specified in PSEL.CSN	Input	Not applicable	
SCK	As specified in PSEL.SCK	Input	Not applicable	
MOSI	As specified in PSEL.MOSI	Input	Not applicable	
MISO	As specified in PSEL.MISO	Input	Not applicable	Emulates that the SPI slave is not selected.

32.5 Registers

Table 72: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40003000	SPIS	SPIS0	SPI slave 0	
0x40004000	SPIS	SPIS1	SPI slave 1	
0x40023000	SPIS	SPIS2	SPI slave 2	

Table 73: Register Overview

Register	Offset	Description	
TASKS_ACQUIRE	0x024	Acquire SPI semaphore	
TASKS_RELEASE	0x028	Release SPI semaphore, enabling the SPI slave to acquire it	
EVENTS_END	0x104	Granted transaction completed	
EVENTS_ENDRX	0x110	End of RXD buffer reached	
EVENTS_ACQUIRED	0x128	Semaphore acquired	
SHORTS	0x200	Shortcut register	
INTENSET	0x304	Enable interrupt	
INTENCLR	0x308	Disable interrupt	
SEMSTAT	0x400	Semaphore status register	
STATUS	0x440	Status from last transaction	
ENABLE	0x500	Enable SPI slave	
PSELSCK	0x508	Pin select for SCK	Deprecated
PSELMISO	0x50C	Pin select for MISO	Deprecated
PSELMOSI	0x510	Pin select for MOSI	Deprecated
PSELCSN	0x514	Pin select for CSN	Deprecated
PSEL.SCK	0x508	Pin select for SCK	
PSEL.MISO	0x50C	Pin select for MISO signal	
PSEL.MOSI	0x510	Pin select for MOSI signal	
PSEL.CSN	0x514	Pin select for CSN signal	
RXDPTR	0x534	RXD data pointer	Deprecated
MAXRX	0x538	Maximum number of bytes in receive buffer	Deprecated
AMOUNTRX	0x53C	Number of bytes received in last granted transaction	Deprecated
RXD.PTR	0x534	RXD data pointer	
RXD.MAXCNT	0x538	Maximum number of bytes in receive buffer	
RXD.AMOUNT	0x53C	Number of bytes received in last granted transaction	
TXDPTR	0x544	TXD data pointer	Deprecated
MAXTX	0x548	Maximum number of bytes in transmit buffer	Deprecated
AMOUNTTX	0x54C	Number of bytes transmitted in last granted transaction	Deprecated
TXD.PTR	0x544	TXD data pointer	
TXD.MAXCNT	0x548	Maximum number of bytes in transmit buffer	
TXD.AMOUNT	0x54C	Number of bytes transmitted in last granted transaction	
CONFIG	0x554	Configuration register	

Register	Offset	Description
DEF	0x55C	Default character. Character clocked out in case of an ignored transaction.
ORC	0x5C0	Over-read character

32.5.1 SHORTS

Address offset: 0x200

Shortcut register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	END_ACQUIRE			Shortcut between END event and ACQUIRE task
					See EVENTS_END and TASKS_ACQUIRE
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

32.5.2 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	END			Write '1' to Enable interrupt for END event
					See EVENTS_END
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	ENDRX			Write '1' to Enable interrupt for ENDRX event
					See EVENTS_ENDRX
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	ACQUIRED			Write '1' to Enable interrupt for ACQUIRED event
					See EVENTS_ACQUIRED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

32.5.3 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	END			Write '1' to Disable interrupt for END event
					See EVENTS_END
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	ENDRX			Write '1' to Disable interrupt for ENDRX event

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									C						B			A	
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
			Disabled	0	Disable SPI slave
			Enabled	2	Enable SPI slave

32.5.7 PSELSCK (Deprecated)

Address offset: 0x508

Pin select for SCK

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF			**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	PSELSCK		[0..31]	Pin number configuration for SPI SCK signal
			Disconnected	0xFFFFFFFF	Disconnect

32.5.8 PSELMISO (Deprecated)

Address offset: 0x50C

Pin select for MISO

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF			**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	PSELMISO		[0..31]	Pin number configuration for SPI MISO signal
			Disconnected	0xFFFFFFFF	Disconnect

32.5.9 PSELMOSI (Deprecated)

Address offset: 0x510

Pin select for MOSI

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF			**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	PSELMOSI		[0..31]	Pin number configuration for SPI MOSI signal
			Disconnected	0xFFFFFFFF	Disconnect

32.5.10 PSELCSN (Deprecated)

Address offset: 0x514

Pin select for CSN

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF			**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	PSELCSN		[0..31]	Pin number configuration for SPI CSN signal
			Disconnected	0xFFFFFFFF	Disconnect

32.5.11 PSEL.SCK

Address offset: 0x508

Pin select for SCK

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											B																			A	A	A	A	A
Reset 0xFFFFFFFF			**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	PIN		[0..31]	Pin number
B	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

32.5.12 PSEL.MISO

Address offset: 0x50C

Pin select for MISO signal

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											B																			A	A	A	A	A
Reset 0xFFFFFFFF			**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	PIN		[0..31]	Pin number
B	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

32.5.13 PSEL.MOSI

Address offset: 0x510

Pin select for MOSI signal

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											B																			A	A	A	A	A
Reset 0xFFFFFFFF			**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	PIN		[0..31]	Pin number
B	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

32.5.14 PSEL.CSN

Address offset: 0x514

Pin select for CSN signal

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											B																			A	A	A	A	A
Reset 0xFFFFFFFF			**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	PIN		[0..31]	Pin number
B	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

32.5.15 RXDPTR (Deprecated)

Address offset: 0x534

RXD data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	RXDPTR			RXD data pointer

32.5.16 MAXRX (Deprecated)

Address offset: 0x538

Maximum number of bytes in receive buffer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	MAXRX			Maximum number of bytes in receive buffer

32.5.17 AMOUNTRX (Deprecated)

Address offset: 0x53C

Number of bytes received in last granted transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	R	AMOUNTRX			Number of bytes received in the last granted transaction

32.5.18 RXD.PTR

Address offset: 0x534

RXD data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	PTR			RXD data pointer

32.5.19 RXD.MAXCNT

Address offset: 0x538

Maximum number of bytes in receive buffer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	MAXCNT			Maximum number of bytes in receive buffer

32.5.20 RXD.AMOUNT

Address offset: 0x53C

Number of bytes received in last granted transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	R	AMOUNT			Number of bytes received in the last granted transaction

32.5.21 TXDPTR (Deprecated)

Address offset: 0x544

TXD data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TXDPTR			TXD data pointer

32.5.22 MAXTX (Deprecated)

Address offset: 0x548

Maximum number of bytes in transmit buffer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	MAXTX			Maximum number of bytes in transmit buffer

32.5.23 AMOUNTTX (Deprecated)

Address offset: 0x54C

Number of bytes transmitted in last granted transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	R	AMOUNTTX			Number of bytes transmitted in last granted transaction

32.5.24 TXD.PTR

Address offset: 0x544

TXD data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	PTR			TXD data pointer

32.5.25 TXD.MAXCNT

Address offset: 0x548

Maximum number of bytes in transmit buffer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	MAXCNT			Maximum number of bytes in transmit buffer

32.5.26 TXD.AMOUNT

Address offset: 0x54C

Number of bytes transmitted in last granted transaction

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	R	AMOUNT			Number of bytes transmitted in last granted transaction

32.5.27 CONFIG

Address offset: 0x554

Configuration register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	ORDER			Bit order
			MsbFirst	0	Most significant bit shifted out first
			LsbFirst	1	Least significant bit shifted out first
B	RW	CPHA			Serial clock (SCK) phase
			Leading	0	Sample on leading edge of clock, shift serial data on trailing edge
			Trailing	1	Sample on trailing edge of clock, shift serial data on leading edge
C	RW	CPOL			Serial clock (SCK) polarity
			ActiveHigh	0	Active high
			ActiveLow	1	Active low

32.5.28 DEF

Address offset: 0x55C

Default character. Character clocked out in case of an ignored transaction.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	DEF			Default character. Character clocked out in case of an ignored transaction.

32.5.29 ORC

Address offset: 0x5C0

Over-read character

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	ORC			Over-read character. Character clocked out after an over-read of the transmit buffer.

32.6 Electrical specification

32.6.1 SPIS slave interface electrical specifications

Symbol	Description	Min.	Typ.	Max.	Units
f_{SPIS}	Bit rates for SPIS[27]			8[28]	Mbps
$I_{SPIS,2Mbps}$	Run current for SPIS, 2 Mbps		45		µA
$I_{SPIS,8Mbps}$	Run current for SPIS, 8 Mbps		45		µA
$I_{SPIS,IDLE}$	Idle current for SPIS (STARTed, no CSN activity)		1		µA
$t_{SPIS,LP,START}$	Time from RELEASE task to ready to receive/transmit (CSN active), Low power mode		$t_{SPIS,CL,STAR}$ + t_{START_HFIN}		µs
$t_{SPIS,CL,START}$	Time from RELEASE task to receive/transmit (CSN active), Constant latency mode		0.125		µs

32.6.2 Serial Peripheral Interface Slave (SPIS) timing specifications

Symbol	Description	Min.	Typ.	Max.	Units
$t_{SPIS,CSCKIN,8Mbps}$	SCK input period at 8Mbps		125		ns
$t_{SPIS,CSCKIN,4Mbps}$	SCK input period at 4Mbps		250		ns
$t_{SPIS,CSCKIN,2Mbps}$	SCK input period at 2Mbps		500		ns
$t_{SPIS,RFSCKIN}$	SCK input rise/fall time			30	ns
$t_{SPIS,WHSCKIN}$	SCK input high time	30			ns
$t_{SPIS,WLSCKIN}$	SCK input low time	30			ns
$t_{SPIS,SUCSN,LP}$	CSN to CLK setup time, Low power mode	$t_{SPIS,SUCSN,}$ + t_{START_HFIN}			ns
$t_{SPIS,SUCSN,CL}$	CSN to CLK setup time, Constant latency mode	1000			ns
$t_{SPIS,HCSN}$	CLK to CSN hold time	2000			ns
$t_{SPIS,ASO}$	CSN to MISO driven[a]			1000	ns
$t_{SPIS,DISSO}$	CSN to MISO disabled[a]			68	ns
$t_{SPIS,CWH}$	CSN inactive time	300			ns
$t_{SPIS,VSO}$	CLK edge to MISO valid			19	ns
$t_{SPIS,HSO}$	MISO hold time after CLK edge	18[29]			ns
$t_{SPIS,SUSI}$	MOSI to CLK edge setup time	59			ns
$t_{SPIS,HSI}$	CLK edge to MOSI hold time	20			ns

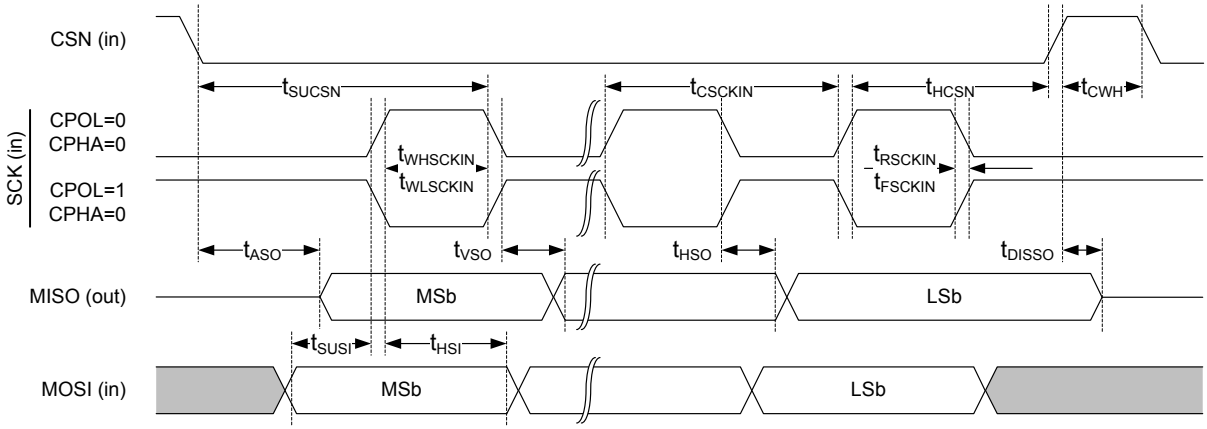


Figure 75: SPIS timing diagram

27 Higher bit rates may require GPIOs to be set as High Drive, see GPIO chapter for more details.

28 The actual maximum data rate depends on the master's CLK to MISO and MOSI setup and hold timings.

a At 25pF load, including GPIO capacitance, see GPIO spec.

29 This is to ensure compatibility to SPI masters sampling MISO on the same edge as MOSI is output

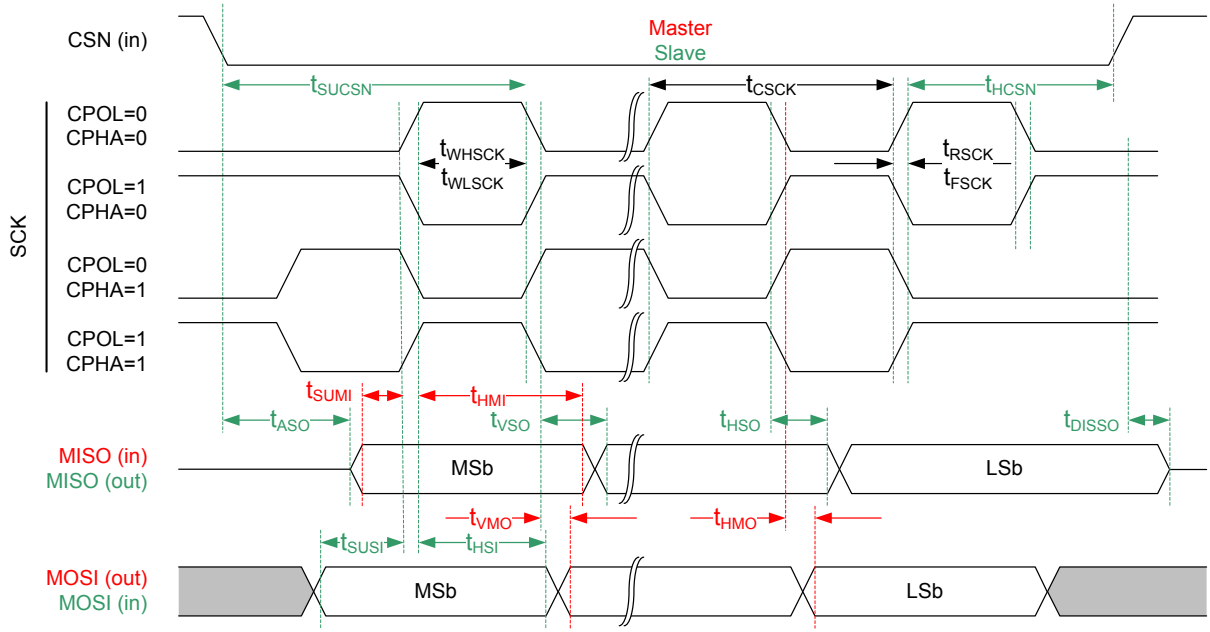


Figure 76: Common SPIM and SPIS timing diagram

33 TWIM — I2C compatible two-wire interface master with EasyDMA

TWI master with EasyDMA (TWIM) is a two-wire half-duplex master which can communicate with multiple slave devices connected to the same bus

Listed here are the main features for TWIM:

- I2C compatible
- 100 kbps, 250 kbps, or 400 kbps
- Support for clock stretching
- EasyDMA

The two-wire interface can communicate with a bi-directional wired-AND bus with two lines (SCL, SDA). The protocol makes it possible to interconnect up to 127 individually addressable devices. TWIM is not compatible with CBUS.

The GPIOs used for each two-wire interface line can be chosen from any GPIO on the device and are independently configurable. This enables great flexibility in device pinout and efficient use of board space and signal routing.

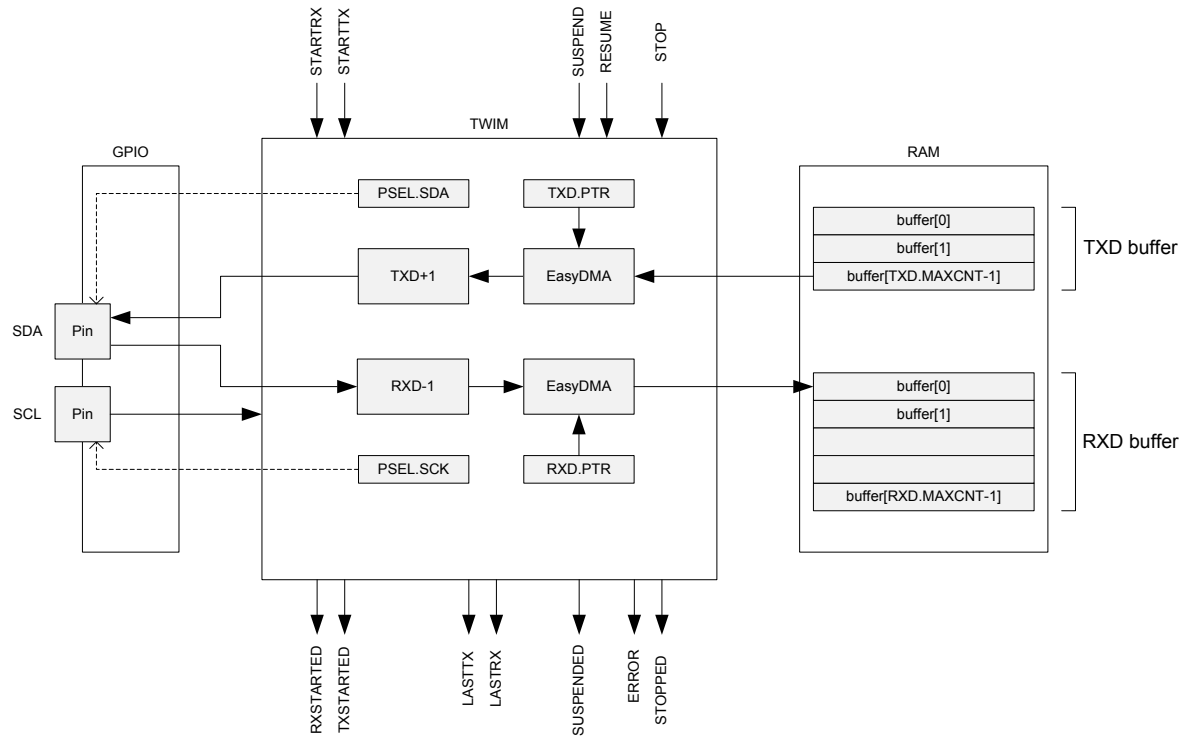


Figure 77: TWI master with EasyDMA

A typical TWI setup consists of one master and one or more slaves. For an example, see *Figure 78: A typical TWI setup comprising one master and three slaves* on page 306. This TWIM is only able to operate as a single master on the TWI bus. Multi-master bus configuration is not supported.

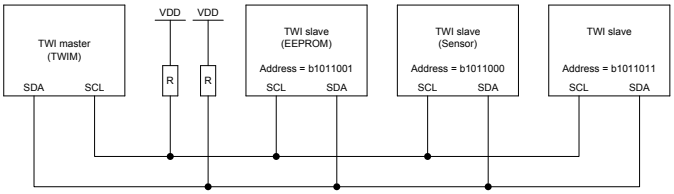


Figure 78: A typical TWI setup comprising one master and three slaves

This TWI master supports clock stretching performed by the slaves. The TWI master is started by triggering the STARTTX or STARTRX tasks, and stopped by triggering the STOP task. The TWI master will generate a STOPPED event when it has stopped following a STOP task.

The TWI master cannot get stopped while it is suspended, so the STOP task has to be issued after the TWI master has been resumed.

After the TWI master is started, the STARTTX task or the STARTRX task should not be triggered again before the TWI master has stopped, i.e. following a LASTRX, LASTTX or STOPPED event.

If a NACK is clocked in from the slave, the TWI master will generate an ERROR event.

33.1 Shared resources

The TWI master shares registers and other resources with other peripherals that have the same ID as the TWI master. Therefore, you must disable all peripherals that have the same ID as the TWI master before the TWI master can be configured and used.

Disabling a peripheral that has the same ID as the TWI master will not reset any of the registers that are shared with the TWI master. It is therefore important to configure all relevant registers explicitly to secure that the TWI master operates correctly.

The Instantiation table in *Instantiation* on page 24 shows which peripherals have the same ID as the TWI.

33.2 EasyDMA

The TWI master implements EasyDMA for reading and writing to and from the RAM.

If the TXD.PTR and the RXD.PTR are not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See *Memory* on page 23 for more information about the different memory regions.

The .PTR and .MAXCNT registers are double-buffered. They can be updated and prepared for the next RX/TX transmission immediately after having received the RXSTARTED/TXSTARTED event.

The STOPPED event indicates that EasyDMA has finished accessing the buffer in RAM.

33.2.1 EasyDMA list

EasyDMA supports one list type.

The supported list type is:

- Array list

EasyDMA array list

The EasyDMA array list can be represented by the data structure ArrayList_type.

For illustration, see the code example below. This data structure includes only a buffer with size equal to Channel.MAXCNT. EasyDMA will use the Channel.MAXCNT register to determine when the buffer is full. Replace 'Channel' by the specific data channel you want to use, for instance 'NRF_SPIM->RXD', 'NRF_SPIM->TXD', 'NRF_TWIM->RXD', etc.

The Channel.MAXCNT register cannot be specified larger than the actual size of the buffer. If Channel.MAXCNT is specified larger than the size of the buffer, the EasyDMA channel may overflow the buffer.

This array list does not provide a mechanism to explicitly specify where the next item in the list is located. Instead, it assumes that the list is organized as a linear array where items are located one after the other in RAM.

```
#define BUFFER_SIZE  4

typedef struct ArrayList
{
  uint8_t buffer[BUFFER_SIZE];
} ArrayList_type;

ArrayList_type MyArrayList[3];

//replace 'Channel' below by the specific data channel you want to use,
//         for instance 'NRF_SPIM->RXD', 'NRF_TWIM->RXD', etc.
Channel.MAXCNT = BUFFER_SIZE;
Channel.PTR = &MyArrayList;
```

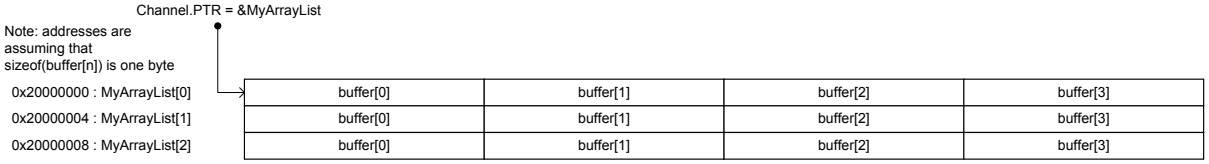


Figure 79: EasyDMA array list

33.3 Master write sequence

A TWI master write sequence is started by triggering the STARTTX task. After the STARTTX task has been triggered, the TWI master will generate a start condition on the TWI bus, followed by clocking out the address and the READ/WRITE bit set to 0 (WRITE=0, READ=1).

The address must match the address of the slave device that the master wants to write to. The READ/WRITE bit is followed by an ACK/NACK bit (ACK=0 or NACK=1) generated by the slave.

After receiving the ACK bit, the TWI master will clock out the data bytes found in the transmit buffer located in RAM at the address specified in the TXD.PTR register. Each byte clocked out from the master will be followed by an ACK/NACK bit clocked in from the slave.

A typical TWI master write sequence is illustrated in *Figure 80: TWI master writing data to a slave* on page 308. Occurrence 2 in the figure illustrates clock stretching performed by the TWI master following a SUSPEND task.

A SUSPENDED event indicates that the SUSPEND task has taken effect; this event can be used to synchronize the software.

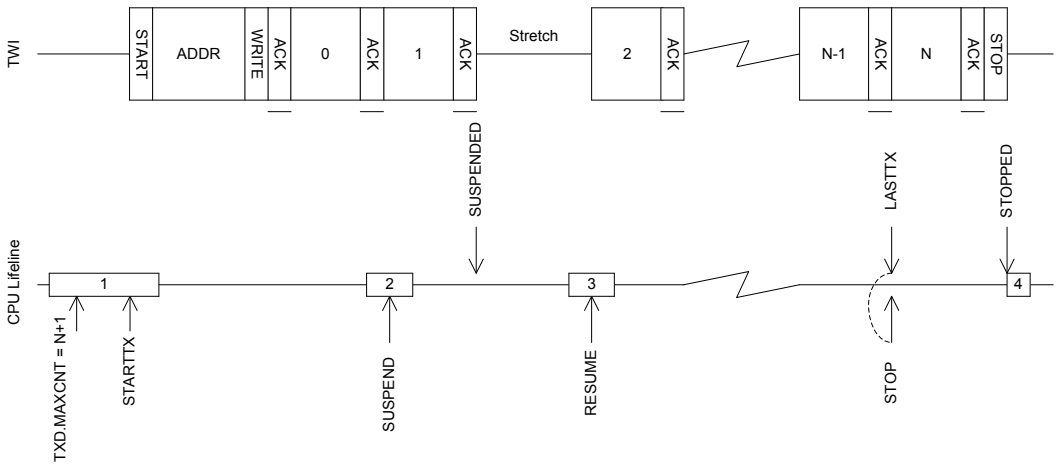


Figure 80: TWI master writing data to a slave

The TWI master will generate a LASTTX event when it starts to transmit the last byte, this is illustrated in *Figure 80: TWI master writing data to a slave* on page 308

The TWI master is stopped by triggering the STOP task, this task should be triggered during the transmission of the last byte to secure that the TWI will stop as fast as possible after sending the last byte. It is safe to use the shortcut between LASTTX and STOP to accomplish this.

Note that the TWI master does not stop by itself when the whole RAM buffer has been sent, or when an error occurs. The STOP task must be issued, through the use of a local or PPI shortcut, or in software as part of the error handler.

The TWI master cannot get stopped while it is suspended, so the STOP task has to be issued after the TWI master has been resumed.

33.4 Master read sequence

A TWI master read sequence is started by triggering the STARTRX task. After the STARTRX task has been triggered the TWI master will generate a start condition on the TWI bus, followed by clocking out the address and the READ/WRITE bit set to 1 (WRITE = 0, READ = 1). The address must match the address of the slave device that the master wants to read from. The READ/WRITE bit is followed by an ACK/NACK bit (ACK=0 or NACK = 1) generated by the slave.

After having sent the ACK bit the TWI slave will send data to the master using the clock generated by the master.

Data received will be stored in RAM at the address specified in the RXD.PTR register. The TWI master will generate an ACK after all but the last byte received from the slave. The TWI master will generate a NACK after the last byte received to indicate that the read sequence shall stop.

A typical TWI master read sequence is illustrated in *Figure 81: The TWI master reading data from a slave* on page 309. Occurrence 2 in the figure illustrates clock stretching performed by the TWI master following a SUSPEND task.

A SUSPENDED event indicates that the SUSPEND task has taken effect; this event can be used to synchronize the software.

The TWI master will generate a LASTRX event when it is ready to receive the last byte, this is illustrated in *Figure 81: The TWI master reading data from a slave* on page 309. If RXD.MAXCNT > 1 the LASTRX event is generated after sending the ACK of the previously received byte. If RXD.MAXCNT = 1 the LASTRX event is generated after receiving the ACK following the address and READ bit.

The TWI master is stopped by triggering the STOP task, this task must be triggered before the NACK bit is supposed to be transmitted. The STOP task can be triggered at any time during the reception of the last byte. It is safe to use the shortcut between LASTRX and STOP to accomplish this.

Note that the TWI master does not stop by itself when the RAM buffer is full, or when an error occurs. The STOP task must be issued, through the use of a local or PPI shortcut, or in software as part of the error handler.

The TWI master cannot get stopped while it is suspended, so the STOP task has to be issued after the TWI master has been resumed.

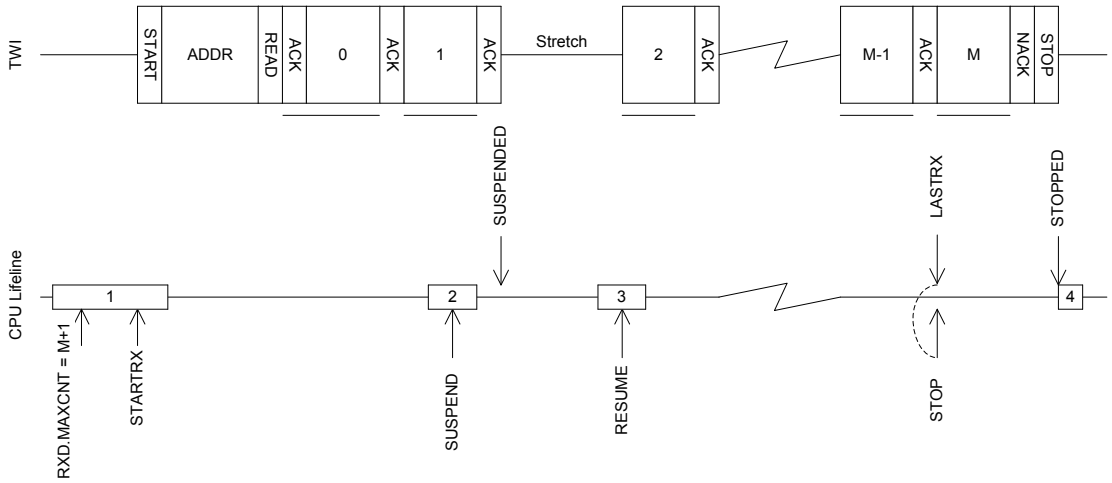


Figure 81: The TWI master reading data from a slave

33.5 Master repeated start sequence

A typical repeated start sequence is one in which the TWI master writes two bytes to the slave followed by reading four bytes from the slave. This example uses shortcuts to perform the simplest type of repeated start sequence, i.e. one write followed by one read. The same approach can be used to perform a repeated start sequence where the sequence is read followed by write.

The figure *Figure 82: A repeated start sequence, where the TWI master writes two bytes followed by reading 4 bytes from the slave* on page 309 illustrates this:

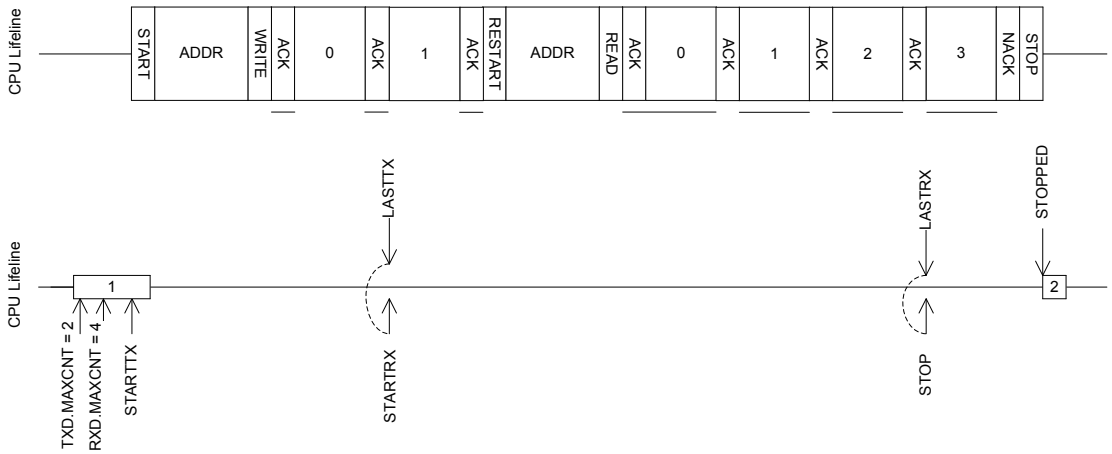


Figure 82: A repeated start sequence, where the TWI master writes two bytes followed by reading 4 bytes from the slave

If a more complex repeated start sequence is needed and the TWI firmware drive is serviced in a low priority interrupt it may be necessary to use the SUSPEND task and SUSPENDED event to guarantee that the correct tasks are generated at the correct time. This is illustrated in *Figure 83: A double repeated start sequence using the SUSPEND task to secure safe operation in low priority interrupts* on page 310.

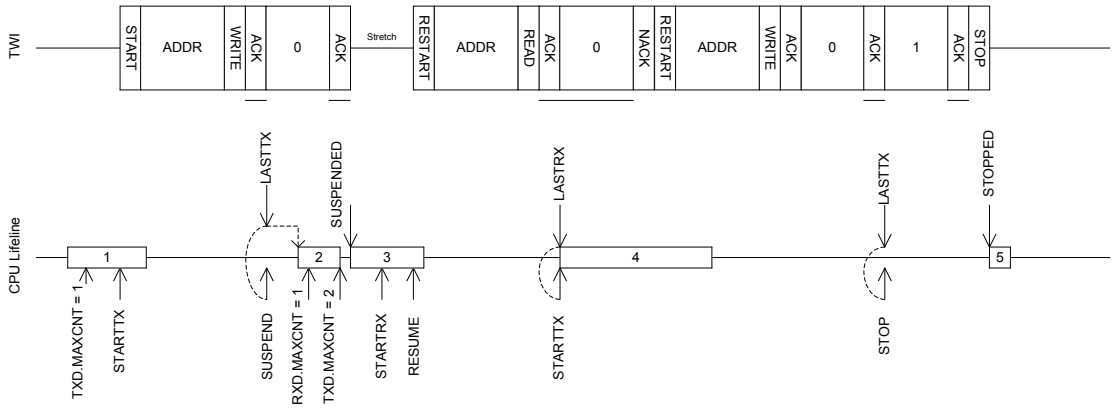


Figure 83: A double repeated start sequence using the SUSPEND task to secure safe operation in low priority interrupts

33.6 Low power

When putting the system in low power and the peripheral is not needed, lowest possible power consumption is achieved by stopping, and then disabling the peripheral.

The STOP task may not be always needed (the peripheral might already be stopped), but if it is sent, software shall wait until the STOPPED event was received as a response before disabling the peripheral through the ENABLE register.

33.7 Master mode pin configuration

The SCL and SDA signals associated with the TWI master are mapped to physical pins according to the configuration specified in the PSEL.SCL and PSEL.SDA registers respectively.

The PSEL.SCL and PSEL.SDA registers and their configurations are only used as long as the TWI master is enabled, and retained only as long as the device is in ON mode. When the peripheral is disabled, the pins will behave as regular GPIOs, and use the configuration in their respective OUT bit field and PIN_CNF[n] register. PSEL.SCL, PSEL.SDA must only be configured when the TWI master is disabled.

To secure correct signal levels on the pins used by the TWI master when the system is in OFF mode, and when the TWI master is disabled, these pins must be configured in the GPIO peripheral as described in *Table 74: GPIO configuration before enabling peripheral* on page 310.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behavior.

Table 74: GPIO configuration before enabling peripheral

TWI master signal	TWI master pin	Direction	Output value	Drive strength
SCL	As specified in PSEL.SCL	Input	Not applicable	S0D1
SDA	As specified in PSEL.SDA	Input	Not applicable	S0D1

33.8 Registers

Table 75: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40003000	TWIM	TWIM0	Two-wire interface master 0	
0x40004000	TWIM	TWIM1	Two-wire interface master 1	

Table 76: Register Overview

Register	Offset	Description
TASKS_STARTRX	0x000	Start TWI receive sequence
TASKS_STARTTX	0x008	Start TWI transmit sequence
TASKS_STOP	0x014	Stop TWI transaction. Must be issued while the TWI master is not suspended.
TASKS_SUSPEND	0x01C	Suspend TWI transaction
TASKS_RESUME	0x020	Resume TWI transaction
EVENTS_STOPPED	0x104	TWI stopped
EVENTS_ERROR	0x124	TWI error
EVENTS_SUSPENDED	0x148	Last byte has been sent out after the SUSPEND task has been issued, TWI traffic is now suspended.
EVENTS_RXSTARTED	0x14C	Receive sequence started
EVENTS_TXSTARTED	0x150	Transmit sequence started
EVENTS_LASTRX	0x15C	Byte boundary, starting to receive the last byte
EVENTS_LASTTX	0x160	Byte boundary, starting to transmit the last byte
SHORTS	0x200	Shortcut register
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ERRORSRC	0x4C4	Error source
ENABLE	0x500	Enable TWIM
PSEL.SCL	0x508	Pin select for SCL signal
PSEL.SDA	0x50C	Pin select for SDA signal
FREQUENCY	0x524	TWI frequency
RXD.PTR	0x534	Data pointer
RXD.MAXCNT	0x538	Maximum number of bytes in receive buffer
RXD.AMOUNT	0x53C	Number of bytes transferred in the last transaction
RXD.LIST	0x540	EasyDMA list type
TXD.PTR	0x544	Data pointer
TXD.MAXCNT	0x548	Maximum number of bytes in transmit buffer
TXD.AMOUNT	0x54C	Number of bytes transferred in the last transaction
TXD.LIST	0x550	EasyDMA list type
ADDRESS	0x588	Address used in the TWI transfer

33.8.1 SHORTS

Address offset: 0x200

Shortcut register

Bit number																															
Id																							F		D	C	B	A			
Reset 0x00000000																							**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	LASTTX_STARTRX			Shortcut between LASTTX event and STARTRX task See *EVENTS_LASTTX* and *TASKS_STARTRX*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
B	RW	LASTTX_SUSPEND			Shortcut between LASTTX event and SUSPEND task See *EVENTS_LASTTX* and *TASKS_SUSPEND*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
C	RW	LASTTX_STOP			Shortcut between LASTTX event and STOP task See *EVENTS_LASTTX* and *TASKS_STOP*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
D	RW	LASTRX_STARTTX			Shortcut between LASTRX event and STARTTX task See *EVENTS_LASTRX* and *TASKS_STARTTX*
			Disabled	0	Disable shortcut

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																							F		D	C	B	A							
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Enabled	1																															
F	RW	LASTRX_STOP																																	
			Disabled	0																															
			Enabled	1																															

33.8.2 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											J	I			H	G	F									D							A	
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	STOPPED																																	
			Disabled	0																															
			Enabled	1																															
D	RW	ERROR																																	
			Disabled	0																															
			Enabled	1																															
F	RW	SUSPENDED																																	
			Disabled	0																															
			Enabled	1																															
G	RW	RXSTARTED																																	
			Disabled	0																															
			Enabled	1																															
H	RW	TXSTARTED																																	
			Disabled	0																															
			Enabled	1																															
I	RW	LASTRX																																	
			Disabled	0																															
			Enabled	1																															
J	RW	LASTTX																																	
			Disabled	0																															
			Enabled	1																															

33.8.3 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											J	I			H	G	F									D								A	
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	STOPPED																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											J	I			H	G	F									D								A		
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
F	RW	SUSPENDED																																	
				See *EVENTS_SUSPENDED*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
G	RW	RXSTARTED																																	
				See *EVENTS_RXSTARTED*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
H	RW	TXSTARTED																																	
				See *EVENTS_TXSTARTED*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
I	RW	LASTRX																																	
				See *EVENTS_LASTRX*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
J	RW	LASTTX																																	
				See *EVENTS_LASTTX*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															

33.8.5 ERRORSRC

Address offset: 0x4C4

Error source

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	OVERRUN																																	
				A new byte was received before previous byte got transferred into RXD buffer. (Previous data is lost)																																
			NotReceived	0																															
			Received	1																															
B	RW	ANACK																																	
			NotReceived	0																															
			Received	1																															
C	RW	DNACK																																	
			NotReceived	0																															
			Received	1																															

33.8.6 ENABLE

Address offset: 0x500

Enable TWIM

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	ENABLE			Enable or disable TWIM
			Disabled	0	Disable TWIM
			Enabled	6	Enable TWIM

33.8.7 PSEL.SCL

Address offset: 0x508

Pin select for SCL signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	PIN		[0..31]	Pin number
B	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

33.8.8 PSEL.SDA

Address offset: 0x50C

Pin select for SDA signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	PIN		[0..31]	Pin number
B	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

33.8.9 FREQUENCY

Address offset: 0x524

TWI frequency

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x04000000				**0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	FREQUENCY			TWI master clock frequency
			K100	0x01980000	100 kbps
			K250	0x04000000	250 kbps
			K400	0x06400000	400 kbps

33.8.10 RXD.PTR

Address offset: 0x534

Data pointer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PTR																																	
				Data pointer																															

33.8.11 RXD.MAXCNT

Address offset: 0x538

Maximum number of bytes in receive buffer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	MAXCNT		[1..255]																															
				Maximum number of bytes in receive buffer																															

33.8.12 RXD.AMOUNT

Address offset: 0x53C

Number of bytes transferred in the last transaction

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	AMOUNT																																	
				Number of bytes transferred in the last transaction. In case of NACK error, includes the NACK'ed byte.																																	

33.8.13 RXD.LIST

Address offset: 0x540

EasyDMA list type

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	LIST																																	
				List type																															
			Disabled	0																															
				Disable EasyDMA list																															
			ArrayList	1																															
				Use array list																															

33.8.14 TXD.PTR

Address offset: 0x544

Data pointer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PTR																																	
				Data pointer																															

33.8.15 TXD.MAXCNT

Address offset: 0x548

Maximum number of bytes in transmit buffer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	MAXCNT		[1..255]	Maximum number of bytes in transmit buffer

33.8.16 TXD.AMOUNT

Address offset: 0x54C

Number of bytes transferred in the last transaction

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	R	AMOUNT			Number of bytes transferred in the last transaction. In case of NACK error, includes the NACK'ed byte.

33.8.17 TXD.LIST

Address offset: 0x550

EasyDMA list type

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	LIST			List type
			Disabled	0	Disable EasyDMA list
			ArrayList	1	Use array list

33.8.18 ADDRESS

Address offset: 0x588

Address used in the TWI transfer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	ADDRESS			Address used in the TWI transfer

33.9 Electrical specification

33.9.1 TWIM interface electrical specifications

Symbol	Description	Min.	Typ.	Max.	Units
f_{TWIM}	Bit rates for TWIM[30]	100		400	kbps
$I_{TWIM,100kbps}$	Run current for TWIM, 100 kbps		50		µA
$I_{TWIM,400kbps}$	Run current for TWIM, 400 kbps		50		µA
$t_{TWIM,START,LP}$	Time from STARTRX/STARTTX task to transmission started, Low power mode		$t_{TWIM,START}$ + t_{START_HFIN}		µs
$t_{TWIM,START,CL}$	Time from STARTRX/STARTTX task to transmission started, Constant latency mode		1.5		µs

[30] Higher bit rates or stronger pull-ups may require GPIOs to be set as High Drive, see GPIO chapter for more details.

33.9.2 Two Wire Interface Master (TWIM) timing specifications

Symbol	Description	Min.	Typ.	Max.	Units
$f_{TWIM,SCL,100kbps}$	SCL clock frequency, 100 kbps		100		kHz
$f_{TWIM,SCL,250kbps}$	SCL clock frequency, 250 kbps		250		kHz
$f_{TWIM,SCL,400kbps}$	SCL clock frequency, 400 kbps		400		kHz
t_{TWIM,SU_DAT}	Data setup time before positive edge on SCL – all modes	300			ns
t_{TWIM,HD_DAT}	Data hold time after negative edge on SCL – all modes	500			ns
$t_{TWIM,HD_STA,100kbps}$	TWIM master hold time for START and repeated START condition, 100 kbps	10000			ns
$t_{TWIM,HD_STA,250kbps}$	TWIM master hold time for START and repeated START condition, 250kbps	4000			ns
$t_{TWIM,HD_STA,400kbps}$	TWIM master hold time for START and repeated START condition, 400 kbps	2500			ns
$t_{TWIM,SU_STO,100kbps}$	TWIM master setup time from SCL high to STOP condition, 100 kbps	5000			ns
$t_{TWIM,SU_STO,250kbps}$	TWIM master setup time from SCL high to STOP condition, 250 kbps	2000			ns
$t_{TWIM,SU_STO,400kbps}$	TWIM master setup time from SCL high to STOP condition, 400 kbps	1250			ns
$t_{TWIM,BUF,100kbps}$	TWIM master bus free time between STOP and START conditions, 100 kbps	5800			ns
$t_{TWIM,BUF,250kbps}$	TWIM master bus free time between STOP and START conditions, 250 kbps	2700			ns
$t_{TWIM,BUF,400kbps}$	TWIM master bus free time between STOP and START conditions, 400 kbps	2100			ns

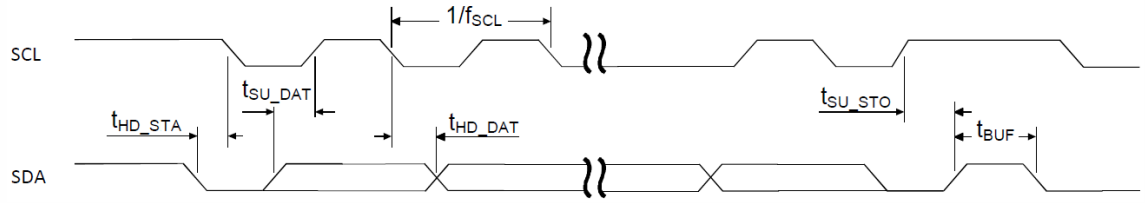


Figure 84: TWIM timing diagram, 1 byte transaction

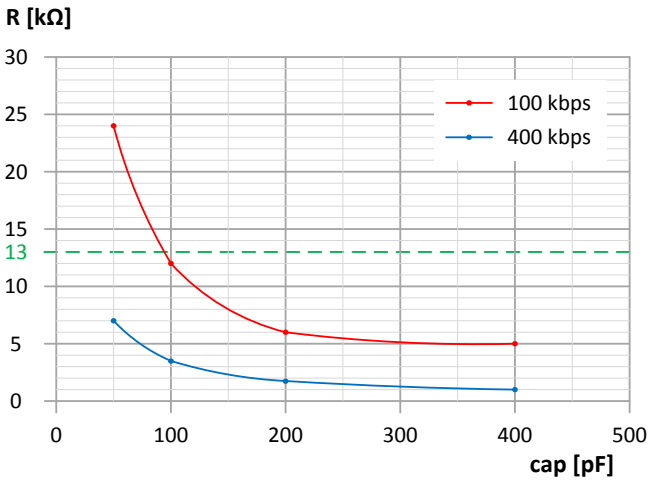


Figure 85: Recommended TWIM pullup value vs. line capacitance

- The I2C specification allows a line capacitance of 400 pF at most.
- The nRF52832 internal pullup has a fixed value of typ. 13 kOhm, see R_{PU} in the GPIO chapter.

34 TWIS — I²C compatible two-wire interface slave with EasyDMA

TWI slave with EasyDMA (TWIS) is compatible with I²C operating at 100 kHz and 400 kHz. The TWI transmitter and receiver implement EasyDMA.

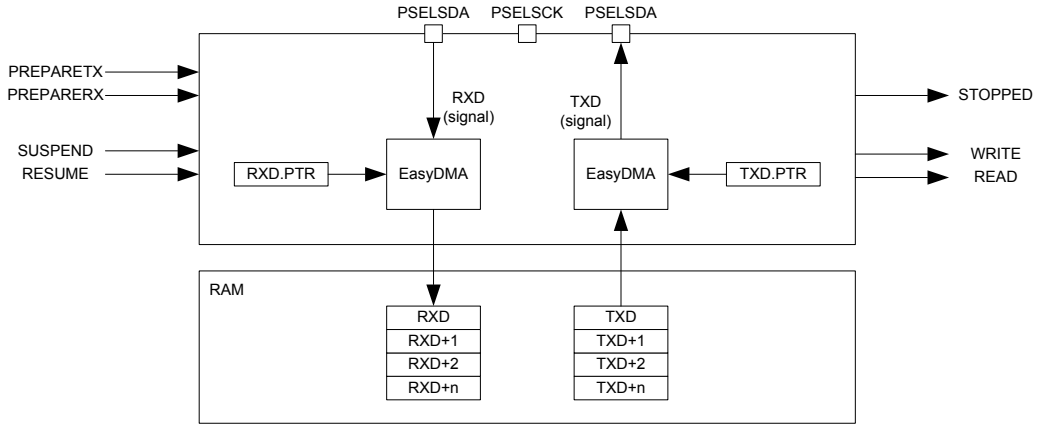


Figure 86: TWI slave with EasyDMA

A typical TWI setup consists of one master and one or more slaves. For an example, see *Figure 87: A typical TWI setup comprising one master and three slaves* on page 319. TWIS is only able to operate with a single master on the TWI bus.

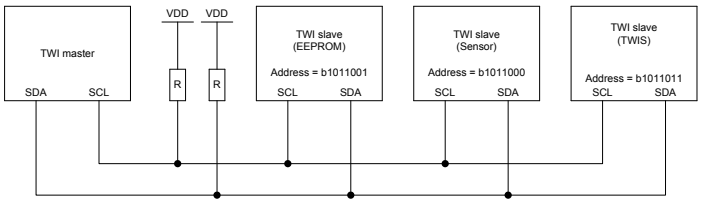


Figure 87: A typical TWI setup comprising one master and three slaves

The TWI slave state machine is illustrated in *Figure 88: TWI slave state machine* on page 320 and *Table 77: TWI slave state machine symbols* on page 320 is explaining the different symbols used in the state machine.

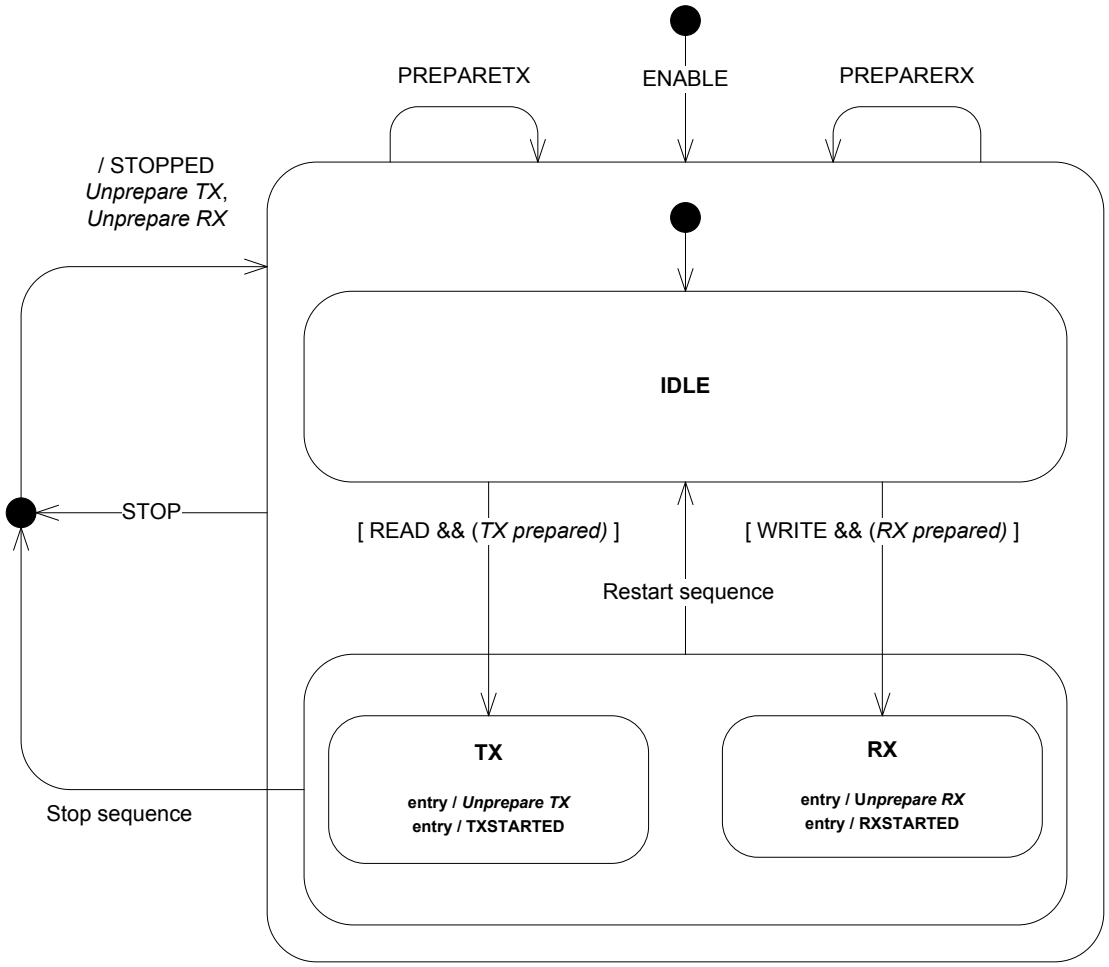


Figure 88: TWI slave state machine

Table 77: TWI slave state machine symbols

Symbol	Type	Description
ENABLE	Register	The TWI slave has been enabled via the *ENABLE* register
PREPARETX	Task	The *TASKS_PREPARETX* task has been triggered
STOP	Task	The *TASKS_STOP* task has been triggered
PREPARERX	Task	The *TASKS_PREPARERX* task has been triggered
STOPPED	Event	The *EVENTS_STOPPED* event was generated
RXSTARTED	Event	The *EVENTS_RXSTARTED* event was generated
TXSTARTED	Event	The *EVENTS_TXSTARTED* event was generated
TX prepared	Internal	Internal flag indicating that a *TASKS_PREPARETX* task has been triggered. This flag is not visible to the user.
RX prepared	Internal	Internal flag indicating that a *TASKS_PREPARERX* task has been triggered. This flag is not visible to the user.
Unprepare TX	Internal	Clears the internal 'TX prepared' flag until next *TASKS_PREPARETX* task.
Unprepare RX	Internal	Clears the internal 'RX prepared' flag until next *TASKS_PREPARERX* task.
Stop sequence	TWI protocol	A TWI stop sequence was detected
Restart sequence	TWI protocol	A TWI restart sequence was detected

The TWI slave supports clock stretching performed by the master.

The TWI slave operates in a low power mode while waiting for a TWI master to initiate a transfer. As long as the TWI slave is not addressed, it will remain in this low power mode.

To secure correct behaviour of the TWI slave, PSEL.SCL, PSEL.SDA, CONFIG and the ADDRESS[n] registers, must be configured prior to enabling the TWI slave through the ENABLE register. Similarly, changing these settings must be performed while the TWI slave is disabled. Failing to do so may result in unpredictable behaviour.

34.1 Shared resources

The TWI slave shares registers and other resources with other peripherals that have the same ID as the TWI slave.

Therefore, you must disable all peripherals that have the same ID as the TWI slave before the TWI slave can be configured and used. Disabling a peripheral that has the same ID as the TWI slave will not reset any of the registers that are shared with the TWI slave. It is therefore important to configure all relevant registers explicitly to secure that the TWI slave operates correctly.

The Instantiation table in *Instantiation* on page 24 shows which peripherals have the same ID as the TWI slave.

34.2 EasyDMA

The TWI slave implements EasyDMA for reading and writing to and from the RAM.

The STOPPED event indicates that EasyDMA has finished accessing the buffer in RAM.

If the TXD.PTR and the RXD.PTR are not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See *Memory* on page 23 for more information about the different memory regions.

34.3 TWI slave responding to a read command

Before the TWI slave can respond to a read command the TWI slave must be configured correctly and enabled via the ENABLE register. When enabled the TWI slave will be in its IDLE state where it will consume I_{IDLE} .

A read command is started when the TWI master generates a start condition on the TWI bus, followed by clocking out the address and the READ/WRITE bit set to 1 (WRITE=0, READ=1). The READ/WRITE bit is followed by an ACK/NACK bit (ACK=0 or NACK=1) response from the TWI slave.

The TWI slave is able to listen for up to two addresses at the same time. Which addresses to listen for is configured in the ADDRESS registers and the CONFIG register.

The TWI slave will only acknowledge (ACK) the read command if the address presented by the master matches one of the addresses the slave is configured to listen for. The TWI slave will generate a READ event when it acknowledges the read command.

The TWI slave is only able to detect a read command from the IDLE state.

The TWI slave will set an internal 'TX prepared' flag when the PREPARETX task is triggered.

When the read command is received the TWI slave will enter the TX state if the internal 'TX prepared' flag is set.

If the internal 'TX prepared' flag is not set when the read command is received, the TWI slave will stretch the master's clock until the PREPARETX task is triggered and the internal 'TX prepared' flag is set.

The TWI slave will generate the TXSTARTED event and clear the 'TX prepared' flag ('unprepare TX') when it enters the TX state. In this state the TWI slave will send the data bytes found in the transmit buffer to the master using the master's clock. The TWI slave will consume I_{TX} in this mode.

The TWI slave will go back to the IDLE state if the TWI slave receives a restart command when it is in the TX state.

The TWI slave is stopped when it receives the stop condition from the TWI master. A STOPPED event will be generated when the transaction has stopped. The TWI slave will clear the 'TX prepared' flag ('unprepare TX') and go back to the IDLE state when it has stopped.

The transmit buffer is located in RAM at the address specified in the TXD.PTR register. The TWI slave will only be able to send TXD.MAXCNT bytes from the transmit buffer for each transaction. If the TWI master

forces the slave to send more than TXD.MAXCNT bytes, the slave will send the byte specified in the ORC register to the master instead. If this happens, an ERROR event will be generated.

The EasyDMA configuration registers, see TXD.PTR etc., are latched when the TXSTARTED event is generated.

The TWI slave can be forced to stop by triggering the STOP task. A STOPPED event will be generated when the TWI slave has stopped. The TWI slave will clear the 'TX prepared' flag and go back to the IDLE state when it has stopped, see also *Terminating an ongoing TWI transaction* on page 324.

Each byte sent from the slave will be followed by an ACK/NACK bit sent from the master. The TWI master will generate a NACK following the last byte that it wants to receive to tell the slave to release the bus so that the TWI master can generate the stop condition. The TXD.AMOUNT register can be queried after a transaction to see how many bytes were sent.

A typical TWI slave read command response is illustrated in *Figure 89: The TWI slave responding to a read command* on page 322. Occurrence 2 in the figure illustrates clock stretching performed by the TWI slave following a SUSPEND task.

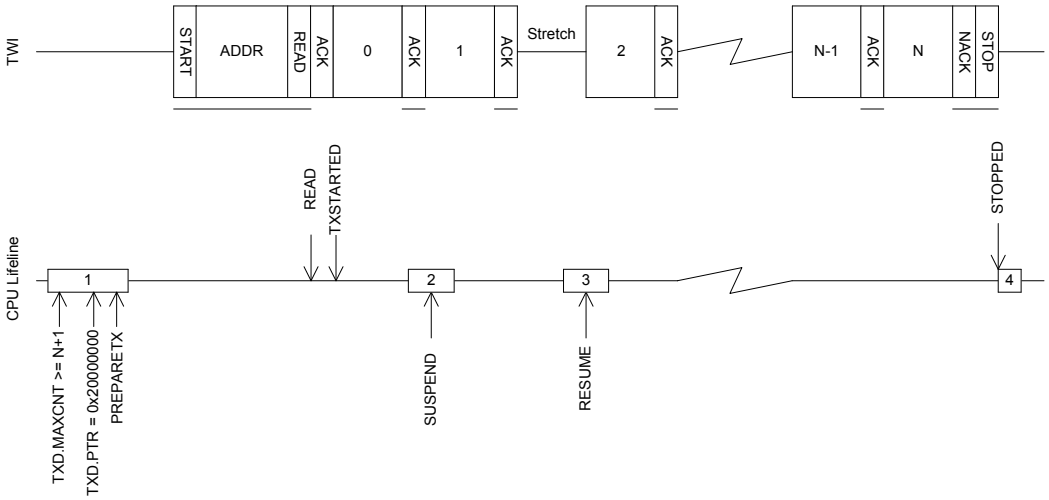


Figure 89: The TWI slave responding to a read command

34.4 TWI slave responding to a write command

Before the TWI slave can respond to a write command the TWI slave must be configured correctly and enabled via the ENABLE register. When enabled the TWI slave will be in its IDLE state where it will consume I_{IDLE}.

A write command is started when the TWI master generates a start condition on the TWI bus, followed by clocking out the address and the READ/WRITE bit set to 0 (WRITE=0, READ=1). The READ/WRITE bit is followed by an ACK/NACK bit (ACK=0 or NACK=1) response from the slave.

The TWI slave is able to listen for up to two addresses at the same time. Which addresses to listen for is configured in the ADDRESS registers and the CONFIG register.

The TWI slave will only acknowledge (ACK) the write command if the address presented by the master matches one of the addresses the slave is configured to listen for. The TWI slave will generate a WRITE event if it acknowledges the write command.

The TWI slave is only able to detect a write command from the IDLE state.

The TWI slave will set an internal 'RX prepared' flag when the PREPARERX task is triggered.

When the write command is received the TWI slave will enter the RX state if the internal 'RX prepared' flag is set.

If the internal 'RX prepared' flag is not set when the write command is received, the TWI slave will stretch the master's clock until the PREPARERX task is triggered and the internal 'RX prepared' flag is set.

The TWI slave will generate the RXSTARTED event and clear the internal 'RX prepared' flag ('unprepare RX') when it enters the RX state. In this state the TWI slave will be able to receive the bytes sent by the TWI master. The TWI slave will consume I_{RX} in this mode.

The TWI slave will go back to the IDLE state if the TWI slave receives a restart command when it is in the RX state.

The TWI slave is stopped when it receives the stop condition from the TWI master. A STOPPED event will be generated when the transaction has stopped. The TWI slave will clear the internal 'RX prepared' flag ('unprepare RX') and go back to the IDLE state when it has stopped.

The receive buffer is located in RAM at the address specified in the TXD.PTR register. The TWI slave will only be able to receive as many bytes as specified in the RXD.MAXCNT register. If the TWI master tries to send more bytes to the slave than the slave is able to receive,these bytes will be discarded and the bytes will be NACKed by the slave. If this happens, an ERROR event will be generated.

The EasyDMA configuration registers, see RXD.PTR etc., are latched when the RXSTARTED event is generated.

The TWI slave can be forced to stop by triggering the STOP task. A STOPPED event will be generated when the TWI slave has stopped. The TWI slave will clear the internal 'RX prepared' flag and go back to the IDLE state when it has stopped, see also *Terminating an ongoing TWI transaction* on page 324.

The TWI slave will generate an ACK after every byte received from the master. The RXD.AMOUNT register can be queried after a transaction to see how many bytes were received.

A typical TWI slave write command response is illustrated in *Figure 90: The TWI slave responding to a write command* on page 323. Occurrence 2 in the figure illustrates clock stretching performed by the TWI slave following a SUSPEND task.

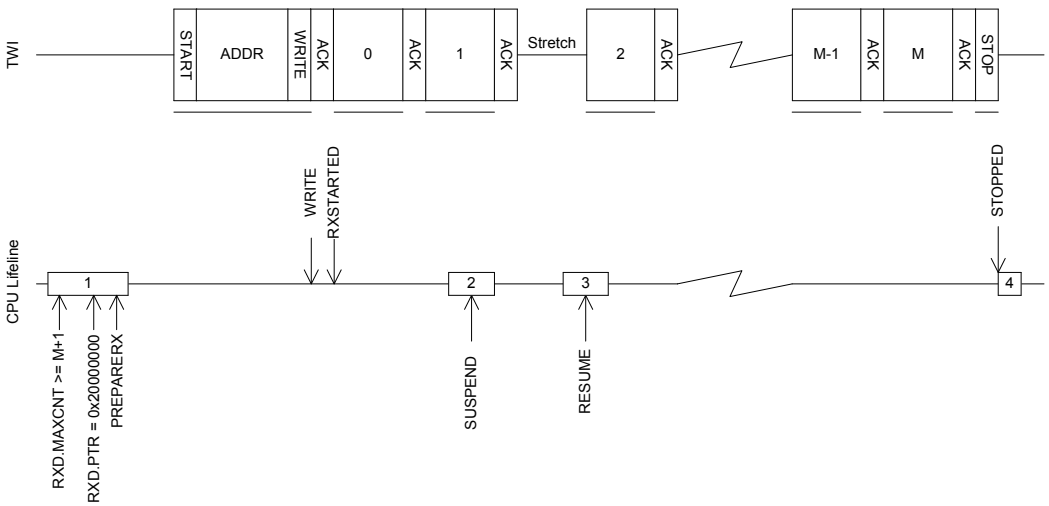


Figure 90: The TWI slave responding to a write command

34.5 Master repeated start sequence

An example of a repeated start sequence is one in which the TWI master writes two bytes to the slave followed by reading four bytes from the slave.

This is illustrated in *Figure 91: A repeated start sequence, where the TWI master writes two bytes followed by reading four bytes from the slave* on page 324.

It is here assumed that the receiver does not know in advance what the master wants to read, and that this information is provided in the first two bytes received in the write part of the repeated start sequence. To guarantee that the CPU is able to process the received data before the TWI slave starts to reply to the read command, the SUSPEND task is triggered via a shortcut from the READ event generated when the read command is received. When the CPU has processed the incoming data and prepared the correct data response, the CPU will resume the transaction by triggering the RESUME task.

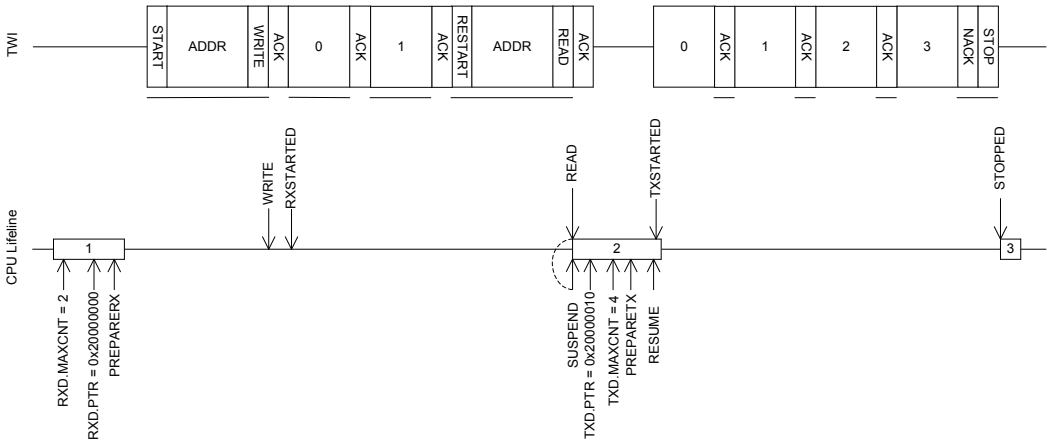


Figure 91: A repeated start sequence, where the TWI master writes two bytes followed by reading four bytes from the slave

34.6 Terminating an ongoing TWI transaction

In some situations, e.g. if the external TWI master is not responding correctly, it may be required to terminate an ongoing transaction.

This can be achieved by triggering the STOP task. In this situation a STOPPED event will be generated when the TWI has stopped independent of whether or not a STOP condition has been generated on the TWI bus. The TWI slave will release the bus when it has stopped and go back to its IDLE state.

34.7 Low power

When putting the system in low power and the peripheral is not needed, lowest possible power consumption is achieved by stopping, and then disabling the peripheral.

The STOP task may not be always needed (the peripheral might already be stopped), but if it is sent, software shall wait until the STOPPED event was received as a response before disabling the peripheral through the ENABLE register.

34.8 Slave mode pin configuration

The SCL and SDA signals associated with the TWI slave are mapped to physical pins according to the configuration specified in the PSEL.SCL and PSEL.SDA registers respectively.

The PSEL.SCL and PSEL.SDA registers and their configurations are only used as long as the TWI slave is enabled, and retained only as long as the device is in ON mode. When the peripheral is disabled, the pins will behave as regular GPIOs, and use the configuration in their respective OUT bit field and PIN_CNF[n] register. PSEL.SCL and PSEL.SDA must only be configured when the TWI slave is disabled.

To secure correct signal levels on the pins used by the TWI slave when the system is in OFF mode, and when the TWI slave is disabled, these pins must be configured in the GPIO peripheral as described in *Table 78: GPIO configuration before enabling peripheral* on page 324.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behavior.

Table 78: GPIO configuration before enabling peripheral

TWI slave signal	TWI slave pin	Direction	Output value	Drive strength
SCL	As specified in PSEL.SCL	Input	Not applicable	S0D1
SDA	As specified in PSEL.SDA	Input	Not applicable	S0D1

34.9 Registers

Table 79: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40003000	TWIS	TWIS0	Two-wire interface slave 0	
0x40004000	TWIS	TWIS1	Two-wire interface slave 1	

Table 80: Register Overview

Register	Offset	Description
TASKS_STOP	0x014	Stop TWI transaction
TASKS_SUSPEND	0x01C	Suspend TWI transaction
TASKS_RESUME	0x020	Resume TWI transaction
TASKS_PREPARERX	0x030	Prepare the TWI slave to respond to a write command
TASKS_PREPARETX	0x034	Prepare the TWI slave to respond to a read command
EVENTS_STOPPED	0x104	TWI stopped
EVENTS_ERROR	0x124	TWI error
EVENTS_RXSTARTED	0x14C	Receive sequence started
EVENTS_TXSTARTED	0x150	Transmit sequence started
EVENTS_WRITE	0x164	Write command received
EVENTS_READ	0x168	Read command received
SHORTS	0x200	Shortcut register
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ERRORSRC	0x4D0	Error source
MATCH	0x4D4	Status register indicating which address had a match
ENABLE	0x500	Enable TWIS
PSEL.SCL	0x508	Pin select for SCL signal
PSEL.SDA	0x50C	Pin select for SDA signal
RXD.PTR	0x534	RXD Data pointer
RXD.MAXCNT	0x538	Maximum number of bytes in RXD buffer
RXD.AMOUNT	0x53C	Number of bytes transferred in the last RXD transaction
TXD.PTR	0x544	TXD Data pointer
TXD.MAXCNT	0x548	Maximum number of bytes in TXD buffer
TXD.AMOUNT	0x54C	Number of bytes transferred in the last TXD transaction
ADDRESS[0]	0x588	TWI slave address 0
ADDRESS[1]	0x58C	TWI slave address 1
CONFIG	0x594	Configuration register for the address match mechanism
ORC	0x5C0	Over-read character. Character sent out in case of an over-read of the transmit buffer.

34.9.1 SHORTS

Address offset: 0x200

Shortcut register

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		B	A													
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	WRITE_SUSPEND			Shortcut between WRITE event and SUSPEND task See *EVENTS_WRITE* and *TASKS_SUSPEND*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
B	RW	READ_SUSPEND			Shortcut between READ event and SUSPEND task See *EVENTS_READ* and *TASKS_SUSPEND*

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																					B	A													
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
			Disabled	0																						Disable shortcut									
			Enabled	1																						Enable shortcut									

34.9.2 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id									H	G					F	E										B							A		
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	RW	STOPPED																								Enable or disable interrupt for STOPPED event See *EVENTS_STOPPED*									
			Disabled	0																						Disable									
			Enabled	1																						Enable									
B	RW	ERROR																								Enable or disable interrupt for ERROR event See *EVENTS_ERROR*									
			Disabled	0																						Disable									
			Enabled	1																						Enable									
E	RW	RXSTARTED																								Enable or disable interrupt for RXSTARTED event See *EVENTS_RXSTARTED*									
			Disabled	0																						Disable									
			Enabled	1																						Enable									
F	RW	TXSTARTED																								Enable or disable interrupt for TXSTARTED event See *EVENTS_TXSTARTED*									
			Disabled	0																						Disable									
			Enabled	1																						Enable									
G	RW	WRITE																								Enable or disable interrupt for WRITE event See *EVENTS_WRITE*									
			Disabled	0																						Disable									
			Enabled	1																						Enable									
H	RW	READ																								Enable or disable interrupt for READ event See *EVENTS_READ*									
			Disabled	0																						Disable									
			Enabled	1																						Enable									

34.9.3 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id									H	G					F	E										B							A		
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	RW	STOPPED																								Write '1' to Enable interrupt for STOPPED event See *EVENTS_STOPPED*									
			Set	1																						Enable									
			Disabled	0																						Read: Disabled									
			Enabled	1																						Read: Enabled									
B	RW	ERROR																								Write '1' to Enable interrupt for ERROR event									

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id									H	G					F	E										B								A	
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
																										See *EVENTS_ERROR*									
			Set	1																						Enable									
			Disabled	0																						Read: Disabled									
			Enabled	1																						Read: Enabled									
E	RW	RXSTARTED																								Write '1' to Enable interrupt for RXSTARTED event									
																										See *EVENTS_RXSTARTED*									
			Set	1																						Enable									
			Disabled	0																						Read: Disabled									
			Enabled	1																						Read: Enabled									
F	RW	TXSTARTED																								Write '1' to Enable interrupt for TXSTARTED event									
																										See *EVENTS_TXSTARTED*									
			Set	1																						Enable									
			Disabled	0																						Read: Disabled									
			Enabled	1																						Read: Enabled									
G	RW	WRITE																								Write '1' to Enable interrupt for WRITE event									
																										See *EVENTS_WRITE*									
			Set	1																						Enable									
			Disabled	0																						Read: Disabled									
			Enabled	1																						Read: Enabled									
H	RW	READ																								Write '1' to Enable interrupt for READ event									
																										See *EVENTS_READ*									
			Set	1																						Enable									
			Disabled	0																						Read: Disabled									
			Enabled	1																						Read: Enabled									

34.9.4 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id									H	G					F	E										B								A	
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	RW	STOPPED																								Write '1' to Disable interrupt for STOPPED event									
																										See *EVENTS_STOPPED*									
			Clear	1																						Disable									
			Disabled	0																						Read: Disabled									
			Enabled	1																						Read: Enabled									
B	RW	ERROR																								Write '1' to Disable interrupt for ERROR event									
																										See *EVENTS_ERROR*									
			Clear	1																						Disable									
			Disabled	0																						Read: Disabled									
			Enabled	1																						Read: Enabled									
E	RW	RXSTARTED																								Write '1' to Disable interrupt for RXSTARTED event									
																										See *EVENTS_RXSTARTED*									
			Clear	1																						Disable									
			Disabled	0																						Read: Disabled									
			Enabled	1																						Read: Enabled									
F	RW	TXSTARTED																								Write '1' to Disable interrupt for TXSTARTED event									
																										See *EVENTS_TXSTARTED*									
			Clear	1																						Disable									

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id									H	G					F	E										B								A	
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
			Disabled	0																						Read: Disabled									
			Enabled	1																						Read: Enabled									
G	RW	WRITE																								Write '1' to Disable interrupt for WRITE event See *EVENTS_WRITE*									
			Clear	1																						Disable									
			Disabled	0																						Read: Disabled									
			Enabled	1																						Read: Enabled									
H	RW	READ																								Write '1' to Disable interrupt for READ event See *EVENTS_READ*									
			Clear	1																						Disable									
			Disabled	0																						Read: Disabled									
			Enabled	1																						Read: Enabled									

34.9.5 ERRORSRC

Address offset: 0x4D0

Error source

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																C	B		A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	RW	OVERFLOW																								RX buffer overflow detected, and prevented									
			NotDetected	0																						Error did not occur									
			Detected	1																						Error occurred									
B	RW	DNACK																								NACK sent after receiving a data byte									
			NotReceived	0																						Error did not occur									
			Received	1																						Error occurred									
C	RW	OVERREAD																								TX buffer over-read detected, and prevented									
			NotDetected	0																						Error did not occur									
			Detected	1																						Error occurred									

34.9.6 MATCH

Address offset: 0x4D4

Status register indicating which address had a match

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	R	MATCH		[0..1]																						Which of the addresses in {ADDRESS} matched the incoming address									

34.9.7 ENABLE

Address offset: 0x500

Enable TWIS

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	RW	ENABLE																								Enable or disable TWIS									
			Disabled	0																						Disable TWIS									

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
			Enabled	9	Enable TWIS

34.9.8 PSEL.SCL

Address offset: 0x508

Pin select for SCL signal

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			B A A A A A
Reset 0xFFFFFFFF			**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	PIN		[0..31]	Pin number
B	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

34.9.9 PSEL.SDA

Address offset: 0x50C

Pin select for SDA signal

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			B A A A A A
Reset 0xFFFFFFFF			**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	PIN		[0..31]	Pin number
B	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

34.9.10 RXD.PTR

Address offset: 0x534

RXD Data pointer

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	PTR			RXD Data pointer

34.9.11 RXD.MAXCNT

Address offset: 0x538

Maximum number of bytes in RXD buffer

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A A A A A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	MAXCNT			Maximum number of bytes in RXD buffer

34.9.12 RXD.AMOUNT

Address offset: 0x53C

Number of bytes transferred in the last RXD transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	R	AMOUNT			Number of bytes transferred in the last RXD transaction

34.9.13 TXD.PTR

Address offset: 0x544

TXD Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	PTR			TXD Data pointer

34.9.14 TXD.MAXCNT

Address offset: 0x548

Maximum number of bytes in TXD buffer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	MAXCNT			Maximum number of bytes in TXD buffer

34.9.15 TXD.AMOUNT

Address offset: 0x54C

Number of bytes transferred in the last TXD transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	R	AMOUNT			Number of bytes transferred in the last TXD transaction

34.9.16 ADDRESS[0]

Address offset: 0x588

TWI slave address 0

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	ADDRESS			TWI slave address

34.9.17 ADDRESS[1]

Address offset: 0x58C

TWI slave address 1

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	ADDRESS			TWI slave address

34.9.18 CONFIG

Address offset: 0x594

Configuration register for the address match mechanism

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		B	A
Reset 0x00000001				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																														
A	RW	ADDRESS0			Enable or disable address matching on ADDRESS[0]																																
			Disabled	0	Disabled																																
			Enabled	1	Enabled																																
B	RW	ADDRESS1			Enable or disable address matching on ADDRESS[1]																																
			Disabled	0	Disabled																																
			Enabled	1	Enabled																																

34.9.19 ORC

Address offset: 0x5C0

Over-read character. Character sent out in case of an over-read of the transmit buffer.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																														
A	RW	ORC			Over-read character. Character sent out in case of an over-read of the transmit buffer.																																

34.10 Electrical specification

34.10.1 TWIS slave interface electrical specifications

Symbol	Description	Min.	Typ.	Max.	Units
f_{TWIS}	Bit rates for TWIS[31]	100		400	kbps
$I_{TWIS,100kbps}$	Run current for TWIS (Average current to receive and transfer a byte to RAM), 100 kbps		45		µA
$I_{TWIS,400kbps}$	Run current for TWIS (Average current to receive and transfer a byte to RAM), 400 kbps		45		µA
$I_{TWIS,IDLE}$	Idle current for TWIS		1		µA
$t_{TWIS,START,LP}$	Time from PREPARERX/PREPARETX task to ready to receive/ transmit, Low power mode		$t_{TWIS,START,}$ + t_{START_HFIN}		µs
$t_{TWIS,START,CL}$	Time from PREPARERX/PREPARETX task to ready to receive/ transmit, Constant latency mode		1.5		µs

34.10.2 TWIS slave timing specifications

Symbol	Description	Min.	Typ.	Max.	Units
$f_{TWIS,SCL,400kbps}$	SCL clock frequency, 400 kbps			400	kHz
t_{TWIS,SU_DAT}	Data setup time before positive edge on SCL – all modes	300			ns
t_{TWIS,HD_DAT}	Data hold time after negative edge on SCL – all modes	500			ns
$t_{TWIS,HD_STA,100kbps}$	TWI slave hold time from for START condition (SDA low to SCL low), 100 kbps	5200			ns
$t_{TWIS,HD_STA,400kbps}$	TWI slave hold time from for START condition (SDA low to SCL low), 400 kbps	1300			ns

[31] Higher bit rates or stronger pull-ups may require GPIOs to be set as High Drive, see GPIO chapter for more details.

Symbol	Description	Min.	Typ.	Max.	Units
$t_{TWIS,SU_STO,100kbps}$	TWI slave setup time from SCL high to STOP condition, 100 kbps	5200			ns
$t_{TWIS,SU_STO,400kbps}$	TWI slave setup time from SCL high to STOP condition, 400 kbps	1300			ns
$t_{TWIS,BUF,100kbps}$	TWI slave bus free time between STOP and START conditions, 100 kbps		4700		ns
$t_{TWIS,BUF,400kbps}$	TWI slave bus free time between STOP and START conditions, 400 kbps		1300		ns

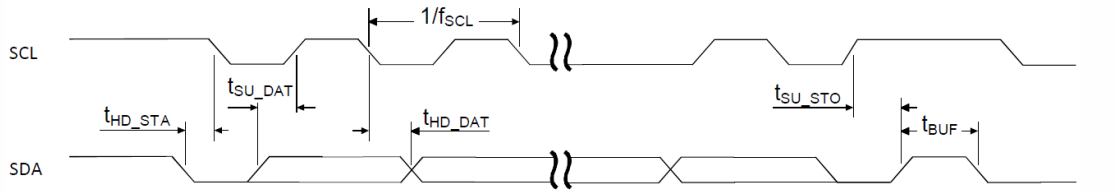


Figure 92: TWIS timing diagram, 1 byte transaction

35 UARTE — Universal asynchronous receiver/ transmitter with EasyDMA

The Universal asynchronous receiver/transmitter with EasyDMA (UARTE) offers fast, full-duplex, asynchronous serial communication with built-in flow control (CTS, RTS) support in hardware at a rate up to 1 Mbps, and EasyDMA data transfer from/to RAM.

Listed here are the main features for UARTE:

- Full-duplex operation
- Automatic hardware flow control
- Parity checking and generation for the 9th data bit
- EasyDMA
- Up to 1 Mbps baudrate
- Return to IDLE between transactions supported (when using HW flow control)
- One stop bit
- Least significant bit (LSB) first

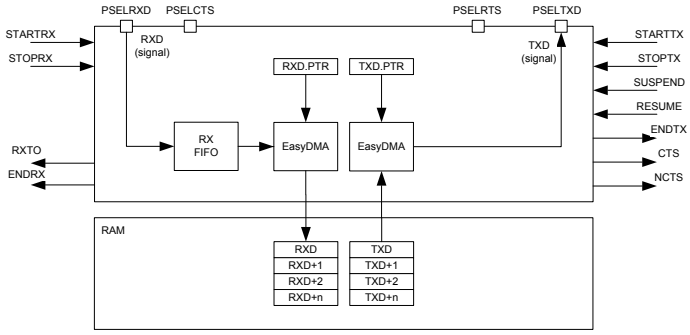


Figure 93: UARTE configuration

The GPIOs used for each UART interface can be chosen from any GPIO on the device and are independently configurable. This enables great flexibility in device pinout and efficient use of board space and signal routing.

35.1 Shared resources

The UARTE shares registers and other resources with other peripherals that have the same ID as the UARTE.

Therefore, you must disable all peripherals that have the same ID as the UARTE before the UARTE can be configured and used. Disabling a peripheral that has the same ID as the UARTE will not reset any of the registers that are shared with the UARTE. It is therefore important to configure all relevant UARTE registers explicitly to ensure that it operates correctly.

See the Instantiation table in *Instantiation* on page 24 for details on peripherals and their IDs.

35.2 EasyDMA

The UARTE implements EasyDMA for reading and writing to and from the RAM.

If the TXD.PTR and the RXD.PTR are not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See *Memory* on page 23 for more information about the different memory regions.

The .PTR and .MAXCNT registers are double-buffered. They can be updated and prepared for the next RX/TX transmission immediately after having received the RXSTARTED/TXSTARTED event.

The ENDRX/ENDTX event indicates that EasyDMA has finished accessing respectively the RX/TX buffer in RAM.

35.3 Transmission

The first step of a DMA transmission is storing bytes in the transmit buffer and configuring EasyDMA. This is achieved by writing the initial address pointer to TXD.PTR, and the number of bytes in the RAM buffer to TXD.MAXCNT. The UARTE transmission is started by triggering the STARTTX task.

After each byte has been sent over the TXD line, a TXDRDY event will be generated.

When all bytes in the TXD buffer, as specified in the TXD.MAXCNT register, have been transmitted, the UARTE transmission will end automatically and an ENDTX event will be generated.

A UARTE transmission sequence is stopped by triggering the STOPTX task, a TXSTOPPED event will be generated when the UARTE transmitter has stopped.

If the ENDTX event has not already been generated when the UARTE transmitter has come to a stop, the UARTE will generate the ENDTX event explicitly even though all bytes in the TXD buffer, as specified in the TXD.MAXCNT register, have not been transmitted.

If flow control is enabled, a transmission will be automatically suspended when CTS is deactivated and resumed when CTS is activated again, as illustrated in *Figure 94: UARTE transmission* on page 334.
A byte that is in transmission when CTS is deactivated will be fully transmitted before the transmission is suspended.

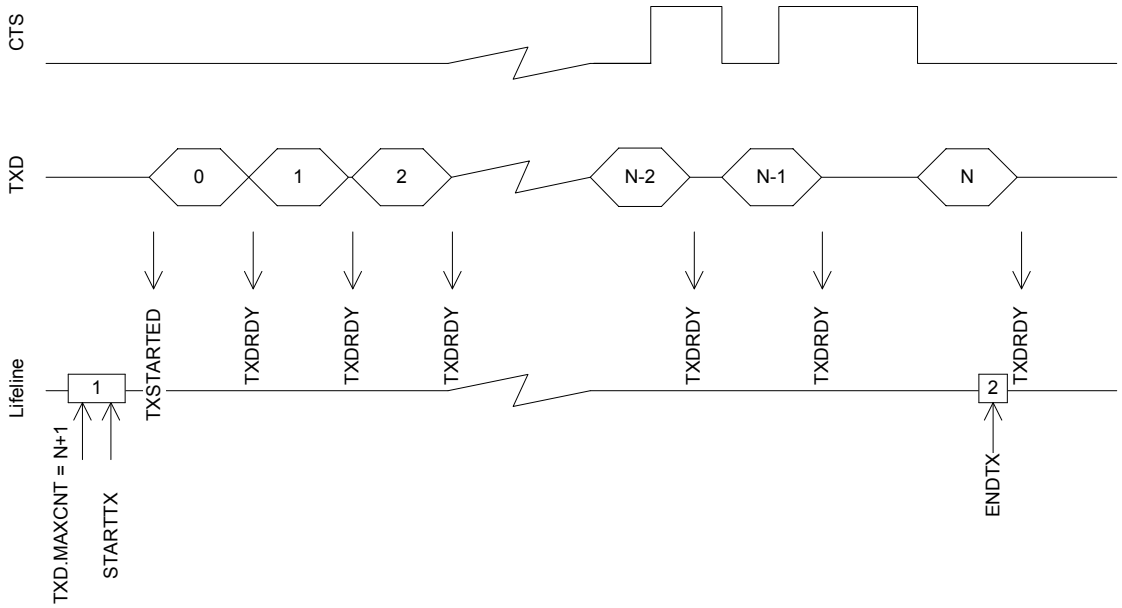


Figure 94: UARTE transmission

The UARTE transmitter will be in its lowest activity level, and consume the least amount of energy, when it is stopped, i.e. before it is started via STARTTX or after it has been stopped via STOPTX and the TXSTOPPED event has been generated. See *POWER — Power supply* on page 78 for more information about power modes.

35.4 Reception

The UARTE receiver is started by triggering the STARTRX task. The UARTE receiver is using EasyDMA to store incoming data in an RX buffer in RAM.

The RX buffer is located at the address specified in the RXD.PTR register. The RXD.PTR register is double-buffered and it can be updated and prepared for the next STARTRX task immediately after the RXSTARTED event is generated. The size of the RX buffer is specified in the RXD.MAXCNT register and the UARTE will generate an ENDRX event when it has filled up the RX buffer, see *Figure 95: UARTE reception* on page 335.

For each byte received over the RXD line, an RXDRDY event will be generated. This event is likely to occur before the corresponding data has been transferred to Data RAM.

The RXD.AMOUNT register can be queried following an ENDRX event to see how many new bytes have been transferred to the RX buffer in RAM since the previous ENDRX event.

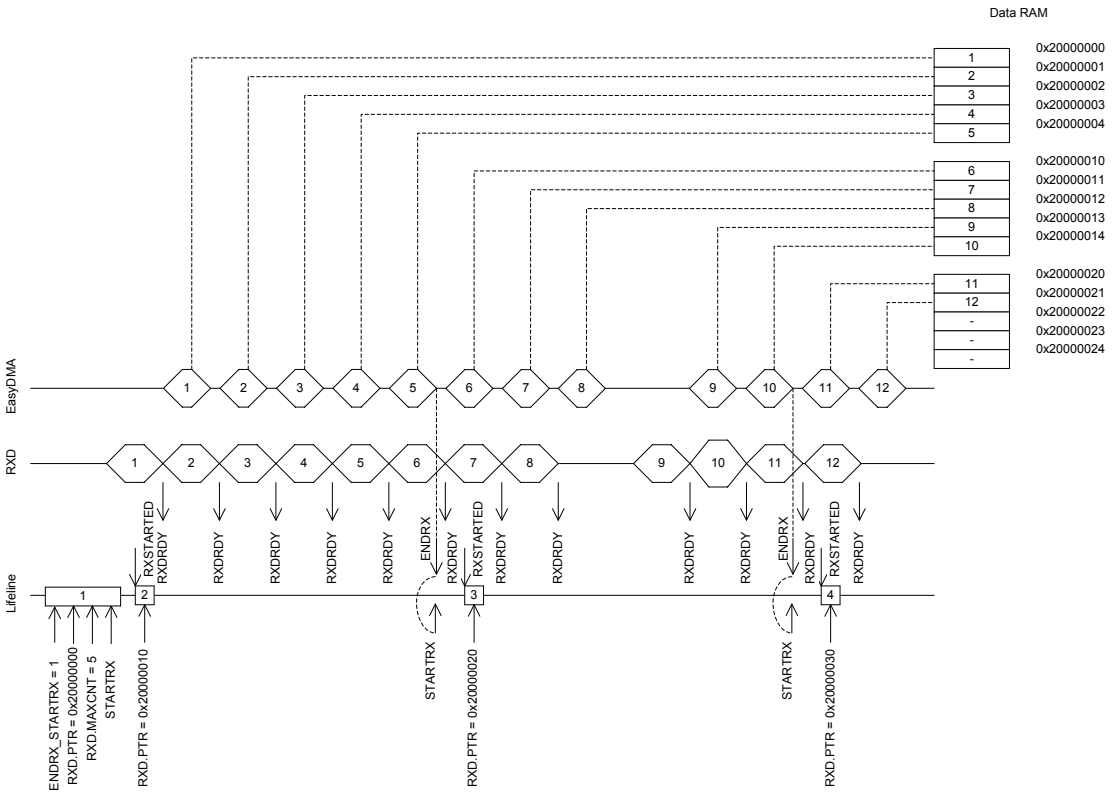


Figure 95: UARTE reception

The UARTE receiver is stopped by triggering the STOPRX task. An RXTO event is generated when the UARTE has stopped. The UARTE will make sure that an impending ENDRX event will be generated before the RXTO event is generated. This means that the UARTE will guarantee that no ENDRX event will be generated after RXTO, unless the UARTE is restarted or a FLUSHRX command is issued after the RXTO event is generated.

> **Important:** If the ENDRX event has not already been generated when the UARTE receiver has come to a stop, which implies that all pending content in the RX FIFO has been moved to the RX buffer, the UARTE will generate the ENDRX event explicitly even though the RX buffer is not full. In this scenario the ENDRX event will be generated before the RXTO event is generated.

To be able to know how many bytes have actually been received into the RX buffer, the CPU can read the RXD.AMOUNT register following the ENDRX event or the RXTO event.

The UARTE is able to receive up to four bytes after the STOPRX task has been triggered as long as these are sent in succession immediately after the RTS signal is deactivated. This is possible because after the RTS is deactivated the UARTE is able to receive bytes for an extended period equal to the time it takes to send 4 bytes on the configured baud rate.

After the RXTO event is generated the internal RX FIFO may still contain data, and to move this data to RAM the FLUSHRX task must be triggered. To make sure that this data does not overwrite data in the RX buffer, the RX buffer should be emptied or the RXD.PTR should be updated before the FLUSHRX task is triggered.

To make sure that all data in the RX FIFO is moved to the RX buffer, the RXD.MAXCNT register must be set to RXD.MAXCNT > 4, see *Figure 96: UARTE reception with forced stop via STOPRX* on page 336. The UARTE will generate the ENDRX event after completing the FLUSHRX task even if the RX FIFO was empty or if the RX buffer does not get filled up. To be able to know how many bytes have actually been received into the RX buffer in this case, the CPU can read the RXD.AMOUNT register following the ENDRX event.

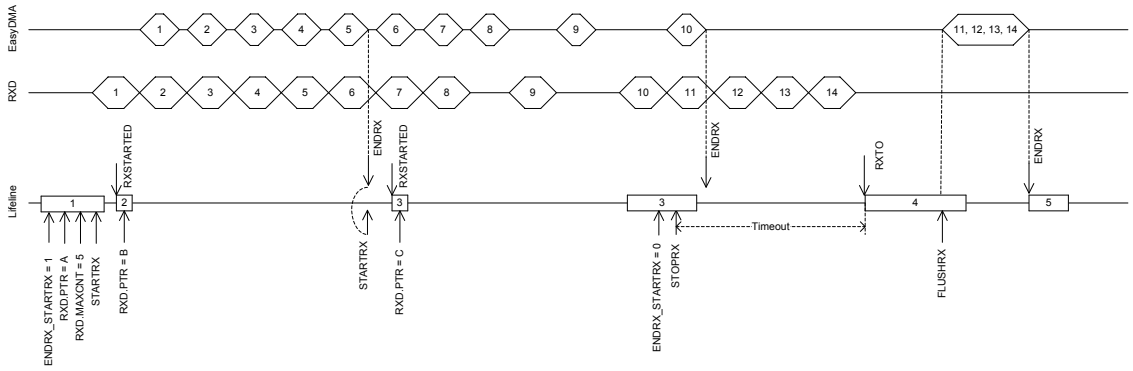


Figure 96: UARTE reception with forced stop via STOPRX

If HW flow control is enabled the RTS signal will be deactivated when the receiver is stopped via the STOPRX task or when the UARTE is only able to receive four more bytes in its internal RX FIFO.

With flow control disabled, the UARTE will function in the same way as when the flow control is enabled except that the RTS line will not be used. This means that no signal will be generated when the UARTE has reached the point where it is only able to receive four more bytes in its internal RX FIFO. Data received when the internal RX FIFO is filled up, will be lost.

The UARTE receiver will be in its lowest activity level, and consume the least amount of energy, when it is stopped, i.e. before it is started via STARTRX or after it has been stopped via STOPRX and the RXTO event has been generated. See *POWER — Power supply* on page 78 for more information about power modes.

35.5 Error conditions

An ERROR event, in the form of a framing error, will be generated if a valid stop bit is not detected in a frame. Another ERROR event, in the form of a break condition, will be generated if the RXD line is held active low for longer than the length of a data frame. Effectively, a framing error is always generated before a break condition occurs.

An ERROR event will not stop reception. If the error was a parity error, the received byte will still be transferred into Data RAM, and so will following incoming bytes. If there was a framing error (wrong stop bit), that specific byte will NOT be stored into Data RAM, but following incoming bytes will.

35.6 Using the UARTE without flow control

If flow control is not enabled, the interface will behave as if the CTS and RTS lines are kept active all the time.

35.7 Parity configuration

When parity is enabled, the parity will be generated automatically from the even parity of TXD and RXD for transmission and reception respectively.

35.8 Low power

When putting the system in low power and the peripheral is not needed, lowest possible power consumption is achieved by stopping, and then disabling the peripheral.

The STOPTX and STOPRX tasks may not be always needed (the peripheral might already be stopped), but if STOPTX and/or STOPRX is sent, software shall wait until the TXSTOPPED and/or RXTO event is received in response, before disabling the peripheral through the ENABLE register.

35.9 Pin configuration

The different signals RXD, CTS (Clear To Send, active low), RTS (Request To Send, active low), and TXD associated with the UARTE are mapped to physical pins according to the configuration specified in the PSEL.RXD, PSEL.CTS, PSEL.RTS, and PSEL.TXD registers respectively.

The PSEL.RXD, PSEL.CTS, PSEL.RTS, and PSEL.TXD registers and their configurations are only used as long as the UARTE is enabled, and retained only for the duration the device is in ON mode. PSEL.RXD, PSEL.RTS, PSEL.RTS and PSEL.TXD must only be configured when the UARTE is disabled.

To secure correct signal levels on the pins by the UARTE when the system is in OFF mode, the pins must be configured in the GPIO peripheral as described in *Table 81: GPIO configuration before enabling peripheral* on page 337.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behavior.

Table 81: GPIO configuration before enabling peripheral

UARTE signal	UARTE pin	Direction	Output value
RXD	As specified in PSEL.RXD	Input	Not applicable
CTS	As specified in PSEL.CTS	Input	Not applicable
RTS	As specified in PSEL.RTS	Output	1
TXD	As specified in PSEL.TXD	Output	1

35.10 Registers

Table 82: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40002000	UARTE	UARTE0	Universal Asynchronous Receiver/ Transmitter with EasyDMA	

Table 83: Register Overview

Register	Offset	Description
TASKS_STARTRX	0x000	Start UART receiver
TASKS_STOPRX	0x004	Stop UART receiver
TASKS_STARTTX	0x008	Start UART transmitter
TASKS_STOPTX	0x00C	Stop UART transmitter
TASKS_FLUSHRX	0x02C	Flush RX FIFO into RX buffer
EVENTS_CTS	0x100	CTS is activated (set low). Clear To Send.
EVENTS_NCTS	0x104	CTS is deactivated (set high). Not Clear To Send.
EVENTS_RXDRDY	0x108	Data received in RXD (but potentially not yet transferred to Data RAM)
EVENTS_ENDRX	0x110	Receive buffer is filled up
EVENTS_TXDRDY	0x11C	Data sent from TXD
EVENTS_ENDTX	0x120	Last TX byte transmitted
EVENTS_ERROR	0x124	Error detected
EVENTS_RXTO	0x144	Receiver timeout
EVENTS_RXSTARTED	0x14C	UART receiver has started
EVENTS_TXSTARTED	0x150	UART transmitter has started
EVENTS_TXSTOPPED	0x158	Transmitter stopped
SHORTS	0x200	Shortcut register
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt

Register	Offset	Description
ERRORSRC	0x480	Error source
ENABLE	0x500	Enable UART
PSEL.RTS	0x508	Pin select for RTS signal
PSEL.TXD	0x50C	Pin select for TXD signal
PSEL.CTS	0x510	Pin select for CTS signal
PSEL.RXD	0x514	Pin select for RXD signal
BAUDRATE	0x524	Baud rate. Accuracy depends on the HFCLK source selected.
RXD.PTR	0x534	Data pointer
RXD.MAXCNT	0x538	Maximum number of bytes in receive buffer
RXD.AMOUNT	0x53C	Number of bytes transferred in the last transaction
TXD.PTR	0x544	Data pointer
TXD.MAXCNT	0x548	Maximum number of bytes in transmit buffer
TXD.AMOUNT	0x54C	Number of bytes transferred in the last transaction
CONFIG	0x56C	Configuration of parity and hardware flow control

35.10.1 SHORTS

Address offset: 0x200

Shortcut register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																													D	C					
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																														
C	RW	ENDRX_STARTRX			Shortcut between ENDRX event and STARTRX task See EVENTS_ENDRX and TASKS_STARTRX																																
			Disabled	0	Disable shortcut																																
			Enabled	1	Enable shortcut																																
D	RW	ENDRX_STOPRX			Shortcut between ENDRX event and STOPRX task See EVENTS_ENDRX and TASKS_STOPRX																																
			Disabled	0	Disable shortcut																																
			Enabled	1	Enable shortcut																																

35.10.2 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id													L		J	I		H								G	F	E			D		C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																														
A	RW	CTS			Enable or disable interrupt for CTS event See EVENTS_CTS																																
			Disabled	0	Disable																																
			Enabled	1	Enable																																
B	RW	NCTS			Enable or disable interrupt for NCTS event See EVENTS_NCTS																																
			Disabled	0	Disable																																
			Enabled	1	Enable																																
C	RW	RXDRDY			Enable or disable interrupt for RXDRDY event See EVENTS_RXDRDY																																
			Disabled	0	Disable																																
			Enabled	1	Enable																																
D	RW	ENDRX			Enable or disable interrupt for ENDRX event See EVENTS_ENDRX																																

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id													L		J	I		H								G	F	E			D		C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**								**Description**																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							
E	RW	TXDRDY										Enable or disable interrupt for TXDRDY event See *EVENTS_TXDRDY*																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							
F	RW	ENDTX										Enable or disable interrupt for ENDTX event See *EVENTS_ENDTX*																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							
G	RW	ERROR										Enable or disable interrupt for ERROR event See *EVENTS_ERROR*																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							
H	RW	RXTO										Enable or disable interrupt for RXTO event See *EVENTS_RXTO*																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							
I	RW	RXSTARTED										Enable or disable interrupt for RXSTARTED event See *EVENTS_RXSTARTED*																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							
J	RW	TXSTARTED										Enable or disable interrupt for TXSTARTED event See *EVENTS_TXSTARTED*																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							
L	RW	TXSTOPPED										Enable or disable interrupt for TXSTOPPED event See *EVENTS_TXSTOPPED*																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							

35.10.3 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id													L		J	I		H								G	F	E			D		C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**								**Description**																							
A	RW	CTS										Write '1' to Enable interrupt for CTS event See *EVENTS_CTS*																							
			Set	1								Enable																							
			Disabled	0								Read: Disabled																							
			Enabled	1								Read: Enabled																							
B	RW	NCTS										Write '1' to Enable interrupt for NCTS event See *EVENTS_NCTS*																							
			Set	1								Enable																							
			Disabled	0								Read: Disabled																							
			Enabled	1								Read: Enabled																							
C	RW	RXDRDY										Write '1' to Enable interrupt for RXDRDY event																							

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id													L		J	I		H								G	F	E			D		C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**								**Description**																							
												See *EVENTS_RXDRDY*																							
			Set	1								Enable																							
			Disabled	0								Read: Disabled																							
			Enabled	1								Read: Enabled																							
D	RW	ENDRX										Write '1' to Enable interrupt for ENDRX event																							
												See *EVENTS_ENDRX*																							
			Set	1								Enable																							
			Disabled	0								Read: Disabled																							
			Enabled	1								Read: Enabled																							
E	RW	TXDRDY										Write '1' to Enable interrupt for TXDRDY event																							
												See *EVENTS_TXDRDY*																							
			Set	1								Enable																							
			Disabled	0								Read: Disabled																							
			Enabled	1								Read: Enabled																							
F	RW	ENDTX										Write '1' to Enable interrupt for ENDTX event																							
												See *EVENTS_ENDTX*																							
			Set	1								Enable																							
			Disabled	0								Read: Disabled																							
			Enabled	1								Read: Enabled																							
G	RW	ERROR										Write '1' to Enable interrupt for ERROR event																							
												See *EVENTS_ERROR*																							
			Set	1								Enable																							
			Disabled	0								Read: Disabled																							
			Enabled	1								Read: Enabled																							
H	RW	RXTO										Write '1' to Enable interrupt for RXTO event																							
												See *EVENTS_RXTO*																							
			Set	1								Enable																							
			Disabled	0								Read: Disabled																							
			Enabled	1								Read: Enabled																							
I	RW	RXSTARTED										Write '1' to Enable interrupt for RXSTARTED event																							
												See *EVENTS_RXSTARTED*																							
			Set	1								Enable																							
			Disabled	0								Read: Disabled																							
			Enabled	1								Read: Enabled																							
J	RW	TXSTARTED										Write '1' to Enable interrupt for TXSTARTED event																							
												See *EVENTS_TXSTARTED*																							
			Set	1								Enable																							
			Disabled	0								Read: Disabled																							
			Enabled	1								Read: Enabled																							
L	RW	TXSTOPPED										Write '1' to Enable interrupt for TXSTOPPED event																							
												See *EVENTS_TXSTOPPED*																							
			Set	1								Enable																							
			Disabled	0								Read: Disabled																							
			Enabled	1								Read: Enabled																							

35.10.4 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id													L		J	I		H								G	F	E			D		C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	CTS																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	NCTS																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	RXDRDY																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	ENDRX																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	TXDRDY																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
F	RW	ENDTX																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
G	RW	ERROR																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
H	RW	RXTO																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
I	RW	RXSTARTED																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
J	RW	TXSTARTED																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
L	RW	TXSTOPPED																																	

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id												L		J	I		H								G	F	E			D		C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**														**Description**									
																									See *EVENTS_TXSTOPPED*									
			Clear								1														Disable									
			Disabled								0														Read: Disabled									
			Enabled								1														Read: Enabled									

35.10.5 ERRORSRC

Address offset: 0x480

Error source

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															D	C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**														**Description**									
A	RW	OVERRUN																							Overrun error									
																									A start bit is received while the previous data still lies in RXD. (Previous data is lost.)									
			NotPresent								0														Read: error not present									
			Present								1														Read: error present									
B	RW	PARITY																							Parity error									
																									A character with bad parity is received, if HW parity check is enabled.									
			NotPresent								0														Read: error not present									
			Present								1														Read: error present									
C	RW	FRAMING																							Framing error occurred									
																									A valid stop bit is not detected on the serial data input after all bits in a character have been received.									
			NotPresent								0														Read: error not present									
			Present								1														Read: error present									
D	RW	BREAK																							Break condition									
																									The serial data input is '0' for longer than the length of a data frame. (The data frame length is 10 bits without parity bit, and 11 bits with parity bit.).									
			NotPresent								0														Read: error not present									
			Present								1														Read: error present									

35.10.6 ENABLE

Address offset: 0x500

Enable UART

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**														**Description**									
A	RW	ENABLE																							Enable or disable UARTE									
			Disabled								0														Disable UARTE									
			Enabled								8														Enable UARTE									

35.10.7 PSEL.RTS

Address offset: 0x508

Pin select for RTS signal

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				B																											A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN		[0..31]																															
B	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

35.10.8 PSEL.TXD

Address offset: 0x50C

Pin select for TXD signal

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				B																											A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN		[0..31]																															
B	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

35.10.9 PSEL.CTS

Address offset: 0x510

Pin select for CTS signal

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				B																											A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN		[0..31]																															
B	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

35.10.10 PSEL.RXD

Address offset: 0x514

Pin select for RXD signal

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				B																											A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN		[0..31]																															
B	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

35.10.11 BAUDRATE

Address offset: 0x524

Baud rate. Accuracy depends on the HFCLK source selected.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x04000000				**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	BAUDRATE																																	
			Baud1200	0x0004F000																															
			Baud2400	0x0009D000																															
			Baud4800	0x0013B000																															
			Baud9600	0x00275000																															
			Baud14400	0x003AF000																															
			Baud19200	0x004EA000																															
			Baud28800	0x0075C000																															
			Baud38400	0x009D0000																															
			Baud57600	0x00EB0000																															
			Baud76800	0x013A9000																															
			Baud115200	0x01D60000																															
			Baud230400	0x03B00000																															
			Baud250000	0x04000000																															
			Baud460800	0x07400000																															
			Baud921600	0x0F000000																															
			Baud1M	0x10000000																															

35.10.12 RXD.PTR

Address offset: 0x534

Data pointer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PTR																																	

35.10.13 RXD.MAXCNT

Address offset: 0x538

Maximum number of bytes in receive buffer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																											A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	MAXCNT																																	

35.10.14 RXD.AMOUNT

Address offset: 0x53C

Number of bytes transferred in the last transaction

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																											A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	AMOUNT																																	

35.10.15 TXD.PTR

Address offset: 0x544

Data pointer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PTR																																	
				Data pointer																															

35.10.16 TXD.MAXCNT

Address offset: 0x548

Maximum number of bytes in transmit buffer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	MAXCNT																																	
				Maximum number of bytes in transmit buffer																															

35.10.17 TXD.AMOUNT

Address offset: 0x54C

Number of bytes transferred in the last transaction

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	AMOUNT																																	
				Number of bytes transferred in the last transaction																															

35.10.18 CONFIG

Address offset: 0x56C

Configuration of parity and hardware flow control

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																B	B	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	HWFC																																	
			Disabled	0																															
			Enabled	1																															
B	RW	PARITY																																	
			Excluded	0x0																															
			Included	0x7																															

35.11 Electrical specification

35.11.1 UARTE electrical specification

Symbol	Description	Min.	Typ.	Max.	Units
f_{UARTE}	Baud rate for UARTE[32].			1000	kbps
$I_{UARTE1M}$	Run current at max baud rate.		55		µA
$I_{UARTE115k}$	Run current at 115200 bps.		55		µA
$I_{UARTE1k2}$	Run current at 1200 bps.		55		µA
$I_{UARTE,IDLE}$	Idle current for UARTE (STARTed, no XXX activity)		1		µA
$t_{UARTE,CTSH}$	CTS high time	1			µs

[32] Higher baud rates may require GPIOs to be set as High Drive, see GPIO chapter for more details.

Symbol	Description	Min.	Typ.	Max.	Units
$t_{UARTE,START,LP}$	Time from STARTRX/STARTTX task to transmission started, low power mode		$t_{UARTE,STAR}$ + t_{START_HFIN}		µs
$t_{UARTE,START,CL}$	Time from STARTRX/STARTTX task to transmission started, constant latency mode		1		µs

36 QDEC — Quadrature decoder

The Quadrature decoder (QDEC) provides buffered decoding of quadrature-encoded sensor signals. It is suitable for mechanical and optical sensors.

The sample period and accumulation are configurable to match application requirements. The QDEC provides the following:

- Decoding of digital waveform from off-chip quadrature encoder.
- Sample accumulation eliminating hard real-time requirements to be enforced on application.
- Optional input de-bounce filters.
- Optional LED output signal for optical encoders.

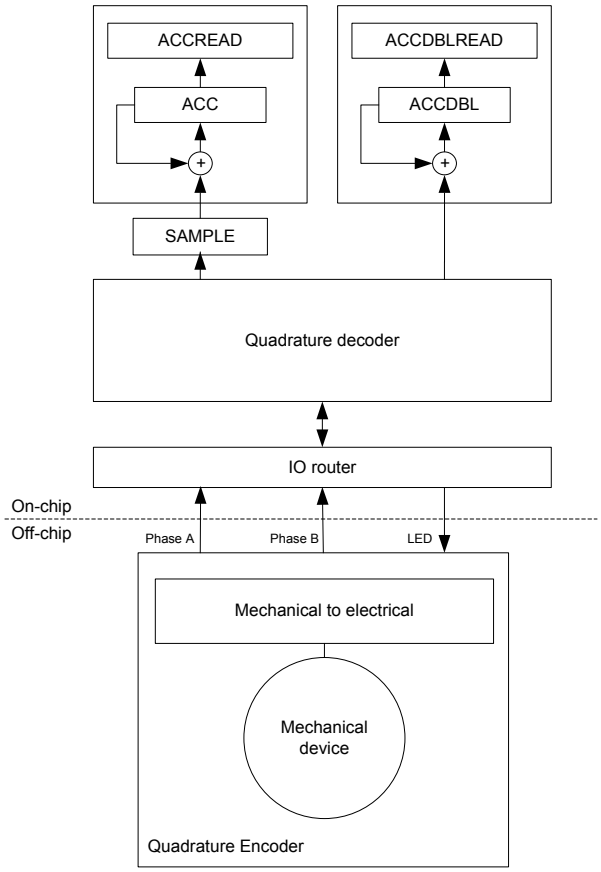


Figure 97: Quadrature decoder configuration

36.1 Sampling and decoding

The QDEC decodes the output from an incremental motion encoder by sampling the QDEC phase input pins (A and B).

The off-chip quadrature encoder is an incremental motion encoder outputting two waveforms, phase A and phase B. The two output waveforms are always 90 degrees out of phase, meaning that one always changes level before the other. The direction of movement is indicated by which of these two waveforms that changes level first. Invalid transitions may occur, that is when the two waveforms switch simultaneously. This may occur if the wheel rotates too fast relative to the sample rate set for the decoder.

The QDEC decodes the output from the off-chip encoder by sampling the QDEC phase input pins (A and B) at a fixed rate as specified in the SAMPLEPER register.

If the SAMPLEPER value needs to be changed, the QDEC shall be stopped using the STOP task. SAMPLEPER can be then changed upon receiving the STOPPED event, and QDEC can be restarted using the START task. Failing to do so may result in unpredictable behaviour.

It is good practice to change other registers (LEDPOL, REPORTPER, DBFEN and LEDPRE) only when the QDEC is stopped.

When started, the decoder continuously samples the two input waveforms and decodes these by comparing the current sample pair (n) with the previous sample pair (n-1).

The decoding of the sample pairs is described in the table below.

Table 84: Sampled value encoding

Previous sample pair(n - 1)		Current samples pair(n)		SAMPLE register	ACC operation	ACCDBL operation	Description
A	B	A	B				
0	0	0	0	0	No change	No change	No movement
0	0	0	1	1	Increment	No change	Movement in positive direction
0	0	1	0	-1	Decrement	No change	Movement in negative direction
0	0	1	1	2	No change	Increment	Error: Double transition
0	1	0	0	-1	Decrement	No change	Movement in negative direction
0	1	0	1	0	No change	No change	No movement
0	1	1	0	2	No change	Increment	Error: Double transition
0	1	1	1	1	Increment	No change	Movement in positive direction
1	0	0	0	1	Increment	No change	Movement in positive direction
1	0	0	1	2	No change	Increment	Error: Double transition
1	0	1	0	0	No change	No change	No movement
1	0	1	1	-1	Decrement	No change	Movement in negative direction
1	1	0	0	2	No change	Increment	Error: Double transition
1	1	0	1	-1	Decrement	No change	Movement in negative direction
1	1	1	0	1	Increment	No change	Movement in positive direction
1	1	1	1	0	No change	No change	No movement

36.2 LED output

The LED output follows the sample period, and the LED is switched on a given period before sampling and switched off immediately after the inputs are sampled. The period the LED is switched on before sampling is given in the LEDPRE register.

The LED output pin polarity is specified in the LEDPOL register.

For using off-chip mechanical encoders not requiring a LED, the LED output can be disabled by writing value 'Disconnected' to the CONNECT field of the PSEL.LED register. In this case the QDEC will not acquire access to a LED output pin and the pin can be used for other purposes by the CPU.

36.3 Debounce filters

Each of the two-phase inputs have digital debounce filters.

When enabled through the DBFEN register, the filter inputs are sampled at a fixed 1 MHz frequency during the entire sample period (which is specified in the SAMPLEPER register), and the filters require all of the samples within this sample period to equal before the input signal is accepted and transferred to the output of the filter.

As a result, only input signal with a steady state longer than twice the period specified in SAMPLEPER are guaranteed to pass through the filter, and any signal with a steady state shorter than SAMPLEPER will always be suppressed by the filter. (This is assumed that the frequency during the debounce period never exceeds 500 kHz (as required by the Nyquist theorem when using a 1 MHz sample frequency).

The LED will always be ON when the debounce filters are enabled, as the inputs in this case will be sampled continuously.

Note that when when the debounce filters are enabled, displacements reported by the QDEC peripheral are delayed by one SAMPLEPER period.

36.4 Accumulators

The quadrature decoder contains two accumulator registers, ACC and ACCDBL, that accumulate respectively valid motion sample values and the number of detected invalid samples (double transitions).

The ACC register will accumulate all valid values (1/-1) written to the SAMPLE register. This can be useful for preventing hard real-time requirements from being enforced on the application. When using the ACC register the application does not need to read every single sample from the SAMPLE register, but can instead fetch the ACC register whenever it fits the application. The ACC register will always hold the relative movement of the external mechanical device since the previous clearing of the ACC register. Sample values indicating a double transition (2) will not be accumulated in the ACC register.

An ACCOF event will be generated if the ACC receives a SAMPLE value that would cause the register to overflow or underflow. Any SAMPLE value that would cause an ACC overflow or underflow will be discarded, but any samples not causing the ACC to overflow or underflow will still be accepted.

The accumulator ACCDBL accumulates the number of detected double transitions since the previous clearing of the ACCDBL register.

The ACC and ACCDBL registers can be cleared by the READCLRACC and subsequently read using the ACCREAD and ACCDBLREAD registers.

The ACC register can be separately cleared by the RDCLRACC and subsequently read using the ACCREAD registers.

The ACCDBL register can be separately cleared by the RDCLRDBL and subsequently read using the ACCDBLREAD registers.

The REPORTPER register allows automating the capture of several samples before it can send out a REPORTRDY event in case a non-null displacement has been captured and accumulated, and a DBLRDY event in case one or more double-displacements have been captured and accumulated. The REPORTPER field in this register selects after how many samples the accumulators contents are evaluated to send (or not) REPORTRDY and DBLRDY events.

Using the RDCLRACC task (manually sent upon receiving the event, or using the DBLRDY_RDCLRACC shortcut), ACCREAD can then be read.

In case at least one double transition has been captured and accumulated, a DBLRDY event is sent. Using the RDCLRDBL task (manually sent upon receiving the event, or using the DBLRDY_RDCLRDBL shortcut), ACCDBLREAD can then be read.

36.5 Output/input pins

The QDEC uses a three-pin interface to the off-chip quadrature encoder.

These pins will be acquired when the QDEC is enabled in the ENABLE register. The pins acquired by the QDEC cannot be written by the CPU, but they can still be read by the CPU.

The pin numbers to be used for the QDEC are selected using the PSEL.n registers.

36.6 Pin configuration

The Phase A, Phase B, and LED signals are mapped to physical pins according to the configuration specified in the PSEL.A, PSEL.B, and PSEL.LED registers respectively.

If the CONNECT field value 'Disconnected' is specified in any of these registers, the associated signal will not be connected to any physical pin. The PSEL.A, PSEL.B, and PSEL.LED registers and their configurations are only used as long as the QDEC is enabled, and retained only as long as the device is in

ON mode. When the peripheral is disabled, the pins will behave as regular GPIOs, and use the configuration in their respective OUT bit field and PIN_CNF[n] register.

To secure correct behavior in the QDEC, the pins used by the QDEC must be configured in the GPIO peripheral as described in *Table 85: GPIO configuration before enabling peripheral* on page 350 before enabling the QDEC. This configuration must be retained in the GPIO for the selected IOs as long as the QDEC is enabled.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behavior.

Table 85: GPIO configuration before enabling peripheral

QDEC signal	QDEC pin	Direction	Output value	Comment
Phase A	As specified in PSEL.A	Input	Not applicable	
Phase B	As specified in PSEL.B	Input	Not applicable	
LED	As specified in PSEL.LED	Input	Not applicable	

36.7 Registers

Table 86: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40012000	QDEC	QDEC	Quadrature decoder	

Table 87: Register Overview

Register	Offset	Description
TASKS_START	0x000	Task starting the quadrature decoder
TASKS_STOP	0x004	Task stopping the quadrature decoder
TASKS_READCLRACC	0x008	Read and clear ACC and ACCDBL
TASKS_RDCLRACC	0x00C	Read and clear ACC
TASKS_RDCLRDBL	0x010	Read and clear ACCDBL
EVENTS_SAMPLERDY	0x100	Event being generated for every new sample value written to the SAMPLE register
EVENTS_REPORTRDY	0x104	Non-null report ready
EVENTS_ACCOF	0x108	ACC or ACCDBL register overflow
EVENTS_DBLRDY	0x10C	Double displacement(s) detected
EVENTS_STOPPED	0x110	QDEC has been stopped
SHORTS	0x200	Shortcut register
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ENABLE	0x500	Enable the quadrature decoder
LEDPOL	0x504	LED output pin polarity
SAMPLEPER	0x508	Sample period
SAMPLE	0x50C	Motion sample value
REPORTPER	0x510	Number of samples to be taken before REPORTRDY and DBLRDY events can be generated
ACC	0x514	Register accumulating the valid transitions
ACCREAD	0x518	Snapshot of the ACC register, updated by the READCLRACC or RDCLRACC task
PSEL.LED	0x51C	Pin select for LED signal
PSEL.A	0x520	Pin select for A signal
PSEL.B	0x524	Pin select for B signal
DBFEN	0x528	Enable input debounce filters
LEDPRE	0x540	Time period the LED is switched ON prior to sampling
ACCDBL	0x544	Register accumulating the number of detected double transitions
ACCDBLREAD	0x548	Snapshot of the ACCDBL, updated by the READCLRACC or RDCLRDBL task

36.7.1 SHORTS

Address offset: 0x200

Shortcut register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	REPORTRDY_READCLRACC			Shortcut between REPORTRDY event and READCLRACC task See *EVENTS_REPORTRDY* and *TASKS_READCLRACC*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
B	RW	SAMPLERDY_STOP			Shortcut between SAMPLERDY event and STOP task See *EVENTS_SAMPLERDY* and *TASKS_STOP*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
C	RW	REPORTRDY_RDCLRACC			Shortcut between REPORTRDY event and RDCLRACC task See *EVENTS_REPORTRDY* and *TASKS_RDCLRACC*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
D	RW	REPORTRDY_STOP			Shortcut between REPORTRDY event and STOP task See *EVENTS_REPORTRDY* and *TASKS_STOP*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
E	RW	DBLRDY_RDCLRDBL			Shortcut between DBLRDY event and RDCLRDBL task See *EVENTS_DBLRDY* and *TASKS_RDCLRDBL*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
F	RW	DBLRDY_STOP			Shortcut between DBLRDY event and STOP task See *EVENTS_DBLRDY* and *TASKS_STOP*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
G	RW	SAMPLERDY_READCLRACC			Shortcut between SAMPLERDY event and READCLRACC task See *EVENTS_SAMPLERDY* and *TASKS_READCLRACC*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

36.7.2 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	SAMPLERDY			Write '1' to Enable interrupt for SAMPLERDY event See *EVENTS_SAMPLERDY*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	REPORTRDY			Write '1' to Enable interrupt for REPORTRDY event See *EVENTS_REPORTRDY*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	ACCOF			Write '1' to Enable interrupt for ACCOF event See *EVENTS_ACCOF*
			Set	1	Enable

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Disabled	0																															
			Enabled	1																															
D	RW	DBLRDY																																	
				See *EVENTS_DBLRDY*																															
			Set	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	STOPPED																																	
				See *EVENTS_STOPPED*																															
			Set	1																															
			Disabled	0																															
			Enabled	1																															

36.7.3 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	SAMPLERDY																																	
				See *EVENTS_SAMPLERDY*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	REPORTRDY																																	
				See *EVENTS_REPORTRDY*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	ACCOF																																	
				See *EVENTS_ACCOF*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	DBLRDY																																	
				See *EVENTS_DBLRDY*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	STOPPED																																	
				See *EVENTS_STOPPED*																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															

36.7.4 ENABLE

Address offset: 0x500

Enable the quadrature decoder

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
A	RW	ENABLE			Enable or disable the quadrature decoder When enabled the decoder pins will be active. When disabled the quadrature decoder pins are not active and can be used as GPIO .
			Disabled	0	Disable
			Enabled	1	Enable

36.7.5 LEDPOL

Address offset: 0x504

LED output pin polarity

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
A	RW	LEDPOL			LED output pin polarity
			ActiveLow	0	Led active on output pin low
			ActiveHigh	1	Led active on output pin high

36.7.6 SAMPLEPER

Address offset: 0x508

Sample period

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
A	RW	SAMPLEPER			Sample period. The SAMPLE register will be updated for every new sample
			128us	0	128 us
			256us	1	256 us
			512us	2	512 us
			1024us	3	1024 us
			2048us	4	2048 us
			4096us	5	4096 us
			8192us	6	8192 us
			16384us	7	16384 us
			32ms	8	32768 us
			65ms	9	65536 us
			131ms	10	131072 us

36.7.7 SAMPLE

Address offset: 0x50C

Motion sample value

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
A	R	SAMPLE		[-1..2]	Last motion sample

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
					The value is a 2's complement value, and the sign gives the direction of the motion. The value '2' indicates a double transition.																															

36.7.8 REPORTPER

Address offset: 0x510

Number of samples to be taken before REPORTRDY and DBLRDY events can be generated

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	REPORTPER			Specifies the number of samples to be accumulated in the ACC register before the REPORTRDY and DBLRDY events can be generated The report period in [us] is given as: RPUS = SP * RP Where RPUS is the report period in [us/report], SP is the sample period in [us/sample] specified in SAMPLEPER, and RP is the report period in [samples/report] specified in REPORTPER .																													
			10Smpl	0	10 samples / report																													
			40Smpl	1	40 samples / report																													
			80Smpl	2	80 samples / report																													
			120Smpl	3	120 samples / report																													
			160Smpl	4	160 samples / report																													
			200Smpl	5	200 samples / report																													
			240Smpl	6	240 samples / report																													
			280Smpl	7	280 samples / report																													
			1Smpl	8	1 sample / report																													

36.7.9 ACC

Address offset: 0x514

Register accumulating the valid transitions

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	R	ACC		[-1024..1023]	Register accumulating all valid samples (not double transition) read from the SAMPLE register Double transitions (SAMPLE = 2) will not be accumulated in this register. The value is a 32 bit 2's complement value. If a sample that would cause this register to overflow or underflow is received, the sample will be ignored and an overflow event (ACCOF) will be generated. The ACC register is cleared by triggering the READCLRACC or the RDCLRACC task.																													

36.7.10 ACCREAD

Address offset: 0x518

Snapshot of the ACC register, updated by the READCLRACC or RDCLRACC task

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	ACCREAD		[-1024..1023]																															
				Snapshot of the ACC register.																															
				The ACCREAD register is updated when the READCLRACC or RDCLRACC task is triggered																															

36.7.11 PSEL.LED

Address offset: 0x51C

Pin select for LED signal

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				B																											A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN		[0..31]																															
				Pin number																															
B	RW	CONNECT																																	
				Connection																															
			Disconnected	1																															
				Disconnect																															
			Connected	0																															
				Connect																															

36.7.12 PSEL.A

Address offset: 0x520

Pin select for A signal

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				B																											A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN		[0..31]																															
				Pin number																															
B	RW	CONNECT																																	
				Connection																															
			Disconnected	1																															
				Disconnect																															
			Connected	0																															
				Connect																															

36.7.13 PSEL.B

Address offset: 0x524

Pin select for B signal

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				B																											A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN		[0..31]																															
				Pin number																															
B	RW	CONNECT																																	
				Connection																															
			Disconnected	1																															
				Disconnect																															
			Connected	0																															
				Connect																															

36.7.14 DBFEN

Address offset: 0x528

Enable input debounce filters

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	DBFEN			Enable input debounce filters
			Disabled	0	Debounce input filters disabled
			Enabled	1	Debounce input filters enabled

36.7.15 LEDPRE

Address offset: 0x540

Time period the LED is switched ON prior to sampling

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A A A A A A A
Reset 0x00000010				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	LEDPRE		[1..511]	Period in us the LED is switched on prior to sampling

36.7.16 ACCDBL

Address offset: 0x544

Register accumulating the number of detected double transitions

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	R	ACCDBL		[0..15]	Register accumulating the number of detected double or illegal transitions. (SAMPLE = 2). When this register has reached its maximum value the accumulation of double / illegal transitions will stop. An overflow event (ACCOF) will be generated if any double or illegal transitions are detected after the maximum value was reached. This field is cleared by triggering the READCLRACC or RDCLRDBL task.

36.7.17 ACCDBLREAD

Address offset: 0x548

Snapshot of the ACCDBL, updated by the READCLRACC or RDCLRDBL task

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	R	ACCDBLREAD		[0..15]	Snapshot of the ACCDBL register. This field is updated when the READCLRACC or RDCLRDBL task is triggered.

36.8 Electrical specification

36.8.1 QDEC Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
I_{QDEC}	Run current		5		µA
t_{SAMPLE}	Time between sampling signals from quadrature decoder	128		131072	µs
t_{LED}	Time from LED is turned on to signals are sampled	0		511	µs

37 SAADC — Successive approximation analog-to-digital converter

The ADC is a differential successive approximation register (SAR) analog-to-digital converter.

Listed here are the main features of SAADC:

- 8/10/12-bit resolution, 14-bit resolution with oversampling
- Up to eight input channels
  - One channel per single-ended input and two channels per differential input
  - Scan mode can be configured with both single-ended channels and differential channels.
- Full scale input range (0 to VDD)
- Sampling triggered via a task from software or a PPI channel for full flexibility on sample frequency source from low power 32.768kHz RTC or more accurate 1/16MHz Timers
- One-shot conversion mode to sample a single channel
- Scan mode to sample a series of channels in sequence. Sample delay between channels is $t_{ack} + t_{conv}$ which may vary between channels according to user configuration of t_{ack}.
- Support for direct sample transfer to RAM using EasyDMA
- Interrupts on single sample and full buffer events
- Samples stored as 16-bit 2’s complement values for differential and single-ended sampling
- Continuous sampling without the need of an external timer
- Internal resistor string
- Limit checking on the fly

37.1 Shared resources

The ADC can coexist with COMP and other peripherals using one of `AIN0-AIN7`, provided these are assigned to different pins.

It is not recommended to select the same analog input pin for both modules.

37.2 Overview

The ADC supports up to eight external analog input channels, depending on package variant. It can be operated in a one-shot mode with sampling under software control, or a continuous conversion mode with a programmable sampling rate.

The analog inputs can be configured as eight single-ended inputs, four differential inputs or a combination of these. Each channel can be configured to select `AIN0` to `AIN7` pins, or the `VDD` pin. Channels can be sampled individually in one-shot or continuous sampling modes, or, using scan mode, multiple channels can be sampled in sequence. Channels can also be oversampled to improve noise performance.

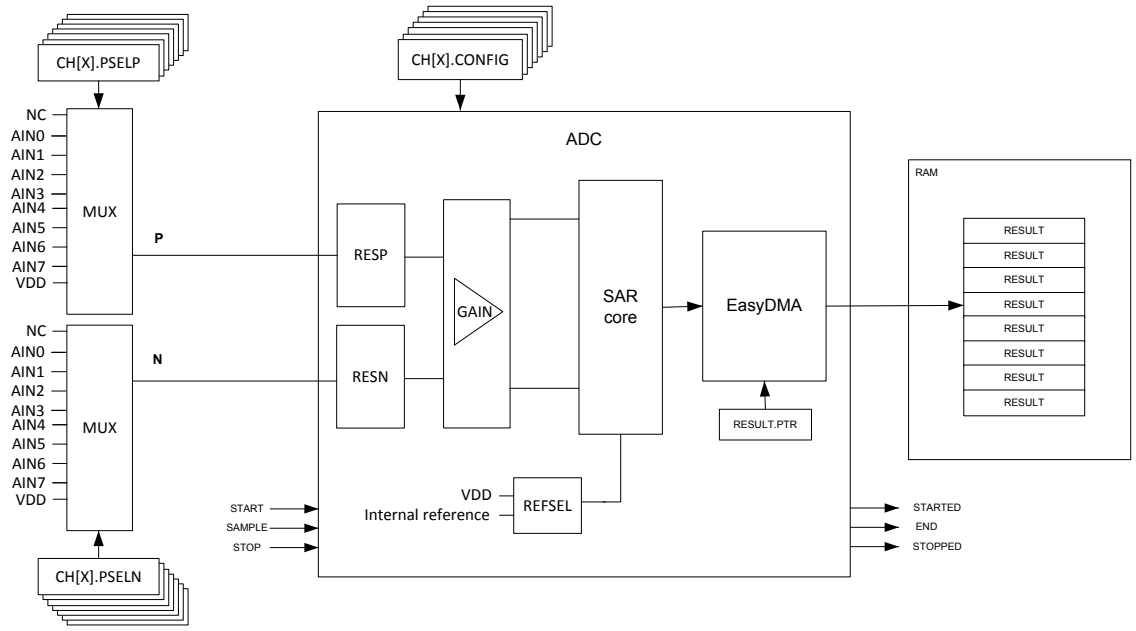


Figure 98: Simplified ADC block diagram

Internally, the ADC is always a differential analog-to-digital converter, but by default it is configured with single-ended input in the MODE field of the CH[n].CONFIG register. In single-ended mode, the negative input will be shorted to ground internally.

The assumption in single-ended mode is that the internal ground of the ADC is the same as the external ground that the measured voltage is referred to. The ADC is thus sensitive to ground bounce on the PCB in single-ended mode. If this is a concern we recommend using differential measurement.

37.3 Digital output

The output result of the ADC depends on the settings in the CH[n].CONFIG and RESOLUTION registers as follows:

$$RESULT = [V(P) - V(N)] * GAIN/REFERENCE * 2^{(RESOLUTION - m)}$$

where

V(P)

is the voltage at input P

V(N)

is the voltage at input N

GAIN

is the selected gain setting

REFERENCE

is the selected reference voltage

and m=0 if CONFIG.MODE=SE, or m=1 if CONFIG.MODE=Diff.

The result generated by the ADC will deviate from the expected due DC errors like offset, gain, differential non-linearity (DNL), and integral non-linearity (INL). See *Electrical specification* for details on these parameters. The result can also vary due to AC errors like non-linearities in the GAIN block, settling errors due to high source impedance and sampling jitter. For battery measurement the DC errors are most noticeable.

The ADC has a wide selection of gains controlled in the GAIN field of the CH[n].CONFIG register. If CH[n].CONFIG.REFSEL=0, the input range of the ADC core is nominally ±0.6 V differential and the input must be scaled accordingly.

The ADC has a temperature dependent offset. If the ADC is to operate over a large temperature range, we recommend running CALIBRATEOFFSET at regular intervals, a CALIBRATEDONE event will be fired when the calibration is complete

37.4 Analog inputs and channels

Up to eight analog input channels, CH[n](n=0..7), can be configured.

See *Shared resources* on page 357 for shared input with comparators.

Any one of the available channels can be enabled for the ADC to operate in one-shot mode. If more than one CH[n] is configured, the ADC enters scan mode.

An analog input is selected as a positive converter input if CH[n].PSELP is set, setting CH[n].PSELP also enables the particular channel.

An analog input is selected as a negative converter input if CH[n].PSELN is set. The CH[n].PSELN register will have no effect unless differential mode is enabled, see MODE field in CH[n].CONFIG register.

If more than one of the CH[n].PSELP registers is set, the device enters scan mode. Input selections in scan mode are controlled by the CH[n].PSELP and CH[n].PSELN registers, where CH[n].PSELN is only used if the particular scan channel is specified as differential, see MODE field in CH[n].CONFIG register.

> **Important:** Channels selected for COMP cannot be used at the same time for ADC sampling, though channels not selected for use by these blocks can be used by the ADC.

Table 88: Legal connectivity CH[n] vs. analog input

Channel input	Source	Connectivity
CH[n].PSELP	AIN0...AIN7	Yes(any)
CH[n].PSELP	VDD	Yes
CH[n].PSELN	AIN0...AIN7	Yes(any)
CH[n].PSELN	VDD	Yes

37.5 Operation modes

The ADC input configuration supports one-shot mode, continuous mode and scan mode.

Scan mode and oversampling cannot be combined.

37.5.1 One-shot mode

One-shot operation is configured by enabling only one of the available channels defined by CH[n].PSELP, CH[n].PSELN, and CH[n].CONFIG registers.

Upon a SAMPLE task, the ADC starts to sample the input voltage. The CH[n].CONFIG.TACQ controls the acquisition time.

A DONE event signals that one sample has been taken.

In this mode, the RESULTDONE event has the same meaning as DONE when no oversampling takes place. Note that both events may occur before the actual value has been transferred into RAM by EasyDMA. For more information, see *EasyDMA* on page 361.

37.5.2 Continuous mode

Continuous sampling can be achieved by using the internal timer in the ADC, or triggering the SAMPLE task from one of the general purpose timers through the PPI.

Care shall be taken to ensure that the sample rate fulfils the following criteria, depending on how many channels are active:

$f_{SAMPLE} < 1/[t_{ACQ} + t_{conv}]$

The SAMPLERATE register can be used as a local timer instead of triggering individual SAMPLE tasks. When SAMPLERATE.MODE is set to Timers, it is sufficient to trigger SAMPLE task only once in order to start the SAADC and triggering the STOP task will stop sampling. The SAMPLERATE.CC field controls the sample rate.

The SAMPLERATE timer mode cannot be combined with SCAN mode, and only one channel can be enabled in this mode.

A DONE event signals that one sample has been taken.

In this mode, the RESULTDONE event has the same meaning as DONE when no oversampling takes place. Note that both events may occur before the actual value has been transferred into RAM by EasyDMA.

37.5.3 Oversampling

An accumulator in the ADC can be used to average noise on the analog input. In general, oversampling improves the signal-to-noise ratio (SNR). Oversampling, however, does not improve the integral non-linearity (INL), or differential non-linearity (DNL).

Oversampling and scan should not be combined, since oversampling and scan will average over input channels.

The accumulator is controlled in the OVERSAMPLE register. The SAMPLE task must be set $2^{OVERSAMPLE}$ number of times before the result is written to RAM. This can be achieved by:

- Configuring a fixed sampling rate using the local timer or a general purpose timer and PPI to trigger a SAMPLE task
- Triggering SAMPLE $2^{OVERSAMPLE}$ times from software
- Enabling BURST mode

CH[n].CONFIG.BURST can be enabled to avoid setting SAMPLE task $2^{OVERSAMPLE}$ times. With BURST = 1 the ADC will sample the input $2^{OVERSAMPLE}$ times as fast as it can (actual timing: $<(t_{ACQ}+t_{CONV})\times 2^{OVERSAMPLE}$). Thus, for the user it will just appear like the conversion took a bit longer time, but other than that, it is similar to one-shot mode. Scan mode can be combined with BURST=1, if burst is enabled on all channels.

A DONE event signals that one sample has been taken.

In this mode, the RESULTDONE event signals that enough conversions have taken place for an oversampled result to get transferred into RAM. Note that both events may occur before the actual value has been transferred into RAM by EasyDMA.

37.5.4 Scan mode

A channel is considered enabled if CH[n].PSELP is set. If more than one channel, CH[n], is enabled, the ADC enters scan mode.

In scan mode, one SAMPLE task will trigger one conversion per enabled channel. The time it takes to sample all channels is:

```
Total time < Sum(CH[x].tACQ+tCONV), x=0..enabled channels
```

A DONE event signals that one sample has been taken.

In this mode, the RESULTDONE event signals has the same meaning as DONE when no oversampling takes place. Note that both events may occur before the actual values have been transferred into RAM by EasyDMA.

Figure 99: Example of RAM placement (even RESULT.MAXCNT), channels 1, 2 and 5 enabled on page 361 provides an example of results placement in Data RAM, with an even RESULT.MAXCNT. In this example, channels 1, 2 and 5 are enabled, all others are disabled.

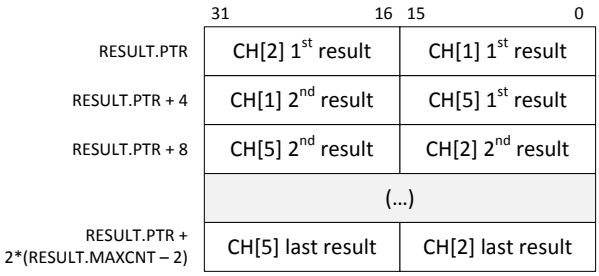


Figure 99: Example of RAM placement (even RESULT.MAXCNT), channels 1, 2 and 5 enabled

Figure 100: Example of RAM placement (odd RESULT.MAXCNT), channels 1, 2 and 5 enabled on page 361 provides an example of results placement in Data RAM, with an odd RESULT.MAXCNT. In this example, channels 1, 2 and 5 are enabled, all others are disabled. The last 32-bit word is populated only with one 16-bit result.

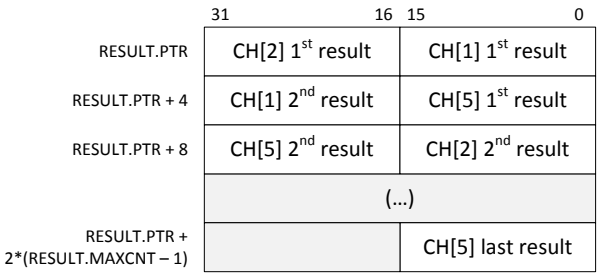


Figure 100: Example of RAM placement (odd RESULT.MAXCNT), channels 1, 2 and 5 enabled

37.6 EasyDMA

After configuring RESULT.PTR and RESULT.MAXCNT, the ADC resources are started by triggering the START task. The ADC is using EasyDMA to store results in a Result buffer in RAM.

The Result buffer is located at the address specified in the RESULT.PTR register. The RESULT.PTR register is double-buffered and it can be updated and prepared for the next START task immediately after the STARTED event is generated. The size of the Result buffer is specified in the RESULT.MAXCNT register and the ADC will generate an END event when it has filled up the Result buffer, see *Figure 101: ADC* on page 362. Results are stored in little-endian byte order in Data RAM. Every sample will be sign extended to 16 bit before stored in the Result buffer.

The ADC is stopped by triggering the STOP task. The STOP task will terminate an ongoing sampling. The ADC will generate a STOPPED event when it has stopped. If the ADC is already stopped when the STOP task is triggered, the STOPPED event will still be generated.

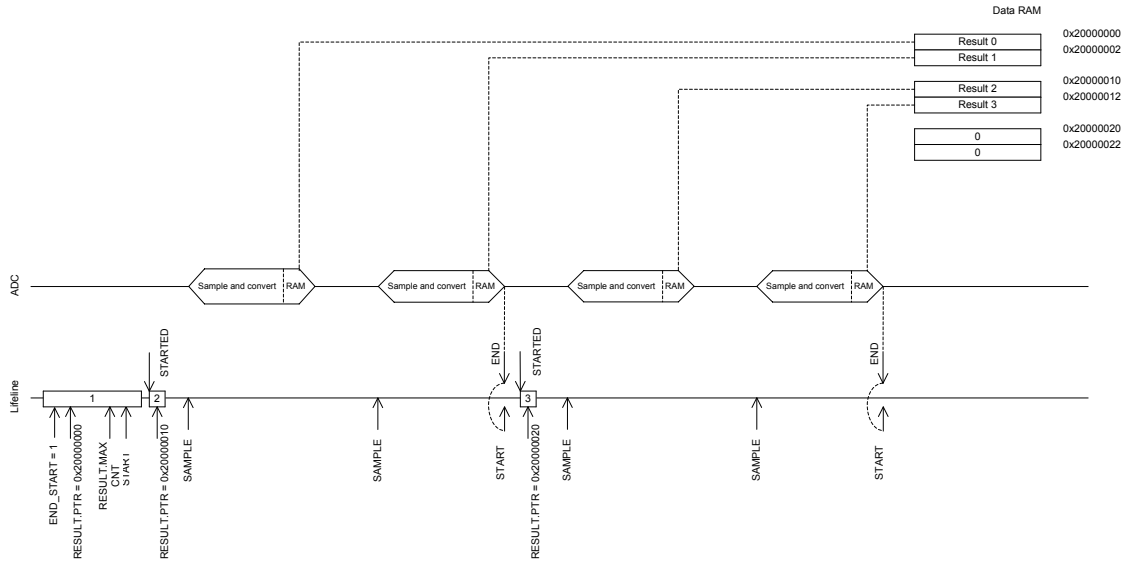


Figure 101: ADC

If the RESULT.PTR is not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See *Memory* on page 23 for more information about the different memory regions.

The EasyDMA will have finished accessing the RAM when the END or STOPPED event has been generated.

The RESULT.AMOUNT register can be read following an END event or a STOPPED event to see how many results have been transferred to the Result buffer in RAM since the START task was triggered.

In Scan mode, the size of the Result buffer must be large enough to have room for a minimum one result from each of the enabled channels. To secure this, RESULT.MAXCNT must be specified to RESULT.MAXCNT >= "number of channels enabled". See *Scan mode* on page 360 for more information about Scan mode.

37.7 Resistor ladder

The ADC has an internal resistor string for positive and negative input.

See *Figure 102: Resistor ladder for positive input (negative input is equivalent, using RESN instead of RESP)* on page 363. The resistors are controlled in the CH[n].CONFIG.RESP and CH[n].CONFIG.RESN registers.

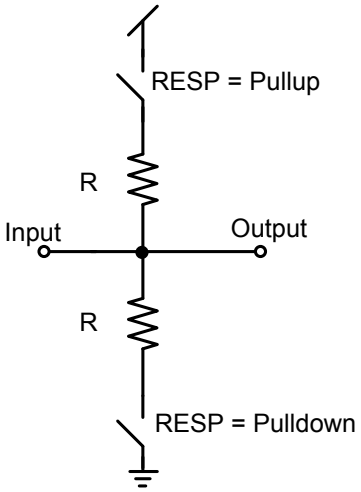


Figure 102: Resistor ladder for positive input (negative input is equivalent, using RESN instead of RESP)

37.8 Reference

The ADC can use two different references, controlled in the REFSEL field of the CH[n].CONFIG register.

These are:

- Internal reference
- VDD as reference

The internal reference results in an input range of ±0.6 V on the ADC core. VDD as reference results in an input range of ±VDD/4 on the ADC core. The gain block can be used to change the effective input range of the ADC.

```
Input range = (+- 0.6 V or +-VDD/4)/Gain
```

For example, choosing VDD as reference, single ended input (grounded negative input), and a gain of 1/4 the input range will be:

```
Input range = (VDD/4)/(1/4) = VDD
```

With internal reference, single ended input (grounded negative input), and a gain of 1/6 the input range will be:

```
Input range = (0.6 V)/(1/6) = 3.6 V
```

The AIN0-AIN7 inputs cannot exceed VDD, or be lower than VSS.

37.9 Acquisition time

To sample the input voltage, the ADC connects a capacitor to the input.

For illustration, see *Figure 103: Simplified ADC sample network* on page 364. The acquisition time indicates how long the capacitor is connected, see TACQ field in CH[n].CONFIG register. The required acquisition time depends on the source (R_{source}) resistance. For high source resistance the acquisition time should be increased, see *Table 89: Acquisition time* on page 364.

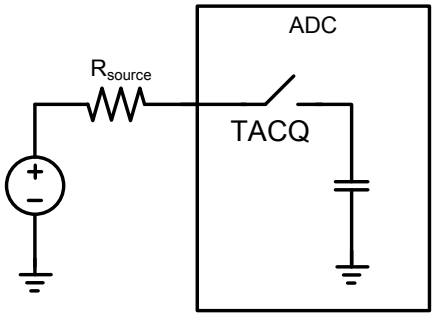


Figure 103: Simplified ADC sample network

Table 89: Acquisition time

TACQ [µs]	Maximum source resistance [kOhm]
3	10
5	40
10	100
15	200
20	400
40	800

37.10 Limits event monitoring

A channel can be event monitored by configuring limit register CH[n].LIMIT.

If the conversion result is higher than the defined high limit, or lower than the defined low limit, the appropriate event will get fired.

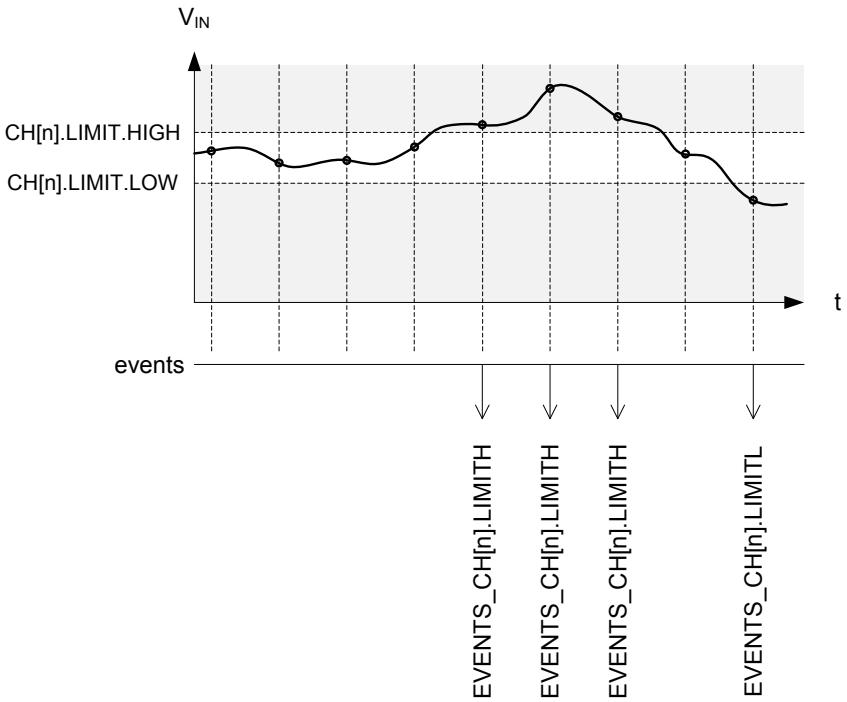


Figure 104: Example of limits monitoring on channel 'n'

Note that when setting the limits, CH[n].LIMIT.HIGH shall always be higher than or equal to CH[n].LIMIT.LOW . In other words, an event can be fired only when the input signal has been sampled

outside of the defined limits. It is not possible to fire an event when the input signal is inside a defined range by swapping high and low limits.

The comparison to limits always takes place, there is no need to enable it. If comparison is not required on a channel, the software shall simply ignore the related events. In that situation, the value of the limits registers is irrelevant, so it does not matter if CH[n].LIMIT.LOW is lower than CH[n].LIMIT.HIGH or not.

37.11 Registers

Table 90: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40007000	SAADC	SAADC	Analog to digital converter	

Table 91: Register Overview

Register	Offset	Description
TASKS_START	0x000	Start the ADC and prepare the result buffer in RAM
TASKS_SAMPLE	0x004	Take one ADC sample, if scan is enabled all channels are sampled
TASKS_STOP	0x008	Stop the ADC and terminate any on-going conversion
TASKS_CALIBRATEOFFSE	0x00C	Starts offset auto-calibration
EVENTS_STARTED	0x100	The ADC has started
EVENTS_END	0x104	The ADC has filled up the Result buffer
EVENTS_DONE	0x108	A conversion task has been completed. Depending on the mode, multiple conversions might be needed for a result to be transferred to RAM.
EVENTS_RESULTDONE	0x10C	A result is ready to get transferred to RAM.
EVENTS_CALIBRATEDON	0x110	Calibration is complete
EVENTS_STOPPED	0x114	The ADC has stopped
EVENTS_CH[0].LIMITH	0x118	Last results is equal or above CH[0].LIMIT.HIGH
EVENTS_CH[0].LIMITL	0x11C	Last results is equal or below CH[0].LIMIT.LOW
EVENTS_CH[1].LIMITH	0x120	Last results is equal or above CH[1].LIMIT.HIGH
EVENTS_CH[1].LIMITL	0x124	Last results is equal or below CH[1].LIMIT.LOW
EVENTS_CH[2].LIMITH	0x128	Last results is equal or above CH[2].LIMIT.HIGH
EVENTS_CH[2].LIMITL	0x12C	Last results is equal or below CH[2].LIMIT.LOW
EVENTS_CH[3].LIMITH	0x130	Last results is equal or above CH[3].LIMIT.HIGH
EVENTS_CH[3].LIMITL	0x134	Last results is equal or below CH[3].LIMIT.LOW
EVENTS_CH[4].LIMITH	0x138	Last results is equal or above CH[4].LIMIT.HIGH
EVENTS_CH[4].LIMITL	0x13C	Last results is equal or below CH[4].LIMIT.LOW
EVENTS_CH[5].LIMITH	0x140	Last results is equal or above CH[5].LIMIT.HIGH
EVENTS_CH[5].LIMITL	0x144	Last results is equal or below CH[5].LIMIT.LOW
EVENTS_CH[6].LIMITH	0x148	Last results is equal or above CH[6].LIMIT.HIGH
EVENTS_CH[6].LIMITL	0x14C	Last results is equal or below CH[6].LIMIT.LOW
EVENTS_CH[7].LIMITH	0x150	Last results is equal or above CH[7].LIMIT.HIGH
EVENTS_CH[7].LIMITL	0x154	Last results is equal or below CH[7].LIMIT.LOW
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
STATUS	0x400	Status
ENABLE	0x500	Enable or disable ADC
CH[0].PSELP	0x510	Input positive pin selection for CH[0]
CH[0].PSELN	0x514	Input negative pin selection for CH[0]
CH[0].CONFIG	0x518	Input configuration for CH[0]
CH[0].LIMIT	0x51C	High/low limits for event monitoring a channel
CH[1].PSELP	0x520	Input positive pin selection for CH[1]
CH[1].PSELN	0x524	Input negative pin selection for CH[1]
CH[1].CONFIG	0x528	Input configuration for CH[1]
CH[1].LIMIT	0x52C	High/low limits for event monitoring a channel
CH[2].PSELP	0x530	Input positive pin selection for CH[2]

Register	Offset	Description
CH[2].PSELN	0x534	Input negative pin selection for CH[2]
CH[2].CONFIG	0x538	Input configuration for CH[2]
CH[2].LIMIT	0x53C	High/low limits for event monitoring a channel
CH[3].PSELP	0x540	Input positive pin selection for CH[3]
CH[3].PSELN	0x544	Input negative pin selection for CH[3]
CH[3].CONFIG	0x548	Input configuration for CH[3]
CH[3].LIMIT	0x54C	High/low limits for event monitoring a channel
CH[4].PSELP	0x550	Input positive pin selection for CH[4]
CH[4].PSELN	0x554	Input negative pin selection for CH[4]
CH[4].CONFIG	0x558	Input configuration for CH[4]
CH[4].LIMIT	0x55C	High/low limits for event monitoring a channel
CH[5].PSELP	0x560	Input positive pin selection for CH[5]
CH[5].PSELN	0x564	Input negative pin selection for CH[5]
CH[5].CONFIG	0x568	Input configuration for CH[5]
CH[5].LIMIT	0x56C	High/low limits for event monitoring a channel
CH[6].PSELP	0x570	Input positive pin selection for CH[6]
CH[6].PSELN	0x574	Input negative pin selection for CH[6]
CH[6].CONFIG	0x578	Input configuration for CH[6]
CH[6].LIMIT	0x57C	High/low limits for event monitoring a channel
CH[7].PSELP	0x580	Input positive pin selection for CH[7]
CH[7].PSELN	0x584	Input negative pin selection for CH[7]
CH[7].CONFIG	0x588	Input configuration for CH[7]
CH[7].LIMIT	0x58C	High/low limits for event monitoring a channel
RESOLUTION	0x5F0	Resolution configuration
OVERSAMPLE	0x5F4	Oversampling configuration. OVERSAMPLE should not be combined with SCAN. The RESOLUTION is applied before averaging, thus for high OVERSAMPLE a higher RESOLUTION should be used.
SAMPLERATE	0x5F8	Controls normal or continuous sample rate
RESULT.PTR	0x62C	Data pointer
RESULT.MAXCNT	0x630	Maximum number of buffer words to transfer
RESULT.AMOUNT	0x634	Number of buffer words transferred since last START

37.11.1 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	STARTED			Enable or disable interrupt for STARTED event See *EVENTS_STARTED*
			Disabled	0	Disable
			Enabled	1	Enable
B	RW	END			Enable or disable interrupt for END event See *EVENTS_END*
			Disabled	0	Disable
			Enabled	1	Enable
C	RW	DONE			Enable or disable interrupt for DONE event See *EVENTS_DONE*
			Disabled	0	Disable
			Enabled	1	Enable
D	RW	RESULTDONE			Enable or disable interrupt for RESULTDONE event See *EVENTS_RESULTDONE*
			Disabled	0	Disable
			Enabled	1	Enable

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id														V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
E	RW	CALIBRATEDONE																																	
			Disabled	0																															
			Enabled	1																															
F	RW	STOPPED																																	
			Disabled	0																															
			Enabled	1																															
G	RW	CH0LIMITH																																	
			Disabled	0																															
			Enabled	1																															
H	RW	CH0LIMITL																																	
			Disabled	0																															
			Enabled	1																															
I	RW	CH1LIMITH																																	
			Disabled	0																															
			Enabled	1																															
J	RW	CH1LIMITL																																	
			Disabled	0																															
			Enabled	1																															
K	RW	CH2LIMITH																																	
			Disabled	0																															
			Enabled	1																															
L	RW	CH2LIMITL																																	
			Disabled	0																															
			Enabled	1																															
M	RW	CH3LIMITH																																	
			Disabled	0																															
			Enabled	1																															
N	RW	CH3LIMITL																																	
			Disabled	0																															
			Enabled	1																															
O	RW	CH4LIMITH																																	
			Disabled	0																															
			Enabled	1																															
P	RW	CH4LIMITL																																	
			Disabled	0																															
			Enabled	1																															
Q	RW	CH5LIMITH																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id														V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Disabled	0																															
			Enabled	1																															
R	RW	CH5LIMITL																																	
				See *EVENTS_CH[5].LIMITL*																															
			Disabled	0																															
			Enabled	1																															
S	RW	CH6LIMITH																																	
				See *EVENTS_CH[6].LIMITH*																															
			Disabled	0																															
			Enabled	1																															
T	RW	CH6LIMITL																																	
				See *EVENTS_CH[6].LIMITL*																															
			Disabled	0																															
			Enabled	1																															
U	RW	CH7LIMITH																																	
				See *EVENTS_CH[7].LIMITH*																															
			Disabled	0																															
			Enabled	1																															
V	RW	CH7LIMITL																																	
				See *EVENTS_CH[7].LIMITL*																															
			Disabled	0																															
			Enabled	1																															

37.11.2 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id														V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	STARTED																																	
				See *EVENTS_STARTED*																															
			Set	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	END																																	
				See *EVENTS_END*																															
			Set	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	DONE																																	
				See *EVENTS_DONE*																															
			Set	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	RESULTDONE																																	
				See *EVENTS_RESULTDONE*																															
			Set	1																															
			Disabled	0																															
			Enabled	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id														V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
E	RW	CALIBRATEDONE																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
F	RW	STOPPED																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
G	RW	CH0LIMITH																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
H	RW	CH0LIMITL																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
I	RW	CH1LIMITH																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
J	RW	CH1LIMITL																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
K	RW	CH2LIMITH																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
L	RW	CH2LIMITL																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
M	RW	CH3LIMITH																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
N	RW	CH3LIMITL																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
O	RW	CH4LIMITH																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id														V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id														V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Disabled	0																															
			Enabled	1																															
B	RW	END																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id														V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Enabled	1																															
L	RW	CH2LIMITL																																	
				See *EVENTS_CH[2].LIMITL*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
M	RW	CH3LIMITH																																	
				See *EVENTS_CH[3].LIMITH*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
N	RW	CH3LIMITL																																	
				See *EVENTS_CH[3].LIMITL*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
O	RW	CH4LIMITH																																	
				See *EVENTS_CH[4].LIMITH*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
P	RW	CH4LIMITL																																	
				See *EVENTS_CH[4].LIMITL*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
Q	RW	CH5LIMITH																																	
				See *EVENTS_CH[5].LIMITH*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
R	RW	CH5LIMITL																																	
				See *EVENTS_CH[5].LIMITL*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
S	RW	CH6LIMITH																																	
				See *EVENTS_CH[6].LIMITH*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
T	RW	CH6LIMITL																																	
				See *EVENTS_CH[6].LIMITL*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
U	RW	CH7LIMITH																																	
				See *EVENTS_CH[7].LIMITH*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id														V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
V	RW	CH7LIMITL																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															

Write '1' to Disable interrupt for CH[7].LIMITL event

See *EVENTS_CH[7].LIMITL*

Disable

Read: Disabled

Read: Enabled

37.11.4 STATUS

Address offset: 0x400

Status

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	STATUS																																	
			Ready	0																															
			Busy	1																															

Status

ADC is ready. No on-going conversion.

ADC is busy. Conversion in progress.

37.11.5 ENABLE

Address offset: 0x500

Enable or disable ADC

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	ENABLE																																	
			Disabled	0																															
			Enabled	1																															

Enable or disable ADC

Disable ADC

Enable ADC

When enabled, the ADC will acquire access to the analog input pins specified in the CH[n].PSELP and CH[n].PSELN registers.

37.11.6 CH[0].PSELP

Address offset: 0x510

Input positive pin selection for CH[0]

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSELP																																	
			NC	0																															
			AnalogInput0	1																															
			AnalogInput1	2																															
			AnalogInput2	3																															
			AnalogInput3	4																															
			AnalogInput4	5																															
			AnalogInput5	6																															
			AnalogInput6	7																															
			AnalogInput7	8																															
			VDD	9																															

Analog positive input channel

Not connected

AIN0

AIN1

AIN2

AIN3

AIN4

AIN5

AIN6

AIN7

VDD

37.11.7 CH[0].PSELN

Address offset: 0x514

Input negative pin selection for CH[0]

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSELN																																	
			NC	0																															
			AnalogInput0	1																															
			AnalogInput1	2																															
			AnalogInput2	3																															
			AnalogInput3	4																															
			AnalogInput4	5																															
			AnalogInput5	6																															
			AnalogInput6	7																															
			AnalogInput7	8																															
			VDD	9																															

37.11.8 CH[0].CONFIG

Address offset: 0x518

Input configuration for CH[0]

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											G				F		E	E	E				D		C	C	C			B	B			A	A
Reset 0x00020000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	RESP																																	
			Bypass	0																															
			Pulldown	1																															
			Pullup	2																															
			VDD1_2	3																															
B	RW	RESN																																	
			Bypass	0																															
			Pulldown	1																															
			Pullup	2																															
			VDD1_2	3																															
C	RW	GAIN																																	
			Gain1_6	0																															
			Gain1_5	1																															
			Gain1_4	2																															
			Gain1_3	3																															
			Gain1_2	4																															
			Gain1	5																															
			Gain2	6																															
			Gain4	7																															
D	RW	REFSEL																																	
			Internal	0																															
			VDD1_4	1																															
E	RW	TACQ																																	
			3us	0																															
			5us	1																															
			10us	2																															
			15us	3																															
			20us	4																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											G				F		E	E	E				D		C	C	C			B	B			A	A
Reset 0x00020000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			40us	5																															
F	RW	MODE																																	
			SE	0																															
			Diff	1																															
G	RW	BURST																																	
			Disabled	0																															
			Enabled	1																															

37.11.9 CH[0].LIMIT

Address offset: 0x51C

High/low limits for event monitoring a channel

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x7FFF8000				**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	LOW		[-32768 to +32767]																															
B	RW	HIGH		[-32768 to +32767]																															

37.11.10 CH[1].PSELP

Address offset: 0x520

Input positive pin selection for CH[1]

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSELP																																	
			NC	0																															
			AnalogInput0	1																															
			AnalogInput1	2																															
			AnalogInput2	3																															
			AnalogInput3	4																															
			AnalogInput4	5																															
			AnalogInput5	6																															
			AnalogInput6	7																															
			AnalogInput7	8																															
			VDD	9																															

37.11.11 CH[1].PSELN

Address offset: 0x524

Input negative pin selection for CH[1]

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSELN																																	
			NC	0																															
			AnalogInput0	1																															

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																														A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
			AnalogInput1	2																														
			AnalogInput2	3																														
			AnalogInput3	4																														
			AnalogInput4	5																														
			AnalogInput5	6																														
			AnalogInput6	7																														
			AnalogInput7	8																														
			VDD	9																														

37.11.12 CH[1].CONFIG

Address offset: 0x528

Input configuration for CH[1]

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id										G				F		E	E	E				D		C	C	C			B	B			A	A
Reset 0x00020000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	RESP																																
			Bypass	0																														
			Pulldown	1																														
			Pullup	2																														
			VDD1_2	3																														
B	RW	RESN																																
			Bypass	0																														
			Pulldown	1																														
			Pullup	2																														
			VDD1_2	3																														
C	RW	GAIN																																
			Gain1_6	0																														
			Gain1_5	1																														
			Gain1_4	2																														
			Gain1_3	3																														
			Gain1_2	4																														
			Gain1	5																														
			Gain2	6																														
			Gain4	7																														
D	RW	REFSEL																																
			Internal	0																														
			VDD1_4	1																														
E	RW	TACQ																																
			3us	0																														
			5us	1																														
			10us	2																														
			15us	3																														
			20us	4																														
			40us	5																														
F	RW	MODE																																
			SE	0																														
			Diff	1																														
G	RW	BURST																																
			Disabled	0																														

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id										G				F		E	E	E				D		C	C	C			B	B			A	A
Reset 0x00020000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
			Enabled	1																														

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																														A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																							
			VDD								9																							

37.11.16 CH[2].CONFIG

Address offset: 0x538

Input configuration for CH[2]

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id										G				F		E	E	E				D		C	C	C			B	B			A	A
Reset 0x00020000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																							
A	RW	RESP																																Positive channel resistor control
			Bypass								0																							Bypass resistor ladder
			Pulldown								1																							Pull-down to GND
			Pullup								2																							Pull-up to VDD
			VDD1_2								3																							Set input at VDD/2
B	RW	RESN																																Negative channel resistor control
			Bypass								0																							Bypass resistor ladder
			Pulldown								1																							Pull-down to GND
			Pullup								2																							Pull-up to VDD
			VDD1_2								3																							Set input at VDD/2
C	RW	GAIN																																Gain control
			Gain1_6								0																							1/6
			Gain1_5								1																							1/5
			Gain1_4								2																							1/4
			Gain1_3								3																							1/3
			Gain1_2								4																							1/2
			Gain1								5																							1
			Gain2								6																							2
			Gain4								7																							4
D	RW	REFSEL																																Reference control
			Internal								0																							Internal reference (0.6 V)
			VDD1_4								1																							VDD/4 as reference
E	RW	TACQ																																Acquisition time, the time the ADC uses to sample the input voltage
			3us								0																							3 us
			5us								1																							5 us
			10us								2																							10 us
			15us								3																							15 us
			20us								4																							20 us
			40us								5																							40 us
F	RW	MODE																																Enable differential mode
			SE								0																							Single ended, PSELN will be ignored, negative input to ADC shorted to GND
			Diff								1																							Differential
G	RW	BURST																																Enable burst mode
			Disabled								0																							Burst mode is disabled (normal operation)
			Enabled								1																							Burst mode is enabled. SAADC takes 2^OVERSAMPLE number of samples as fast as it can, and sends the average to Data RAM.

37.11.17 CH[2].LIMIT

Address offset: 0x53C

High/low limits for event monitoring a channel

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x7FFF8000				**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	LOW		[-32768 to +32767]																															
B	RW	HIGH		[-32768 to +32767]																															

37.11.18 CH[3].PSELP

Address offset: 0x540

Input positive pin selection for CH[3]

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSELP																																	
			NC	0																															
			AnalogInput0	1																															
			AnalogInput1	2																															
			AnalogInput2	3																															
			AnalogInput3	4																															
			AnalogInput4	5																															
			AnalogInput5	6																															
			AnalogInput6	7																															
			AnalogInput7	8																															
			VDD	9																															

37.11.19 CH[3].PSELN

Address offset: 0x544

Input negative pin selection for CH[3]

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSELN																																	
			NC	0																															
			AnalogInput0	1																															
			AnalogInput1	2																															
			AnalogInput2	3																															
			AnalogInput3	4																															
			AnalogInput4	5																															
			AnalogInput5	6																															
			AnalogInput6	7																															
			AnalogInput7	8																															
			VDD	9																															

37.11.20 CH[3].CONFIG

Address offset: 0x548

Input configuration for CH[3]

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											G				F		E	E	E			D		C	C	C			B	B			A	A
Reset 0x00020000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	RESP																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											G				F		E	E	E				D		C	C	C			B	B			A	A
Reset 0x00020000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Bypass	0																															
			Pulldown	1																															
			Pullup	2																															
			VDD1_2	3																															
B	RW	RESN																																	
			Bypass	0																															
			Pulldown	1																															
			Pullup	2																															
			VDD1_2	3																															
C	RW	GAIN																																	
			Gain1_6	0																															
			Gain1_5	1																															
			Gain1_4	2																															
			Gain1_3	3																															
			Gain1_2	4																															
			Gain1	5																															
			Gain2	6																															
			Gain4	7																															
D	RW	REFSEL																																	
			Internal	0																															
			VDD1_4	1																															
E	RW	TACQ																																	
			3us	0																															
			5us	1																															
			10us	2																															
			15us	3																															
			20us	4																															
			40us	5																															
F	RW	MODE																																	
			SE	0																															
			Diff	1																															
G	RW	BURST																																	
			Disabled	0																															
			Enabled	1																															

37.11.21 CH[3].LIMIT

Address offset: 0x54C

High/low limits for event monitoring a channel

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x7FFF8000				**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	LOW		[-32768 to +32767]																															
B	RW	HIGH		[-32768 to +32767]																															

37.11.22 CH[4].PSELP

Address offset: 0x550

Input positive pin selection for CH[4]

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																														A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																							**Description**
A	RW	PSELP																																Analog positive input channel
			NC								0																							Not connected
			AnalogInput0								1																							AIN0
			AnalogInput1								2																							AIN1
			AnalogInput2								3																							AIN2
			AnalogInput3								4																							AIN3
			AnalogInput4								5																							AIN4
			AnalogInput5								6																							AIN5
			AnalogInput6								7																							AIN6
			AnalogInput7								8																							AIN7
			VDD								9																							VDD

37.11.23 CH[4].PSELN

Address offset: 0x554

Input negative pin selection for CH[4]

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																														A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																							**Description**
A	RW	PSELN																																Analog negative input, enables differential channel
			NC								0																							Not connected
			AnalogInput0								1																							AIN0
			AnalogInput1								2																							AIN1
			AnalogInput2								3																							AIN2
			AnalogInput3								4																							AIN3
			AnalogInput4								5																							AIN4
			AnalogInput5								6																							AIN5
			AnalogInput6								7																							AIN6
			AnalogInput7								8																							AIN7
			VDD								9																							VDD

37.11.24 CH[4].CONFIG

Address offset: 0x558

Input configuration for CH[4]

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id										G				F		E	E	E				D		C	C	C			B	B			A	A
Reset 0x00020000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																							**Description**
A	RW	RESP																																Positive channel resistor control
			Bypass								0																							Bypass resistor ladder
			Pulldown								1																							Pull-down to GND
			Pullup								2																							Pull-up to VDD
			VDD1_2								3																							Set input at VDD/2
B	RW	RESN																																Negative channel resistor control
			Bypass								0																							Bypass resistor ladder
			Pulldown								1																							Pull-down to GND
			Pullup								2																							Pull-up to VDD
			VDD1_2								3																							Set input at VDD/2
C	RW	GAIN																																Gain control
			Gain1_6								0																							1/6
			Gain1_5								1																							1/5

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											G				F		E	E	E				D		C	C	C			B	B			A	A
Reset 0x00020000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Gain1_4	2																															
			Gain1_3	3																															
			Gain1_2	4																															
			Gain1	5																															
			Gain2	6																															
			Gain4	7																															
D	RW	REFSEL																																	
			Internal	0																															
			VDD1_4	1																															
E	RW	TACQ																																	
			3us	0																															
			5us	1																															
			10us	2																															
			15us	3																															
			20us	4																															
			40us	5																															
F	RW	MODE																																	
			SE	0																															
			Diff	1																															
G	RW	BURST																																	
			Disabled	0																															
			Enabled	1																															

37.11.25 CH[4].LIMIT

Address offset: 0x55C

High/low limits for event monitoring a channel

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x7FFF8000				**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	LOW		[-32768 to +32767]																															
B	RW	HIGH		[-32768 to +32767]																															

37.11.26 CH[5].PSELP

Address offset: 0x560

Input positive pin selection for CH[5]

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSELP																																	
			NC	0																															
			AnalogInput0	1																															
			AnalogInput1	2																															
			AnalogInput2	3																															
			AnalogInput3	4																															
			AnalogInput4	5																															
			AnalogInput5	6																															

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																														A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**							**Value**																								
			AnalogInput6							7																								
			AnalogInput7							8																								
			VDD							9																								

37.11.27 CH[5].PSELN

Address offset: 0x564

Input negative pin selection for CH[5]

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																														A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**							**Value**																								
A	RW	PSELN																																
			NC							0																								
			AnalogInput0							1																								
			AnalogInput1							2																								
			AnalogInput2							3																								
			AnalogInput3							4																								
			AnalogInput4							5																								
			AnalogInput5							6																								
			AnalogInput6							7																								
			AnalogInput7							8																								
			VDD							9																								

37.11.28 CH[5].CONFIG

Address offset: 0x568

Input configuration for CH[5]

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id										G				F		E	E	E				D		C	C	C			B	B			A	A
Reset 0x00020000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**							**Value**																								
A	RW	RESP																																
			Bypass							0																								
			Pulldown							1																								
			Pullup							2																								
			VDD1_2							3																								
B	RW	RESN																																
			Bypass							0																								
			Pulldown							1																								
			Pullup							2																								
			VDD1_2							3																								
C	RW	GAIN																																
			Gain1_6							0																								
			Gain1_5							1																								
			Gain1_4							2																								
			Gain1_3							3																								
			Gain1_2							4																								
			Gain1							5																								
			Gain2							6																								
			Gain4							7																								
D	RW	REFSEL																																
			Internal							0																								

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											G				F		E	E	E				D		C	C	C			B	B			A	A
Reset 0x00020000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			VDD1_4	1																															
E	RW	TACQ																																	
			3us	0																															
			5us	1																															
			10us	2																															
			15us	3																															
			20us	4																															
			40us	5																															
F	RW	MODE																																	
			SE	0																															
			Diff	1																															
G	RW	BURST																																	
			Disabled	0																															
			Enabled	1																															

37.11.29 CH[5].LIMIT

Address offset: 0x56C

High/low limits for event monitoring a channel

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x7FFF8000				**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	LOW		[-32768 to +32767]																															
B	RW	HIGH		[-32768 to +32767]																															

37.11.30 CH[6].PSELP

Address offset: 0x570

Input positive pin selection for CH[6]

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSELP																																	
			NC	0																															
			AnalogInput0	1																															
			AnalogInput1	2																															
			AnalogInput2	3																															
			AnalogInput3	4																															
			AnalogInput4	5																															
			AnalogInput5	6																															
			AnalogInput6	7																															
			AnalogInput7	8																															
			VDD	9																															

37.11.31 CH[6].PSELN

Address offset: 0x574

Input negative pin selection for CH[6]

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																														A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														**Description**
A	RW	PSELN																																Analog negative input, enables differential channel
			NC	0																														Not connected
			AnalogInput0	1																														AIN0
			AnalogInput1	2																														AIN1
			AnalogInput2	3																														AIN2
			AnalogInput3	4																														AIN3
			AnalogInput4	5																														AIN4
			AnalogInput5	6																														AIN5
			AnalogInput6	7																														AIN6
			AnalogInput7	8																														AIN7
			VDD	9																														VDD

37.11.32 CH[6].CONFIG

Address offset: 0x578

Input configuration for CH[6]

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id										G				F		E	E	E				D		C	C	C			B	B			A	A
Reset 0x00020000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														**Description**
A	RW	RESP																																Positive channel resistor control
			Bypass	0																														Bypass resistor ladder
			Pulldown	1																														Pull-down to GND
			Pullup	2																														Pull-up to VDD
			VDD1_2	3																														Set input at VDD/2
B	RW	RESN																																Negative channel resistor control
			Bypass	0																														Bypass resistor ladder
			Pulldown	1																														Pull-down to GND
			Pullup	2																														Pull-up to VDD
			VDD1_2	3																														Set input at VDD/2
C	RW	GAIN																																Gain control
			Gain1_6	0																														1/6
			Gain1_5	1																														1/5
			Gain1_4	2																														1/4
			Gain1_3	3																														1/3
			Gain1_2	4																														1/2
			Gain1	5																														1
			Gain2	6																														2
			Gain4	7																														4
D	RW	REFSEL																																Reference control
			Internal	0																														Internal reference (0.6 V)
			VDD1_4	1																														VDD/4 as reference
E	RW	TACQ																																Acquisition time, the time the ADC uses to sample the input voltage
			3us	0																														3 us
			5us	1																														5 us
			10us	2																														10 us
			15us	3																														15 us
			20us	4																														20 us
			40us	5																														40 us
F	RW	MODE																																Enable differential mode
			SE	0																														Single ended, PSELN will be ignored, negative input to ADC shorted to GND
			Diff	1																														Differential

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											G				F		E	E	E				D		C	C	C			B	B			A	A
Reset 0x00020000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
G	RW	BURST																																	
			Disabled	0																															
			Enabled	1																															

37.11.33 CH[6].LIMIT

Address offset: 0x57C

High/low limits for event monitoring a channel

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x7FFF8000				**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	LOW		[-32768 to +32767]																															
B	RW	HIGH		[-32768 to +32767]																															

37.11.34 CH[7].PSELP

Address offset: 0x580

Input positive pin selection for CH[7]

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSELP																																	
			NC	0																															
			AnalogInput0	1																															
			AnalogInput1	2																															
			AnalogInput2	3																															
			AnalogInput3	4																															
			AnalogInput4	5																															
			AnalogInput5	6																															
			AnalogInput6	7																															
			AnalogInput7	8																															
			VDD	9																															

37.11.35 CH[7].PSELN

Address offset: 0x584

Input negative pin selection for CH[7]

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSELN																																	
			NC	0																															
			AnalogInput0	1																															
			AnalogInput1	2																															
			AnalogInput2	3																															
			AnalogInput3	4																															
			AnalogInput4	5																															
			AnalogInput5	6																															

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																														A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																							
			AnalogInput6								7																							
			AnalogInput7								8																							
			VDD								9																							

37.11.36 CH[7].CONFIG

Address offset: 0x588

Input configuration for CH[7]

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id										G				F		E	E	E				D		C	C	C			B	B			A	A
Reset 0x00020000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																							
A	RW	RESP																																
			Bypass								0																							
			Pulldown								1																							
			Pullup								2																							
			VDD1_2								3																							
B	RW	RESN																																
			Bypass								0																							
			Pulldown								1																							
			Pullup								2																							
			VDD1_2								3																							
C	RW	GAIN																																
			Gain1_6								0																							
			Gain1_5								1																							
			Gain1_4								2																							
			Gain1_3								3																							
			Gain1_2								4																							
			Gain1								5																							
			Gain2								6																							
			Gain4								7																							
D	RW	REFSEL																																
			Internal								0																							
			VDD1_4								1																							
E	RW	TACQ																																
			3us								0																							
			5us								1																							
			10us								2																							
			15us								3																							
			20us								4																							
			40us								5																							
F	RW	MODE																																
			SE								0																							
			Diff								1																							
G	RW	BURST																																
			Disabled								0																							
			Enabled								1																							

37.11.37 CH[7].LIMIT

Address offset: 0x58C

High/low limits for event monitoring a channel

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x7FFF8000				**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	LOW		[-32768 to +32767]																															
B	RW	HIGH		[-32768 to +32767]																															

37.11.38 RESOLUTION

Address offset: 0x5F0

Resolution configuration

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																	A	A	A
Reset 0x00000001				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	VAL																																	
			8bit	0																															
			10bit	1																															
			12bit	2																															
			14bit	3																															

37.11.39 OVERSAMPLE

Address offset: 0x5F4

Oversampling configuration. OVERSAMPLE should not be combined with SCAN. The RESOLUTION is applied before averaging, thus for high OVERSAMPLE a higher RESOLUTION should be used.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	OVERSAMPLE																																	
			Bypass	0																															
			Over2x	1																															
			Over4x	2																															
			Over8x	3																															
			Over16x	4																															
			Over32x	5																															
			Over64x	6																															
			Over128x	7																															
			Over256x	8																															

37.11.40 SAMPLERATE

Address offset: 0x5F8

Controls normal or continuous sample rate

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																							B	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	CC		[80..2047]																															
B	RW	MODE																																	
			Task	0																															
			Timers	1																															

37.11.41 RESULT.PTR

Address offset: 0x62C

Data pointer

Bit number																																
Id																																
Reset 0x00000000																																
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																											
A	RW	PTR			Data pointer																											

37.11.42 RESULT.MAXCNT

Address offset: 0x630

Maximum number of buffer words to transfer

Bit number																																
Id																																
Reset 0x00000000																																
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																											
A	RW	MAXCNT			Maximum number of buffer words to transfer																											

37.11.43 RESULT.AMOUNT

Address offset: 0x634

Number of buffer words transferred since last START

Bit number																																
Id																																
Reset 0x00000000																																
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																											
A	R	AMOUNT			Number of buffer words transferred since last START. This register can be read after an END or STOPPED event.																											

37.12 Electrical specification

37.12.1 SAADC Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
DNL	Differential non-linearity, 10-bit resolution	-0.95	<1		LSB
INL	Integral non-linearity, 10-bit resolution		1		LSB
V_{OS}	Differential offset error (calibrated), 10-bit resolution [a]		+-2		LSB
C_{EG}	Gain error temperature coefficient		0.02		%/°C
f_{SAMPLE}	Maximum sampling rate			200	kHz
$t_{ACQ,10k}$	Acquisition time (configurable), source Resistance <= 10kOhm		3		µs
$t_{ACQ,40k}$	Acquisition time (configurable), source Resistance <= 40kOhm		5		µs
$t_{ACQ,100k}$	Acquisition time (configurable), source Resistance <= 100kOhm		10		µs
$t_{ACQ,200k}$	Acquisition time (configurable), source Resistance <= 200kOhm		15		µs
$t_{ACQ,400k}$	Acquisition time (configurable), source Resistance <= 400kOhm		20		µs
$t_{ACQ,800k}$	Acquisition time (configurable), source Resistance <= 800kOhm		40		µs
t_{CONV}	Conversion time		<2		µs
$I_{ADC,CONV}$	ADC current during ACQuisition and CONVersion		700		µA
$I_{ADC,IDLE}$	Idle current, when not sampling, excluding clock sources and regulator base currents[33]		<5		µA

[a] Digital output code at zero volt differential input.

[33] When t_{ACQ} is 10us or longer, and if DC/DC is active, it will be allowed to work in refresh mode if no other resource is requiring a high quality power supply from 1V3. If t_{ACQ} is smaller than 10us and DC/DC is active,

Symbol	Description	Min.	Typ.	Max.	Units
$E_{G1/6}$	Error[b] for Gain = 1/6	-3		3	%
$E_{G1/4}$	Error[b] for Gain = 1/4	-3		3	%
$E_{G1/2}$	Error[b] for Gain = 1/2	-3		4	%
E_{G1}	Error[b] for Gain = 1	-3		4	%
C_{SAMPLE}	Sample and hold capacitance at maximum gain[34]		2.5		pF
R_{INPUT}	Input resistance		>1		MΩ
E_{NOB}	Effective number of bits, differential mode, 12-bit resolution, 1/1 gain, 3 µs acquisition time, crystal HFCLK, 200 ksps		9		Bit
S_{NDR}	Peak signal to noise and distortion ratio, differential mode, 12-bit resolution, 1/1 gain, 3 µs acquisition time, crystal HFCLK, 200 ksps		56		dB
S_{FDR}	Spurious free dynamic range, differential mode, 12-bit resolution, 1/1 gain, 3 µs acquisition time, crystal HFCLK, 200 ksps		70		dBc
R_{LADDER}	Ladder resistance		160		kΩ

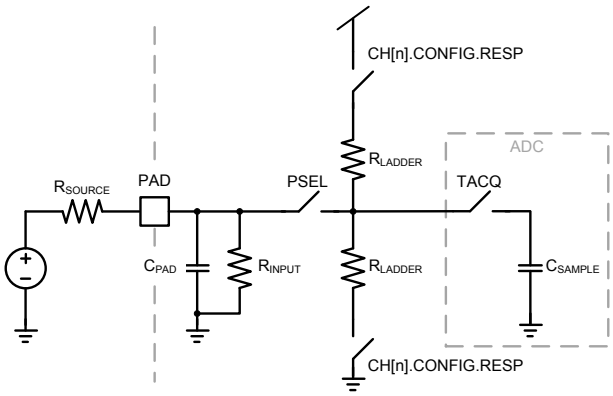


Figure 105: Model of SAADC input (one channel)

Note: SAADC average current calculation for a given application is based on the sample period, conversion and acquisition time (t_{conv} and t_{ACQ}) and conversion and idle current ($I_{ADC,CONV}$ and $I_{ADC,IDLE}$). For example, sampling at 4kHz gives a sample period of 250µs. The average current consumption would then be:

$$I_{AVERAGE} = \left(\frac{\left(t_{CONV} + t_{ACQ}\right)}{250}\right)\left(I_{ADC,CONV}\right) + \left(\frac{250 - \left(t_{CONV} + t_{ACQ}\right)}{250}\right)\left(I_{ADC,IDLE}\right)$$

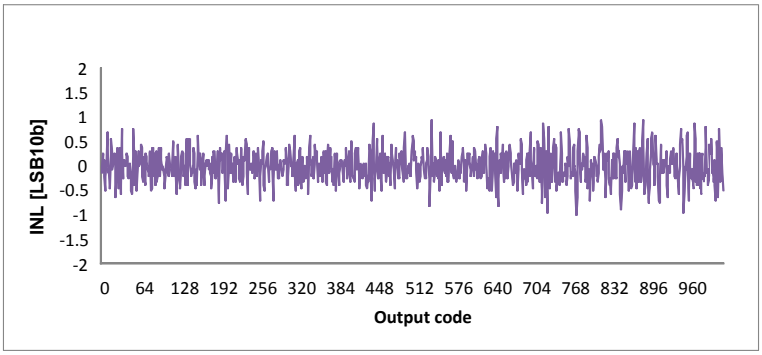


Figure 106: ADC INL vs Output Code

refresh mode will not be allowed, and it will remain in normal mode from the START task to the STOPPED event. So depending on t_{ACQ} and other resources' needs, the appropriate base current needs to be taken into account.

b Does not include temperature drift

34 Maximum gain corresponds to highest capacitance.

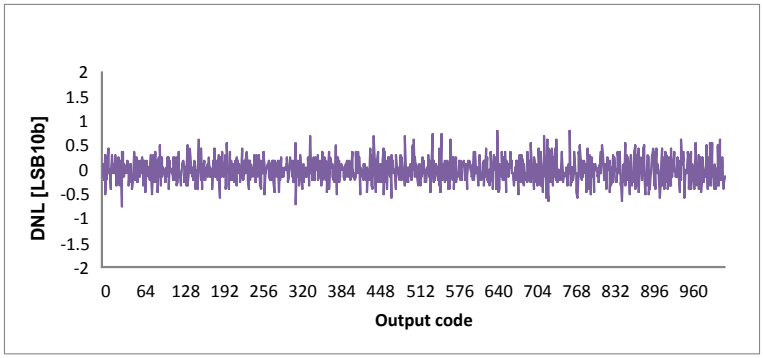


Figure 107: ADC DNL vs Output Code

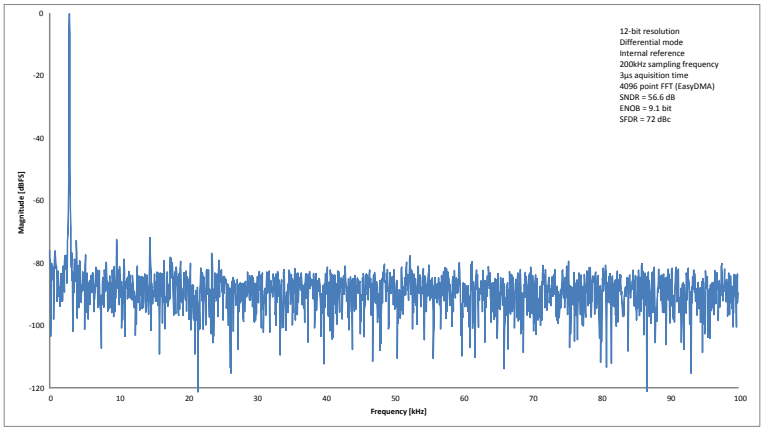


Figure 108: FFT of a 2.8 kHz sine at 200 ksps ()

37.13 Performance factors

Clock jitter, affecting sample timing accuracy, and circuit noise can affect ADC performance.

Jitter can be between START tasks or from START task to acquisition. START timer accuracy and startup times of regulators and references will contribute to variability. Sources of circuit noise may include CPU activity and the DC/DC regulator. Best ADC performance is achieved using START timing based on the TIMER module, HFXO clock source, and Constant Latency mode.

38 COMP — Comparator

The comparator (COMP) compares an input voltage (VIN+) against a second input voltage (VIN-). VIN+ can be derived from an analog input pin (AIN0-AIN7). VIN- can be derived from multiple sources depending on the operation mode of the comparator.

Main features of the comparator are:

- Input range from 0 V to VDD
- Single-ended mode
  - Fully flexible hysteresis using a 64-level reference ladder
- Differential mode
  - Configurable 50 mV hysteresis
- Reference inputs (VREF):
  - VDD
  - External reference from AIN0 to AIN7 (between 0 V and VDD)
  - Internal references 1.2 V, 1.8 V and 2.4 V
- Three speed/power consumption modes: low-power, normal and high-speed
- Single-pin capacitive sensor support
- Event generation on output changes
  - UP event on VIN- > VIN+
  - DOWN event on VIN- < VIN+
  - CROSS event on VIN+ and VIN- crossing
  - READY event on core and internal reference (if used) ready

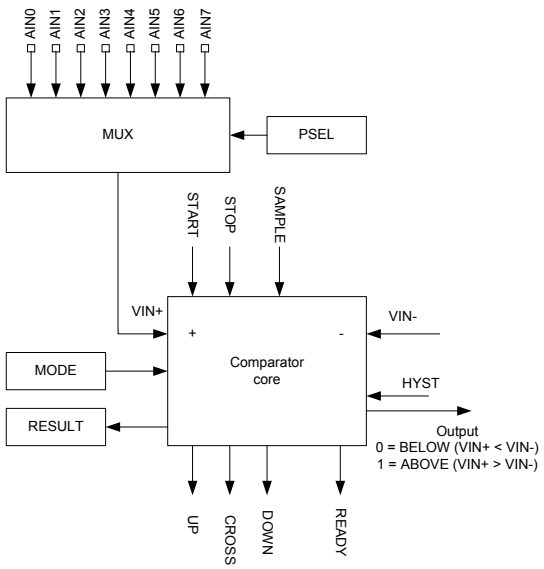


Figure 109: Comparator overview

Once enabled (using the *ENABLE* register), the comparator is started by triggering the START task and stopped by triggering the STOP task. After a start-up time of $t_{COMP,START}$, the comparator will generate a READY event to indicate that it is ready for use and that its output is correct. When the COMP module is started, events will be generated every time VIN+ crosses VIN-.

38 Operation modes

The comparator can be configured to operate in two main operation modes, differential mode and single-ended mode. See the *MODE* register for more information. In both operation modes, the comparator can operate in different speed and power consumption modes (low-power, normal and high-speed). High-speed mode will consume more power compared to low-power mode, and low-power mode will result in slower response time compared to high-speed mode.

Use the *PSEL* register to select any of the AIN0-AIN7 pins as VIN+ input, irregardless of the operation mode selected for the comparator. The source of VIN- depends on which operation mode is used:

- Differential mode: Derived directly from AIN0 to AIN7
- Single-ended mode: Derived from VREF. VREF can be derived from VDD, AIN0-AIN7 or internal 1.2 V, 1.8 V and 2.4 V references.

The selected analog pins will be acquired by the comparator once it is enabled.

An optional hysteresis on VIN+ and VIN- can be enabled when the module is used in differential mode through the *HYST* register. In single-ended mode, VUP and VDOWN thresholds can be set to implement a hysteresis using the reference ladder (see *Figure 112: Comparator in single-ended mode* on page 395). This hysteresis is in the order of magnitude of 50 mV, and shall prevent noise on the signal to create unwanted events. See *Figure 113: Hysteresis example where VIN+ starts below VUP* on page 395 for illustration of the effect of an active hysteresis on a noisy input signal.

An upward crossing will generate an UP event and a downward crossing will generate a DOWN event. The CROSS event will be generated every time there is a crossing, independent of direction.

The immediate value of the comparator can be sampled to *RESULT* register by triggering the SAMPLE task.

38.1 Differential mode

In differential mode, the reference input VIN- is derived directly from one of the AINx pins.

Before enabling the comparator via the *ENABLE* register, the following registers must be configured for the differential mode:

- *PSEL*
- *MODE*
- *EXTREFSEL*

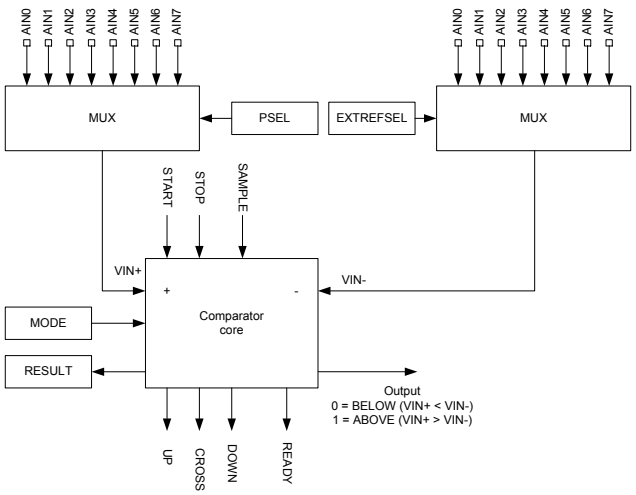


Figure 110: Comparator in differential mode

Restriction: Depending on the device, not all the analog inputs may be available for each MUX. See definitions for *PSEL* and *EXTREFSEL* for more information about which analog pins are available on a particular device.

When *HYST* register is turned on while in this mode, the output of the comparator (and associated events) will change from ABOVE to BELOW whenever VIN+ becomes lower than VIN- - ($V_{DIFFHYST}$ / 2). It will also change from BELOW to ABOVE whenever VIN+ becomes higher than VIN- + ($V_{DIFFHYST}$ / 2). This behavior is illustrated in *Figure 111: Hysteresis enabled in differential mode* on page 394.

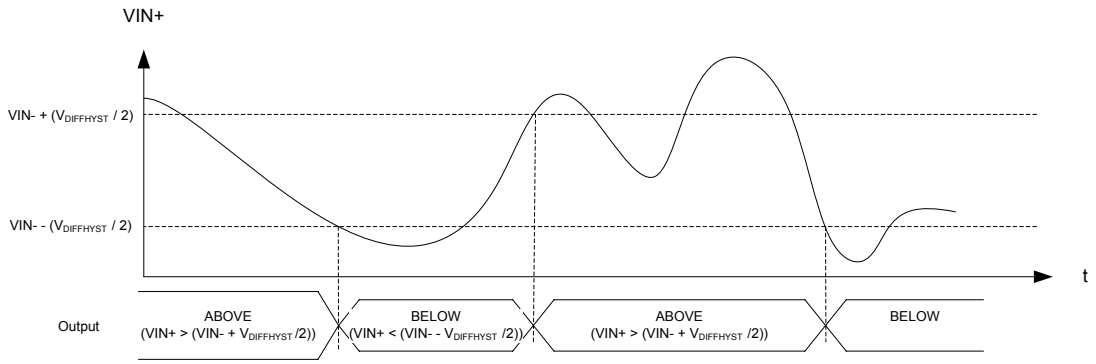


Figure 111: Hysteresis enabled in differential mode

38.2 Single-ended mode

In single-ended mode, VIN- is derived from the reference ladder.

Before enabling the comparator via the *ENABLE* register, the following registers must be configured for the single-ended mode:

- *PSEL*
- *MODE*
- *REFSEL*
- *EXTREFSEL*
- *TH*

The reference ladder uses the reference voltage (VREF) to derive two new voltage references, VUP and VDOWN. VUP and VDOWN are configured using THUP and THDOWN respectively in the *TH* register. VREF can be derived from any of the available reference sources, configured using the *EXTREFSEL* and *REFSEL* registers as illustrated in *Figure 112: Comparator in single-ended mode* on page 395. When AREF is selected in the *REFSEL* register, the *EXTREFSEL* register is used to select one of the AIN0-AIN7 analog input pins as reference input. The selected analog pins will be acquired by the comparator once it is enabled.

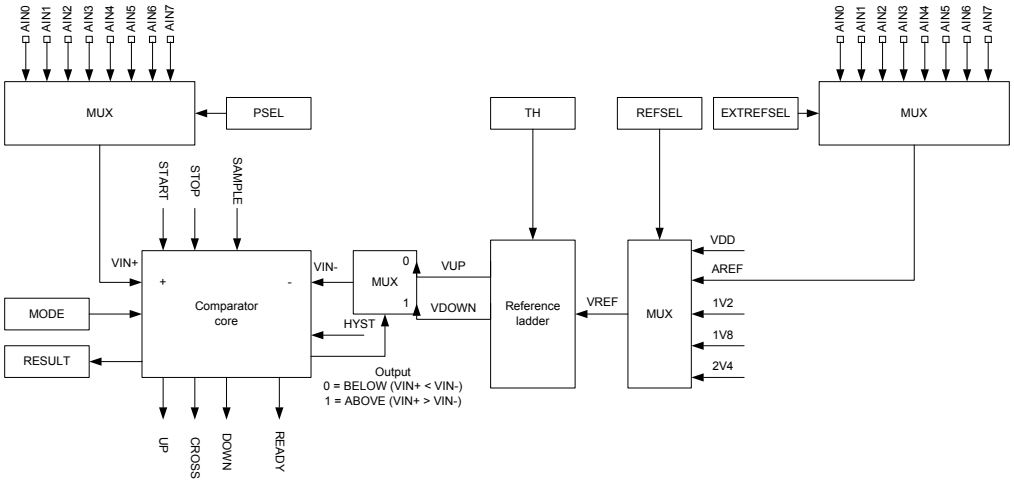


Figure 112: Comparator in single-ended mode

> **Restriction:** Depending on the device, not all the analog inputs may be available for each MUX. See definitions for *PSEL* and *EXTREFSEL* for more information about which analog pins are available on a particular device.

When the comparator core detects that VIN+ > VIN-, i.e. ABOVE as per the *RESULT* register, VIN- will switch to VDOWN. When VIN+ falls below VIN- again, VIN- will be switched back to VUP. By specifying VUP larger than VDOWN, a hysteresis can be generated as illustrated in *Figure 113: Hysteresis example where VIN+ starts below VUP* on page 395 and *Figure 114: Hysteresis example where VIN+ starts above VUP* on page 396.

Writing to *HYST* has no effect in single-ended mode, and the content of this register is ignored.

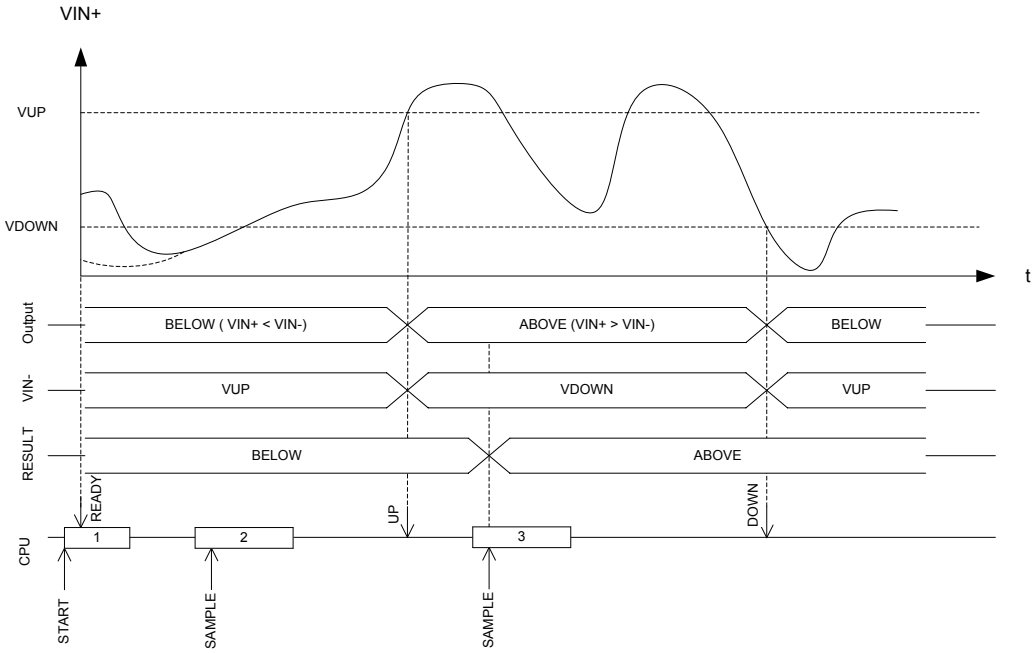


Figure 113: Hysteresis example where VIN+ starts below VUP

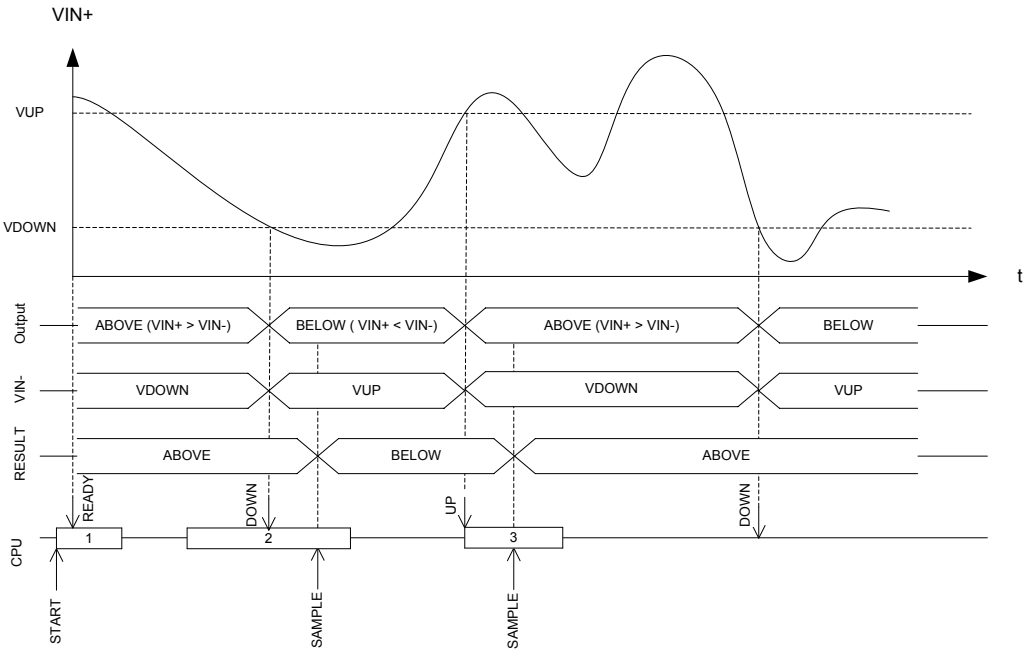


Figure 114: Hysteresis example where VIN+ starts above VUP

38.3 Registers

Table 92: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40013000	COMP	COMP	General purpose comparator	

Table 93: Register Overview

Register	Offset	Description
TASKS_START	0x000	Start comparator
TASKS_STOP	0x004	Stop comparator
TASKS_SAMPLE	0x008	Sample comparator value
EVENTS_READY	0x100	COMP is ready and output is valid
EVENTS_DOWN	0x104	Downward crossing
EVENTS_UP	0x108	Upward crossing
EVENTS_CROSS	0x10C	Downward or upward crossing
SHORTS	0x200	Shortcut register
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
RESULT	0x400	Compare result
ENABLE	0x500	COMP enable
PSEL	0x504	Pin select
REFSEL	0x508	Reference source select for single-ended mode
EXTREFSEL	0x50C	External reference select
TH	0x530	Threshold configuration for hysteresis unit
MODE	0x534	Mode configuration
HYST	0x538	Comparator hysteresis enable
ISOURCE	0x53C	Current source select on analog input

38.3.1 SHORTS

Address offset: 0x200

Shortcut register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	READY_SAMPLE																																	

Enable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	READY																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	DOWN																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	UP																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	CROSS																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															

38.3.4 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	READY																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	DOWN																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	UP																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	CROSS																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															

38.3.5 RESULT

Address offset: 0x400

Compare result

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	RESULT																																	
			Below	0																															
			Above	1																															

38.3.6 ENABLE

Address offset: 0x500

COMP enable

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	ENABLE																																	
			Disabled	0																															
			Enabled	2																															

38.3.7 PSEL

Address offset: 0x504

Pin select

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSEL																																	
			AnalogInput0	0																															
			AnalogInput1	1																															
			AnalogInput2	2																															
			AnalogInput3	3																															
			AnalogInput4	4																															
			AnalogInput5	5																															
			AnalogInput6	6																															
			AnalogInput7	7																															

38.3.8 REFSEL

Address offset: 0x508

Reference source select for single-ended mode

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																	A	A	A
Reset 0x00000004				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	REFSEL																																	
			Int1V2	0																															
			Int1V8	1																															
			Int2V4	2																															
			VDD	4																															
			ARef	7																															

38.3.9 EXTREFSEL

Address offset: 0x50C

External reference select

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	EXTREFSEL																																	
			AnalogReference0	0																															
			AnalogReference1	1																															
			AnalogReference2	2																															
			AnalogReference3	3																															
			AnalogReference4	4																															
			AnalogReference5	5																															
			AnalogReference6	6																															
			AnalogReference7	7																															

38.3.10 TH

Address offset: 0x530

Threshold configuration for hysteresis unit

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																						B	B	B	B	B	B		A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	THDOWN		[63:0]																															
B	RW	THUP		[63:0]																															

38.3.11 MODE

Address offset: 0x534

Mode configuration

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																											B							A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	SP																																	
			Low	0																															
			Normal	1																															
			High	2																															
B	RW	MAIN																																	
			SE	0																															
			Diff	1																															

38.3.12 HYST

Address offset: 0x538

Comparator hysteresis enable

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	HYST																																	
			NoHyst	0																															
			Hyst50mV	1																															

38.3.13 ISOURCE

Address offset: 0x53C

Current source select on analog input

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																														
A	RW	ISOURCE			Comparator hysteresis																														
			Off	0	Current source disabled																														
			Ien2mA5	1	Current source enabled (+/- 2.5 uA)																														
			Ien5mA	2	Current source enabled (+/- 5 uA)																														
			Ien10mA	3	Current source enabled (+/- 10 uA)																														

38.4 Electrical specification

38.4.1 COMP Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
$I_{COMP,LP}$	Core run current in low power mode		2		µA
$I_{COMP,N}$	Core run current in normal mode		5		µA
$I_{COMP,HS}$	Core run current in high-speed mode		10		µA
$t_{PROPDLY,LP}$	Propagation delay, low-power mode[a]		0.6		µS
$t_{PROPDLY,N}$	Propagation delay, normal mode[a]		0.2		µS
$t_{PROPDLY,HS}$	Propagation delay, high-speed mode[a]		0.1		µS
$V_{DIFFHYST}$	Optional hysteresis applied to differential input		30		mV
$V_{VDD-VREF}$	Required difference between VDD and a selected VREF, VDD > VREF	0.3			V
I_{INT_REF}	Current used by the internal bandgap reference when selected as source for VREF		13		µA
$t_{INT_REF,START}$	Startup time for the internal bandgap reference		50	80	µS
E_{INT_REF}	Internal bandgap reference error	-3		3	%
R_{LADDER}	Reference ladder resistance, I_{LADDER} = VREF / R_{LADDER}		550		kΩ
$V_{INPUTOFFSET}$	Input offset	-10		10	mV
$D_{NLLADDER}$	Differential non-linearity of reference ladder		<0.1		LSB
$t_{COMP,START}$	Startup time for the comparator core		3		µS

Total comparator run current must be calculated from the I_{COMP}, I_{INT_REF}, and I_{LADDER} values for a given reference voltage.

[a] Propagation delay is with 10 mV overdrive.

39 LPCOMP — Low power comparator

LPCOMP compares an input voltage against a reference voltage.

Listed here are the main features of LPCOMP:

- 0 - VDD input range
- Ultra low power
- Eight input options (AIN0 to AIN7)
- Reference voltage options:
  - Two external analog reference inputs, or
  - 15-level internal reference ladder (VDD/16)
- Optional hysteresis enable on input
- Wakeup source from OFF mode

In System ON, the LPCOMP can generate separate events on rising and falling edges of a signal, or sample the current state of the pin as being above or below the selected reference. The block can be configured to use any of the analog inputs on the device. Additionally, the low power comparator can be used as an analog wakeup source from System OFF or System ON. The comparator threshold can be programmed to a range of fractions of the supply voltage.

Restriction: LPCOMP cannot be used (STARTed) at the same time as COMP. Only one comparator can be used at a time.

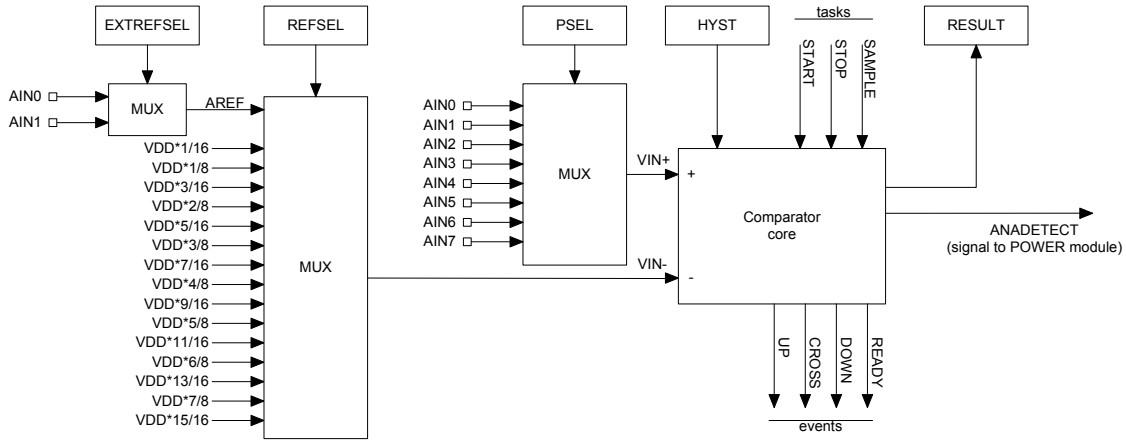


Figure 115: Low power comparator

The wakeup comparator (LPCOMP) compares an input voltage (VIN+), which comes from an analog input pin selected via the PSEL register against a reference voltage (VIN-) selected via the *REFSEL* on page 407 and *EXTREFSEL* registers.

The *PSEL*, *REFSEL*, and *EXTREFSEL* registers must be configured before the LPCOMP is enabled through the *ENABLE* register.

The *HYST* register allows enabling an optional hysteresis in the comparator core. This hysteresis is in the order of magnitude of 50 mV, and shall prevent noise on the signal to create unwanted events. See *Figure 116: Effect of hysteresis on a noisy input signal* on page 403 for illustration of the effect of an active hysteresis on a noisy input signal. It is disabled by default, and shall be configured before enabling LPCOMP as well.

The LPCOMP is started by triggering the START task. After a start-up time of $t_{LPCOMP,STARTUP}$ the LPCOMP will generate a READY event to indicate that the comparator is ready to use and the output of the LPCOMP is correct. The LPCOMP will generate events every time VIN+ crosses VIN-. More specifically, every time VIN+ rises above VIN- (upward crossing) an UP event is generated along with a CROSS event. Every time VIN+ falls below VIN- (downward crossing), a DOWN event is generated along with a CROSS event. When

hysteresis is enabled, the upward crossing level becomes (VIN- + VHYST/2), and the downward crossing level becomes (VIN- - VHYST/2).

The LPCOMP is stopped by triggering the STOP task.

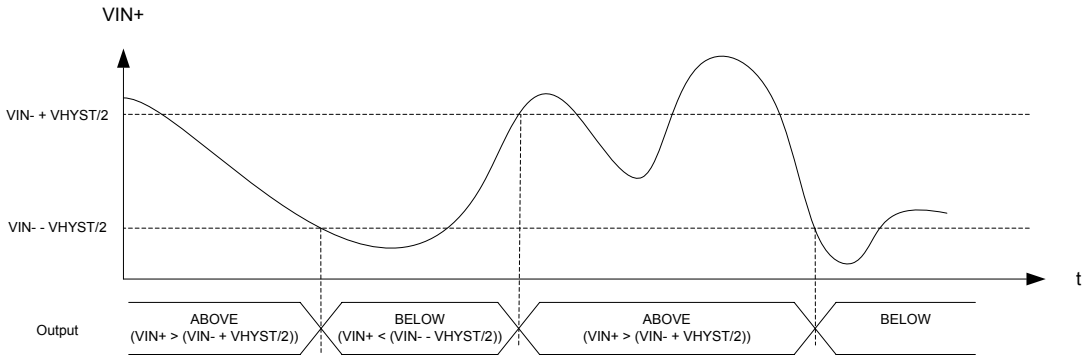


Figure 116: Effect of hysteresis on a noisy input signal

LPCOMP will be operational in both System ON and System OFF mode when it is enabled through the ENABLE register. See *POWER — Power supply* on page 78 for more information about power modes. Note that it is not allowed to go to System OFF when a READY event is pending to be generated.

All LPCOMP registers, including *ENABLE*, are classified as retained registers when the LPCOMP is enabled. However, when the device wakes up from System OFF, all LPCOMP registers will be reset.

The LPCOMP can wake up the system from System OFF by asserting the ANADETECT signal. The ANADETECT signal can be derived from any of the event sources that generate the UP, DOWN and CROSS events. In case of wakeup from System OFF, no events will be generated, only the ANADETECT signal. See the ANADETECT register (*ANADETECT* on page 407) for more information on how to configure the ANADETECT signal.

The immediate value of the LPCOMP can be sampled to *RESULT* on page 406 by triggering the SAMPLE task.

See *RESETREAS* on page 85 for more information on how to detect a wakeup from LPCOMP.

39.1 Shared resources

The LPCOMP shares resources with other peripherals.

The LPCOMP shares analog resources with SAADC and COMP. While it is possible to use SAADC at the same time as COMP or LPCOMP, COMP and LPCOMP are mutually exclusive: enabling one will automatically disable the other. In addition, when using SAADC and COMP or LPCOMP simultaneously, it is not possible to select the same analog input pin for both modules.

The LPCOMP peripheral shall not be disabled (by writing to the ENABLE register) before the peripheral has been stopped. Failing to do so may result in unpredictable behaviour.

39.2 Pin configuration

You can use the LPCOMP.PSEL register to select one of the analog input pins, `AIN0` through `AIN7`, as the analog input pin for the LPCOMP.

See *GPIO — General purpose input/output* on page 111 for more information about the pins. Similarly, you can use *EXTREFSEL* on page 407 to select one of the analog reference input pins, `AIN0` and `AIN1`, as input for AREF in case AREF is selected in *EXTREFSEL* on page 407. The selected analog pins will be acquired by the LPCOMP when it is enabled through *ENABLE* on page 406.

39.3 Registers

Table 94: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40013000	LPCOMP	LPCOMP	Low power comparator	

Table 95: Register Overview

Register	Offset	Description
TASKS_START	0x000	Start comparator
TASKS_STOP	0x004	Stop comparator
TASKS_SAMPLE	0x008	Sample comparator value
EVENTS_READY	0x100	LPCOMP is ready and output is valid
EVENTS_DOWN	0x104	Downward crossing
EVENTS_UP	0x108	Upward crossing
EVENTS_CROSS	0x10C	Downward or upward crossing
SHORTS	0x200	Shortcut register
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
RESULT	0x400	Compare result
ENABLE	0x500	Enable LPCOMP
PSEL	0x504	Input pin select
REFSEL	0x508	Reference select
EXTREFSEL	0x50C	External reference select
ANADETECT	0x520	Analog detect configuration
HYST	0x538	Comparator hysteresis enable

39.3.1 SHORTS

Address offset: 0x200

Shortcut register

Bit number																															
Id																												E	D	C	B	A
Reset 0x00000000																												**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	READY_SAMPLE			Shortcut between READY event and SAMPLE task See *EVENTS_READY* and *TASKS_SAMPLE*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
B	RW	READY_STOP			Shortcut between READY event and STOP task See *EVENTS_READY* and *TASKS_STOP*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
C	RW	DOWN_STOP			Shortcut between DOWN event and STOP task See *EVENTS_DOWN* and *TASKS_STOP*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
D	RW	UP_STOP			Shortcut between UP event and STOP task See *EVENTS_UP* and *TASKS_STOP*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
E	RW	CROSS_STOP			Shortcut between CROSS event and STOP task See *EVENTS_CROSS* and *TASKS_STOP*
			Disabled	0	Disable shortcut

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Enabled	1																															

Enable shortcut

39.3.2 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	READY																																	

Write '1' to Enable interrupt for READY event

See *EVENTS_READY*

			Set	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	DOWN																																	

Write '1' to Enable interrupt for DOWN event

See *EVENTS_DOWN*

			Set	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	UP																																	

Write '1' to Enable interrupt for UP event

See *EVENTS_UP*

			Set	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	CROSS																																	

Write '1' to Enable interrupt for CROSS event

See *EVENTS_CROSS*

			Set	1																															
			Disabled	0																															
			Enabled	1																															

Read: Enabled

39.3.3 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	READY																																	

Write '1' to Disable interrupt for READY event

See *EVENTS_READY*

			Clear	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	DOWN																																	

Write '1' to Disable interrupt for DOWN event

See *EVENTS_DOWN*

			Clear	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	UP																																	

Write '1' to Disable interrupt for UP event

See *EVENTS_UP*

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	CROSS																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															

39.3.4 RESULT

Address offset: 0x400

Compare result

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	RESULT																																	
			Bellow	0																															
			Below	0																															
			Above	1																															

39.3.5 ENABLE

Address offset: 0x500

Enable LPCOMP

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	ENABLE																																	
			Disabled	0																															
			Enabled	1																															

39.3.6 PSEL

Address offset: 0x504

Input pin select

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																A	A	A	
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSEL																																	
			AnalogInput0	0																															
			AnalogInput1	1																															
			AnalogInput2	2																															
			AnalogInput3	3																															
			AnalogInput4	4																															
			AnalogInput5	5																															
			AnalogInput6	6																															
			AnalogInput7	7																															

39.3.7 REFSEL

Address offset: 0x508

Reference select

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																A	A	A	A
Reset 0x00000004				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	REFSEL																																	
			Ref1_8Vdd	0																															
			Ref2_8Vdd	1																															
			Ref3_8Vdd	2																															
			Ref4_8Vdd	3																															
			Ref5_8Vdd	4																															
			Ref6_8Vdd	5																															
			Ref7_8Vdd	6																															
			ARef	7																															
			Ref1_16Vdd	8																															
			Ref3_16Vdd	9																															
			Ref5_16Vdd	10																															
			Ref7_16Vdd	11																															
			Ref9_16Vdd	12																															
			Ref11_16Vdd	13																															
			Ref13_16Vdd	14																															
			Ref15_16Vdd	15																															

39.3.8 EXTREFSEL

Address offset: 0x50C

External reference select

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	EXTREFSEL																																	
			AnalogReference0	0																															
			AnalogReference1	1																															

39.3.9 ANADETECT

Address offset: 0x520

Analog detect configuration

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	ANADETECT																																	
			Cross	0																															
			Up	1																															
			Down	2																															

39.3.10 HYST

Address offset: 0x538

Comparator hysteresis enable

Bit number																																
Id																																A
Reset 0x00000000					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																											
A	RW	HYST			Comparator hysteresis enable																											
			NoHyst	0	Comparator hysteresis disabled																											
			Hyst50mV	1	Comparator hysteresis disabled (typ. 50 mV)																											

39.4 Electrical specification

39.4.1 LPCOMP Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
I_{LPC}	Run current for low power comparator		0.5		µA
$t_{LPCANADET}$	Time from VIN crossing (>=50mV above threshold) to ANADETECT signal generated.		5		µs
$E_{REFLADDER}$	Error in reference ladder threshold voltage	-30		30	mV
V_{HYST}	Optional hysteresis		30		mV

40 WDT — Watchdog timer

A countdown watchdog timer using the low-frequency clock source (LFCLK) offers configurable and robust protection against application lock-up.

The watchdog timer is started by triggering the START task.

The watchdog can be paused during long CPU sleep periods for low power applications and when the debugger has halted the CPU. The watchdog is implemented as a down-counter that generates a TIMEOUT event when it wraps over after counting down to 0. When the watchdog timer is started through the START task, the watchdog counter is loaded with the value specified in the CRV register. This counter is also reloaded with the value specified in the CRV register when a reload request is granted.

The watchdog's timeout period is given by:

```
timeout [s] = ( CRV + 1 ) / 32768
```

When started, the watchdog will automatically force the 32.768 kHz RC oscillator on as long as no other 32.768 kHz clock source is running and generating the 32.768 kHz system clock, see chapter *CLOCK — Clock control* on page 101.

40.1 Reload criteria

The watchdog has eight separate reload request registers, which shall be used to request the watchdog to reload its counter with the value specified in the CRV register. To reload the watchdog counter, the special value 0x6E524635 needs to be written to all enabled reload registers.

One or more RR registers can be individually enabled through the RREN register.

40.2 Temporarily pausing the watchdog

By default, the watchdog will be active counting down the down-counter while the CPU is sleeping and when it is halted by the debugger. It is however possible to configure the watchdog to automatically pause while the CPU is sleeping as well as when it is halted by the debugger.

40.3 Watchdog reset

A TIMEOUT event will automatically lead to a watchdog reset.

See *Reset* on page 82 for more information about reset sources. If the watchdog is configured to generate an interrupt on the TIMEOUT event, the watchdog reset will be postponed with two 32.768 kHz clock cycles after the TIMEOUT event has been generated. Once the TIMEOUT event has been generated, the impending watchdog reset will always be effectuated.

The watchdog must be configured before it is started. After it is started, the watchdog's configuration registers, which comprise registers CRV, RREN, and CONFIG, will be blocked for further configuration.

The watchdog can be reset from several reset sources, see *Reset behavior* on page 83.

When the device starts running again, after a reset, or waking up from OFF mode, the watchdog configuration registers will be available for configuration again.

40.4 Registers

Table 96: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40010000	WDT	WDT	Watchdog timer	

Table 97: Register Overview

Register	Offset	Description
TASKS_START	0x000	Start the watchdog
EVENTS_TIMEOUT	0x100	Watchdog timeout
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
RUNSTATUS	0x400	Run status
REQSTATUS	0x404	Request status
CRV	0x504	Counter reload value
RREN	0x508	Enable register for reload request registers
CONFIG	0x50C	Configuration register
RR[0]	0x600	Reload request 0
RR[1]	0x604	Reload request 1
RR[2]	0x608	Reload request 2
RR[3]	0x60C	Reload request 3
RR[4]	0x610	Reload request 4
RR[5]	0x614	Reload request 5
RR[6]	0x618	Reload request 6
RR[7]	0x61C	Reload request 7

40.4.1 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																														
A	RW	TIMEOUT			Write '1' to Enable interrupt for TIMEOUT event See *EVENTS_TIMEOUT*																																
			Set	1	Enable																														
			Disabled	0	Read: Disabled																														
			Enabled	1	Read: Enabled																														

40.4.2 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																														
A	RW	TIMEOUT			Write '1' to Disable interrupt for TIMEOUT event See *EVENTS_TIMEOUT*																																
			Clear	1	Disable																														
			Disabled	0	Read: Disabled																														
			Enabled	1	Read: Enabled																														

40.4.3 RUNSTATUS

Address offset: 0x400

Run status

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	RUNSTATUS																																	
			NotRunning	0																															
			Running	1																															

40.4.4 REQSTATUS

Address offset: 0x404

Request status

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												H	G	F	E	D	C	B	A
Reset 0x00000001				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	RR0																																	
			DisabledOrRequested	0																															
			EnabledAndUnrequested	1																															
B	R	RR1																																	
			DisabledOrRequested	0																															
			EnabledAndUnrequested	1																															
C	R	RR2																																	
			DisabledOrRequested	0																															
			EnabledAndUnrequested	1																															
D	R	RR3																																	
			DisabledOrRequested	0																															
			EnabledAndUnrequested	1																															
E	R	RR4																																	
			DisabledOrRequested	0																															
			EnabledAndUnrequested	1																															
F	R	RR5																																	
			DisabledOrRequested	0																															
			EnabledAndUnrequested	1																															
G	R	RR6																																	
			DisabledOrRequested	0																															
			EnabledAndUnrequested	1																															
H	R	RR7																																	
			DisabledOrRequested	0																															
			EnabledAndUnrequested	1																															

40.4.5 CRV

Address offset: 0x504

Counter reload value

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	CRV		[0x0000000F..0xFFFFFFFF																															
				clock																															

40.4.6 RREN

Address offset: 0x508

Enable register for reload request registers

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												H	G	F	E	D	C	B	A
Reset 0x00000001				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	RR0																																	
			Disabled	0																															
			Enabled	1																															
B	RW	RR1																																	
			Disabled	0																															
			Enabled	1																															
C	RW	RR2																																	
			Disabled	0																															
			Enabled	1																															
D	RW	RR3																																	
			Disabled	0																															
			Enabled	1																															
E	RW	RR4																																	
			Disabled	0																															
			Enabled	1																															
F	RW	RR5																																	
			Disabled	0																															
			Enabled	1																															
G	RW	RR6																																	
			Disabled	0																															
			Enabled	1																															
H	RW	RR7																																	
			Disabled	0																															
			Enabled	1																															

40.4.7 CONFIG

Address offset: 0x50C

Configuration register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																C			A
Reset 0x00000001				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	SLEEP																																	
			Pause	0																															
			Run	1																															
C	RW	HALT																																	
			Pause	0																															
			Run	1																															

40.4.8 RR[0]

Address offset: 0x600

Reload request 0

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	W	RR																																	
			Reload	0x6E524635																															

40.4.9 RR[1]

Address offset: 0x604

Reload request 1

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	W	RR																																	
			Reload	0x6E524635																															

40.4.10 RR[2]

Address offset: 0x608

Reload request 2

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	W	RR																																	
			Reload	0x6E524635																															

40.4.11 RR[3]

Address offset: 0x60C

Reload request 3

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	W	RR																																	
			Reload	0x6E524635																															

40.4.12 RR[4]

Address offset: 0x610

Reload request 4

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	W	RR																																	
			Reload	0x6E524635																															

40.4.13 RR[5]

Address offset: 0x614

Reload request 5

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	W	RR																								Reload request register									
			Reload	0x6E524635																						Value to request a reload of the watchdog timer									

40.4.14 RR[6]

Address offset: 0x618

Reload request 6

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	W	RR																								Reload request register									
			Reload	0x6E524635																						Value to request a reload of the watchdog timer									

40.4.15 RR[7]

Address offset: 0x61C

Reload request 7

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																						**Description**									
A	W	RR																								Reload request register									
			Reload	0x6E524635																						Value to request a reload of the watchdog timer									

40.5 Electrical specification

40.5.1 Watchdog Timer Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
I_{WDT}	Run current for watchdog timer		0.3	2	µA
t_{WDT}	Time out interval	458 µs		36 h	

41 SWI — Software interrupts

A set of interrupts have been reserved for use as software interrupts.

41.1 Registers

Table 98: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40014000	SWI	SWI0	Software interrupt 0	
0x40015000	SWI	SWI1	Software interrupt 1	
0x40016000	SWI	SWI2	Software interrupt 2	
0x40017000	SWI	SWI3	Software interrupt 3	
0x40018000	SWI	SWI4	Software interrupt 4	
0x40019000	SWI	SWI5	Software interrupt 5	

42 NFCT — Near field communication tag

The NFCT peripheral (referred to as the 'NFC peripheral' from now on) supports communication signal interface type A and 106 kbps bit rate from the NFC Forum.

With appropriate software, the NFC peripheral can be used to emulate the listening device NFC-A as specified by the NFC Forum.

Listed here are the main features for the NFC peripheral:

- NFC-A listen mode operation
  - 13.56 MHz input frequency
  - Bit rate 106 kbps
- Wake-on-field low power field detection (SENSE) mode
- Frame assemble and disassemble for the NFC-A frames specified by the NFC Forum
- Programmable frame timing controller
- Integrated automatic collision resolution, CRC and parity functions

42.1 Overview

The NFC peripheral is an implementation of an NFC Forum compliant listening device NFC-A.

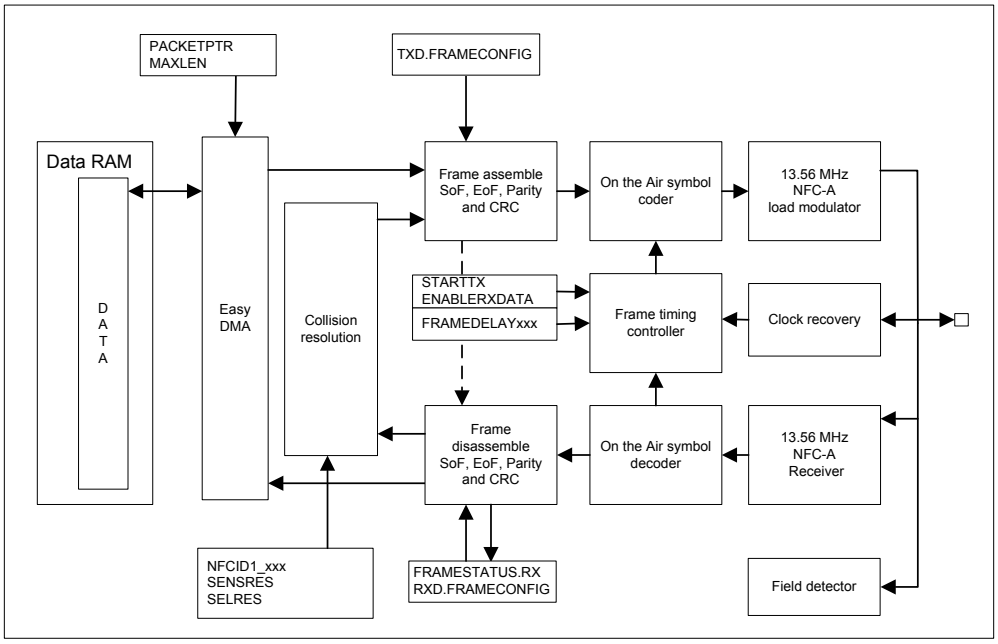


Figure 117: NFC block diagram

The NFC peripheral contains a 13.56 MHz AM receiver and a 13.56 MHz load modulator compatible with the NFC-A technology defined in the NFC Forum with 106 kbps data rate.

The received frames will be automatically disassembled and the data part of the frame transferred to RAM. When transmitting, the frame data will be transferred directly from RAM and transmitted with configurable frame type and delay timing. The system will be notified by an event whenever a complete frame is received or sent.

It also supports the collision detection and resolution ("anticollision") as defined by the NFC Forum.

Wake-on-field is supported in SENSE mode while the device is either in System OFF or System ON mode. When the antenna enters an NFC field, an event will be triggered notifying the system to activate the NFC functionality for incoming frames. In System ON, if the energy detected at the antenna increases beyond a threshold value, the module will generate a FIELDDETECTED event. The module will generate a FIELDLOST event when the quality or strength of the field no longer support NFC communication. Please refer to *NFCT Electrical Specification* on page 435 for the Low Power Field Detect threshold values.

In system OFF, the NFC Low Power Field Detect function can wake the system up through a reset. The NFC bit in register *RESETREAS* on page 85 will be set as cause of the wake-up.

If the system is put into system OFF mode while a field is already present, the NFC Low Power Field Detect function will wake the system up right away and generate a reset.

Note that as a consequence of reset, NFC is disabled, so the reset handler will have to activate NFC again and set it up properly.

The HFXO must be running before the NFC peripheral goes into ACTIVATED state. Note that the NFC peripheral calibration is automatically done on ACTIVATE task. The HFXO can be turned off when the NFC peripheral goes into SENSE mode. The shortcut FIELDDETECTED_ACTIVATE can be used when the HFXO is already running while in SENSE mode.

Outgoing data will be collected from RAM with the EasyDMA function and assembled according to the TXD.FRAMECONFIG register. Incoming data will be disassembled according to the RXD.FRAMECONFIG register and the data section in the frame will be written to RAM via the EasyDMA function.

The NFC peripheral includes a frame timing controller that can be used to accurately control the inter-frame delay between the incoming frame and a corresponding outgoing frame. It also includes optional CRC functionality.

The NFC peripheral has a set of different states. The module can change state by triggering a task, or when specific operations are finalized. Events and tasks allow software to keep track of and change the current state.

See *Figure 117: NFC block diagram* on page 416 and *Figure 118: NFC state diagram* on page 418 for more information.

Notes:

- FIELDLOST event will not be reflected in the state machine (for instance by going back to the DISABLE state), it is up to software to decide on the actions to take when a field lost occurs.
- FIELDLOST event is not generated in SENSE mode.
- FIELDDETECTED event is generated only on the transition from FIELDLOST event to energy detected by the NFC peripheral. So, sending SENSE task while field is still present does not generate FIELDDETECTED event.
- If the FIELDDETECTED event is cleared before sending the ACTIVATE task, then the FIELDDETECTED event shows up again after sending the ACTIVATE task. The shortcut FIELDDETECTED_ACTIVATE can be used to avoid this condition.

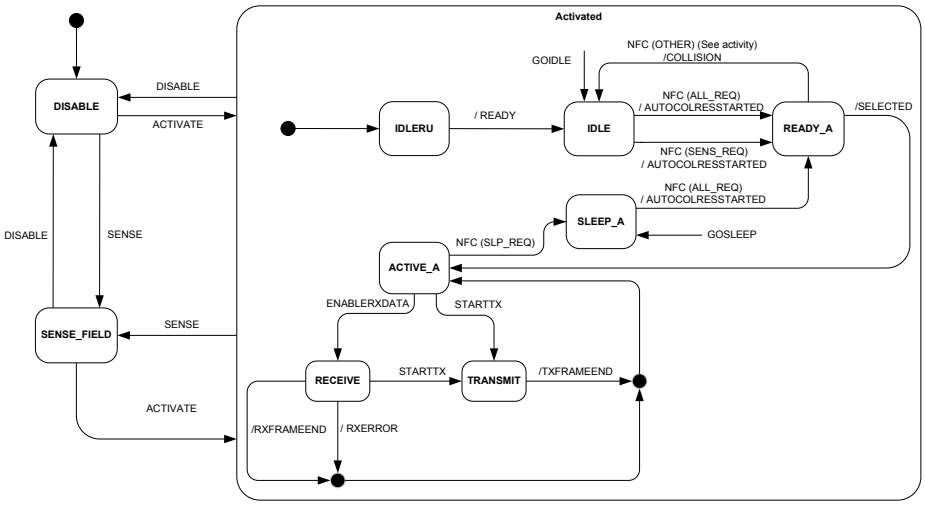


Figure 118: NFC state diagram

42.2 Pin configuration

NFC uses two pins to connect the antenna.

These pins are shared with GPIOs, and the PROTECT field in the NFCPINS register in *UICR* defines the usage of these pins and their protection level against excessive voltages. The content of the NFCPINS register is reloaded at every reset.

When NFCPINS.PROTECT=NFC, a protection circuit will be enabled on the dedicated pins, preventing the chip from being damaged in the presence of a strong NFC field. The GPIO function will be disabled on those pins as well.

When NFCPINS.PROTECT=Disabled, the device will not be protected against strong NFC field damages caught by a connected NFC antenna, and the NFCT peripheral will not operate as expected, as it will never leave the DISABLE state.

The pins dedicated to the NFC antenna function will have some limitation when the pins are configured for normal GPIO operation. The pin capacitance will be higher on those (refer to C_{PAD_NFC} in the *GPIO Electrical Specification* on page 154 below), and some increased leakage current between the two pins is to be expected if they are used in GPIO mode, and are driven to different logical values. To save power the two pins should always be set to the same logical value whenever entering one of the device power saving modes. Please refer to I_{NFC_LEAK} in *GPIO Electrical Specification* on page 154 for details.

42.3 EasyDMA

The NFC peripheral implements EasyDMA for reading and writing of data packets from and to the Data RAM without CPU involvement.

The NFC EasyDMA utilizes one pointer called PACKETPTR for receiving and transmitting packets.

The EasyDMA can either read or write between the NFC peripheral and the RAM, but not at the same time. The event RXFRAMESTART indicates that the EasyDMA has started writing to the RAM for a receive frame and the event RXFRAMEND indicates that the EasyDMA has completed writing to the RAM. Similarly, the event TXFRAMESTART indicates that the EasyDMA has started reading from the RAM for a transmit frame and the event TXFRAMEND indicates that the EasyDMA has completed reading from the RAM. If a transmit and a receive operation is issued at the same time, the transmit operation would be prioritized.

Starting a transmit operation while the EasyDMA has already started writing a receive frame to the RAM will result in unpredictable behavior. Starting an EasyDMA operation whilst there is an ongoing EasyDMA operation may result in unpredictable behavior. It is recommended to wait for the TXFRAMEEND or

RXFRAMEND event for the respective ongoing transmit or receive before starting a new receive or transmit operation.

The MAXLEN register determines the maximum number of bytes that can be read from or written to the RAM. This feature can be used to secure that the NFC peripheral does not overwrite, or read beyond, the RAM assigned to a packet. Note that if the RXD.AMOUNT or TXD.AMOUNT register indicates longer data packets than set in MAXLEN, the frames sent to or received from the physical layer will be incomplete. In RX, the OVERRUN bit in the FRAMESTATUS.RX register will be set and an RXERROR event will be triggered in that situation.

Note that RXD.AMOUNT and TXD.AMOUNT define a frame length in bytes and bits excluding SoF, EoF and parity, but including CRC for RXD.AMOUNT only, make sure to take potential additional bits into account when setting MAXLEN.

Only sending task ENABLERXDATA ensures that a new value in PACKETPTR pointing to the RX buffer in Data RAM is taken into account.

If PACKETPTR is not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See Chapter *Memory* on page 23 for more information about the different memory regions.

The NFC peripherals normally do alternative receive and transmit frames. So, to prepare for the next frame, the PACKETPTR, MAXLEN, TXD.FRAMECONFIG and TXD.AMOUNT can be updated while the receive is in progress, and, similarly, the PACKETPTR, MAXLEN and RXD.FRAMECONFIG can be updated while the transmit is in progress. They can be updated and prepared for the next NFC frame immediately after the STARTED event of the current frame has been received. Updating the TXD.FRAMECONFIG and TXD.AMOUNT during the current transmit frame or updating RXD.FRAMECONFIG during current receive frame may cause unpredictable behaviour.

In accordance with *NFC Forum, NFC Digital Protocol Technical Specification*, the least a significant bit from the least significant byte is sent on air first. The bytes are stored in increasing order, starting at the lowest address in the EasyDMA buffer in RAM.

42.4 Collision resolution

The NFC peripheral implements an automatic collision resolution function as defined by the NFC Forum.

The SENSRES and SELRES registers need to be programmed upfront in order for the collision resolution to behave correctly. Depending on the NFCIDSIZE field in SENSRES, the following registers also need to be programmed upfront:

- NFCID1_LAST if NFCID1SIZE=NFCID1Single (ID = 4 bytes);
- NFCID1_2ND_LAST and NFCID1_LAST if NFCID1SIZE=NFCID1Double (ID = 7 bytes);
- NFCID1_3RD_LAST, NFCID1_2ND_LAST and NFCID1_LAST if NFCID1SIZE=NFCID1Triple (ID = 10 bytes);

Table 99: NFCID1 byte allocation (top sent first on air) on page 419 explains the position of the ID bytes in NFCID1_3RD_LAST, NFCID1_2ND_LAST and NFCID1_LAST, depending on the ID size, and as compared to the definition used in the *NFC Forum, NFC Digital Protocol Technical Specification*.

Table 99: NFCID1 byte allocation (top sent first on air)

	ID = 4 bytes	ID = 7 bytes	ID = 10 bytes
NFCID1_Q			$nfcid1_0$
NFCID1_R			$nfcid1_1$
NFCID1_S			$nfcid1_2$
NFCID1_T		$nfcid1_0$	$nfcid1_3$
NFCID1_U		$nfcid1_1$	$nfcid1_4$
NFCID1_V		$nfcid1_2$	$nfcid1_5$
NFCID1_W	$nfcid1_0$	$nfcid1_3$	$nfcid1_6$
NFCID1_X	$nfcid1_1$	$nfcid1_4$	$nfcid1_7$
NFCID1_Y	$nfcid1_2$	$nfcid1_5$	$nfcid1_8$
NFCID1_Z	$nfcid1_3$	$nfcid1_6$	$nfcid1_9$

Automatic collision resolution is enabled by default.

The hardware implementation can handle the states from IDLE to ACTIVE_A automatically as defined in the *NFC Forum, NFC Activity Technical Specification*, and the other states are to be handled by software. The software keeps track of the state through events. The collision resolution will trigger an AUTOCOLRESSTARTED event when it has started. Reaching the ACTIVE_A state is indicated by the SELECTED event.

If collision resolution fails, a COLLISION event is triggered. Note that errors occurring during automatic collision resolution may also cause ERROR and/or RXERROR events to be generated. Also, other events may get generated. It is recommended that the software ignores any event except COLLISION, SELECTED and FIELDLOST during automatic collision resolution. Software shall also make sure that any unwanted SHORT or PPI shortcut are disabled during automatic collision resolution.

A pre-defined set of registers, NFC.TAGHEADER0..3, containing a valid NFCID1 value, is available in *FICR*, and can be used by software to populate the NFCID1_3RD_LAST, NFCID1_2ND_LAST and NFCID1_LAST registers. Refer to the release notes of the NFC stack for more details on the format.

The automatic collision resolution will be restarted, if the packets are received with CRC or parity errors while in ACTIVE_A state.

The SLP_REQ is automatically handled by the NFC peripheral. However, this results in an ERROR event (with FRAMEDELAYTIMEOUT cause in ERRORSTATUS) since the SLP_REQ has no response. This error must be ignored until the SELECTED event is triggered and this error should be cleared by the software when the SELECTED event is triggered.

42.5 Frame timing controller

The NFC peripheral includes a frame timing controller that continuously keeps track of the number of the 13.56 MHz RF-carrier clock periods since the end of the EoF of the last received frame.

The NFC peripheral can be programmed to send a responding frame within a time window or at an exact count of RF carrier periods. In case of FRAMEDELAYMODE = Window a STARTTX task triggered before the frame timing controller counter is equal to FRAMEDELAYMIN will force the transmission to halt until the counter is equal to FRAMEDELAYMIN. If the counter is within FRAMEDELAYMIN and FRAMEDELAYMAX when the STARTTX task is triggered, the peripheral will start the transmission straight away. In case of FRAMEDELAYMODE = ExactVal, a STARTTX task, triggered before the frame delay counter is equal to FRAMEDELAYMAX, will halt the actual transmission start until the counter is equal to FRAMEDELAYMAX.

In case of FRAMEDELAYMODE = WindowGrid, the behaviour is similar to the FRAMEDELAYMODE = Window, but the actual transmission between FRAMEDELAYMIN and FRAMEDELAYMAX starts on a bit grid as defined for NFC-A Listen frames (slot duration of 128 RF carrier periods).

The FRAMEDELAYMIN and FRAMEDELAYMAX values shall only be updated before the STARTTX task is triggered. Failing to do so may cause unpredictable behaviour. An ERROR event (with FRAMEDELAYTIMEOUT cause in ERRORSTATUS) will be asserted if the frame timing controller counter reaches FRAMEDELAYMAX without any STARTTX task triggered. This may happen even when the response is not required as per *NFC Forum, NFC Digital Protocol Technical Specification*. Any commands handled by the automatic collision resolution that don't involve a response being generated may also result in an ERROR event (with FRAMEDELAYTIMEOUT cause in ERRORSTATUS).

The frame timing controller operation is illustrated in *Figure 119: Frame timing controller (FRAMEDELAYMODE=Window)* on page 421. The frame timing controller automatically adjusts the frame timing counter based on the last received data bit according to NFC-A technology in the *NFC Forum, NFC Digital Protocol Technical Specification*.

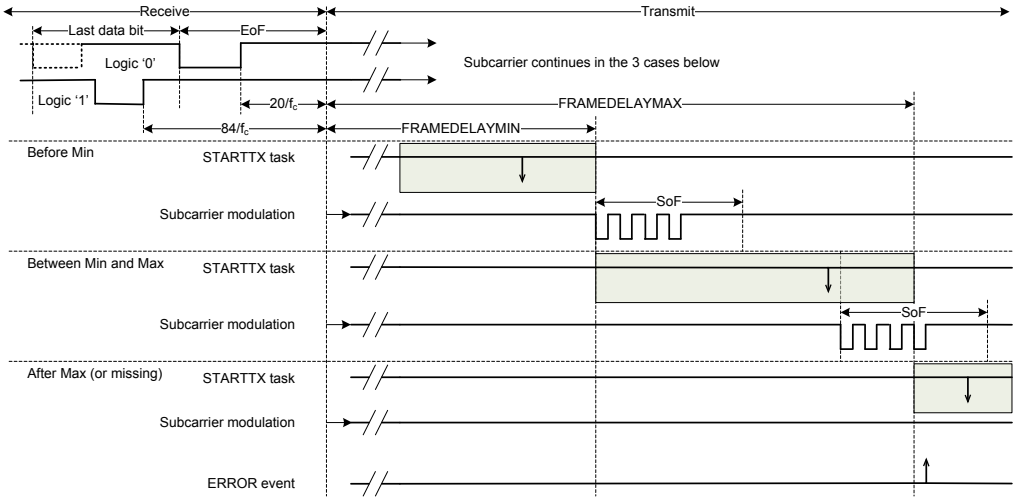


Figure 119: Frame timing controller (FRAMEDELAYMODE=Window)

42.6 Frame assembler

The NFC peripheral implements a frame assembler in hardware.

When the NFC peripheral is in the ACTIVE_A state, the software can decide to enter RX or TX mode. For RX, see *Frame disassembler* on page 422. For TX, the software must indicate the address of the source buffer in Data RAM and its size through programming the PACKETPTR and MAXCNT registers respectively, then issuing a TXSTART task.

MAXCNT must be set so that it matches the size of the frame to be sent.

The STARTED event indicates that the PACKETPTR and MAXCNT registers have been captured by the frame assembler's EasyDMA.

When asserting the STARTTX task, the frame assembler module will start reading TXD.AMOUNT.TXDATABYTES bytes (plus one additional byte if TXD.AMOUNT.TXDATABITS > 0) from the RAM position set by the PACKETPTR.

The NFC peripheral transmits the data as read from RAM, adding framing and the CRC calculated on the fly. The NFC peripheral will take (8*TXD.AMOUNT.TXDATABYTES + TXD.AMOUNT.TXDATABITS) bits and assemble a frame according to settings in TXD.FRAMECONFIG. Both short frames, standard frames and bit oriented SDD frames as specified in the *NFC Forum, NFC Digital Protocol Technical Specification* can be assembled by correct setting of the TXD.FRAMECONFIG register.

The bytes will be transmitted on air in the same order as they are read from RAM with a rising bit order within each byte (least significant bit first). That is, b0 will be transmitted on air before b1, and so on. The bits read from RAM will be coded into symbols as defined in the *NFC Forum, NFC Digital Protocol Technical Specification*.

> **Important:** Some NFC Forum documents, such as *NFC Forum, NFC Digital Protocol Technical Specification*, define bit numbering in a byte from b1 (LSB) to b8 (MSB), while most other technical documents from the NFC Forum, and also the Nordic Semiconductor documentation, traditionally numbers them from b0 to b7. The present document uses the b0 to b7 numbering scheme. Be aware of this when comparing with the *NFC Forum, NFC Digital Protocol Technical Specification* to others.

The frame assembler can be configured in TXD.FRAMECONFIG to add Start of Frame (SoF) symbol, calculate and add parity bits, and calculate and add CRC to the data read from RAM when assembling the frame. The total frame will then be longer than what is defined by TXD.AMOUNT.TXDATABYTES and TXDATABITS. DISCARDMODE will select if the first bits in the first byte read from RAM or the last bits in the last byte read from RAM will be discarded if TXD.AMOUNT.TXDATABITS are not equal to zero. Note that if TXD.FRAMECONFIG.PARITY = Parity and TXD.FRAMECONFIG.DISCARDMODE=DiscardStart, a parity bit will be included after the non-complete first byte. No parity will be added after a non-complete last byte.

The Frame Assemble operation is illustrated in *Figure 120: Frame assemble* on page 422 for different settings in TXD.FRAMECONFIG. All shaded bits fields are added by the frame assembler. Some of these bits are optional and appearances are configured in TXD.FRAMECONFIG. Please note that the frames illustrated do not necessarily comply with the NFC specification. The figure is only to illustrate the behavior of the NFC peripheral.

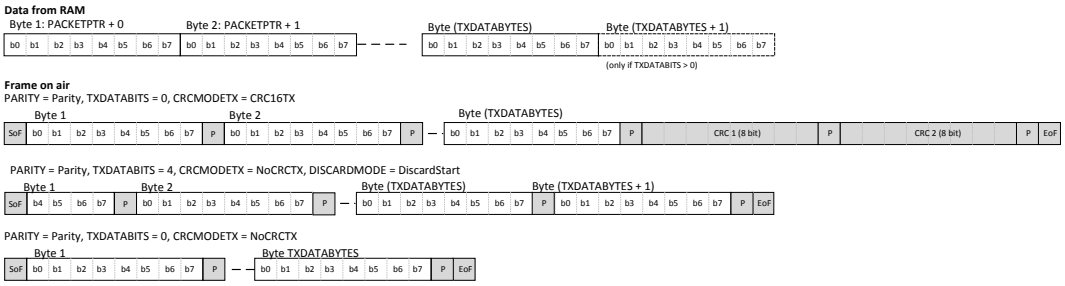


Figure 120: Frame assemble

The accurate timing for transmitting the frame on air is set using the frame timing controller settings.

42.7 Frame disassembler

The NFC peripheral implements a frame disassembler in hardware.

When the NFC peripheral is in the ACTIVE_A state, the software can decide to enter RX or TX mode. For TX, see *Frame assembler* on page 421. For RX, the software must indicate the address of the destination buffer in Data RAM and its size through programming the PACKETPTR and MAXCNT registers respectively, then issuing a ENABLERXDATA task.

The STARTED event indicates that the PACKETPTR and MAXCNT registers have been captured by the frame disassembler's EasyDMA.

When an incoming frame starts, the RXFRAMESTART event will get issued and data will be written to the buffer in Data RAM. The frame disassembler will verify and remove on the fly any parity bits and SoF and End of Frame (EoF) symbols based on RXD.FRAMECONFIG register configuration. It will, however, verify and transfer the CRC bytes into RAM, if the CRC is was enabled through RXD.FRAMECONFIG.

When an EoF symbol is detected, the NFC peripheral will assert the RXFRAMEEND event and write the RXD.AMOUNT register to indicate numbers of received bytes and bits in the data packet. The module does not interpret the content of the data received from the remote NFC device, except for SoF, EoF, parity and CRC checking, as described above. The Frame disassemble operation is illustrated in *Figure 121: Frame disassemble illustration* on page 422.

Per NFC specification, the time between end of frame to the next start of frame can be as short as 86 µs, so care must be taken that PACKETPTR and MAXCNT are ready and ENABLERXDATA is issued on time after the end of previous frame. The use of a PPI shortcut from TXFRAMEEND to ENABLERXDATA is recommended.

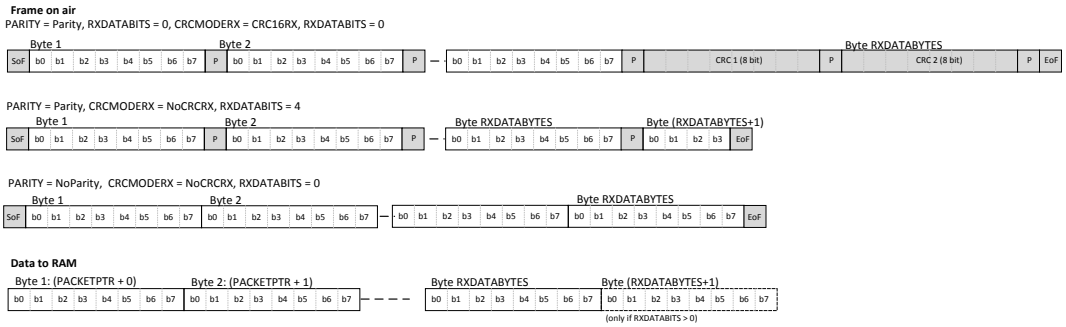


Figure 121: Frame disassemble illustration

42.8 Antenna interface

In ACTIVATED state, an amplitude regulator will adjust the voltage swing on the antenna pins to a value that is within the V_{swing} limit.

Refer to *NFCT Electrical Specification* on page 435.

42.9 NFCT antenna recommendations

The NFCT antenna coil must be connected differential between NFC1 and NFC2 pins of the device.

Two external capacitors should be used to tune the resonance of the antenna circuit to 13.56 MHz.

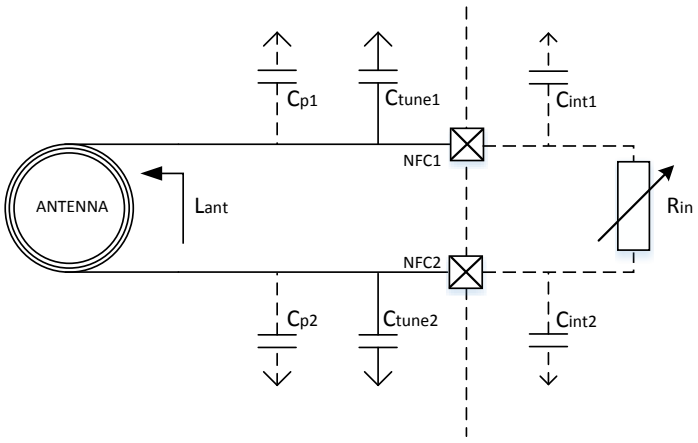


Figure 122: NFCT antenna recommendations

The required tuning capacitor value is given by the below equations:

$$C'_{tune} = \frac{1}{(2\pi \cdot 13.56\ MHz)^2 \cdot L_{ant}} \quad where\ C'_{tune} = \frac{1}{2} \cdot (C_p + C_{\text{int}} + C_{tune})$$

$$and\ C_{tune1} = C_{tune2} = C_{tune} \qquad C_{p1} = C_{p2} = C_p \qquad C_{\text{int1}} = C_{\text{int2}} = C_{\text{int}}$$

$$C_{tune} = \frac{2}{(2\pi \cdot 13.56\ MHz)^2 \cdot L_{ant}} - C_p - C_{\text{int}}$$

An antenna inductance of L_{ant} = 2 µH will give tuning capacitors in the range of 130 pF on each pin. For good performance, match the total capacitance on NFC1 and NFC2.

42.10 Battery protection

If the antenna is exposed to a strong NFC field, current may flow in the opposite direction on the supply due to parasitic diodes and ESD structures.

If the battery used does not tolerate return current, a series diode must be placed between the battery and the device in order to protect the battery.

42.11 References

NFC Forum, NFC Analog Specification version 1.0, www.nfc-forum.org

NFC Forum, NFC Digital Protocol Technical Specification version 1.1, www.nfc-forum.org

NFC Forum, NFC Activity Technical Specification version 1.1, www.nfc-forum.org

42.12 Registers

Table 100: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40005000	NFCT	NFCT	Near Field Communication Tag	

Table 101: Register Overview

Register	Offset	Description
TASKS_ACTIVATE	0x000	Activate NFC peripheral for incoming and outgoing frames, change state to activated
TASKS_DISABLE	0x004	Disable NFC peripheral
TASKS_SENSE	0x008	Enable NFC sense field mode, change state to sense mode
TASKS_STARTTX	0x00C	Start transmission of a outgoing frame, change state to transmit
TASKS_ENABLERXDATA	0x01C	Initializes the EasyDMA for receive.
TASKS_GOIDLE	0x024	Force state machine to IDLE state
TASKS_GOSLEEP	0x028	Force state machine to SLEEP_A state
EVENTS_READY	0x100	The NFC peripheral is ready to receive and send frames
EVENTS_FIELDDETECTED	0x104	Remote NFC field detected
EVENTS_FIELDLOST	0x108	Remote NFC field lost
EVENTS_TXFRAMESTART	0x10C	Marks the start of the first symbol of a transmitted frame
EVENTS_TXFRAMEEND	0x110	Marks the end of the last transmitted on-air symbol of a frame
EVENTS_RXFRAMESTART	0x114	Marks the end of the first symbol of a received frame
EVENTS_RXFRAMEEND	0x118	Received data have been checked (CRC, parity) and transferred to RAM, and EasyDMA has ended accessing the RX buffer
EVENTS_ERROR	0x11C	NFC error reported. The ERRORSTATUS register contains details on the source of the error.
EVENTS_RXERROR	0x128	NFC RX frame error reported. The FRAMESTATUS.RX register contains details on the source of the error.
EVENTS_ENDRX	0x12C	RX buffer (as defined by PACKETPTR and MAXLEN) in Data RAM full.
EVENTS_ENDTX	0x130	Transmission of data in RAM has ended, and EasyDMA has ended accessing the TX buffer
EVENTS_AUTOCOLRESST	0x138	Auto collision resolution process has started
EVENTS_COLLISION	0x148	NFC Auto collision resolution error reported.
EVENTS_SELECTED	0x14C	NFC Auto collision resolution successfully completed
EVENTS_STARTED	0x150	EasyDMA is ready to receive or send frames.
SHORTS	0x200	Shortcut register
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ERRORSTATUS	0x404	NFC Error Status register
FRAMESTATUS.RX	0x40C	Result of last incoming frames
CURRENTLOADCTRL	0x430	Current value driven to the NFC Load Control
FIELDPRESENT	0x43C	Indicates the presence or not of a valid field
FRAMEDELAYMIN	0x504	Minimum frame delay
FRAMEDELAYMAX	0x508	Maximum frame delay
FRAMEDELAYMODE	0x50C	Configuration register for the Frame Delay Timer
PACKETPTR	0x510	Packet pointer for TXD and RXD data storage in Data RAM
MAXLEN	0x514	Size of allocated for TXD and RXD data storage buffer in Data RAM
TXD.FRAMECONFIG	0x518	Configuration of outgoing frames
TXD.AMOUNT	0x51C	Size of outgoing frame
RXD.FRAMECONFIG	0x520	Configuration of incoming frames

Register	Offset	Description
RXD.AMOUNT	0x524	Size of last incoming frame
NFCID1_LAST	0x590	Last NFCID1 part (4, 7 or 10 bytes ID)
NFCID1_2ND_LAST	0x594	Second last NFCID1 part (7 or 10 bytes ID)
NFCID1_3RD_LAST	0x598	Third last NFCID1 part (10 bytes ID)
SENSRES	0x5A0	NFC-A SENS_RES auto-response settings
SELRES	0x5A4	NFC-A SEL_RES auto-response settings

42.12.1 SHORTS

Address offset: 0x200

Shortcut register

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	FIELDDETECTED_ACTIVATE			Shortcut between FIELDDETECTED event and ACTIVATE task See *EVENTS_FIELDDETECTED* and *TASKS_ACTIVATE*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
B	RW	FIELDLOST_SENSE			Shortcut between FIELDLOST event and SENSE task See *EVENTS_FIELDLOST* and *TASKS_SENSE*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

42.12.2 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id												T	S	R				N		M	L	K			H	G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	READY			Enable or disable interrupt for READY event See *EVENTS_READY*
			Disabled	0	Disable
			Enabled	1	Enable
B	RW	FIELDDETECTED			Enable or disable interrupt for FIELDDETECTED event See *EVENTS_FIELDDETECTED*
			Disabled	0	Disable
			Enabled	1	Enable
C	RW	FIELDLOST			Enable or disable interrupt for FIELDLOST event See *EVENTS_FIELDLOST*
			Disabled	0	Disable
			Enabled	1	Enable
D	RW	TXFRAMESTART			Enable or disable interrupt for TXFRAMESTART event See *EVENTS_TXFRAMESTART*
			Disabled	0	Disable
			Enabled	1	Enable
E	RW	TXFRAMEEND			Enable or disable interrupt for TXFRAMEEND event See *EVENTS_TXFRAMEEND*
			Disabled	0	Disable
			Enabled	1	Enable
F	RW	RXFRAMESTART			Enable or disable interrupt for RXFRAMESTART event

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id															T	S	R				N		M	L	K			H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id															T	S	R				N		M	L	K			H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Set	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	FIELDDETECTED																																	
				See *EVENTS_FIELDDETECTED*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	FIELDLOST																																	
				See *EVENTS_FIELDLOST*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	TXFRAMESTART																																	
				See *EVENTS_TXFRAMESTART*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	TXFRAMEEND																																	
				See *EVENTS_TXFRAMEEND*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
F	RW	RXFRAMESTART																																	
				See *EVENTS_RXFRAMESTART*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
G	RW	RXFRAMEEND																																	
				See *EVENTS_RXFRAMEEND*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
H	RW	ERROR																																	
				See *EVENTS_ERROR*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
K	RW	RXERROR																																	
				See *EVENTS_RXERROR*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
L	RW	ENDRX																																	
				See *EVENTS_ENDRX*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
M	RW	ENDTX																																	
				See *EVENTS_ENDTX*																																	
			Set	1																															

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			T S R N M L K H G F E D C B A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
N	RW	AUTOCOLRESSTARTED			Write '1' to Enable interrupt for AUTOCOLRESSTARTED event See *EVENTS_AUTOCOLRESSTARTED*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
R	RW	COLLISION			Write '1' to Enable interrupt for COLLISION event See *EVENTS_COLLISION*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
S	RW	SELECTED			Write '1' to Enable interrupt for SELECTED event See *EVENTS_SELECTED*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
T	RW	STARTED			Write '1' to Enable interrupt for STARTED event See *EVENTS_STARTED*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

42.12.4 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			T S R N M L K H G F E D C B A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
A	RW	READY			Write '1' to Disable interrupt for READY event See *EVENTS_READY*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	FIELDDETECTED			Write '1' to Disable interrupt for FIELDDETECTED event See *EVENTS_FIELDDETECTED*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	FIELDLOST			Write '1' to Disable interrupt for FIELDLOST event See *EVENTS_FIELDLOST*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	TXFRAMESTART			Write '1' to Disable interrupt for TXFRAMESTART event See *EVENTS_TXFRAMESTART*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id															T	S	R				N		M	L	K			H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
E	RW	TXFRAMEEND																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
F	RW	RXFRAMESTART																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
G	RW	RXFRAMEEND																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
H	RW	ERROR																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
K	RW	RXERROR																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
L	RW	ENDRX																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
M	RW	ENDTX																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
N	RW	AUTOCOLRESSTARTED																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
R	RW	COLLISION																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
S	RW	SELECTED																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
T	RW	STARTED																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id															T	S	R				N		M	L	K			H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**											**Description**																				
															See *EVENTS_STARTED*																					
			Clear	1											Disable																					
			Disabled	0											Read: Disabled																					
			Enabled	1											Read: Enabled																					

42.12.5 ERRORSTATUS

Address offset: 0x404

NFC Error Status register

Write a bit to '1' to clear it. Writing '0' has no effect.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																D	C		A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**											**Description**																				
A	RW	FRAMEDELAYTIMEOUT													No STARTTX task triggered before expiration of the time set in FRAMEDELAYMAX																				
C	RW	NFCFIELDTOOSTRONG													Field level is too high at max load resistance																				
D	RW	NFCFIELDTOOWEAK													Field level is too low at min load resistance																				

42.12.6 FRAMESTATUS.RX

Address offset: 0x40C

Result of last incoming frames

Write a bit to '1' to clear it. Writing '0' has no effect.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																C	B		A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**											**Description**																				
A	RW	CRCERROR													No valid End of Frame detected																				
			CRCCorrect	0											Valid CRC detected																				
			CRCError	1											CRC received does not match local check																				
B	RW	PARITYSTATUS													Parity status of received frame																				
			ParityOK	0											Frame received with parity OK																				
			ParityError	1											Frame received with parity error																				
C	RW	OVERRUN													Overrun detected																				
			NoOverrun	0											No overrun detected																				
			Overrun	1											Overrun error																				

42.12.7 CURRENTLOADCTRL

Address offset: 0x430

Current value driven to the NFC Load Control

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																														A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**											**Description**																				
A	R	CURRENTLOADCTRL													Current value driven to the NFC Load Control																				

42.12.8 FIELDPRESENT

Address offset: 0x43C

Indicates the presence or not of a valid field

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	FIELDPRESENT																																	
			NoField	0																															
			FieldPresent	1																															
B	R	LOCKDETECT																																	
			NotLocked	0																															
			Locked	1																															

42.12.9 FRAMEDELAYMIN

Address offset: 0x504

Minimum frame delay

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000480				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	FRAMEDELAYMIN																																	

42.12.10 FRAMEDELAYMAX

Address offset: 0x508

Maximum frame delay

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00001000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	FRAMEDELAYMAX																																	

42.12.11 FRAMEDELAYMODE

Address offset: 0x50C

Configuration register for the Frame Delay Timer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		A	A
Reset 0x00000001				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	FRAMEDELAYMODE																																	
			FreeRun	0																															
			Window	1																															
			ExactVal	2																															
			WindowGrid	3																															

42.12.12 PACKETPTR

Address offset: 0x510

Packet pointer for TXD and RXD data storage in Data RAM

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PTR																																	
				Packet pointer for TXD and RXD data storage in Data RAM. This address is a byte aligned RAM address.																																	

42.12.13 MAXLEN

Address offset: 0x514

Size of allocated for TXD and RXD data storage buffer in Data RAM

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																											A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	MAXLEN		[0..257]																															
				Size of allocated for TXD and RXD data storage buffer in Data RAM																																	

42.12.14 TXD.FRAMECONFIG

Address offset: 0x518

Configuration of outgoing frames

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															D		C	B	A
Reset 0x00000017				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PARITY																																	
				Adding parity or not in the frame																																	
			NoParity	0																																	
				Parity is not added in TX frames																																	
			Parity	1																																	
				Parity is added TX frames																																	
B	RW	DISCARDMODE																																	
				Discarding unused bits in start or at end of a Frame																																	
			DiscardEnd	0																																	
				Unused bits is discarded at end of frame																																	
			DiscardStart	1																																	
				Unused bits is discarded at start of frame																																	
C	RW	SOF																																	
				Adding SoF or not in TX frames																																	
			NoSoF	0																																	
				Start of Frame symbol not added																																	
			SoF	1																																	
				Start of Frame symbol added																																	
D	RW	CRCMODETX																																	
				CRC mode for outgoing frames																																	
			NoCRCTX	0																																	
				CRC is not added to the frame																																	
			CRC16TX	1																																	
				16 bit CRC added to the frame based on all the data read from RAM that is used in the frame																																	

42.12.15 TXD.AMOUNT

Address offset: 0x51C

Size of outgoing frame

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																								B	B	B	B	B	B	B	B	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	TXDATABITS		[0..7]																															
				Number of bits in the last or first byte read from RAM that shall be included in the frame (excluding parity bit). The DISCARDMODE field in FRAMECONFIG.TX selects if unused bits is discarded at the start or at the end of a frame. A value of 0 data bytes and 0 data bits is invalid.																																	
B	RW	TXDATABYTES		[0..257]																															
				Number of complete bytes that shall be included in the frame, excluding CRC, parity and framing																																	

42.12.16 RXD.FRAMECONFIG

Address offset: 0x520

Configuration of incoming frames

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															C		B		A
Reset 0x00000015				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**0**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PARITY																																	
			NoParity	0																															
			Parity	1																															
B	RW	SOF																																	
			NoSoF	0																															
			SoF	1																															
C	RW	CRCMODERX																																	
			NoCRCRX	0																															
			CRC16RX	1																															

Id	RW	Field	Value Id	Value	Description
A	RW	PARITY			Parity expected or not in RX frame
			NoParity	0	Parity is not expected in RX frames
			Parity	1	Parity is expected in RX frames
B	RW	SOF			SoF expected or not in RX frames
			NoSoF	0	Start of Frame symbol is not expected in RX frames
			SoF	1	Start of Frame symbol is expected in RX frames
C	RW	CRCMODERX			CRC mode for incoming frames
			NoCRCRX	0	CRC is not expected in RX frames
			CRC16RX	1	Last 16 bits in RX frame is CRC, CRC is checked and CRCSTATUS updated

42.12.17 RXD.AMOUNT

Address offset: 0x524

Size of last incoming frame

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																								B	B	B	B	B	B	B	B	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	RXDATABITS																																	
B	R	RXDATABYTES																																	

Id	RW	Field	Value Id	Value	Description
A	R	RXDATABITS			Number of bits in the last byte in the frame, if less than 8 (including CRC, but excluding parity and SoF/EoF framing). Frames with 0 data bytes and less than 7 data bits are invalid and are not received properly.
B	R	RXDATABYTES			Number of complete bytes received in the frame (including CRC, but excluding parity and SoF/EoF framing)

42.12.18 NFCID1_LAST

Address offset: 0x590

Last NFCID1 part (4, 7 or 10 bytes ID)

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				D	D	D	D	D	D	D	D	C	C	C	C	C	C	C	C	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A
Reset 0x00006363				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**0**	**0**	**1**	**1**	**0**	**1**	**1**	**0**	**0**	**0**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	NFCID1_Z																																	
B	RW	NFCID1_Y																																	
C	RW	NFCID1_X																																	
D	RW	NFCID1_W																																	

Id	RW	Field	Value Id	Value	Description
A	RW	NFCID1_Z			NFCID1 byte Z (very last byte sent)
B	RW	NFCID1_Y			NFCID1 byte Y
C	RW	NFCID1_X			NFCID1 byte X
D	RW	NFCID1_W			NFCID1 byte W

42.12.19 NFCID1_2ND_LAST

Address offset: 0x594

Second last NFCID1 part (7 or 10 bytes ID)

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id												C	C	C	C	C	C	C	C	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**								**Description**																							
A	RW	NFCID1_V										NFCID1 byte V																							
B	RW	NFCID1_U										NFCID1 byte U																							
C	RW	NFCID1_T										NFCID1 byte T																							

42.12.20 NFCID1_3RD_LAST

Address offset: 0x598

Third last NFCID1 part (10 bytes ID)

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id												C	C	C	C	C	C	C	C	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**								**Description**																							
A	RW	NFCID1_S										NFCID1 byte S																							
B	RW	NFCID1_R										NFCID1 byte R																							
C	RW	NFCID1_Q										NFCID1 byte Q																							

42.12.21 SENSRES

Address offset: 0x5A0

NFC-A SENS_RES auto-response settings

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																				E	E	E	E	D	D	D	D	C	C	B	A	A	A	A	A
Reset 0x00000001				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**								**Description**																							
A	RW	BITFRAMESDD										Bit frame SDD as defined by the b5:b1 of byte 1 in SENS_RES response in the NFC Forum, NFC Digital Protocol Technical Specification																							
			SDD00000	0								SDD pattern 00000																							
			SDD00001	1								SDD pattern 00001																							
			SDD00010	2								SDD pattern 00010																							
			SDD00100	4								SDD pattern 00100																							
			SDD01000	8								SDD pattern 01000																							
			SDD10000	16								SDD pattern 10000																							
B	RW	RFU5										Reserved for future use. Shall be 0.																							
C	RW	NFCIDSIZE										NFCID1 size. This value is used by the Auto collision resolution engine.																							
			NFCID1Single	0								NFCID1 size: single (4 bytes)																							
			NFCID1Double	1								NFCID1 size: double (7 bytes)																							
			NFCID1Triple	2								NFCID1 size: triple (10 bytes)																							
D	RW	PLATFCONFIG										Tag platform configuration as defined by the b4:b1 of byte 2 in SENS_RES response in the NFC Forum, NFC Digital Protocol Technical Specification																							
E	RW	RFU74										Reserved for future use. Shall be 0.																							

42.12.22 SELRES

Address offset: 0x5A4

NFC-A SEL_RES auto-response settings

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																									E	D	D	C	C	B	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	RFU10			Reserved for future use. Shall be 0.
B	RW	CASCADE			Cascade bit (controlled by hardware, write has no effect)
			Complete	0	NFCID1 complete
			NotComplete	1	NFCID1 not complete
C	RW	RFU43			Reserved for future use. Shall be 0.
D	RW	PROTOCOL			Protocol as defined by the b7:b6 of SEL_RES response in the NFC Forum, NFC Digital Protocol Technical Specification
E	RW	RFU7			Reserved for future use. Shall be 0.

42.13 Electrical specification

42.13.1 NFCT Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
f_c	Frequency of operation		13.56		MHz
C_{MI}	Carrier modulation index	95			%
DR	Data Rate		106		kbps
f_s	Modulation sub-carrier frequency		f_c/16		MHz
V_{swing}	Peak differential Input voltage swing on NFC1 and NFC2			VDD	Vp
V_{sense}	Peak differential Field detect threshold level on NFC1-NFC2[35]		1.0		Vp
I_{sense}	Current in SENSE STATE		100		nA
$I_{activated}$	Current in ACTIVATED STATE		480		µA
R_{in_min}	Minimum input resistance when regulating voltage swing			40	Ω
R_{in_max}	Maximum input resistance when regulating voltage swing	1.0			kΩ
$R_{in_loadmod}$	Input resistance when load modulating	8		22	Ω
I_{max}	Maximum input current on NFC pins			80	mA

42.13.2 NFCT Timing Parameters

Symbol	Description	Min.	Typ.	Max.	Units
$t_{activate}$	Time from task_ACTIVATE in SENSE or DISABLE state to ACTIVATE_A or IDLE state[36]			500	us
t_{sense}	Time from remote field is present in SENSE mode to FIELDDETECTED event is asserted			20	us

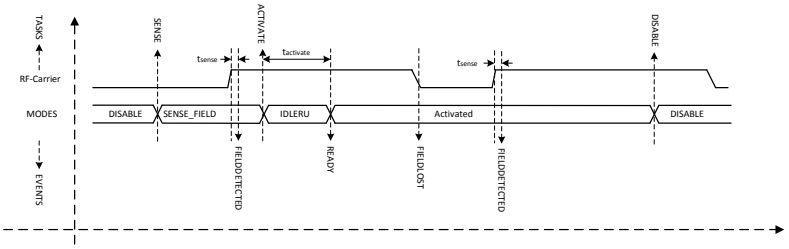


Figure 123: NFCT timing parameters (Shortcuts for FIELDDETECTED and FIELDLOST are disabled)

[35] Input is high impedance in sense mode

[36] Does not account for voltage supply and oscillator startup times

43 PDM — Pulse density modulation interface

The pulse density modulation (PDM) module enables input of pulse density modulated signals from external audio frontends, for example, digital microphones. The PDM module generates the PDM clock and supports single-channel or dual-channel (Left and Right) data input. Data is transferred directly to RAM buffers using EasyDMA.

Listed here are the main features for PDM:

- Up to two PDM microphones configured as a Left/Right pair using the same data input
- 16 kHz output sample rate, 16-bit samples
- EasyDMA support for sample buffering
- HW decimation filters

The PDM module illustrated in *Figure 124: PDM module* on page 436 is interfacing up to two digital microphones with the PDM interface. It implements EasyDMA, which relieves real-time requirements associated with controlling the PDM slave from a low priority CPU execution context. It also includes all the necessary digital filter elements to produce PCM samples. The PDM module allows continuous audio streaming.

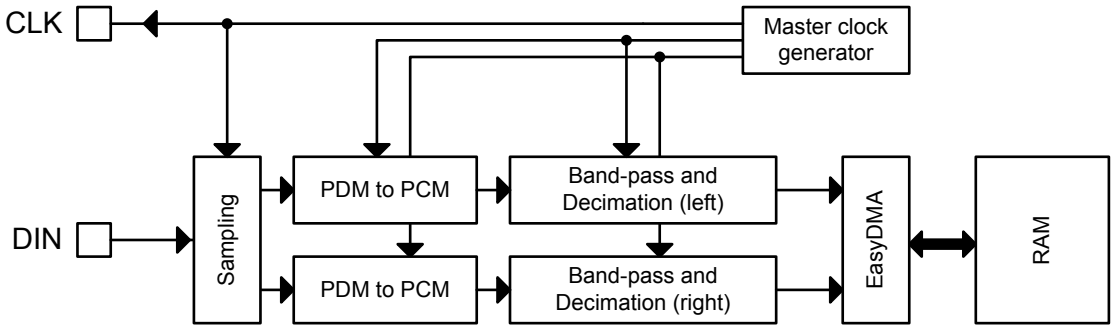


Figure 124: PDM module

43.1 Master clock generator

The FREQ field in the master clock's PDMCLKCTRL register allows adjusting the PDM clock's frequency.

The master clock generator does not add any jitter to the HFCLK source chosen. It is recommended (but not mandatory) to use the Xtal as HFCLK source.

43.2 Module operation

By default, bits from the left PDM microphone are sampled on PDM_CLK falling edge, bits for the right are sampled on the rising edge of PDM_CLK, resulting in two bitstreams. Each bitstream is fed into a digital filter which converts the PDM stream into 16-bit PCM samples, and filters and down-samples them to reach the appropriate sample rate.

The EDGE field in the MODE register allows swapping Left and Right, so that Left will be sampled on rising edge, and Right on falling.

The PDM module uses EasyDMA to store the samples coming out from the filters into one buffer in RAM.

Depending on the mode chosen in the OPERATION field in the MODE register, memory either contains alternating left and right 16-bit samples (Stereo), or only left 16-bit samples (Mono).

To ensure continuous PDM sampling, it is up to the application to update the EasyDMA destination address pointer as the previous buffer is filled.

The continuous transfer can be started or stopped by sending the START and STOP tasks. STOP becomes effective after the current frame has finished transferring, which will generate the STOPPED event. The STOPPED event indicates that all activity in the module are finished, and that the data is available in RAM (EasyDMA has finished transferring as well). Attempting to restart before receiving the STOPPED event may result in unpredictable behaviour.

43.3 Decimation filter

In order to convert the incoming data stream into PCM audio samples, a decimation filter is included in the PDM interface module.

The input of the filter is the two-channel PDM serial stream (with left channel on clock high, right channel on clock low), its output is 2 × 16-bit PCM samples at a sample rate 64 times lower than the PDM clock rate.

The filter stage of each channel is followed by a digital volume control, to attenuate or amplify the output samples in a range of -20 dB to +20 dB around the default (reset) setting, defined by $G_{PDM,default}$. The gain is controlled by the GAINL and GAINR registers.

As an example, if the goal is to achieve 2500 RMS output samples (16 bit) with a 1 kHz 90 dBA signal into a -26 dBFS sensitivity PDM microphone, the user will have to sum the PDM module's default gain ($G_{PDM,default}$) and the gain introduced by the microphone and acoustic path of his implementation (an attenuation would translate into a negative gain), and adjust GAINL and GAINR by this amount. Assuming that only the PDM module influences the gain, GAINL and GAINR must be set to $-G_{PDM,default}$ dB to achieve the requirement.

With $G_{PDM,default}$=3.2 dB, and as GAINL and GAINR are expressed in 0.5 dB steps, the closest value to program would be 3.0 dB, which can be calculated as:

```
GAINL = GAINR = (DefaultGain - (2 * 3))
```

Remember to check that the resulting values programmed into GAINL and GAINR fall within MinGain and MaxGain.

43.4 EasyDMA

Samples will be written directly to RAM, and EasyDMA must be configured accordingly.

The address pointer for the EasyDMA channel is set in SAMPLE.PTR register. If the destination address set in SAMPLE.PTR is not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See *Memory* on page 23 for more information about the different memory regions.

DMA supports Stereo (Left+Right 16-bit samples) and Mono (Left only) data transfer, depending on setting in the OPERATION field in the MODE register. The samples are stored little endian.

Table 102: DMA sample storage

MODE.OPERATION	Bits per sample	Result stored per RAM word	Physical RAM allocated (32 bit words)	Result boundary indexes in RAM	Note
Stereo	32 (2x16)	L+R	ceil(SAMPLE.MAXCNT/2)	R0=[31:16]; L0=[15:0]	Default
Mono	16	2xL	ceil(SAMPLE.MAXCNT/2)	L1=[31:16]; L0=[15:0]	

The destination buffer in RAM consists of one block, the size of which is set in SAMPLE.MAXCNT register. Format is number of 16-bit samples. The physical RAM allocated is always:

```
(RAM allocation, in bytes) = SAMPLE.MAXCNT * 2;
```

(but the mapping of the samples depends on MODE.OPERATION.

If OPERATION=Stereo, RAM will contain a succession of Left and Right samples.

If OPERATION=Mono, RAM will contain a succession of mono samples.

For a given value of SAMPLE.MAXCNT, the buffer in RAM can contain half the stereo sampling time as compared to the mono sampling time.

The PDM acquisition can be started by the START task, after the SAMPLE.PTR and SAMPLE.MAXCNT registers have been written. When starting the module, it will take some time for the filters to start outputting valid data. Transients from the PDM microphone itself may also occur. The first few samples (typically around 50) might hence contain invalid values or transients. It is therefore advised to discard the first few samples after a PDM start.

As soon as the STARTED event is received, the firmware can write the next SAMPLE.PTR value (this register is double-buffered), to ensure continuous operation.

When the buffer in RAM is filled with samples, an END event is triggered. The firmware can start processing the data in the buffer. Meanwhile, the PDM module starts acquiring data into the new buffer pointed to by SAMPLE.PTR, and sends a new STARTED event, so that the firmware can update SAMPLE.PTR to the next buffer address.

43.5 Hardware example

Connect the microphone clock to CLK, and data to DIN.

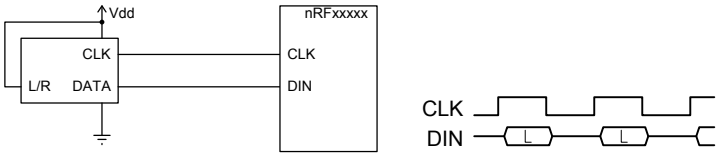


Figure 125: Example of a single PDM microphone, wired as left

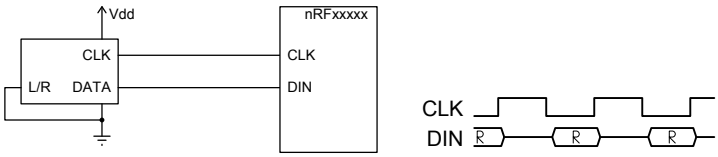


Figure 126: Example of a single PDM microphone, wired as right

Note that in a single-microphone (mono) configuration, depending on the microphone's implementation, either the left or the right channel (sampled at falling or rising CLK edge respectively) will contain reliable data. If two microphones are used, one of them has to be set as left, the other as right (L/R pin tied high or to GND on the respective microphone). It is strongly recommended to use two microphones of exactly the same brand and type so that their timings in left and right operation match.

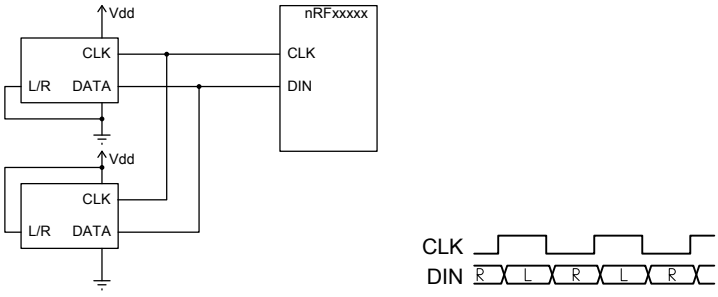


Figure 127: Example of two PDM microphones

43.6 Pin configuration

The CLK and DIN signals associated to the PDM module are mapped to physical pins according to the configuration specified in the PSEL.CLK and PSEL.DIN registers respectively. If the CONNECT field in any PSEL register is set to Disconnected, the associated PDM module signal will not be connected to the required physical pins, and will not operate properly.

The PSEL.CLK and PSEL.DIN registers and their configurations are only used as long as the PDM module is enabled, and retained only as long as the device is in System ON mode. See *POWER — Power supply* on page 78 for more information about power modes. When the peripheral is disabled, the pins will behave as regular GPIOs, and use the configuration in their respective OUT bit field and PIN_CNF[n] register.

To ensure correct behaviour in the PDM module, the pins used by the PDM module must be configured in the GPIO peripheral as described in *Table 103: GPIO configuration before enabling peripheral* on page 439 before enabling the PDM module. This is to ensure that the pins used by the PDM module are driven correctly if the PDM module itself is temporarily disabled or the device temporarily enters System OFF. This configuration must be retained in the GPIO for the selected I/Os as long as the PDM module is supposed to be connected to an external PDM circuit.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behaviour.

Table 103: GPIO configuration before enabling peripheral

PDM signal	PDM pin	Direction	Output value	Comment
CLK	As specified in PSEL.CLK	Output	0	
DIN	As specified in PSEL.DIN	Input	Not applicable	

43.7 Registers

Table 104: Instances

Base address	Peripheral	Instance	Description	Configuration
0x4001D000	PDM	PDM	Pulse Density Modulation (Digital Microphone Interface)	

Table 105: Register Overview

Register	Offset	Description
TASKS_START	0x000	Starts continuous PDM transfer
TASKS_STOP	0x004	Stops PDM transfer
EVENTS_STARTED	0x100	PDM transfer has started
EVENTS_STOPPED	0x104	PDM transfer has finished
EVENTS_END	0x108	The PDM has written the last sample specified by SAMPLE.MAXCNT (or the last sample after a STOP task has been received) to Data RAM
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ENABLE	0x500	PDM module enable register
PDMCLKCTRL	0x504	PDM clock generator control
MODE	0x508	Defines the routing of the connected PDM microphones' signals
GAINL	0x518	Left output gain adjustment
GAINR	0x51C	Right output gain adjustment
PSEL.CLK	0x540	Pin number configuration for PDM CLK signal
PSEL.DIN	0x544	Pin number configuration for PDM DIN signal
SAMPLE.PTR	0x560	RAM address pointer to write samples to with EasyDMA
SAMPLE.MAXCNT	0x564	Number of samples to allocate memory for in EasyDMA mode

43.7.1 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	STARTED			Enable or disable interrupt for STARTED event See *EVENTS_STARTED*
			Disabled	0	Disable
			Enabled	1	Enable
B	RW	STOPPED			Enable or disable interrupt for STOPPED event See *EVENTS_STOPPED*
			Disabled	0	Disable
			Enabled	1	Enable
C	RW	END			Enable or disable interrupt for END event See *EVENTS_END*
			Disabled	0	Disable
			Enabled	1	Enable

43.7.2 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	STARTED			Write '1' to Enable interrupt for STARTED event See *EVENTS_STARTED*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	STOPPED			Write '1' to Enable interrupt for STOPPED event See *EVENTS_STOPPED*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	END			Write '1' to Enable interrupt for END event See *EVENTS_END*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

43.7.3 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	STARTED			Write '1' to Disable interrupt for STARTED event See *EVENTS_STARTED*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	STOPPED			Write '1' to Disable interrupt for STOPPED event

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															

43.7.7 GAINL

Address offset: 0x518

Left output gain adjustment

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																													A	A	A	A	A	A	A
Reset 0x00000028				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**0**	**0**	**0**	
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	GAINL																																	
			MinGain	0x00																															
			DefaultGain	0x28																															
			MaxGain	0x50																															

43.7.8 GAINR

Address offset: 0x51C

Right output gain adjustment

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																													A	A	A	A	A	A	A
Reset 0x00000028				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	GAINR																																	
			MinGain	0x00																															
			DefaultGain	0x28																															
			MaxGain	0x50																															

43.7.9 PSEL.CLK

Address offset: 0x540

Pin number configuration for PDM CLK signal

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				B																												A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN		[0..31]																															
B	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

43.7.10 PSEL.DIN

Address offset: 0x544

Pin number configuration for PDM DIN signal

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				B																											A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN		[0..31]																															
B	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

43.7.11 SAMPLE.PTR

Address offset: 0x560

RAM address pointer to write samples to with EasyDMA

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	SAMPLEPTR																																	

43.7.12 SAMPLE.MAXCNT

Address offset: 0x564

Number of samples to allocate memory for in EasyDMA mode

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	BUFFSIZE		[0..32767]																															

43.8 Electrical specification

43.8.1 PDM Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
$I_{PDM,stereo}$	PDM module active current, stereo operation[37]		1.4		mA
$f_{PDM,CLK}$	PDM clock speed		1.032		MHz
$t_{PDM,JITTER}$	Jitter in PDM clock output			20	ns
$T_{dPDM,CLK}$	PDM clock duty cycle	40	50	60	%
$t_{PDM,DATA}$	Decimation filter delay			5	ms
$t_{PDM,cv}$	Allowed clock edge to data valid			125	ns
$t_{PDM,ci}$	Allowed (other) clock edge to data invalid	0			ns
$t_{PDM,s}$	Data setup time at $f_{PDM,CLK}$=1.024 MHz	65			ns
$t_{PDM,h}$	Data hold time at $f_{PDM,CLK}$=1.024 MHz	0			ns
$G_{PDM,default}$	Default (reset) absolute gain of the PDM module		3.2		dB

[37] Average current including PDM and DMA transfers, excluding clock and power supply base currents

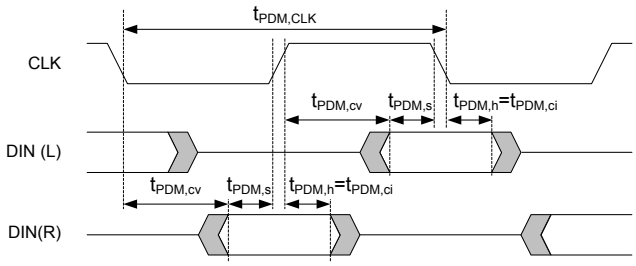


Figure 128: PDM timing diagram

44 I2S — Inter-IC sound interface

The I2S (Inter-IC Sound) module, supports the original two-channel I2S format, and left or right-aligned formats. It implements EasyDMA for sample transfer directly to and from RAM without CPU intervention.

The I2S peripheral has the following main features:

- Master and Slave mode
- Simultaneous bi-directional (TX and RX) audio streaming
- Original I2S and left- or right-aligned format
- 8, 16 and 24-bit sample width
- Low-jitter Master Clock generator
- Various sample rates

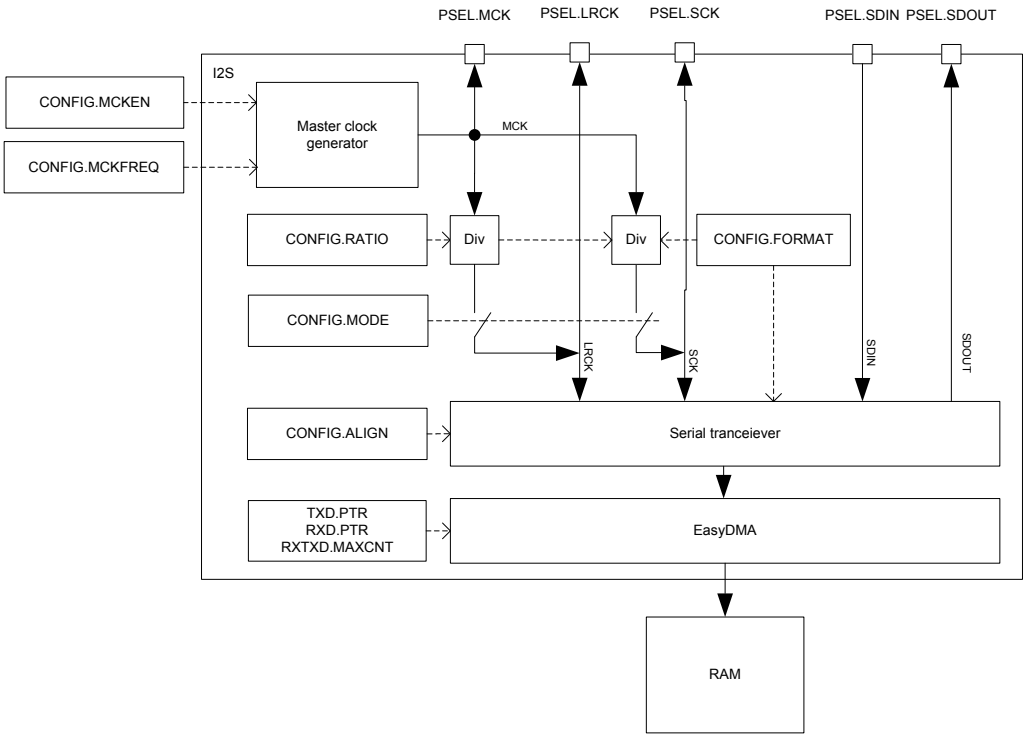


Figure 129: I2S master

44.1 Mode

The I2S protocol specification defines two modes of operation, Master and Slave.

The I2S mode decides which of the two sides (Master or Slave) shall provide the clock signals LRCK and SCK, and these signals are always supplied by the Master to the Slave.

44.2 Transmitting and receiving

The I2S module supports both transmission (TX) and reception (RX) of serial data. In both cases the serial data is shifted synchronously to the clock signals SCK and LRCK.

TX data is written to the SDOUT pin on the falling edge of SCK, and RX data is read from the SDIN pin on the rising edge of SCK. The most significant bit (MSB) is always transmitted first.

TX and RX are available in both Master and Slave modes and can be enabled/disabled independently in the *CONFIG.TXEN* on page 455 and *CONFIG.RXEN* on page 455.

Transmission and/or reception is started by triggering the START task. When started and transmission is enabled (in *CONFIG.TXEN* on page 455), the TXPTRUPD event will be generated for every *RXTXD.MAXCNT* on page 458 number of transmitted data words (containing one or more samples). Similarly, when started and reception is enabled (in *CONFIG.RXEN* on page 455), the RXPTRUPD event will be generated for every *RXTXD.MAXCNT* on page 458 received data words.

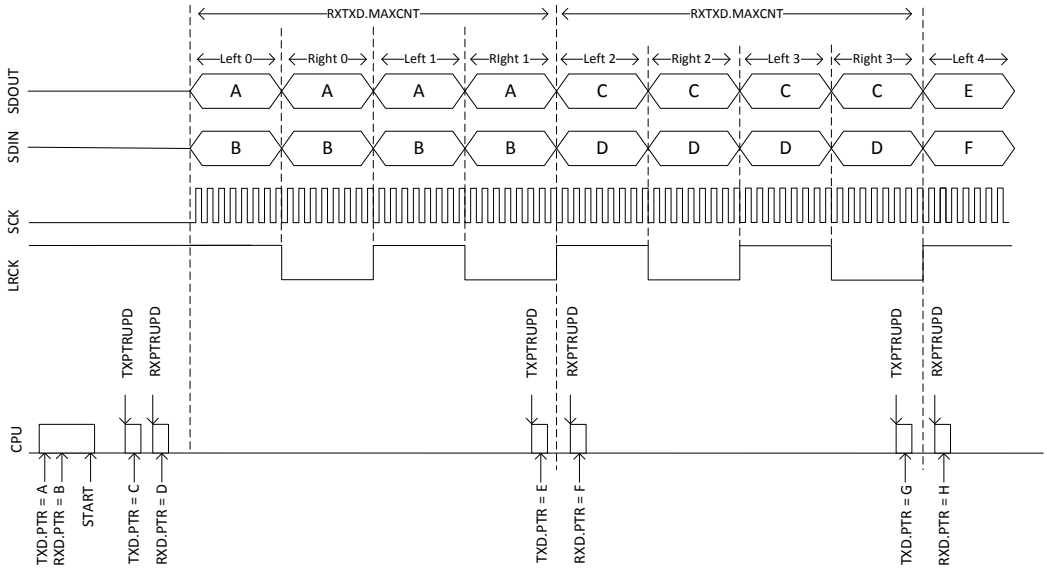


Figure 130: Transmitting and receiving. CONFIG.FORMAT = Aligned, CONFIG.SWIDTH = 8Bit, CONFIG.CHANNELS = Stereo, RXTXD.MAXCNT = 1.

44.3 Left right clock (LRCK)

The Left Right Clock (LRCK), often referred to as "word clock", "sample clock" or "word select" in I2S context, is the clock defining the frames in the serial bit streams sent and received on SDOUT and SDIN, respectively.

In I2S mode, each frame contains one left and right sample pair, with the left sample being transferred during the low half period of LRCK followed by the right sample being transferred during the high period of LRCK.

In Aligned mode, each frame contains one left and right sample pair, with the left sample being transferred during the high half period of LRCK followed by the right sample being transferred during the low period of LRCK.

Consequently, the LRCK frequency is equivalent to the audio sample rate.

When operating in Master mode, the LRCK is generated from the MCK, and the frequency of LRCK is then given as:

```
LRCK = MCK / CONFIG.RATIO
```

LRCK always toggles around the falling edge of the serial clock SCK.

44.4 Serial clock (SCK)

The serial clock (SCK), often referred to as the serial bit clock, pulses once for each data bit being transferred on the serial data lines SDIN and SDOUT.

When operating in Master mode the SCK is generated from the MCK, and the frequency of SCK is then given as:

```
SCK = 2 * LRCK * CONFIG.SWIDTH
```

The falling edge of the SCK falls on the toggling edge of LRCK.

When operating in Slave mode SCK is provided by the external I2S master.

44.5 Master clock (MCK)

The master clock (MCK) is the clock from which LRCK and SCK are derived when operating in Master mode.

The MCK is generated by an internal MCK generator. This generator always needs to be enabled when in Master mode, but the generator can also be enabled when in Slave mode. Enabling the generator when in slave mode can be useful in the case where the external Master is not able to generate its own master clock.

The MCK generator is enabled/disabled in the register *CONFIG.MCKEN* on page 456, and the generator is started or stopped by the START or STOP tasks.

In Master mode the LRCK and the SCK frequencies are closely related, as both are derived from MCK and set indirectly through *CONFIG.RATIO* on page 456 and *CONFIG.SWIDTH* on page 457.

When configuring these registers, the user is responsible for fulfilling the following requirements:

1. SCK frequency can never exceed the MCK frequency, which can be formulated as:

```
CONFIG.RATIO >= 2 * CONFIG.SWIDTH
```

2. The MCK/LRCK ratio shall be a multiple of 2 * CONFIG.SWIDTH, which can be formulated as:

```
Integer = (CONFIG.RATIO / (2 * CONFIG.SWIDTH))
```

The MCK signal can be routed to an output pin (specified in PSEL.MCK) to supply external I2S devices that require the MCK to be supplied from the outside.

When operating in Slave mode, the I2S module does not use the MCK and the MCK generator does not need to be enabled.

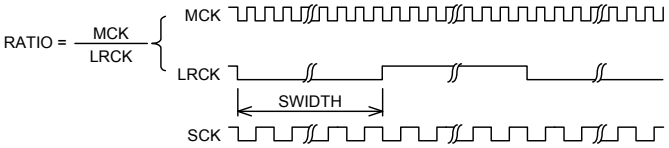


Figure 131: Relation between RATIO, MCK and LRCK.

Table 106: Configuration examples

Desired LRCK [Hz]	CONFIG.SWIDTH	CONFIG.RATIO	CONFIG.MCKFREQ	MCK [Hz]	LRCK [Hz]	LRCK error [%]
16000	16Bit	32X	32MDIV63	507936.5	15873.0	-0.8
16000	16Bit	64X	32MDIV31	1032258.1	16129.0	0.8
16000	16Bit	256X	32MDIV8	4000000.0	15625.0	-2.3
32000	16Bit	32X	32MDIV31	1032258.1	32258.1	0.8
32000	16Bit	64X	32MDIV16	2000000.0	31250.0	-2.3
32000	16Bit	256X	32MDIV4	8000000.0	31250.0	-2.3
44100	16Bit	32X	32MDIV23	1391304.3	43478.3	-1.4
44100	16Bit	64X	32MDIV11	2909090.9	45454.5	3.1
44100	16Bit	256X	32MDIV3	10666666.7	41666.7	-5.5

44.6 Width, alignment and format

The CONFIG.SWIDTH register primarily defines the sample width of the data written to memory. In master mode, it then also sets the amount of bits per frame. In Slave mode it controls padding/trimming if required. Left, right, transmitted, and received samples always have the same width. The CONFIG.FORMAT register specifies the position of the data frames with respect to the LRCK edges in both Master and Slave modes.

When using I2S format, the first bit in a half-frame (containing one left or right sample) gets sampled on the second rising edge of the SCK after a LRCK edge. When using Aligned mode, the first bit in a half-frame gets sampled on the first rising edge of SCK following a LRCK edge.

For data being received on SDIN the sample value can be either right or left-aligned inside a half-frame, as specified in *CONFIG.ALIGN* on page 457. *CONFIG.ALIGN* on page 457 affects only the decoding of the incoming samples (SDIN), while the outgoing samples (SDOUT) are always left-aligned (or justified).

When using left-alignment, each half-frame starts with the MSB of the sample value (both for data being sent on SDOUT and received on SDIN).

When using right-alignment, each half-frame of data being received on SDIN ends with the LSB of the sample value, while each half-frame of data being sent on SDOUT starts with the MSB of the sample value (same as for left-alignment).

In Master mode, the size of a half-frame (in number of SCK periods) equals the sample width (in number of bits), and in this case the alignment setting does not care as each half-frame in any case will start with the MSB and end with the LSB of the sample value.

In slave mode, however, the sample width does not need to equal the frame size. This means you might have extra or fewer SCK pulses per half-frame than what the sample width specified in *CONFIG.SWIDTH* requires.

In the case where we use **left-alignment** and the number of SCK pulses per half-frame is **higher** than the sample width, the following will apply:

- For data received on SDIN, all bits after the LSB of the sample value will be discarded.
- For data sent on SDOUT, all bits after the LSB of the sample value will be 0.

In the case where we use **left-alignment** and the number of SCK pulses per frame is **lower** than the sample width, the following will apply:

- Data sent and received on SDOUT and SDIN will be truncated with the LSBs being removed first.

In the case where we use **right-alignment** and the number of SCK pulses per frame is **higher** than the sample width, the following will apply:

- For data received on SDIN, all bits before the MSB of the sample value will be discarded.
- For data sent on SDOUT, all bits after the LSB of the sample value will be 0 (same behavior as for left-alignment).

In the case where we use **right-alignment** and the number of SCK pulses per frame is **lower** than the sample width, the following will apply:

- Data received on SDIN will be sign-extended to "sample width" number of bits before being written to memory.
- Data sent on SDOUT will be truncated with the LSBs being removed first (same behavior as for left-alignment).

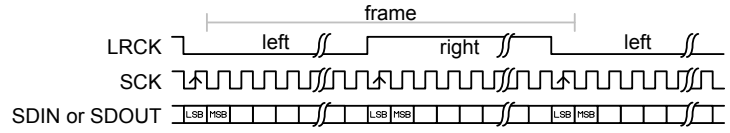


Figure 132: I2S format. CONFIG.SWIDTH equalling half-frame size.

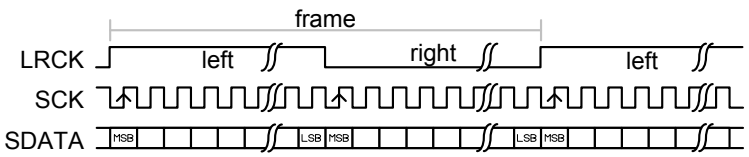


Figure 133: Aligned format. CONFIG.SWIDTH equalling half-frame size.

44.7 EasyDMA

The I2S module implements EasyDMA for accessing internal Data RAM without CPU intervention.

The source and destination pointers for the TX and RX data are configured in *TXD.PTR* on page 458 and *RXD.PTR* on page 458. The memory pointed to by these pointers will only be read or written when TX or RX are enabled in *CONFIG.TXEN* on page 455 and *CONFIG.RXEN* on page 455.

The addresses written to the pointer registers *TXD.PTR* on page 458 and *RXD.PTR* on page 458 are double-buffered in hardware, and these double buffers are updated for every *RXTXD.MAXCNT* on page 458 words (containing one or more samples) read/written from/to memory. The events TXPTRUPD and RXPTRUPD are generated whenever the TXD.PTR and RXD.PTR are transferred to these double buffers.

If *TXD.PTR* on page 458 is not pointing to the Data RAM region when transmission is enabled, or *RXD.PTR* on page 458 is not pointing to the Data RAM region when reception is enabled, an EasyDMA transfer may result in a HardFault and/or memory corruption. See *Memory* on page 23 for more information about the different memory regions.

Due to the nature of I2S, where the number of transmitted samples always equals the number of received samples (at least when both TX and RX are enabled), one common register *RXTXD.MAXCNT* on page 458 is used for specifying the sizes of these two memory buffers. The size of the buffers is specified in a number of 32-bit words. Such a 32-bit memory word can either contain four 8-bit samples, two 16-bit samples or one right-aligned 24-bit sample sign extended to 32 bit.

In stereo mode (CONFIG.CHANNELS=Stereo), the samples are stored as "left and right sample pairs" in memory. Figure *Figure 134: Memory mapping for 8 bit stereo. CONFIG.SWIDTH = 8Bit, CONFIG.CHANNELS = Stereo.* on page 449, *Figure 136: Memory mapping for 16 bit stereo. CONFIG.SWIDTH = 16Bit, CONFIG.CHANNELS = Stereo.* on page 450 and *Figure 138: Memory mapping for 24 bit stereo. CONFIG.SWIDTH = 24Bit, CONFIG.CHANNELS = Stereo.* on page 450 show how the samples are mapped to memory in this mode. The mapping is valid for both RX and TX.

In mono mode (CONFIG.CHANNELS=Left or Right), RX sample from only one channel in the frame is stored in memory, the other channel sample is ignored. Illustrations *Figure 135: Memory mapping for 8 bit mono. CONFIG.SWIDTH = 8Bit, CONFIG.CHANNELS = Left.* on page 450, *Figure 137: Memory mapping for 16 bit mono, left channel only. CONFIG.SWIDTH = 16Bit, CONFIG.CHANNELS = Left.* on page 450 and *Figure 139: Memory mapping for 24 bit mono, left channel only. CONFIG.SWIDTH = 24Bit, CONFIG.CHANNELS = Left.* on page 451 show how RX samples are mapped to memory in this mode.

For TX, the same outgoing sample read from memory is transmitted on both left and right in a frame, resulting in a mono output stream.

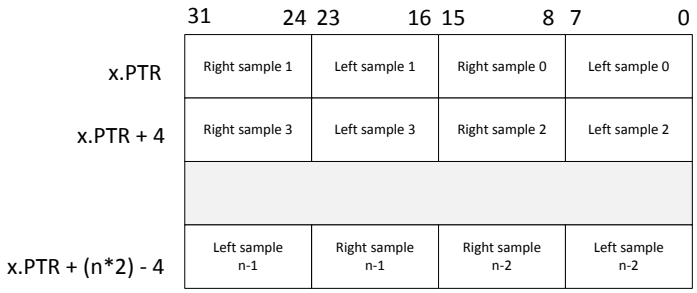


Figure 134: Memory mapping for 8 bit stereo. CONFIG.SWIDTH = 8Bit, CONFIG.CHANNELS = Stereo.

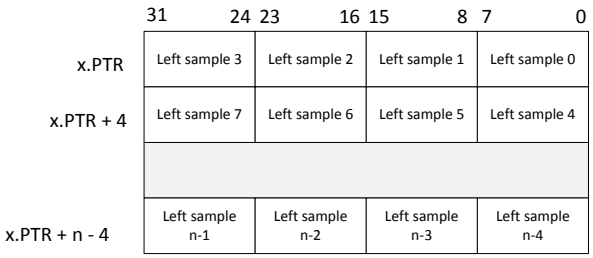


Figure 135: Memory mapping for 8 bit mono. CONFIG.SWIDTH = 8Bit, CONFIG.CHANNELS = Left.

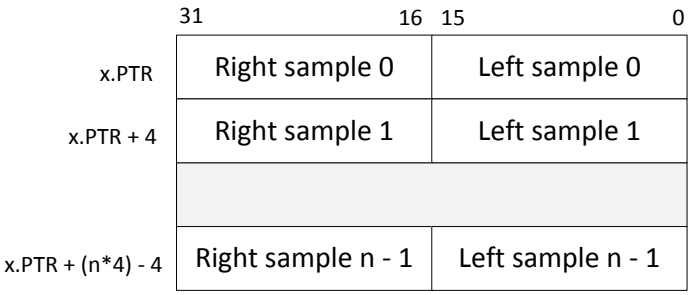


Figure 136: Memory mapping for 16 bit stereo. CONFIG.SWIDTH = 16Bit, CONFIG.CHANNELS = Stereo.

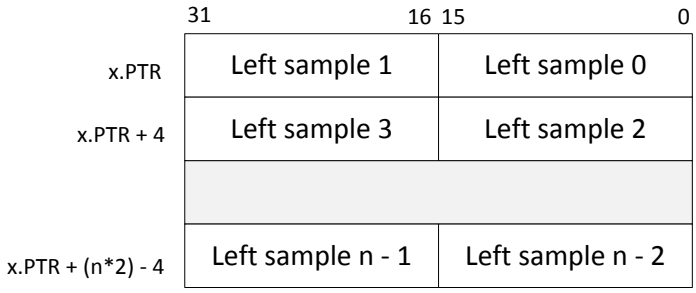


Figure 137: Memory mapping for 16 bit mono, left channel only. CONFIG.SWIDTH = 16Bit, CONFIG.CHANNELS = Left.

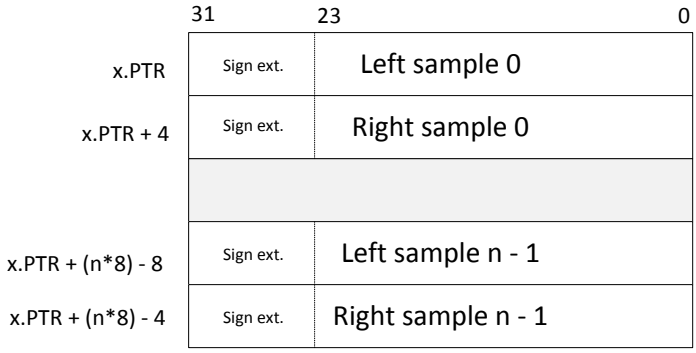


Figure 138: Memory mapping for 24 bit stereo. CONFIG.SWIDTH = 24Bit, CONFIG.CHANNELS = Stereo.

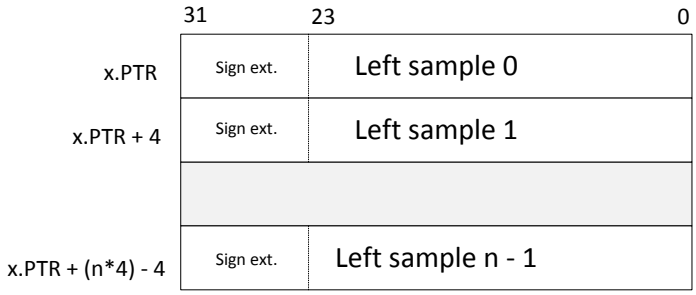


Figure 139: Memory mapping for 24 bit mono, left channel only. CONFIG.SWIDTH = 24Bit, CONFIG.CHANNELS = Left.

44.8 Module operation

Described here is a typical operating procedure for the I2S module.

1. Configure the I2S module using the CONFIG registers

```
// Enable reception
NRF_I2S->CONFIG.RXEN = (I2S_CONFIG_RXEN_RXEN_Enabled <<
                                        I2S_CONFIG_RXEN_RXEN_Pos);
// Enable transmission
NRF_I2S->CONFIG.TXEN = (I2S_CONFIG_TXEN_TXEN_Enabled <<
                                        I2S_CONFIG_TXEN_TXEN_Pos);
// Enable MCK generator
NRF_I2S->CONFIG.MCKEN = (I2S_CONFIG_MCKEN_MCKEN_Enabled <<
                                        I2S_CONFIG_MCKEN_MCKEN_Pos);
// MCKFREQ = 4 MHz
NRF_I2S->CONFIG.MCKFREQ = I2S_CONFIG_MCKFREQ_MCKFREQ_32MDIV8 <<
                                        I2S_CONFIG_MCKFREQ_MCKFREQ_Pos;
// Ratio = 256
NRF_I2S->CONFIG.RATIO = I2S_CONFIG_RATIO_RATIO_256X <<
                                        I2S_CONFIG_RATIO_RATIO_Pos;
// MCKFREQ = 4 MHz and Ratio = 256 gives sample rate = 15.625 ks/s
// Sample width = 16 bit
NRF_I2S->CONFIG.SWIDTH = I2S_CONFIG_SWIDTH_SWIDTH_16Bit <<
                                        I2S_CONFIG_SWIDTH_SWIDTH_Pos;
// Alignment = Left
NRF_I2S->CONFIG.ALIGN = I2S_CONFIG_ALIGN_ALIGN_Left <<
                                        I2S_CONFIG_ALIGN_ALIGN_Pos;
// Format = I2S
NRF_I2S->CONFIG.FORMAT = I2S_CONFIG_FORMAT_FORMAT_I2S <<
                                        I2S_CONFIG_FORMAT_FORMAT_Pos;
// Use stereo
NRF_I2S->CONFIG.CHANNELS = I2S_CONFIG_CHANNELS_CHANNELS_Stereo <<
                                        I2S_CONFIG_CHANNELS_CHANNELS_Pos;
```

2. Map IO pins using the PINSEL registers

```
// MCK routed to pin 0
NRF_I2S->PSEL.MCK = (0 << I2S_PSEL_MCK_PIN_Pos) |
                    (I2S_PSEL_MCK_CONNECT_Connected <<
                                                I2S_PSEL_MCK_CONNECT_Pos);
// SCK routed to pin 1
NRF_I2S->PSEL.SCK = (1 << I2S_PSEL_SCK_PIN_Pos) |
                    (I2S_PSEL_SCK_CONNECT_Connected <<

 I2S_PSEL_SCK_CONNECT_Pos);
// LRCK routed to pin 2
NRF_I2S->PSEL.LRCK = (2 << I2S_PSEL_LRCK_PIN_Pos) |
                     (I2S_PSEL_LRCK_CONNECT_Connected <<
```

```
 I2S_PSEL_LRCK_CONNECT_Pos);
// SDOUT routed to pin 3
NRF_I2S->PSEL.SDOUT = (3 << I2S_PSEL_SDOUT_PIN_Pos) |
                      (I2S_PSEL_SDOUT_CONNECT_Connected <<

 I2S_PSEL_SDOUT_CONNECT_Pos);
// SDIN routed on pin 4
NRF_I2S->PSEL.SDIN = (4 << I2S_PSEL_SDIN_PIN_Pos) |
                     (I2S_PSEL_SDIN_CONNECT_Connected <<

 I2S_PSEL_SDIN_CONNECT_Pos);
```

3. Configure TX and RX data pointers using the TXD, RXD and RXTXD registers

```
NRF_I2S->TXD.PTR = my_tx_buf;
NRF_I2S->RXD.PTR = my_rx_buf;
NRF_I2S->TXD.MAXCNT = MY_BUF_SIZE;
```

4. Enable the I²S module using the ENABLE register

```
NRF_I2S->ENABLE = 1;
```

5. Start audio streaming using the START task

```
NRF_I2S->TASKS_START = 1;
```

6. Handle received and transmitted data when receiving the TXPTRUPD and RXPTRUPD events

```
if(NRF_I2S->EVENTS_TXPTRUPD  != 0)
{
    NRF_I2S->TXD.PTR = my_next_tx_buf;
    NRF_I2S->EVENTS_TXPTRUPD = 0;
}

if(NRF_I2S->EVENTS_RXPTRUPD != 0)
{
    NRF_I2S->RXD.PTR = my_next_rx_buf;
    NRF_I2S->EVENTS_RXPTRUPD = 0;
}
```

44.9 Pin configuration

The MCK, SCK, LRCK, SDIN and SDOUT signals associated with the I²S module are mapped to physical pins according to the pin numbers specified in the PSEL.x registers.

These pins are acquired whenever the I²S module is enabled through the register *ENABLE* on page 455.

When a pin is acquired by the I²S module, the direction of the pin (input or output) will be configured automatically, and any pin direction setting done in the GPIO module will be overridden. The directions for the various I²S pins are shown below in *Table 107: GPIO configuration before enabling peripheral (master mode)* on page 452 and *Table 108: GPIO configuration before enabling peripheral (slave mode)* on page 453.

To secure correct signal levels on the pins when the system is in OFF mode, and when the I²S module is disabled, these pins must be configured in the GPIO peripheral directly.

Table 107: GPIO configuration before enabling peripheral (master mode)

I²S signal	I²S pin	Direction	Output value	Comment
MCK	As specified in PSEL.MCK	Output	0	
LRCK	As specified in PSEL.LRCK	Output	0	

I2S signal	I2S pin	Direction	Output value	Comment
SCK	As specified in PSEL.SCK	Output	0	
SDIN	As specified in PSEL.SDIN	Input	Not applicable	
SDOUT	As specified in PSEL.SDOUT	Output	0	

Table 108: GPIO configuration before enabling peripheral (slave mode)

I2S signal	I2S pin	Direction	Output value	Comment
MCK	As specified in PSEL.MCK	Output	0	
LRCK	As specified in PSEL.LRCK	Input	Not applicable	
SCK	As specified in PSEL.SCK	Input	Not applicable	
SDIN	As specified in PSEL.SDIN	Input	Not applicable	
SDOUT	As specified in PSEL.SDOUT	Output	0	

44.10 Registers

Table 109: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40025000	I2S	I2S	Inter-IC Sound Interface	

Table 110: Register Overview

Register	Offset	Description
TASKS_START	0x000	Starts continuous I2S transfer. Also starts MCK generator when this is enabled.
TASKS_STOP	0x004	Stops I2S transfer. Also stops MCK generator. Triggering this task will cause the {event:STOPPED} event to be generated.
EVENTS_RXPTRUPD	0x104	The RXD.PTR register has been copied to internal double-buffers. When the I2S module is started and RX is enabled, this event will be generated for every RXTXD.MAXCNT words that are received on the SDIN pin.
EVENTS_STOPPED	0x108	I2S transfer stopped.
EVENTS_TXPTRUPD	0x114	The TDX.PTR register has been copied to internal double-buffers. When the I2S module is started and TX is enabled, this event will be generated for every RXTXD.MAXCNT words that are sent on the SDOUT pin.
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ENABLE	0x500	Enable I2S module.
CONFIG.MODE	0x504	I2S mode.
CONFIG.RXEN	0x508	Reception (RX) enable.
CONFIG.TXEN	0x50C	Transmission (TX) enable.
CONFIG.MCKEN	0x510	Master clock generator enable.
CONFIG.MCKFREQ	0x514	Master clock generator frequency.
CONFIG.RATIO	0x518	MCK / LRCK ratio.
CONFIG.SWIDTH	0x51C	Sample width.
CONFIG.ALIGN	0x520	Alignment of sample within a frame.
CONFIG.FORMAT	0x524	Frame format.
CONFIG.CHANNELS	0x528	Enable channels.
RXD.PTR	0x538	Receive buffer RAM start address.
TXD.PTR	0x540	Transmit buffer RAM start address.
RXTXD.MAXCNT	0x550	Size of RXD and TXD buffers.
PSEL.MCK	0x560	Pin select for MCK signal.
PSEL.SCK	0x564	Pin select for SCK signal.
PSEL.LRCK	0x568	Pin select for LRCK signal.
PSEL.SDIN	0x56C	Pin select for SDIN signal.
PSEL.SDOUT	0x570	Pin select for SDOUT signal.

44.10.1 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																													F				C	B	
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																		**Description**					
B	RW	RXPTRUPD																											Enable or disable interrupt for RXPTRUPD event See *EVENTS_RXPTRUPD*					
			Disabled								0																		Disable					
			Enabled								1																		Enable					
C	RW	STOPPED																											Enable or disable interrupt for STOPPED event See *EVENTS_STOPPED*					
			Disabled								0																		Disable					
			Enabled								1																		Enable					
F	RW	TXPTRUPD																											Enable or disable interrupt for TXPTRUPD event See *EVENTS_TXPTRUPD*					
			Disabled								0																		Disable					
			Enabled								1																		Enable					

44.10.2 INTENSET

Address offset: 0x304

Enable interrupt

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																													F				C	B	
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																		**Description**					
B	RW	RXPTRUPD																											Write '1' to Enable interrupt for RXPTRUPD event See *EVENTS_RXPTRUPD*					
			Set								1																		Enable					
			Disabled								0																		Read: Disabled					
			Enabled								1																		Read: Enabled					
C	RW	STOPPED																											Write '1' to Enable interrupt for STOPPED event See *EVENTS_STOPPED*					
			Set								1																		Enable					
			Disabled								0																		Read: Disabled					
			Enabled								1																		Read: Enabled					
F	RW	TXPTRUPD																											Write '1' to Enable interrupt for TXPTRUPD event See *EVENTS_TXPTRUPD*					
			Set								1																		Enable					
			Disabled								0																		Read: Disabled					
			Enabled								1																		Read: Enabled					

44.10.3 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																													F				C	B	
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																		**Description**					
B	RW	RXPTRUPD																											Write '1' to Disable interrupt for RXPTRUPD event See *EVENTS_RXPTRUPD*					
			Clear								1																		Disable					
			Disabled								0																		Read: Disabled					
			Enabled								1																		Read: Enabled					
C	RW	STOPPED																											Write '1' to Disable interrupt for STOPPED event					

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																													F				C	B	
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																		**Description**						
																													See *EVENTS_STOPPED*						
			Clear								1																		Disable						
			Disabled								0																		Read: Disabled						
			Enabled								1																		Read: Enabled						
F	RW	TXPTRUPD																											Write '1' to Disable interrupt for TXPTRUPD event See *EVENTS_TXPTRUPD*						
			Clear								1																		Disable						
			Disabled								0																		Read: Disabled						
			Enabled								1																		Read: Enabled						

44.10.4 ENABLE

Address offset: 0x500

Enable I2S module.

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																		**Description**						
A	RW	ENABLE																											Enable I2S module.						
			Disabled								0																		Disable						
			Enabled								1																		Enable						

44.10.5 CONFIG.MODE

Address offset: 0x504

I2S mode.

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																		**Description**						
A	RW	MODE																											I2S mode.						
			Master								0																		Master mode. SCK and LRCK generated from internal master clcok (MCK) and output on pins defined by PSEL.xxx.						
			Slave								1																		Slave mode. SCK and LRCK generated by external master and received on pins defined by PSEL.xxx						

44.10.6 CONFIG.RXEN

Address offset: 0x508

Reception (RX) enable.

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																		**Description**						
A	RW	RXEN																											Reception (RX) enable.						
			Disabled								0																		Reception disabled and now data will be written to the RXD.PTR address.						
			Enabled								1																		Reception enabled.						

44.10.7 CONFIG.TXEN

Address offset: 0x50C

Transmission (TX) enable.

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		A
Reset 0x00000001			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
A	RW	TXEN																									Transmission (TX) enable.							
			Disabled								0																Transmission disabled and now data will be read from the RXD.TXD address.							
			Enabled								1																Transmission enabled.							

44.10.8 CONFIG.MCKEN

Address offset: 0x510

Master clock generator enable.

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		A
Reset 0x00000001			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
A	RW	MCKEN																									Master clock generator enable.							
			Disabled								0																Master clock generator disabled and PSEL.MCK not connected(available as GPIO).							
			Enabled								1																Master clock generator running and MCK output on PSEL.MCK.							

44.10.9 CONFIG.MCKFREQ

Address offset: 0x514

Master clock generator frequency.

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x20000000			**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
A	RW	MCKFREQ																									Master clock generator frequency.							
			32MDIV2								0x80000000																32 MHz / 2 = 16.0 MHz							
			32MDIV3								0x50000000																32 MHz / 3 = 10.6666667 MHz							
			32MDIV4								0x40000000																32 MHz / 4 = 8.0 MHz							
			32MDIV5								0x30000000																32 MHz / 5 = 6.4 MHz							
			32MDIV6								0x28000000																32 MHz / 6 = 5.3333333 MHz							
			32MDIV8								0x20000000																32 MHz / 8 = 4.0 MHz							
			32MDIV10								0x18000000																32 MHz / 10 = 3.2 MHz							
			32MDIV11								0x16000000																32 MHz / 11 = 2.9090909 MHz							
			32MDIV15								0x11000000																32 MHz / 15 = 2.1333333 MHz							
			32MDIV16								0x10000000																32 MHz / 16 = 2.0 MHz							
			32MDIV21								0x0C000000																32 MHz / 21 = 1.5238095							
			32MDIV23								0x0B000000																32 MHz / 23 = 1.3913043 MHz							
			32MDIV30								0x08800000																32 MHz / 30 = 1.0666667 MHz							
			32MDIV31								0x08400000																32 MHz / 31 = 1.0322581 MHz							
			32MDIV32								0x08000000																32 MHz / 32 = 1.0 MHz							
			32MDIV42								0x06000000																32 MHz / 42 = 0.7619048 MHz							
			32MDIV63								0x04100000																32 MHz / 63 = 0.5079365 MHz							
			32MDIV125								0x020C0000																32 MHz / 125 = 0.256 MHz							

44.10.10 CONFIG.RATIO

Address offset: 0x518

MCK / LRCK ratio.

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															A	A	A	A
Reset 0x00000006			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	RATIO																																
			32X	0																														
			48X	1																														
			64X	2																														
			96X	3																														
			128X	4																														
			192X	5																														
			256X	6																														
			384X	7																														
			512X	8																														

44.10.11 CONFIG.SWIDTH

Address offset: 0x51C

Sample width.

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																	A	A
Reset 0x00000001			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	SWIDTH																																
			8Bit	0																														
			16Bit	1																														
			24Bit	2																														

44.10.12 CONFIG.ALIGN

Address offset: 0x520

Alignment of sample within a frame.

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	ALIGN																																
			Left	0																														
			Right	1																														

44.10.13 CONFIG.FORMAT

Address offset: 0x524

Frame format.

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																		A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	FORMAT																																
			I2S	0																														
			Aligned	1																														

44.10.14 CONFIG.CHANNELS

Address offset: 0x528

Enable channels.

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
A	RW	CHANNELS			Enable channels.
			Stereo	0	Stereo.
			Left	1	Left only.
			Right	2	Right only.

44.10.15 RXD.PTR

Address offset: 0x538

Receive buffer RAM start address.

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
A	RW	PTR			Receive buffer Data RAM start address. When receiving, words containing samples will be written to this address. This address is a word aligned Data RAM address.

44.10.16 TXD.PTR

Address offset: 0x540

Transmit buffer RAM start address.

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
A	RW	PTR			Transmit buffer Data RAM start address. When transmitting, words containing samples will be fetched from this address. This address is a word aligned Data RAM address.

44.10.17 RXTXD.MAXCNT

Address offset: 0x550

Size of RXD and TXD buffers.

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			A A A A A A A A A A A A A A A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
A	RW	MAXCNT			Size of RXD and TXD buffers in number of 32 bit words.

44.10.18 PSEL.MCK

Address offset: 0x560

Pin select for MCK signal.

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			C A A A A
Reset 0xFFFFFFFF			**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

Id	RW	Field	Value Id	Value	Description
A	RW	PIN		[0..31]	Pin number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

44.10.19 PSEL.SCK

Address offset: 0x564

Pin select for SCK signal.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				C																											A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN		[0..31]																															
C	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

44.10.20 PSEL.LRCK

Address offset: 0x568

Pin select for LRCK signal.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				C																											A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN		[0..31]																															
C	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

44.10.21 PSEL.SDIN

Address offset: 0x56C

Pin select for SDIN signal.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				C																											A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN		[0..31]																															
C	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

44.10.22 PSEL.SDOUT

Address offset: 0x570

Pin select for SDOUT signal.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				C																											A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PIN		[0..31]																															
C	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

44.11 Electrical specification

44.11.1 I2S timing specification

Symbol	Description	Min.	Typ.	Max.	Units
t_{S_SDIN}	SDIN setup time before SCK rising	20			ns
t_{H_SDIN}	SDIN hold time after SCK rising	15			ns
t_{S_SDOUT}	SDOUT setup time after SCK falling	40			ns
t_{H_SDOUT}	SDOUT hold time before SCK falling	6			ns
t_{SCK_LRCK}	SCLK falling to LRCK edge	-5	0	5	ns
f_{MCK}	MCK frequency			4000	kHz
f_{LRCK}	LRCK frequency			48	kHz
f_{SCK}	SCK frequency			2000	kHz
DC_{CK}	Clock duty cycle (MCK, LRCK, SCK)	45		55	%

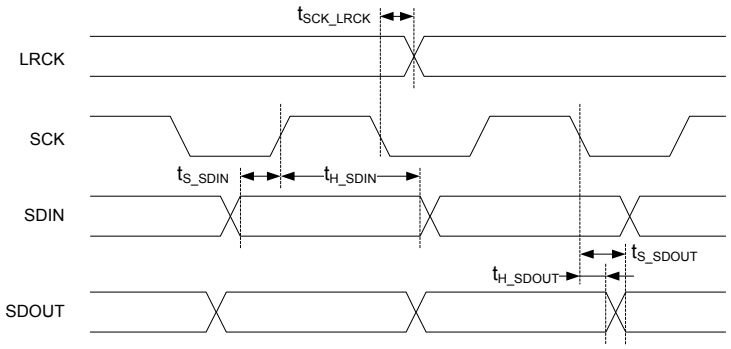


Figure 140: I2S timing diagram

45 MWU — Memory watch unit

The Memory watch unit (MWU) can be used to generate events when a memory region is accessed by the CPU. The MWU can be configured to trigger events for access to Data RAM and Peripheral memory segments. The MWU allows an application developer to generate memory access events during development for debugging or during production execution for failure detection and recovery.

Listed here are the main features for MWU:

- Six memory regions, four user-configurable and two fixed regions in peripheral address space
- Flexible configuration of regions with START and END addresses
- Generate events on CPU read and/or write to a defined region of Data RAM or peripheral memory address space
- Programmable maskable or non-maskable (NMI) interrupt on events
- Peripheral interfaces can be watched for read and write access using subregions of the two fixed memory regions

Table 111: Memory regions

Memory region	START address	END address
REGION[0..3]	Configurable	Configurable
PREGION[0]	0x40000000	0x4001FFFF
PREGION[1]	0x40020000	0x4003FFFF

Each MWU region is defined by a start address and an end address, configured by the START and END registers respectively. These addresses are byte aligned and inclusive. The END register value has to be greater or equal to the START register value. Each region is associated with a pair of events that indicate that either a write access or a read access from the CPU has been detected inside the region.

For regions containing subregions (see below), a set of status registers PERREGION[0..1].SUBSTATWA and PERREGION[0..1].SUBSTATRA indicate which subregion(s) caused the EVENT_PREGION[0..1].WA and EVENT_PREGION[0..1].RA respectively.

The MWU is only able to detect memory accesses in the Data RAM and Peripheral memory segments from the CPU, see *Memory* on page 23 for more information about the different memory segments. EasyDMA accesses are not monitored by the MWU. The MWU requires two HCLK cycles to detect and generate the event.

The peripheral regions, PREGION[0...1], are divided into 32 equally sized subregions, SR[0...31]. All subregions are excluded in the main region by default, and any can be included by specifying them in the SUBS register. When a subregion is excluded from the main region, the memory watch mechanism will not trigger any events when that subregion is accessed.

Subregions in PREGION[0..1] cannot be individually configured for read or write access watch. Watch configuration is only possible for a region as a whole. The PRGNiRA and PRGNiWA (i=0..1) fields in the REGIONEN register control watching read and write access.

REGION[0..3] can be individually enabled for read and/or write access watching through their respective RGNiRA and RGNiWA (i=0..3) fields in the REGIONEN register.

REGIONENSET and REGIONENCLR allow respectively enabling and disabling one or multiple REGIONs or PREGIONs watching in a single write access.

45.1 Registers

Table 112: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40020000	MWU	MWU	Memory Watch Unit	

Table 113: Register Overview

Register	Offset	Description
EVENTS_REGION[0].WA	0x100	Write access to region 0 detected
EVENTS_REGION[0].RA	0x104	Read access to region 0 detected
EVENTS_REGION[1].WA	0x108	Write access to region 1 detected
EVENTS_REGION[1].RA	0x10C	Read access to region 1 detected
EVENTS_REGION[2].WA	0x110	Write access to region 2 detected
EVENTS_REGION[2].RA	0x114	Read access to region 2 detected
EVENTS_REGION[3].WA	0x118	Write access to region 3 detected
EVENTS_REGION[3].RA	0x11C	Read access to region 3 detected
EVENTS_PREGION[0].WA	0x160	Write access to peripheral region 0 detected
EVENTS_PREGION[0].RA	0x164	Read access to peripheral region 0 detected
EVENTS_PREGION[1].WA	0x168	Write access to peripheral region 1 detected
EVENTS_PREGION[1].RA	0x16C	Read access to peripheral region 1 detected
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
NMIEN	0x320	Enable or disable non-maskable interrupt
NMIENSET	0x324	Enable non-maskable interrupt
NMIENCLR	0x328	Disable non-maskable interrupt
PERREGION[0].SUBSTAT	0x400	Source of event/interrupt in region 0, write access detected while corresponding subregion was enabled for watching
PERREGION[0].SUBSTATI	0x404	Source of event/interrupt in region 0, read access detected while corresponding subregion was enabled for watching
PERREGION[1].SUBSTAT	0x408	Source of event/interrupt in region 1, write access detected while corresponding subregion was enabled for watching
PERREGION[1].SUBSTATI	0x40C	Source of event/interrupt in region 1, read access detected while corresponding subregion was enabled for watching
REGIONEN	0x510	Enable/disable regions watch
REGIONENSET	0x514	Enable regions watch
REGIONENCLR	0x518	Disable regions watch
REGION[0].START	0x600	Start address for region 0
REGION[0].END	0x604	End address of region 0
REGION[1].START	0x610	Start address for region 1
REGION[1].END	0x614	End address of region 1
REGION[2].START	0x620	Start address for region 2
REGION[2].END	0x624	End address of region 2
REGION[3].START	0x630	Start address for region 3
REGION[3].END	0x634	End address of region 3
PREGION[0].START	0x6C0	Reserved for future use
PREGION[0].END	0x6C4	Reserved for future use
PREGION[0].SUBS	0x6C8	Subregions of region 0
PREGION[1].START	0x6D0	Reserved for future use
PREGION[1].END	0x6D4	Reserved for future use
PREGION[1].SUBS	0x6D8	Subregions of region 1

45.1.1 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	RW	REGION0WA																										Enable or disable interrupt for REGION[0].WA event							
																												See *EVENTS_REGION[0].WA*							
			Disabled	0																								Disable							

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**								**Description**																							
			Enabled	1								Enable																							
B	RW	REGION0RA										Enable or disable interrupt for REGION[0].RA event See *EVENTS_REGION[0].RA*																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							
C	RW	REGION1WA										Enable or disable interrupt for REGION[1].WA event See *EVENTS_REGION[1].WA*																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							
D	RW	REGION1RA										Enable or disable interrupt for REGION[1].RA event See *EVENTS_REGION[1].RA*																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							
E	RW	REGION2WA										Enable or disable interrupt for REGION[2].WA event See *EVENTS_REGION[2].WA*																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							
F	RW	REGION2RA										Enable or disable interrupt for REGION[2].RA event See *EVENTS_REGION[2].RA*																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							
G	RW	REGION3WA										Enable or disable interrupt for REGION[3].WA event See *EVENTS_REGION[3].WA*																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							
H	RW	REGION3RA										Enable or disable interrupt for REGION[3].RA event See *EVENTS_REGION[3].RA*																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							
I	RW	PREGION0WA										Enable or disable interrupt for PREGION[0].WA event See *EVENTS_PREGION[0].WA*																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							
J	RW	PREGION0RA										Enable or disable interrupt for PREGION[0].RA event See *EVENTS_PREGION[0].RA*																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							
K	RW	PREGION1WA										Enable or disable interrupt for PREGION[1].WA event See *EVENTS_PREGION[1].WA*																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							
L	RW	PREGION1RA										Enable or disable interrupt for PREGION[1].RA event See *EVENTS_PREGION[1].RA*																							
			Disabled	0								Disable																							
			Enabled	1								Enable																							

45.1.2 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	RW	REGION0WA																										Write '1' to Enable interrupt for REGION[0].WA event See *EVENTS_REGION[0].WA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
B	RW	REGION0RA																										Write '1' to Enable interrupt for REGION[0].RA event See *EVENTS_REGION[0].RA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
C	RW	REGION1WA																										Write '1' to Enable interrupt for REGION[1].WA event See *EVENTS_REGION[1].WA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
D	RW	REGION1RA																										Write '1' to Enable interrupt for REGION[1].RA event See *EVENTS_REGION[1].RA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
E	RW	REGION2WA																										Write '1' to Enable interrupt for REGION[2].WA event See *EVENTS_REGION[2].WA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
F	RW	REGION2RA																										Write '1' to Enable interrupt for REGION[2].RA event See *EVENTS_REGION[2].RA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
G	RW	REGION3WA																										Write '1' to Enable interrupt for REGION[3].WA event See *EVENTS_REGION[3].WA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
H	RW	REGION3RA																										Write '1' to Enable interrupt for REGION[3].RA event See *EVENTS_REGION[3].RA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
I	RW	PREGION0WA																										Write '1' to Enable interrupt for PREGION[0].WA event See *EVENTS_PREGION[0].WA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
J	RW	PREGION0RA																										Write '1' to Enable interrupt for PREGION[0].RA event See *EVENTS_PREGION[0].RA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
K	RW	PREGION1WA																										Write '1' to Enable interrupt for PREGION[1].WA event							

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
																												See *EVENTS_PREGION[1].WA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
L	RW	PREGION1RA																										Write '1' to Enable interrupt for PREGION[1].RA event							
																												See *EVENTS_PREGION[1].RA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							

45.1.3 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	RW	REGION0WA																										Write '1' to Disable interrupt for REGION[0].WA event							
																												See *EVENTS_REGION[0].WA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
B	RW	REGION0RA																										Write '1' to Disable interrupt for REGION[0].RA event							
																												See *EVENTS_REGION[0].RA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
C	RW	REGION1WA																										Write '1' to Disable interrupt for REGION[1].WA event							
																												See *EVENTS_REGION[1].WA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
D	RW	REGION1RA																										Write '1' to Disable interrupt for REGION[1].RA event							
																												See *EVENTS_REGION[1].RA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
E	RW	REGION2WA																										Write '1' to Disable interrupt for REGION[2].WA event							
																												See *EVENTS_REGION[2].WA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
F	RW	REGION2RA																										Write '1' to Disable interrupt for REGION[2].RA event							
																												See *EVENTS_REGION[2].RA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
G	RW	REGION3WA																										Write '1' to Disable interrupt for REGION[3].WA event							
																												See *EVENTS_REGION[3].WA*							
			Clear	1																								Disable							

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
H	RW	REGION3RA																										Write '1' to Disable interrupt for REGION[3].RA event See *EVENTS_REGION[3].RA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
I	RW	PREGION0WA																										Write '1' to Disable interrupt for PREGION[0].WA event See *EVENTS_PREGION[0].WA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
J	RW	PREGION0RA																										Write '1' to Disable interrupt for PREGION[0].RA event See *EVENTS_PREGION[0].RA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
K	RW	PREGION1WA																										Write '1' to Disable interrupt for PREGION[1].WA event See *EVENTS_PREGION[1].WA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
L	RW	PREGION1RA																										Write '1' to Disable interrupt for PREGION[1].RA event See *EVENTS_PREGION[1].RA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							

45.1.4 NMIEN

Address offset: 0x320

Enable or disable non-maskable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	RW	REGION0WA																										Enable or disable non-maskable interrupt for REGION[0].WA event See *EVENTS_REGION[0].WA*							
			Disabled	0																								Disable							
			Enabled	1																								Enable							
B	RW	REGION0RA																										Enable or disable non-maskable interrupt for REGION[0].RA event See *EVENTS_REGION[0].RA*							
			Disabled	0																								Disable							
			Enabled	1																								Enable							
C	RW	REGION1WA																										Enable or disable non-maskable interrupt for REGION[1].WA event See *EVENTS_REGION[1].WA*							
			Disabled	0																								Disable							
			Enabled	1																								Enable							

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
D	RW	REGION1RA			Enable or disable non-maskable interrupt for REGION[1].RA event See *EVENTS_REGION[1].RA*
			Disabled	0	Disable
			Enabled	1	Enable
E	RW	REGION2WA			Enable or disable non-maskable interrupt for REGION[2].WA event See *EVENTS_REGION[2].WA*
			Disabled	0	Disable
			Enabled	1	Enable
F	RW	REGION2RA			Enable or disable non-maskable interrupt for REGION[2].RA event See *EVENTS_REGION[2].RA*
			Disabled	0	Disable
			Enabled	1	Enable
G	RW	REGION3WA			Enable or disable non-maskable interrupt for REGION[3].WA event See *EVENTS_REGION[3].WA*
			Disabled	0	Disable
			Enabled	1	Enable
H	RW	REGION3RA			Enable or disable non-maskable interrupt for REGION[3].RA event See *EVENTS_REGION[3].RA*
			Disabled	0	Disable
			Enabled	1	Enable
I	RW	PREGION0WA			Enable or disable non-maskable interrupt for PREGION[0].WA event See *EVENTS_PREGION[0].WA*
			Disabled	0	Disable
			Enabled	1	Enable
J	RW	PREGION0RA			Enable or disable non-maskable interrupt for PREGION[0].RA event See *EVENTS_PREGION[0].RA*
			Disabled	0	Disable
			Enabled	1	Enable
K	RW	PREGION1WA			Enable or disable non-maskable interrupt for PREGION[1].WA event See *EVENTS_PREGION[1].WA*
			Disabled	0	Disable
			Enabled	1	Enable
L	RW	PREGION1RA			Enable or disable non-maskable interrupt for PREGION[1].RA event See *EVENTS_PREGION[1].RA*
			Disabled	0	Disable
			Enabled	1	Enable

45.1.5 NMIENSET

Address offset: 0x324

Enable non-maskable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	RW	REGION0WA																										Write '1' to Enable non-maskable interrupt for REGION[0].WA event See *EVENTS_REGION[0].WA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
B	RW	REGION0RA																										Write '1' to Enable non-maskable interrupt for REGION[0].RA event See *EVENTS_REGION[0].RA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
C	RW	REGION1WA																										Write '1' to Enable non-maskable interrupt for REGION[1].WA event See *EVENTS_REGION[1].WA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
D	RW	REGION1RA																										Write '1' to Enable non-maskable interrupt for REGION[1].RA event See *EVENTS_REGION[1].RA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
E	RW	REGION2WA																										Write '1' to Enable non-maskable interrupt for REGION[2].WA event See *EVENTS_REGION[2].WA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
F	RW	REGION2RA																										Write '1' to Enable non-maskable interrupt for REGION[2].RA event See *EVENTS_REGION[2].RA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
G	RW	REGION3WA																										Write '1' to Enable non-maskable interrupt for REGION[3].WA event See *EVENTS_REGION[3].WA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
H	RW	REGION3RA																										Write '1' to Enable non-maskable interrupt for REGION[3].RA event See *EVENTS_REGION[3].RA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
I	RW	PREGION0WA																										Write '1' to Enable non-maskable interrupt for PREGION[0].WA event See *EVENTS_PREGION[0].WA*							

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
J	RW	PREGION0RA																										Write '1' to Enable non-maskable interrupt for PREGION[0].RA event See *EVENTS_PREGION[0].RA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
K	RW	PREGION1WA																										Write '1' to Enable non-maskable interrupt for PREGION[1].WA event See *EVENTS_PREGION[1].WA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
L	RW	PREGION1RA																										Write '1' to Enable non-maskable interrupt for PREGION[1].RA event See *EVENTS_PREGION[1].RA*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							

45.1.6 NMIENCLR

Address offset: 0x328

Disable non-maskable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	RW	REGION0WA																										Write '1' to Disable non-maskable interrupt for REGION[0].WA event See *EVENTS_REGION[0].WA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
B	RW	REGION0RA																										Write '1' to Disable non-maskable interrupt for REGION[0].RA event See *EVENTS_REGION[0].RA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
C	RW	REGION1WA																										Write '1' to Disable non-maskable interrupt for REGION[1].WA event See *EVENTS_REGION[1].WA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
D	RW	REGION1RA																										Write '1' to Disable non-maskable interrupt for REGION[1].RA event See *EVENTS_REGION[1].RA*							

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
E	RW	REGION2WA																										Write '1' to Disable non-maskable interrupt for REGION[2].WA event See *EVENTS_REGION[2].WA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
F	RW	REGION2RA																										Write '1' to Disable non-maskable interrupt for REGION[2].RA event See *EVENTS_REGION[2].RA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
G	RW	REGION3WA																										Write '1' to Disable non-maskable interrupt for REGION[3].WA event See *EVENTS_REGION[3].WA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
H	RW	REGION3RA																										Write '1' to Disable non-maskable interrupt for REGION[3].RA event See *EVENTS_REGION[3].RA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
I	RW	PREGION0WA																										Write '1' to Disable non-maskable interrupt for PREGION[0].WA event See *EVENTS_PREGION[0].WA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
J	RW	PREGION0RA																										Write '1' to Disable non-maskable interrupt for PREGION[0].RA event See *EVENTS_PREGION[0].RA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
K	RW	PREGION1WA																										Write '1' to Disable non-maskable interrupt for PREGION[1].WA event See *EVENTS_PREGION[1].WA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
L	RW	PREGION1RA																										Write '1' to Disable non-maskable interrupt for PREGION[1].RA event See *EVENTS_PREGION[1].RA*							
			Clear	1																								Disable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							

45.1.7 PERREGION[0].SUBSTATWA

Address offset: 0x400

Source of event/interrupt in region 0, write access detected while corresponding subregion was enabled for watching

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	SR0			Subregion 0 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
B	RW	SR1			Subregion 1 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
C	RW	SR2			Subregion 2 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
D	RW	SR3			Subregion 3 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
E	RW	SR4			Subregion 4 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
F	RW	SR5			Subregion 5 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
G	RW	SR6			Subregion 6 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
H	RW	SR7			Subregion 7 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
I	RW	SR8			Subregion 8 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
J	RW	SR9			Subregion 9 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
K	RW	SR10			Subregion 10 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
L	RW	SR11			Subregion 11 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
M	RW	SR12			Subregion 12 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
N	RW	SR13			Subregion 13 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
O	RW	SR14			Subregion 14 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
P	RW	SR15			Subregion 15 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
Q	RW	SR16			Subregion 16 in region 0 (write '1' to clear)

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			NoAccess	0																															
			Access	1																															
R	RW	SR17																																	
			NoAccess	0																															
			Access	1																															
S	RW	SR18																																	
			NoAccess	0																															
			Access	1																															
T	RW	SR19																																	
			NoAccess	0																															
			Access	1																															
U	RW	SR20																																	
			NoAccess	0																															
			Access	1																															
V	RW	SR21																																	
			NoAccess	0																															
			Access	1																															
W	RW	SR22																																	
			NoAccess	0																															
			Access	1																															
X	RW	SR23																																	
			NoAccess	0																															
			Access	1																															
Y	RW	SR24																																	
			NoAccess	0																															
			Access	1																															
Z	RW	SR25																																	
			NoAccess	0																															
			Access	1																															
a	RW	SR26																																	
			NoAccess	0																															
			Access	1																															
b	RW	SR27																																	
			NoAccess	0																															
			Access	1																															
c	RW	SR28																																	
			NoAccess	0																															
			Access	1																															
d	RW	SR29																																	
			NoAccess	0																															
			Access	1																															
e	RW	SR30																																	
			NoAccess	0																															
			Access	1																															
f	RW	SR31																																	
			NoAccess	0																															
			Access	1																															

45.1.8 PERREGION[0].SUBSTATRA

Address offset: 0x404

Source of event/interrupt in region 0, read access detected while corresponding subregion was enabled for watching

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	SR0																																	
			NoAccess	0																															
			Access	1																															
B	RW	SR1																																	
			NoAccess	0																															
			Access	1																															
C	RW	SR2																																	
			NoAccess	0																															
			Access	1																															
D	RW	SR3																																	
			NoAccess	0																															
			Access	1																															
E	RW	SR4																																	
			NoAccess	0																															
			Access	1																															
F	RW	SR5																																	
			NoAccess	0																															
			Access	1																															
G	RW	SR6																																	
			NoAccess	0																															
			Access	1																															
H	RW	SR7																																	
			NoAccess	0																															
			Access	1																															
I	RW	SR8																																	
			NoAccess	0																															
			Access	1																															
J	RW	SR9																																	
			NoAccess	0																															
			Access	1																															
K	RW	SR10																																	
			NoAccess	0																															
			Access	1																															
L	RW	SR11																																	
			NoAccess	0																															
			Access	1																															
M	RW	SR12																																	
			NoAccess	0																															
			Access	1																															
N	RW	SR13																																	
			NoAccess	0																															
			Access	1																															
O	RW	SR14																																	
			NoAccess	0																															
			Access	1																															
P	RW	SR15																																	
			NoAccess	0																															
			Access	1																															
Q	RW	SR16																																	
			NoAccess	0																															
			Access	1																															
R	RW	SR17																																	
			NoAccess	0																															
			Access	1																															
S	RW	SR18																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			NoAccess	0																															
			Access	1																															
T	RW	SR19																																	
			NoAccess	0																															
			Access	1																															
U	RW	SR20																																	
			NoAccess	0																															
			Access	1																															
V	RW	SR21																																	
			NoAccess	0																															
			Access	1																															
W	RW	SR22																																	
			NoAccess	0																															
			Access	1																															
X	RW	SR23																																	
			NoAccess	0																															
			Access	1																															
Y	RW	SR24																																	
			NoAccess	0																															
			Access	1																															
Z	RW	SR25																																	
			NoAccess	0																															
			Access	1																															
a	RW	SR26																																	
			NoAccess	0																															
			Access	1																															
b	RW	SR27																																	
			NoAccess	0																															
			Access	1																															
c	RW	SR28																																	
			NoAccess	0																															
			Access	1																															
d	RW	SR29																																	
			NoAccess	0																															
			Access	1																															
e	RW	SR30																																	
			NoAccess	0																															
			Access	1																															
f	RW	SR31																																	
			NoAccess	0																															
			Access	1																															

45.1.9 PERREGION[1].SUBSTATWA

Address offset: 0x408

Source of event/interrupt in region 1, write access detected while corresponding subregion was enabled for watching

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	SR0																																	
			NoAccess	0																															
			Access	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
B	RW	SR1																																	
			NoAccess	0																															
			Access	1																															
C	RW	SR2																																	
			NoAccess	0																															
			Access	1																															
D	RW	SR3																																	
			NoAccess	0																															
			Access	1																															
E	RW	SR4																																	
			NoAccess	0																															
			Access	1																															
F	RW	SR5																																	
			NoAccess	0																															
			Access	1																															
G	RW	SR6																																	
			NoAccess	0																															
			Access	1																															
H	RW	SR7																																	
			NoAccess	0																															
			Access	1																															
I	RW	SR8																																	
			NoAccess	0																															
			Access	1																															
J	RW	SR9																																	
			NoAccess	0																															
			Access	1																															
K	RW	SR10																																	
			NoAccess	0																															
			Access	1																															
L	RW	SR11																																	
			NoAccess	0																															
			Access	1																															
M	RW	SR12																																	
			NoAccess	0																															
			Access	1																															
N	RW	SR13																																	
			NoAccess	0																															
			Access	1																															
O	RW	SR14																																	
			NoAccess	0																															
			Access	1																															
P	RW	SR15																																	
			NoAccess	0																															
			Access	1																															
Q	RW	SR16																																	
			NoAccess	0																															
			Access	1																															
R	RW	SR17																																	
			NoAccess	0																															
			Access	1																															
S	RW	SR18																																	
			NoAccess	0																															
			Access	1																															
T	RW	SR19																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			NoAccess	0																															
			Access	1																															
U	RW	SR20																																	
			NoAccess	0																															
			Access	1																															
V	RW	SR21																																	
			NoAccess	0																															
			Access	1																															
W	RW	SR22																																	
			NoAccess	0																															
			Access	1																															
X	RW	SR23																																	
			NoAccess	0																															
			Access	1																															
Y	RW	SR24																																	
			NoAccess	0																															
			Access	1																															
Z	RW	SR25																																	
			NoAccess	0																															
			Access	1																															
a	RW	SR26																																	
			NoAccess	0																															
			Access	1																															
b	RW	SR27																																	
			NoAccess	0																															
			Access	1																															
c	RW	SR28																																	
			NoAccess	0																															
			Access	1																															
d	RW	SR29																																	
			NoAccess	0																															
			Access	1																															
e	RW	SR30																																	
			NoAccess	0																															
			Access	1																															
f	RW	SR31																																	
			NoAccess	0																															
			Access	1																															

45.1.10 PERREGION[1].SUBSTATRA

Address offset: 0x40C

Source of event/interrupt in region 1, read access detected while corresponding subregion was enabled for watching

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	SR0																																	
			NoAccess	0																															
			Access	1																															
B	RW	SR1																																	
			NoAccess	0																															
			Access	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
C	RW	SR2																																	
			NoAccess	0																															
			Access	1																															
D	RW	SR3																																	
			NoAccess	0																															
			Access	1																															
E	RW	SR4																																	
			NoAccess	0																															
			Access	1																															
F	RW	SR5																																	
			NoAccess	0																															
			Access	1																															
G	RW	SR6																																	
			NoAccess	0																															
			Access	1																															
H	RW	SR7																																	
			NoAccess	0																															
			Access	1																															
I	RW	SR8																																	
			NoAccess	0																															
			Access	1																															
J	RW	SR9																																	
			NoAccess	0																															
			Access	1																															
K	RW	SR10																																	
			NoAccess	0																															
			Access	1																															
L	RW	SR11																																	
			NoAccess	0																															
			Access	1																															
M	RW	SR12																																	
			NoAccess	0																															
			Access	1																															
N	RW	SR13																																	
			NoAccess	0																															
			Access	1																															
O	RW	SR14																																	
			NoAccess	0																															
			Access	1																															
P	RW	SR15																																	
			NoAccess	0																															
			Access	1																															
Q	RW	SR16																																	
			NoAccess	0																															
			Access	1																															
R	RW	SR17																																	
			NoAccess	0																															
			Access	1																															
S	RW	SR18																																	
			NoAccess	0																															
			Access	1																															
T	RW	SR19																																	
			NoAccess	0																															
			Access	1																															
U	RW	SR20																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			NoAccess	0																															
			Access	1																															
V	RW	SR21																																	
			NoAccess	0																															
			Access	1																															
W	RW	SR22																																	
			NoAccess	0																															
			Access	1																															
X	RW	SR23																																	
			NoAccess	0																															
			Access	1																															
Y	RW	SR24																																	
			NoAccess	0																															
			Access	1																															
Z	RW	SR25																																	
			NoAccess	0																															
			Access	1																															
a	RW	SR26																																	
			NoAccess	0																															
			Access	1																															
b	RW	SR27																																	
			NoAccess	0																															
			Access	1																															
c	RW	SR28																																	
			NoAccess	0																															
			Access	1																															
d	RW	SR29																																	
			NoAccess	0																															
			Access	1																															
e	RW	SR30																																	
			NoAccess	0																															
			Access	1																															
f	RW	SR31																																	
			NoAccess	0																															
			Access	1																															

45.1.11 REGIONEN

Address offset: 0x510

Enable/disable regions watch

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	RGN0WA																																	
			Disable	0																															
			Enable	1																															
B	RW	RGN0RA																																	
			Disable	0																															
			Enable	1																															
C	RW	RGN1WA																																	
			Disable	0																															
			Enable	1																															
D	RW	RGN1RA																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
			Disable	0																								Disable read access watch in this region							
			Enable	1																								Enable read access watch in this region							
E	RW	RGN2WA																										Enable/disable write access watch in region[2]							
			Disable	0																								Disable write access watch in this region							
			Enable	1																								Enable write access watch in this region							
F	RW	RGN2RA																										Enable/disable read access watch in region[2]							
			Disable	0																								Disable read access watch in this region							
			Enable	1																								Enable read access watch in this region							
G	RW	RGN3WA																										Enable/disable write access watch in region[3]							
			Disable	0																								Disable write access watch in this region							
			Enable	1																								Enable write access watch in this region							
H	RW	RGN3RA																										Enable/disable read access watch in region[3]							
			Disable	0																								Disable read access watch in this region							
			Enable	1																								Enable read access watch in this region							
I	RW	PRGN0WA																										Enable/disable write access watch in PREGION[0]							
			Disable	0																								Disable write access watch in this PREGION							
			Enable	1																								Enable write access watch in this PREGION							
J	RW	PRGN0RA																										Enable/disable read access watch in PREGION[0]							
			Disable	0																								Disable read access watch in this PREGION							
			Enable	1																								Enable read access watch in this PREGION							
K	RW	PRGN1WA																										Enable/disable write access watch in PREGION[1]							
			Disable	0																								Disable write access watch in this PREGION							
			Enable	1																								Enable write access watch in this PREGION							
L	RW	PRGN1RA																										Enable/disable read access watch in PREGION[1]							
			Disable	0																								Disable read access watch in this PREGION							
			Enable	1																								Enable read access watch in this PREGION							

45.1.12 REGIONENSET

Address offset: 0x514

Enable regions watch

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	RW	RGN0WA																										Enable write access watch in region[0]							
			Set	1																								Enable write access watch in this region							
			Disabled	0																								Write access watch in this region is disabled							
			Enabled	1																								Write access watch in this region is enabled							
B	RW	RGN0RA																										Enable read access watch in region[0]							
			Set	1																								Enable read access watch in this region							
			Disabled	0																								Read access watch in this region is disabled							
			Enabled	1																								Read access watch in this region is enabled							
C	RW	RGN1WA																										Enable write access watch in region[1]							
			Set	1																								Enable write access watch in this region							
			Disabled	0																								Write access watch in this region is disabled							
			Enabled	1																								Write access watch in this region is enabled							
D	RW	RGN1RA																										Enable read access watch in region[1]							
			Set	1																								Enable read access watch in this region							
			Disabled	0																								Read access watch in this region is disabled							
			Enabled	1																								Read access watch in this region is enabled							
E	RW	RGN2WA																										Enable write access watch in region[2]							
			Set	1																								Enable write access watch in this region							
			Disabled	0																								Write access watch in this region is disabled							

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
			Enabled	1																								Write access watch in this region is enabled							
F	RW	RGN2RA																										Enable read access watch in region[2]							
			Set	1																								Enable read access watch in this region							
			Disabled	0																								Read access watch in this region is disabled							
			Enabled	1																								Read access watch in this region is enabled							
G	RW	RGN3WA																										Enable write access watch in region[3]							
			Set	1																								Enable write access watch in this region							
			Disabled	0																								Write access watch in this region is disabled							
			Enabled	1																								Write access watch in this region is enabled							
H	RW	RGN3RA																										Enable read access watch in region[3]							
			Set	1																								Enable read access watch in this region							
			Disabled	0																								Read access watch in this region is disabled							
			Enabled	1																								Read access watch in this region is enabled							
I	RW	PRGN0WA																										Enable write access watch in PREGION[0]							
			Set	1																								Enable write access watch in this PREGION							
			Disabled	0																								Write access watch in this PREGION is disabled							
			Enabled	1																								Write access watch in this PREGION is enabled							
J	RW	PRGN0RA																										Enable read access watch in PREGION[0]							
			Set	1																								Enable read access watch in this PREGION							
			Disabled	0																								Read access watch in this PREGION is disabled							
			Enabled	1																								Read access watch in this PREGION is enabled							
K	RW	PRGN1WA																										Enable write access watch in PREGION[1]							
			Set	1																								Enable write access watch in this PREGION							
			Disabled	0																								Write access watch in this PREGION is disabled							
			Enabled	1																								Write access watch in this PREGION is enabled							
L	RW	PRGN1RA																										Enable read access watch in PREGION[1]							
			Set	1																								Enable read access watch in this PREGION							
			Disabled	0																								Read access watch in this PREGION is disabled							
			Enabled	1																								Read access watch in this PREGION is enabled							

45.1.13 REGIONENCLR

Address offset: 0x518

Disable regions watch

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	RW	RGN0WA																										Disable write access watch in region[0]							
			Clear	1																								Disable write access watch in this region							
			Disabled	0																								Write access watch in this region is disabled							
			Enabled	1																								Write access watch in this region is enabled							
B	RW	RGN0RA																										Disable read access watch in region[0]							
			Clear	1																								Disable read access watch in this region							
			Disabled	0																								Read access watch in this region is disabled							
			Enabled	1																								Read access watch in this region is enabled							
C	RW	RGN1WA																										Disable write access watch in region[1]							
			Clear	1																								Disable write access watch in this region							
			Disabled	0																								Write access watch in this region is disabled							
			Enabled	1																								Write access watch in this region is enabled							
D	RW	RGN1RA																										Disable read access watch in region[1]							
			Clear	1																								Disable read access watch in this region							
			Disabled	0																								Read access watch in this region is disabled							
			Enabled	1																								Read access watch in this region is enabled							

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
E	RW	RGN2WA																										Disable write access watch in region[2]							
			Clear	1																								Disable write access watch in this region							
			Disabled	0																								Write access watch in this region is disabled							
			Enabled	1																								Write access watch in this region is enabled							
F	RW	RGN2RA																										Disable read access watch in region[2]							
			Clear	1																								Disable read access watch in this region							
			Disabled	0																								Read access watch in this region is disabled							
			Enabled	1																								Read access watch in this region is enabled							
G	RW	RGN3WA																										Disable write access watch in region[3]							
			Clear	1																								Disable write access watch in this region							
			Disabled	0																								Write access watch in this region is disabled							
			Enabled	1																								Write access watch in this region is enabled							
H	RW	RGN3RA																										Disable read access watch in region[3]							
			Clear	1																								Disable read access watch in this region							
			Disabled	0																								Read access watch in this region is disabled							
			Enabled	1																								Read access watch in this region is enabled							
I	RW	PRGN0WA																										Disable write access watch in PREGION[0]							
			Clear	1																								Disable write access watch in this PREGION							
			Disabled	0																								Write access watch in this PREGION is disabled							
			Enabled	1																								Write access watch in this PREGION is enabled							
J	RW	PRGN0RA																										Disable read access watch in PREGION[0]							
			Clear	1																								Disable read access watch in this PREGION							
			Disabled	0																								Read access watch in this PREGION is disabled							
			Enabled	1																								Read access watch in this PREGION is enabled							
K	RW	PRGN1WA																										Disable write access watch in PREGION[1]							
			Clear	1																								Disable write access watch in this PREGION							
			Disabled	0																								Write access watch in this PREGION is disabled							
			Enabled	1																								Write access watch in this PREGION is enabled							
L	RW	PRGN1RA																										Disable read access watch in PREGION[1]							
			Clear	1																								Disable read access watch in this PREGION							
			Disabled	0																								Read access watch in this PREGION is disabled							
			Enabled	1																								Read access watch in this PREGION is enabled							

45.1.14 REGION[0].START

Address offset: 0x600

Start address for region 0

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	RW	START																										Start address for region							

45.1.15 REGION[0].END

Address offset: 0x604

End address of region 0

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	RW	END																										End address of region.							

45.1.16 REGION[1].START

Address offset: 0x610

Start address for region 1

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	START			Start address for region

45.1.17 REGION[1].END

Address offset: 0x614

End address of region 1

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	END			End address of region.

45.1.18 REGION[2].START

Address offset: 0x620

Start address for region 2

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	START			Start address for region

45.1.19 REGION[2].END

Address offset: 0x624

End address of region 2

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	END			End address of region.

45.1.20 REGION[3].START

Address offset: 0x630

Start address for region 3

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	START			Start address for region

45.1.21 REGION[3].END

Address offset: 0x634

End address of region 3

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	END			End address of region.

45.1.22 PREGION[0].START

Address offset: 0x6C0

Reserved for future use

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	R	START			Reserved for future use

45.1.23 PREGION[0].END

Address offset: 0x6C4

Reserved for future use

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	R	END			Reserved for future use

45.1.24 PREGION[0].SUBS

Address offset: 0x6C8

Subregions of region 0

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id	f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	SR0			Include or exclude subregion 0 in region
			Exclude	0	Exclude
			Include	1	Include
B	RW	SR1			Include or exclude subregion 1 in region
			Exclude	0	Exclude
			Include	1	Include
C	RW	SR2			Include or exclude subregion 2 in region
			Exclude	0	Exclude
			Include	1	Include
D	RW	SR3			Include or exclude subregion 3 in region
			Exclude	0	Exclude
			Include	1	Include
E	RW	SR4			Include or exclude subregion 4 in region
			Exclude	0	Exclude
			Include	1	Include
F	RW	SR5			Include or exclude subregion 5 in region
			Exclude	0	Exclude
			Include	1	Include
G	RW	SR6			Include or exclude subregion 6 in region
			Exclude	0	Exclude
			Include	1	Include

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
H	RW	SR7																																	
			Exclude	0																															
			Include	1																															
I	RW	SR8																																	
			Exclude	0																															
			Include	1																															
J	RW	SR9																																	
			Exclude	0																															
			Include	1																															
K	RW	SR10																																	
			Exclude	0																															
			Include	1																															
L	RW	SR11																																	
			Exclude	0																															
			Include	1																															
M	RW	SR12																																	
			Exclude	0																															
			Include	1																															
N	RW	SR13																																	
			Exclude	0																															
			Include	1																															
O	RW	SR14																																	
			Exclude	0																															
			Include	1																															
P	RW	SR15																																	
			Exclude	0																															
			Include	1																															
Q	RW	SR16																																	
			Exclude	0																															
			Include	1																															
R	RW	SR17																																	
			Exclude	0																															
			Include	1																															
S	RW	SR18																																	
			Exclude	0																															
			Include	1																															
T	RW	SR19																																	
			Exclude	0																															
			Include	1																															
U	RW	SR20																																	
			Exclude	0																															
			Include	1																															
V	RW	SR21																																	
			Exclude	0																															
			Include	1																															
W	RW	SR22																																	
			Exclude	0																															
			Include	1																															
X	RW	SR23																																	
			Exclude	0																															
			Include	1																															
Y	RW	SR24																																	
			Exclude	0																															
			Include	1																															
Z	RW	SR25																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Exclude	0																															
			Include	1																															
a	RW	SR26																																	
			Exclude	0																															
			Include	1																															
b	RW	SR27																																	
			Exclude	0																															
			Include	1																															
c	RW	SR28																																	
			Exclude	0																															
			Include	1																															
d	RW	SR29																																	
			Exclude	0																															
			Include	1																															
e	RW	SR30																																	
			Exclude	0																															
			Include	1																															
f	RW	SR31																																	
			Exclude	0																															
			Include	1																															

45.1.25 PREGION[1].START

Address offset: 0x6D0

Reserved for future use

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	START																																	

Reserved for future use

45.1.26 PREGION[1].END

Address offset: 0x6D4

Reserved for future use

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	END																																	

45.1.27 PREGION[1].SUBS

Address offset: 0x6D8

Subregions of region 1

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	SR0																																	
			Exclude	0																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Include	1																															
B	RW	SR1																																	
			Exclude	0																															
			Include	1																															
C	RW	SR2																																	
			Exclude	0																															
			Include	1																															
D	RW	SR3																																	
			Exclude	0																															
			Include	1																															
E	RW	SR4																																	
			Exclude	0																															
			Include	1																															
F	RW	SR5																																	
			Exclude	0																															
			Include	1																															
G	RW	SR6																																	
			Exclude	0																															
			Include	1																															
H	RW	SR7																																	
			Exclude	0																															
			Include	1																															
I	RW	SR8																																	
			Exclude	0																															
			Include	1																															
J	RW	SR9																																	
			Exclude	0																															
			Include	1																															
K	RW	SR10																																	
			Exclude	0																															
			Include	1																															
L	RW	SR11																																	
			Exclude	0																															
			Include	1																															
M	RW	SR12																																	
			Exclude	0																															
			Include	1																															
N	RW	SR13																																	
			Exclude	0																															
			Include	1																															
O	RW	SR14																																	
			Exclude	0																															
			Include	1																															
P	RW	SR15																																	
			Exclude	0																															
			Include	1																															
Q	RW	SR16																																	
			Exclude	0																															
			Include	1																															
R	RW	SR17																																	
			Exclude	0																															
			Include	1																															
S	RW	SR18																																	
			Exclude	0																															
			Include	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
T	RW	SR19																																	
			Exclude	0																															
			Include	1																															
U	RW	SR20																																	
			Exclude	0																															
			Include	1																															
V	RW	SR21																																	
			Exclude	0																															
			Include	1																															
W	RW	SR22																																	
			Exclude	0																															
			Include	1																															
X	RW	SR23																																	
			Exclude	0																															
			Include	1																															
Y	RW	SR24																																	
			Exclude	0																															
			Include	1																															
Z	RW	SR25																																	
			Exclude	0																															
			Include	1																															
a	RW	SR26																																	
			Exclude	0																															
			Include	1																															
b	RW	SR27																																	
			Exclude	0																															
			Include	1																															
c	RW	SR28																																	
			Exclude	0																															
			Include	1																															
d	RW	SR29																																	
			Exclude	0																															
			Include	1																															
e	RW	SR30																																	
			Exclude	0																															
			Include	1																															
f	RW	SR31																																	
			Exclude	0																															
			Include	1																															

46 EGU — Event generator unit

The Event generator unit (EGU) provides support for inter-layer signaling. This means support for atomic triggering of both CPU execution and hardware tasks from both firmware (by CPU) and hardware (by PPI). This feature can, for instance, be used for triggering CPU execution at a lower priority execution from a higher priority execution, or to handle a peripheral's ISR execution at a lower priority for some of its events. However, triggering any priority from any priority is possible.

Listed here are the main EGU features:

- Enables SW triggering of interrupts
- 6 EGU instances – separate interrupt vectors
- Up to 16 separate event flags per interrupt for multiplexing

The EGU implements a set of tasks which can individually be triggered to generate the corresponding event, i.e., the corresponding event for TASKS_TRIGGER[n] is EVENTS_TRIGGERED[n].

Table 114: EGU configuration

EGU instance	Number of event flags
0-5	16

46.1 Registers

Table 115: Instances

Base address	Peripheral	Instance	Description	Configuration
0x40014000	EGU	EGU0	Event Generator Unit 0	
0x40015000	EGU	EGU1	Event Generator Unit 1	
0x40016000	EGU	EGU2	Event Generator Unit 2	
0x40017000	EGU	EGU3	Event Generator Unit 3	
0x40018000	EGU	EGU4	Event Generator Unit 4	
0x40019000	EGU	EGU5	Event Generator Unit 5	

Table 116: Register Overview

Register	Offset	Description
TASKS_TRIGGER[0]	0x000	Trigger 0 for triggering the corresponding TRIGGERED[0] event
TASKS_TRIGGER[1]	0x004	Trigger 1 for triggering the corresponding TRIGGERED[1] event
TASKS_TRIGGER[2]	0x008	Trigger 2 for triggering the corresponding TRIGGERED[2] event
TASKS_TRIGGER[3]	0x00C	Trigger 3 for triggering the corresponding TRIGGERED[3] event
TASKS_TRIGGER[4]	0x010	Trigger 4 for triggering the corresponding TRIGGERED[4] event
TASKS_TRIGGER[5]	0x014	Trigger 5 for triggering the corresponding TRIGGERED[5] event
TASKS_TRIGGER[6]	0x018	Trigger 6 for triggering the corresponding TRIGGERED[6] event
TASKS_TRIGGER[7]	0x01C	Trigger 7 for triggering the corresponding TRIGGERED[7] event
TASKS_TRIGGER[8]	0x020	Trigger 8 for triggering the corresponding TRIGGERED[8] event
TASKS_TRIGGER[9]	0x024	Trigger 9 for triggering the corresponding TRIGGERED[9] event
TASKS_TRIGGER[10]	0x028	Trigger 10 for triggering the corresponding TRIGGERED[10] event
TASKS_TRIGGER[11]	0x02C	Trigger 11 for triggering the corresponding TRIGGERED[11] event
TASKS_TRIGGER[12]	0x030	Trigger 12 for triggering the corresponding TRIGGERED[12] event
TASKS_TRIGGER[13]	0x034	Trigger 13 for triggering the corresponding TRIGGERED[13] event
TASKS_TRIGGER[14]	0x038	Trigger 14 for triggering the corresponding TRIGGERED[14] event
TASKS_TRIGGER[15]	0x03C	Trigger 15 for triggering the corresponding TRIGGERED[15] event
EVENTS_TRIGGERED[0]	0x100	Event number 0 generated by triggering the corresponding TRIGGER[0] task
EVENTS_TRIGGERED[1]	0x104	Event number 1 generated by triggering the corresponding TRIGGER[1] task
EVENTS_TRIGGERED[2]	0x108	Event number 2 generated by triggering the corresponding TRIGGER[2] task
EVENTS_TRIGGERED[3]	0x10C	Event number 3 generated by triggering the corresponding TRIGGER[3] task
EVENTS_TRIGGERED[4]	0x110	Event number 4 generated by triggering the corresponding TRIGGER[4] task

Register	Offset	Description
EVENTS_TRIGGERED[5]	0x114	Event number 5 generated by triggering the corresponding TRIGGER[5] task
EVENTS_TRIGGERED[6]	0x118	Event number 6 generated by triggering the corresponding TRIGGER[6] task
EVENTS_TRIGGERED[7]	0x11C	Event number 7 generated by triggering the corresponding TRIGGER[7] task
EVENTS_TRIGGERED[8]	0x120	Event number 8 generated by triggering the corresponding TRIGGER[8] task
EVENTS_TRIGGERED[9]	0x124	Event number 9 generated by triggering the corresponding TRIGGER[9] task
EVENTS_TRIGGERED[10]	0x128	Event number 10 generated by triggering the corresponding TRIGGER[10] task
EVENTS_TRIGGERED[11]	0x12C	Event number 11 generated by triggering the corresponding TRIGGER[11] task
EVENTS_TRIGGERED[12]	0x130	Event number 12 generated by triggering the corresponding TRIGGER[12] task
EVENTS_TRIGGERED[13]	0x134	Event number 13 generated by triggering the corresponding TRIGGER[13] task
EVENTS_TRIGGERED[14]	0x138	Event number 14 generated by triggering the corresponding TRIGGER[14] task
EVENTS_TRIGGERED[15]	0x13C	Event number 15 generated by triggering the corresponding TRIGGER[15] task
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt

46.1.1 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	TRIGGERED0			Enable or disable interrupt for TRIGGERED[0] event See *EVENTS_TRIGGERED[0]*
			Disabled	0	Disable
			Enabled	1	Enable
B	RW	TRIGGERED1			Enable or disable interrupt for TRIGGERED[1] event See *EVENTS_TRIGGERED[1]*
			Disabled	0	Disable
			Enabled	1	Enable
C	RW	TRIGGERED2			Enable or disable interrupt for TRIGGERED[2] event See *EVENTS_TRIGGERED[2]*
			Disabled	0	Disable
			Enabled	1	Enable
D	RW	TRIGGERED3			Enable or disable interrupt for TRIGGERED[3] event See *EVENTS_TRIGGERED[3]*
			Disabled	0	Disable
			Enabled	1	Enable
E	RW	TRIGGERED4			Enable or disable interrupt for TRIGGERED[4] event See *EVENTS_TRIGGERED[4]*
			Disabled	0	Disable
			Enabled	1	Enable
F	RW	TRIGGERED5			Enable or disable interrupt for TRIGGERED[5] event See *EVENTS_TRIGGERED[5]*
			Disabled	0	Disable
			Enabled	1	Enable
G	RW	TRIGGERED6			Enable or disable interrupt for TRIGGERED[6] event See *EVENTS_TRIGGERED[6]*
			Disabled	0	Disable
			Enabled	1	Enable
H	RW	TRIGGERED7			Enable or disable interrupt for TRIGGERED[7] event See *EVENTS_TRIGGERED[7]*
			Disabled	0	Disable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
			Enabled	1	Enable
I	RW	TRIGGERED8			Enable or disable interrupt for TRIGGERED[8] event See *EVENTS_TRIGGERED[8]*
			Disabled	0	Disable
			Enabled	1	Enable
J	RW	TRIGGERED9			Enable or disable interrupt for TRIGGERED[9] event See *EVENTS_TRIGGERED[9]*
			Disabled	0	Disable
			Enabled	1	Enable
K	RW	TRIGGERED10			Enable or disable interrupt for TRIGGERED[10] event See *EVENTS_TRIGGERED[10]*
			Disabled	0	Disable
			Enabled	1	Enable
L	RW	TRIGGERED11			Enable or disable interrupt for TRIGGERED[11] event See *EVENTS_TRIGGERED[11]*
			Disabled	0	Disable
			Enabled	1	Enable
M	RW	TRIGGERED12			Enable or disable interrupt for TRIGGERED[12] event See *EVENTS_TRIGGERED[12]*
			Disabled	0	Disable
			Enabled	1	Enable
N	RW	TRIGGERED13			Enable or disable interrupt for TRIGGERED[13] event See *EVENTS_TRIGGERED[13]*
			Disabled	0	Disable
			Enabled	1	Enable
O	RW	TRIGGERED14			Enable or disable interrupt for TRIGGERED[14] event See *EVENTS_TRIGGERED[14]*
			Disabled	0	Disable
			Enabled	1	Enable
P	RW	TRIGGERED15			Enable or disable interrupt for TRIGGERED[15] event See *EVENTS_TRIGGERED[15]*
			Disabled	0	Disable
			Enabled	1	Enable

46.1.2 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**
A	RW	TRIGGERED0			Write '1' to Enable interrupt for TRIGGERED[0] event See *EVENTS_TRIGGERED[0]*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	TRIGGERED1			Write '1' to Enable interrupt for TRIGGERED[1] event See *EVENTS_TRIGGERED[1]*
			Set	1	Enable

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																				P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Disabled	0																															
			Enabled	1																															
C	RW	TRIGGERED2																																	
				See *EVENTS_TRIGGERED[2]*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	TRIGGERED3																																	
				See *EVENTS_TRIGGERED[3]*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	TRIGGERED4																																	
				See *EVENTS_TRIGGERED[4]*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
F	RW	TRIGGERED5																																	
				See *EVENTS_TRIGGERED[5]*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
G	RW	TRIGGERED6																																	
				See *EVENTS_TRIGGERED[6]*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
H	RW	TRIGGERED7																																	
				See *EVENTS_TRIGGERED[7]*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
I	RW	TRIGGERED8																																	
				See *EVENTS_TRIGGERED[8]*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
J	RW	TRIGGERED9																																	
				See *EVENTS_TRIGGERED[9]*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
K	RW	TRIGGERED10																																	
				See *EVENTS_TRIGGERED[10]*																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
L	RW	TRIGGERED11																																	
				See *EVENTS_TRIGGERED[11]*																																	
			Set	1																															
			Disabled	0																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																				P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Enabled	1																															
M	RW	TRIGGERED12																																	
				See *EVENTS_TRIGGERED[12]*																																
			Set	1																															
			Disabled	0																															
			Enabled	1																															
N	RW	TRIGGERED13																																	
				See *EVENTS_TRIGGERED[13]*																																
			Set	1																															
			Disabled	0																															
			Enabled	1																															
O	RW	TRIGGERED14																																	
				See *EVENTS_TRIGGERED[14]*																																
			Set	1																															
			Disabled	0																															
			Enabled	1																															
P	RW	TRIGGERED15																																	
				See *EVENTS_TRIGGERED[15]*																																
			Set	1																															
			Disabled	0																															
			Enabled	1																															

46.1.3 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																				P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	TRIGGERED0																																	
				See *EVENTS_TRIGGERED[0]*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	TRIGGERED1																																	
				See *EVENTS_TRIGGERED[1]*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	TRIGGERED2																																	
				See *EVENTS_TRIGGERED[2]*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	TRIGGERED3																																	
				See *EVENTS_TRIGGERED[3]*																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	TRIGGERED4																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																				P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**								**Value**																**Description**							
			Clear								1																Disable							
			Disabled								0																Read: Disabled							
			Enabled								1																Read: Enabled							
P	RW	TRIGGERED15																									Write '1' to Disable interrupt for TRIGGERED[15] event See *EVENTS_TRIGGERED[15]*							
			Clear								1																Disable							
			Disabled								0																Read: Disabled							
			Enabled								1																Read: Enabled							

46.2 Electrical specification

46.2.1 EGU Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
$t_{EGU,EVT}$	Latency between setting an EGU event flag and the system setting an interrupt		1		cycles

47 PWM — Pulse width modulation

The PWM module enables the generation of pulse width modulated signals on GPIO. The module implements an up or up-and-down counter with four PWM channels that drive assigned GPIOs.

Three PWM modules can provide up to 12 PWM channels with individual frequency control in groups of up to four channels. Furthermore, a built-in decoder and EasyDMA capabilities make it possible to manipulate the PWM duty cycles without CPU intervention. Arbitrary duty-cycle sequences are read from Data RAM and can be chained to implement ping-pong buffering or repeated into complex loops.

Listed here are the main features of one PWM module:

- Fixed PWM base frequency with programmable clock divider
- Up to four PWM channels with individual polarity and duty-cycle values
- Edge or center-aligned pulses across PWM channels
- Multiple duty-cycle arrays (sequences) defined in Data RAM
- Autonomous and glitch-free update of duty cycle values directly from memory through EasyDMA
- Change of polarity, duty-cycle, and base frequency possibly on every PWM period
- Data RAM sequences can be repeated or connected into loops

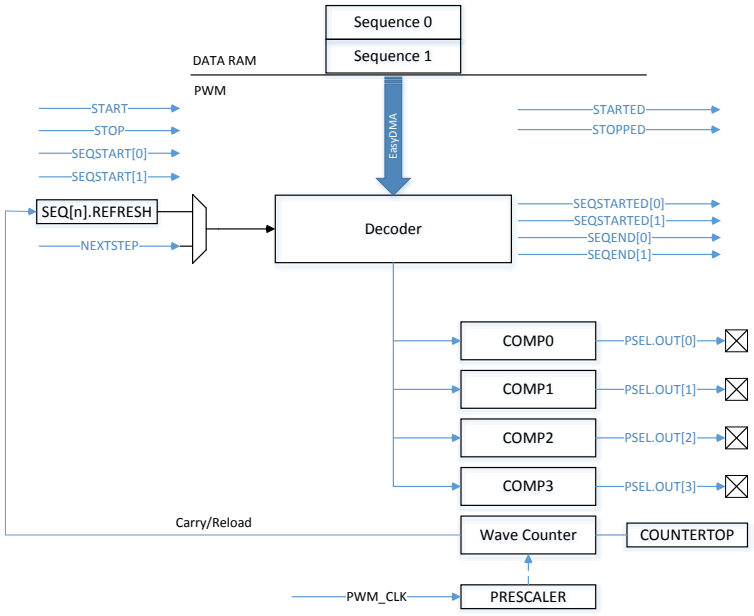


Figure 141: PWM Module

47.1 Wave counter

The wave counter is responsible for generating the pulses at a duty-cycle that depends on the compare values, and at a frequency that depends on COUNTERTOP.

There is one common 15-bit counter with four compare channels. Thus, all four channels will share the same period (PWM frequency), but can have individual duty-cycle and polarity. The polarity is set by the value read from RAM (see *Figure 144: Decoder memory access modes* on page 498), while the MODE register controls if the counter counts up, or up and down. The timer top value is controlled by the COUNTERTOP register. This register value in conjunction with the selected PRESCALER of the PWM_CLK will result in a given PWM period. A COUNTERTOP value smaller than the compare setting will result in a state where no PWM edges are generated. Respectively, OUT[n] is held high, given that the polarity is set to FallingEdge. All the compare registers are internal and can only be configured through the decoder presented later.

COUNTERTOP can be safely written at any time. It will get sampled following a START task. If DECODER.LOAD is anything else than WaveForm, it will also get sampled following a STARTSEQ[n] task,

and when loading a new value from RAM during a sequence playback. If DECODER.LOAD=WaveForm, the register value is ignored, and taken from RAM instead (see *Decoder with EasyDMA* on page 498 below).

Figure 142: PWM up counter example - FallingEdge polarity on page 496 shows the counter operating in up (MODE=PWM_MODE_Up) mode with three PWM channels with the same frequency but different duty cycle. The counter is automatically reset to zero when COUNTERTOP is reached and OUT[n] will invert. OUT[n] is held low if the compare value is 0 and held high respectively if set to COUNTERTOP given that the polarity is set to FallingEdge. Running in up counter mode will result in pulse widths that are edge-aligned. See the code example below:

```
uint16_t pwm_seq[4] = {PWM_CH0_DUTY, PWM_CH1_DUTY, PWM_CH2_DUTY,
 PWM_CH3_DUTY};
NRF_PWM0->PSEL.OUT[0] = (first_pin << PWM_PSEL_OUT_PIN_Pos) |
                        (PWM_PSEL_OUT_CONNECT_Connected <<
                                                PWM_PSEL_OUT_CONNECT_Pos);
NRF_PWM0->PSEL.OUT[1] = (second_pin << PWM_PSEL_OUT_PIN_Pos) |
                        (PWM_PSEL_OUT_CONNECT_Connected <<
                                                PWM_PSEL_OUT_CONNECT_Pos);
NRF_PWM0->ENABLE      = (PWM_ENABLE_ENABLE_Enabled << PWM_ENABLE_ENABLE_Pos);
NRF_PWM0->MODE        = (PWM_MODE_UPDOWN_Up << PWM_MODE_UPDOWN_Pos);
NRF_PWM0->PRESCALER   = (PWM_PRESCALER_PRESCALER_DIV_1 <<
                                                PWM_PRESCALER_PRESCALER_Pos);
NRF_PWM0->COUNTERTOP  = (16000 << PWM_COUNTERTOP_COUNTERTOP_Pos); //1 msec
NRF_PWM0->LOOP        = (PWM_LOOP_CNT_Disabled << PWM_LOOP_CNT_Pos);
NRF_PWM0->DECODER   = (PWM_DECODER_LOAD_Individual << PWM_DECODER_LOAD_Pos) |
                      (PWM_DECODER_MODE_RefreshCount << PWM_DECODER_MODE_Pos);
NRF_PWM0->SEQ[0].PTR  = ((uint32_t)(pwm_seq) << PWM_SEQ_PTR_PTR_Pos);
NRF_PWM0->SEQ[0].CNT  = ((sizeof(pwm_seq) / sizeof(uint16_t)) <<
                                                PWM_SEQ_CNT_CNT_Pos);
NRF_PWM0->SEQ[0].REFRESH  = 0;
NRF_PWM0->SEQ[0].ENDDELAY = 0;
NRF_PWM0->TASKS_SEQSTART[0] = 1;
```

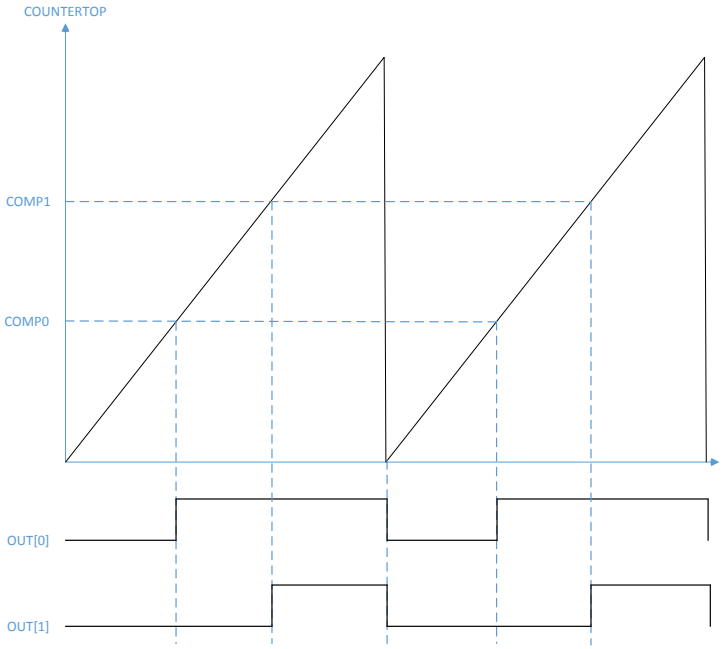


Figure 142: PWM up counter example - FallingEdge polarity

In up counting mode, the following formula can be used to compute PWM period and step size:

PWM period: $T_{PWM(Up)} = T_{PWM_CLK} * COUNTERTOP$

Step width/Resolution: $T_{steps} = T_{PWM_CLK}$

Figure 143: PWM up-and-down counter example on page 497 shows the counter operating in up and down mode with (MODE=PWM_MODE_UpAndDown) two PWM channels with the same frequency but different duty cycle and output polarity. The counter starts decrementing to zero when COUNTERTOP is reached and will invert the OUT[n] when compare value is hit for the second time. This results in a set of pulses that are center- aligned.

```
uint16_t pwm_seq[4] = {PWM_CH0_DUTY, PWM_CH1_DUTY, PWM_CH2_DUTY,
 PWM_CH3_DUTY};
NRF_PWM0->PSEL.OUT[0] = (first_pin << PWM_PSEL_OUT_PIN_Pos) |
                        (PWM_PSEL_OUT_CONNECT_Connected <<
                                                PWM_PSEL_OUT_CONNECT_Pos);
NRF_PWM0->PSEL.OUT[1] = (second_pin << PWM_PSEL_OUT_PIN_Pos) |
                        (PWM_PSEL_OUT_CONNECT_Connected <<
                                                PWM_PSEL_OUT_CONNECT_Pos);
NRF_PWM0->ENABLE      = (PWM_ENABLE_ENABLE_Enabled << PWM_ENABLE_ENABLE_Pos);
NRF_PWM0->MODE        = (PWM_MODE_UPDOWN_UpAndDown << PWM_MODE_UPDOWN_Pos);
NRF_PWM0->PRESCALER   = (PWM_PRESCALER_PRESCALER_DIV_1 <<
                                                PWM_PRESCALER_PRESCALER_Pos);
NRF_PWM0->COUNTERTOP  = (16000 << PWM_COUNTERTOP_COUNTERTOP_Pos); //1 msec
NRF_PWM0->LOOP        = (PWM_LOOP_CNT_Disabled << PWM_LOOP_CNT_Pos);
NRF_PWM0->DECODER   = (PWM_DECODER_LOAD_Individual << PWM_DECODER_LOAD_Pos) |
                      (PWM_DECODER_MODE_RefreshCount << PWM_DECODER_MODE_Pos);
NRF_PWM0->SEQ[0].PTR  = ((uint32_t)(pwm_seq) << PWM_SEQ_PTR_PTR_Pos);
NRF_PWM0->SEQ[0].CNT  = ((sizeof(pwm_seq) / sizeof(uint16_t)) <<
                                                PWM_SEQ_CNT_CNT_Pos);
NRF_PWM0->SEQ[0].REFRESH  = 0;
NRF_PWM0->SEQ[0].ENDDELAY = 0;
NRF_PWM0->TASKS_SEQSTART[0] = 1;
```

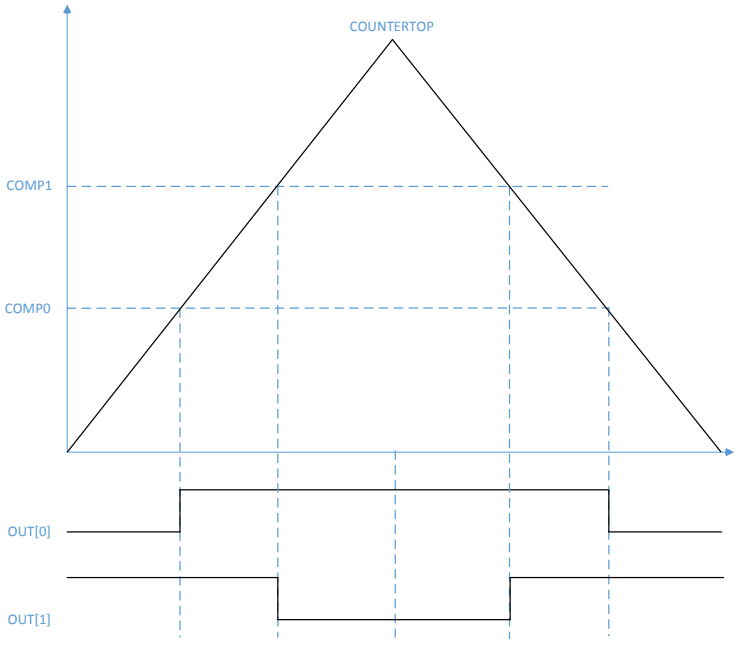


Figure 143: PWM up-and-down counter example

In up-and-down counting modes, the following formula can be used to compute PWM period and step size:
$T_{PWM(Up\ And\ Down)} = T_{PWM_CLK} * 2 * COUNTERTOP$

Step width/Resolution: $T_{steps} = T_{PWM_CLK} * 2$

47.2 Decoder with EasyDMA

The decoder uses EasyDMA to take PWM parameters stored in Data RAM by ways of EasyDMA and updates the internal compare registers of the wave counter based on the mode of operation.

The mentioned PWM parameters are organized into a sequence containing at least one half word (16 bit). Its most significant bit[15] denotes the polarity of the OUT[n] while bit[14:0] is the 15-bit compare value. See below for further details of these RAM defined registers.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																				B	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																														
A	RW	COMPARE			Duty cycle setting - value loaded to internal compare register																														
B	RW	POLARITY			Edge polarity of GPIO.																														
			RisingEdge	0	First edge within the PWM period is rising																														
			FallingEdge	1	First edge within the PWM period is falling																														

The DECODER register controls how the RAM content is interpreted and loaded to the internal compare registers. The LOAD field can be used to control if the RAM values are loaded to all compare channels - or alternatively to update a group or all channels with individual values. *Figure 144: Decoder memory access modes* on page 498 illustrates how the parameters stored in RAM are organized and routed to the various compare channels in the different modes.

A special mode of operation is available when DECODER.LOAD is set to WaveForm. In this mode, up to three PWM channels can be enabled - OUT[0] to OUT[2]. In RAM, four values are loaded at a time: the first, second and third location are used to load the values, and the fourth RAM location is used to load the COUNTERTOP register. This way one can have up to three PWM channels with a frequency base that changes on a per PWM period basis. This mode of operation is useful for arbitrary wave form generation in applications such as LED lighting.

The register SEQ[n].REFRESH=N (one per sequence n=0 or 1) will instruct a new RAM stored pulse width value on every (N+1)th PWM period. Setting the register to zero will result in a new duty cycle update every PWM period as long as the minimum PWM period is observed.

Note that registers SEQ[n].REFRESH and SEQ[n].ENDDELAY are ignored when DECODER.MODE=NextStep . The next value is loaded upon receiving every NEXTSTEP task.

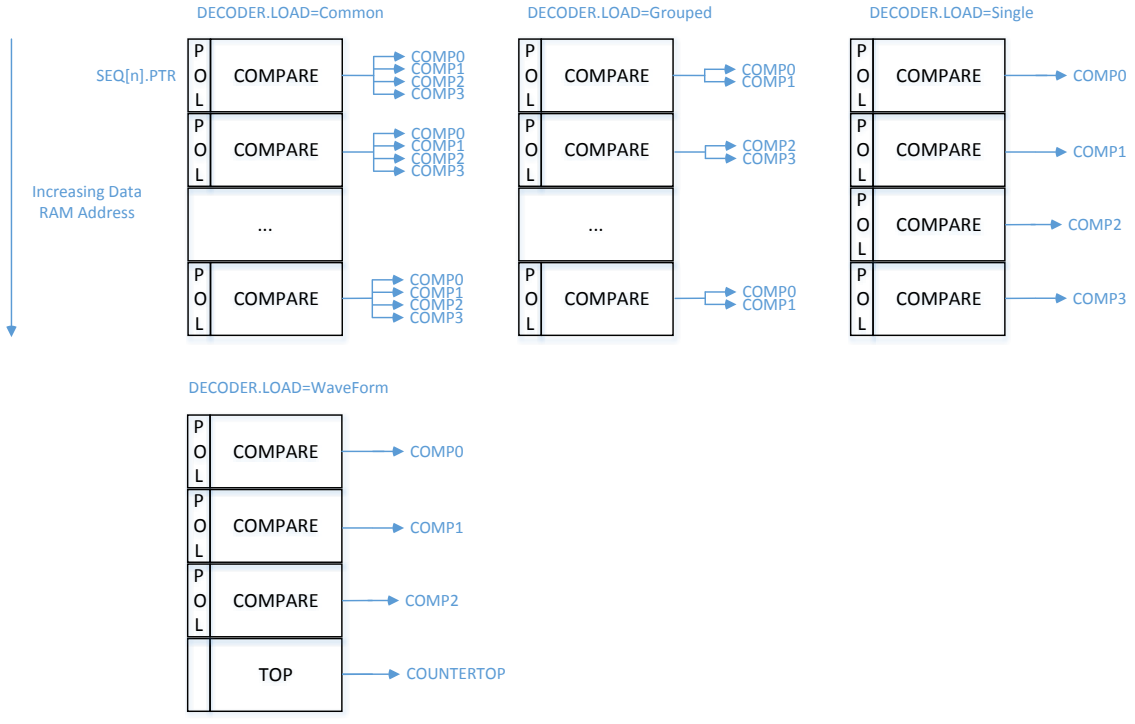


Figure 144: Decoder memory access modes

SEQ[n].PTR is the pointer used to fetch COMPARE values from RAM. If the SEQ[n].PTR is not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See *Memory* on page 23 for more information about the different memory regions.

After the SEQ[n].PTR is set to the desired RAM location, the SEQ[n].CNT register must be set to the number of 16-bit half words in the sequence. It is important to observe that the Grouped and Single modes require one half word per group or one half word per channel respectively, and thus increases RAM size occupation. If PWM generation was not running yet at that point, sending the SEQSTART[n] task will load the first value from RAM, then start the PWM generation. A SEQSTARTED[n] event is generated as soon as the EasyDMA has read the first PWM parameter from RAM and the wave counter has started executing it. When LOOP.CNT=0, sequence n=0 or 1 is played back once. After the last value in the sequence has been loaded and started executing, a SEQEND[n] event is generated. The PWM generation will then continue with the last loaded value. See *Figure 145: Simple sequence example* on page 500 for an example of such simple playback.

To completely stop the PWM generation and force the associated pins to a defined state, a STOP task can be fired at any time. A STOPPED event is generated when the PWM generation has stopped at the end of currently running PWM period, and the pins go into their idle state as defined in GPIO->OUT. PWM generation can then only be restarted through a SEQSTART[n] task. SEQSTART[n] will resume PWM generation after having loaded the first value from the RAM buffer defined in the SEQ[n].PTR register.

The table below provides indication of when specific registers get sampled by the hardware. Care should be taken when updating these registers to avoid values to be applied earlier than expected.

Table 117: When to safely update PWM registers

Register	Taken into account by hardware	Recommended (safe) update
SEQ[n].PTR	When sending the SEQSTART[n] task	After having received the SEQSTARTED[n] event
SEQ[n].CNT	When sending the SEQSTART[n] task	After having received the SEQSTARTED[n] event
SEQ[0].ENDDELAY	When sending the SEQSTART[0] task	Before starting sequence [0] through a SEQSTART[0] task
	Every time a new value from sequence [0] has been loaded from RAM and gets applied to the Wave Counter (indicated by the PWMPERIODEND event)	When no more value from sequence [0] gets loaded from RAM (indicated by the SEQEND[0] event)
		At any time during sequence [1] (which starts when the SEQSTARTED[1] event is fired)
SEQ[1].ENDDELAY	When sending the SEQSTART[1] task	Before starting sequence [1] through a SEQSTART[1] task
	Every time a new value from sequence [1] has been loaded from RAM and gets applied to the Wave Counter (indicated by the PWMPERIODEND event)	When no more value from sequence [1] gets loaded from RAM (indicated by the SEQEND[1] event)
		At any time during sequence [0] (which starts when the SEQSTARTED[0] event is fired)
SEQ[0].REFRESH	When sending the SEQSTART[0] task	Before starting sequence [0] through a SEQSTART[0] task
	Every time a new value from sequence [0] has been loaded from RAM and gets applied to the Wave Counter (indicated by the PWMPERIODEND event)	At any time during sequence [1] (which starts when the SEQSTARTED[1] event is fired)
SEQ[1].REFRESH	When sending the SEQSTART[1] task	Before starting sequence [1] through a SEQSTART[1] task
	Every time a new value from sequence [1] has been loaded from RAM and gets applied to the Wave Counter (indicated by the PWMPERIODEND event)	At any time during sequence [0] (which starts when the SEQSTARTED[0] event is fired)
COUNTERTOP	In DECODER.LOAD=WaveForm: this register is ignored.	Before starting PWM generation through a SEQSTART[n] task
	In all other LOAD modes: at the end of current PWM period (indicated by the PWMPERIODEND event)	After a STOP task has been issued, and the STOPPED event has been received.
MODE	Immediately	Before starting PWM generation through a SEQSTART[n] task
		After a STOP task has been issued, and the STOPPED event has been received.
DECODER	Immediately	Before starting PWM generation through a SEQSTART[n] task
		After a STOP task has been issued, and the STOPPED event has been received.
PRESCALER	Immediately	Before starting PWM generation through a SEQSTART[n] task
		After a STOP task has been issued, and the STOPPED event has been received.
LOOP	Immediately	Before starting PWM generation through a SEQSTART[n] task
		After a STOP task has been issued, and the STOPPED event has been received.
PSEL.OUT[n]	Immediately	Before enabling the PWM instance through the ENABLE register

Important: SEQ[n].REFRESH and SEQ[n].ENDDELAY are ignored at the end of a complex sequence, indicated by a LOOPSDONE event. The reason for this is that the last value loaded from RAM is maintained until further action from software (restarting a new sequence, or stopping PWM generation).

Figure 145: Simple sequence example on page 500 depicts the source code used for configuration and timing details in a sequence where only sequence 0 is used and only run once with a new PWM duty cycle for each period.

```
NRF_PWM0->PSEL.OUT[0] = (first_pin << PWM_PSEL_OUT_PIN_Pos) |
                        (PWM_PSEL_OUT_CONNECT_Connected <<
                                                  PWM_PSEL_OUT_CONNECT_Pos);
NRF_PWM0->ENABLE      = (PWM_ENABLE_ENABLE_Enabled << PWM_ENABLE_ENABLE_Pos);
NRF_PWM0->MODE        = (PWM_MODE_UPDOWN_Up << PWM_MODE_UPDOWN_Pos);
NRF_PWM0->PRESCALER   = (PWM_PRESCALER_PRESCALER_DIV_1 <<
                                                  PWM_PRESCALER_PRESCALER_Pos);
NRF_PWM0->COUNTERTOP  = (16000 << PWM_COUNTERTOP_COUNTERTOP_Pos); //1 msec
NRF_PWM0->LOOP        = (PWM_LOOP_CNT_Disabled << PWM_LOOP_CNT_Pos);
NRF_PWM0->DECODER   = (PWM_DECODER_LOAD_Common << PWM_DECODER_LOAD_Pos) |
                      (PWM_DECODER_MODE_RefreshCount << PWM_DECODER_MODE_Pos);
NRF_PWM0->SEQ[0].PTR  = ((uint32_t)(seq0_ram) << PWM_SEQ_PTR_PTR_Pos);
NRF_PWM0->SEQ[0].CNT  = ((sizeof(seq0_ram) / sizeof(uint16_t)) <<
                                                  PWM_SEQ_CNT_CNT_Pos);
NRF_PWM0->SEQ[0].REFRESH  = 0;
NRF_PWM0->SEQ[0].ENDDELAY = 0;
NRF_PWM0->TASKS_SEQSTART[0] = 1;
```

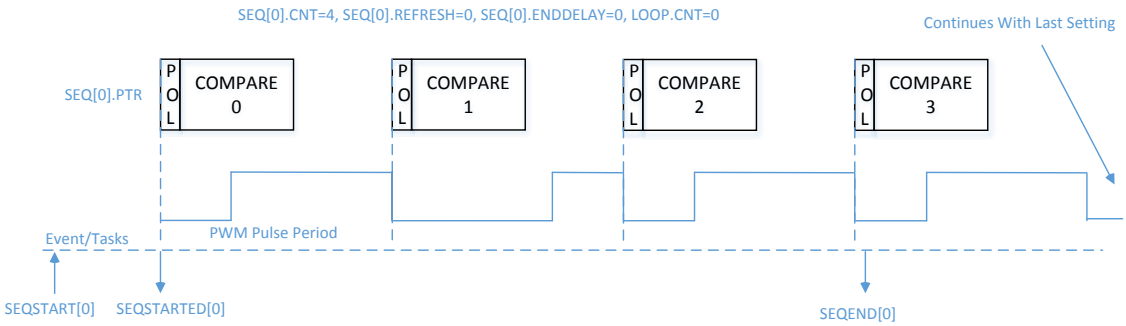


Figure 145: Simple sequence example

A more complex example is shown in *Figure 146: Example using two sequences* on page 501, where LOOP.CNT>0 . In this case, an automated playback takes place, consisting of SEQ[0], delay 0, SEQ[1], delay 1, then again SEQ[0], etc. The user can choose to start a complex playback with SEQ[0] or SEQ[1] through sending the SEQSTART[0] or SEQSTART[1] task.

The complex playback always ends with delay 1.

The two sequences 0 and 1 are defined with address of values tables in Data RAM (pointed by SEQ[n].PTR) and respective buffer size (SEQ[n].CNT). The rate at which a new value is loaded is defined individually for each sequence by SEQ[n].REFRESH . The chaining of sequence 1 following sequence 0 is implicit, the LOOP.CNT register allows the chaining of sequence 1 to sequence 0 for a determined number of times. In other words, it allows to repeat a complex sequence a number of times in a fully automated way.

In the example below, sequence 0 is defined with SEQ[0].REFRESH set to one - that means that a new PWM duty cycle is pushed every second PWM period. This complex sequence is started with the SEQSTART[0] task, so SEQ[0] is played first. Since SEQ[0].ENDDELAY=1 there will be one PWM period delay between last period on sequence 0 and the first period on sequence 1. Since SEQ[1].ENDDELAY=0 there is no delay 1, so SEQ[0] would be started immediately after the end of SEQ[1]. However, as LOOP.CNT is one, the playback stops after having played only once SEQ[1], and both SEQEND[1] and LOOPSDONE are generated (their order is not guaranteed in this case).

```
NRF_PWM0->PSEL.OUT[0] = (first_pin << PWM_PSEL_OUT_PIN_Pos) |
                        (PWM_PSEL_OUT_CONNECT_Connected <<
                                                  PWM_PSEL_OUT_CONNECT_Pos);
NRF_PWM0->ENABLE      = (PWM_ENABLE_ENABLE_Enabled << PWM_ENABLE_ENABLE_Pos);
NRF_PWM0->MODE        = (PWM_MODE_UPDOWN_Up << PWM_MODE_UPDOWN_Pos);
NRF_PWM0->PRESCALER   = (PWM_PRESCALER_PRESCALER_DIV_1 <<
                                                  PWM_PRESCALER_PRESCALER_Pos);
NRF_PWM0->COUNTERTOP  = (16000 << PWM_COUNTERTOP_COUNTERTOP_Pos); //1 msec
NRF_PWM0->LOOP        = (1 << PWM_LOOP_CNT_Pos);
NRF_PWM0->DECODER   = (PWM_DECODER_LOAD_Common << PWM_DECODER_LOAD_Pos) |
                      (PWM_DECODER_MODE_RefreshCount << PWM_DECODER_MODE_Pos);
NRF_PWM0->SEQ[0].PTR  = ((uint32_t)(seq0_ram) << PWM_SEQ_PTR_PTR_Pos);
NRF_PWM0->SEQ[0].CNT  = ((sizeof(seq0_ram) / sizeof(uint16_t)) <<
                                                  PWM_SEQ_CNT_CNT_Pos);
NRF_PWM0->SEQ[0].REFRESH  = 1;
NRF_PWM0->SEQ[0].ENDDELAY = 1;
NRF_PWM0->SEQ[1].PTR  = ((uint32_t)(seq1_ram) << PWM_SEQ_PTR_PTR_Pos);
NRF_PWM0->SEQ[1].CNT  = ((sizeof(seq1_ram) / sizeof(uint16_t)) <<
                                                  PWM_SEQ_CNT_CNT_Pos);
NRF_PWM0->SEQ[1].REFRESH  = 0;
NRF_PWM0->SEQ[1].ENDDELAY = 0;
NRF_PWM0->TASKS_SEQSTART[0] = 1;
```

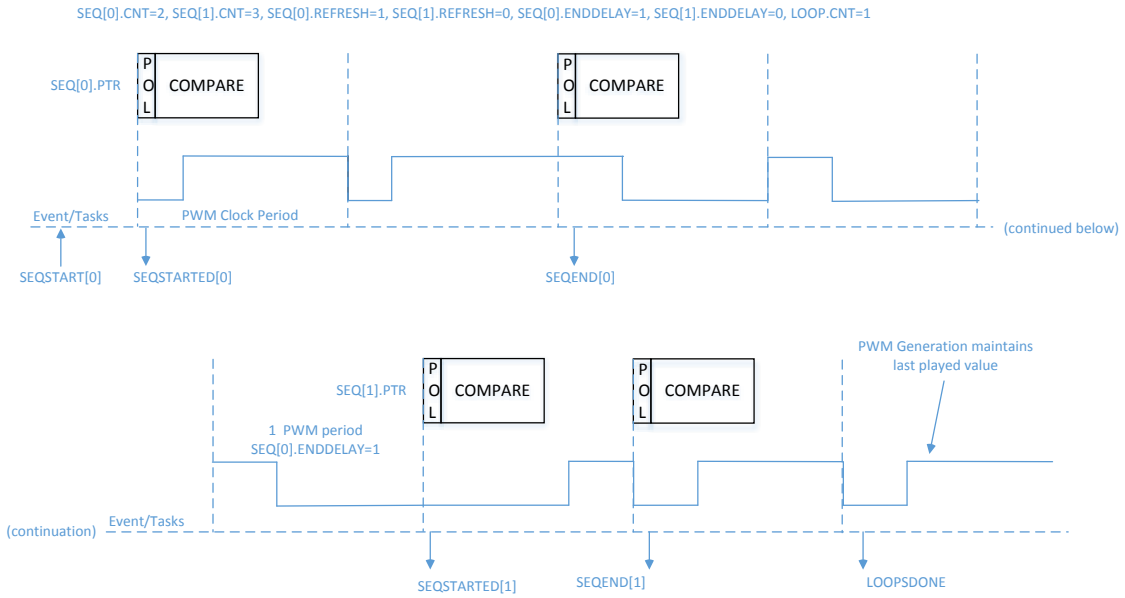


Figure 146: Example using two sequences

The decoder can also be configured to asynchronously load a new PWM duty cycle. If the DECODER.MODE register is set to NextStep - then the NEXTSTEP task will cause an update of the internal compare registers on the next PWM period.

The figures below provide an overview of each part of an arbitrary sequence, in various modes (LOOP.CNT=0 and LOOP.CNT>0). In particular are represented:

- Initial and final duty cycle on the PWM output(s)
- Chaining of SEQ[0] and SEQ[1] if LOOP.CNT>0
- Influence of registers on the sequence
- Events fired during a sequence
- DMA activity (loading of next value and applying it to the output(s))

Note that the single-shot example applies also to SEQ[1], only SEQ[0] is represented for simplicity.

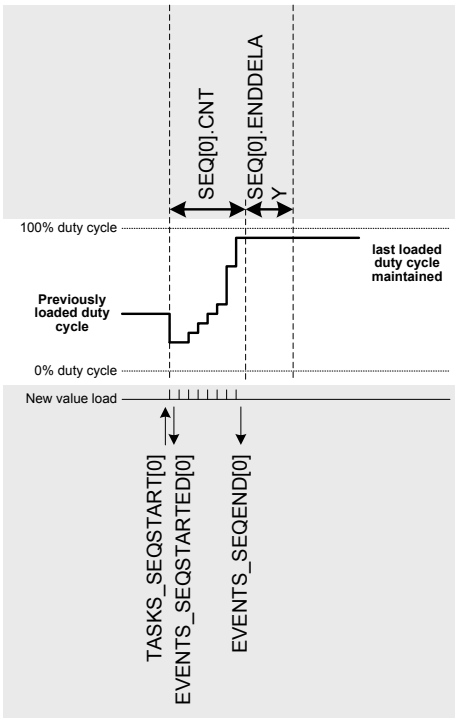


Figure 147: Single shot (LOOP.CNT=0)

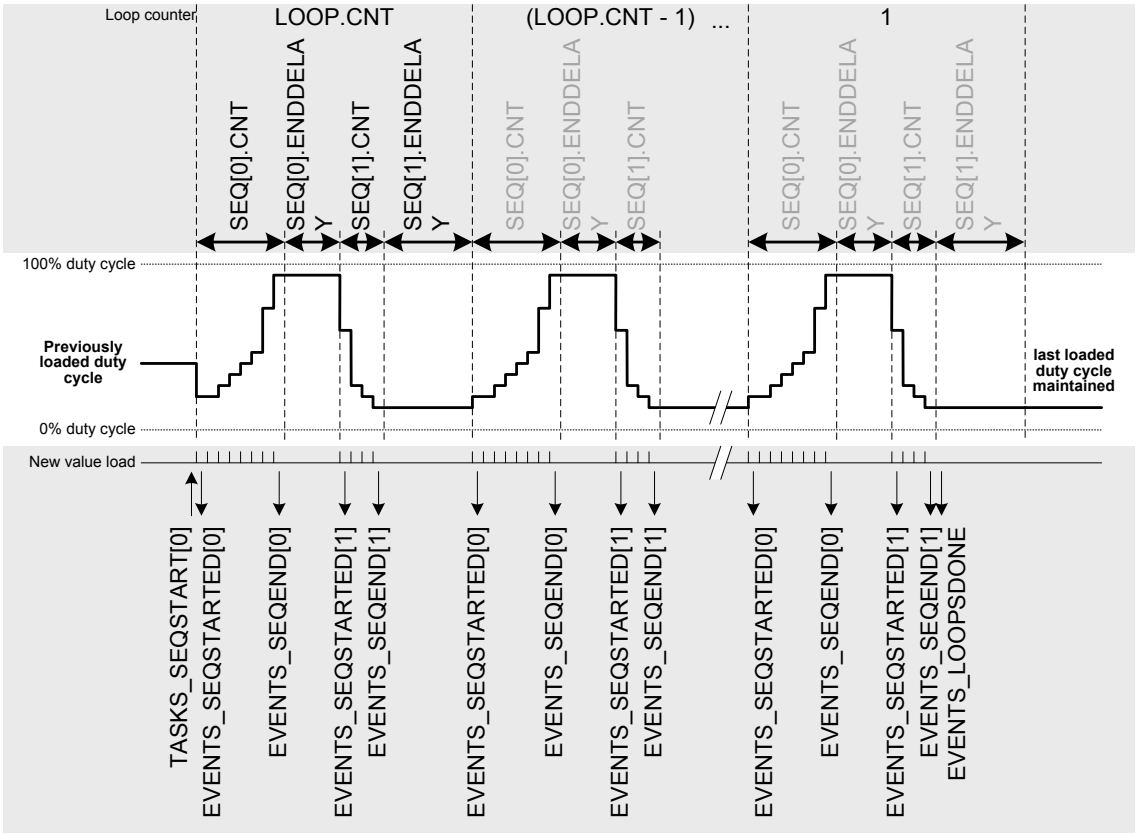


Figure 148: Complex sequence (LOOP.CNT>0) starting with SEQ[0]

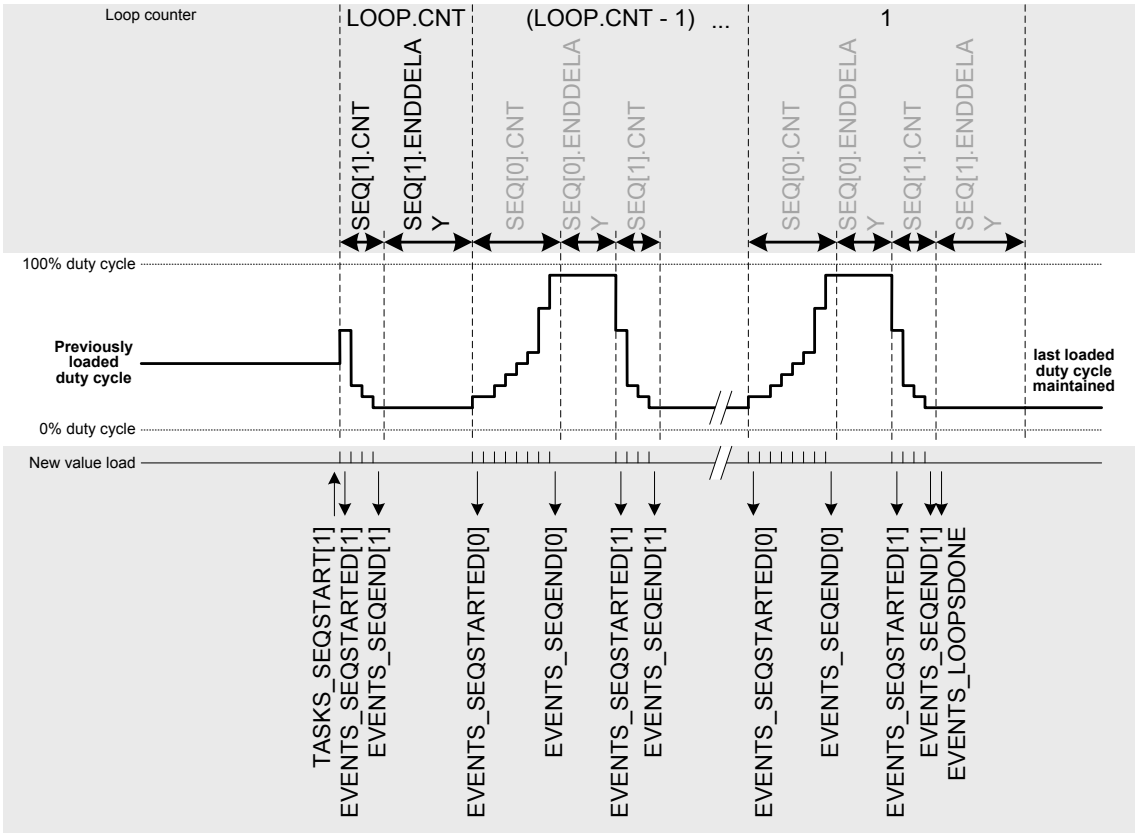


Figure 149: Complex sequence (LOOP.CNT>0) starting with SEQ[1]

Note that if a sequence is in use in a simple or complex sequence, it must have a length of SEQ[n].CNT > 0 .

47.3 Limitations

The previous compare value will be repeated if the PWM period is selected to be shorter than the time it takes for the EasyDMA to fetch from RAM and update the internal compare registers.

This is to ensure a glitch-free operation even if very short PWM periods are chosen.

47.4 Pin configuration

The OUT[n] (n=0..3) signals associated to each channel of the PWM module are mapped to physical pins according to the configuration specified in the respective PSEL.OUT[n] registers. If a PSEL.OUT[n].CONNECT is set to Disconnected, the associated PWM module signal will not be connected to any physical pins.

The PSEL.OUT[n] registers and their configurations are only used as long as the PWM module is enabled and PWM generation is active (wave counter started), and retained only as long as the device is in System ON mode, see *POWER* chapter for more information about power modes.

To ensure correct behaviour in the PWM module, the pins used by the PWM module must be configured in the GPIO peripheral as described in *Table 118: Recommended GPIO configuration before starting PWM generation* on page 504 before enabling the PWM module. The pins' idle state is defined by the OUT registers in the GPIO module. This is to ensure that the pins used by the PWM module are driven correctly, if PWM generation is stopped through a STOP task, the PWM module itself is temporarily disabled, or the device temporarily enters System OFF. This configuration must be retained in the GPIO for the selected IOs as long as the PWM module is supposed to be connected to an external PWM circuit.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behaviour.

Table 118: Recommended GPIO configuration before starting PWM generation

PWM signal	PWM pin	Direction	Output value	Comment
OUT[n]	As specified in PSEL.OUT[n] (n=0..3)	Output	0	Idle state defined in GPIO->OUT

47.5 Registers

Table 119: Instances

Base address	Peripheral	Instance	Description	Configuration
0x4001C000	PWM	PWM0	Pulse Width Modulation Unit 0	
0x40021000	PWM	PWM1	Pulse Width Modulation Unit 1	
0x40022000	PWM	PWM2	Pulse Width Modulation Unit 2	

Table 120: Register Overview

Register	Offset	Description
TASKS_STOP	0x004	Stops PWM pulse generation on all channels at the end of current PWM period, and stops sequence playback
TASKS_SEQSTART[0]	0x008	Loads the first PWM value on all enabled channels from sequence 0, and starts playing that sequence at the rate defined in SEQ[0]REFRESH and/or DECODER.MODE. Causes PWM generation to start it was not running.
TASKS_SEQSTART[1]	0x00C	Loads the first PWM value on all enabled channels from sequence 1, and starts playing that sequence at the rate defined in SEQ[1]REFRESH and/or DECODER.MODE. Causes PWM generation to start it was not running.
TASKS_NEXTSTEP	0x010	Steps by one value in the current sequence on all enabled channels if DECODER.MODE=NextStep. Does not cause PWM generation to start it was not running.
EVENTS_STOPPED	0x104	Response to STOP task, emitted when PWM pulses are no longer generated
EVENTS_SEQSTARTED[0]	0x108	First PWM period started on sequence 0
EVENTS_SEQSTARTED[1]	0x10C	First PWM period started on sequence 1
EVENTS_SEQEND[0]	0x110	Emitted at end of every sequence 0, when last value from RAM has been applied to wave counter
EVENTS_SEQEND[1]	0x114	Emitted at end of every sequence 1, when last value from RAM has been applied to wave counter
EVENTS_PWMPERIODEN	0x118	Emitted at the end of each PWM period
EVENTS_LOOPSDONE	0x11C	Concatenated sequences have been played the amount of times defined in LOOP.CNT
SHORTS	0x200	Shortcut register
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ENABLE	0x500	PWM module enable register
MODE	0x504	Selects operating mode of the wave counter
COUNTERTOP	0x508	Value up to which the pulse generator counter counts
PRESCALER	0x50C	Configuration for PWM_CLK
DECODER	0x510	Configuration of the decoder
LOOP	0x514	Amount of playback of a loop
SEQ[0].PTR	0x520	Beginning address in Data RAM of this sequence
SEQ[0].CNT	0x524	Amount of values (duty cycles) in this sequence
SEQ[0].REFRESH	0x528	Amount of additional PWM periods between samples loaded into compare register
SEQ[0].ENDDELAY	0x52C	Time added after the sequence
SEQ[1].PTR	0x540	Beginning address in Data RAM of this sequence
SEQ[1].CNT	0x544	Amount of values (duty cycles) in this sequence
SEQ[1].REFRESH	0x548	Amount of additional PWM periods between samples loaded into compare register
SEQ[1].ENDDELAY	0x54C	Time added after the sequence
PSEL.OUT[0]	0x560	Output pin select for PWM channel 0
PSEL.OUT[1]	0x564	Output pin select for PWM channel 1
PSEL.OUT[2]	0x568	Output pin select for PWM channel 2
PSEL.OUT[3]	0x56C	Output pin select for PWM channel 3

47.5.1 SHORTS

Address offset: 0x200

Shortcut register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	SEQEND0_STOP																																	
			Disabled	0																															
			Enabled	1																															
B	RW	SEQEND1_STOP																																	
			Disabled	0																															
			Enabled	1																															
C	RW	LOOPSDONE_SEQSTART0																																	
			Disabled	0																															
			Enabled	1																															
D	RW	LOOPSDONE_SEQSTART1																																	
			Disabled	0																															
			Enabled	1																															
E	RW	LOOPSDONE_STOP																																	
			Disabled	0																															
			Enabled	1																															

47.5.2 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												H	G	F	E	D	C	B	
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
B	RW	STOPPED																																	
			Disabled	0																															
			Enabled	1																															
C	RW	SEQSTARTED0																																	
			Disabled	0																															
			Enabled	1																															
D	RW	SEQSTARTED1																																	
			Disabled	0																															
			Enabled	1																															
E	RW	SEQEND0																																	
			Disabled	0																															
			Enabled	1																															
F	RW	SEQEND1																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												H	G	F	E	D	C	B	
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
																												See *EVENTS_SEQEND[1]*							
			Disabled	0																								Disable							
			Enabled	1																								Enable							
G	RW	PWMPERIODEND																										Enable or disable interrupt for PWMPERIODEND event See *EVENTS_PWMPERIODEND*							
			Disabled	0																								Disable							
			Enabled	1																								Enable							
H	RW	LOOPSDONE																										Enable or disable interrupt for LOOPSDONE event See *EVENTS_LOOPSDONE*							
			Disabled	0																								Disable							
			Enabled	1																								Enable							

47.5.3 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												H	G	F	E	D	C	B	
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
B	RW	STOPPED																										Write '1' to Enable interrupt for STOPPED event See *EVENTS_STOPPED*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
C	RW	SEQSTARTED0																										Write '1' to Enable interrupt for SEQSTARTED[0] event See *EVENTS_SEQSTARTED[0]*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
D	RW	SEQSTARTED1																										Write '1' to Enable interrupt for SEQSTARTED[1] event See *EVENTS_SEQSTARTED[1]*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
E	RW	SEQEND0																										Write '1' to Enable interrupt for SEQEND[0] event See *EVENTS_SEQEND[0]*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
F	RW	SEQEND1																										Write '1' to Enable interrupt for SEQEND[1] event See *EVENTS_SEQEND[1]*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
G	RW	PWMPERIODEND																										Write '1' to Enable interrupt for PWMPERIODEND event See *EVENTS_PWMPERIODEND*							
			Set	1																								Enable							
			Disabled	0																								Read: Disabled							
			Enabled	1																								Read: Enabled							
H	RW	LOOPSDONE																										Write '1' to Enable interrupt for LOOPSDONE event							

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			H G F E D C B
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
					See *EVENTS_LOOPSDONE*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

47.5.4 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			H G F E D C B
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
B	RW	STOPPED			Write '1' to Disable interrupt for STOPPED event See *EVENTS_STOPPED*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	SEQSTARTED0			Write '1' to Disable interrupt for SEQSTARTED[0] event See *EVENTS_SEQSTARTED[0]*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	SEQSTARTED1			Write '1' to Disable interrupt for SEQSTARTED[1] event See *EVENTS_SEQSTARTED[1]*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	SEQEND0			Write '1' to Disable interrupt for SEQEND[0] event See *EVENTS_SEQEND[0]*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	SEQEND1			Write '1' to Disable interrupt for SEQEND[1] event See *EVENTS_SEQEND[1]*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	PWMPERIODEND			Write '1' to Disable interrupt for PWMPERIODEND event See *EVENTS_PWMPERIODEND*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	LOOPSDONE			Write '1' to Disable interrupt for LOOPSDONE event See *EVENTS_LOOPSDONE*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

47.5.5 ENABLE

Address offset: 0x500

PWM module enable register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															**Description**
A	RW	ENABLE																																	Enable or disable PWM module
			Disabled	0																															Disabled
			Enabled	1																															Enable

47.5.6 MODE

Address offset: 0x504

Selects operating mode of the wave counter

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															**Description**
A	RW	UPDOWN																																	Selects up or up and down as wave counter mode
			Up	0																															Up counter - edge aligned PWM duty-cycle
			UpAndDown	1																															Up and down counter - center aligned PWM duty cycle

47.5.7 COUNTERTOP

Address offset: 0x508

Value up to which the pulse generator counter counts

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x000003FF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															**Description**
A	RW	COUNTERTOP		[3..32767]																															Value up to which the pulse generator counter counts. This register is ignored when DECODER.MODE=WaveForm and only values from RAM will be used.

47.5.8 PRESCALER

Address offset: 0x50C

Configuration for PWM_CLK

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															**Description**
A	RW	PRESCALER																																	Pre-scaler of PWM_CLK
			DIV_1	0																															Divide by 1 (16MHz)
			DIV_2	1																															Divide by 2 (8MHz)
			DIV_4	2																															Divide by 4 (4MHz)
			DIV_8	3																															Divide by 8 (2MHz)
			DIV_16	4																															Divide by 16 (1MHz)
			DIV_32	5																															Divide by 32 (500kHz)
			DIV_64	6																															Divide by 64 (250kHz)
			DIV_128	7																															Divide by 128 (125kHz)

47.5.9 DECODER

Address offset: 0x510

Configuration of the decoder

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																										B							A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	LOAD																																
				How a sequence is read from RAM and spread to the compare register																														
			Common	0																														
				1st half word (16-bit) used in all PWM channels 0..3																														
			Grouped	1																														
				1st half word (16-bit) used in channel 0..1; 2nd word in channel 2..3																														
			Individual	2																														
				1st half word (16-bit) in ch.0; 2nd in ch.1; ...; 4th in ch.3																														
			WaveForm	3																														
				1st half word (16-bit) in ch.0; 2nd in ch.1; ...; 4th in COUNTERTOP																														
B	RW	MODE																																
				Selects source for advancing the active sequence																														
			RefreshCount	0																														
				SEQ[n].REFRESH is used to determine loading internal compare registers																														
			NextStep	1																														
				NEXTSTEP task causes a new value to be loaded to internal compare registers																														

47.5.10 LOOP

Address offset: 0x514

Amount of playback of a loop

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																			A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	CNT																																
				Amount of playback of pattern cycles																														
			Disabled	0																														
				Looping disabled (stop at the end of the sequence)																														

47.5.11 SEQ[0].PTR

Address offset: 0x520

Beginning address in Data RAM of this sequence

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	PTR																																
				Beginning address in Data RAM of this sequence																														

47.5.12 SEQ[0].CNT

Address offset: 0x524

Amount of values (duty cycles) in this sequence

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	CNT																																
				Amount of values (duty cycles) in this sequence																														
			Disabled	0																														
				Sequence is disabled, and shall not be started as it is empty																														

47.5.13 SEQ[0].REFRESH

Address offset: 0x528

Amount of additional PWM periods between samples loaded into compare register

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000001			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**							**Description**																							
A	RW	CNT									Amount of additional PWM periods between samples loaded into compare register (load every REFRESH.CNT+1 PWM periods)																							
			Continuous	0							Update every PWM period																							

47.5.14 SEQ[0].ENDDELAY

Address offset: 0x52C

Time added after the sequence

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**							**Description**																							
A	RW	CNT									Time added after the sequence in PWM periods																							

47.5.15 SEQ[1].PTR

Address offset: 0x540

Beginning address in Data RAM of this sequence

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**							**Description**																							
A	RW	PTR									Beginning address in Data RAM of this sequence																							

47.5.16 SEQ[1].CNT

Address offset: 0x544

Amount of values (duty cycles) in this sequence

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**							**Description**																							
A	RW	CNT									Amount of values (duty cycles) in this sequence																							
			Disabled	0							Sequence is disabled, and shall not be started as it is empty																							

47.5.17 SEQ[1].REFRESH

Address offset: 0x548

Amount of additional PWM periods between samples loaded into compare register

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id											A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000001			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**							**Description**																							
A	RW	CNT									Amount of additional PWM periods between samples loaded into compare register (load every REFRESH.CNT+1 PWM periods)																							
			Continuous	0							Update every PWM period																							

47.5.18 SEQ[1].ENDDELAY

Address offset: 0x54C

Time added after the sequence

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id												A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**								**Description**																							
A	RW	CNT										Time added after the sequence in PWM periods																							

47.5.19 PSEL.OUT[0]

Address offset: 0x560

Output pin select for PWM channel 0

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				C																											A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**								**Description**																							
A	RW	PIN		[0..31]								Pin number																							
C	RW	CONNECT										Connection																							
			Disconnected	1								Disconnect																							
			Connected	0								Connect																							

47.5.20 PSEL.OUT[1]

Address offset: 0x564

Output pin select for PWM channel 1

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				C																											A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**								**Description**																							
A	RW	PIN		[0..31]								Pin number																							
C	RW	CONNECT										Connection																							
			Disconnected	1								Disconnect																							
			Connected	0								Connect																							

47.5.21 PSEL.OUT[2]

Address offset: 0x568

Output pin select for PWM channel 2

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				C																											A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**								**Description**																							
A	RW	PIN		[0..31]								Pin number																							
C	RW	CONNECT										Connection																							
			Disconnected	1								Disconnect																							
			Connected	0								Connect																							

47.5.22 PSEL.OUT[3]

Address offset: 0x56C

Output pin select for PWM channel 3

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				C																											A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**								**Description**																							
A	RW	PIN		[0..31]								Pin number																							

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			C																											A	A	A	A	A
Reset 0xFFFFFFFF			**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**								**Value**																							
C	RW	CONNECT																																
			Disconnected								1																							
			Connected								0																							

47.6 Electrical specification

47.6.1 PWM Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
$I_{PWM,16MHz}$	PWM run current, Prescaler set to DIV_1 (16 MHz), excluding DMA and GPIO		200		µA
$I_{PWM,8MHz}$	PWM run current, Prescaler set to DIV_2 (8 MHz), excluding DMA and GPIO		150		µA
$I_{PWM,125kHz}$	PWM run current, Prescaler set to DIV_128 (125 kHz), excluding DMA and GPIO		150		µA

48 SPI — Serial peripheral interface master

The SPI master provides a simple CPU interface which includes a TXD register for sending data and an RXD register for receiving data. This section is added for legacy support for now.

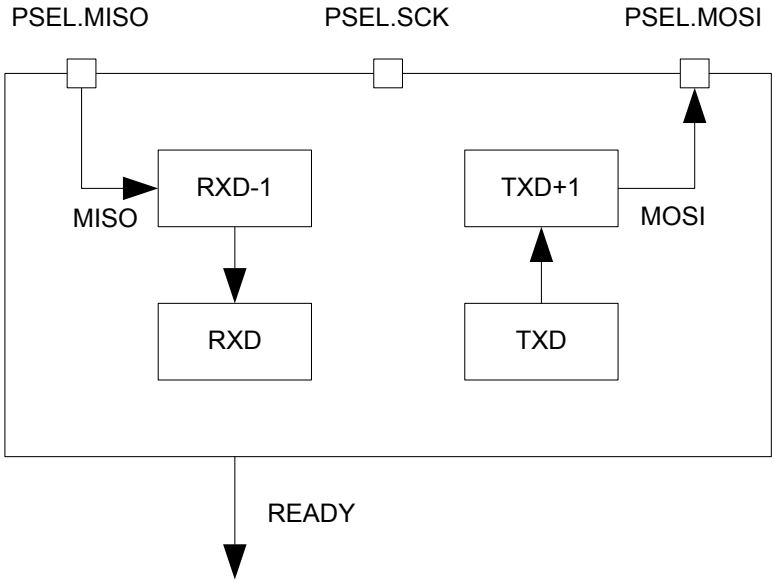


Figure 150: SPI master

RXD-1 and TXD+1 illustrate the double buffered version of RXD and TXD respectively.

48.1 Functional description

The TXD and RXD registers are double-buffered to enable some degree of uninterrupted data flow in and out of the SPI master.

The SPI master does not implement support for chip select directly. Therefore, the CPU must use available GPIOs to select the correct slave and control this independently of the SPI master. The SPI master supports SPI modes 0 through 3.

Table 121: SPI modes

Mode	Clock polarity CPOL	Clock phase CPHA
SPI_MODE	0 (Leading)	0 (Active High)
SPI_MODE	0 (Leading)	1 (Active Low)
SPI_MODE	1 (Trailing)	0 (Active High)
SPI_MODE	1 (Trailing)	1 (Active Low)

48.1.1 SPI master mode pin configuration

The different signals SCK, MOSI, and MISO associated with the SPI master are mapped to physical pins.

This mapping is according to the configuration specified in the PSELSCK, PSELMOSI, and PSELMISO registers respectively. If a value of 0xFFFFFFFF is specified in any of these registers, the associated SPI master signal is not connected to any physical pin. The PSELSCK, PSELMOSI, and PSELMISO registers and their configurations are only used as long as the SPI master is enabled, and retained only as long as the device is in ON mode. PSELSCK, PSELMOSI, and PSELMISO must only be configured when the SPI master is disabled.

To secure correct behavior in the SPI, the pins used by the SPI must be configured in the GPIO peripheral as described in *Table 122: GPIO configuration* on page 514 prior to enabling the SPI. The SCK must

always be connected to a pin, and that pin's input buffer must always be connected for the SPI to work. This configuration must be retained in the GPIO for the selected IOs as long as the SPI is enabled.

Only one peripheral can be assigned to drive a particular GPIO pin at a time, failing to do so may result in unpredictable behavior.

Table 122: GPIO configuration

SPI master signal	SPI master pin	Direction	Output value
SCK	As specified in PSELSCK	Output	Same as CONFIG.CPOL
MOSI	As specified in PSELMOSI	Output	0
MISO	As specified in PSELMISO	Input	Not applicable

48.1.2 Shared resources

The SPI shares registers and other resources with other peripherals that have the same ID as the SPI. Therefore, the user must disable all peripherals that have the same ID as the SPI before the SPI can be configured and used.

Disabling a peripheral that has the same ID as the SPI will not reset any of the registers that are shared with the SPI. It is therefore important to configure all relevant SPI registers explicitly to secure that it operates correctly.

See the Instantiation table in *Instantiation* on page 24 for details on peripherals and their IDs.

48.1.3 SPI master transaction sequence

An SPI master transaction is started by writing the first byte, which is to be transmitted by the SPI master, to the TXD register.

Since the transmitter is double buffered, the second byte can be written to the TXD register immediately after the first one. The SPI master will then send these bytes in the order they are written to the TXD register.

The SPI master is a synchronous interface, and for every byte that is sent, a different byte will be received at the same time; this is illustrated in *Figure 151: SPI master transaction* on page 515. Bytes that are received will be moved to the RXD register where the CPU can extract them by reading the register. The RXD register is double buffered in the same way as the TXD register, and a second byte can therefore be received at the same time as the first byte is being extracted from RXD by the CPU. The SPI master will generate a READY event every time a new byte is moved to the RXD register. The double buffered byte will be moved from RXD-1 to RXD as soon as the first byte is extracted from RXD. The SPI master will stop when there are no more bytes to send in TXD and TXD+1.

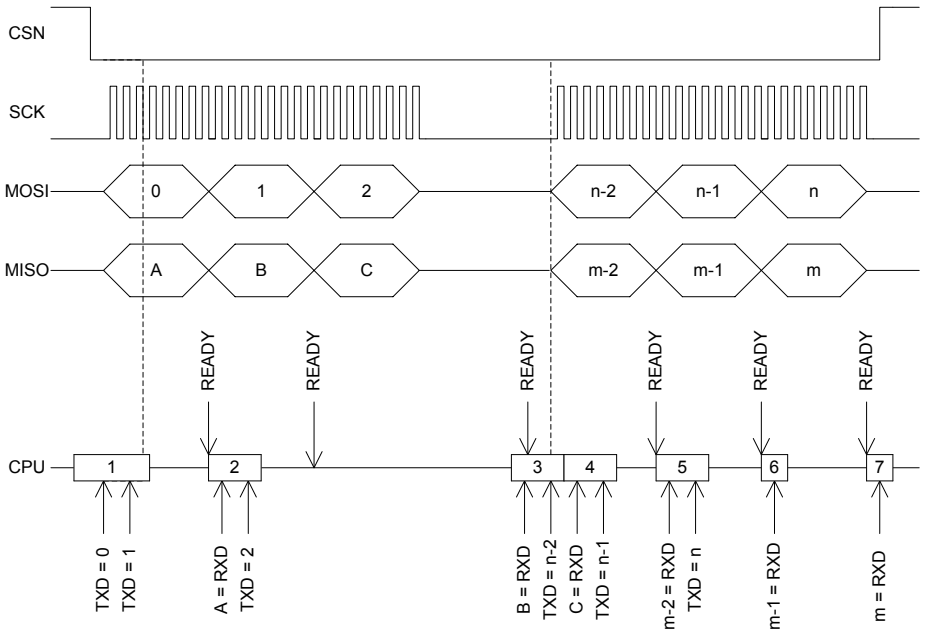


Figure 151: SPI master transaction

The READY event of the third byte transaction is delayed until B is extracted from RXD in occurrence number 3 on the horizontal lifeline. The reason for this is that the third event is generated first when C is moved from RXD-1 to RXD after B is read.

The SPI master will move the incoming byte to the RXD register after a short delay following the SCK clock period of the last bit in the byte. This also means that the READY event will be delayed accordingly, see *Figure 152: SPI master transaction* on page 515. Therefore, it is important that you always clear the READY event, even if the RXD register and the data that is being received is not used.

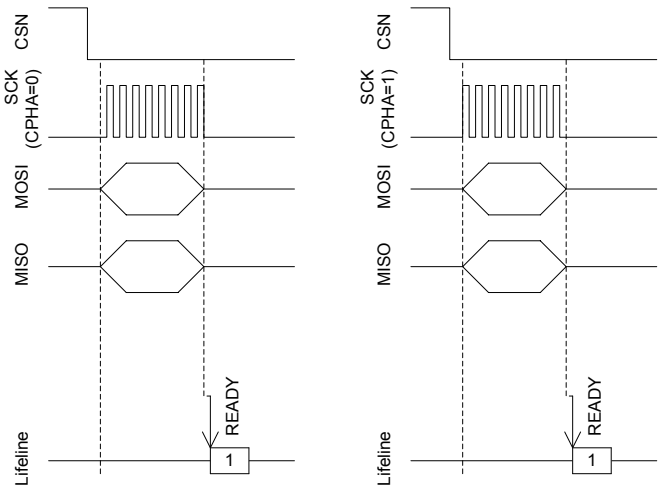


Figure 152: SPI master transaction

48.2 Registers

Table 123: Instances

Base address	Peripheral	Instance	Description	Configuration	
0x40003000	SPI	SPI0	SPI master 0		Deprecated
0x40004000	SPI	SPI1	SPI master 1		Deprecated
0x40023000	SPI	SPI2	SPI master 2		Deprecated

Table 124: Register Overview

Register	Offset	Description	
EVENTS_READY	0x108	TXD byte sent and RXD byte received	
INTENSET	0x304	Enable interrupt	
INTENCLR	0x308	Disable interrupt	
ENABLE	0x500	Enable SPI	
PSELSCK	0x508	Pin select for SCK	Deprecated
PSELMOSI	0x50C	Pin select for MOSI	Deprecated
PSELMISO	0x510	Pin select for MISO	Deprecated
PSEL.SCK	0x508	Pin select for SCK	
PSEL.MOSI	0x50C	Pin select for MOSI	
PSEL.MISO	0x510	Pin select for MISO	
RXD	0x518	RXD register	
TXD	0x51C	TXD register	
FREQUENCY	0x524	SPI frequency	
CONFIG	0x554	Configuration register	

48.2.1 INTENSET

Address offset: 0x304

Enable interrupt

Bit number																															
Id																														A	
Reset 0x00000000																														**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	READY			Write '1' to Enable interrupt for READY event
					See *EVENTS_READY*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

48.2.2 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number																															
Id																														A	
Reset 0x00000000																														**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	READY			Write '1' to Disable interrupt for READY event
					See *EVENTS_READY*
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

48.2.3 ENABLE

Address offset: 0x500

Enable SPI

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	ENABLE																																	
			Disabled	0																															
			Enabled	1																															

48.2.4 PSELSCK (Deprecated)

Address offset: 0x508

Pin select for SCK

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSELSCK		[0..31]																															
			Disconnected	0xFFFFFFFF																															

48.2.5 PSELMOSI (Deprecated)

Address offset: 0x50C

Pin select for MOSI

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSELMOSI		[0..31]																															
			Disconnected	0xFFFFFFFF																															

48.2.6 PSELMISO (Deprecated)

Address offset: 0x510

Pin select for MISO

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSELMISO		[0..31]																															
			Disconnected	0xFFFFFFFF																															

48.2.7 PSEL.SCK

Address offset: 0x508

Pin select for SCK

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSELSCK		[0..31]																															
			Disconnected	0xFFFFFFFF																															

48.2.8 PSEL.MOSI

Address offset: 0x50C

Pin select for MOSI

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSELMOSI		[0..31]																															
			Disconnected	0xFFFFFFFF																															

48.2.9 PSEL.MISO

Address offset: 0x510

Pin select for MISO

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSELMISO		[0..31]																															
			Disconnected	0xFFFFFFFF																															

48.2.10 RXD

Address offset: 0x518

RXD register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	R	RXD																																	

48.2.11 TXD

Address offset: 0x51C

TXD register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	TXD																																	

48.2.12 FREQUENCY

Address offset: 0x524

SPI frequency

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x04000000				**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	FREQUENCY																																	
			K125	0x02000000																															
			K250	0x04000000																															
			K500	0x08000000																															
			M1	0x10000000																															

Bit number																																
Id																																
Reset 0x04000000																																
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																											
			M2	0x20000000	2 Mbps																											
			M4	0x40000000	4 Mbps																											
			M8	0x80000000	8 Mbps																											

48.2.13 CONFIG

Address offset: 0x554

Configuration register

Bit number																																
Id																																
Reset 0x00000000																																
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																											
A	RW	ORDER			Bit order																											
			MsbFirst	0	Most significant bit shifted out first																											
			LsbFirst	1	Least significant bit shifted out first																											
B	RW	CPHA			Serial clock (SCK) phase																											
			Leading	0	Sample on leading edge of clock, shift serial data on trailing edge																											
			Trailing	1	Sample on trailing edge of clock, shift serial data on leading edge																											
C	RW	CPOL			Serial clock (SCK) polarity																											
			ActiveHigh	0	Active high																											
			ActiveLow	1	Active low																											

48.3 Electrical specification

48.3.1 SPI master interface

Symbol	Description	Min.	Typ.	Max.	Units
f_{SPI}	Bit rates for SPI[38]			8[39]	Mbps
$I_{SPI,2Mbps}$	Run current for SPI, 2 Mbps			50	µA
$I_{SPI,8Mbps}$	Run current for SPI, 8 Mbps			50	µA
$I_{SPI,IDLE}$	Idle current for SPI (STARTed, no CSN activity)		<1		µA
$t_{SPI,START,LP}$	Time from writing TXD register to transmission started, low power mode		$t_{SPI,START,CL}$ + t_{START_HFIN}		µs
$t_{SPI,START,CL}$	Time from writing TXD register to transmission started, constant latency mode		1		µs

48.3.2 Serial Peripheral Interface (SPI) Master timing specifications

Symbol	Description	Min.	Typ.	Max.	Units
$t_{SPI,CSCK,8Mbps}$	SCK period at 8Mbps		125		ns
$t_{SPI,CSCK,4Mbps}$	SCK period at 4Mbps		250		ns
$t_{SPI,CSCK,2Mbps}$	SCK period at 2Mbps		500		ns
$t_{SPI,RSCK,LD}$	SCK rise time, low drive[a]			$t_{RF,25pF}$	
$t_{SPI,RSCK,HD}$	SCK rise time, high drive[a]			$t_{HRF,25pF}$	
$t_{SPI,FSCK,LD}$	SCK fall time, low drive[a]			$t_{RF,25pF}$	
$t_{SPI,FSCK,HD}$	SCK fall time, high drive[a]			$t_{HRF,25pF}$	

38 Higher bit rates may require GPIOs to be set as High Drive, see GPIO chapter for more details.

39 The actual maximum data rate depends on the slave's CLK to MISO and MOSI setup and hold timings.

a At 25pF load, including GPIO capacitance, see GPIO spec.

Symbol	Description	Min.	Typ.	Max.	Units
$t_{SPI,WHSCK}$	SCK high time[a]	(0.5*t_{CSCK}) – t_{RSCK}			
$t_{SPI,WLSCK}$	SCK low time[a]	(0.5*t_{CSCK}) – t_{FSCK}			
$t_{SPI,SUMI}$	MISO to CLK edge setup time	19			ns
$t_{SPI,HMI}$	CLK edge to MISO hold time	18			ns
$t_{SPI,VMO}$	CLK edge to MOSI valid			59	ns
$t_{SPI,HMO}$	MOSI hold time after CLK edge	20			ns

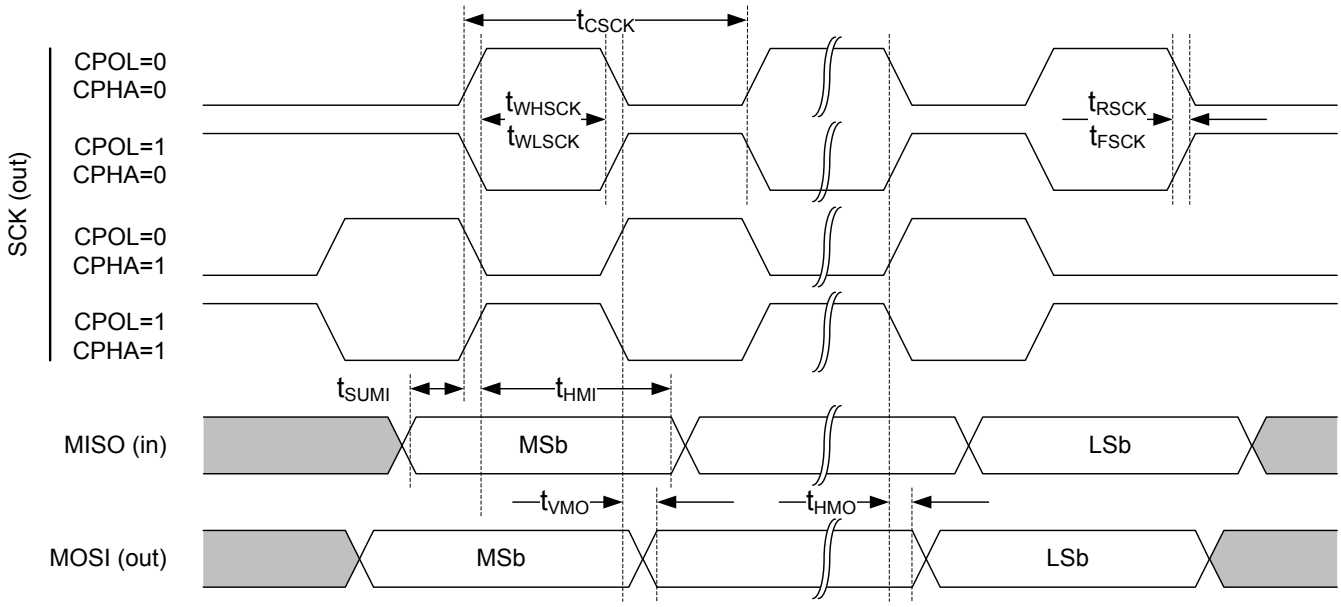


Figure 153: SPI master timing diagram

49 TWI — I2C compatible two-wire interface

The TWI master is compatible with I2C operating at 100 kHz and 400 kHz.

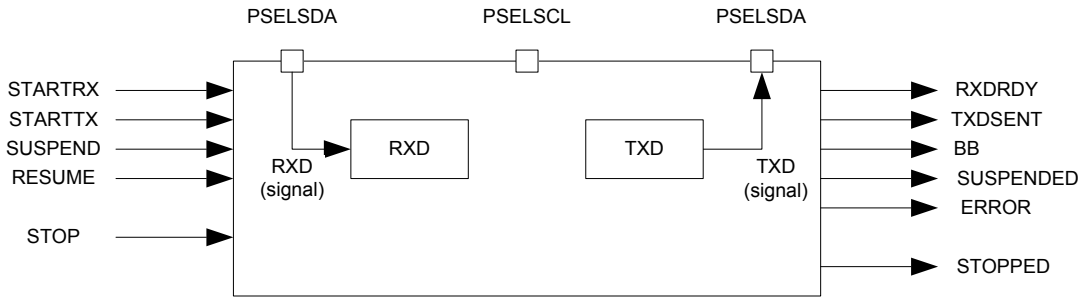


Figure 154: TWI master's main features

49.1 Functional description

This TWI master is not compatible with CBUS. The TWI transmitter and receiver are single buffered.

See, *Figure 154: TWI master's main features* on page 521.

A TWI setup comprising one master and three slaves is illustrated in *Figure 155: A typical TWI setup comprising one master and three slaves* on page 521. This TWI master is only able to operate as the only master on the TWI bus.

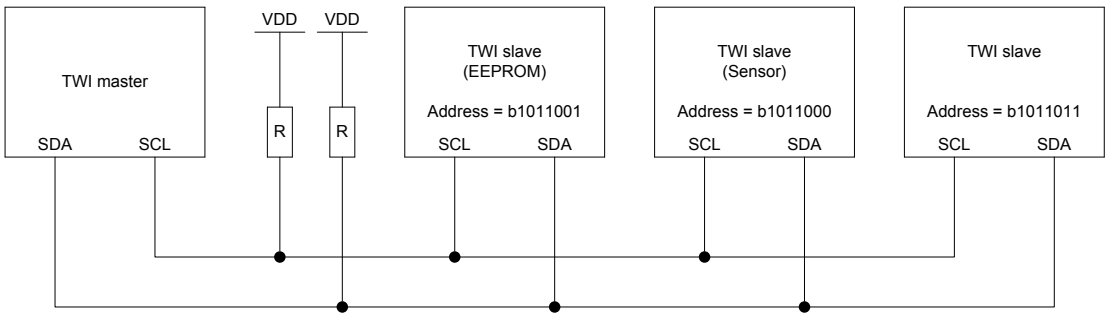


Figure 155: A typical TWI setup comprising one master and three slaves

This TWI master supports clock stretching performed by the slaves. The TWI master is started by triggering the STARTTX or STARTRX tasks, and stopped by triggering the STOP task.

If a NACK is clocked in from the slave, the TWI master will generate an ERROR event.

49.2 Master mode pin configuration

The different signals SCL and SDA associated with the TWI master are mapped to physical pins according to the configuration specified in the PSELSCL and PSELSDA registers respectively.

If a value of 0xFFFFFFFF is specified in any of these registers, the associated TWI master signal is not connected to any physical pin. The PSELSCL and PSELSDA registers and their configurations are only used

as long as the TWI master is enabled, and retained only as long as the device is in ON mode. PSELSCL and PSELSDA must only be configured when the TWI is disabled.

To secure correct signal levels on the pins used by the TWI master when the system is in OFF mode, and when the TWI master is disabled, these pins must be configured in the GPIO peripheral as described in *Table 125: GPIO configuration* on page 522.

Only one peripheral can be assigned to drive a particular GPIO pin at a time, failing to do so may result in unpredictable behavior.

Table 125: GPIO configuration

TWI master signal	TWI master pin	Direction	Drive strength	Output value
SCL	As specified in PSELSCL	Input	S0D1	Not applicable
SDA	As specified in PSELSDA	Input	S0D1	Not applicable

49.3 Shared resources

The TWI shares registers and other resources with other peripherals that have the same ID as the TWI.

Therefore, you must disable all peripherals that have the same ID as the TWI before the TWI can be configured and used. Disabling a peripheral that has the same ID as the TWI will not reset any of the registers that are shared with the TWI. It is therefore important to configure all relevant TWI registers explicitly to secure that it operates correctly.

The Instantiation table in *Instantiation* on page 24 shows which peripherals have the same ID as the TWI.

49.4 Master write sequence

A TWI master write sequence is started by triggering the STARTTX task. After the STARTTX task has been triggered, the TWI master will generate a start condition on the TWI bus, followed by clocking out the address and the READ/WRITE bit set to 0 (WRITE=0, READ=1).

The address must match the address of the slave device that the master wants to write to. The READ/WRITE bit is followed by an ACK/NACK bit (ACK=0 or NACK=1) generated by the slave.

After receiving the ACK bit, the TWI master will clock out the data bytes that are written to the TXD register. Each byte clocked out from the master will be followed by an ACK/NACK bit clocked in from the slave. A TXDSENT event will be generated each time the TWI master has clocked out a TXD byte, and the associated ACK/NACK bit has been clocked in from the slave.

The TWI master transmitter is single buffered, and a second byte can only be written to the TXD register after the previous byte has been clocked out and the ACK/NACK bit clocked in, that is, after the TXDSENT event has been generated.

If the CPU is prevented from writing to TXD when the TWI master is ready to clock out a byte, the TWI master will stretch the clock until the CPU has written a byte to the TXD register.

A typical TWI master write sequence is illustrated in *Figure 156: The TWI master writing data to a slave* on page 523. Occurrence 3 in the figure illustrates delayed processing of the TXDSENT event associated with TXD byte 1. In this scenario the TWI master will stretch the clock to prevent writing erroneous data to the slave.

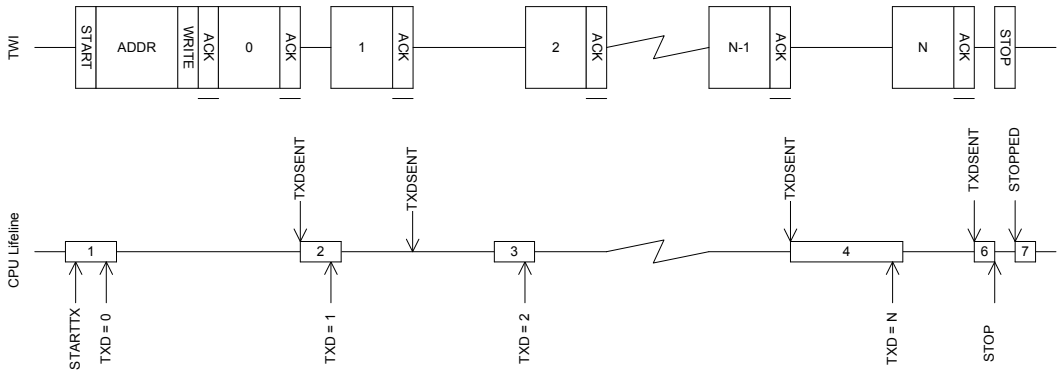


Figure 156: The TWI master writing data to a slave

The TWI master write sequence is stopped when the STOP task is triggered whereupon the TWI master will generate a stop condition on the TWI bus.

49.5 Master read sequence

A TWI master read sequence is started by triggering the STARTRX task. After the STARTRX task has been triggered the TWI master will generate a start condition on the TWI bus, followed by clocking out the address and the READ/WRITE bit set to 1 (WRITE = 0, READ = 1).

The address must match the address of the slave device that the master wants to read from. The READ/WRITE bit is followed by an ACK/NACK bit (ACK=0 or NACK = 1) generated by the slave.

After having sent the ACK bit the TWI slave will send data to the master using the clock generated by the master.

The TWI master will generate a RXDRDY event every time a new byte is received in the RXD register.

After receiving a byte, the TWI master will delay sending the ACK/NACK bit by stretching the clock until the CPU has extracted the received byte, that is, by reading the RXD register.

The TWI master read sequence is stopped by triggering the STOP task. This task must be triggered before the last byte is extracted from RXD to ensure that the TWI master sends a NACK back to the slave before generating the stop condition.

A typical TWI master read sequence is illustrated in *Figure 157: The TWI master reading data from a slave* on page 524. Occurrence 3 in this figure illustrates delayed processing of the RXDRDY event associated with RXD byte B. In this scenario the TWI master will stretch the clock to prevent the slave from overwriting the contents of the RXD register.

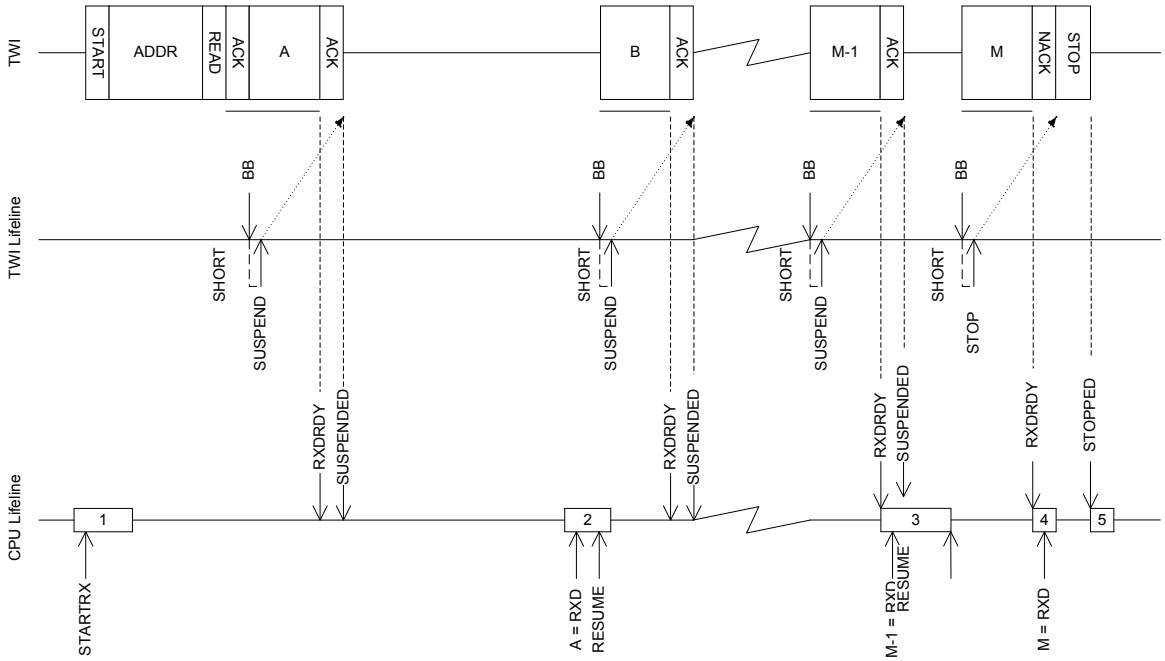


Figure 157: The TWI master reading data from a slave

49.6 Master repeated start sequence

A typical repeated start sequence is one in which the TWI master writes one byte to the slave followed by reading M bytes from the slave. Any combination and number of transmit and receive sequences can be combined in this fashion. Only one shortcut to STOP can be enabled at any given time.

The figure below illustrates a repeated start sequence where the TWI master writes one byte, followed by reading M bytes from the slave without performing a stop in-between.

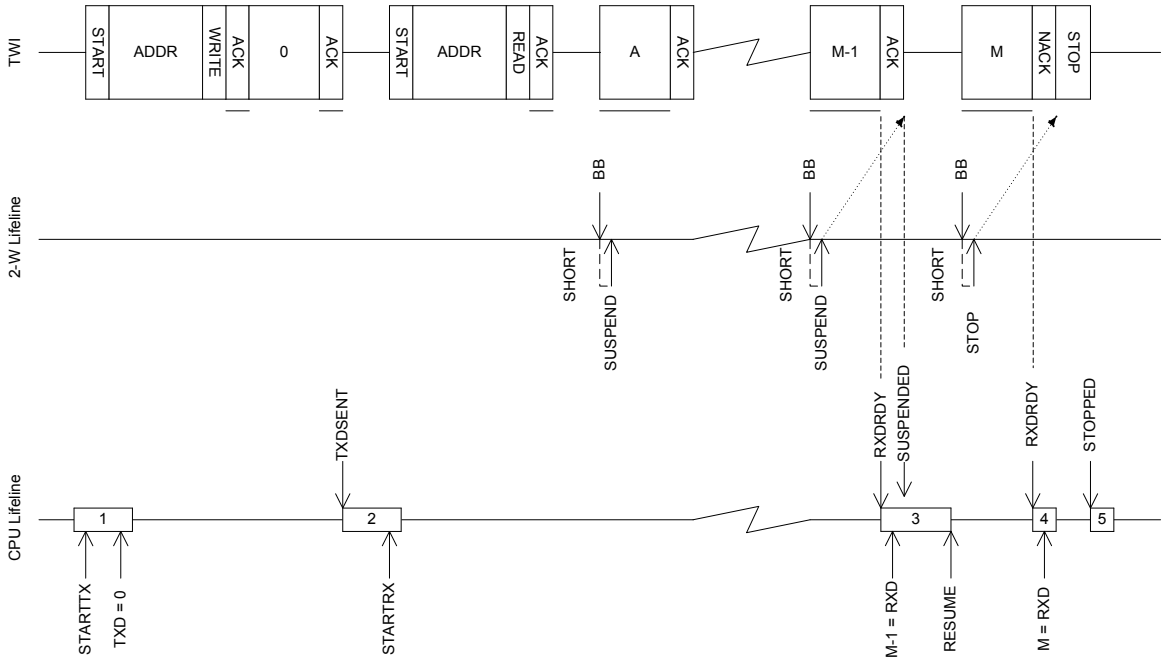


Figure 158: A repeated start sequence, where the TWI master writes one byte, followed by reading M bytes from the slave without performing a stop in-between

To generate a repeated start after a read sequence, a second start task must be triggered instead of the STOP task, that is, STARTRX or STARTTX. This start task must be triggered before the last byte is extracted from RXD to ensure that the TWI master sends a NACK back to the slave before generating the repeated start condition.

49.7 Low power

When putting the system in low power and the peripheral is not needed, lowest possible power consumption is achieved by stopping, and then disabling the peripheral.

The STOP task may not be always needed (the peripheral might already be stopped), but if it is sent, software shall wait until the STOPPED event was received as a response before disabling the peripheral through the ENABLE register.

49.8 Registers

Table 126: Instances

Base address	Peripheral	Instance	Description	Configuration	
0x40003000	TWI	TWI0	Two-wire interface master 0		Deprecated
0x40004000	TWI	TWI1	Two-wire interface master 1		Deprecated

Table 127: Register Overview

Register	Offset	Description
TASKS_STARTRX	0x000	Start TWI receive sequence
TASKS_STARTTX	0x008	Start TWI transmit sequence
TASKS_STOP	0x014	Stop TWI transaction
TASKS_SUSPEND	0x01C	Suspend TWI transaction
TASKS_RESUME	0x020	Resume TWI transaction
EVENTS_STOPPED	0x104	TWI stopped
EVENTS_RXDREADY	0x108	TWI RXD byte received
EVENTS_TXDSENT	0x11C	TWI TXD byte sent
EVENTS_ERROR	0x124	TWI error
EVENTS_BB	0x138	TWI byte boundary, generated before each byte that is sent or received
EVENTS_SUSPENDED	0x148	TWI entered the suspended state
SHORTS	0x200	Shortcut register
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ERRORSRC	0x4C4	Error source
ENABLE	0x500	Enable TWI
PSELSCL	0x508	Pin select for SCL
PSELSDA	0x50C	Pin select for SDA
RXD	0x518	RXD register
TXD	0x51C	TXD register
FREQUENCY	0x524	TWI frequency
ADDRESS	0x588	Address used in the TWI transfer

49.8.1 SHORTS

Address offset: 0x200

Shortcut register

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	BB_SUSPEND			Shortcut between BB event and SUSPEND task

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			B A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
					See *EVENTS_BB* and *TASKS_SUSPEND*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
B	RW	BB_STOP			Shortcut between BB event and STOP task See *EVENTS_BB* and *TASKS_STOP*
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

49.8.2 INTENSET

Address offset: 0x304

Enable interrupt

Bit number			31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id			F E D C B A
Reset 0x00000000			**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

Id	RW	Field	Value Id	Value	Description
A	RW	STOPPED			Write '1' to Enable interrupt for STOPPED event See *EVENTS_STOPPED*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	RXDREADY			Write '1' to Enable interrupt for RXDREADY event See *EVENTS_RXDREADY*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	TXDSENT			Write '1' to Enable interrupt for TXDSENT event See *EVENTS_TXDSENT*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	ERROR			Write '1' to Enable interrupt for ERROR event See *EVENTS_ERROR*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	BB			Write '1' to Enable interrupt for BB event See *EVENTS_BB*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	SUSPENDED			Write '1' to Enable interrupt for SUSPENDED event See *EVENTS_SUSPENDED*
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

49.8.3 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id															F					E					D		C						B	A	
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	STOPPED			Write '1' to Disable interrupt for STOPPED event See *EVENTS_STOPPED*																															
			Clear	1	Disable																													
			Disabled	0	Read: Disabled																													
			Enabled	1	Read: Enabled																													
B	RW	RXDREADY			Write '1' to Disable interrupt for RXDREADY event See *EVENTS_RXDREADY*																															
			Clear	1	Disable																													
			Disabled	0	Read: Disabled																													
			Enabled	1	Read: Enabled																													
C	RW	TXDSENT			Write '1' to Disable interrupt for TXDSENT event See *EVENTS_TXDSENT*																															
			Clear	1	Disable																													
			Disabled	0	Read: Disabled																													
			Enabled	1	Read: Enabled																													
D	RW	ERROR			Write '1' to Disable interrupt for ERROR event See *EVENTS_ERROR*																															
			Clear	1	Disable																													
			Disabled	0	Read: Disabled																													
			Enabled	1	Read: Enabled																													
E	RW	BB			Write '1' to Disable interrupt for BB event See *EVENTS_BB*																															
			Clear	1	Disable																													
			Disabled	0	Read: Disabled																													
			Enabled	1	Read: Enabled																													
F	RW	SUSPENDED			Write '1' to Disable interrupt for SUSPENDED event See *EVENTS_SUSPENDED*																															
			Clear	1	Disable																													
			Disabled	0	Read: Disabled																													
			Enabled	1	Read: Enabled																													

49.8.4 ERRORSRC

Address offset: 0x4C4

Error source

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**	**Description**																													
A	RW	OVERRUN			Overrun error A new byte was received before previous byte got read by software from the RXD register. (Previous data is lost)																															
			NotPresent	0	Read: no overrun occured																													
			Present	1	Read: overrun occured																													
			Clear	1	Write: clear error on writing '1'																													
B	RW	ANACK			NACK received after sending the address (write '1' to clear)																													
			NotPresent	0	Read: error not present																													
			Present	1	Read: error present																													
			Clear	1	Write: clear error on writing '1'																													
C	RW	DNACK			NACK received after sending a data byte (write '1' to clear)																													
			NotPresent	0	Read: error not present																													

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																C	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
			Present	1																														
			Clear	1																														

49.8.5 ENABLE

Address offset: 0x500

Enable TWI

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	ENABLE																																
			Disabled	0																														
			Enabled	5																														

49.8.6 PSELSCL

Address offset: 0x508

Pin select for SCL

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF			**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	PSELSCL		[0..31]																														
			Disconnected	0xFFFFFFFF																														

49.8.7 PSELSDA

Address offset: 0x50C

Pin select for SDA

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF			**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	RW	PSELSDA		[0..31]																														
			Disconnected	0xFFFFFFFF																														

49.8.8 RXD

Address offset: 0x518

RXD register

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																											A	A	A	A	A	A	A	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																														
A	R	RXD																																

49.8.9 TXD

Address offset: 0x51C

TXD register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	RW	TXD																										TXD register							

49.8.10 FREQUENCY

Address offset: 0x524

TWI frequency

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x04000000				**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	RW	FREQUENCY																										TWI master clock frequency							
			K100	0x01980000																								100 kbps							
			K250	0x04000000																								250 kbps							
			K400	0x06680000																								400 kbps (actual rate 410.256 kbps)							

49.8.11 ADDRESS

Address offset: 0x588

Address used in the TWI transfer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	RW	ADDRESS																										Address used in the TWI transfer							

49.9 Electrical specification

49.9.1 TWI interface electrical specifications

Symbol	Description	Min.	Typ.	Max.	Units
f_{TWI}	Bit rates for TWI[40]	100		400	kbps
$I_{TWI,100kbps}$	Run current for TWI, 100 kbps		50		µA
$I_{TWI,400kbps}$	Run current for TWI, 400 kbps		50		µA
$t_{TWI,START,LP}$	Time from STARTRX/STARTTX task to transmission started, Low power mode		$t_{TWI,START,C}$ + t_{START_HFIN}		µs
$t_{TWI,START,CL}$	Time from STARTRX/STARTTX task to transmission started, Constant latency mode		1.5		µs

49.9.2 Two Wire Interface (TWI) timing specifications

Symbol	Description	Min.	Typ.	Max.	Units
$f_{TWI,SCL,100kbps}$	SCL clock frequency, 100 kbps		100		kHz
$f_{TWI,SCL,250kbps}$	SCL clock frequency, 250 kbps		250		kHz
$f_{TWI,SCL,400kbps}$	SCL clock frequency, 400 kbps		400		kHz
t_{TWI,SU_DAT}	Data setup time before positive edge on SCL – all modes	300			ns
t_{TWI,HD_DAT}	Data hold time after negative edge on SCL – all modes	500			ns
$t_{TWI,HD_STA,100kbps}$	TWI master hold time for START and repeated START condition, 100 kbps	10000			ns

[40] Higher bit rates or stronger pull-ups may require GPIOs to be set as High Drive, see GPIO chapter for more details.

Symbol	Description	Min.	Typ.	Max.	Units
$t_{TWI,HD_STA,250kbps}$	TWI master hold time for START and repeated START condition, 250kbps	4000			ns
$t_{TWI,HD_STA,400kbps}$	TWI master hold time for START and repeated START condition, 400 kbps	2500			ns
$t_{TWI,SU_STO,100kbps}$	TWI master setup time from SCL high to STOP condition, 100 kbps	5000			ns
$t_{TWI,SU_STO,250kbps}$	TWI master setup time from SCL high to STOP condition, 250 kbps	2000			ns
$t_{TWI,SU_STO,400kbps}$	TWI master setup time from SCL high to STOP condition, 400 kbps	1250			ns
$t_{TWI,BUF,100kbps}$	TWI master bus free time between STOP and START conditions, 100 kbps	5800			ns
$t_{TWI,BUF,250kbps}$	TWI master bus free time between STOP and START conditions, 250 kbps	2700			ns
$t_{TWI,BUF,400kbps}$	TWI master bus free time between STOP and START conditions, 400 kbps	2100			ns

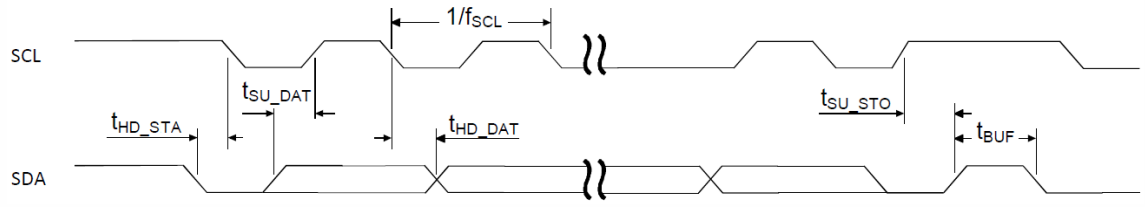


Figure 159: TWI timing diagram, 1 byte transaction

50 UART — Universal asynchronous receiver/ transmitter

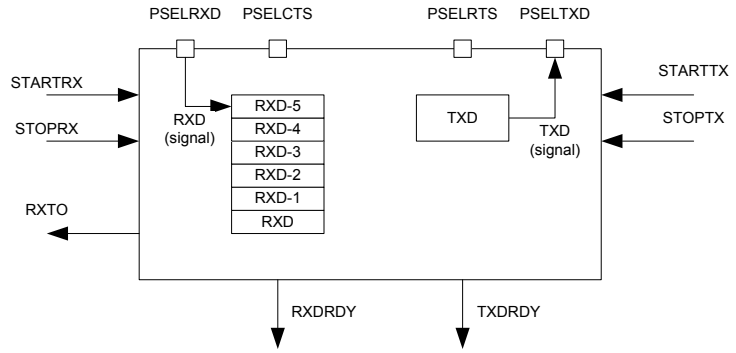


Figure 160: UART configuration

50.1 Functional description

Listed here are the main features of UART.

The UART implements support for the following features:

- Full-duplex operation
- Automatic flow control
- Parity checking and generation for the 9th data bit

As illustrated in *Figure 160: UART configuration* on page 531, the UART uses the TXD and RXD registers directly to transmit and receive data. The UART uses one stop bit.

50.2 Pin configuration

The different signals RXD, CTS (Clear To Send, active low), RTS (Request To Send, active low), and TXD associated with the UART are mapped to physical pins according to the configuration specified in the PSELRXD, PSELCTS, PSELRTS, and PSELTXD registers respectively.

If a value of 0xFFFFFFFF is specified in any of these registers, the associated UART signal will not be connected to any physical pin. The PSELRXD, PSELCTS, PSELRTS, and PSELTXD registers and their configurations are only used as long as the UART is enabled, and retained only for the duration the device is in ON mode. PSELRXD, PSELCTS, PSELRTS and PSELTXD must only be configured when the UART is disabled.

To secure correct signal levels on the pins by the UART when the system is in OFF mode, the pins must be configured in the GPIO peripheral as described in *Pin configuration* on page 531.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behavior.

Table 128: GPIO configuration

UART pin	Direction	Output value
RXD	Input	Not applicable
CTS	Input	Not applicable
RTS	Output	1
TXD	Output	1

50.3 Shared resources

The UART shares registers and other resources with other peripherals that have the same ID as the UART.

Therefore, you must disable all peripherals that have the same ID as the UART before the UART can be configured and used. Disabling a peripheral that has the same ID as the UART will not reset any of the registers that are shared with the UART. It is therefore important to configure all relevant UART registers explicitly to ensure that it operates correctly.

See the Instantiation table in *Instantiation* on page 24 for details on peripherals and their IDs.

50.4 Transmission

A UART transmission sequence is started by triggering the STARTTX task.

Bytes are transmitted by writing to the TXD register. When a byte has been successfully transmitted the UART will generate a TXDRDY event after which a new byte can be written to the TXD register. A UART transmission sequence is stopped immediately by triggering the STOPTX task.

If flow control is enabled a transmission will be automatically suspended when CTS is deactivated and resumed when CTS is activated again, as illustrated in *Figure 161: UART transmission* on page 532.
A byte that is in transmission when CTS is deactivated will be fully transmitted before the transmission is suspended. For more information, see *Suspending the UART* on page 533.

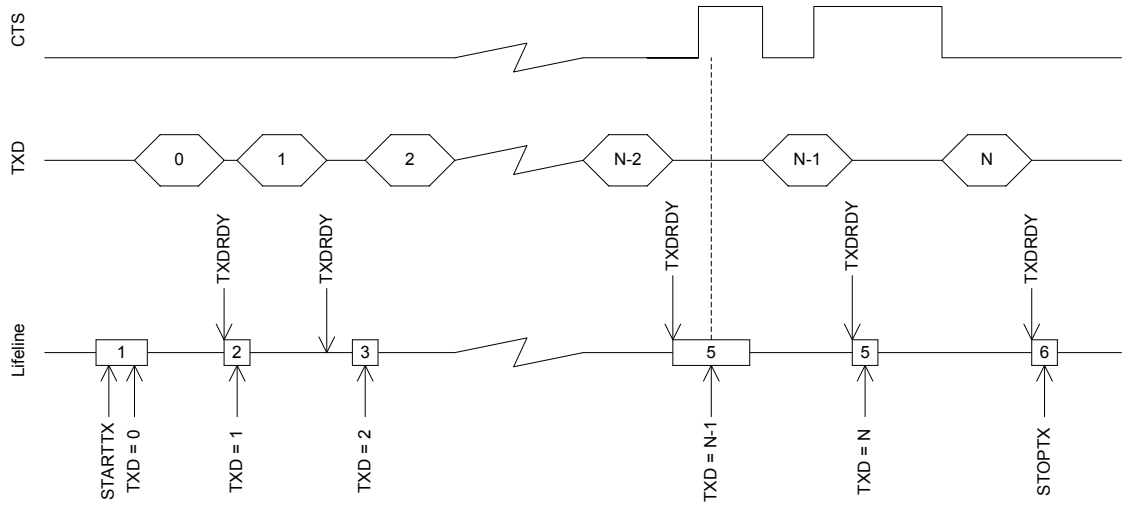


Figure 161: UART transmission

50.5 Reception

A UART reception sequence is started by triggering the STARTRX task.

The UART receiver chain implements a FIFO capable of storing six incoming RXD bytes before data is overwritten. Bytes are extracted from this FIFO by reading the RXD register. When a byte is extracted from the FIFO a new byte pending in the FIFO will be moved to the RXD register. The UART will generate an RXDRDY event every time a new byte is moved to the RXD register.

When flow control is enabled, the UART will deactivate the RTS signal when there is only space for four more bytes in the receiver FIFO. The counterpart transmitter is therefore able to send up to four bytes after the RTS signal is deactivated before data is being overwritten. To prevent overwriting data in the FIFO, the counterpart UART transmitter must therefore make sure to stop transmitting data within four bytes after the RTS line is deactivated.

The RTS signal will first be activated again when the FIFO has been emptied, that is, when all bytes in the FIFO have been read by the CPU, see *Figure 162: UART reception* on page 533.

The RTS signal will also be deactivated when the receiver is stopped through the STOPRX task as illustrated in *Figure 162: UART reception* on page 533. The UART is able to receive four to five additional bytes if they are sent in succession immediately after the RTS signal has been deactivated. This is possible because the UART is, even after the STOPRX task is triggered, able to receive bytes for an extended period of time dependent on the configured baud rate. The UART will generate a receiver timeout event (RXTO) when this period has elapsed.

To prevent loss of incoming data the RXD register must only be read one time following every RXDRDY event.

To secure that the CPU can detect all incoming RXDRDY events through the RXDRDY event register, the RXDRDY event register must be cleared before the RXD register is read. The reason for this is that the UART is allowed to write a new byte to the RXD register, and therefore can also generate a new event, immediately after the RXD register is read (emptied) by the CPU.

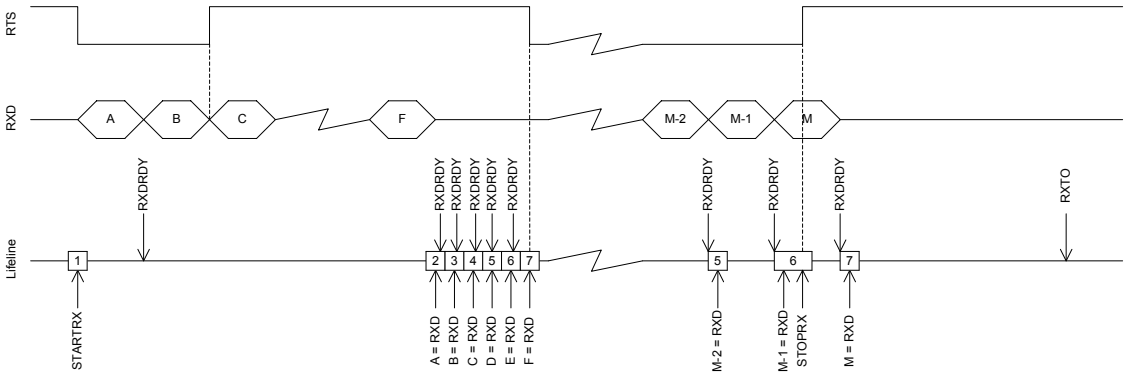


Figure 162: UART reception

As indicated in occurrence 2 in the figure, the RXDRDY event associated with byte B is generated first after byte A has been extracted from RXD.

50.6 Suspending the UART

The UART can be suspended by triggering the SUSPEND task.

SUSPEND will affect both the UART receiver and the UART transmitter, i.e. the transmitter will stop transmitting and the receiver will stop receiving. UART transmission and reception can be resumed, after being suspended, by triggering STARTTX and STARTRX respectively.

Following a SUSPEND task, an ongoing TXD byte transmission will be completed before the UART is suspended.

When the SUSPEND task is triggered, the UART receiver will behave in the same way as it does when the STOPRX task is triggered.

50.7 Error conditions

An ERROR event, in the form of a framing error, will be generated if a valid stop bit is not detected in a frame. Another ERROR event, in the form of a break condition, will be generated if the RXD line is held active low for longer than the length of a data frame. Effectively, a framing error is always generated before a break condition occurs.

50.8 Using the UART without flow control

If flow control is not enabled, the interface will behave as if the CTS and RTS lines are kept active all the time.

50.9 Parity configuration

When parity is enabled, the parity will be generated automatically from the even parity of TXD and RXD for transmission and reception respectively.

50.10 Registers

Table 129: Instances

Base address	Peripheral	Instance	Description	Configuration	
0x40002000	UART	UART0	Universal Asynchronous Receiver/ Transmitter		Deprecated

Table 130: Register Overview

Register	Offset	Description
TASKS_STARTRX	0x000	Start UART receiver
TASKS_STOPRX	0x004	Stop UART receiver
TASKS_STARTTX	0x008	Start UART transmitter
TASKS_STOPTX	0x00C	Stop UART transmitter
TASKS_SUSPEND	0x01C	Suspend UART
EVENTS_CTS	0x100	CTS is activated (set low). Clear To Send.
EVENTS_NCTS	0x104	CTS is deactivated (set high). Not Clear To Send.
EVENTS_RXDRDY	0x108	Data received in RXD
EVENTS_TXDRDY	0x11C	Data sent from TXD
EVENTS_ERROR	0x124	Error detected
EVENTS_RXTO	0x144	Receiver timeout
SHORTS	0x200	Shortcut register
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ERRORSRC	0x480	Error source
ENABLE	0x500	Enable UART
PSELRTS	0x508	Pin select for RTS
PSELTXD	0x50C	Pin select for TXD
PSELCTS	0x510	Pin select for CTS
PSELRXD	0x514	Pin select for RXD
RXD	0x518	RXD register
TXD	0x51C	TXD register
BAUDRATE	0x524	Baud rate
CONFIG	0x56C	Configuration of parity and hardware flow control

50.10.1 SHORTS

Address offset: 0x200

Shortcut register

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												B	A			
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Id	RW	Field	Value Id	Value	Description
A	RW	CTS_STARTRX			Shortcut between CTS event and STARTRX task

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															B	A			
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
			Disabled	0																															
			Enabled	1																															
B	RW	NCTS_STOPRX																																	
			Disabled	0																															
			Enabled	1																															

50.10.2 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		F							E		D					C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	CTS																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	NCTS																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	RXDRDY																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	TXDRDY																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	ERROR																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
F	RW	RXTO																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															

50.10.3 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																		F								E		D					C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	CTS																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	NCTS																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	RXDRDY																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	TXDRDY																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	ERROR																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
F	RW	RXTO																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															

50.10.4 ERRORSRC

Address offset: 0x480

Error source

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	OVERRUN																																	
			NotPresent	0																															
			Present	1																															
B	RW	PARITY																																	
			NotPresent	0																															
			Present	1																															
C	RW	FRAMING																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
				A valid stop bit is not detected on the serial data input after all bits in a character have been received.																																
			NotPresent	0																																
			Present	1																																
D	RW	BREAK																																		
				The serial data input is '0' for longer than the length of a data frame. (The data frame length is 10 bits without parity bit, and 11 bits with parity bit.).																																
			NotPresent	0																																
			Present	1																																

50.10.5 ENABLE

Address offset: 0x500

Enable UART

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																																A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	ENABLE																																	
			Disabled	0																																
			Enabled	4																																

50.10.6 PSELRTS

Address offset: 0x508

Pin select for RTS

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSELRTS		[0..31]																															
			Disconnected	0xFFFFFFFF																															

50.10.7 PSELTXD

Address offset: 0x50C

Pin select for TXD

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																															
A	RW	PSELTXD		[0..31]																															
			Disconnected	0xFFFFFFFF																															

50.10.8 PSELCTS

Address offset: 0x510

Pin select for CTS

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	RW	PSELCTS		[0..31]																								Pin number configuration for UART CTS signal							
			Disconnected	0xFFFFFFFF																								Disconnect							

50.10.9 PSELRXD

Address offset: 0x514

Pin select for RXD

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	RW	PSELRXD		[0..31]																								Pin number configuration for UART RXD signal							
			Disconnected	0xFFFFFFFF																								Disconnect							

50.10.10 RXD

Address offset: 0x518

RXD register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	R	RXD																										RX data received in previous transfers, double buffered							

50.10.11 TXD

Address offset: 0x51C

TXD register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	W	TXD																										TX data to be transferred							

50.10.12 BAUDRATE

Address offset: 0x524

Baud rate

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x04000000				**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**																								**Description**							
A	RW	BAUDRATE																										Baud rate							
			Baud1200	0x0004F000																								1200 baud (actual rate: 1205)							
			Baud2400	0x0009D000																								2400 baud (actual rate: 2396)							
			Baud4800	0x0013B000																								4800 baud (actual rate: 4808)							
			Baud9600	0x00275000																								9600 baud (actual rate: 9598)							
			Baud14400	0x003B0000																								14400 baud (actual rate: 14414)							
			Baud19200	0x004EA000																								19200 baud (actual rate: 19208)							
			Baud28800	0x0075F000																								28800 baud (actual rate: 28829)							
			Baud38400	0x009D5000																								38400 baud (actual rate: 38462)							
			Baud57600	0x00EBF000																								57600 baud (actual rate: 57762)							

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id			A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x04000000			**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**			**Value**																**Description**												
			Baud76800			0x013A9000																76800 baud (actual rate: 76923)												
			Baud115200			0x01D7E000																115200 baud (actual rate: 115942)												
			Baud230400			0x03AFB000																230400 baud (actual rate: 231884)												
			Baud250000			0x04000000																250000 baud												
			Baud460800			0x075F7000																460800 baud (actual rate: 470588)												
			Baud921600			0x0EBED000																921600 baud (actual rate: 941176)												
			Baud1M			0x10000000																1Mega baud												

50.10.13 CONFIG

Address offset: 0x56C

Configuration of parity and hardware flow control

Bit number			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id																															B	B	B	A
Reset 0x00000000			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**			**Value**																**Description**												
A	RW	HWFC																				Hardware flow control												
			Disabled			0																Disabled												
			Enabled			1																Enabled												
B	RW	PARITY																				Parity												
			Excluded			0x0																Exclude parity bit												
			Included			0x7																Include parity bit												

50.11 Electrical specification

50.11.1 UART electrical specification

Symbol	Description	Min.	Typ.	Max.	Units
f_{UART}	Baud rate for UART[41].			1000	kbps
I_{UART1M}	Run current at max baud rate.		55		µA
$I_{UART115k}$	Run current at 115200 bps.		55		µA
$I_{UART1k2}$	Run current at 1200 bps.		55		µA
$I_{UART,IDLE}$	Idle current for UART		1		µA
$t_{UART,CTSH}$	CTS high time	1			µs
$t_{UART,START,LP}$	Time from STARTRX/STARTTX task to transmission started, low power mode		$t_{UART,START}$ + t_{START_HFIN}		µs
$t_{UART,START,CL}$	Time from STARTRX/STARTTX task to transmission started, constant latency mode		1		µs

[41] Higher baud rates may require GPIOs to be set as High Drive, see GPIO chapter for more details.

51 Mechanical specifications

The mechanical specifications for the packages show the dimensions in millimeters.

51.1 QFN48 6 x 6 mm package

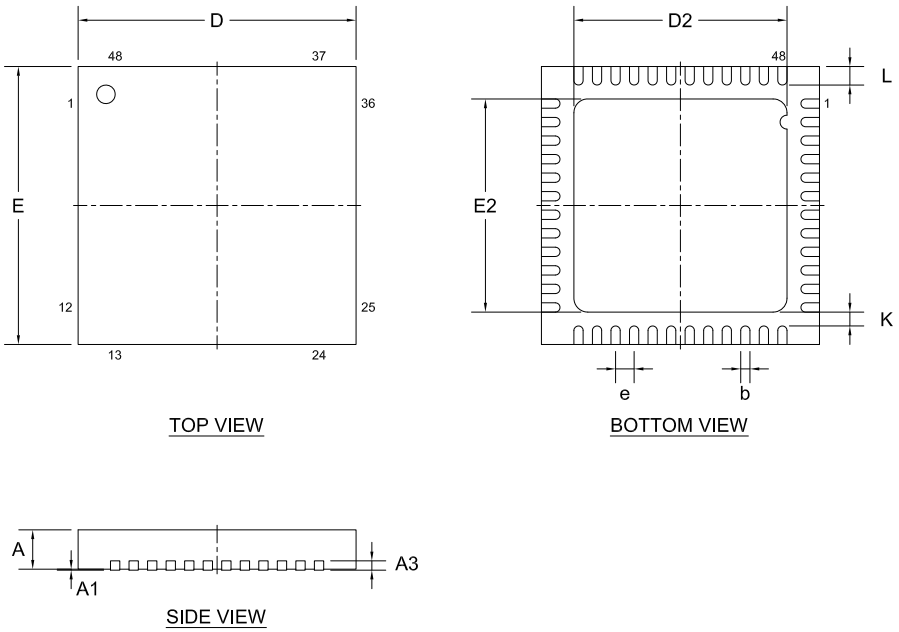


Figure 163: QFN48 6 x 6 mm package

Table 131: QFN48 dimensions in millimeters

Package	A	A1	A3	b	D, E	D2, E2	e	K	L	
	0.80	0.00		0.15		4.50		0.20	0.35	**Min.**
QFN48 (6x6)	0.85	0.02	0.2	0.20	6.0	4.60	0.4		0.40	**Nom.**
	0.90	0.05		0.25		4.70			0.45	**Max.**

51.2 WLCSP package

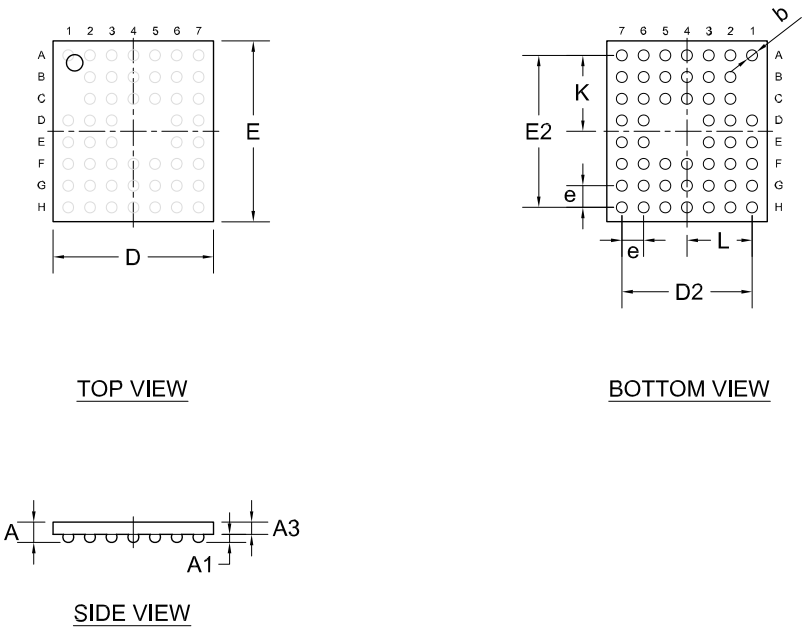


Figure 164: WLCSP package

Table 132: WLCSP packet dimensions in millimeters

Package	A	A1	A3	b	D	E	D2	E2	e	K	L	
	0.351	0.13		0.19								**Min.**
WLCSP (3.0 × 3.2)	0.375	0.15	0.225	0.20	2.956	3.226	2.4	2.8	0.4	1.4	1.2	**Nom.**
	0.399	0.17		0.25								**Max.**

52 Ordering information

This chapter contains information on IC marking, ordering codes, and container sizes.

52.1 IC marking

The nRF52832 IC package is marked like described below.

N	5	2	8	3	2
<P	P>	<V	V>	<H>	<P>
<Y	Y>	<W	W>	<L	L>

Figure 165: Package marking

52.2 Box labels

Here are the box labels used for the nRF52832.

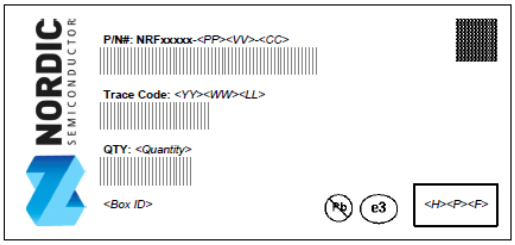


Figure 166: Inner box label

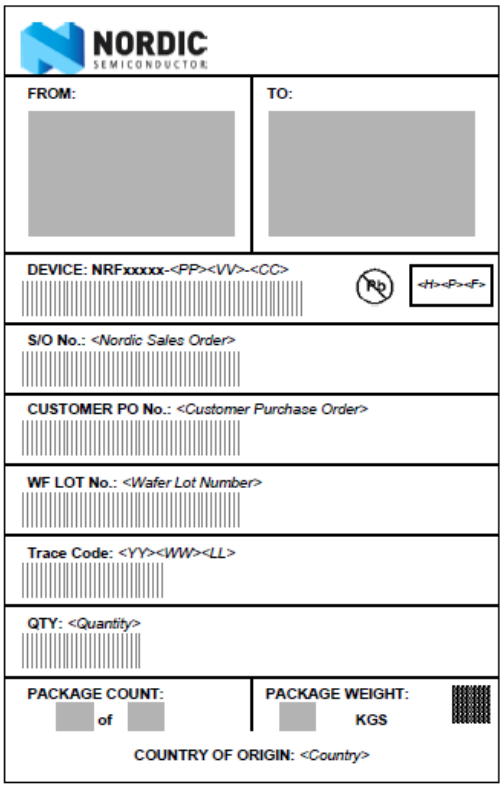


Figure 167: Outer box label

52.3 Order code

Here are the nRF52832 order codes and definitions.

n	R	F	5	2	8	3	2	-	<P	P>	<V	V>	-	<C	C>

Figure 168: Order code

Table 133: Abbreviations

Abbreviation	Definition and implemented codes
N52/nRF52	nRF52 Series product
832	Part code
<PP>	Package variant code
<VV>	Function variant code
<H><P><F>	Build code
	H - Hardware version code
	P - Production configuration code (production site, etc.)
	F - Firmware version code (only visible on shipping container label)
<YY><WW><LL>	Tracking code
	YY - Year code
	WW - Assembly week number
	LL - Wafer lot code
<CC>	Container code

52.4 Code ranges and values

Defined here are the nRF52832 code ranges and values.

Table 134: Package variant codes

<PP>	Package	Size (mm)	Pin/Ball count	Pitch (mm)
QF	QFN	6 x 6	48	0.4
CI	WLCSP	3.0 x 3.2	50	0.4

Table 135: Function variant codes

<VV>	Flash (kB)	RAM (kB)
AA	512	64
AB	256	32

Table 136: Hardware version codes

<H>	Description
[A . . Z]	Hardware version/revision identifier (incremental)

Table 137: Production configuration codes

<P>	Description
[0 . . 9]	Production device identifier (incremental)
[A . . Z]	Engineering device identifier (incremental)

Table 138: Production version codes

<F>	Description
[A . . N, P . . Z]	Version of preprogrammed firmware
[0]	Delivered without preprogrammed firmware

Table 139: Year codes

<YY>	Description
[15 . . 99]	Production year: 2015 to 2099

Table 140: Week codes

<WW>	Description
[1 . . 52]	Week of production

Table 141: Lot codes

<LL>	Description
[AA . . ZZ]	Wafer production lot identifier

Table 142: Container codes

<CC>	Description
R7	7" Reel
R	13" Reel
T	Tray

52.5 Product options

Defined here are the nRF52832 product options.

Table 143: nRF52832 order codes

Order code	Minimum ordering quantity (MOQ)	Comment
nRF52832-QFAA-R7	1000	Availability to be announced.
nRF52832-QFAA-R	3000	
nRF52832-QFAA-T	490	
nRF52832-CIAA-R7	1500	
nRF52832-CIAA-R	7000	
nRF52832-QFAB-R	3000	
nRF52832-QFAB-R7	1000	
nRF52832-QFAB-T	490	

Table 144: Development tools order code

Order code	Description
nRF52-DK	nRF52 Development Kit

53 Reference circuitry

To ensure good RF performance when designing PCBs, it is highly recommended to use the PCB layouts and component values provided by Nordic Semiconductor.

Documentation for the different package reference circuits, including Altium Designer files, PCB layout files, and PCB production files can be downloaded from *Reference layout nRF52 Series*.

53.1 Schematic QFAA and QFAB QFN48 with internal LDO setup

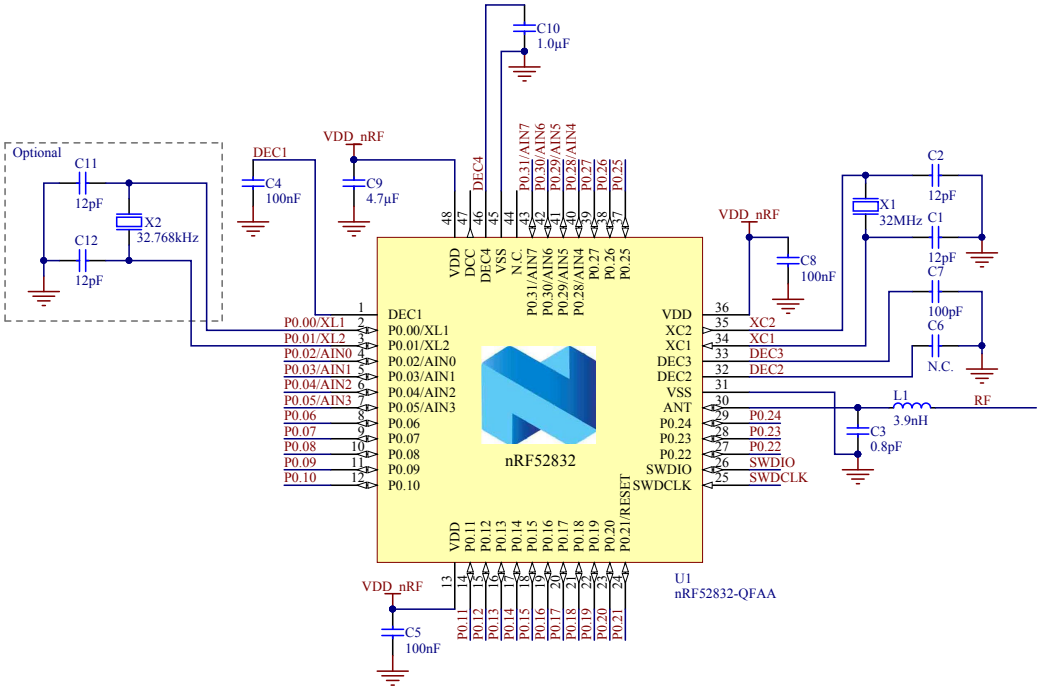


Figure 169: QFAA and QFAB QFN48 with internal LDO setup

For PCB reference layouts, see *Reference layout nRF52 Series*.

Table 145: Bill of material for QFAA and QFAB QFN48 with internal LDO setup

Designator	Value	Description	Footprint
C1, C2, C11, C12	12 pF	Capacitor, NP0, ±2%	0402
C3	0.8 pF	Capacitor, NP0, ±5%	0402
C4, C5, C8	100 nF	Capacitor, X7R, ±10%	0402
C6	N.C.	Not mounted	0402
C7	100 pF	Capacitor, NP0, ±5%	0402
C9	4.7 µF	Capacitor, X5R, ±10%	0603
C10	1.0 µF	Capacitor, X7R, ±10%	0603
L1	3.9 nH	High frequency chip inductor ±5%	0402
U1	nRF52832-QFAA and nRF52832-QFAB	Multi-protocol *Bluetooth* low energy, ANT, and 2.4 GHz proprietary system on chip	QFN-48
X1	32 MHz	XTAL SMD 2016, 32 MHz, CI=8 pF, total tol. ±40 ppm	XTAL_2016
X2	32.768 kHz	XTAL SMD 3215, 32.768 kHz, CI=9 pF, total tol. ±50 ppm	XTAL_3215

53.2 Schematic QFAA and QFAB QFN48 with DC/DC regulator setup

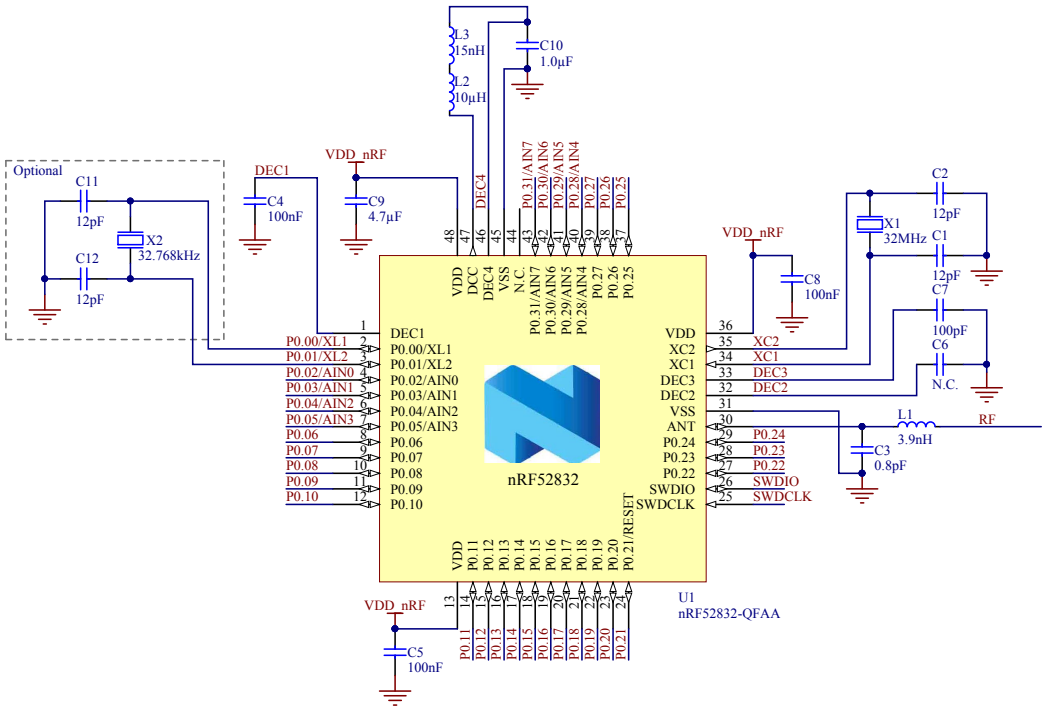


Figure 170: QFAA and QFAB QFN48 with DC/DC regulator setup

For PCB reference layouts, see *Reference layout nRF52 Series*.

Table 146: Bill of material for QFAA and QFAB QFN48 with DC/DC regulator setup

Designator	Value	Description	Footprint
C1, C2, C11, C12	12 pF	Capacitor, NP0, ±2%	0402
C3	0.8 pF	Capacitor, NP0, ±5%	0402
C4, C5, C8	100 nF	Capacitor, X7R, ±10%	0402
C6	N.C.	Not mounted	0402
C7	100 pF	Capacitor, NP0, ±5%	0402
C9	4.7 µF	Capacitor, X5R, ±10%	0603
C10	1.0 µF	Capacitor, X7R, ±10%	0603
L1	3.9 nH	High frequency chip inductor ±5%	0402
L2	10 µH	Chip inductor, IDC,min = 50 mA, ±20%	0603
L3	15 nH	High frequency chip inductor ±10%	0402
U1	nRF52832-QFAA and nRF52832-QFAB	Multi-protocol *Bluetooth* low energy, ANT, and 2.4 GHz proprietary system on chip	QFN-48
X1	32 MHz	XTAL SMD 2016, 32 MHz, CI=8 pF, total tol. ±40 ppm	XTAL_2016
X2	32.768 kHz	XTAL SMD 3215, 32.768 kHz, CI=9 pF, total tol. ±50 ppm	XTAL_3215

53.3 Schematic QFAA and QFAB QFN48 with DC/DC regulator and NFC setup

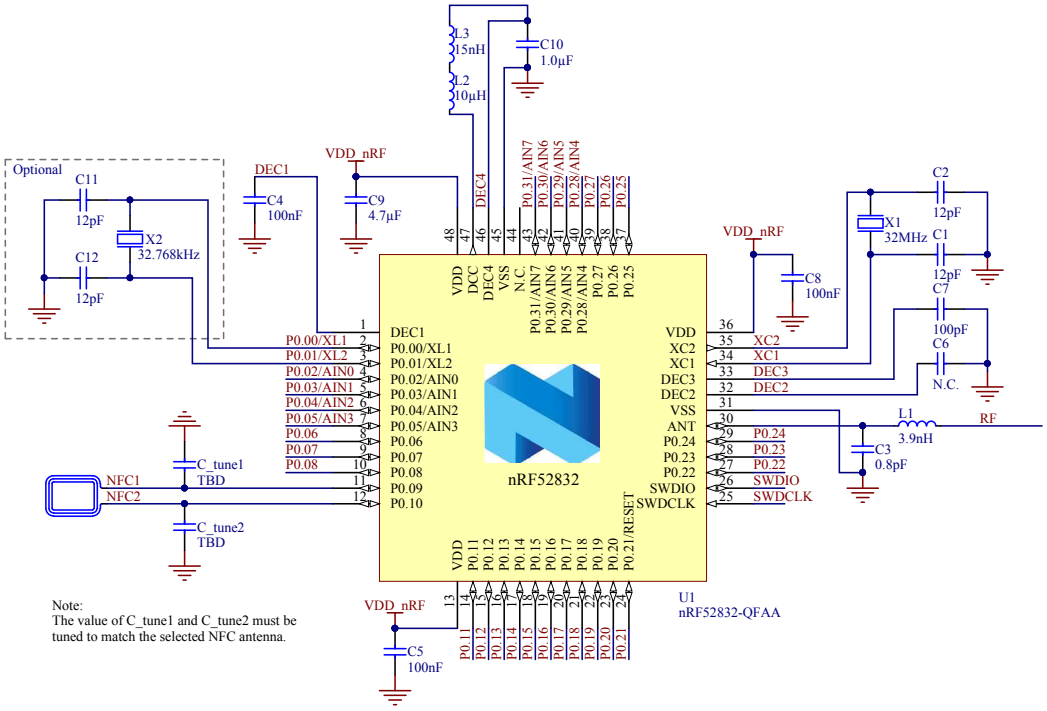


Figure 171: QFAA and QFAB QFN48 with DC/DC regulator and NFC setup

For PCB reference layouts, see *Reference layout nRF52 Series*.

Table 147: Bill of material for QFAA and QFAB QFN48 with DC/DC converter and NFC setup

Designator	Value	Description	Footprint
C1, C2, C11, C12	12 pF	Capacitor, NP0, ±2%	0402
C3	0.8 pF	Capacitor, NP0, ±5%	0402
C4, C5, C8	100 nF	Capacitor, X7R, ±10%	0402
C6	N.C.	Not mounted	0402
C7	100 pF	Capacitor, NP0, ±5%	0402
C9	4.7 µF	Capacitor, X5R, ±10%	0603
C10	1.0 µF	Capacitor, X7R, ±10%	0603
C_{tune1}, C_{tune2}	TBD pF	Capacitor, NP0, ±5%	0402
L1	3.9 nH	High frequency chip inductor ±5%	0402
L2	10 µH	Chip inductor, IDC,min = 50 mA, ±20%	0603
L3	15 nH	High frequency chip inductor ±10%	0402
U1	nRF52832-QFAA and nRF52832-QFAB	Multi-protocol *Bluetooth* low energy, ANT, and 2.4 GHz proprietary system on chip	QFN-48
X1	32 MHz	XTAL SMD 2016, 32 MHz, CI=8 pF, Total Tol: ±40 ppm	XTAL_2016
X2	32.768 kHz	XTAL SMD 3215, 32.768 kHz, CI=9 pF, ±50 ppm	XTAL_3215

53.4 Schematic CIAA WLCSP with internal LDO setup

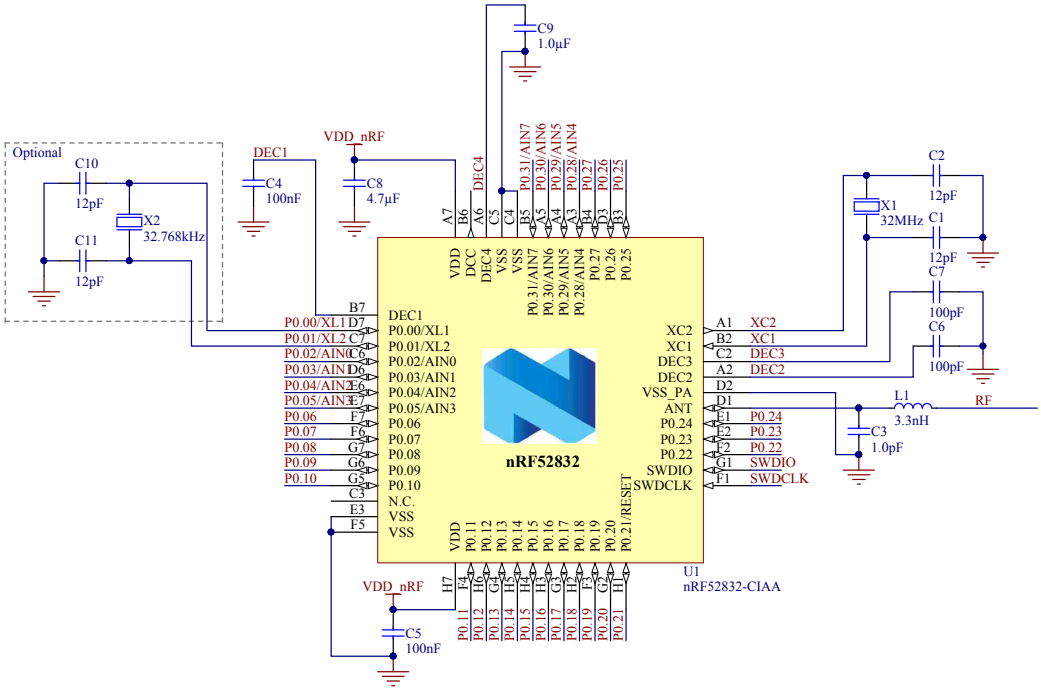


Figure 172: CIAA WLCSP with internal LDO setup

For PCB reference layouts, see *Reference layout nRF52 Series*.

Table 148: Bill of material for CIAA WLCSP with internal LDO setup

Designator	Value	Description	Footprint
C1, C2, C10, C11	12 pF	Capacitor, NP0, ±2%	0201
C3	1.0 pF	Capacitor, NP0, ±5%	0201
C4, C5	100 nF	Capacitor, X7R, ±10%	0201
C6, C7	100 pF	Capacitor, NP0, ±5%	0201
C8	4.7 µF	Capacitor, X5R, ±10%	0603
C9	1.0 µF	Capacitor, X5R, ±5%	0402
L1	3.3 nH	High frequency chip inductor ±5%	0201
U1	nRF52832-CIAA	Multi-protocol *Bluetooth* low energy, ANT, and 2.4 GHz proprietary system on chip	WLCSP_C50
X1	32 MHz	XTAL SMD 2016, 32 MHz, Cl=8 pF, total tol. ±40 ppm	XTAL_2016
X2	32.768 kHz	XTAL SMD 2012, 32.768 kHz, Cl=9 pF, ±50 ppm	XTAL_2012

53.5 Schematic CIAA WLCSP with DC/DC regulator setup

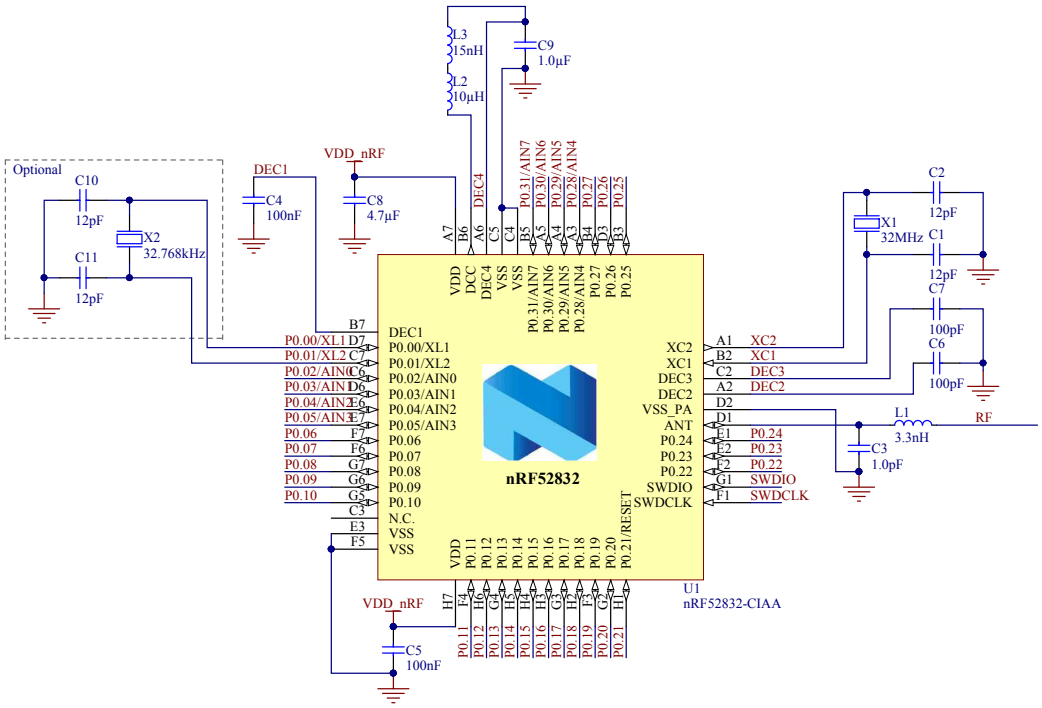


Figure 173: CIAA WLCSP with DC/DC regulator setup

For PCB reference layouts, see *Reference layout nRF52 Series*.

Table 149: Bill of material for CIAA WLCSP with DC/DC regulator setup

Designator	Value	Description	Footprint
C1, C2, C10, C11	12 pF	Capacitor, NP0, ±2%	0201
C3	1.0 pF	Capacitor, NP0, ±5%	0201
C4, C5	100 nF	Capacitor, X7R, ±10%	0201
C6, C7	100 pF	Capacitor, NP0, ±5%	0201
C8	4.7 µF	Capacitor, X5R, ±10%	0603
C9	1.0 µF	Capacitor, X5R, ±5%	0402
L1	3.3 nH	High frequency chip inductor ±5%	0201
L2	10 µH	Chip inductor, IDC,min = 50 mA, ±20%	0603
L3	15 nH	High frequency chip inductor ±10%	0402
U1	nRF52832-CIAA	Multi-protocol *Bluetooth* low energy, ANT, and 2.4 GHz proprietary system on chip	WLCSP_C50
X1	32 MHz	XTAL SMD 2016, 32 MHz, Cl=8 pF, Total Tol: ±40 ppm	XTAL_2016
X2	32.768 kHz	XTAL SMD 2012, 32.768 kHz, Cl=9 pF, ±50 ppm	XTAL_2012

53.6 Schematic CIAA WLCSP with DC/DC regulator and NFC setup

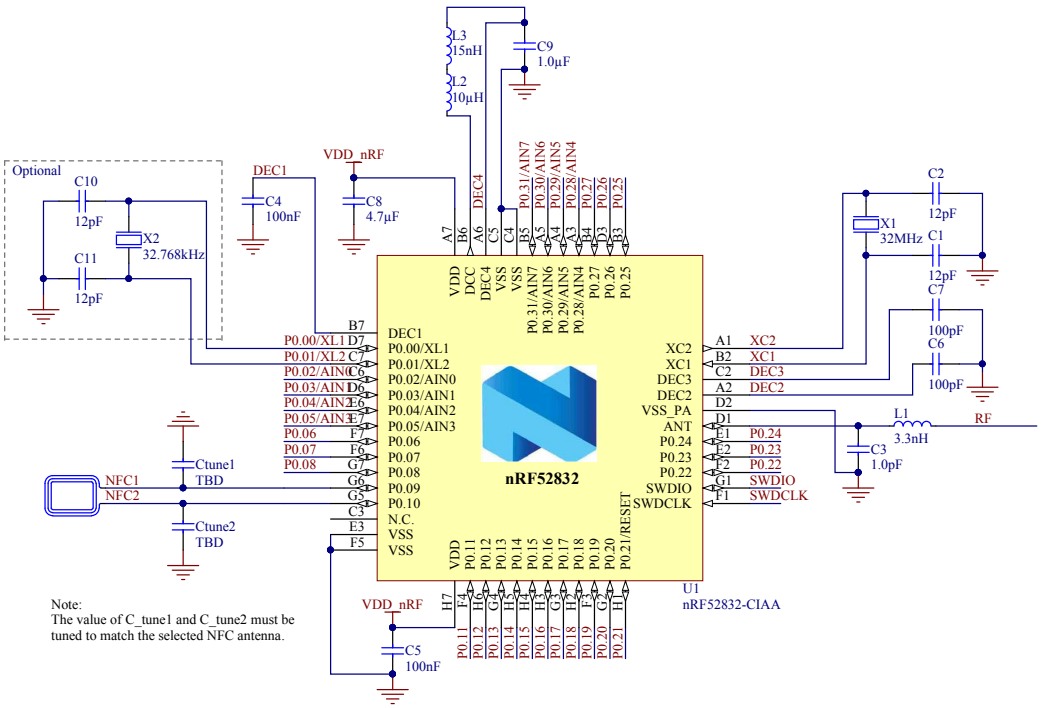


Figure 174: CIAA WLCSP with DC/DC regulator and NFC setup

For PCB reference layouts, see *Reference layout nRF52 Series*.

Table 150: Bill of material for CIAA WLCSP with DC/DC converter and NFC setup

Designator	Value	Description	Footprint
C1, C2, C10, C11	12 pF	Capacitor, NP0, ±2%	0201
C3	1.0 pF	Capacitor, NP0, ±5%	0201
C4, C5	100 nF	Capacitor, X7R, ±10%	0201
C6, C7	100 pF	Capacitor, NP0, ±5%	0201
C8	4.7 µF	Capacitor, X5R, ±10%	0603
C9	1.0 µF	Capacitor, X5R, ±5%	0402
C_{tune1}, C_{tune2}	TBD pF	Capacitor, NP0, ±5%	0201
L1	3.3 nH	High frequency chip inductor ±5%	0201
L2	10 µH	Chip inductor, IDC,min = 50 mA, ±20%	0603
L3	15 nH	High frequency chip inductor ±10%	0402
U1	nRF52832-CIAA	Multi-protocol *Bluetooth* low energy, ANT, and 2.4 GHz proprietary system on chip	WLCSP_C50
X1	32 MHz	XTAL SMD 2016, 32 MHz, Cl=8 pF, Total Tol: ±40 ppm	XTAL_2016
X2	32.768 kHz	XTAL SMD 2012, 32.768 kHz, Cl=9 pF, ±50 ppm	XTAL_2012

53.7 PCB guidelines

A well-designed PCB is necessary to achieve good RF performance. A poor layout can lead to loss in performance or functionality.

A qualified RF layout for the IC and its surrounding components, including matching networks, can be downloaded from *Reference layout nRF52 Series*.

To ensure optimal performance, it is essential that you follow the schematics and layout references closely. Especially in the case of the antenna matching circuitry (components between device pin ANT and the antenna), any changes to the layout can change the behavior, resulting in degradation of RF performance or a need to change component values. All the reference circuits are designed for use with a 50 ohm single end antenna.

A PCB with a minimum of two layers, including a ground plane, is recommended for optimal performance. On PCBs with more than two layers, put a keep-out area on the inner layers directly below the antenna

matching circuitry (components between device pin ANT and the antenna) to reduce the stray capacitances that influence RF performance.

A matching network is needed between the RF pin ANT and the antenna, to match the antenna impedance (normally 50 ohm) to the optimum RF load impedance for the chip. For optimum performance, the impedance for the matching network should be set as described in the recommended package reference circuitry in *Reference circuitry* on page 545 above.

The DC supply voltage should be decoupled as close as possible to the VDD pins with high performance RF capacitors. See the schematics for recommended decoupling capacitor values. The supply voltage for the chip should be filtered and routed separately from the supply voltages of any digital circuitry.

Long power supply lines on the PCB should be avoided. All device grounds, VDD connections, and VDD bypass capacitors must be connected as close as possible to the IC. For a PCB with a topside RF ground plane, the VSS pins should be connected directly to the ground plane. For a PCB with a bottom ground plane, the best technique is to have via holes as close as possible to the VSS pads. A minimum of one via hole should be used for each VSS pin.

Fast switching digital signals should not be routed close to the crystal or the power supply lines. Capacitive loading of fast switching digital output lines should be minimized in order to avoid radio interference.

53.8 PCB layout example

The PCB layout shown below is a reference layout for the QFN package with internal LDO setup.

> **Important:** Pay attention to how the capacitor C3 is grounded. It is not directly connected to the ground plane, but grounded via VSS pin 31. This is done to create additional filtering of harmonic components.

For all available reference layouts, see *Reference layout nRF52 Series*.

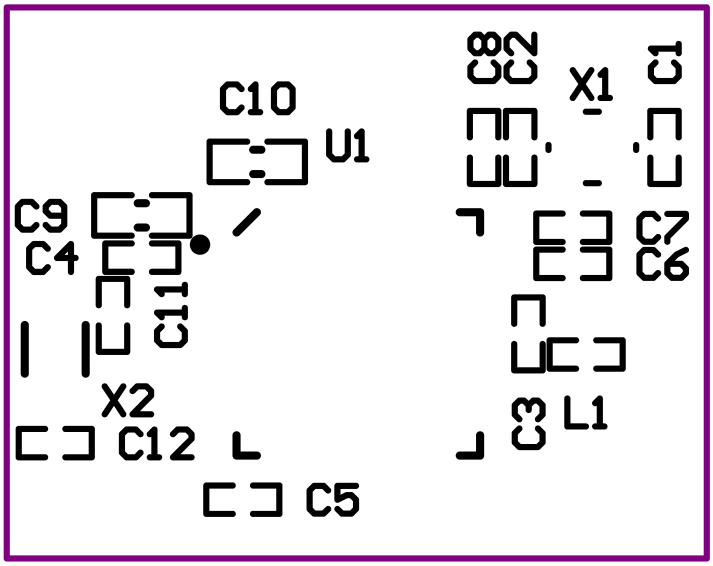


Figure 175: Top silk layer

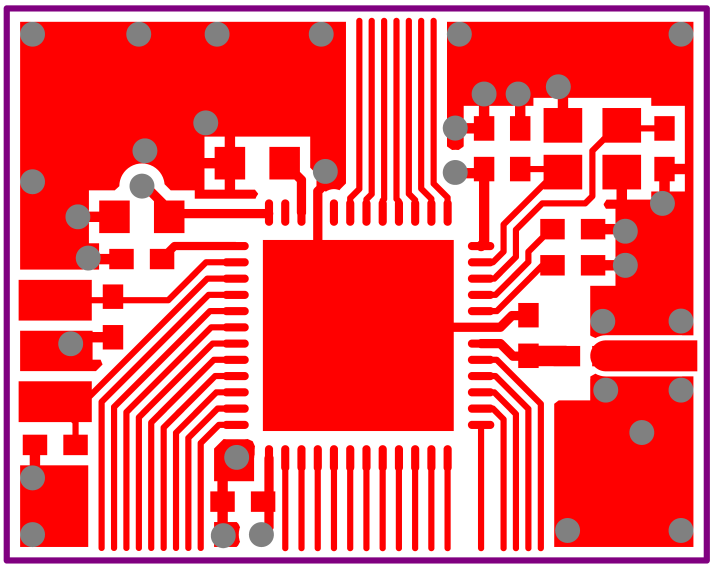


Figure 176: Top layer

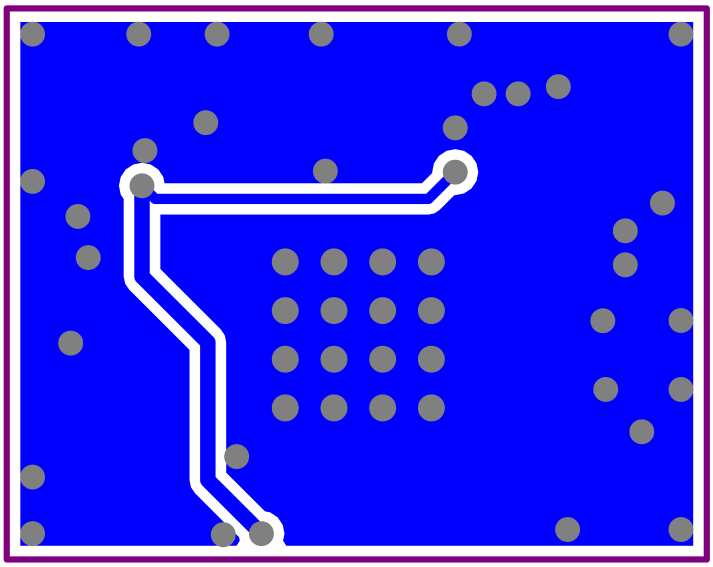


Figure 177: Bottom layer

Important: No components in bottom layer.

54 Liability disclaimer

Nordic Semiconductor ASA reserves the right to make changes without further notice to the product to improve reliability, function or design. Nordic Semiconductor ASA does not assume any liability arising out of the application or use of any product or circuits described herein.

54.1 RoHS and REACH statement

Nordic Semiconductor products meet the requirements of Directive 2011/65/EU of the European Parliament and of the Council on the Restriction of Hazardous Substances (RoHS 2) and the requirements of the REACH regulation (EC 1907/2006) on Registration, Evaluation, Authorization and Restriction of Chemicals.

The SVHC (Substances of Very High Concern) candidate list is continually being updated. Complete hazardous substance reports, material composition reports and latest version of Nordic's REACH statement can be found on our website *www.nordicsemi.com*.

54.2 Life support applications

Nordic Semiconductor products are not designed for use in life support appliances, devices, or systems where malfunction of these products can reasonably be expected to result in personal injury.

Nordic Semiconductor ASA customers using or selling these products for use in such applications do so at their own risk and agree to fully indemnify Nordic Semiconductor ASA for any damages resulting from such improper use or sale.